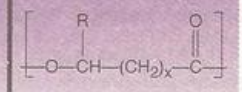
Polysaccharides, Polyhydroxyalkanoates, and Lipids



Microorganisms, in the presence of excess supply of carbon (glucose) are capable of producing storage compounds. These storage compounds are polysaccharides and polyhydroxyalkanoates (from some microorganisms), and lipids (from some yeasts and fungi).

In general, the synthesis of storage compounds by microorganisms is higher on restricting their growth, by limiting the supply of one or more essential nutrients (other than carbon source).

POLYSACCHARIDES

The microorganisms can produce large amounts of polysaccharides in the presence of surplus carbon source. Some of these polysaccharides (e.g. glycogen) serve as storage compounds. The polysaccharides excreted by the cells, referred to as ***exopolysaccharides***, are of commercial importance. The exopolysaccharides may be found in association with the cells or may remain in the medium.

The microbial polysaccharides may be neutral (e.g. dextran, scleroglucan) or acidic (xanthan, gellan) in nature. Acidic polysaccharides possessing ionized groups such as carboxyl, which can function as polyelectrolytes, are commercially more important.

Applications of microbial polysaccharides

Microbial polysaccharides have immense commercial importance. They are employed in the stabilization of foods, and production of several industrial and pharmaceutical compounds. The ***commercial value of a polysaccharide is based on its ability to modify the flow characteristics of solutions*** (technically known as ***rheology***). Polysaccharides can increase the viscosity and, are therefore useful as thickening and gelling agents.

Microbial polysaccharides are of great importance in ***oil industry***. By conventional methods, only 50% of the oil can be extracted. And the rest is either trapped in the rock or too viscous to be pumped out. It is now possible to recover such oils also by a technique called ***microbial enhanced oil recovery (MEOR)***. This can be done by injecting surfactants and viscosity decreasing biological agents (i.e. the microbial polysaccharides e.g. xanthan and emulsan).

Production of microbial polysaccharides

The synthesis of polysaccharides favourably occurs in the excess supply of carbon substrate in the growth medium while limiting nitrogen supply. A carbon/nitrogen ratio of around 10 : 1 is considered to be favourable for optimal polysaccharide synthesis. The production process is

mostly carried out by ***batch culture fermentation***. By manipulating the nutrient supply, differential synthesis of polysaccharides can be achieved. By limiting nitrogen supply in the medium, mostly neutral polysaccharides are produced. When metal ions are limited, acidic polysaccharides are mainly synthesized. Molecular oxygen supply of around 90% saturation is ideal for good growth and polysaccharide synthesis.

Biosynthesis of polysaccharides : Micro-organisms are capable of producing a large number of polysaccharides. The pathways for their biosynthesis are comparable to the processess that occur for the formation bacterial cell wall. It is estimated that there are well over 100 enzymatic reactions, directly or indirectly involved in the synthesis of polysaccharides. ***Starting with glucose, appropriate sugars*** (by transforming glucose to others) ***are incorporated in the formation of polysaccharides***.

Recovery of polysaccharides : As the polysaccharide production increases, there occurs a marked increase in viscosity of the culture broth. The polysaccharides can be precipitated by salts, acids or organic solvents, and recovered by employing appropriate techniques.

Microbial polysaccharides versus plant polysaccharides

There is a lot of competition between microbial and plant polysaccharides for industrial applications. Production of plant polysaccharides is relatively cheap, although it is uncontrolled and occurs for a short period in a year. In contrast, production of microbial polysaccharides is well controlled and can be ***continued throughout the year***. However, fermentation processes for manufacture of cheap (from plant sources) polysaccharides is not advisable.

GENERAL FEATURES OF MICROBIAL POLYSACCHARIDES

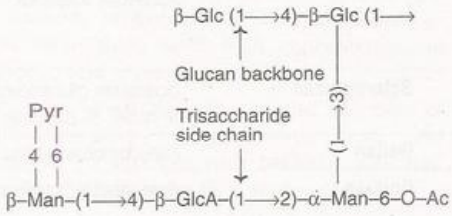
Of the several microbial polysaccharides, around 20 are of industrial importance. As already stated, the ***commercial value*** of a polysaccharide is mostly dependent on its ***rheological properties*** i.e. its ability to modify the flow characteristics of solutions.

A selected list of commercially important polysaccharides, the microorganisms used for their production, and their applications are given in the ***Table 30.1***. Some of the important features of individual microbial polysaccharides are briefly described hereunder.

XANTHAN

Xanthan or more frequently referred to as ***xanthan gum*** was the first polysaccharide available commercially. It is a well studied and most widely used ***hexopolysaccharide***.

Chemistry : Xanthan has a molecular weight in the range of 2–15 $\times$ 10^4 daltons. The basic repeating unit of xanthan is a pentasaccharide containing glucose (Glc), mannose (Man) and glucuronic acid (GlcA) with acetate (Ac) and pyruvate (Pyr) as depicted below.



Basically, xanthan is a branched polymer with β (1→4) linked glucan (glucose polymer) backbone bound to a trisaccharide (Man, GlcA, Man) side chain on alternate glucose residues. The mannose has either acetate or pyruvate groups. The number of acetate or pyruvate molecules in xanthan is variable and is dependent on the bacterial strain used. The culture conditions and the recovery processes also influence the quantities of pyruvate and acetate residues. It is believed that the viscosity of xanthan gum is influenced by the contents of pyruvate and acetate.

Applications : Xanthan gum is used as a ***food additive for the preparation of soft foods*** (ice cream, cheese). It is also used in oil industry for enhancing oil recovery. Further, xanthan is useful for the preparation of tooth pastes and water based paints.

Biosynthesis : For the biosynthesis of xanthan, the monomers are bound to a carrier lipid molecule and then transferred to a growing polymer chain. The activated monosaccharide nucleotides (e.g. uridine diphosphate glucose, UDP-glucose) supply energy for the formation of glycosidic bonds

Table 30.1 Commercially important microbial polysaccharides and their applications

Polysaccharide	*Producing organism(s)*	*Application(s)*
Xanthan	*Xanthomonas campestris*	As a food additive for stabilization, gelling and viscosity control, i.e. for the preparation of soft foods e.g. ice cream, cheese. In oil industry for enhanced oil recovery. In the preparation of toothpastes, and water based paints.
Dextran	*Leuconostoc mesenteroides,* *Acetobacter* sp, *Streptococcus mutans*	Blood plasma expander Used in the preventionn of thrombosis, and in wound dressing (as adsorbent). In the laboratory for chromatographic and other techniques involved in purification. As a foodstuff.
Alginate	*Pseudomonas aeruginosa* *Azobacter vinelandii*	In food industry as thickening and gelling agent. Alginate beads are employed in immobilization of cells and enzymes. Used as ion-exchange agent.
Scleroglucan	*Sclerotium glucanicum* *S. rolfsii, S. delphinii*	Used for stabilizing latex paints, printing inks, and drilling muds.
Gellan	*Pseudomonas elodea*	In food industry as a thickner and solidifying agent.
Polluan	*Aureobasidium pollulans*	Being a biodegradable polysaccharide, it is used in food coating and packaging.
Curdlan	*Alcaligenes faecalis*	As a gelling agent in cooked foods (forms a strong gel above 55°C) Useful for immobilization of enzymes.
Emulsan	*Acinetobacter calcoaceticus* *Arthrobacter* sp	In oil industry for enhanced recovery. For cleaning of oil spills.

between adjacent units. The biosynthesis of other exopolysaccharides is comparable with that of xanthan. Dextran synthesis, however is much simpler as described later.

Production : Xanthan is commercially produced by the Gram-negative bacterium, *Xanthomonas campestris*. The culture medium usually consists of 4-5% carbohydrate (glucose, sucrose, corn starch hydrolysate), 0.05-0.1% nitrogen (ammonium nitrate, urea, yeast extract) and salts. The pH is maintained around 7.0, and the fermentation is carried out by batch culture for 2-3 days. Xanthan in the culture broth is precipitated by isopropanol or methanol. These agents also kill the microorganisms. The precipitated xanthan can be dried and used for commercial purposes.

Genetic engineering of *Xanthomonas campestris* for xanthan production

The wild type *X. campestris* can efficiently utilize glucose, sucrose or starch as a carbon source. They are however, unable to use lactose as a carbon substrate.

Whey is a byproduct obtained in the manufacture of cheese. Disposal of large quantities of whey is a major problem in dairy industry. Fortunately, whey is rich in lactose, besides containing small quantities of proteins, vitamins and minerals. Atempts are made to use whey in fermentation industries.

Genetically engineered *X. campestris* have been developed ***that can utilize lactose (from whey)*** for the production of xanthan. For this purpose,

the *E. coli* ***laczy*** genes (encoding the enzyme β-galactosidase and lactose permease respectively) were cloned under the transcriptional control of *X. campestris* bacteriophage promoter. This construct was first introduced into *E. coli*, and then transferred to *X. campestris*. The genetically engineered strains of *X. campestris* expressed the genes and produced high quantities of the enzymes β-galactosidase and lactose permease. These new strains utilize lactose or whey very efficiently for the industrial production of xanthan. This is a good example of successfully ***converting a waste product (whey) into a commercially important and valuable product (a biopolymer namely xanthan gum)***.

DEXTRAN

Chemically, dextrans are ***glucans*** (polymers of glucose) containing 1 → 6 glycosidic linkages. Some dextrans also have α 1→2, α 1→3 and α 1→4 linkages. The molecular weights of dextrans are in the range of 15,000–500,000.

Applications : Dextrans are used as ***blood plasma expanders, for the prevention of thrombosis*** and in wound dressing. In addition, dextrans are useful in the laboratory analytical techniques for purification of biomolecules.

Production : Dextrans can be produced by a wide range of Gram-positive and Gram-negative bacteria e.g. *Leuconostoc mesenteroides* and *Streptococcus mutans*. In contrast to other exopolysaccharides (which are synthesized within the cells), dextrans are produced ***by extracellular enzyme*** in the medium. The enzyme is ***dextransucrase*** (a transglucosidase) which acts on sucrose and brings about polymerisation of glucose residues, and simultaneously liberates free fructose into the medium.

The commercial production is carried out by using lactic acid bacterium, *L. mesenteroides* by a ***batch fermentation process***. Besides sucrose, the culture medium contains organic nitrogen source and inorganic phosphate. The crude dextran produced is precipitated by alcohol and then subjected to acid hydrolysis. In recent years, the alcohol precipitated polymeric dextran is subjected to enzymatic hydrolysis by using ***exo-*** or ***endo-α dextranases*** to get dextrans of desired molecular weight. The resultant dextrans can be fractionated and dried.

It is also possible to use a cell free system for the production of dextrans. The extracellular enzyme dextrasucrase can transform sucrose into dextran in a cell-free nutrient solution. This reaction is optimum at pH 5.0-5.5 and temperature 25–30°C.

ALGINATE

Alginate is a linear ***polymer composed of mannuronic acid and glucuronic acid*** (both of them being uronic acids) in a proportion ranging from 4 : 1 to 20 : 1. Some of the mannuronic acid residues are acetylated. Alginate is commercially produced by Gram-negative bacteria, *Pseudomonas aeruginosa* and *Azobacter vinelandii*.

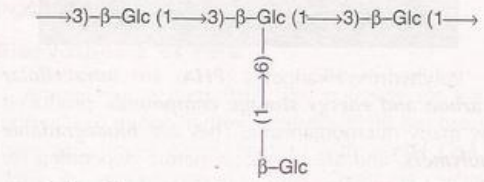
The type of organism used and the culture conditions determine the relative proportion of mannuronic acid and glucuronic acid residues and the degree of acetylation in alginate. Alginates with high contents of mannuronic acid are elastic in nature while those with high concentration of glucuronic acid are strong and brittle.

Algal (seaweed) alginates are also polymers of mannuronic acid and glucuronic acid, and comparable in structure with bacterial alginates. However, algal alginates lack acetylation.

For commercial purposes, seaweed alginates are more commonly used than bacterial alginates. This is mainly because bacterial alginates are relatively unstable and get easily degraded. Alginates are ***useful as thickening agents in food industry***, and for immobilization of cells and enzymes.

SCLEROGLUCAN

Scleroglucan is a ***glucose polymer*** (***glucomer***). It is a neutral polysaccharide with β 1→3 glucan backbone and single glucose (Glc) residue branches (β 1→6 linkage). The branching occurs at a regular sequence at every third glucose unit in the polymer backbone chain.



Scleroglucan is a fungal heoxpolysaccharide. It is commercially produced by *Sclerotium glucanicum, S. rolfsii* and *S. delphinii*. Scleroglucan is ***useful for stabilizing latex paints***, printing inks and drilling muds.

GELLAN

Gellan is ***a linear heteropolysaccharide***. The repeating unit of gellan is composed of two glucose, one glucuronic acid and one rhamnose molecules. Gellan is produced by *Pseudomonas elodea*.

A deacetylated gellan which forms firm and brittle gels under the trade name ***Gelrite*** has been developed by a reputed company in USA (Kalco Inc).

Gellan is used in food industry. Even at a low concentration, it is a thickner.

POLLULAN

Pollulan is an α-glucose polymer (α-glucan) with α 1→4, and a few α, 1→6 glycosidic bonds. Pollulan is produced by using the fungus, *Aureobasidium pollulans*. It is estimated that about 70% of glucose (the substrate) is converted to pollulan during fermentation, although the time taken is rather long (5-7 days).

Pollulan is mainly used in food coating and packaging.

CURDLAN

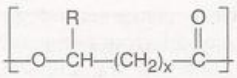
Curdlan is a β-glucose polymer (β-glucan). The glucose residues are held together by β 1→3 glycosidic bonds. The exopolysaccharide curdlan is commercially produced by employing *Alcaligenes faecalis*. Curdlan-like polysaccharides are also produced by other microorganisms such as *Agrobacterium rhizogenes* and *Rhizobium trifolii*.

Curdlan forms strong gels when heated to above 55°C. Therefore, it is used as a gelling agent for cooked foods. In addition, curdlan is also employed for immobilization of enzymes.

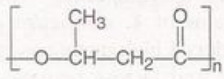
POLYHYDROXYALKANOATES

Polyhydroxyalkanoates (***PHA***) are ***intracellular carbon and energy storage compounds***, produced by many microorganisms. They are ***biodegradable polymers***, and are elastic in nature depending on the polymer composition. PHA are well suited for the synthesis of plastics, the biodegradable packing materials.

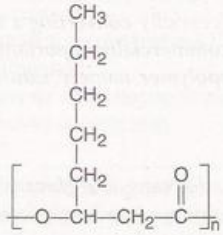
The conventional plastics, made from coal or oil, are not biodegradable. They survive hundreds of years and are a major source of environmental pollution, often resulting in ecological imbalance. A heavy demand for biodegradable plastic materials exists. There are some attempts to chemically synthesize biodegradable polyesters e.g. polylactic acid and polyglycolic acid. The production of polyhydroxyalkanoates by fermentation is preferred for use as ***biodegradable plastics***.



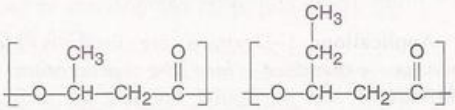
(A) General structure of a monomer in PHA



(B) Polyhydroxybutyrate



(C) PHA with 3-hydroxyoctanoate



(D) 3-Hydroxybutyrate-co-3-hydroxyvalerate

Fig. 30.1 : Structures of polyhydroxyalkanoates (PHA).

PHA-CHEMISTRY AND PROPERTIES

PHA serve as ***lipid reserve materials in bacteria***. Their function in bacteria is comparable to that of fats and oils in yeasts and fungi. The granules of PHA, stored within the cells are clearly visible under electron microscope. Some bacteria may accumulate huge quantities (upto 80% of dry weight) of PHA.

Polyhydroxyalkanoates are linear polyester polymers composed of hydroxyacid monomers (***Fig. 30.1A***). The most commonly found monomers

are 3 hydroxy acids with a carbon length ranging from C_3 to C_{14}.

Homopolymer PHA-polyhydroxybutyrate

The most common PHA is ***polyhydroxybutyrate*** which is frequently referred to as ***PHB***. It is a polyester with 3-hydroxybutyrate as the repeating unit (***Fig. 30.1B***). PHB is the homopolymer PHA. PHB, as such, is rather hard and inflexible.

Being a high molecular weight compound, the accumulation of PHB in huge quantities also does not significantly effect the osmotic pressure within the cell. The reserve carbon compound PHB can be oxidized to carbon dioxide and water, releasing large amount of energy. Bacteria require energy, although they are not growing, to maintain pH gradient and concentration gradient of several compounds. This energy, referred to as ***maintenance energy*** essential for the survival of the cells, is met by the reserve material PHB.

Heteropolymers of PHA

Majority of polyhydroxyalkanoates, with the exception of PHB, contain two or more different monomers and are referred to as heteropolymers. These heteropolymers are usually composed of a random sequence of monomers, and not different monomers, in different chains. Besides 3-hydroxy acids, several other hydroxy acids are found in the structure of PHA. e.g. 4-hydroxybutyrate (4HB). In fact, it has been found that the bacteria are capable of incorporating more than 100 different hydroxylated monomers into PHA. This depends on the organism and the nature of the carbon source supplied during the accumulation of the polymer.

The properties of PHA mostly depend on the nature of the monomers it contains. In general, PHA with longer side chains (e.g. 3-hydroxy-octanoate containing PHA) and heteropolymeric PHA are more flexible and soft.

3-Hydroxybutyrate-co-3-hydroxyvalerate

As already stated, by selecting a specific organism and by manipulating the composition of the medium, the chemical construction of the PHA can be altered. Thus, when the medium contains glucose and propionic acid, the organism *Ralstonia*

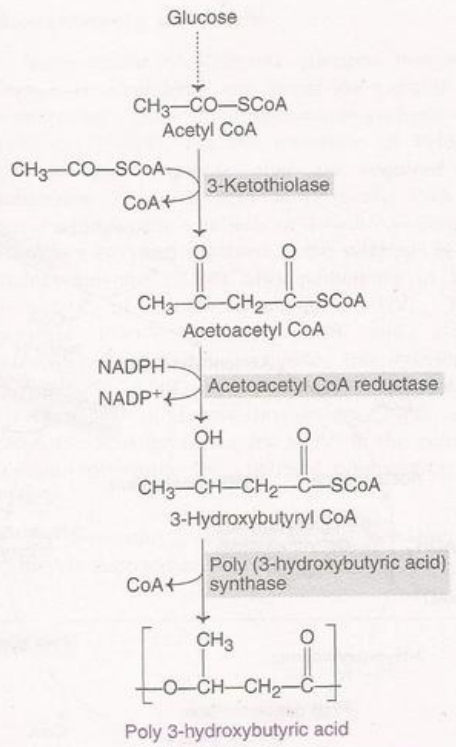


Fig. 30.2 : Biosynthesis of polyhydroxybutyric acid (PHB).

eutrophia produces a copolymer of **3-*hydroxy-butyrate*** and **3-*hydroxyvalerate*** (abbreviated PHB/V). PHB as such is hard and brittle but the presence 3-hydroxyvalerate monomers makes PHB/V flexible and stronger. The properties of PHB/V are similar to those of polypropylene, and therefore it is commercially more useful.

POLYHYDROXYBUTYRATE (PHB)

Biosynthesis of PHB

Among the several PHA, the pathway for the biosynthesis of polyhydroxybutyrate (PHB) has been thoroughly investigated. Starting with ***acetyl CoA***, PHB is synthesized in three reaction steps (***Fig. 30.2***). Acetyl CoA is converted acetoacetyl CoA by the enzyme 3-ketothiolase which is then reduced to 3-hydroxybutyryl CoA by acetoacetyl CoA reductase. The reducing equivalents are

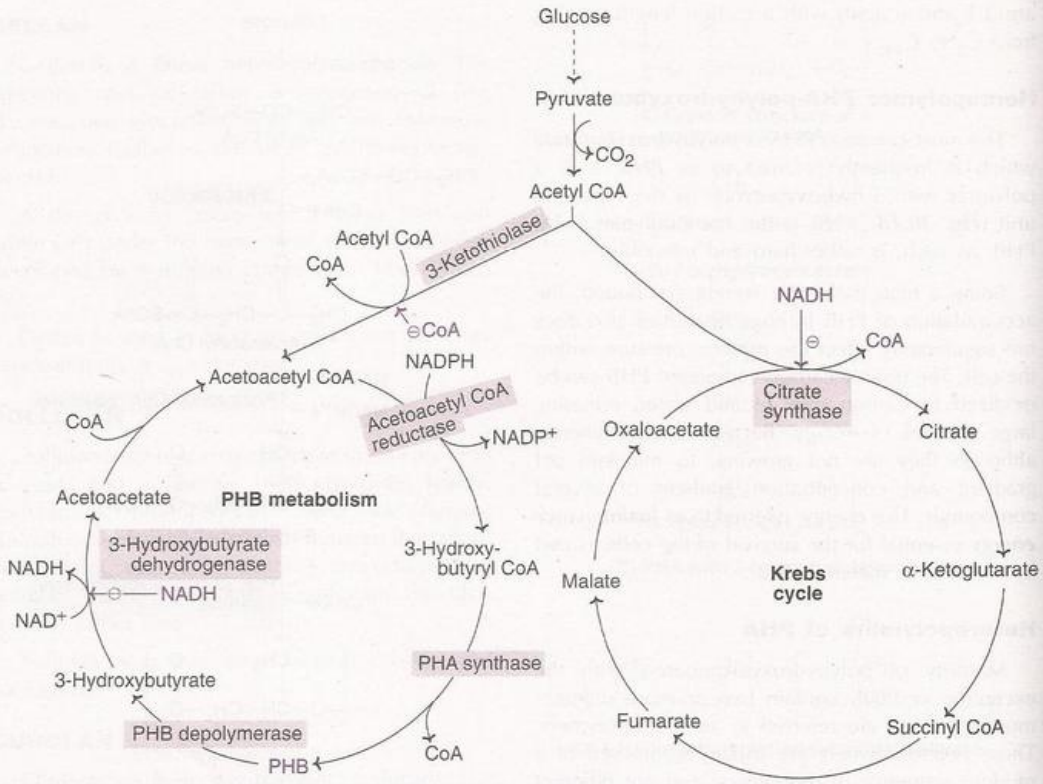


Fig. 30.3 : Regulation of polyhydroxybutyrate (PHB) metabolism.

supplied by NADPH. The enzyme PHA synthase is responsible for the addition of 3-hydroxybutyrate residues to the growing PHB chain.

Regulation of PHB biosynthesis

The outline pathways concerned with the biosynthesis and degradation of PHB in *Ralstonia eutrophia* are depicted in ***Fig. 30.3***. The ***enzymes of PHB biosynthesis*** are ***constitutive*** in nature. Thus the enzyme machinery for PHB synthesis is present all the time in the cell. When the growth of the microorganism is restricted by limiting an essential nutrient, PHB synthesis rapidly occurs.

During the active growth phase, acetyl CoA gets oxidized through the Krebs cycle. Further, free coenzyme A inhibits 3-ketothiolase and therefore very little PHB is synthesized. When the growth ceases or is restricted by limiting a nutrient, the operation of Krebs cycle (i.e. oxidation of acetyl CoA) decreases. This is mainly due to the inhibition of citrate synthase by NADH. A decrease in the concentration of coenzyme A (as citrate is not formed from acetyl CoA, CoA is low) relieves the inhibition of 3-ketothiolase. This enables the diversion of acetyl CoA for the production of PHB. The supply of reducing equivalents (NADPH) required by the enzyme acetoacetyl CoA reductase also regulates the synthesis of PHB.

PHB, the storage energy reserve compound, can be degraded to acetyl CoA and metabolised via Krebs cycle. This occurs when the organism is deprived of energy supplying carbon sources. PHB is degraded by PHB depolymerase to form 3-hydroxybutyrate. NADH is the inhibitor of the subsequent reaction, catalysed by the enzyme 3-hydroxybutyrate dehydrogenase i.e. formation of acetoacetate from 3-hydroxybutyrate. The biosynthesis and breakdown of PHB form a cyclic process, as depicted in ***Fig. 30.3***.

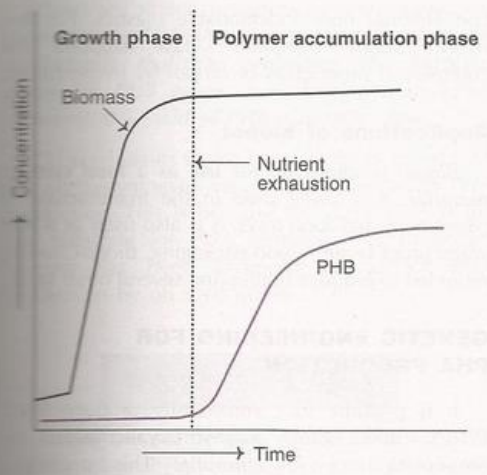


Fig. 30.4 : Production of polyhydroxybutyrate (PHB).

Production of PHB

Polyhydroxybutyrate is mostly manufactured by ***batch culture***. PHB production occurs when there is an excess supply of carbon source, and limitation of some other essential nutrient such as nitrogen, phosphorus or sulfur source. The production/ accumulation of PHB is depicted in ***Fig. 30.4***. There are two distinct phases—a ***growth phase*** and a ***polymer accumulation phase***. As the growth phase ceases, due to nutrient exhaustion, synthesis of polymer (PHB) commences. It is also possible to produce PHB by restricting the oxygen supply to aerobic bacteria.

Applications of PHB

PHB can be implanted in the human body without rejection. This is because PHB does not produce any immune response and thus it is ***biocompatible***. PHB has several medical applications e.g. as durable bone implants, for wound dressings. Attempts to use PHB as degradable sutures and other implants has not met with success due to very slow degradation of PHB.

BIOSYNTHESIS OF OTHER PHA

The biosynthesis of other PHA is quite comparable to that described for PHB. Some important features with regard to the synthesis of two important PHA are described.

Biosynthesis of PHB/V

Some strains of *Ralstonia eutropha* (formerly known as *Alcaligenes eutrophus*) are capable of synthesizing poly (hydroxybutyrate-*co*-hydroxy-valerate) (PHB/V). For the formation of PHB/V, ***glucose and propionic acid are required as substrates***. Propionic acid as propionyl CoA is responsible for the synthesis of 3-hydroxyvalerate. The three enzymes involved in the synthesis of 3-hydroxybutyrate (3-HB) also participate in the formation of 3-hydroxyvalerate (3-HV). The polymer PHB/V contains 3-HB and 3-HV monomers in a random sequence (no individual polymers of 3-HB or 3-HV exist). The relative concentrations of glucose (precursor for 3-HB) and propionic acid (precursor for 3-HV) in the culture medium determine the chemical composition of PHB/V.

The biosynthetic pathway for the formation of PHB/V is depicted in ***Fig. 30.5***.

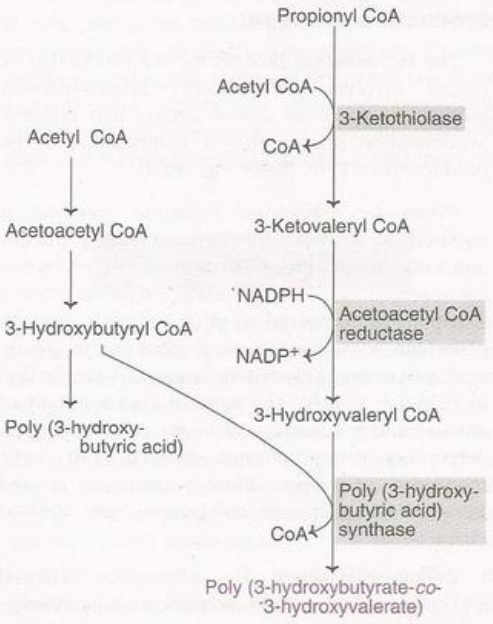


Fig. 30.5 : Biosynthesis of poly (3-hydroxybutyrate-co-3-hydroxyvalerate) (Note : The same 3 enzymes are required for the synthesis of poly 3-hydroxybutyric acid; For details refer Fig. 30.2).

Biosynthesis of PHA with long side chains

It is generally observed that when a PHA producing organism is grown on a carbon source containing ***n*** carbons, 3-hydroxy monomers with ***n-2*** or ***n + 2*** carbons are produced. This suggests the involvement of β-oxidation of fatty acids (which sequentially removes 2-carbon fragment) in the biosynthesis of PHA.

The organism *Pseudomonas oleovorans* can synthesize PHA when organic acids and alkanes are supplied in the medium as carbon sources. For instance, PHA with high content of 3-hydroxyoctanoate can be produced from *n*-octane or *n*-octanoic acid.

BIOPOL-A BIODEGRADABLE PLASTIC

Biopol was the trade name used by ICI plc in UK for the biodegradable plastics composed of PHB and PHB/V. The Gram-negative soil bacterium, ***Ralstonia eutropha*** was employed for the manufacture of biopol.

Production of biopol

The fermentation process for the production of *biopol* involves two distinct phases-biomass production phase or growth phase, and polymer accumulation phase. This is comparable to the production of PHB (Refer ***Fig. 30.4***).

When the bacterium *Ralstonia eutropha* is cultured on a simple medium containing glucose and salts, growth phase is observed. By restricting the essential nutrient phosphate (preferred since it is expensive compared to other nutrients), growth phase can be stopped. Now, glucose and propionic acid are continuously fed for large scale production of PHA i.e. PHB/V. The relative concentration of glucose and propionic acid in the culture medium determines the proportion of 3-HB and 3-HV monomers in *biopol*. *Biopol* containing around 10% 3-HV is found to possess the desired characteristics.

Recovery of biopol **:** The cells can be disrupted and subjected to solubilisation of components other than *biopol*. The polymer can be washed and recovered by centrifugation.

Limitations in the production of biopol : The manufacture of *biopol* by fermentation is very expensive when compared to the production of conventional non-biodegradable plastics. For this reason, many companies have stopped the commercial production of *biopol* by fermentation.

Applications of biopol

Biopol is ***approved for use as a food contact material***. It is being used in the manufacture of paper cups and food trays. It is also used as a thin water proof layer in food packaging. *Biopol*, can be moulded to produce bottles and several other items.

GENETIC ENGINEERING FOR PHA PRODUCTION

It is possible to commercially produce PHB, PHB/V and other polyhydroxyalkanoates by fermentation of *R. eutropha*. This organism, however, grows very slowly and utilizes only a limited number of carbon sources for growth. For these reasons, the production of PHA by *R. eutropha* is expensive.

Fortunately, the genes responsible for the synthesis of important types of PHA by *R. eutropha* have been characterized and cloned. These ***genes have been transferred to E. coli*** (a bacterium that does not normally synthesize PHA). The resultant transformants, *E. coli* cells, grow rapidly and produced large quantities of PHB. But the major limitation is that about half of the *E. coli* cells lose the plasmid constructs in about 50 generations. This poses a big problem, particularly for large scale continuous cultures. In recent years, some workers have tried to genetically manipulate and produce stable plasmids with some success.

Recombinant strains of *E. coli* have been used for the production of PHB, PHB/V and other types of PHA. There are some other ***advantages of producing PHA by E. coli*** instead of *R. eutropha*. The polymers are produced in a crystalline state by *E. coli* (they are in amorphous state in *R. eutropha*), and their ***extraction and recovery are much easier***.

PHA production by plants

The major limitation of producing PHA by fermentation is the cost factor. Approximately, it is ten times more expensive to produce PHA by bacteria compared to the manufacture of petrochemical plastics.

Plants are attractive sources for a less expensive production of PHA. A ***transgenic plant of Arabidopsis thaliana, containing the bacterial genes for PHB genes, was developed*** in 1992. However, the ***yield of PHB*** was rather ***low***.

Oil seed plants which have plenty of acetyl CoA (for oil biosynthesis) are attractive sources for PHA production. This is because the acetyl CoA is the key substrate for PHA synthesis. No significant success has been achieved so far for PHA production by oil seed plants.

MICROBIAL LIPIDS

Among the microorganisms, the eukaryotes can accumulate triacylglycerols. In contrast, prokaryotes predominantly accumulate polyhydroxyalkanoates (as described above).

Microorganisms as oil factories

In general, commercial oils (and fats) used in foodstuffs and cosmotics are obtained from plants and animals. Thus, the traditional sources of oils are plants and animals. In recent years, microbial production of oils is gaining importance. There is a trend to use ***microorganisms as oil factories*** for the following reasons.

1. Plant oils contain residues of pesticides and herbicides that are used in the cultivation of oil seed plants.

2. Fats and fatty acids from animal sources are unacceptable by some sections of the society on moral and/or religious grounds. Further, animal fats have the risk of transfer of disease-causing agents (e.g. prions) to humans.

3. The eukaryotic microorganisms are capable of producing a wide range of polyunsaturated fatty acids. Certain microorganisms can synthesize huge quantities of a single fatty acid which is quite advantageous for large scale production and easy recovery of a particular fatty acid.

4. Microbially produced oils which are devoid of harmful contaminants will be acceptable to the public on moral and ethical grounds also.

PRODUCTION OF SINGLE-CELL OILS

Oils rich in a particular fatty acid are regarded as single-cell oils. So far, researchers have been successful in producing single-cell oils rich in three polyunsaturated fatty acids. These developments have occurred due to the interest taken by multinational pharmaceutical companies.

Linolenic acid : Single-cell oil rich in linolenic acid was the first oil produced by the fungus *Mucor circinelloides*. Its production however, was later discontinued due to cost factor.

Arachidonic acid : The soil fungus, *Mortierella alpina* has been used for the production of oils rich in arachidonic acid.

Docosahexaenoic acid : The marine alga, *Crypthecodinium cohnii* has been employed for commercial production of decosahexaenoic acid (DHA). Microbial DHA is used for inclusion, in baby foods.

Limitations in the production of single-cell oils

The fermentation technology used for the production of single-cell oils is very expensive.

It is visualised that the day may not be far off to transfer the genes producing single cell oils from microorganisms to plants. These transgenic plants can produce large quantities of single-cell oils more cheaply. But these oils will be genetically engineered, and therefore, it is doubtful whether they will have wide-spread public acceptance.

MICROBIAL RUBBER

Natural rubber is produced by a large number of plants. Chemically, it is a ***biopolymer composed of cis-1, 4-polyisoprene***. The biosynthetic pathway for rubber has been worked out. About 17 enzymatic reactions occur for the formation of rubber. The substrates are simple sugars. The final step involves the polymerisation of isopentenyl pyrophosphate on to allylic pyrophosphate, catalysed by the enzyme rubber polymerase.

Attempts are on to ***synthesize rubber in genetically engineered microorgnaisms***. For this purpose, the rubber producing plant *Hevea brasiliensis* was selected and a cDNA library by using mRNA was contructed. It may be possible in the near future, to synthesize rubber from genetically manipulated microorganisms.

MICROBIAL ADHESIVE BIOPOLYMER

The biopolymer, adhesive protein produced by the blue mussel *Mytilus edulis* is very strong and water proof. It is useful for the mussel to attach itself to different surfaces. As the adhesive is secreted, it gets randomly cross-linked, hence it cannot be sequenced. Fortunately, the immediate precursor for the formation of adhesive protein has been isolated and characterized. It is a 130-kDa protein rich in certain amino acids like serine threonine, tyrosine, lysine and proline. Most of the proline and tyrosine residues are hydroxylated. Further, biochemical analysis revealed that this precursor protein contains repeating units of a decapeptide.

The cDNA for the precursor adhesive protein has been constructed by using mRNA. This cDNA was introduced into yeast cells and ***adhesive protein molecules were synthesized by the recombinant yeast.***

The gene encoding the repeating unit of decapeptide has been identified and chemically synthesized. This gene was incorported into *E. coli* cells for synthesizing adhesive protein. But there is a limitation-*E. coli* cannot carry out post-translational reactions i.e. hydroxylation of the adhesive protein. This deficiency can be overcome by carrying out *in vitro* hydroxylation. It is possible to hydroxylate amino acids by employing bacterial enzymes (e.g. tyrosinase) in the presence of vitamin C.

It is expected that adhesive biopolymers will be soon produced for widespread use in dentistry and medicine.

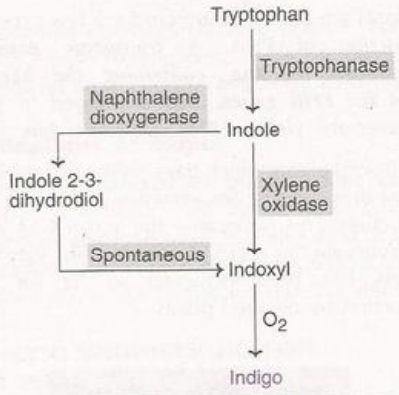


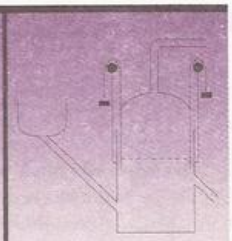
Fig. 30.6 : *Biosynthesis of indigo from tryptophan by genetically engineered* E. coli.

MICROBIAL PRODUCTION OF INDIGO

Indigo is a ***blue pigment***, extensively used ***to dye cotton and wool***. There is a heavy demand for indigo worldover, and it is at present the largest selling dye in the world. Originally, indigo was isolated from plants. Now, it is being synthesized by using the hazardous compounds such a aniline, formaldehyde and cyanide.

In the recent years, it has been possible to synthesize indigo from tryptophan, by genetically engineered *E. coli*. The biosynthetic pathway for the production of indigo is given in ***Fig. 30.6***. However, a large scale commercial manufacture of indigo by genetically engineered *E. coli* is yet to commence.

Biomass and Bioenergy



The capacity to do work is referred to as energy. Energy may be considered as a form of matter which is interconvertible. The modern man is mostly dependent on three sources for his energy needs—***coal***, ***natural gas and oils***, collectively referred to as ***fossil fuels*** or ***fossil energy sources***. The fear of depletion of global fossil fuels has forced man to look for suitable alternative energy sources such as solar, hydro, tidal and wind power, and more recently nuclear energy. In addition to these, advances in biotechnology have helped to fruitfully utilize the ***energy from biological systems***.

BIOMASS

Biomass is the ***total cellular and organic mass, produced by the living organisms***. It is the ***primary product of photosynthesis*** and is a ***good source of energy*** i.e. bioenergy. Broadly speaking, biomass represents all forms of matter derived from biological activities. These include plants and agricultural products, microorganisms, animal wastes and manure. The term biomass is also used to collectively describe the waste materials produced in food and agricultural industries. Besides being a good source of energy, biomass is important for the production of several commercially important products. Thus, biomass is appropriately regarded as a renewable source of energy which can be directly converted to energy or energy carrier compounds by various means.

In most developed countries, biomass is utilized for the production of industrial and commercial products (ethanol, oils, methane, single-cell protein). In contrast, in the developing countries (India, Latin America, Africa), a major part of the biomass is directly used as a source of energy (as firewood).

It is estimated that the annual net yield of plant biomass is around 175 billion tons of dry matter (125 billion tons on land and 50 billion tons on oceans). Forests significantly contribute to the production of land based biomass (around 45%). Agricultural crops on the cultivated land account for about 6% of the plant biomass. The agricultural biomass products (cereals, pulses, oils, animal feed etc.) adequately meet requirements of foods for humans and animals, besides other basic needs (fuels, chemicals etc.). From the chemical point of view, about ***50% of the land produced biomass*** is in the complex form of ***lignocellulose***.

Fossil fuels-derivatives of biomass

The modern society is dependent of on the non-renewable sources of energy namely oil, gas and coal. These fossil fuels are actually derivatives of ancient biomass. It took millions of years for the fossil fuels to be deposited beneath the earth and oceans. However, in just within a century of exploration, the major fuel reserves (particularly gas and oil) are depleted, and at the present rate, they are not likely to last long. As such, there exists

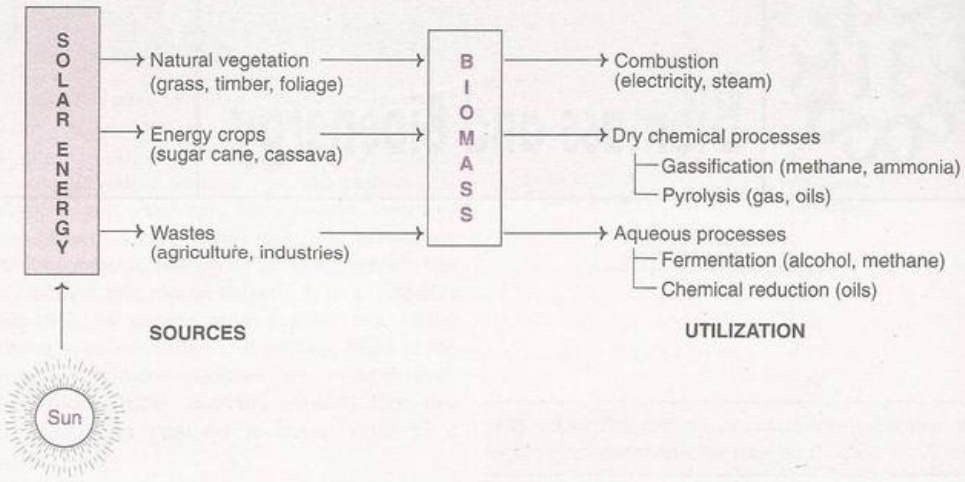


Fig. 31.1 : An overview of the sources and utilization of biomass.

an ***energy crisis*** throughout the world. Consequently, researchers continue their search for alternate and renewable sources of energy.

Photosynthesis-the ultimate source of energy

Photosynthetic organisms are the ultimate sources for trapping the solar energy. In the presence of photosynthetic pigment chlorophyll, carbon dioxide is converted into complex carbohydrates with the evolution of oxygen.

$$CO_2 + H_2O \xrightarrow[\text{Light}]{\text{Chlorophyll}} (CH_2O) + O_2$$

In the reactions that follow later, solar energy is trapped into molecules such as fat and proteins, besides other complex carbohydrates (cellulose, hemicellulose, lignin).

Photosynthetic organisms are the true solar energy converters. It is estimated that at present more than 10 times more energy is generated by photosynthesis annually than consumed by the world's population. Unfortunately, the role of photosynthesis is not well recognized to solve the present day problem of energy crisis. This, despite the fact that it is only the photosynthetically produced biomass that is available today in the form of fossil fuels.

The biomass produced by photosynthesis can be appropriately utilized for the production of fuels (alcohol, methane) and various other commercial products.

Chemical nature of biomass

The plant biomass is mainly composed of ***cellulose, hemicellulose, lignin, starch,*** proteins, water soluble (sugars, amino acids) and fat soluble (oils, pigments) compounds. In fact, majority of these constituents are present in the plant cell walls. There is a wide variation in the chemical composition of the biomass, depending on the source. For instance, the biomass obtained from sugar cane and beet sugar is rich in sugars while the biomass of potato and topioca is rich in starch. On the other hand, cotton has high content of cellulose.

The chemical nature of biomass derived from industrial and municipal wastes is highly variable which mostly depends on the sources that contribute to the biomass.

SOURCES AND UTILIZATION OF BIOMASS

The major sources of biomass are natural vegetation, energy crops, and agricultural, industrial and urban organic wastes (***Fig. 31.1***). Their production in turn is dependent on the solar energy.

The natural vegetation (growing natural forests and aquatic weeds) significantly contributes to biomass. Wood-rich plants are grown in many countries (particularly developing countries) to generate fire for cooking and other purposes. In recent years, well planned and organized plantations are carried out in some countries to produce biomass to meet energy demands. For instance, sugar cane and cassava plantations in Brazil and Australia are used for ethanol production. Plants rich in lignocellulose are grown in America and Sweden which are useful for the production of liquid fuels (ethanol, methanol).

Agricultural, industrial and municipal wastes were earlier considered as useless and discarded. But in the recent past, many countries have developed methods for converting these ***wastes into biofuels and commercially important products***. The successfully used agricultural wastes include straw, bagasse, bran, cotton wastes. Among the industrial wastes, molasses, whey, distillery wastes and sewage are the important ones.

As already stated, the biomass is utilized for the production of biofuels and various other compounds. The technique mainly depends on the chemical nature and moisture content of biomass.

Combustion : Low moisture containing biomass (wood, straw, bran) can be ***directly burnt*** by a process referred to as ***combustion*** to generate electricity.

Dry chemical processes : The biomass with little moisture content can be subjected to various dry chemical processes-pyrolysis, gassification to produce methanol, oil and ammonia biomass.

Aqueous processes : The biomass with high water content is used in aqueous processes such as fermentation to produce ethanol, oils and methane.

An overview of the sources and utilization of biomass is depicted in ***Fig. 31.1***. The biotechnological strategies for the production of biofuels (alcohol, hydrogen) from biomass are briefly described in the following pages.

PRODUCTION OF ALCOHOL FROM BIOMASS

Alcohol, chemically ethanol (C_2H_5OH) has been produced by fermentation for thousands of years. Although the developed countries these days prefer to manufacture ethanol by chemical means, the developing countries continue to produce it by microbial fermentation. ***Alcohol is the liquid fuel*** which is mostly ***produced from the biomass***. The ***raw materials*** (biomass) used for alcohol production include ***starchy materials*** (wheat, rice, maize, potato) and ***cellulosic materials*** (wood, agricultural wastes).

The reader must refer the Chapter 23 for full details on alcohol fermentation.

PRODUCTION OF BIOGAS FROM BIOMASS

Biogas is a mixture of gases composed of ***methane*** (50–80%), ***carbon dioxide*** (15–40%), ***nitrogen*** (4%) and hydrogen sulfide (1%) with traces of hydrogen, oxygen and carbon monoxide. There are many other common names for biogas—***gobar gas***, ***sewage gas***, ***klar gas*** and ***sludge gas***. Biogas is a ***gaseous fuel*** and serves as a good source of energy for various purposes.

1. It can be ***used for cooking purposes*** (on combustion).

2. Gobar gas can generate electricity.

3. It can be purified to yield good grade methane. Methane gas is extensively used as a fuel for domestic and industrial purposes. It is employed for the generation of electrical, mechanical and heat energy.

It is estimated that under ideal conditions, 10 kg of biomass can produce 3 m^3 of biogas. This biogas can provide 3 hour cooking, 3 hour lighting or 24 hour refrigeration (with suitable equipment). The calorific value of biogas (with 80% methane) is around 8,500 cal/m^3.

Biogas production in different countries

Biogas production significantly contributes to world's energy source. China has the largest number of biogas or gobar gas systems, with an estimated 7 million units. Government of China encourages and offers subsidies for construction of biogas units. As a result, the cost of a biogas plant is cheaper than a bicycle in China!

Governments in many developing countries also encourage installation of biogas plants. In India, Government provides 25% subsidy, besides encouraging banks to offer loans for construction of biogas plants. They are becoming popular in rural areas. Pakistan also has a good number of gobar gas plants.

Substrates for biogas

The usual substrates for biogas production are the waste products of animal husbandry, industries, agriculture and municipalities. In India and other developing countries, ***cattle dung (gobar) is most commonly used***. The major raw material for biogas in China is pig dung.

The concentration of organic dry matter or total solids (TS) is useful for grading the industrial, agricultural and municipal wastes. Thus, low grade (< 1% TS), medium grade (1-5% TS), high grade (5–20% TS) and solid (20–40% TS) wastes are available. Solid or high grade wastes are preferred as substrates for biogas production.

In general, most of the substrates used in biogas plants contain adequate quantities of almost all the essential nutrients required for microbial growth. If necessary, nitrogen, phosphorus and trace elements are added. A carbon nitrogen ratio (C/N) less than 40 : 1 is preferred for optimal biogas formation.

Water hyacinth (*Eichornia crasipes*) an aquatic weed with huge biomass is a good source (raw material) for methane production. With high C/N ratio and low lignin content, the yield of methane is high. Another aquatic weed ***Azolla*** is equally important for methane generation.

Microbial production of methane (biogas)

Methane is the most abundant constituent of biogas. It can also be directly used for various domestic and industrial purposes. The microbial generation of methane, appropriately referred to as ***methanogenesis*** from biomass occurs in four phases (***Fig. 31.2***).

1. **Hydrolytic phase** : Certain facultative anaerobic bacteria hydrolyse the complex organic materials of the biomass (cellulose, starch, proteins, lipids) to low molecular weight soluble products and some organic acids.

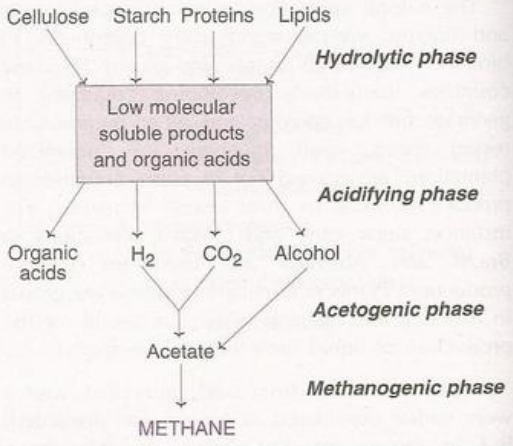


Fig. 31.2 : Microbial production of methane (methanogenesis from biomass).

2. **Acidifying phase** : This phase is characterized by more formation of organic acids, besides H_2, CO_2, and alcohol.

3. **Acetogenic phase** : Acetogenic bacteria convert alcohol into acetate. These bacteria also generate acetate from H_2 and CO_2.

4. **Methanogenic phase** : This is the actual phase of methane gas formation. The methanogenic bacteria (e.g. *Methanobacterium omelianskii, M. formicicum, M. bryantii, Methanosarcina barkeri*) convert acetate, and CO_2 and H_2 into methane.

$$CH_3COOH \longrightarrow CH_4 + CO_2$$

$$4H_2 + CO_2 \longrightarrow CH_4 + 2H_2O$$

Some other substrates like formate and methanol can also be converted to methane.

$$4HCOOH \longrightarrow CH_4 + 3CO_2 + 2H_2O$$

$$4CH_3OH \longrightarrow 3CH_4 + CO_2 + 2H_2O$$

The overall reaction of methane formation from glucose as the starting material may be represented as follows.

$$C_6H_{12}O_6 \longrightarrow 3CH_4 + 3CO_2$$

The complex polysaccharides particularly lignin and cellulose due to their inefficient conversion, limit methane production. In the normal process of methanogenesis, approximately 50% of the complex polysaccharides contribute to methanogenesis.

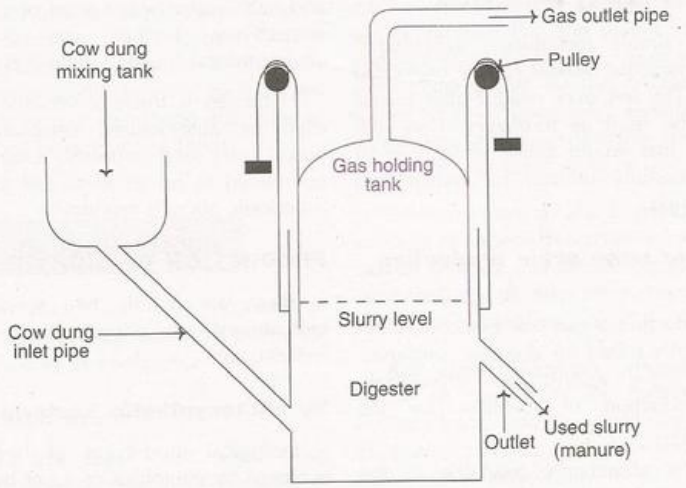


Fig. 31.3 : Diagrammatic representation of a biogas plant (gobar gas plant).

PROCESS OF BIOGAS PRODUCTION (BIOGAS PLANT)

Biogas poduction from biomass is an ***anaerobic process***. The anaerobic digestion is usually carried out by using air tight cylindrical tanks which are referred to as ***anaerobic digesters***. A digester may be made up of concrete bricks and cement or steel, usually built underground. The digester has an inlet attached to a mixing tank for feeding cow dung (***Fig. 31.3***). The methanogenic bacteria from another digester are also added with cow dung. The digester is attached to a movable gas holding or storage tank with a gas outlet. The ***used slurry*** (spent cow dung) comes out from the digester through an outlet. This can be used ***as a manure***. The anaerobic digester described above, is a low technology gobar gas plant, commonly used for domestic purposes in rural areas in India. The process of digestion usually takes about 2-3 weeks when cow dung is used as the substrate.

Landfill sites for methane production

Landfill sites are low cost ***digesters built underground for the digestion of solid wastes*** (of industries and municipalities). As the anaerobic digestion of solid organic material occurs, methane gas is generated. It can be recovered by boring gas wells into the top of the landfill. (For more details Refer Chapter 58).

Factors affecting biogas (methane) production

The factors affecting methane production, with special reference to biogas plant, are briefly described.

1. **Temperature and pH :** The ideal temperature is 30–40°C, while the pH is 6-8, for good yield.

2. **Slurry composition :** The ratio between solid and water composition in the slurry should be around 1 : 1. A carbon nitrogen ratio of 30 : 1 in the slurry results in optimal methane production. Good mixing and solubilization of the organic constituents is required.

3. **Anaerobic conditions :** The digester should be completely airtight, so as to create suitable anaerobic conditions.

4. **Presence of inhibitors :** Ammonium sulfate and antibiotics inhibit methane production. Agricultural wastes, pig and chicken manure (generating ammonia) and wastes from paper (rich in sulfate) inhibit biogas production.

Advantages of biogas production

By using a simple technology, agricultural, industrial and municipal ***wastes can be converted into a biofuel***. The ***left over residue*** after biogas formation can be used ***as fertilizers***. Thus, the waste materials that would cause environmental pollution are fruitfully utilized for biogas and fertilizer production.

Limitations for large scale production of methane

Although production of methane as a constituent of biogas is ideally suited for domestic purposes. Many people argue against the large scale commercial production of methane for the following reasons.

1. Methane is abundantly available in the natural oil and gas fields (produced by the same mechanism of methanogenesis, over a period of years).

2. Microbial production of methane is more expensive than its isolation from the natural gas.

3. Methane production by gassification of coal is more economical than its production from biomass.

4. Being a gaseous fuel, it is quite difficult as well as expensive to store, transport and distribute methane.

5. Methane is unsuitable for use as a fuel in automobiles. This is because it is very difficult to convert the gaseous methane into liquid state.

Despite the limitations listed above, production of methane from a wide range of biodegradable materials (particularly the wastes) is still attractive. This is due to the fact that the biomass used for methane generation is renewable, in contrast to the permanent depletion of naturally produced methane (in the gas and oil fields).

HYDROGEN — A NEW BIOFUEL

Hydrogen is a simple molecule which can be easily collected, stored (as a gas or liquid) and transported. It is highly combustible and can be ***used as a fuel or for the production of electricity***. Hydrogen, on mixing with oxygen, provides around 30,000 calories per gram as compared to 11,000 and 8,000 calories per gram of gasoline and coal respectively. Further, use of hydrogen is environmental friendly, since it is a non-pollutant.

Hydrogen is truely a versatile fuel. It can be used for automobiles, aeroplanes, helicopters, buses, cars and scooters. Liquid hydrogen is considered to be an ideal fuel for subsonic and supersonic aircrafts worldover.

PRODUCTION OF BIOHYDROGEN

There are mainly two ways of generating biohydrogen — by photosynthetic bacteria and by fermentation.

By photosynthetic bacteria

Biological production of hydrogen can be achieved by photolysis of water by photosynthetic algae and bacteria, a phenomenon referred to as ***biophotolysis***. Certain microalgae, and cyanobacteria (e.g. *Chlorella, Chlamydomonas, Scenedesmus, Microcystis, Oscillatoria, Aneboena*) can generate molecular hydrogen. Water is the source of raw material.

$$H_2O \xrightarrow{\text{Photolysis}} O_2 + H^+ + e^-$$

$$H^+ + H^+ \xrightarrow{\text{Hydrogenase}} H_2$$

The action of hydrogenase can be inhibited by creating oxygen pressure. This condition favours release of free hydrogen.

Isolated chloroplasts along with the bacterial enzyme hydrogenase have also been used for production of hydrogen.

By fermentation

It is possible to produce hydrogen from glucose, by bacterial action. However, the yield is less and uneconomical. Hydrogen can also be generated by anaerobic fermentation, by a process comparable to that of methane production. This is also not economical, besides being low in efficiency. Photosynthetic bacterium *Rhodospirillium* can be used to produce hydrogen from organic wastes.

By legume crops

The leguminous plants convert N_2 to NH_3 and H_2. This reaction is catalysed by the enzyme ***nitrogenase***. In the normal circumstances, this H_2

gas, a byproduct of nitrogen metabolism is lost in the soil. It is estimated that from a soybean crop in one hectare field, about 30 billion m^3 hydrogen is generated and lost annually. As such, there are no methods available to trap such huge quantities of hydrogen produced in agricultural fields.

ENERGY-RICH CROPS

Some of the plants are ***very efficient in converting CO_2 into biomass*** and such plants are collectively referred to as energy-rich crops.

Sugar and starch crops

Certain plants like sugar cane, sugar beet, cereals and tuber crops produce high quantities of starch and fermentation sugars. These crops supply energy-rich foods and feeds. Such plants are useful for production of ***biofuels***, particularly ethanol often referred to as ***bioethanol***.

Wood-rich plants

Some plants grow very fast and they serve as good suppliers of wood. e.g. *Eucalyptus, Butea, Melia, Casurina*. These plants are important sources of firewood. It is estimated that approximately 50% of the total wood harvested annually is utilized for the purpose of firewood. Wood is also useful for the supply of pulp for paper manufacture.

Petroleum plants

There are certain plants which can ***accumulate high molecular weight hydrocarbons***. They are referred to as ***petro-crops*** or ***gasoline plantations***. The products of these hydrocarbon-rich plants can serve as good ***substitutes of conventional petroleum*** and petroleum products.

The ***rubber plant*** (*Hevea rubber*), grown in South-East Asia is the principal source of rubber. Rubber is collected in the form of latex from the stems of trees. This plant meets about one third of the total world's demand of rubber. However, the rubber produced from petroleum is preferred for use in automobilies and planes, due to low-cost and high elasticity. Besides *Hevea rubber*, there are some other plants for the production of natural rubber e.g. *Parthenium agrentatum* (guayule) *Taraxacum koksaghyz* (Russian dandelion) grown in Mexico and some parts of USA.

Euphorbia lathyrus and *E. terucalli* contain high contents of terpenoids (complex hydrocarbons) that can be directly converted to gasoline/petrol. It is estimated that *E. terucelli* can yield about 5–10 barrels of oil/acre/year.

Aak plant (*Calotropis procera*) secretes latex which is very rich in hydrocarbons. These hydrocarbons, and the yield are comparable to *Euphorbia lathyrus*, and they also serve as good substitutes of petroleum.

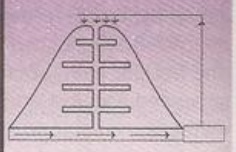
For obvious reasons, the cultivation of petroleum plants is encouraged throughout the world.

THE FUTURE OF BIOMASS

Besides its utility for the generation biofuels (alcohol, methane), biomass is also used for the ***production of butanol, acetone, single-cell protein and many other products*** (Chapter 29).

As such, the contribution of biomass to the world's requirements of energy is very low. It is around 5% in the U.S.A., and may be a little higher in the developing countries. However, ***being a renewable source of energy, biomass will have immense value in future***. This is particularly true as the world's non-renewable fuels (gas and oil) get depleted. There is a growing realisation on the fuel value of biomass. In the coming years, biomass production and utilization strategies will be fully exploited. In addition, further improvements in the biotechnological processes for better management and utilization of industrial, agricultural and domestic wastes will also solve the problem of world energy crisis.

32 Microbial Mining and Metal Biotechnology



Soil microorganisms are very closely involved as catalytic agents in many geological processes. These include mineral formation, mineral degradation, sedimentation and geochemical cycling.

In recent years, a new discipline of mineral science namely ***biohydrometallurgy*** or ***microbial mining*** (***mining with microbes***) is rapidly growing. Broadly speaking, biohydrometallurgy deals with the ***application of biotechnology in mining industry***. In fact, microorganisms can be successfully used for the extraction of metals (e.g., copper, zinc, cobalt, lead, uranium) from low grade ores. Mining with microbes is both economical and environmental friendly.

The term metal is used to any substance that is hard, possessing silvery lusture, and is a good conductor of heat and electricity. Some of the metals, however, are relatively soft, malleable and ductile e.g. sulfur. An ore is a naturally occurring solid mineral aggregate from which one or more minerals can be recovered by processing.

Majority of microorganisms can interact with metals. The metals can be recovered by the microorganisms by two processes.

1. **Bioleaching or microbial leaching :** This broadly involves the extraction or solubilization of minerals from the ores by the microorganisms.

2. **Biosorption :** It deals with the microbial cell surface adsorption of metals from the mine wastes or dilute mixtures.

BIOLEACHING

In ***microbial leaching*** (bioleaching), metals can be extracted from large quantities of low grade ores. Although recovery of metals (e.g. copper) from the drainage water of mines has been known for centuries, the involvement of microbes in this process was recognized about 40 years ago.

The bacteria which are naturally associated with the rocks can lead to bioleaching by one of the following ways.

1. ***Direct action*** of bacteria on the ore to extract metal.

2. Bacteria produce certain substances such as sulfuric acid and ferric iron which extract the metal (***indirect action***).

In practice, ***both the methods may work together*** for efficient recovery of metals.

Organisms for bioleaching

The ***most commonly used*** microorganisms for bioleaching are ***Thiobacillus ferrooxidans*** and ***Thiobacillus thiooxidans***.

Thiobacillus ferrooxidans is a rod-shaped, motile, non-spore forming, Gram-negative bacterium. It derives energy for growth from the oxidation of iron or sulfur. This bacterium is capable of oxidising ferrous iron (Fe^{2+}) to ferric form (Fe^{3+}), and converting sulfur (soluble or

insoluble sulfides, thiosulfate, elemental sulfur) to sulfate (SO_4^{2-}). *Thiobacillus thiooxidans* is comparable with *T. ferrooxidams*, and grows mostly on sulfur compounds.

Several studies indicate that the two bacteria *T. ferrooxidans* and *T. thiooxidans*, when put together, work synergistically and improve the extraction of metals from the ores.

Besides the above two bacteria, there are other microorganisms involved in the process of bioleaching. A selected few of them are briefly described below.

Sulfolobus acidocaldarius and ***S. brierlevi*** are thermophilic and acidophilic bacteria which can grow in acidic hot springs (>60°C). These bacteria can be used to extract copper and molybdenum respectively from chalcopyrite ($CuFeS_2$) and molybdenite (MoS_2).

A combination of two bacteria *Leptospirillum ferrooxidans* and *Thiobacillus organoparpus* can effectively degrade pyrite (FeS_2) and chalcopyrite ($CuFeS_2$). The individual organisms alone are of no use in extracting metals.

Pseudomonas aeruginosa can be employed in mining low grade uranium (0.02%) ore. This organism has been shown to accumulate about 100 mg uranium per one liter solution in less than ten seconds. Another organism, *Rhizopus arrhizus* is also effective for extracting uranium from waste water.

Certain ***fungi*** have also found use in bioleaching. Thus, ***Aspergillus niger*** can extract copper and nickel while *Aspergillus oryzae* is used for extracting gold.

As already stated, among the various microorganisms, *T. ferrooxidans* and *T. thiooxidans* are the most widely used in bioleaching. The utilization of many of the other organisms is still at the experimental stage.

Mechanism of bioleaching

The mechanism of bioleaching is rather complex and not well understood. The chemical transformation of metals by microorganisms may occur by direct or indirect bioleaching.

Direct bioleaching : In this process, there is a ***direct enzymatic attack on the minerals*** (which are susceptible to oxidation) by the microorganisms. For instance, certain bacteria (e.g., *T. ferrooxidans*) can transfer electrons (coupled with ATP production) from iron or sulfur to oxygen. That is these organisms can obtain energy from the oxidation of Fe^{2+} to Fe^{3+} or from the oxidation of sulfur and reduced sulfur compounds to sulfate as illustrated below.

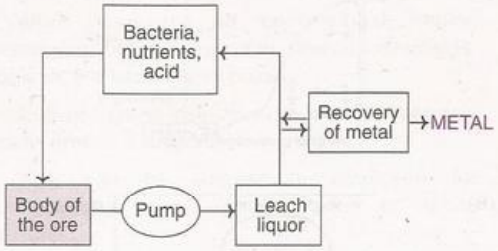


Fig. 32.1 : A flow diagram for microbial bioleaching– a general picture.

$$4FeSO_4 + 2H_2SO_4 + O_2 \longrightarrow 2Fe_2(SO_4)_3 + 2H_2O$$

$$2S^0 + 3O_2 + 2H_2O \longrightarrow 2H_2SO_4$$

$$2FeS_2 + 7O_2 + 2H_2O \longrightarrow 2FeSO_4 + 2H_2SO_4$$

As is evident from the third reaction given above, iron is extracted in the soluble form the iron ore pyrite (FeS_2).

Indirect bioleaching : In this indirect method, the ***bacteria produce strong oxidizing agents*** such as ferric iron and sulfuric acid on oxidation of soluble iron or soluble sulfur respectively. Ferric iron or sulfuric acid, being powerful oxidizing agents ***react with metals and extract them***. For indirect bioleaching, acidic environment is absolutely essential in order to keep ferric iron and other metals in solution. It is possible to continuously maintain acidic environment by the oxidation of iron, sulfur, metal sulfides or by dissolution of carbonate ions.

COMMERCIAL PROCESS OF BIOLEACHING

The naturally occurring mineral leaching is very slow. The microbial bioleaching process can be optimized by creating ideal conditions—temperature, pH, and nutrient, O_2 and CO_2 supply etc. A diagrammatic representation of general bioleaching process is depicted in ***Fig. 32.1***. The desired microorganisms with nutrients, acid etc, are pumped into the ore bed. The microorganisms

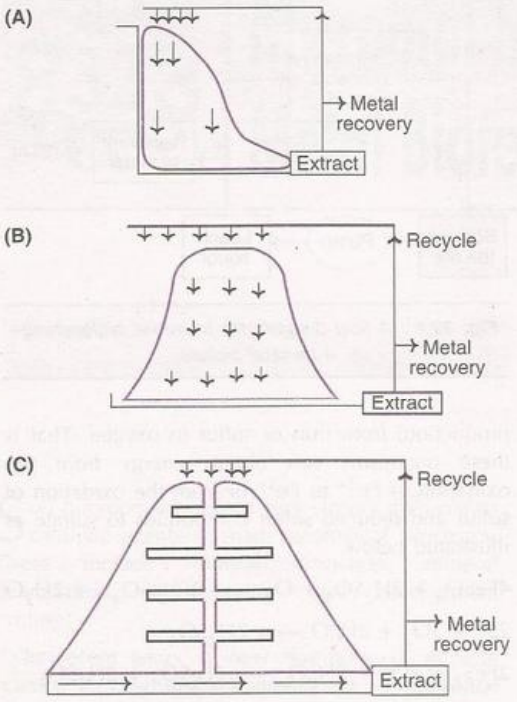


Fig. 32.2 : Commercial bioleaching processes (A) Slope leaching (B) Heap leaching (C) In situ leaching.

grow and produce more acid. The extracted ***leach liquor*** is processed for the metal recovery. The leach liquor can be recycled again and again for further metal extraction.

In commercial bioleaching, three methods are commonly used-slope leaching, heap leaching and *in situ* leaching (***Fig. 32.2***).

Slope leaching : The ore is finally ground and dumped in large piles down a mountainside (***Fig. 32.2A***). This ore is then subjected to continuous sprinkling of water containing the desired microorganism (*T. ferrooxidans*). The water collected at the bottom is used for metal extraction. The water can be recycled for regeneration of bacteria.

Heap leaching : In this case, the ore is arranged in large heaps (***Fig. 32.2B***) and subjected to treatments as in slope leaching.

In situ* leaching :** The ore, in its original natural place is subjected to leaching (Fig. 32.2C***). Water containing the microorganisms is pumped through drilled passages. In most cases, the permeability of rock is increased by subsurface blasting of the rock. As the acidic water seeps through the rock, it collects at the bottom which is used for metal extraction. This water can be recycled and reused.

Selected examples of microbial bioleaching are briefly described below.

BIOLEACHING OF COPPER

Copper ores (chalcopyrite, covellite and chalcocite) are mostly composed of other metals, besides copper. For instance, chalcopyrite mainly contains 26% copper, 26% iron, 33% sulfur and 2.5% zinc.

Bioleaching of copper ore (chalcopyrite) is widely used in many countries. This is ***carried out by the microorganism Thiobacillus ferrooxidans*** which oxidises insoluble chalcopyrite ($CuFeS_2$) and converts it into soluble copper sulfate ($CuSO_4$). Sulfuric acid, a byproduct formed in this reaction, maintains acidic environment (low pH) required for growth of the microorganisms.

Copper leaching is usually carried out by ***heap*** and ***in situ process*** (details given above). As the copper-containing solution (i.e., copper in the dissolved state) comes out, copper can be precipitated and the water is recycled, after adjusting the pH to around 2.

Extraction of copper by bioleaching is very common since the technique is efficient, besides being economical. It is estimated that about 5% of the world's copper production is obtained via microbial leaching. In the USA alone, at least 10% of the copper is produced by bioleaching process.

BIOLEACHING OF URANIUM

Bioleaching is the method of choice ***for the large-scale production uranium from its ores***. Uranium bioleaching is widely used in India, USA, Canada and several other countries. It is possible to recover uranium from low grade ores (0.01 to 0.5% uranium) and low grade nuclear wastes.

In situ bioleaching technique is commonly used for extracting uranium. In the technique employed, the insoluble tetravalent uranium is oxidized (in the presence of hot H_2SO_4/Fe^{3+} solution) to soluble hexavalent uranium sulfate.

$$UO_2 + Fe_2(SO_4)_3 \longrightarrow UO_2SO_4 + 2FeSO_4$$

Bioleaching of uranium is an indirect process since the microbial action is on the iron oxidant, and not directly on the uranium. The organism *Thiobacillus ferrooxidans* is capable of producing sulfuric acid and ferric sulfate from the pyrite (FeS_2) within the uranium ore.

For optimal extraction of uranium by bioleaching, the ideal conditions are temperature 45–50°C, pH 1.5-3.5, and CO_2 around 0.2% of the incoming air.

The soluble form of uranium from the leach liquor can be extracted into organic solvents (e.g., tributyl phosphate) which can be precipitated and then recovered.

Heap leaching process is sometimes preferred instead of the ***in situ*** technique. This is because the ***recovery of uranium in much higher*** with heap leaching.

BIOLEACHING OF OTHER METALS

Besides copper and uranium, bioleaching technique is also used for extraction of other metals such as ***nickel, gold, silver, cobalt, molybdenum*** and antimony. It may be noted that removal of iron is desirable prior to the actual process of leaching for other metals. This can be done by using the organism *Thiobacillus ferrooxidans* which can precipitate iron under aerobic conditions.

Bioleaching is also useful for the removal of certain impurities from the metal rich ores. For instance, the microorganisms such as *Rhizobium* sp and *Bradyrhizobium* sp can remove silica from bauxite (aluminium ore).

Bioleaching in desulfurization of coal

The process of ***removal of sulfur containing pyrite*** (FeS_2) ***from high sulfur coal*** by microorganisms is referred to as ***biodesulfurization***. High sulfur coal, when used in thermal power stations, emits sulfur dioxide (SO_2) that causes environmental pollution.

By using the microorganisms *Thiobacillus ferrooxidans* and *T. thiooxidans*, the pyrite which contains most of the sulfur (80–90%) can be removed. Thus, by employing bioleaching, high sulfur coal can be fruitfully utilized in an environment friendly manner. In addition, this approach is quite economical also.

ADVANTAGES OF BIOLEACHING

When compared to conventional mining techniques, bioleaching offers several advantages. Some of them are listed below.

1. Bioleaching can ***recover metals from low grade ores*** in a cost-effective manner.
2. It can be successfully employed for concentrating metals from wastes or dilute mixtures.
3. Bioleaching is ***environmental friendly***, since it does not cause any pollution (which is the case with conventional mining techniques).
4. It can be used to produce refined and expensive metals which otherwise may not be possible.
5. Bioleaching is a simple process with low cost technology.
6. It is ideally suited for the developing countries.

The major ***limitation*** or disadvantage ***of bioleaching*** is the ***slowness of the biological process***. This problem can, however, be solved by undertaking an in depth research to make the process faster, besides increasing the efficiency.

BIOSORPTION

Biosorption primarily deals with the ***microbial cell surface adsorption of metals*** from the mine wastes or dilute mixtures. The microorganisms can be used as ***biosorbents*** or ***bioaccumulators*** of metals. The process of biosorption performs two important functions.

1. Removal of toxic metals from the industrial effluents.
2. Recovery of valuable but toxic metals.

Both the above processes are concerned with a reduction in environmental poisoning/pollution.

A wide range of microorganisms (bacteria, algae, yeasts, moulds) are employed in biosorption. In fact, some workers have developed ***biosorbent-based granules for waste water/industrial effluent treatment, and metal recovery***.

In general, the microbial cell membranes are negatively charged due to the presence of carboxyl

(COO^-), hydroxyl (OH^-) phosphoryl (PO_4^{3-}) and sulfhydryl (HS^-) groups. This enables the positively charged metal ions (from solutions) to be adsorbed on to the microbial surfaces.

The different groups of microorganisms used in biosorption processes are briefly described below.

Bacteria

Several bacteria and actinomycetes ***adsorb and accumulate metals*** such as mercury, cadmium, lead, zinc, nickel, cobalt and uranium. For example, *Rhodospirullum* sp can accumulate Cd, Pb and Hg. *Bacillus circulans* can adsorb metals such as Cu, Cd, Co, and Zn. By use of electron microscopy, deposition of metals on the bacterial cell walls was recorded. It appears that the cell wall composition plays a key role in the metal adsorption.

Fungi

There is a large scale production of fungal biomass in many fermentation industries. This biomass can be utilized ***for metal biosorption from industrial effluents***. Immobilized fungal biomass is more effective in biosorption due to increased density, mechanical strength and resistance to chemical environment. Further, immobilized biomass can be reused after suitable processing.

The fungus *Rhizopus arrhizus* can adsorb several metallic cations e.g. uranium, thorium. *Pencillium lapidorum, P. spimulosum* are useful for the biosorption of metals such as Hg, Zn, Pb, Cu. Several fungi were tried with some degree of success to selectively adsorb uranium e.g. *Aspergillus niger, A. oryzae, Mucor haemalis, Penicillium chrysogenum.*

Edible mushrooms were also found to adsorb certain metals. For instance, fruit bodies of *Agaricus bisporus* can take up mercury while *Pleurotus sajor-caju can* adsorb lead and cadmium.

Many yeasts, commonly used in fermentation industries, are capable of adsorbing and accumulating metals. For instance, *Saccharomyces cerevisae* and *Sporobolomyces salmonicolour* can respectively adsorb mercury and zinc.

Algae

Several species of algae (fresh water or marine) can ***serve as bioaccumulators of metals***. For instance, *Chlorella vulgaris* and *C. regularis* can accumulate certain metals like Pb, Hg, Cu, Mo and U. The green algae *Hydrodictyon reticulatum* adsorbs and accumulates high quantities of Pb, Fe and Mn.

Some workers are in fact trying to use marine algae (e.g., *Luminaria, Ulva, Codium* sp) as bioaccumulators to reduce the metal pollution in rivers.

Higher plants in control of metal pollution

Besides the microorganisms described above, there are some higher aquatic plants (i.e., aquatic macrophytes) that can accumulate potential toxic wastes including many metals. ***Water hyacinth*** (*Eichornia crassipes*), ***duck weeds*** (*Spirodel* sp), ***water lettuce*** (*Pistia stratiotes*) and certain ferns (*Salvinia* sp) are important in the control of metal pollution. For more details, the reader must refer Chapter 54.

MICROBIAL RECOVERY OF PETROLEUM

By the conventional technique used in the oil fields, approximately one-third of the oil can be recovered. However, the oil recovery can be enhanced by using solvents, and surfactants. ***Certain polymers produced by microorganisms*** (e.g., ***xanthan gum***), ***when added to oil wells are capable of increasing oil recovery***. Xanthan gum can pass through small pore spaces and promote the release of more trapped oil.

In recent years, oil technologists are trying to directly use microorganisms *in situ* for increasing the oil recovery. In this process, there is no formalised use of a bioreactor. The natural geological site itself is the bioreactor. This allows the water and microorganisms to flow over the ore which are collected after seepage and outflow. The microorganisms, through surfactant production, gas formation or by other microbial activities reduce the viscosity of the oil so as to enhance its recovery. However, the success in this direction is not very commendable.

Continued further research may one day help to use microbes for commercial release of oil from oil wells or tar sands.

SECTION VI

ANIMAL CELL/ ANIMAL BIOTECHNOLOGY

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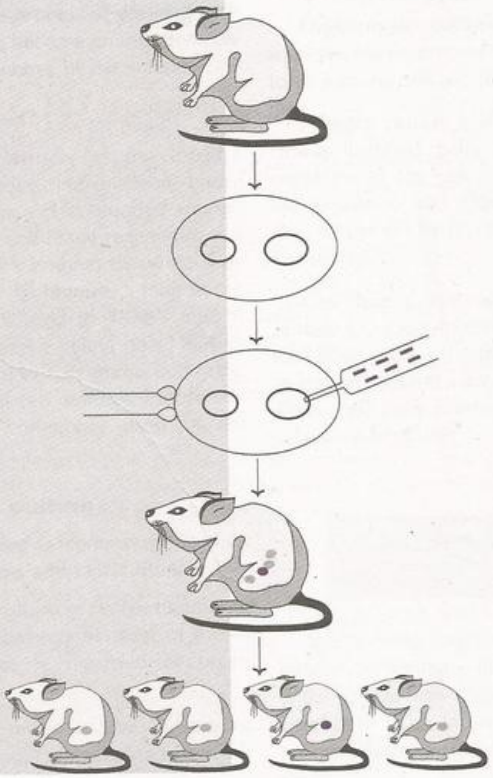
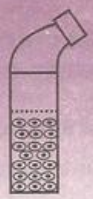


Animal Cell Culture — Fundamentals, Facilities and Applications



Animal cell culture basically involves the *in vitro* (in the laboratory) maintenance and propagation of animal cells in a suitable nutrient media. Thus, ***culturing is a process of growing cells artificially***. Cell culture has become an indispensible technology in various branches of life sciences.

Historical background

It was in 1907, Ross Harrison first developed a frog tissue culture technique. He probably chose frog for two reasons—being a cold-blooded animal, no incubation is required and tissue regeneration is fast in frog. In 1940's chick embryo tissue became a favourite for culture techniques. Interest in culturing human tissues started in 1950's after it was demonstrated (by HeLa; Gey) that human tumor cells could give rise to continuous cell lines. Among the various animal cell cultures, mouse cell cultures are the most commonly used in the laboratory.

Terminology in cell culture

The term ***tissue culture*** is commonly used to include both organ culture and cell culture.

Organ culture : The culture of native tissue (i.e. undisaggregated tissue) that retains most of the *in vivo* histological features is regarded as organ culture.

Cell culture : This refers to the culture of dispersed (or disaggregated) cells obtained from the original tissue, or from a cell line.

Histotypic culture : The culturing of the cells for their reaggregation to form a tissue—like structure represents histotypic culture.

Organotypic culture : This culture technique involves the recombination of different cell types to form a more defined tissue or an organ.

Primary culture : The culture produced by the freshly isolated cells or tissues taken from an organism is the primary culture. These cell are heterogenous and slow growing, and represent the tissue of their origin with regard to their properties.

Cell line : The subculturing of the primary culture gives rise to cell lines. The term ***continuous cell lines*** implies the ***indefinite growth of the cells*** in the subsequent subculturing. On the other hand, ***finite cell lines*** represent the death of cells after several subcultures.

FACILITIES FOR ANIMAL CELL CULTURE

While designing the laboratory for animal cell culture technology, utmost care should be taken with regard to the maintenance of aspetic conditions. The facilities required with regard to infrastructure and equipment are listed in the next page.

MINIMAL REQUIREMENTS FOR CELL CULTURE

INFRASTRUCTURE

- Clean and quite sterile area
- Preparation facilities
- Animal house
- Microbiology laboratory
- Storage facilities (for glassware, chemicals, liquids, small equipment).

EQUIPMENT

Laminar-flow, sterilizer, incubator, refrigerator and freezer (–20°C), balance, CO_2 cylinder, centrifuge, inverted microscope, water purifier, hemocytometer, liquid nitrogen freezer, slow-cooling device (for freezing cells), pipette washer, deep washing sink.

Besides the basic and minimal requirements listed above, there are many more facilities that may be beneficial or useful for tissue cultures. These include air-conditioned rooms, containment room for biohazard work, phase-contrast microscope, fluorescence microscope, confocal microscope, osmometer, high capacity centrifuge and time lapse video equipment.

CULTURE VESSELS

In the tissue culture technology, the cells attach to the surface of a vessel which ***serves as the substrate***, and grow. Hence there is a lot of importance attached to the nature of the materials used and the quality of the culture vessels.

The term ***anchorage dependent cells*** is used when the cells require an attachment for their growth. On the other hand, some cells undergo transformation, and become ***anchorage independent***.

Materials used for culture vessels

Glass : Although glass was the original substrate used for culturing, its use is almost discontinued now. This is mainly because of the availability of more suitable and alternate substrates.

Disposable plastics : Synthetic plastic materials with good consistency and optical properties are now in use to provide uniform and reproducible cultures. The most commonly used plastics are polystyrene, polyvinyl chloride (PVC), polycarbonate, metinex and thermonex (TPX).

Types of culture vessels

The following are the common types of culture vessels.

- Multiwell plates
- Petridishes
- Flasks
- Stirrer bottles.

The actual choice of selecting a culture vessel depends on several factors.

1. The way cells grow in culture—monolayer or suspension.
2. The quantity of the cells required.
3. The frequency of sampling for the desired work.
4. The purpose for which the cells are grown.
5. The cost factor.

In general, for ***monolayer cultures***, the cell yield is almost proportional to the surface area of the culture vessel. The flasks are usually employed for this purpose.

Any type of culture vessel can be used to grow suspension cultures. It is necessary to slowly and continuously agitate the suspended cells in the vessel.

Treatment of culture vessel surfaces

For improving the attachment of cells to the surfaces, and for efficient growth, some devices have been developed. It is a common observation that the growth of the culture cells is better on the surfaces for second seeding. This is attributed to ***matrix coating of the surfaces*** due to the accumulation of certain compounds like collagen and fibronectin released by the cells of the previous culture.

There are now commercially available matrices (e.g. matrigel, pronectin, cell-tak).

Feeder layers : Some of the tissue cultures require the support of metabolic products from living cells e.g. mouse embryo fibroblasts. In this

case, the growing fibroblasts release certain products which when fed to new cells enhance their growth.

Alternate substrates as substitutes of culture vessels

In recent years, certain alternatives for culture vessels have been developed. The important alternative artificial substrates are microcarriers and metallic substrates.

Microcarriers : They are in bead form and are made up of collagen, gelatin, polyacrylamide and polystyrene. Microcarriers are mostly used for the propagation of anchorage-dependent cells in suspension.

Metallic substrates : Certain types of cells could be successfully grown on some metallic surfaces or even on the stainless steel discs. For instance, fibroblasts were grown on palledium.

USE OF NON-ADHESIVE SUBSTRATES IN TISSUE CULTURE

The growth of anchorage independent cells can be carried out by plating cells on non-adhesive substrate like ***agar, agarose and methyl cellulose***. In this situation, as the cell growth occurs, the parent and daughter cells get immobilized and form a colony, although they are non-adhesive.

CONTAMINATION, ASEPTIC CONDITIONS, AND STERILIZATION

There are several ***routes of contamination*** in the tissue culture laboratory (***Table 33.1***). These include the various ***materials*** (glassware, pipettes), ***equipment*** (incubators, refrigerators, laminar-flow hoods), ***reagents*** (media, solutions), ***contaminated cell lines*** and ***poor techniques***.

The routes of contamination are mostly associated with the laboratory environment, and operating techniques.

Types of microbial contamination

Several species of bacteria, yeasts, fungi, molds and mycoplasmas, besides viruses are responsible for contamination. Major problems of contamination are linked to the repeated recurrence of a single species. Despite utmost care taken, no laboratory can claim to be totally free from contamination. It is necessary to continuously monitor for contamination and eliminate the same at the earliest.

Table 33.1 Major routes of contamination in a tissue culture laboratory

Equipment and facilities
Laminar-flow hoods
Dry incubators
CO_2 incubators
Humidified incubators
Wooden furniture, benches
Other instruments
Glassware and reagents
Pipettes
Screw caps
Culture glasses
Media bottles
Media and various solutions
Biological materials
Infected tissue samples
Cell lines
Operating techniques
Operator hands, hair, clothing, breathing
Work spaces
Pipetting, dispensing
Operating manipulations

ASEPTIC CONDITIONS

Maintenance of proper aseptic conditions is necessary ***to eliminate various contaminants (due to different microorganisms and viruses)***. The following measures are suggested for minimizing contamination, and maintenance of aseptic conditions.

- Strict adherence to standard sterile techniques and code of practices.

- Checking of reagents and media for sterility before use.
- Checking of cultures by eyes, and microscopes (phase contrast) every time they are used.
- Use of media and separate bottles for each cell line is advised.
- Maintenance of clean and tidy conditions at work places.
- Personal hygiene of the staff is very important.

STERILIZATION

The sterilization procedures are designed to kill the microorganisms, besides destroying the spores. There are three major devices for sterilization.

1. Dry heat
2. Moist heat (autoclave)
3. Filters.

In the ***Table 33.2***, the sterilization of major equipment, apparatus and liquids is given.

Sterilization by dry heat : This is carried out at a minimum temperature of 160°C for about one hour.

Sterilization by moist heat : Certain fluids and perishable items can be sterilized in an autoclave at 121°C for 15–20 minutes. For effective moist heat sterilization, it is necessary that the steam penetrates to all the parts of the sterilizing materials.

Sterilization by filters : The use of filters for sterilization of liquids often becomes necessary, since the constituents of these liquids may get destroyed at higher temperatures (dry heat or moist heat).

Sterile filtration is a novel technique for heat-labile solutions. The size of micropores of the filters is 0.1-0.2 µm. Filters, made from several materials are in use. These materials include nylon, cellulose acetate, cellulose nitrate, polycarbonate, polyethersulfone (PES) and ceramics. The filters are made in different designs-disc filters, cartridges and hollow fiber.

In fact, many commercial companies (e.g. Millipore, Durapore) supply ***reusable*** and ***disposable filters***, designed for different purposes of sterilization.

TABLE 33.2 Sterilization of major equipment, apparatus and liquids used in tissue culture

Sterilization device	*Items sterilized*
I *For equipment and apparatus*	
Dry heat	Glass slides
	Pipettes
	Ampoules (glass)
	Pasteur pipettes
	Instruments
	Test tubes
Autoclave	Ampoules (plastic)
	Apparatus with silicone tubing
	Filters (reusable)
	Glass bottles with screw caps
	Glass syringes
	Magnetic stirrer bases
	Screw caps
	Stoppers (rubber silicone)
II *For liquids and nutrients*	
Autoclave	Salt solutions
	Glucose–20%
	Agar
	Bacto–peptone
	Glycerol
	Lactalbumin hydrolysate
	Phenol red
	Tryptose
	HEPES
	EDTA
	Water
Filter	Serum
	Amino acids
	Vitamins
	Antibiotics
	Bovine serum albumin
	Collagenase
	Glutamine
	Drugs
	NaOH
	Trypsin
	Transferrin

ADVANTAGES AND LIMITATIONS OF TISSUE CULTURE

ADVANTAGES OF TISSUE CULTURE

Tissue culture technique has a wide range of applications. The most important advantages of this technique are listed below.

1. Control of physico-chemical environment-pH, temperature, dissolved gases (O_2 and CO_2), osmolarity.

2. ***Regulation of physiological conditions***-nutrient concentration, cell to cell interactions, hormonal control.

3. The cultured cell lines become homogenous (i.e. cells are identical) after one or two subcultures. This is in contrast to the heterogenous cells of tissue samples. ***The homogenous cells are highly useful for a wide range of purposes.***

4. It is easy to characterize cells for cytological and immunological studies.

5. Cultured cells can be stored in liquid nitrogen for several years.

6. Due to direct access and contact to the cells, biological studies can be carried out more conveniently. The main advantage is the low quantities of the reagents required in contrast to *in vivo* studies where most of the reagents (more than 90% in some cases) are lost by distribution to various tissues, and excretion.

7. Utility of tissue cultures will drastically reduce the use of animals for various experiments.

Limitations of tissue culture

There are several limitations of tissue culture, some of them are given below.

1. Need of expertise and technical skill for the development, and regular use of tissue culture.

2. ***Cost factor*** is a major limitation. Establishment of infrastructure, equipment and other facilites are expensive.

3. It is estimated that the cost of production of cells is about 10 times higher than direct use of animal tissues.

4. Control of the environmental factors (pH, temperature, dissolved gases, disposal of biohazards) is not easy.

5. The native *in vivo* cells exist in a three-dimensional geometry while in *in vitro* tissue culture, the propagation of cells occurs on a two dimensional substrate. Due to this, the ***cell to cell interactive characters are lost.***

6. The cell lines may represent one or two types of cells from the native tissue while others may go unrepresented.

7. Tissue culture techniques are associated with the differentiation i.e. loss of the characters of the tissue cells from which they were originally isolated.

8. This happens due to adaptation and selection processes while culturing.

9. Continuous cell lines may result in genetic instability of the cells. This may ultimately lead to heterogeneity of cells.

10. The components of homeostatic *in vivo* regulation (nervous system, endocrine system, metabolic integration) are lacking in *in vitro* cultures. Addition of hormones and growth factors has been started recently.

APPLICATIONS OF ANIMAL CELL CULTURES

There is a widespread concern that extensive use of animals for laboratory experiments is not morally and ethically justifiable. ***Animal welfare groups worldover are increasingly criticising the use of animals.*** Some research workers these days prefer to ***utilize animal cell cultures wherever possible*** for various studies. The major applications of laboratory animal cell cultures are given in ***Table 33.3***, and listed below.

- Studies on intracellular activity e.g. cell cycle and differentiation, metabolisms.
- Elucidation of intracellular flux e.g. hormonal receptors, signal transduction.
- Studies related to cell to cell interaction e.g. cell adhesion and motility, metabolic cooperation.
- Evalution of environmental interactions e.g. cytotoxicity, mutagenesis.

Table 33.3 Summary of the applications of animal cell cultures

Category	*Applications*
Intracellular activity	Studies related to cell cycle and differentiation, transcription, translation, energy metabolism, drug metabolism.
Intracellular flux	Studies involving hormonal receptors, metabolites, signal transduction, membrane trafficking.
Cell to cell interaction	Studies dealing with cell adhesion and motility, matrix interaction, morphogenesis, paracrine control, metabolic cooperation.
Environmental interaction	Studies related to drug actions, infections, cytotoxicity, mutagenesis, carcinogenesis.
Genetics	Studies dealing with genetic analysis, transfection, transformation, immortalization, senescence.
Cell products	Wide range of applications of the cellular products formed (Refer ***Table 33.4***) e.g. vaccines, hormones, interferons etc.

- Studies dealing with genetics e.g. genetic analysis, immortalization, senescence.
- Laboratory production of medical/pharmaceutical compounds for wide range of applications e.g. vaccines, interferons, hormones.

There are however, several limitations on the use of animal cell cultures. This is mostly due to the differences that exist between the *in vivo* and *in vitro* systems, and the validity of the studies conducted in the laboratory.

MEDICAL / PHARMACEUTICAL PRODUCTS OF ANIMAL CELL CULTURES

The most important application of animal cell cultures is the production of a wide range of commercial compounds for medical and pharmaceutical use. The commercial production of several biological compounds by the microorganisms is described in Chapters 23–30.

A selected list of animal cell culture products of commercial importance is given in ***Table 33.4***.

Production of vaccines

Monkey kidney or chick embryo cells, or recently human diploid cells are in use for the production of vaccines. The vaccine manufacture in animal cell cultures is rather complex with risk of contamination, and safety aspect. For these reasons, production of vaccines by recombinant DNA technology employing bacteria or yeasts is preferred (Refer Chapter 16).

Production of high value therapeutics

Many human proteins with high therapeutic potential are often in short supply e.g. tissue plasminogen activator, clotting factors (VIII and IX), erythropoietin. There is a major limitation to produce human proteins that undergo post-translational modifications (glycosylation, carboxylation etc.) in bacteria and yeasts. This is due to the fact that these organisms do not possess the machinery to perform post-translational changes. However, pharmaceutical proteins that do not require post-translational modifications can be produced by bacteria or yeasts e.g. insulin, albumin, growth hormone.

Animal cell cultures (particularly mammalian cell cultures) are useful for the production of many pharmaceutically/medically important proteins (***Table 33.4***). These include the following.

- Plasminogen
- Interferons
- Blood clotting factors
- Hormones
- Monoclonal antibodies
- Erythropoietin.

Purification of pharmaceutical products

As the desired product is produced in the cell culture medium, its purification, isolation and storage (collectively referred to as downstream processing) assumes significance. The final product for therapeutic applications is expected to satisfy the following criteria.

- The product should have a stable structure with optimal activity.

Table 33.4 Selected examples of animal cell culture products (proteins) of medical/pharmaceutical importance

Product(s)	*Application(s)*
Vaccines	
Polio vaccines	Poliomyelitis prophylaxis
Measles vaccine	Measles prophylaxis
Rabies vaccine	Rabies prophylaxis
Malaria vaccines	Malaria prophylaxis
HIV vaccine	AIDS prophylaxis and treatment
Plasminogen activators	
Tissue-type plasminogen activator Urokinase-type plasminogen activator Recombinant plasminogen activator	Acute myocardial infarction, pulmonary embolism, deep vein thrombosis, acute stroke.
Interferons	
Interferon-α	Anticancer, immunomodulator
Interferon-β	Anticancer, antiviral
Interferon-γ	Anticancer, immunomodulator
Blood clotting factors	
Factors VII, VIII, IX and X	Hemophilia, as blood clotting agents.
Hormones	
Human growth hormone	Growth retardation in children
Somatotropin	Chronic renal insufficiency
Follicle stimulating hormone	Treatment of infertility
Human chorionic gonadotropin	Treatment of infertility
Monoclonal antibodies	
Anti-lipopolysaccharide	Treatment of sepsis
Human B-cell Lymphomas	Treatment of B-cell lymphoma
Anti-fibrin 99	Diagnosis of blood clot by imaging
Tcm-FAb (breast)	Diagnosis of breast cancer
Others	
Erythropoietin	Antianaemic agent
Interleukin-2	Anticancer, HIV treatment
Tumor necrosis factor	Anticancer
Granulocyte stimulating factor	Anticancer
Carcinoembryonic antigen	Diagnosis and monitoring of cancer patients.

- The product should be free from other biomolecules that may interfere with its activity and/or cause immunological complications.
- It should be free from all pathogens including viruses.

GENETIC ENGINEERING OF ANIMAL CELLS AND THEIR APPLICATIONS

It is now possible to genetically modify the animal (mammalian) cells to introduce the genes needed for the production of a specific protein or improve the characteristics of a cell line. The following methods are used to introduce foreign DNA into mammalian cells.

1. Electroporation
2. Lipofection
3. Microinjection
4. Fusion of mammalian cells with bacteria or viruses.

As the foreign DNA gets integrated into the mammalian cellular genome, the gene expresses to produce the desired protein. It is however, necessary to select the best producing recombinant cells by conventional methods using selectable marker genes. The following selectable markers are used for choosing the transfected cells.

- Viral thymidine kinase
- Bacterial dihydrofolate reductase
- Bacterial neomycin phosphotransferase

It has been possible to overproduce several proteins in mammalian cells through genetic manipulations e.g., tissue plasminogen activator, erythropoietin, interleukin-2, interferon-β, clotting factors VIII and IX, tumor necrosis factors.

The ***recombinant mammalian cells are conveniently used for the production of monoclonal antibodies*** which have wide range of applications (Refer Chapter 17).

RISKS IN A TISSUE CULTURE LABORATORY AND SAFETY

There are several risks associated with tissue culture technology. Most of the accidents that occur in culture laboratories are ***due to negligence and casual approach while dealing with biological and radiological samples***, besides improper maintenance

Table 33.5 Risks in a tissue culture laboratory

Category	*Contributing factor(s)*
Maintenance risks	Age and condition of various equipment, leakage of disposals.
Personnel risks	Inadequate training, lack of concentration and interest.
Physical risks	Electric shocks, fire, intense cold.
Chemical risks	Toxicity due to poisons, carcinogens, mutagens, irritants, allergens.
Biohazards	Pathogenic organisms, viruses, genetic manipulations, culture cells and DNA (quality and quantity).
Radioisotope risks	Energy emission and its penetration, ionization.

of the laboratory. A broad categorization of risks and the contributory factors is given in ***Table 33.5***.

Safety regulations

Some of the developed countries have formulated general safety regulations ***to minimize the risks associated with tissue culture laboratories***. Selected examples :

1. "Biosafety in microbiological and biomedical laboratories", U. S. Department of Health and Human Sciences (1993).

2. "Safe working and the prevention of infection in clinical laboratories" U.K. Health Services Advisory Committee (1991).

Some of the general precautions for the safety of a tissue culture laboratory are listed here.

- Strict adherence to recommendations of regulatory bodies.
- Periodical meetings and discussions of local safety committees.
- Regular monitoring of the laboratories.
- Periodical training of the personnel through seminars and workshops.
- Print and make the standard operating procedures (SOPs) available to all staff.
- Good record keeping.
- Limited access to the laboratory (only for the trained personnel and selected visitors).
- Appropriate waste disposal system for biohazards, radioactive wastes, toxins and corrosives.

BIOHAZARDS

The accidents or the risks associated with the biological materials are regarded as biohazards or ***biological hazards***. There are two main systems that contribute to the occurrence of biohazards (***Table 33.6***).

1. The direct sources of the biological materials.

2. The processes or operations involved in their handling.

Control of biohazards

Biohazards can be controlled to a large extent by strict adherence to the regulatory guidelines and maintenance programmes. Some important aspects are listed.

- Microbiological safety cabinet or biohazard wood with pathogen trap filters have been developed.
- Vertical laminar-flow hood (instead of horizontal laminor-flow hood) is recently in use. This minimizes the direct exposure of the operator to the samples/processes.
- Pathogen containing samples are treated in separate rooms with separate facilities (centrifuge, incubator, cell counting etc.).
- Sterilization of all wastes, solid glassware etc. and their proper disposal.
- Facilities for change of clothing while entering and leaving the rooms.
- Strict adherence to the access of designated personnel to the culture rooms.

Table 33.6 Sources that contribute to biohazards

Biological material(s)
Tissue samples and cultures with human pathogens.
Human cells infected with viruses (including retroviruses)
Cells subjected to various genetic manipulations.
Operating processes
Preparation of the media.
Development of primary cultures, cell lines and other laboratory works.

Culture Media for Animal Cells

MCDB 202
MCDB 402
MCDB 110
MCDB 131
MCDB 170
WAJC 404
LHC 9
MCDB 202
MCDB 402

The selection of an appropriate growth medium for the *in vitro* cultivation of cells is an important and essential step. The mammalian cells of an organ in the body receive nutrients from blood circulation. For culturing these cells *in vitro*, ***it is expected that they should be provided with the components similar to those present in blood.*** In general, the choice of the medium mostly depends on the type of the cells to be cultured, and the purpose of the culture (growth, differentiation, production of desired products). The culture media may be natural or artificial.

NATURAL MEDIA

In the early years, the natural media obtained from various biological sources were used.

Body fluids : Plasma, serum, lymph, amniotic fluid, ascitic and pleural fluids, aqueous humour from eyes and insect hemolymph were in common use. These fluids were tested for sterility and toxicity before their utility.

Tissue extracts : Among the tissue extracts, ***chick embryo extract*** was the most commonly employed. The extracts of liver, spleen, bone marrow and leucocytes were also used as culture media.

Some workers still prefer natural media for organ culture.

ARTIFICIAL MEDIA

The artificial media (containing partly defined components) have been in use for cell culture since 1950. The minimal criteria needed for choosing a medium for animal cell cultures are listed below.

- The medium should provide all the nutrients to the cells.
- Maintain the physiological pH around 7.0 with adequate buffering.
- The medium must be sterile, and isotonic to the cells.

The basis for the cell culture media was the ***balanced salt solution*** which was originally ***used to create a physiological pH and osmolarity*** required to maintain cells *in vitro*. For promoting growth and proliferation of cells, various constituents (glucose, amino acids, vitamins, growth factors, antibiotics etc.) were added, and several media developed. Addition of serum to the various media is a common practice. However, some workers in recent years have started using serum-free media.

The physicochemical properties of media required for tissue cultures are briefly described. This is followed by a brief account on balanced salt solutions, commonly used culture media and the serum-free media.

PHYSICOCHEMICAL PROPERTIES OF CULTURE MEDIA

The culture media is expected to possess certain physicochemical properties (pH, O_2, CO_2, buffering, osmolarity, viscosity, temperature etc.) to support good growth and proliferation of the cultured cells.

pH

Most of the cells can grow at a pH ***in the range of 7.0-7.4***, although there are slight variations depending on the type of cells (i.e. cell lines). The indicator ***phenol red*** is most commonly used for visible detection of pH of the media. Its colouration at the different pH is shown below.

At pH 7.4 — Red
At pH 7.0 — Orange
At pH 6.5 — Yellow
At pH 7.8 — Purple

CO_2, bicarbonate and buffering

Carbon dioxide in the medium is in a dissolved state, the concentration of which depends on the atmospheric CO_2 tension and temperature. CO_2 in the medium exists as carbonic acid (H_2CO_3), and bicarbonate (HCO_3^-) and H^+ ions as shown below.

$$CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow H^+ + HCO_3^-$$

As is evident from the above equation, the concentrations of CO_2, HCO_3^- and pH are interrelated. By increasing the atmospheric CO_2, the pH will be reduced making the medium acidic.

Addition of sodium bicarbonate (as a component of bicarbonate buffer) neutralizes bicarbonate ions.

$$NaHCO_3 \longleftrightarrow Na^+ + HCO_3^-$$

In fact, the commercially available media contain a recommended concentration of bicarbonate, and CO_2 tension for the required pH. ***In recent years HEPES*** (hydroxyethyl piperazine 2-sulfonic acid) ***buffer which is more efficient than bicarbonate buffer is being used in the culture media***. However, bicarbonate buffer is preferred by most workers because of the low cost, less toxicity and nutritional benefit to the medium. This is in contrast to HEPES which is expensive, besides being toxic to the cells.

The presence of pyruvate in the medium results in the increased endogenous production of CO_2 by the cells. This is advantageous since the dependence on the exogenous supply of CO_2 and HCO_3^- will be less. In such a case, the buffering can be achieved by high concentration of amino acids.

Oxygen

A great majority of cells *in vivo* are dependent on the O_2 supply for aerobic respiration. This is in fact made possible by a continuous supply of O_2 to the tissues by hemoglobin.

The cultured cells mostly rely on the dissolved O_2 in the medium which may be toxic at high concentration due to the generation of free radicals. Therefore, it is absolutely necessary to supply adequate quantities of O_2 so that the cellular requirements are met, avoiding toxic effects. Some workers add free-radical scavengers (glutathione, mercaptoethanol) to nullify the toxicity. Addition of selenium to the medium is also advocated to reduce O_2 toxicity. This is because selenium is a cofactor for the synthesis of glutathione.

In general, the glycolysis occurring in cultured cells is more anaerobic when compared to *in vivo* cells. Since the depth of the culture medium influences the rate of O_2 diffusion, it is advisable to keep the depth of the medium in the range 2-5 mm.

Temperature

In general, the optimal temperature for a given cell culture is ***dependent on the body temperature of the organism, serving as the source of the cells***. Accordingly, for cells obtained from humans and warm blooded animals, the optimal temperature is 37°C. *In vitro* cells cannot tolerate higher temperature and most of them die if the temperature goes beyond 40°C. It is therefore absolutely necessary to maintain a constant temperature (± 0.5°C) for reproducible results.

If the cells are obtained from birds, the optimal temperature is slightly higher (38.5°C) for culturing. For cold blooded animals (poikiltherms) that do not regulate their body heat (e.g. cold-water fish), the culture temperature may be in the range of 15–25°C.

Besides directly influencing growth of cells, temperature also affects the solubility of CO_2 i.e. higher temperature enhances solubility.

Osmolality

In general, the osmolality for most of the cultured cells (from different organisms) is in the range of 260–320 mosm/kg. This is ***comparable to the osmolality of human plasma*** (290 mosm/kg). Once an osmolality is selected for a culture medium, it should be maintained at that level (with an allowance of ± 10 mosm/kg).

Table 34.1 Composition (g/l) of balanced salt solutions (BSS)

Ingradient	*Earle's BSS*	*Hank's BSS*
NaCl	6.68	8.0
KCl	0.4	0.4
$CaCl_2$ (anhydrous)	0.02	0.14
$MgSO_4.7H_2O$	0.2	0.1
$NaHCO_3$	2.2	0.35
$NaH_3PO_4.H_2O$	0.14	—
$Na_2HPO_4.7H_2O$	—	0.09
KH_2PO_4	—	0.06
D-Glucose	1.0	1.0
Phenol red	0.01	0.01
HEPES, Na salt (buffer)	13.02	2.08

Whenever there is an addition of acids, bases, drugs etc. to the medium, the osmolality gets affected. The instrument osmometer is employed for measuring osmolalities in the laboratory.

BALANCED SALT SOLUTIONS

The balanced salt solutions (***BSS***) are ***primarily composed of inorganic salts***. Sometimes, sodium bicarbonate, glucose and HEPES buffer may also be added to BSS. ***Phenol red serves as a pH indicator***. The important functions of balanced salt solutions are listed hereunder.

- Supply essential inorganic ions.
- Provide the requisite pH.
- Maintain the desired osmolality.
- Supply energy from glucose.

In fact, balanced salt solutions form the basis for the preparation of complete media with the requisite additions. Further, BSS is also useful for a short period (up to 4 hours) incubation of cells.

The composition of two most widely used BSS namely Earle's BSS and Hank's BSS is given in ***Table 34.1***.

COMPLETE CULTURE MEDIA

In the early years, balanced salt solutions were supplemented with various nutrients (amino acids, vitamins, serum etc.) to promote proliferation of cells in culture. Eagle was a pioneer in media formulation. He determined (during 1950–60) the nutrient requirements for mammalian cell cultures. Many developments in media preparation have occurred since then. There are more than a dozen media now available for different types of cultures. Some of them are stated below.

EMEM—Eagle's minimal essential medium

DMEM—Dulbecco's modification of Eagle's medium

GMEM—Glasgow's modification of Eagle's medium

RPMI 1630 and RPMI 1640—Media from Rosewell Park Memorial Institute.

The other important culture media are Ham's F10, and F12, TC 199 and CMRL 1060.

The detailed composition of three commonly used media namely Eagle's MEM, RPMI 1640 and Ham's F12 is given in ***Table 34.2***. The ***complete media, in general, contains*** a large number of components ***amino acids, vitamins, salts, glucose, other organic supplements, growth factors and hormones, and antibiotics, besides serum***. Depending on the medium, the quality and quantity of the ingradients vary. Some important aspects of the media ingradients are briefly described.

Amino acids

All the ***essential amino acids*** (which cannot be synthesized by the cells) have to be added to the medium. In addition, even the ***non-essential amino acids*** (that can be synthesized by the cells) are also usually added to avoid any limitation of their cellular synthesis. Among the non-essential amino acids, glutamine and/or glutamate are frequently added in good quantities to the media since these amino acids serve as good sources of energy and carbon.

Vitamins

The quality and quantity of vitamins depends on the medium. For instance, Eagle's MEM contains only water soluble vitamins (e.g. B-complex, choline, inositol). The other vitamins are obtained from the serum added. The medium M 199 contains all the fat soluble vitamins (A, D, E and K) also. In general, ***for the media without serum,***

Table 34.2 Composition of three commonly used culture media

Component	Eagle's MEM	RPMI 1640	Ham's F 12
Amino acids			
L-Alanine			8.91
L-Arginine HCl	105	200	211
L-Asparagine H_2O		50	15.0
L-Aspartic acid		20	13.3
L-Cystine	24	50	24.0
L-Glutamic acid		20	14.7
L-Glutamine	292	300	146.2
Glycine		10	7.51
L-Histidine HCl H_2O	31	15	21.0
L-Isoleucine	52	50	3.94
L-Leucine	52	50	13.12
L-Lysine	58	40	36.54
L-Methionine	15	15	4.48
L-Phenylalanine	32	15	4.96
L-Proline		20	34.5
L-Serine		30	10.51
L-Threonine	48	20	11.91
L-Tryptophan	10	5	2.042
L-Tyrosine	36	20	5.43
L-Valine	46	20	11.7
Glutathione (red)		1	
L-Hydroxyproline		20	
Vitamins			
D-Biotin		0.2	0.007
Ca D-pantothenate	1	0.25	0.26
Choline chloride	1	3.0	13.96
Folic acid	1	1.0	1.32
i-Inositol	2		18.02
Nicotinamide	1	35	0.037
p-Aminobenzoic acid		1.0	
Pyridoxine HCl		1	0.062
Pyridoxal HCl	1		
Riboflavin	0.1	0.2	0.038
Thiamine HCl	1	1.0	0.34
Vitamin B_{12}		0.005	1.36
Inorganic salts			
$CaCl_2.2H_2O$	200		44.1
$CaNO_3.4H_2O$		100	
$CuSO_4.5H_2O$			0.0025
$FeSO_4.7H_2O$			0.83
KCl	400	400	223
$MgSO_4.7H_2O$	220	100	133
NaCl	6800	6000	7599
$NaHCO_3$	2000	2000	1176
$Na_2HPO_4.7H_2O$		1512	268
$NaH_2PO_4.2H_2O$	150		
Other components			
D-Glucose	1000	2000	1801
Phenol red		5.0	1.2
Sodium pyruvate			110
Lipoic acid			0.21
Linoleic acid			0.084
Hypoxanthine			4.08
Putrescine 2HCl			0.16

Table 34.2 contd. next column

more vitamins in higher concentrations are required.

Salts

The salts present in the various media are basically those found in ***balanced salt solutions*** (Eagle's BSS and Hank's BSS). The salts contribute to cations (Na^+, K^+, Mg^{2+}, Ca^{2+} etc.) and anions (Cl^-, HCO_3^-, SO_4^{2-}, PO_4^{3-}), and are mainly responsible for the maintenance of osmolality. There are some other important functions of certain ions contributed by the salts.

- Ca^{2+} ions are required for cell adhesion, in signal transduction, besides their involvement in cell proliferation and differentiation.
- Na^+, K^+ and Cl^- ions regulate membrane potential.

- PO_4^{3-}, SO_4^{2-} and HCO_3^- ions are involved in the maintenance of intracellular charge, besides serving as precursors for the production of certain important compounds e.g. PO_4^{3-} is required for ATP synthesis.

Glucose

Majority of culture media contain glucose which ***serves as an important source of energy***. Glucose is degraded in glycolysis to form pyruvate/lactate. These compounds on their further metabolism enter citric acid cycle and get oxidized to CO_2. However, experimental evidence indicates that the contribution of glucose for the operation of citric acid cycle is very low *in vitro* (in culture cells) compared to *in vivo* situation. ***Glutamine*** rather than glucose ***supplies carbon for the operation of citric acid cycle***. And for this reason, the cultured cells require very high content of glutamine.

Hormones and growth factors

For the media with serum, addition of hormones and growth factors is usually not required. They are frequently added to serum-free media.

Other organic supplements

Several additional organic compounds are usually added to the media to support cultures. These include certain proteins, peptides, lipids, nucleosides and citric acid cycle intermediates. For serum-free media, supplementation with these compounds is very useful.

Antibiotics

In the early years, culture media invariably contained antibiotics. The most commonly used antibiotics were ampicillin, penicillin, gentamycin, erythromycin, kanamycin, neomycin and tetracycline. Antibiotics were added to reduce contamination. However, ***with improved aseptic conditions in the present day tissue culture laboratories, the addition of antibiotics is not required***. In fact, the use of antibiotics is associated with several disadvantages.

- Possibility of developing antibiotic-resistant cells in culture.
- May cause antimetabolic effects and hamper proliferation.
- Possibility of hiding several infections temporarily.
- May encourage poor aseptic conditions.

The present recommendation is that for the routine culture of cells, antibiotics should not be added. However, they may be used for the development of primary cultures.

SERUM

Serum is a natural biological fluid, and is rich in various components to support cell proliferation. The major constituents found in different types of sera are listed in ***Table 34.3***. The most commonly used sera are ***calf serum*** (CS), ***fetal bovine serum*** (FBS), ***horse serum*** and human serum. While using human serum, it must be screened for viral diseases (hepatitis B, HIV).

Approximately 5–20% (v/v) of serum is mostly used for supplementing several media. Some of the important features of the serum constituents are briefly described.

Proteins

The *in vitro* functions of serum protein are not very clear. Some of them are involved in promoting cell attachment and growth e.g. fetuin, fibronectin. Proteins ***increase the viscosity*** of the culture medium, besides ***contributing to buffering action***.

Nutrients and metabolites

Serum contains several amino acids, glucose, phospholipids, fatty acids, nucleosides and metabolic intermediates (pyruvic acid, lactic acid etc.). These constituents do contribute to some extent for the nutritional requirements of cells. This may however, be insignificant in complex media with well supplemented nutrients.

Growth factors

There are certain growth factors in the serum that stimulate the proliferation of cells in the culture.

- Platelet-derived growth factor (PDGF).
- Fibroblast growth factor (FGF).
- Epidermal growth factor (EGF).
- Vascular endothelial growth factor (VEGF).
- Insulin-like growth factors (IGF-1, IGF-2).

In fact, almost all these growth factors are commercially available for use in tissue culture.

Table 34.3 Major constituents of serum

Proteins	
	Albumin
	Globulins
	Fetuin
	Fibronectin
	Transferrin
	Protease inhibitors (α_1-antitrypsin)
Amino acids	
	Almost all the 20
Lipids	
	Cholesterol
	Phospholipids
	Fatty acids
Carbohydrates	
	Glucose
	Hexosamine
Other organic compounds	
	Lactic acid
	Pyruvic acid
	Polyamines
	Urea
Vitamins	
	Vitamin A
	Folic acid
Growth factors	
	Epidermal growth factor
	Platelet-derived growth factor
	Fibroblast growth factor
Hormones	
	Hydrocortisone
	Thyroxine
	Triiodothyronine
	Insulin
Inorganics	
	Calcium
	Sodium
	Potassium
	Chlorides
	Iron
	Phosphates
	Zinc
	Selenium

Hormones

Hydrocortisone promotes cell attachment, while insulin facilitates glucose uptake by cells. Growth hormone, in association with somatomedins (IGFs), promotes cell proliferation.

Inhibitors

Serum may also contain cellular growth inhibiting factors. Majority of them are artefacts e.g. bacterial toxins, antibodies. The natural serum also contains a physiological growth inhibitor namely ***transforming growth factor β (TGF-β)***. Most of these growth inhibitory factors may be removed by heat inactivation (at 56°C for 30 minutes).

SELECTION OF MEDIUM AND SERUM

As already stated, there are around a dozen media for the cell cultures. The selection of a particular medium is based on the cell line and the purpose of culturing. For instance, for chick embryo fibroblasts and HeLa cells, EMEM is used. The medium DMEM can be used for the cultivation of neurons. A selected list of cells and cell lines along with the media and sera used is given in ***Table 34.4***. In fact, information on the selection of appropriate medium for a particular cell line is available from literature.

The selection of serum is also based on the type of cells being cultured. The following criteria are taken into consideration while choosing serum.

- Batch to batch variations.
- Quality control.
- Efficiency to promote growth and preservation of cells.
- Sterility.
- Heat inactivation.

In recent years, there is a tendency to discontinue the use of serum, and switch over to more clearly defined media (described later).

SUPPLEMENTATION OF THE MEDIUM WITH TISSUE EXTRACTS

Besides serum, the culture media can also be supplemented with certain tissue extracts and microbial culture extracts. The examples are—chick embryo extract, proteolytic digests of beef heart, bactopeptone, lactalbumin hydrolysate, tryptose.

The ***chick embryo extract*** was found to contain both high molecular weight and low molecular weight compounds that ***support growth and proliferation of cells.***

SERUM-FREE MEDIA

Addition of serum to the culture media has been an age-old practice. However, in recent years, certain serum-free media have been developed. It is worthwhile to know the disadvantages associated with the the use of serum, and the advantages and disadvantages of serum-free media.

Disadvantages of serum in media

Variable composition : There is no uniformity in the composition of the serum. It is highly variable (source, batch, season, collection method, processing). Such differences in the composition significantly influence the cells in culture.

Quality control : To maintain a uniform quality of the serum, special tests have to be performed with each batch of serum, before its use.

TABLE 34.4 A selected list of the cells or cell lines along with the media and serum used for their culture

Cells or cell line	*Medium*	*Serum*
Chick embryo fibroblasts	EMEM	CS
Chinese hamster ovary (CHO)	EMEM, Ham's F12	CS
HeLa cells	EMEM	CS
Human leukemia	RPMI 1640	FB
Mouse leukemia	Fischer's medium, RPMI 1640	FB, HoS
Neurons	DMEM	FB
Mammary epithelium	RPMI 1640, DMEM	FB
Hematopoietic cells	RPMI 1640, Fischer's medium	FB
Skeletal muscle	DMEM, F 12	FB, HoS
Glial cells	MEM, F 12, DMEM	FB
3T3 cells	MEM, DMEM	CS

(EMEM–Eagle's minimal essential medium; RPMI 1640–Medium from to Rosewell Park Memorial Institute; DMEM–Dulbecco's modification of Eagle's medium; CS–Calf serum; FB–Fetal bovine serum; HoS–Horse serum)

Contamination : It is rather difficult to get serum totally free from all pathogens, particularly viruses.

Presence of growth inhibitors : In general, the concentration of growth promoters in the serum is much higher than the inhibitors. But sometimes, the growth inhibitors such as TGF-β may dominate and inhibit cell proliferation.

Availability and cost : There is a dependence on the cattle for the supply of serum. Hence the availability may be restricted on several occasions for political and economic reasons. Further, cost also is another factor for discouraging the use of serum.

Downstream processing : The presence of serum in the culture medium ***interferes with the isolation and purification of cell culture products.*** For this reason, several additional steps may be required for the isolation of the desired product.

ADVANTAGES AND DISADVANTAGES OF SERUM-FREE MEDIA

Advantages

The limitations associated with the use of serum in the media (described above) are eliminated in the serum-free media. In addition, there are two more distinct advantages.

Selection of media with defined composition : The main advantage of serum-free medium is to ***control growth of the cells as desired, with a well defined medium.*** This is in contrast to the use of serum wherein the growth frequently proceeds in an uncontrolled fashion.

Regulation of differentiation : It is possible to use a factor or a set of factors to achieve differentiation of cells with the desired and specialised functions.

Disadvantages

Slow cell proliferation : Most of the serum-free media are not as efficient as serum added media in the growth promotion of cells.

Need for multiple media : A large number of serum-free media need to be developed for different cell lines. This may create some practical difficulties in a laboratory simultaneously handling several cell lines. Another limitation of serum-free medium is that a given medium may not be able to support the different stages of development even

for a given cell line. Hence, sometimes separate media may be required even for the same cell line.

Purity of reagents : The native serum does possess some amount of protective and detoxifying machinary that can offer a cleansing effect on the apparatus and reagents. And therefore, in the absence of serum, pure grade reagents and completely sterile apparatus should be used.

Availability and cost : In general, the serum-free media are costlier than the serum added media. This is mainly due to the fact that many of the pure chemicals added to the serum-free media are themselves expensive. Further, the availability of serum-free media is also another limitation.

DEVELOPMENT OF SERUM-FREE MEDIA

While designing serum-free media, it is desirable to identify the various serum constituents (Refer ***Table 34.3***) and their quantities. The most important constituents of natural serum with reference to their use in cell cultures may be categorized as follows.

- Growth regulatory factors e.g. PDGF, TGF-β.
- Cell adhesion factors e.g. vitamins.
- Essential nutrients e.g. vitamins, metabolites, minerals, fatty acids.
- Hormones e.g. insulin, hydrocortisone.

For replacing the serum and development of serum-free media, several constituents should supplement the media. Some highlights are given below.

Growth factors

A large number of growth factors (nearly 100) that promote *in vitro* cell proliferation and differentiation have been idenfitied. Besides the factors described already (above), some others are listed below.

- Erythropoietin (EPO).
- Eye-derived growth factors (EDGF 1 and EDGF 2).
- Interleukins (IL-1, IL-2).
- Hepatocyte growth factor (HGF).
- Brain-derived neurotrophic factor (BDNF).
- Phytohemagglutinin (PHA).
- Lipopolysaccharide (LPS).

The growth factors may act synergistically or additively with each other or with other factors (e.g. hormones, prostaglandins).

TABLE 34.5 A selected list of cell lines along with serum-free media

Cell line	*Medium*
Chick embryo fibroblasts	MCDB 202
Chinese hamster ovary (CHO)	MCDB 402
Human lung fibroblasts	MCDB 110
Human vascular endothelium	MCDB 131
Mammary epithelium	MCDB 170
Prostatic epithelium	WAJC 404
Bronchial epithelium	LHC 9
Fibroblasts	MCDB 202
3T3 cells	MCDB 402

Almost all the growth factors are now commercially available for the preparation of serum-free media.

Hormones

Growth hormone, insulin and hydrocortisone are the most commonly added hormones into the serum-free media. A combination of steroid hormones-hydrocortisone, estrogen, androgen and progesterone are used in formulating serum-free media for the maintenance of mammary epithelium.

Nutrients

Addition of certain nutrients-choline, ethanol-amine, linoleic acid, iron, copper, selenium etc., are added in most of the serum-free media.

Proteins

Bovine serum albumin (BSA) is the most commonly added protein. It promotes cell survival and growth.

Polyamines

Putrescine is the most widely added polyamine to the serum-free media. Polyamines promote cellular growth and differentiation.

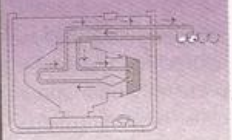
Protease inhibitors

Addition of protease inhibitors (e.g. soy bean trypsin inhibitor) is done to the serum-free media for the trypsin-mediated subcultures.

COMMONLY USED SERUM-FREE MEDIA

Several types of serum-free media have been developed for different cell lines. A selected list of cell lines and the media is given in the ***Table 34.5***.

Cultured Cells – Biology and Characterization



It is an accepted fact that the *in vitro* cultured cells do not possess the same characteristics as the *in vivo* cells. This is mostly due to the changes in the cellular environment.

CHARACTERISTICS OF CULTURED CELLS

Some of the important ***distinguishing properties of cultured cells*** are given below.

1. Cells which do not normally proliferate *in vivo* can be grown and proliferated in cultures.
2. ***Cell to cell interactions*** in the cultured cells are very ***much low***.
3. The three dimensional architecture of the *in vivo* cells is not found in cultured cells.
4. The hormonal and nutritional influence on the cultured cells differs from that on the *in vivo* cells.
5. Cultured cells ***cannot perform*** differentiated and ***specialized functions***.
6. The environment of the cultured cells favours proliferation and spreading of unspecialized cells.

Environmental influence on cultured cells

The environmental factors strongly influence the cells in culture. The major routes through which environmental influence occurs are listed.

- The nature of the substrate or phase in which cells grow. For monolayer cultures, the substrate is a solid (e.g. plastic) while for suspension cultures, it is a liquid.
- The composition of the medium used for culture-nutrients and physicochemical properties.
- Addition of hormones and growth factors.
- The composition of the gas phase.
- The temperature of culture incubation.

The ***biological and other aspects of cultured cells*** with special reference to the following parameters are briefly described.

1. Cell adhesion.
2. Cell proliferation.
3. Cell differentiation.
4. Metabolism of cultured cells.
5. Initiation of cell culture.
6. Evolution and development of cell lines.

CELL ADHESION

Most of the cells obtained from solid tissues grow as adherent monolayers in cultures. The cells, derived from tissue aggregation or subculture, attach to the substrate and then start proliferating. In the early days of culture techniques, slightly negatively charged glasses were used as substrates. In recent years, plastics such as polystyrene, after treatment with electric ion discharge, are in use.

The cell adhesion occurs through cell surface receptors for the molecules in the extracellular matrix. It appears that the cells secrete matrix proteins which spread on the substrate. Then the cells bind to matrix through receptors. It is a common observation that the substrates (glass or plastic) with previous cell culture are conditioned to provide better surface area for adhesion.

Cell adhesion molecules

Three groups of proteins collectively referred to as cell adhesion molecules (***CAMs***) are involved in the cell-cell adhesion and cell-substrate adhesion.

Cell-cell adhesion molecules: These proteins are primarily involved in cell-to-cell interaction between the homologous cells. CAMs are of two types — calcium-dependent ones ***(cadherins)*** and calcium-independent CAMs.

Integrins : These molecules mediate the cell substrate interactions. Integrins possess receptors for matrix molecules such as fibronectin and collagen.

Proteoglycans : These are low affinity transmembrane receptors. Proteoglycans can bind to matrix collagen and growth factors.

Cell adhesion molecules are attached to the cytoskeletons of the cultured cells.

CELL PROLIFERATION

Proliferation of cultured cells occurs through the ***cell cycle***, which has four distinct phases (***Fig. 35.1***)

M phase : In this phase (***M = mitosis***), the two chromatids, which constitute the chromosomes, segregate to daughter cells.

G_1 phase : This ***gap 1 phase*** is highly susceptible to various control processes that determine whether cell should proceed towards DNA synthesis, re-enter the cycle or take the course towards differentiation.

S phase : This phase is characterized by ***DNA synthesis*** wherein DNA replication occurs.

G_2 phase : This is ***gap 2 phase*** that prepares the cell for reentry into mitosis.

The integrity of the DNA, its repair or entry into ***apoptosis*** (programmed cell death) if repair is not possible is determined by two check points-at the beginning of DNA synthesis and in G_2 phase.

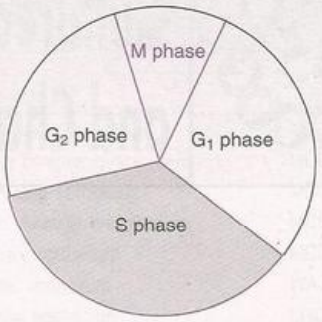


Fig. 35.1 : The cell cycle.

Control of cell proliferation

For the cells in culture, the environmental signals regulate the cell cycle, and thereby the cell proliferation. Low density of the cells in a medium coupled with the presence certain growth factors (e.g. epidermal growth factor, platelet-derived growth factor) allows the cells to enter the cell cycle. On the other hand, high cell density and crowding of cells inhibits the cell cycle and thereby proliferation.

Besides the influence of the environmental factors, certain intracellular factors also regulate the cell cycle. For instance, ***cyclins promote while p53 and Rb gene products inhibit cell cycle.***

CELL DIFFERENTIATION

The various cell culture conditions favour maximum cell proliferation and propagation of cell lines. Among the factors that promote cell proliferation, the following are important.

- Low cell density
- Low Ca^{2+} concentration
- Presence of growth factors

For the process of cell differentiation to occur, the proliferation of cells has to be severely limited or completely abolished. Cell differentiation can be promoted (or induced) by the following factors.

- High cell density.
- High Ca^{2+} concentration.
- Presence of differentiation inducers (e.g. hydrocortisone, nerve growth factor).

As is evident from the above, different and almost opposing conditions are required for cell proliferation, and for cell differentiation. Therefore if cell differentiation is required two distinct sets of conditions are necessary.

1. To optimize cell proliferation.
2. To optimize cell differentiation.

Maintenance of differentiation

It is now recognized that the ***cells retain their native and original functions for long when their three dimensional structures are retained***. This is possible with organ cultures. However, organ cultures cannot be propagated. In recent years, some workers are trying to create three dimensional structures by perfusing monolayer cultures. Further, *in vitro* culturing of cells on or in special matrices (e.g. cellulose, collagen gel, matrix of glycoproteins) also results in cells with three dimensional structures.

Dedifferentiation

Dedifferentiation refers to the ***irreversible loss of specialized properties of cells when they are cultured in vitro***. This happens when the differentiated *in vitro* cells lose their properties (***Fig. 35.2***).

In the ***in vivo situation, a small group of stem cells give rise to progenitor cells that are capable of producing differentiated cell pool*** (***Fig. 35.2A***). On the other hand, in the *in vitro* culture system, progenitor cells are predominantly produced which go on proliferating. Very few of the newly formed cells can form differentiated cells (***Fig. 35.2B***). The net result is a blocked differentiation.

As already described, dedifferentiation implies an irreversible loss of specialized properties of the cells. On the other hand, ***deadaptation*** refers to the reinduction of specialized properties of the cells by creating appropriate conditions.

METABOLISM OF CULTURED CELLS

The metabolism of mammalian cultured cells with special reference to energy aspects is depicted in ***Fig. 35.3***. The cultured cells can use ***glucose or glutamine as the source of energy***. These two compounds also generate important anabolic precursors.

As glucose gets degraded by glycolysis, lactate is mainly produced. This is because oxygen is in limited supply in the normal culture conditions (i.e. atmospheric oxygen and a submerged culture) creating an anaerobic situation. Lactate, secreted into the medium, accumulates. Some amount of pyruvate produced in glycolysis gets oxidized through Krebs cycle. ***A small fraction of glucose*** (4-9%) ***enters pentose phosphate pathway*** to supply ribose 5-phosphate and reducing equivalents (NADPH) for biosynthetic pathways e.g. synthesis of nucleotides.

Glutamine is an important source of energy for the cultured cells. By the action of the enzyme glutaminase, glutamine undergoes deamination to produce glutamate and ammonium ions. Glutamate, on transamination (or oxidative deamination) forms α-ketoglutarate which enters the Krebs cycle. Pyruvate predominantly participates in transamination reaction to produce alanine, which is easily excreted into the medium. In the rapidly growing cultured cells, transamination reaction is a dominant route of glutamine metabolism.

Deamination of glutamine releases free ammonium ions, which are toxic to the cultured cells, limiting their growth. In recent years, dipeptides ***glutamyl-alanine*** or ***glutamyl-glycine*** are being used to minimize the production of ammonia. Further, these dipeptides are more stable in the medium.

As already stated, α-ketoglutarate obtained from glutamine (via glutamate) enters the Krebs cycle and gets oxidized to carbon dioxide and water. For proper operation of Kerbs cycle, balancing of the intermediates of the cycle is required. Two metabolites of Kerbs cycle namely malate and oxaloacetate leave the cycle and get converted respectively to pyruvate and phosphoenol pyruvate. The latter two compounds can reenter the Krebs cycle in the form of acetyl CoA. Thus, the continuity of Kerbs cycle is maintained.

Glucose as well as glutamine gets metabolised by the cultured cells to supply energy in the form of ATP.

INITIATION OF CELL CULTURE

The cell culture can be initiated by the cells derived from a tissue through enzymatic or mechanical treatments. For details on primary culture, the reader must refer Chapter 36. Primary culture is a selective process that finally results in a relatively uniform cell line. The selection occurs by virtue of the capacity of the cells to survive as

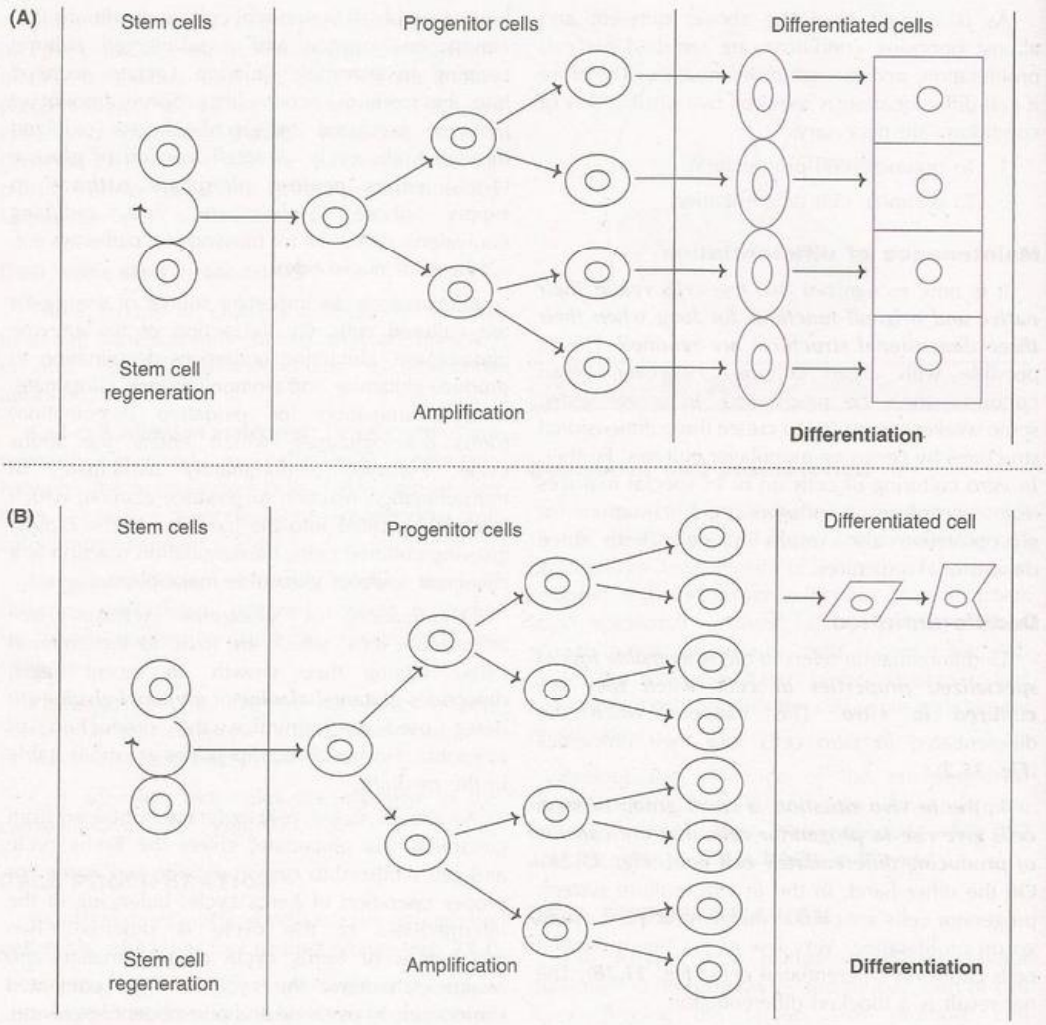


Fig. 35.2 : Differentiation of cells (A) In vivo differentiation of stem cells (B) Blocked differentiation in cultured cells.

monolayer cultures (by adhering to substrates) or as suspension cultures.

Among the cultured cells, some cells can grow and proliferate while some are unable to survive under the culture environment. The cells continue to grow in monolayer cultures, till the availability of the substrate is occupied.

The term ***confluence*** is used when ***the cultured cells make close contact with one another by fully utilizing the available growth area***. For certain cells, which are sensitive to growth limitation due to density, the cells stop growing once confluence is reached. However, the transformed cells are insensitive to confluence and continue to overgrow.

When the culture becomes confluent, the cells possess the following characters.

1. The closest morphological resemblance to the tissue of origin (i.e. parent tissue).

2. The expression of specialized functions of the cells comparable to that of the native cells.

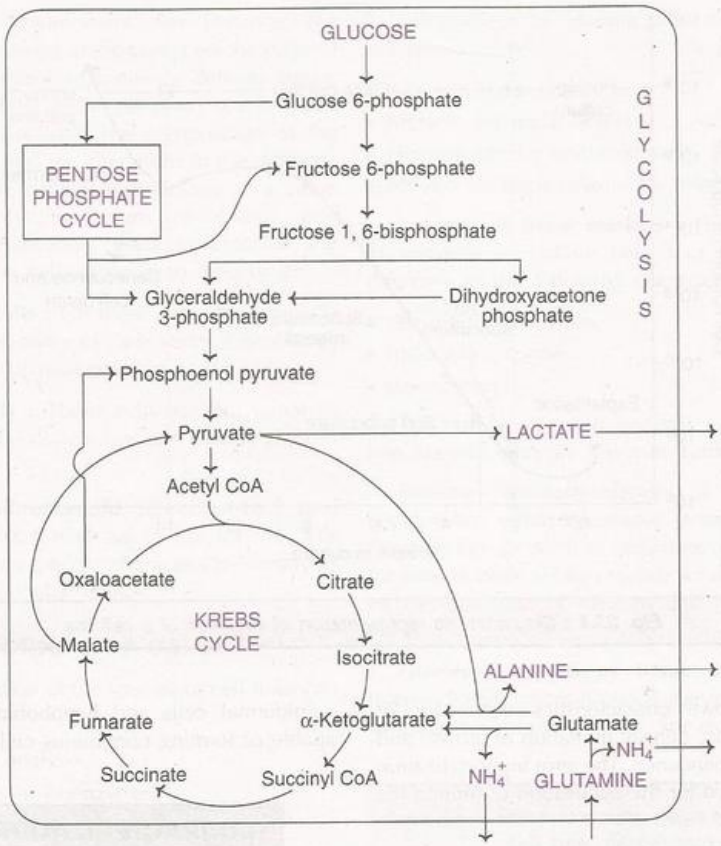


Fig. 35.3 : An outline of the glucose and glutamine metabolism in cultured mammalian cells.

EVOLUTION AND DEVELOPMENT OF CELL LINES

The primary culture grown after the first subculture is referred to as cell line. A given cell line may be propagated by further subculturing. As the subcultures are repeated, the most rapidly proliferating cells dominate while the non-proliferating or slowly proliferating cells will get diluted, and consequently disappear.

Senescence : The genetically determined event of cell divisions for a limited number of times (i.e. population doublings), followed by their ***death in a normal tissue is referred to as senescence.*** However, germ cells and transformed cells are capable of continuously proliferating. In the *in vitro* culture, transformed cells can give rise to continuous cell lines.

The evolution of a continuous cell line is depicted in ***Fig. 35.4***. The cumulative cell number in a culture is represented on Y-axis on a log scale, while the X-axis represents the time in weeks. The time for development of a continuous cell line is variable. For instance, for human diploid fibroblasts, the continuous cell line arises at about 14 weeks while the senescence may occur between 10 to 20 weeks; usually after 30 and 60 cell doublings.

Development of continuous cell lines

Certain alterations in the culture, collectively referred to as ***transformation,*** can give rise to continuous cell lines. Transformation may be spontaneously occurring, chemically or virally-induced. Transformation basically involves an

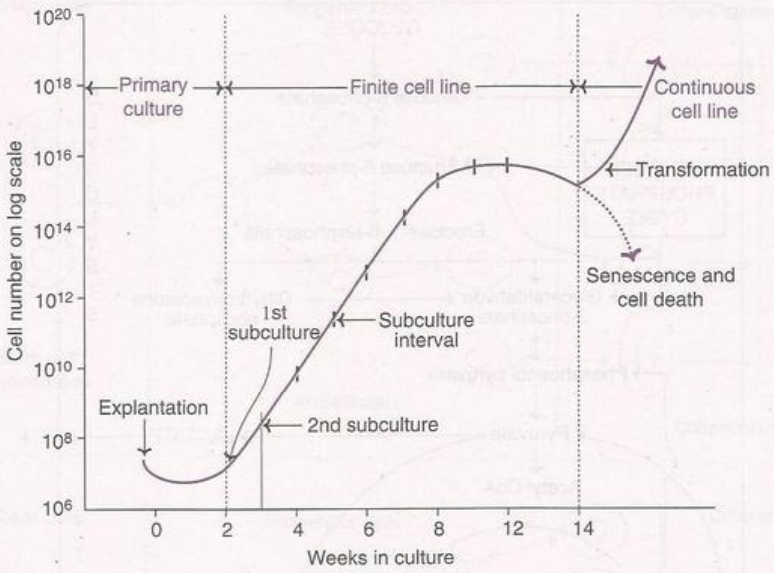


Fig. 35.4 : Diagrammatic representation of evolution of a cell line.

alteration in growth characteristics such as loss of contact inhibition, density limitation of growth and anchorage independence. The term immortalization is frequently used for the acquisition of infinite life span to cultured cells.

Genetic variations : The ability of the cells to grow continuously in cell lines represents genetic variation in the cells. Most often, the deletion or mutation of the p^{53} gene is responsible for continuous proliferation of cells. In the normal cells, the normal p^{53} gene is responsible for the arrest of cell cycle.

Most of the continuous cell lines are ***aneuploid***, possessing chromosome number between diploid and tetraploid value.

Normal cells and continuous cell lines

A great majority of normal cells are not capable of giving rise to continuous cell lines. For instance, normal human fibroblasts go on proliferating for about 50 generations, and then stop dividing. However, they remain viable for about 18 months. And throughout their life span, fibroblasts remain euploid. Chick fibroblasts also behave in a similar fashion.

Epidermal cells and lymphoblastoid cells are capable of forming continuous cell lines.

CHARACTERIZATION OF CULTURED CELLS

Characterization of cultured cells or cell lines is important for dissemination of cell lines through cell banks, and to establish contacts between research laboratories and commercial companies.

Characterization of cell lines with special reference to the following aspects is generally done.

1. Morphology of cells
2. Species of origin.
3. Tissue of origin.
4. Whether cell line is transformed or not.
5. Identification of specific cell lines.

MORPHOLOGY OF CELLS

A simple and direct identification of the cultured cells can be done by observing their morphological characteristics. However, the morphology has to be viewed with caution since it is largely dependent

on the culture environment. For instance, the epithelial cells growing at the center (of the culture) are regular polygonal with clearly defined edges, while those growing at the periphery are irregular and distended (swollen). The composition of the culture medium, and the alterations in the substrate also influence the cellular morphology. In a tissue culture laboratory, the terms fibroblastic and epithelial are commonly used to describe the appearance of the cells rather than their origin.

Fibroblastic cells : For these cells, the length is usually more than twice of their width. Fibroblastic cells are bipolar or multipolar in nature.

Epithelial cells : These cells are polygonal in nature with regular dimensions and usually grow in monolayers.

The terms ***fibroblastoid*** (fibroblast-like) and ***epitheloid*** (epithelial-like) are in use for the cells that do not possess specific characters to identify as fibroblastic or epithelial cells.

SPECIES OF ORIGIN OF CELLS

The identification of the species of cell lines can be done by.

- Chromosomal analysis.
- Electrophoresis of isoenzymes.
- A combination of both these methods.

In recent years, chromosomal identification is being done by employing molecular probes.

IDENTIFICATION OF TISSUE OF ORIGIN

The identification of cell lines with regard to tissue of origin is carried out with reference to the following two characteristics.

1. The lineage to which the cells belong.

2. The status of the cells i.e. stem cells, precursor cells.

Tissue markers for cell line identification

Some of the important tissue or lineage markers for cell line identification are briefly described.

Differentiated products as cell markers : The cultured cells, on complete expression, are capable of producing differentiation markers, which serve as cell markers for identification. Some examples are given below.

- Albumin for hepatocytes.
- Melanin for melanocytes
- Hemoglobin for erythroid cells
- Myosin (or tropomyosin) for muscle cells.

Enzymes as tissue markers : The identification of enzymes in culture cells can be made with reference to the following characters.

- Constitutive enzymes.
- Inducible enzymes.
- Isoenzymes.

The commonly used enzyme markers for cell line identification are given in ***Table 35.1***.

Tyrosine aminotransferase is specific for hepatocytes, while tyrosinase is for melanocytes. Creatine kinase (MM) in serum serves as a marker for muscle cells, while creatine kinase (BB) is used for the detection of neurons and neuroendocrine cells.

Filament proteins as tissue markers : The intermediate filament proteins are very widely used as tissue or lineage markers. For example—

TABLE 35.1 Some common enzyme markers for cell line identification

Enzyme	*Cell type*
Tyrosine aminotransferase	Hepatocytes
Tysosinase	Melanocytes
Glutamyl synthase	Brain (astroglia)
Creatine kinase (isoenzyme MM)	Muscle cells
Creatine kinase (isoenzyme BB)	Neurons, neuroendocrine cells
Non-specific esterase	Macrophages
DOPA-decarboxylase	Neurons
Alkaline phosphatase	Enterocytes, type II pneumocytes
Angiotensin–converting enzyme	Endothelium
Sucrase	Enterocytes
Neuron-specific esterase	Neurons

DOPA–Dihydroxy phenylalanine; MM–Two polypeptide subunits of muscle; BB–Two polypeptide subunits of brain.

Table 35.2 A selected list of antibodies used for the detection of cell types

Antibody	*Cell type*
Cytokeratin	Epithelium
Epithelial membrane antigen	Epithelium
Albumin	Hepatocytes
α-Lactalbumin	Breast epithelium
Carcinoembryonic antigen (CEA)	Colorectal and lung adenocarcinoma
Prostate specific antigen (PSA)	Prostatic epithelium
Intracellular cell adhesion molecule (I-CAM)	T-cells and endothelium
α-Fetoprotein	Fetal hepatocytes
Human chorionic gonadotropin (hCG)	Placental epithelium
Human growth hormone (hGH)	Anterior pituitary
Vimentin	Mesodermal cells
Integrins	All cells
Actin	All cells

- Astrocytes can be detected by glial fibrillary acidic protein (GFAP).
- Muscle cells can be identified by desmin.
- Epithelial and mesothelial cells by cytokeratin.

Cell surface antigens as tissue markers : The antigens of the cultured cells are useful for the detection of tissue or cells of origin. In fact, many antibodies have been developed (commercial kits are available) for the identification cell lines (***Table. 35.2***). These antibodies are raised against cell surface antigens or other proteins.

The antibodies raised against secreted antigen α-fetoprotein serves as a marker for the identification of fetal hepatocytes. Antibodies of cell surface antigens namely integrins can be used for the general detection of cell lines.

TRANSFORMED CELLS

Transformation is the phenomenon of the ***change in phenotype due to the acquirement of new genetic material***. Transformation is associated with promotion of genetic instability. The transformed and cultured cells exhibit alterations in many characters with reference to

- Growth rate
- Mode of growth
- Longevity
- Tumorigenicity
- Specialized product formation.

While characterizing the cell lines, it is necessary to consider the above characters to determine whether the cell line has originated from tumor cells or has undergone transformation in culture.

For more details on transformation, Refer Chapter 6.

IDENTIFICATION OF SPECIFIC CELL LINES

There are many approaches in a culture laboratory to identify specific cell lines.

- Chromosome analysis
- DNA detection
- RNA and protein analysis
- Enzyme activities
- Antigenic markers.

Chromosome analysis

The species and sex from which the cell line is derived can be identified by chromosome analysis. Further, it is also ***possible to distinguish normal and malignant cells by the analysis of chromosomes***. It may be noted that the normal cells contain more stable chromosomes. The important techniques employed with regard to chromosome analysis are briefly described.

Chromosome banding : By this technique, it is possible to identify individual chromosome pairs when there is little morphological difference between them. Chromosome banding can be done by using Giemsa staining.

Chromosome count : A direct count of chromosomes can be done per spread between 50–100 spreads. A camera Lucida attachment or a closed circuit television may be useful.

Chromosome karyotyping : In this technique, the chromosomes are cut, sorted into sequence, and then pasted on to a sheet. The image can be recorded or scanned from the slide. Chromosome karyotyping is time consuming when compared to chromosome counting.

DNA detection

The ***total quantity of DNA per normal cell*** is quite constant, and ***is characteristic to the species of origin***. e.g. normal cell lines from human, chick and hamster fibroblasts. However, the DNA content varies in the normal cell lines of mouse, and also the cell lines obtained from cancerous tissues.

As already stated, most of the transformed cells are aneuploid and heteroploid. DNA analysis is particularly useful for characterization of such cells.

Analysis of DNA can be carried out by ***DNA hybridization and DNA fingerprinting.***

DNA hybridization : The popular ***Southern blotting technique*** (For details, Refer Chapter 7), can be used to detect unique DNA sequences. Specific molecular probes with radioisotope, fluorescent or luminescent labels can be used for this purpose.

The DNA from the desired cell lines is extracted, cut with restriction endonucleases, subjected to electrophoresis, blotted on to nitrocellulose, and then hybridized with a molecular (labeled) probe, or a set of probes. By this approach, specific sequences of DNA in the cell lines can be detected.

DNA fingerprinting : There are certain regions in the DNA of a cell that are not transcribed. These regions, referred to as satellite DNA, have no known functions, and it is believed that they may provide reservoir for genetic evolution. Satellite DNA regions are considered as regions of hypervariability. These regions may be cut with specific restriction endonucleases, and detected by using cDNA probes.

By using electrophoresis and autoradiography, the patterns of satellite DNA variations can be detected. Such patterns referred to as DNA fingerprints are cell line specific.

In recent years, the technique of DNA fingerprinting has become a very popular and a powerful tool to determine the origin of cell lines.

RNA and protein analysis

The phenotype characteristics of a cell line can be detected by gene expression i.e. identification of RNAs and/or proteins. ***mRNAs can be identified by Northern blot technique while proteins can be detected by Western blot technique***.

Enzyme activities

Some of the *in vivo* enzyme activities are lost when the cells are cultured *in vitro*. For instance, arginase activity of the liver cells is lost within a few days of culturing. However, ***certain cell lines express specific enzymes that can be employed for their detection*** e.g. tyrosine aminotransferase for hepatocytes, glutamyl synthase activity for astroglia in brain. For more examples of enzymes useful in cell line detection, refer ***Table 35.1***.

Isoenzymes : The multiple forms of an enzyme catalysing the same reaction are referred to as isoenzymes or isozymes. Isoenzymes differ in many physical and chemical properties—structure, electrophoretic and immunological properties, K_m and V_{max} values.

The isoenzymes can be separated by analytical techniques such as electrophoresis and chromatography. Most frequently, electrophoresis by employing agarose, cellulose acetate, starch and polyacrylamide is used. The crude enzyme is applied at one point on the electrophoretic medium. As the isoenzymes migrate, they distribute in different bands, which can be detected by staining with suitable chromogenic substrates.

Isoenzymes are characteristic to the species or tissues. Isoenzymes of the following enzymes are commonly ***used for cell line detection***.

- Lactate dehydrogenase
- Malate dehydrogenase
- Glucose 6-phosphate dehydrogenase
- Aspartate aminotransferase
- Peptidase B.

Isoenzyme analysis is also useful for the detection of interspecies cross-contamination of cell lines. For instance, contamination of mouse cell line with hamster cell line can be identified by using peptidase B isoenzymes.

Antigenic markers

Cell lines can be characterized by detection of antigenic markers through the use of antibodies. The antigenic markers may be located on the cell surface or secreted by the cells into the culture medium.

Some of the antibodies in common use for the detection of different cell types are given in ***Table 35.2*** (See p. 430).

MEASUREMENT OF GROWTH PARAMETERS OF CULTURED CELLS

Information on the growth state of a given culture is required to :

- Design culture experiments.
- Routine maintenance of culture.
- Measurement of cell proliferation.
- Know the time for subculture.
- Determine the culture response to a particular stimulus or toxin.

Some of the commonly used terms in relation to the measurement of growth of cultured cells are explained.

Population doubling time (PDT) : The time interval for the cell population to double at the middle of the logarithmic (log) phase.

Cell cycle time or generation time : The interval from one point in the cell division to the same point in the cycle, one division later. Thus cell cycle time is measured form one point in the cell cycle until the same point is reached again.

Confluence : It denotes the culture stage wherein all the available substrate (growth area) is utilized, and the cells are in close contact with each other.

Contact inhibition : Inhibition of cell motility and plasma membrane ruffling when the cells are in complete contact with other adjacent cells. This mostly occurs at confluence state, and results in the ceasation of the cell proliferation.

Cell density : The number of cells per ml of the medium.

Saturation density : The density of the cells (cells/ml^2 surface area) in the plateau phase.

GROWTH CYCLE OF CULTURED CELLS

The growth cycle of cultured cells is conventionally represented by three phases — the ***lag phase***, the ***log (exponential) phase*** and the ***plateau phase*** (***Fig. 35.5***). The properties of the cultured cells vary in the phases.

The lag phase

The lag phase represents a period of adaptation during which the cell forms the cell surface and extracellular matrix (lost during trypsinization),

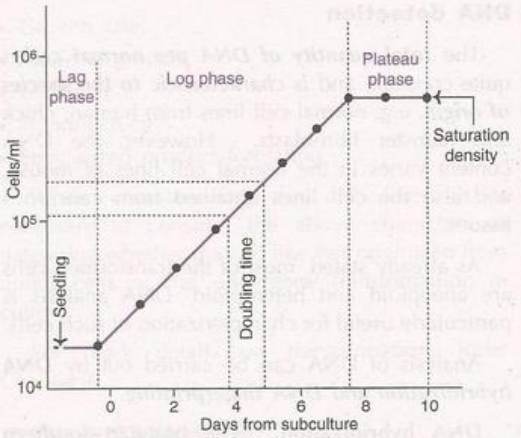


*Fig. 35.5 : Growth curve of cultured cells (**Note :** The cell concentration is expressed in semilog plot).*

attaches to the substrate and spreads out. There is an increased synthesis of certain enzymes (e.g. DNA polymerase) and structural proteins, preparing the cells for proliferation. The production of specialized products disappears which may not reappear until the cell proliferation ceases.

The lag phase represents preparative stage of the cells for proliferation following subculture and reseeding.

The log phase

The log phase is characterized by an exponential growth of cells, following the lag phase. The duration of log phase depends on the cells with reference to :

- Seeding density.
- Growth rate.
- Density after proliferation.

During the log phase, the cultured cells are in the most uniform and reproducible state with high viability. This is an ideal time for sampling.

The log phase terminates after confluence is reached with an addition of one or two population doublings.

The plateau phase

As the cells reach confluence, the growth rate is much reduced, and the proliferation of cultured

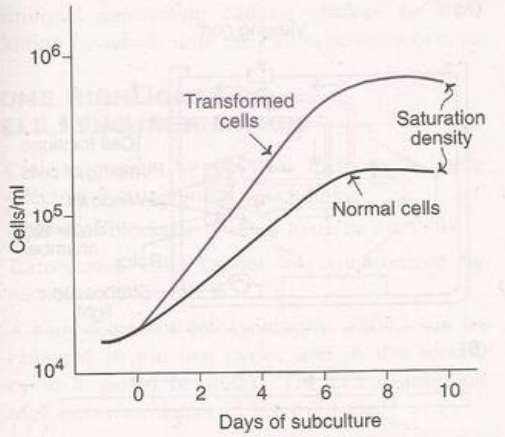


*Fig. 35.6 : Growth curves of transformed and normal cells (**Note** : The cell concentration is expressed in semilog plot).*

cells almost stops. This stage represents plateau or stationary phase, and is characterized by.

- Low motility of cells.
- Reduced ruffling of plasma membrane.
- Cells occupying minimum surface area.
- Contact inhibition.
- Saturation density.
- Depletion of nutrients and growth factors.
- Reduced synthesis of structural proteins.
- Increased formation of specialized products.

The majority of normal cultured cells that form monolayers stop growing as they reach confluence. Some of the cells however, with replenishment of medium continue to grow (at a reduced rate) after confluence, forming multilayers of cells.

The transformed cultured cells usually reach a higher cell density compared to the normal cells in the plateau phase (***Fig. 35.6***).

PLATING EFFICIENCY OF CULTURED CELLS

Plating efficiency, representing colony formation at low cell density, is a measure used for analyzing cell proliferation and survival.

When the cells, at low densities, are cultured in the form of single cell suspensions, they grow as discrete colonies. Plating efficiency is calculated as follows.

$$\text{Plating efficiency} = \frac{\text{No. of colonies formed}}{\text{No. of cells seeded}} \times 100$$

The term ***cloning efficiency*** is used (instead of plating efficiency) when each colony grows from a single cell.

Seeding efficiency representing the survival of cells at higher densities, is calculated as follows.

$$\text{Seeding efficiency} = \frac{\text{No. of cells recovered}}{\text{No. of cells seeded}} \times 100$$

CELL SYNCHRONIZATION

Synchronization literally means to make two or more things happen exactly simultaneously. For instance, two or more watches can be synchronized to show exactly the same time.

The ***cells at different stages of the cell cycle in a culture can be synchronized so that the cells will be at the same phase.*** Cell synchrony is required to study the progression of cells through cell cycle. Several laboratory techniques have been developed to achieve cell synchronization. They are broadly categorized into two groups.

1. Physical fractionation for cell separation.
2. Chemical blockade for cell separation.

CELL SEPARATION BY PHYSICAL MEANS

Physical fractionation or cell separation techniques, based on the following characteristics are in use.

- Cell density.
- Cell size.
- Affinity of antibodies on cell surface epitopes.
- Light scatter or fluorescent emission by labeled cells.

The two commonly used techniques namely centrifugal elutriation and fluorescence-activated cell separation are briefly described hereunder.

Centrifugal elutriation

The physical characteristics — ***cell size and sedimentation velocity are operative*** in the technique of centrifugal elutriation.

Centrifugal elutriator (from Beckman) is an advanced device for increasing the sedimentation rate so that the yield and resolution of cells is better. The cell separation is carried out in a specially designed centrifuge and rotor (***Fig. 35.7***). The cells in the medium are pumped into the separating chamber while the rotor is turning. Due to centrifugal force, the cell will be pushed to the edges. As the medium is then pumped through the chamber in such a way that the centripetal flow is equal to the sedimentation rate of cells. Due to differences in the cells (size, density, cell surface configuration), the cells tend to sediment at different rates, and reach equilibrium at different positions in the chamber. The entire operation in the elutriator can be viewed through the port, as the chamber is illuminated by stroboscopic light. At the equilibrium the flow rate can be increased and the cells can be pumped out, and separated in collecting vessels in different fractions. It is possible to carry out separation of cells in a complete medium, so that the cells can be directly cultured after separation.

Fluorescence-activated cell sorting

Fluorescence-activated cell sorting is a technique for sorting out the cells ***based on*** the differences that can be detected by light scatter (e.g. cell size) ***or fluorescence emission*** (by pretreated DNA, RNA, proteins, antigens).

The procedure involves passing of a single stream of cells through a laser beam so that the scattered light from the cells can be detected and recorded. When the cells are pretreated with a fluorescent stain (e.g. chromomycin A for DNA), the fluorescent emission excited by the laser can be detected.

There are two instruments in use based on the principle of fluorescent-activated cell sorting.

1. **Flow cytometer :** This instrument is capable of sorting out cells (from a population) in different phases of the cell cycle based on the measurements of a combination of cell size and DNA fluorescence.

2. **Fluorescent-activated cell sorter (FACS) :** In this instrument, the emission signals from the cells are measured, and the cells sorted out into collection tubes.

Comparison between physical methods

For separation of a large number of cells, centrifugal elutriator is preferred. On the other hand, fluorescent-activated cell sorting is mostly used to obtain high grade pure fractions of cells from small quantities of cells.

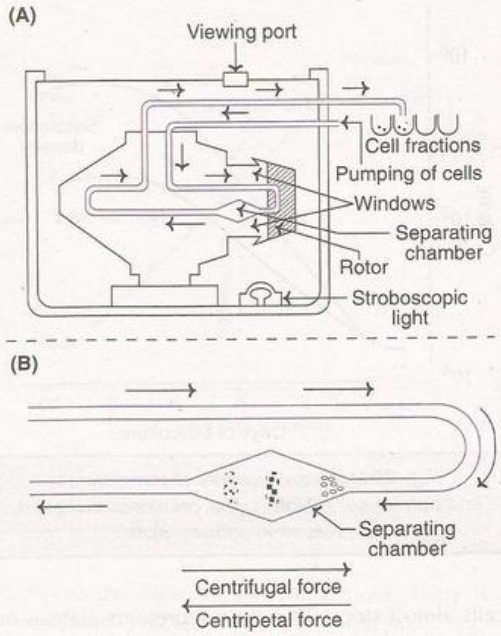


*Fig. 35.7 : (**A**) Diagrammatic view of a centrifugal elutriator, (**B**) Separation chamber of elutriator.*

CELL SEPARATION BY CHEMICAL BLOCKADE

The cells can be separated by blocking metabolic reactions. Two types of metabolic blockades are in use — inhibition of DNA synthesis and nutritional deprivation.

Inhibition of DNA synthesis

During the S phase of cell cycle, DNA synthesis can be inhibited by using inhibitors such as thymidine, aminopterine, hydroxyurea and cytosine arabinoside. The effects of these inhibitors are variable. The cell cycle is predominantly blocked in S phase that results in viable cells.

Nutritional deprivation

Elimination of serum or isoleucine from the culture medium for about 24 hours results in the accumulation of cells at G_1 phase. This effect of

nutritional deprivation can be restored by their addition by which time the cell synchrony occurs.

SOME HIGHLIGHTS OF CELL SYNCHRONIZATION

- Cell separation by physical methods is more effective than chemical procedures.
- Chemical blockade is often toxic to the cells.
- Transformed cells cannot be synchronized by nutritional deprivation.
- A high degree of cell synchrony (>80%) can be obtained in the first cycle, and in the second cycle it would be <60%. The cell distribution may occur randomly in the third cycle.

SENESCENCE AND APOPTOSIS

As the cells grow in culture, they become old due to aging, and they cannot proliferate any more. The ***end of the proliferative life span of cells is referred to as senescence***.

CELLULAR SENESCENCE

The ***growth of the cells*** is usually ***measured*** as ***population doublings (PDs)***. The PDs refer to the number of times the cell population doubles in number during the period of culture and is calculated by the following formula.

$$PD = \frac{\log_{10}(\text{No. of cells harvested}) - \log_{10}(\text{No. of cells seeded})}{\log_{10} 2}$$

The phenomenon of senescence has been mostly studied with human fibroblast cultures. After 30-60 populations doublings, the culture is mainly composed of senescent fibroblasts. These senescent fibroblast are unable to divide in response to mitotic stimuli. It must be noted that the cells do not appear suddenly, but they gradually accumulate and increase in number during the life span of the culture.

The different parameters used for the measurement of cell growth in cultures are listed below.

- Direct measure of cell number.
- Determination of DNA/RNA content.
- Estimation of protein/ATP concentration.

MEASUREMENT OF SENESCENCE

The direct measurement of senescent cells is rather difficult. Some of the indirect measures are

- Loss of metabolic activity
- Lack of labeled precursor (^{3}H-thymidine) incorporation into DNA.
- Certain histochemical techniques.

Senescence-associated β-galactosidase activity assay

There occurs an overexpression of the lysosomal enzyme β-galactosidase at senescence. This enzyme elevation is also associated with an increase in the cell size as the cell enters a permanent non-dividing state.

The number of senescent cells in a culture can be measured by senescence-associated β-galactosidase (SA-β) assay. The assay consists of the following stages.

1. Wash the cells and fix them using a fixative (e.g. para formaldehyde), and wash again.
2. Add the staining solution (X-gal powder in dimethylformamide dissolved in buffer) to the fixed cells and incubate.
3. The senescent cells display a dense blue colour which can be counted.

APOPTOSIS

The process of ***programmed cell death (PCD)*** is referred to as apoptosis. The cell death may be initiated by a specific stimulus or as a result of several signals received from the external environment.

Apoptosis occurs as a result of inherent cellular mechanisms, which finally lead to self destruction. The cell activates a series of molecular events that cause an orderly degradation of the cellular constituents with minimal impact on the neighbouring tissues.

Reasons for *in situ* apoptosis

1. **For proper development :** The formation of fingers and toes of the fetus requires the removal of the tissues between them. This is usually carried out by apoptosis.

2. **Destruction of cells that pose threat to the integrity of the organism :** Programmed cell death is needed to destroy and remove the cells that may

otherwise damage the organisms. Some examples are listed

- Cells with damaged DNA during the course of embryonic development. If they are not destroyed, they may result in birth defects.
- Cells of the immune system, after their appropriate immune function, undergo apoptosis. This is needed to prevent autoimmune diseases e.g. rheumatoid arthritis.
- Cells infected with viruses are destroyed by apoptosis.

3. **Cell destruction due to negative signals :** There are several negative signals within the cells that promote apoptosis. These include accumulation of free radicals, exposure to UV rays, X-rays and chemotherapeutic drugs.

Mechanism of apoptosis

The programmed cell death may occur due to three different mechanisms.

1. Apoptisis due to internal signals.

2. Apoptosis triggered by external signals e.g. tumor necrosis factor-α (TNF-α), lymphotoxin.

3. Apoptosis triggered by reactive oxygen species.

Role of caspases in apoptosis

A group of enzymes namely ***activated proteases*** play a crucial role in the programmed cell death. These proteases are actually ***c***ysteinyl ***a***spartate ***s***pecific ***p***roteina***ses*** or in short, commonly referred to as ***caspases***. There are about ten different types of caspases acting on different substrates ultimately leading to cell death. For instance, capsase I cleaves interleukin 1β.

Inhibition of caspase activities

Since the caspases are closely involved in apoptosis, it is possible to prevent cell death by inhibiting their activities. Certain specific peptides that can inhibit caspases, and thus apoptosis have been identified.

MEASUREMENT OF APOPTOSIS

A simple and easy way of detecting dead or dying cells is the direct microscopic observation. The ***dying cells are rounded with dense bodies*** which can be identified under phase contrast microscope.

The ***cells that have undergone apoptosis contain fragmented chromatin*** which can be detected by conventional staining techniques.

In recent years, more sensitive and reliable techniques have been developed for measuring apoptosis. Some of them are briefly described.

Determination ADP/ATP ratio

Both the growth and apoptosis of cells require ATP. But when there is growth arrest, an elevation of ADP occurs. Thus measuring ADP/ATP ratio will throw light on the dead cells. In fact, some assay systems for measuring ADP/ATP ratios are commercially available.

TUNEL assay

A significant biochemical event for the apoptosis is the activation of endogenous nuclease activity. This enzyme cleaves DNA into fragments with free 3-hydroxyl groups. The newly formed small DNA fragments can be extended by employing the enzyme DNA polymerase. If labeled nucleotides are used for DNA fragment extension, they can be detected.

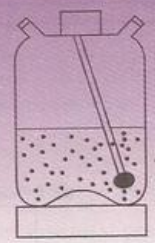
TUNEL is an abbreviation for **T**dT-mediated d**U**TP **n**ick **e**nd-**l**abeling assay. TUNEL is very fast and effective for the determination of DNA fragments formed by endogenous nuclease activity. The apoptotic nuclei can be identified by a fluorescent technique using fluorescein isothiocyanate (FITC) and 4,6-diaminophenylindole.

DNA laddering test

During the course of apoptosis, the genomic DNA is cleaved to mono — and oligonucleosomal DNA fragments. These fragments can be separated by agarose electrophoresis, and detected. The nucleosomal fragments of apoptotic cells give a characteristic ladder pattern on electrophoresis.

Limitations of the test : DNA laddering test is not very specific since several cells that have undergone apoptosis may not show DNA laddering. Further, some cells not subjected to apoptosis may also show DNA ladders, For these reasons, DNA laddering test is coupled with some other test for measurement of apoptosis.

36 Primary Culture and Cell Lines



A primary culture refers to ***the starting culture of cells, tissues or organs, taken directly from an organism***. Thus, the primary culture is the initial culture before the first subculture. The term ***cell line is used for the propagation of cultures*** after the first subculture. Some basic and fundamental aspects of primary culture and cell lines are briefly described.

PRIMARY CELL CULTURE

As already stated, primary culture broadly involves ***the culturing techniques carried following the isolation of the cells, but before the first subculture***. Primary cultures are usually prepared from large tissue masses. Thus, these cultures may contain a variety of differentiated cells e.g. fibroblasts, lymphocytes, macrophages, epithelial cells.

With the experiences of the personnel working in tissue culture laboratories, the following criteria/characteristics are considered for efficient development of primary cultures.

- Embryonic tissues rather than adult tissues are preferred for primary cultures. This is due to the fact that the embryonic cells can be disaggregated easily and yield more viable cells, besides rapidly proliferating *in vitro*.
- The quantity of cells used in the primary culture should be higher since their survival rate is substantially lower (when compared to subcultures).
- The tissues should be processed with minimum damage to cells for use in primary culture. Further, the dead cells should be removed.
- Selection of an appropriate medium (preferably a nutrient rich one) is advisable. For the addition of serum, fetal bovine source is preferred rather than calf or horse serum.
- It is necessary to remove the enzymes used for disaggregation of cells by centrifugation.

TECHNIQUES FOR PRIMARY CULTURE

Among the various techniques devised for the primary culture of isolated tissues, three techniques are most commonly used.

1. Mechanical disaggregation.
2. Enzymatic disaggregation.
3. Primary explant technique.

An outline of these techniques is depicted in ***Fig. 36.1***, and the procedures are briefly described.

MECHANICAL DISAGGREGATION

For the disaggregation of ***soft tissues*** (e.g. spleen, brain, embryonic liver, soft tumors), ***mechanical technique is usually employed***. This technique basically involves careful chopping or slicing of tissue into pieces and collection of spill out cells. The cells can be collected by two ways.

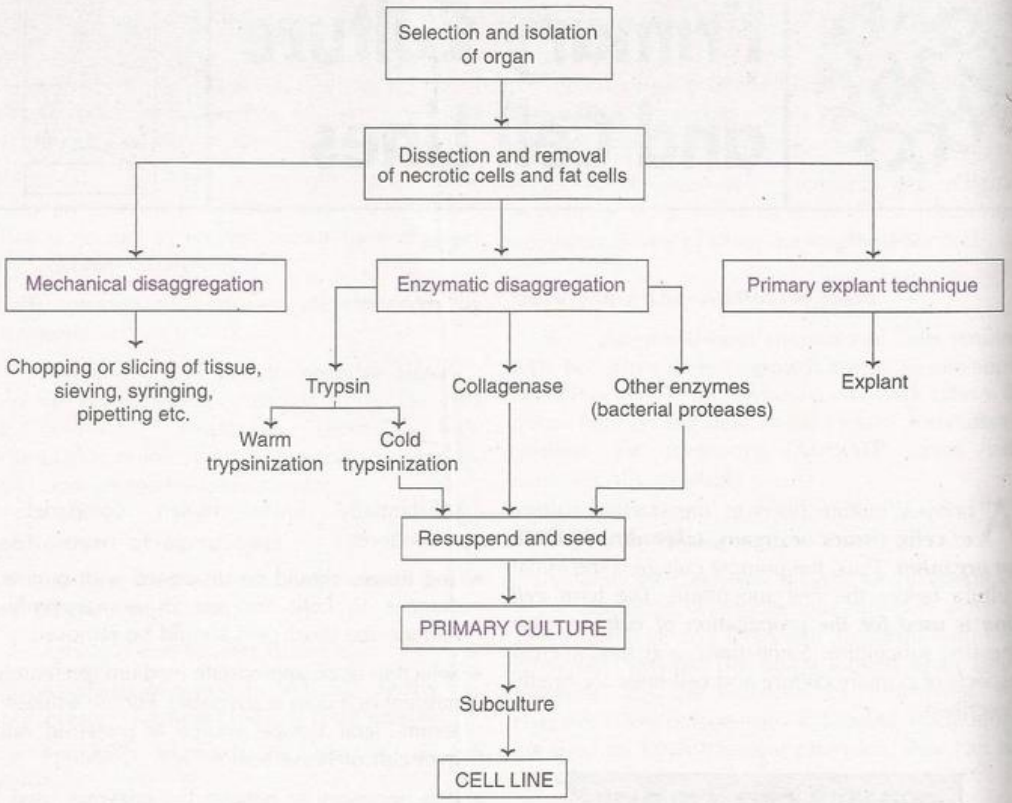


Fig. 36.1 : Different techniques used for primary culture.

- Pressing the tissue pieces through a series of sieves with a gradual reduction in the mesh size.
- Forcing the tissue fragments through a syringe and needle.

Although mechanical disaggregation involves the risk of cell damage, the procedure is less expensive, quick and simple. This technique is particularly useful when the availability of the tissue is in plenty, and the efficiency of the yield is not very crucial. It must however, be noted that the viability of cells obtained from mechanical techniques is much lower than the enzymatic technique.

ENZYMATIC DISAGGREGATION

Enzymatic disaggregation is mostly ***used when high recovery of cells is required from a tissue.*** Disaggregation of embryonic tissues is more efficient with higher yield of cells by use of enzymes. This is due to the presence of less fibrous connective tissue and extracellular matrix. Enzymatic disaggregation can be carried out by using trypsin, collagenase or some other enzymes.

Disaggregation by trypsin

The term ***trypsinization*** is commonly used for disaggregation of tissues by the enzyme, trypsin. Many workers prefer to use crude trypsin rather than pure trypsin for the following reasons.

- The crude trypsin is more effective due to the presence of other proteases
- Cells can tolerate crude trypsin better.
- The residual activity of crude trypsin can be easily neutralized by the serum of the culture media (when serum-free media are used, a trypsin inhibitor can be used for neutralization).

Disaggregation of cells can also be carried out by using pure trypsin which is less toxic and more specific in its action.

The desired tissue is chopped to 2-3 mm pieces and then subjected to disaggregation by trypsin. There are two techniques of trypsinization-warm trypsinization and cold trypsinization (***Fig. 36.2***).

Warm trypsinization (*Fig. 36.2A*) : This method is widely used for disaggregation of cells. The chopped tissue is washed with dissection basal salt solution (DBSS), and then transferred to a flask containing warm trypsin (37° C). The contents are stirred, and at an interval of every thirty minutes, the supernatant containing the dissociated cells can be collected. After removal of trypsin, the cells are dispersed in a suitable medium and preserved (by keeping the vial on ice).

The process of addition of fresh trypsin (to the tissue pieces), incubation and collection of dissociated cells (at 30 minutes intervals) is carried out for about 4 hours. The disaggregated cells are pooled, counted, appropriately diluted and then incubated.

Cold trypsinization (*Fig. 36.2B*) : This technique is more appropriately referred to as ***trypsinization with cold preexposure***. The risk of damage to the cells by prolonged exposure to trypsin at 37°C (in warm trypsinization) can be minimized in this technique.

After chopping and washing, the tissue pieces are kept in a vial (on ice) and soaked with cold trypsin for about 6–24 hours. The trypsin is removed and discarded. However, the tissue pieces contain residual trypsin. These tissue pieces in a medium are incubated at 37°C for 20–30 minutes. The cells get dispersed by repeated pipettings. The dissociated cells can be counted, appropriately diluted and then used.

The ***cold trypsinization method usually results in a higher yield of viable cells with an improved survival of cells*** after 24 hours of incubation. This method does not involve stirring or centrifugation, and can be conveniently adopted in a laboratory. The major limitation of cold trypsinization is that it is not suitable for disaggregation of cells from large quantities of tissues.

Limitations of trypsin disaggregation

Disaggregation by trypsin may damage some cells (e.g. epithelial cells) or it may be almost ineffective for certain tissues (e.g. fibrous connective tissue). Hence other enzymes are also in use for dissociation of cells.

Disaggregation by collagenase

Collagen is the most abundant structural protein in higher animals. It is mainly present in the extracellular matrix of connective tissue and muscle. The enzyme collagenase (usually a crude one contaminated with non-specific proteases) can be effectively used for the disaggregation of several tissues (normal or malignant) that may be sensitive to trypsin. Highly purified grades of collagenase have been tried, but they are less effective when compared to crude collagenase.

The important stages in collagenase disaggregation, depicted in ***Fig. 36.3***, are briefly described hereunder.

The desired tissue suspended in basal salt solution, containing antibiotics is chopped into pieces. These pieces are washed by settling, and then suspended in a complete medium containing collagenase. After incubating for 1-5 days, the tissue pieces are dispersed by pipetting. The clusters of cells are separated by settling. The epithelial cells and fibroblastic cells can be separated.

Collagenase disaggregation has been successfully used for human brain, lung and several other epithelial tissues, besides various human tumors, and other animal tissues. Addition of another enzyme hyaluronidase (acts on carbohydrate residues on cell surfaces) promotes disaggregation. Collagenase in combination with hyaluronidase is found to be very effective for dissociating rat or rabbit liver. This can be done by perfusing the whole organ *in situ*.

Some workers use collagenase in conjunction with trypsin, a formulation developed in chick serum, for disaggregation of certain tissues.

Use of other enzymes in disaggregation

Trypsin and collagenase are the most widely used enzymes for disaggregation. Certain bacterial proteases (e.g. pronase, dispase) have been used with limited success. Besides hyaluronidase (described above), neuraminidase is also used in conjunction with collagenase for effective degradation of cell surface carbohydrates.

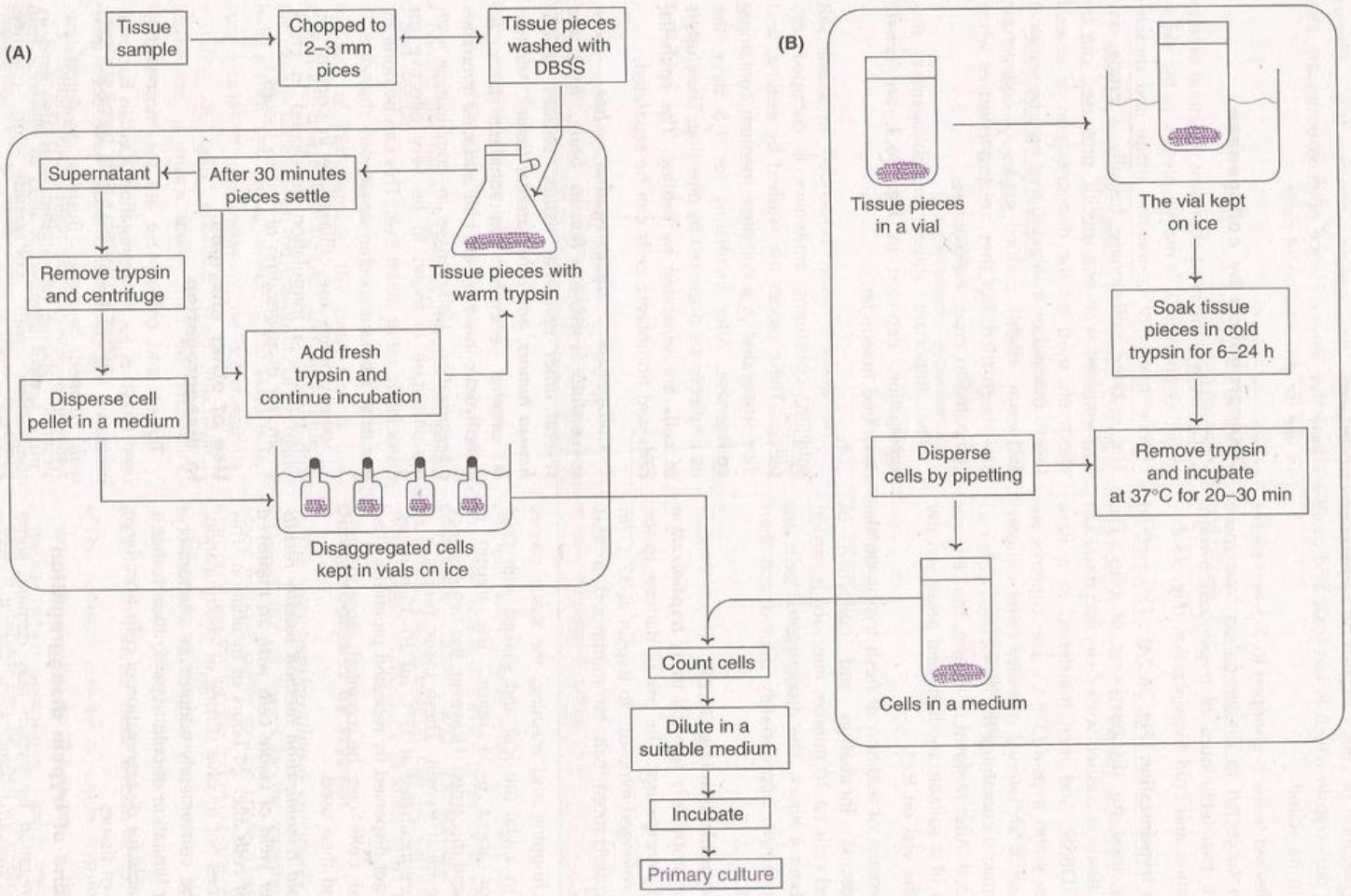


*Fig. 36.2 : Preparation of primary culture by trypsin disaggregation (**A**) Warm trypsinization (**B**) Cold trypsinization (DBSS–Dissection basal salt solution).*

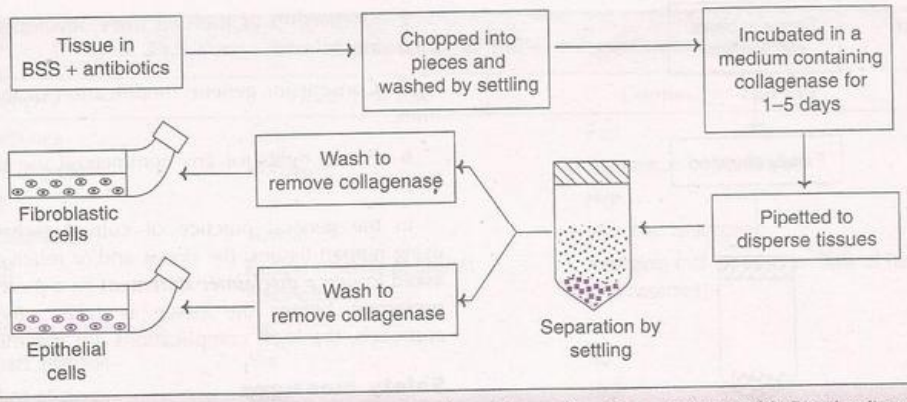


Fig. 36.3 : Important stages in collagenase disaggregation of tissue for primary culture (BSS–Basal salt solution).

PRIMARY EXPLANT TECHNIQUE

The primary explant technique was, in fact the original method, developed by Harrison in 1907. This technique has undergone several modifications, and is still in use. The simplified procedure adopted for primary explant culture is depicted in ***Fig. 36.4***, and briefly described below.

The tissue in basal salt solution is finely chopped, and washed by settlings. The basal salt solution is then removed. The tissue pieces are spread evenly over the growth surface. After addition of appropriate medium, incubation is carried out for 3-5 days. Then the medium is changed at weekly intervals until a substantial out-growth of cells is observed. Now, the explants are removed and transferred to a fresh culture vessel.

The ***primary explant technique is particularly useful for disaggregation of small quantities of tissues*** (e.g. skin biopsies). The other two techniques-mechanical or enzymatic disaggregation however, are not suitable for small amounts of tissues, as there is a risk of losing the cells. The limitation of explant technique is the poor adhesiveness of certain tissues to the growth surface, and the selection of cells in the outgrowth. It is however, observed that the primary explant technique can be used for a majority of embryonic cells e.g. fibroblasts, myoblasts, epithelial cells, glial cells.

SEPARATION OF VIABLE AND NON-VIABLE CELLS

It is a common practice to remove the non-viable cells while the primary culture is prepared from the disaggregated cells. This ***is usually done when the first change of the medium is carried out***. The very few left over non-viable cells get diluted and gradually disappear as the proliferation of viable cells commences.

Sometimes, the non-viable cells from the primary cultures may be removed by centrifugation. The cells are mixed with ficoll and sodium metrizoate, and centrifuged. The dead cells form a pellet at the bottom of the tube.

MEDICAL ETHICS AND SAFETY MEASURES IN CULTURE TECHNIQUES

Since the culture techniques involve the use of animal or human tissues, it is absolutely necessary to follow several safety measures and medical ethics. In fact, in some countries there are established legislation/norms for selection and use of tissues in cultures. For example, in United Kingdom, Animal Experiments (Scientific Procedures) Act of 1986 is followed.

The ***handling of human tissues poses several problems that are not usually encountered with animal tissues***. While dealing with fetal materials and human biopsies, the consent of the patient and/his or her relatives, besides the consent of local ethical committee is required. Further, taking any tissue (even in minute quantities) from human donors requires the full consent of the donor in a prescribed format. The following issues need to be fully considered while dealing with human tissues.

1. The consent of the patient and/or relatives for using tissues for research purposes.

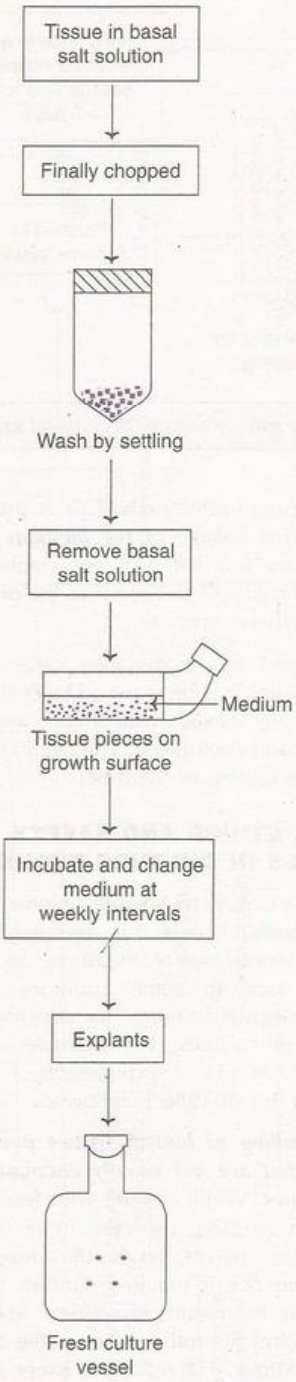


Fig. 36.4 : Primary explant technique for primary culture.

2. Ownership of the cell lines developed and their derivatives.

3. Consent for genetic modification of the cell lines.

6. Patent rights for any commercial use of cell lines.

In the general practice of culture techniques using human tissues, the donor and/or relatives are asked to sign a ***disclaimer statement*** (in a prescribed proforma) before the tissue is taken. By this approach, the legal complications are minimized.

Safety measures

Handling of ***human tissues*** is associated with a heavy ***risk of exposure for various infections***. Therefore, it is absolutely necessary that the human materials are handled in a ***biohazard cabinet.*** The tissues should be screened for various infections such as hepatitis, tuberculosis, HIV, before their use. Further, the media and apparatus, after their use must be autoclaved or disinfected, so that the spread of infections is drastically reduced.

CELL LINES

The development and various other aspects of primary culture are described above. The term cell line refers to the ***propagation of culture after the first subculture.*** In other words, once the primary culture is subcultured, it becomes a cell line. A given cell line contains several cell lineages of either similar or distinct phenotypes.

It is possible to select a particular cell lineage by cloning or physical cell separation or some other selection method. Such a ***cell line derived by selection or cloning is referred to as cell strain.*** Cell strains do not have infinite life, as they die after some divisions.

FINITE CELL LINES

The cells in culture divide only a limited number of times, before their growth rate declines and they eventually die. The ***cell lines with limited culture life spans are referred to as finite cell lines.*** The cells normally divide 20 to 100 times (i.e. is 20-100 population doublings) before extinction. The actual number of doublings depends on the species, cell

TABLE 36.1 Comparison of properties of finite and continuous cell lines

Property	*Finite cell line*	*Continuous cell line*
Growth rate	Slow	Fast
Mode of growth	Monolayer	Suspension or monolayer
Yield	Low	High
Transformation	Normal	Immortal, tumorigenic
Ploidy	Euploid (multiple of haploid chromosomes)	Aneuploid (not an exact multiple of haploid chromosomes)
Anchorage dependence	Yes	No
Contact inhibition	Yes	No
Cloning efficiency	Low	High
Serum requirement	High	Low
Markers	Tissue specific	Chromosomal, antigenic or enzymatic

lineage differences, culture conditions etc. The human cells generally divide 50-100 times, while murine cells divide 30-50 times before dying.

CONTINUOUS CELL LINES

A few cells in culture may acquire a different morphology and get altered. Such cells are capable of growing faster resulting in an independent culture. The progeny derived from these altered cells has unlimited life (unlike the cell strains from which they originated). They are designated as continuous cell lines. The ***continuous cell lines are transformed, immortal and tumorigenic.*** The transformed cells for continuous cell lines may be obtained from normal primary cell cultures (or cells strains) by treating them with chemical carcinogens or by infecting with oncogenic viruses.

In the ***Table. 36.1***, the different properties of finite cell lines and continuous cell lines are compared.

The most commonly used terms while dealing with cell lines are explained below.

Split ratio : The divisor of the dilution ratio of a cell culture at subculture. For instance, when each subculture divided the culture to half, the split ratio is 1 : 2.

Passage number : It is the number of times that the culture has been subcultured.

Generation number : It refers to ***the number of doublings that a cell population has undergone.***

It must be noted that the passage number and generation number are not the same, and they are totally different.

NOMENCLATURE OF CELL LINES

It is a common practice to give codes or designations to cell lines for their identification. For instance, the code ***NHB 2-1*** represents the cell line from ***n***ormal ***h***uman ***b***rain, followed by cell strain (or cell line number) **2** and clone number **I.** The usual practice in a culture laboratory is to maintain a log book or computer database file for each of the cell lines.

While naming the cell lines, it is absolutely necessary to ensure that each cell line designation is unique so that there occurs no confusion when reports are given in literature. Further, at the time of publication, the cell line should be prefixed with a code designating the laboratory from which it was obtained e.g. NCI for National Cancer Institute, WI for Wistar Institute.

Commonly used cell lines

There are thousands of cell lines developed from different laboratories worldover. A selected list of some commonly used cell lines along with their origin, morphology and other characters are given in ***Table. 36.2***.

SELECTION OF CELL LINES

Several factors need to be considered while selecting a cell line. Some of them are briefly described.

Table 36.2 A selected list of commonly used cell lines

Cell line	*Species of origin*	*Tissue of origin*	*Morphology*	*Ploidy*	*Characteristics*
IMR-90	Human	Lung	Fibroblast	Diploid	Susceptible to human viral infections.
3T3-A31	Mouse	Connective tissue	Fibroblast	Aneuploid	Contact inhibited, readily transformed
BHK21-C13	Hamster (Syrian)	Kidney	Fibroblast	Aneuploid	Readily transformable
CHO-k1	Chinese hamster	Ovary	Fibroblast	Diploid	Simple karyotype
NRK49F	Rat	Kidney	Fibroblast	Aneuploid	Induction of suspension growth by TGF-α, β.
BRL 3A	Rat	Liver	Epithelial	Diploid	Produces IGF-2
Vero	Monkey	Kidney	Fibroblast	Aneuploid	Viral substrate and assay
HeLa-S_3	Human	Cervical carcinoma	Epithelial	Aneuploid	Rapid growth, high plating efficiency.
Sk/HEP-I	Human	Hepatoma	Endothelial	Aneuploid	Factor VIII
Caco-2	Human	Colo-rectal carcinoma	Epithelial	Aneuploid with polarised	Forms tight monolayer support.
MCF-7	Human	Breast tumor (effusion)	Epithelial	Aneuploid	Estrogen receptor positive.
Friend	Mouse	Spleen	Suspension	Aneuploid	Hemoglobin, growth hormone.

1. **Species :** In general, non-human cell lines have less risk of biohazards, hence preferred. However, species differences need to be taken into account while extrapolating the data to humans.

2. **Finite or continuous cell lines :** Cultures with continuous cell lines are preferred as they grow faster, easy to clone and maintain, and produce higher yield. But it is doubtful whether the continuous cell lines express the right and appropriate functions of the cells. Therefore, some workers suggest the use of finite cell lines, although it is difficult.

3. **Normal or transformed cells :** The transformed cells are preferred as they are immortalized and grow rapidly.

4. **Availability :** The ready availability of cell lines is also important. Sometimes, it may be necessary to develop a particular cell line in a laboratory.

5. **Growth characteristics :** The following growth parameters need to be considered.

- Population doubling time
- Ability to grow in suspension
- Saturation density (yield per flask)
- Cloning efficiency.

6. **Stability**: The stability of cell line with particular reference to cloning, generation of adequate stock and storage are important.

7. **Phenotypic expression**: It is important that the cell lines possess cells with the right phenotypic expression.

MAINTENANCE OF CELL CULTURES

For the routine and good maintenance of cell lines in culture (primary culture or subculture) the examination of cell morphology and the periodic change of medium are very important.

Cell morphology : The cells in the culture must be examined regularly to check the health status of the cells, the absence of contamination, and any other serious complications (toxins in medium, inadequate nutrients etc).

Replacement of medium : Periodic change of the medium is required for the maintenance of cell lines in culture, whether the cells are proliferating or non-proliferating. ***For the proliferating cells, the***

medium need to be changed more frequently when compared to non-proliferating cells. The time interval between medium changes depends on the rate of cell growth and metabolism. For instance, for rapidly growing transformed cells (e.g. HeLa), the medium needs to be changed twice a week, while for slowly growing non-transformed cells (e.g. IMR-90), the medium may be changed once a week. Further, for rapidly proliferating cells, the subculturing has to be done more frequently than for the slowly growing cells.

The following ***factors*** need to be considered ***for the replacement of the medium***.

1. **Cell concentration :** The cultures with high cell concentration utilize the nutrients in the medium faster than those with low concentration, hence the medium is required to be changed more frequently for the former.

2. **A decrease in pH :** A fall in the pH of the medium is an indication for change of medium. Most of the cells can grow optimally at pH 7.0, and they almost stop growing when the pH falls to 6.5. A further drop in pH (between 6.5 and 6.0), the cells may lose their viability. The rate of fall in pH is generally estimated for each cell line with a chosen medium. If the fall is less than 0.1 pH units per day, there is no harm even if the medium is not immediately changed. But when the fall is 0.4 pH units per day, medium should be changed immediately.

3. **Cell type :** Embryonic cells, transformed cells and continuous cell lines grow rapidly and require more frequent subculturing and change of medium. This is in contrast to normal cells, which grow slowly.

4. **Morphological changes**: Frequent examination of cell morphology is very important in culture techniques. Any deterioration in cell morphology may lead to an irreversible damage to cells. Change of the medium has to be done to completely avoid the risk of cell damage.

SUBCULTURE

Subculture (or passage) refers to the ***transfer of cells from one culture vessel to another culture vessel***. Subculture usually (not always) involves the subdivision of proliferating cells that enables the propagation of a cell line. The term ***passage number*** is used to indicate the number of times a culture has been subcultured.

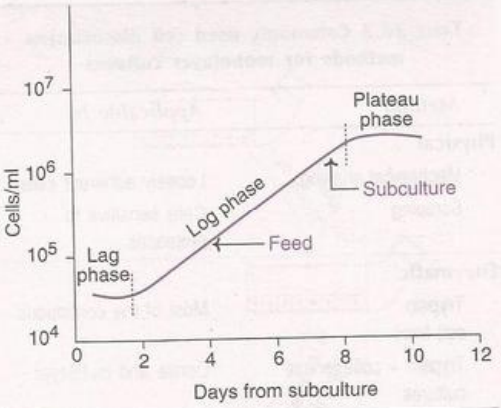


Fig. 36.5 : Depiction of a standard growth of cells in a culture (Note : the cell concentration is expressed in semilog plot).

The standard growth curve of cells in a culture is depicted in ***Fig. 36.5.*** The initial lag phase is followed by exponential or log phase and a plateau phase. During the active growth period in the log phase, the medium must be changed frequently, or else growth ceases. As the cell concentration exceeds the capacity of the medium, the culture has to be divided to subculture.

There are two types of subcultures-monolayer subculture and suspension subculture.

MONOLAYER CULTURES

When the ***bottom of the culture vessel is covered with a continuous layer of cells, usually one cell in thickness***, they are referred to as monolayer cultures. The attachment of cells among themselves and to the substrate (i.e. culture vessel) is mediated through surface glycoproteins (cell adhesion molecules) and calcium ions (Ca^{2+}). For subculturing of monolayer cultures, it is usually necessary to remove the medium and dissociate the cells in the monolayer by degrading the cell adhesion molecules, besides removing Ca^{2+}.

Methods of cell dissociation

There are ***physical*** and ***enzymatic methods*** for dissociation of monolayer cultures (***Table. 36.3***).

Table 36.3 Commonly used cell dissociation methods for monolayer cultures

Method	*Applicable to*
Physical	
Mechanical shaking	Loosely adherent cells
Scraping	Cells sensitive to proteases
Enzymatic	
Trypsin	Most of the continuous cell lines
Trypsin + collagenase	Dense and multilayer cultures
Dispase	Removal of epithelium in sheets
Pronase	Single–cell suspensions

Mechanical shaking and cell scraping are employed for cultures which are loosely adhered, and the use of proteases has to be avoided. Among the enzymes, ***trypsin is the most frequently used***. For certain cell monolayers, which cannot be dissociated by trypsin, other enzymes such as pronase, dispase and collagenase are used. Prior to cell dissociation by enzymes, the monolayers are usually subjected to pretreatment of EDTA for the removal of Ca^{2+}.

Criteria for subculture of monolayers

The subculturing is ideally carried out between the middle of the log phase and the time before they enter plateau phase (***Fig. 36.6***). Subculture of cells should ***not be done when they are in lag phase***. The other important criteria for subculture of monolayers are briefly described.

Culture density : It is advisable to subculture the normal or transformed monolayer cultures, as soon as they reach confluence. Confluence denotes the culture stage wherein all the available growth area is utilized and the cells make close contact with each other.

Medium exhaustion : A drop in pH is usually accompanied by an increase in culture cell density. Thus, when the pH falls, the medium must be changed, followed by subculture.

Scheduled timings of subculture : It is now possible to have specified schedule timings for subculture of each cell line. For a majority of cell cultures, the medium change is usually done after 3-4 days, and subculturing after 7 days.

Purpose of subculture : The purpose for which the cells are required is another important criteria for consideration of subculturing. Generally, if the cells are to be used for any specialized purpose, they have to be subcultured more frequently.

Techniques of monolayer subculture

The subculture of monolayer cells basically consists of the following steps (***Fig. 36.6***)

1. Removal of the medium
2. Brief exposure of the cells to trypsin.
3. Removal of trypsin and dispersion in a medium.
4. Incubation of cells to round up.
5. Resuspension of the cells in a medium for counting and reseeding.
6. Cells reseeded and grown to monolayers.

Cell concentration at subculture : Most of the continuous cell lines are subcultured at a seeding concentration between 1×10^4 and 5×10^4 cells/ml. However for a new culture, subculture has to be started at a high concentration and gradually reduced.

SUSPENSION CULTURES

Majority of continuous cell lines grow as monolayers. Some of the cells which are ***non-adhesive*** e. g. cells of leukemia or certain cells ***which can be mechanically kept in suspension, can be propagated in suspension***. The transformed cells are subcultured by this method. Subculture by suspension is comparable to culturing of bacteria or yeast.

Advantages of cell propagation by suspension

- The process of propagation is faster.
- The lag period is usually shorter.
- Results in homogeneous suspension of cells.
- Treatment with trypsin is not required.
- Scale-up is convenient.
- No need for frequent replacement of the medium.

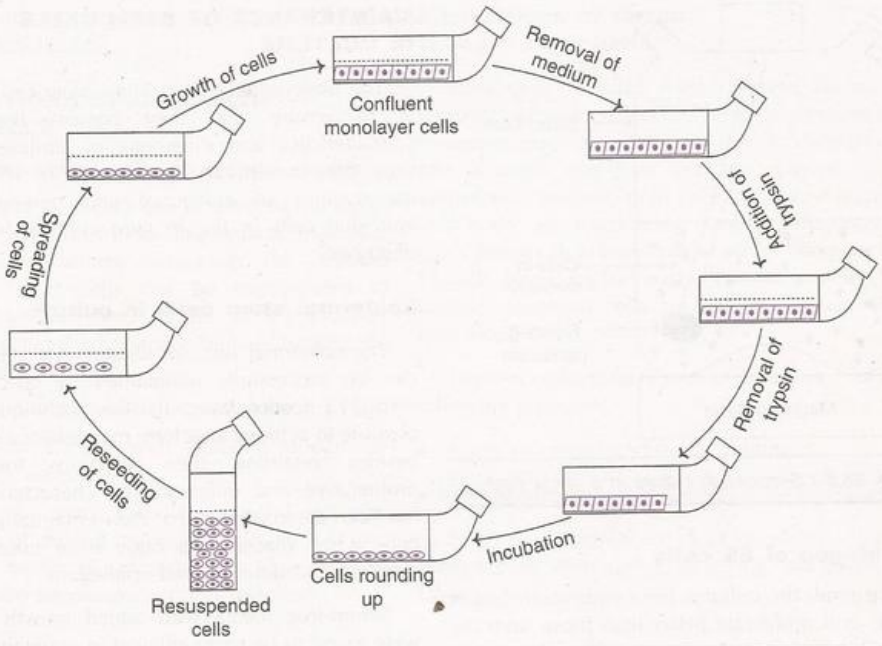


Fig. 36.6 : Technique of monolayer subculture.

- Maintenance is easy.
- Bulk production of cells can be conveniently achieved.

Criteria for suspension subculture

The criteria adopted for suspension subculture are the same as that already described for monolayer subcultures (See p. 446). The following aspects have to be considered.

- Culture density.
- pH change representing medium exhaustion.
- Schedule timings of subculture.
- Purpose of subculture.

Technique of suspension culture

The cells can be suspended in a culture flask (a stirrer flask) containing the desired medium (***Fig. 36.7***). The medium is continuously stirred with a magnetic pendulum rotating at the base of the flask. The cells have to periodically examined for contamination or signs of deterioration.

STEM CELL CULTURES

The ***cells that retain their proliferative capacity throughout life*** are regarded as ***stem cells***. When the stem cells divide, they can generate differentiated cells and/or some more stem cells. These stem cells are capable of regenerating tissue after injury. The lack of tissue-specific differentiation markers is a characteristic feature of stem cells.

EMBRYONIC STEM (ES) CELLS

As the embryonic development occurs, cells of the ***inner mass of embryo*** (i.e. those contributing to future fetus) represent embryonic stem (ES) cells. They continue to divide and remain in an un-differentiated to totipotent state. It has been possible to establish and maintain cell lines for ES cells. The ES cells isolated from mouse blastocyst are the most commonly used in the laboratory.

The most widely used embryonic stem lines are the various 3T3 lines, WI-38, MRC-5 and other human fetal lung fibroblasts.

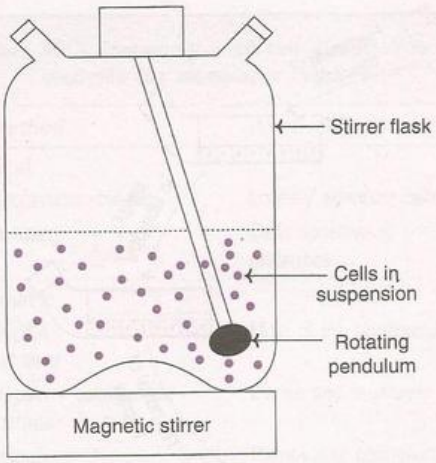


Fig. 36.7 : Suspension culture in a stirrer flask.

Advantages of ES cells

In general, the cultures from embryonic tissues survive, and proliferate better than those from the adult. This is due to the fact that ES cells are less specialized with higher proliferative potential.

Limitations of ES cells

In some cases, the ES cells will be different from the adult cells, and thus there is no guarentee that they will mature to adult-type cells. Therefore, it is necessary to characterize the cells by appropriate methods.

EPITHELIAL STEM CELLS

The epithelial cells (e.g. epidermis lining of gut) are constantly being shed from their outer surface. This cellular loss is compensated by a continuous replacement process. The replacement occurs in a highly organized and a regulated fashion.

It is estimated that in humans the entire outer layer of skin is shed daily. The entire epithelial lining of the mouse gut is replaced once in 3-4 days. This process of shedding and replacement continues throughout life.

The epidermis of the skin has a proliferative compartment containing stem cells and post-mitotic cells, besides some transit amplifying population of cells. The transit amplifying cells, produced from stem cells with limited life span are shed from the epidermis.

MAINTENANCE OF STEM CELLS IN CULTURE

The basic criteria to maintain stem cell *in vitro* is to ensure that they possess the same characteristics and differentiating abilities when they are present in the tissue *in vivo*. The maintenance of epidermal and non-epidermal epithelial cells in the *in vitro* cultures is briefly described.

Epidermal stem cells in culture

The epidermal stem (or keratinocyte stem) cells can be successfully maintained by co-culturing with 3T3 feeder layer. By this technique, it is possible to achieve long term maintenance of cells, besides retaining their capacity for both proliferative and differentiating characteristics. It has been demonstrated that the so maintained stem cells when placed into nude mice could form stratified and differentiated epithelium.

Serum-free media with added growth factors were found to be more efficient in maintaining the epidermal stem cells in culture.

Epithelial cells in culture

Several types of non-epidermal epithelial cells can be grown and maintained in cultures. As in the case with epidermal stem cells, use of feeder layer is advantageous for epithelial cell culture.

Epithelial cells of prostate gland have been successfully grown in suspension cultures in the presence of 3T3 feeder layer. The same method is also used for culturing human breast epithelial cells and colorectal carcinoma cells.

CHARACTERIZATION OF STEM CELLS

Immunological techniques are widely used for the characterization of different populations of stem cells. These techniques are mostly based on immunocytochemistry using fluorescent microscopy or staining technique involving colour reactions.

The cells of the tissues produce specific cell surface and cytoplasmic proteins. The cell surface proteins such as integrins and the members of CD (cluster of differentiation) antigens (e.g. CD_{10}, CD_{31}, CD_{44}) can be used as markers of epithelial cell types. Further, the cytoplasmic proteins of

epithelial cells (of cytokeratin family) are also useful for their identification.

APPLICATIONS OF CULTURED STEM CELLS

Embroynic stem cells in tissue repair

The culture stem cells can be used for the repair of tissues with functional impairment that may occur due to damage or ageing. The cultured embryonic stem cells can be manipulated to produce cultures characteristic of a particular tissue. Thus, there exists a possibility of treating the following diseases.

- Diabetes with pancreatic insulin producing cells.
- Parkinson's disease with cultured dopamine-producing neurons.

Embryonic stem cells are useful for the production of defined transgenic animals. It is also possible to modify ES cell genome by gene targeting using *in vitro* transformation and selection.

Applications of tissue specific stem cells

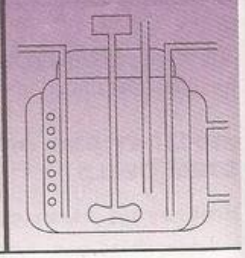
Stem cells, isolated from different tissues of humans and animals, and cultured *in vitro* are less toitpotent than ES cells. They usually differentiate into a single cell type and are referred to as ***unipotent.*** However, stem cells from bone marrow and brain are capable of forming different cell types though to a lesser extent when compared to ES cells. In mouse lacking bone marrow, when the cultured neuronal cells are placed, they develop into blood cells.

Tissue specific culture stem cells are used for the following purposes.

- In surgical repair and tissue grafting.
- In gene therapy.

Some more details on ES cells, with particular reference to stem cell engineering, are given in Chapter 40.

Animal Cell Culture — General Considerations and Scale-up



In the laboratories, small scale cultures of cells in flasks (usually 1-5 litre volume) are done for establishing the cell lines. Such cell cultures are useful for studying the morphology, growth, metabolism etc. Large-scale cultures are required for semi-industrial (100–1,000 l capacity) and ***large-scale industrial*** (5,000–20,000 l capacity) ***use of cells for production of wide range of biologically important compounds*** (e.g. enzymes, antibodies, hormones, interferons, plasminogen activator, interleukins).

The terms ***fermenter*** and ***bioreactor*** are in common use while dealing with the ***industrial use of cells***. A fermenter usually refers to the containment system for the cultivation of prokaryotic cells (bacteria, fungi), while a bioreactor grows the eukaryotic cells (mammalian, insect).

Scale-up refers to the process of developing the culture systems in stages from a laboratory to the industry. Scale-up although tedious, labour intensive and expensive, is required for the production of commercially important products.

For a better understanding of scale-up, certain basic and fundamental concepts of cell culture should be clear.

CELL CULTURE — GENERAL CONSIDERATIONS

There are several parameters that need to be considered for appropriate growth, proliferation and maintenance of cells in culture. A good understanding of these parameters, listed below is also necessary for scale-up.

- Cell quantitation.
- Equipment and medium.
- pH and buffer systems.
- Oxygen.
- Growth kinetics.
- Types of culture processes.
- Other practical considerations.

CELL QUANTITATION

The total number of cells in a culture can be measured by counting in a haemocytometer. It is however, not possible to identify the viable and non-viable cells by this method.

Cell viability : The viability of cells can be detected by use of dyes e.g. tryphan blue. The principle is based on the fact that the dye is permeable to dead cells while the viable cells do not take up dye.

Indirect measurements for cell viability : The viability of cells can be measured by their metabolic activity. Some of the most commonly used parameters are listed.

- Glucose utilization.
- Oxygen consumption.
- Pyruvate production.
- Carbon dioxide formation.

In recent years, many laboratories have started measuring the activity of lactate dehydrogenase (LDH) to detect cell viability. Dead cells release LDH and therefore, this enzyme can be used to quantitatively measure the loss of cell viability.

EQUIPMENT AND MEDIUM

The various aspects of equipment and medium used in culture laboratory are described in Chapters 33 and 34.

Culture vessels : The materials made up of ***glass or stainless steel*** are commonly used for cell cultures. Borosilicate glass (e.g. Pyrex) is preferred as it can better withstand autoclaving for suspension cultures, wherein cell attachment to the surface has to be discouraged; the culture vessels are usually treated with silicone (siliconization).

Medium and nutrients : Appropriate selection of the medium is done based on the nutritional requirements, and the purpose for which the cultured cells are required. ***Eagle's basal medium and minimal essential medium are the most commonly used***. The media may be supplemented with serum.

Additional feeding of certain nutrients is often required as they are quickly utilized and get exhausted. These include glucose, glutamine and cystine.

For ***suspension cultures, media lacking calcium and magnesium are used***, since their absence minimizes the surface attachment.

Non-nutrient medium supplements : Certain non-nutrient compounds are often added to the medium for improvement of cell cultures. Sodium carboxymethyl cellulose addition to medium helps to minimize mechanical damage that may occur due to forced aeration or the forces generated by stirred impeller. Polyglycol (trade name Pluronic F-68) in the medium reduces foaming in stirred and aerated cultures.

pH AND BUFFER SYSTEMS

The ***ideal pH*** for animal cell cultures is ***around 7.4***. A pH below 6.8 inhibits cell growth. The factors that can alter pH include the stability of the medium, type of buffer and its buffering capacity, concentration of glucose and headspace.

The commonly used buffer of the *in vitro* culture carbon dioxide-bicarbonate system (2-5% CO_2 with 10–25 mM $NaHCO_3$) is comparable to the blood buffer. The presence of phosphates in the medium improves the buffering capacity. Some laboratories use HEPES instead of bicarbonate for more efficient buffering.

As glucose is utilized by the cells, pyruvic acid and lactic acid are produced which can alter the pH. If fructose and galactose (instead of glucose) are used, the acid formation is less, but the cell growth is reduced.

OXYGEN

Oxygen ***has to be continuously supplied*** to the medium throughout the life of the culture. This has to be done without causing damage to the cells. Oxygen can be supplied to the cultures in one of the following ways.

Surface aeration : In closed system static cultures, the headspace is used for the supply of oxygen. For instance, in a 1 litre flask with 100 ml medium 900 ml of the space containing air has about 0.27 g of O_2. This O_2 is capable of supporting 10^8 cells for about 450 hours.

Sparging : The process of ***bubbling gas through the culture*** is referred to as sparging. This is an efficient means of O_2 supply, but may often damage the cells due to effects of the bubble on the cell membrane surfaces. Use of higher air bubbles minimizes the damaging effect.

Membrane diffusion : Adequate diffusion of oxygen into the culture can be obtained through silicone tubing which is highly permeable to gases. This approach however, is inconvenient, besides the high cost of silicone tubing.

Medium perfusion : The medium is perfused through an ***oxygenation*** chamber before it enters the culture system. This method ensures good O_2 saturation. Medium perfusion is in fact used in glass bead system and microcarrier systems.

GROWTH KINETICS

The standard pattern of growth of cultured cells follows a lag phase, an exponential (log) phase and a stationary phase. For more details, refer Chapter 35 and ***Fig. 35.5***.

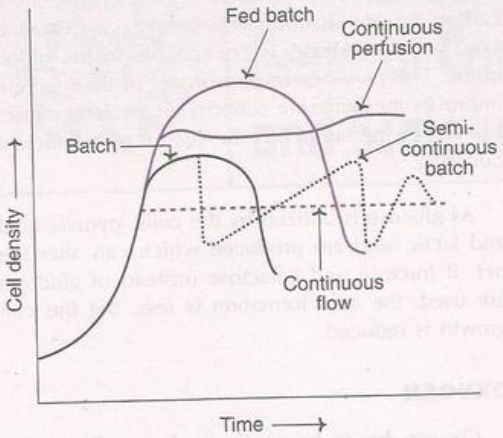


Fig. 37.1 : Comparison of different culture processes.

Growth of cells usually means an increase in cell numbers. However, increase in cell mass may occur without replication.

The following terms are in common use to represent growth of cultured cells.

Specific growth rate : The rate of cell growth per unit amount of biomass.

Doubling time : The time required for a population of cells to double in number or mass.

Degree of multiplication (number of doublings) : The number of times a given inoculum has replicated.

TYPES OF CULTURE PROCESSES

The different culture processes and the growth patterns of cells (represented by cell density) are depicted in ***Fig 37.1***. They are briefly described.

Batch culture

In this technique, when the cells are inoculated into a fixed volume of the medium, they utilize the nutrients and grow, and simultaneously accumulate metabolites. As the nutrients get exhausted, toxic waste products accumulate and the cell multiplication ceases. Further, the cell density drops due to death of the cells.

Batch culture is a standard technique. Several modifications have been made to increase proliferation of cells, besides prolonging their life. The other culture processes described below are the modified batch cultures.

Fed batch culture

There is a gradual addition of fresh medium so that the cell proliferation is much higher than the batch culture. Thus, in the fed batch culture there is an increase in the volume of culture.

Semi-continuous batch culture

A portion of the culture medium is intermittently replaced with an equal volume of fresh medium. The growth pattern of the cells is fluctuating, with a rapid increase in cell density after each replacement of the medium.

Continuous perfusion culture

There is a continuous addition of the medium to the culture and a withdrawal of an equal volume of used cell-free medium. The continuous perfusion process may close or open for circulation of the medium.

Continuous-flow culture

In the continuous-flow culture, a homeostatic condition with no change in the cell numbers, nutrients and metabolites is attained. This is made possible by a balance between the addition of the medium and withdrawal of medium along with cells. This is mostly suitable for suspension cultures.

OTHER PRACTICAL CONSIDERATIONS

Besides the parameters described above, there are several other practical considerations for *in vitro* culture and scale-up. Some important ones are given below.

Culture surface area : The available surface is important for the cells to grow. In general, the culture processes are planned in such a way that the surface area is not a limiting factor.

Inoculation density of cells : As such, there is no set rule for the density of inoculation. However, inoculation with high density is preferred for better growth.

Growth phase of cells : ***Cells in the late exponential (log) phase are most suitable for inoculation.*** The cells at the stationary phase should be avoided since they have either prolonged lag phase or no growth at all.

Stirring rate of culture : The stirring rates of different culture lines are developed in the laboratories. It is usually in the range of 100-500 rpm for most of the cultures.

Temperature of the medium : It is advisable to warm the medium to 37°C before adding to the culture.

SCALE-UP

Scale-up involves the ***development of culture systems in stages from (small scale) laboratory to (large scale) industry***. The methodology adopted to increase the scale of a culture depends on the proliferation of cells and is broadly divided into two categories.

1. Scale-up in suspension.
2. Scale-up in monolayer.

SCALE-UP IN SUSPENSION

Scale-up in suspension is the preferred method as it is simpler. Scale-up of suspension culture primarily involves an increase in the volume of the culture. Small scale generally means the culture capacity less than 2 litres volume (or sometimes 5 litres).

Stirred suspension cultures

It is usually necessary to maintain cell strains in ***stirred suspension cultures, by agitation*** (or stirring) ***of the medium***. The stirring of the culture medium is achieved by a magnet encased in a glass pendulum or by a large surface area paddle. The stirring is usually done at a speed of 30–100 rpm. This is sufficient to prevent sedimentation of cells without creating shear forces that would damage cells.

Static suspension cultures

Some cells can grow in suspension cultures, without stirring or agitation of the medium, and form monolayer cells. However, static suspension cultures are unsuitable for scale-up.

FACTORS IN SCALING-UP

For appropriate scale-up, the physical and chemical requirements of cells have to be satisfied.

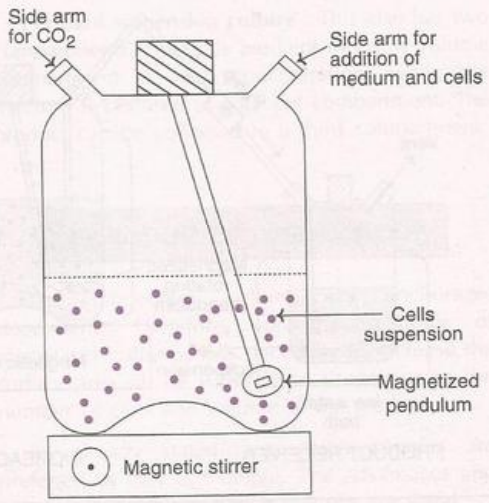


Fig. 37.2 : Diagrammatic representation of a stirrer flask.

Physical parameters

- Configuration of the bioreactor.
- Supply of power.
- Stirring of the medium.

Chemical parameters

- Medium and nutrients.
- Oxygen.
- pH and buffer systems.
- Removal of waste products.

Some of the relevant aspects of the factors in scale-up have already been described under cell culture-general considerations (See p. 450).

The most commonly used ***techniques for stirred suspension cultures*** are briefly described hereunder.

STIRRER CULTURE

A diagrammatic representation of a stirrer flask is shown in ***Fig 37.2***. The size of the stirrer flask is in the range of 2–10 litres. It is fitted with a magnetized rotating pendulum, and two side arms — one for the addition of cells and medium, and the other for the supply of CO_2.

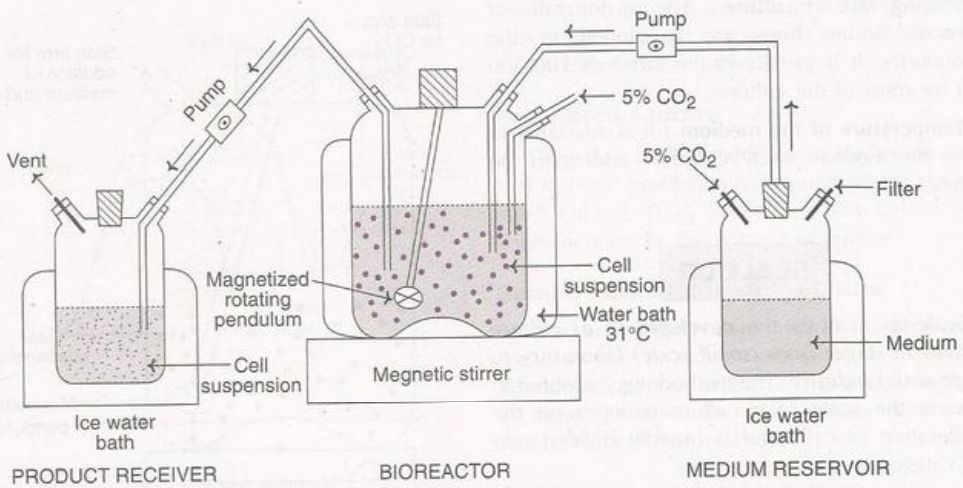


Fig. 37.3 : Suspension culture in a biostat (chemostat).

The stirrer culture vessel is autoclaved (at 15 lb/in^2 for 15 minutes), and is then set up as in ***Fig. 37.2***. The flask is seeded with the culture.

Then medium along with an antifoam agent is added. The flask is connected to CO_2 and stirred at a speed of 60 rpm. The flask is incubated for about 2 hours.

The contents of the small stirrer flask are transferred to a large flask and the entire set up is restarted. Incubation at 37°C is carried out for 4-7 days. The growth of the cells is monitored daily, and the cells are counted. There is a tendency of the cells to enter apoptosis, if the concentration exceeds 1×10^6 cells/ml.

CONTINUOUS FLOW CULTURE

In a continuous flow culture, it is possible to keep the cells at a desired and set concentration, and maintain. This is carried out by a ***biostat or chemostat*** (***Fig 37.3***).

Continuous flow culture consists of growing the cells at the mid-log phase, removal of a measured volume of cells, and replacement by an equal volume of medium. The equipment, specially designed for this purpose has the facility for removal of the cells and addition of medium. The flow rate of the medium addition can be determined from the growth rate of the culture. The medium flow can be regulated by a peristaltic pump. By this technique, it is possible to keep the culture conditions constant rather than to produce large number of cells.

The continuous flow cultures are useful for monitoring metabolic changes in relation to cell density. However, these cultures are more susceptible to contamination.

AIR-LIFT FERMENTER CULTURE

The major limitation of scale-up in suspension culture is inadequate mixing and gas exchange. For small cultures, stirring of the medium is easy, but the problem is with large cultures. The design of fermenter should be such that maximum movement of liquid is achieved with minimum shear to damage the cells.

A diagrammatic representation of an air-lift fermenter is depicted in ***Fig. 37.4***. A 5% CO_2 in air is pumped through the bottom of the fermenter. The bubbles formed move up to agitate and aerate the culture. These bubbles carry a flow of liquid along with them and release at the top which goes to the bottom for recycling.

It is possible to continuously supply O_2 to the culture in this technique. ***Air-lift fermenter culture technique is suitable for fragile animal as well as plant cells***. This fermenter is extensively used in the

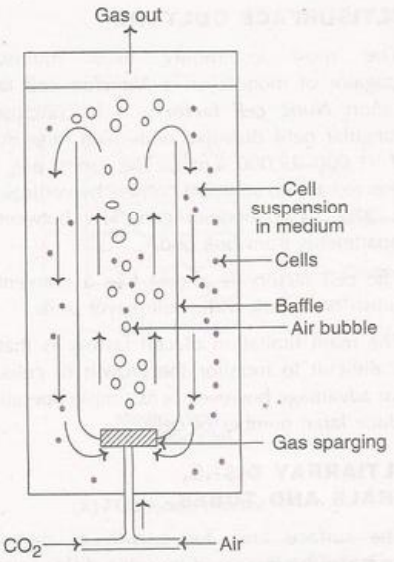


Fig. 37.4 : Diagrammatic representation of airlift fermenter.

biotechnology industry for culture capacities upto 20,000 litres.

NASA bioreactor

NASA (National Aeronautics Space Administration, USA) constructed a bioreactor to grow the cells at zero gravity by slowly rotating the chamber (***Fig. 37.5***). The cells remain stationary and form three-dimensional aggregates and this enhances the product formation. In the NASA bioreactor, there is almost no shear force; hence the cells are not damaged.

As the culture chamber stops its rotations, the cell aggregates sediment and the medium can be replaced.

OTHER SYSTEMS FOR SUSPENSION CULTURE

Rotating chambers : The mixing and aeration of the culture medium can be achieved by 2 or 3 rotating chambers. The chambers are so designed that the cell suspension and mixing are high in one chamber while the product and spent medium remain in the other chamber. These chambers are separated by semipermeable membrane.

Perfused suspension culture : This also has two compartments. The cells are kept in a low-volume compartment at high concentration, while the medium is perfused in adjacent compartment. The product can be collected in a third compartment.

SCALE-UP IN MONOLAYER

The monolayer culture are anchorage-dependent. Therefore, for the scale-up of monolayer cultures, it is necessary to increase the surface area of the substrate in proportion to the number of cells and volume of the medium.

As already stated, suspension cultures are preferred as they are simple. The advantages and disadvantages of monolayer cultures are listed.

Advantages

- Change of medium and washing of cells easy.
- It is easy to perfuse immobilized monolayer cells.

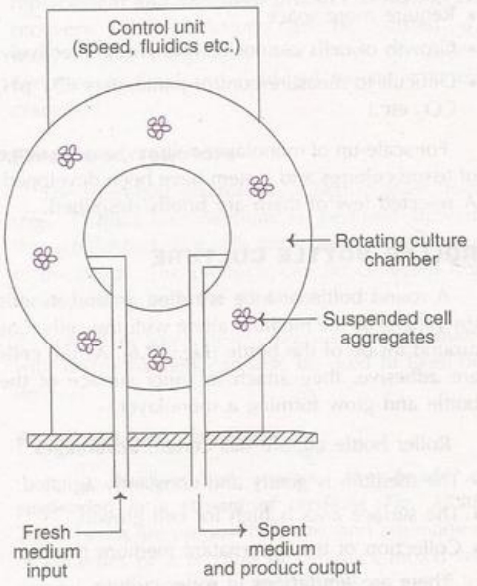


Fig. 37.5 : A bioreactor developed by NASA.

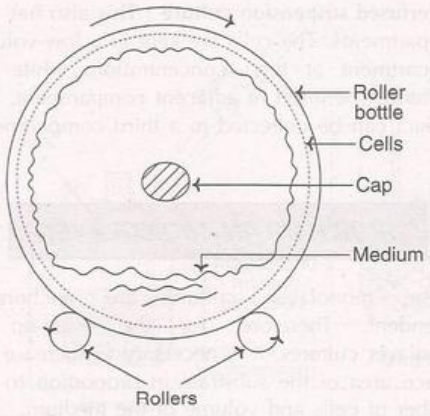


Fig. 37.6 : Roller bottle culture.

- The cell product formation (pharmaceutically important compounds e.g. interferon, antibodies) is much higher.
- The same set up and apparatus can be repeatedly used with different media and cells.

Disadvantages

- Tedious and costly.
- Require more space.
- Growth of cells cannot be monitored effectively.
- Difficult to measure control parameters (O_2, pH, CO_2 etc.)

For scale-up of monolayer cultures, a wide range of tissue cultures and system have been developed. A selected few of them are briefly described.

ROLLER BOTTLE CULTURE

A round bottle or tube is rolled around its axis (by rollers) as the medium along with the cells runs around inside of the bottle (***Fig. 37.6***). As the cells are adhesive, they attach to inner surface of the bottle and grow forming a monolayer.

Roller bottle culture has certain ***advantages***.

- The medium is gently and constantly agitated.
- The surface area is high for cell growth.
- Collection of the supernatant medium is easy.

There are ***limitations*** in roller culture.

- Monitoring of cells is very difficult.
- Investment is rather high.

MULTISURFACE CULTURE

The most commonly used multisurface propagator of monolayer is ***Nunclon cell factory*** (in short ***Nunc cell factory***). It is composed of rectangular petri dish-like units with huge surface area (1,000–25,000 cm^2). The units are interconnected at two adjacent corners by vertical tubes (***Fig. 37.7***). The medium can flow between the compartments from one end.

The cell factory is almost like a conventional petridish or a flask with multiplayer units.

The main limitation of cell factory is that it is very difficult to monitor the growth of cells. The major advantage however, is its simple operation to produce large number of cells.

MULTIARRAY DISKS, SPIRALS AND TUBES

The surface area for growth of monolayer cultures can be increased by using disks, spirals or tubes. They are however, not in common use as their commercial importance is limited.

MICROCARRIER CULTURE

Monolayers can be ***grown on small spherical carriers or microbeads*** (80–300 µm diameter) referred to as microcarriers. The microcarriers are made up of any one the following materials (trade names given in brackets).

- Plastic (acrobeads, bioplas).
- Glass (bioglass, ventreglas).
- Gelatin (ventregel, cytodex-3).
- Collagen (biospex, biospheres)
- Cellulose (DE-52/53).
- DEAE Dextran (cytodex I, dormacell).

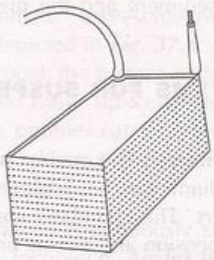


Fig. 37.7 : A diagrammatic representation of Nunc cell factory.

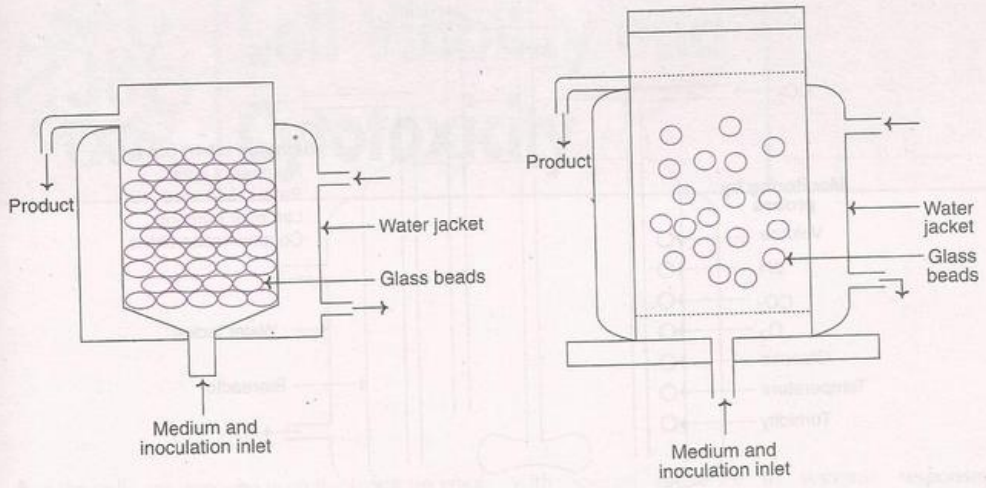


Fig. 37.8 : Fixed and fluidized–bed reactors.

The microbeads ***provide maximum surface area for monolayer cultures***. This actually depends on the size and density of the beads. The cells can grow well on the smooth surface at the solid-liquid interface. However, microcarriers need efficient stirring without grinding the beads. The main advantage with microcarrier culture is that it can be treated as a suspension culture for all practical purposes.

Microcarriers can be cultured in stirrer flask (See ***Fig. 37.2***) or in continuous suspension (See ***Fig. 37.3***). In fact, the suppliers of microcarriers provide the technical literature and other relevant information for setting up a microcarrier culture.

Factors affecting microcarrier culture

- Composition and coating of beads (gelatin and collagen beads are preferred as they can be solubilised by proteases).
- Higher stirring speed is usually required.
- Glass beads are used when the microcarriers need to be recycled.

Analysis of microcarrier culture

The cell counting techniques are difficult to be used for microcarrier cultures. The growth rate can be detected by analyzing DNA or protein.

PERFUSED MONOLAYER CULTURE

The growth surface areas of the monolayer cultures can be ***perfused to facilitate medium replacement and improved product formation and recovery***. The perfusion can be carried out with pumps, oxygenator and other controllers. Perfusion of fixed and fluid-bed reactor is briefly described.

Fixed-bed reactors

The fixed-bed reactor has a bed of glass beads (***Fig. 37.8A***). The medium is perfused upwards through the bed. The cells are grown on the surfaces of the beads. The products can be collected from the top along with the spent medium.

Instead of glass beads, porous ceramic matrix with microchannels can also be used in fixed-bed reactors.

Fluidized-bed reactors

In a fluidized-bed reactor, ***the beads are suspended in a stream of medium*** (***Fig. 37.8B***). These beads are porous in nature, and are made up of ceramics or a mixture of ceramics mixed with natural products such as collagen. They are of low density and float in the medium. The flow rate of the perfused medium is equal to the sedimentation

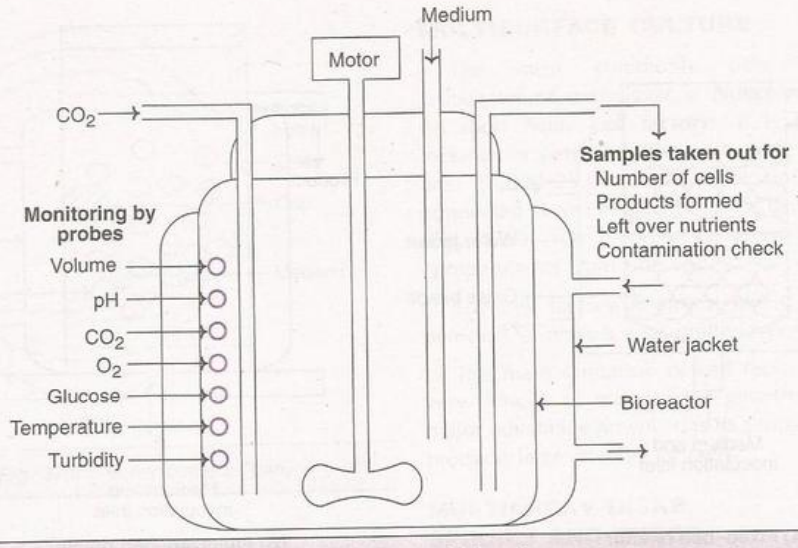


Fig. 37.9 : In situ *monitoring of bioreactor processes.*

rate of the beads. The cells can grow as monolayers on the outer surfaces and inside of the porous beads.

Other perfused monolayer cultures

Membrane perfusion, hollow-fiber perfusion, matrix perfusion and microencapsulation are among the other techniques for perfusion of monolayer cultures.

MONITORING OF CELL GROWTH IN SCALE-UP

Monitoring of the progress of cell growth and the culture systems are very important in scale-up.

MONITORING OF SUSPENSION CULTURES

The progress of suspension cultures can be monitored *in situ* ***by measuring glucose, O_2, CO_2, pH or metabolites produced*** (lactate, ammonia) or specialized products formed (e.g. immunoglobulins by hybridoma cells).

The cell proliferation and rate of biomass formation can also be determined by estimating DNA, protein and ATP.

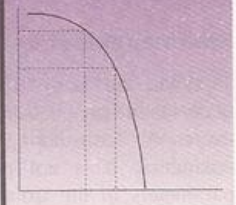
The different parameters for monitoring of a bioreactor for suspension culture are depicted in *Fig 37.9*.

MONITORING OF MONOLAYER CULTURES

It is rather difficult to monitor monolayer cell cultures for scale-up. This is due to the fact that in most of the techniques employed for monolayer cultures, the cells cannot be observed directly to monitor the progress of the culture.

In recent years, nuclear magnetic resonance (NMR) technique is used to assay the contents of culture. The characteristic NMR spectra generated by specific metabolites enables the identification and quantitation of metabolites, besides detecting the progress of cell growth.

Cell Viability and Cytotoxicity



As the cells are removed from the living (*in vivo*) environment and subjected to experimental manipulations in the culture systems (*in vitro*), their viability assumes significance. ***Viability*** of the cells ***represents the capability of their existence, survival and development***.

Many experiments are carried out with cells in the culture rather than using the animal models. This is particularly so with regard to the determination of safety and cytotoxicity of several compounds (pharmaceuticals, cosmotics, anticancer drugs, food additives). *In vitro* testing for cytotoxicity and safety evaluation is in fact cost-effective, besides reducing the use of animals. Studies on ***cytotoxicity*** broadly involve the ***metabolic alterations of the cells, including the death of cells as a result of toxic effects of the compounds***. For instance, in case of anti-cancer drugs, one may look for death of cells, while for cosmetics the metabolic alterations and allergic responses may be more important.

LIMITATION OF *IN VITRO* CYTOTOXICITY STUDIES

Toxicity of a given compound is a complex process as it occurs *in vivo*. This may result in direct damage to cells, alterations in physiological and biochemical functions, inflammatory changes and other systemic effects, not only at the site of application but also at the other sites. Some of the important *in vitro* limitations are briefly described with special reference to systemic responses, metabolism and pharmacokinetics.

Tissue and systemic responses

The cultured cells represent the cells in isolation, and not an integral part of a tissue or an organ. As already stated, the nature of the *in vitro* effect is measured by cell survival or by an altered metabolic effect. On the other hand, the *in vivo* reactions are complicated that may lead to tissue responses (fibrosis, inflammatory reaction) or systemic responses (pyrexia, vascular dilation). Therefore, the *in vitro* cytotoxic responses have to be considered carefully.

In recent years, some attempts are made to create *in vitro* organotypic cultures by assembly of different cell types. Even with this approach, it is not possible to observe all the tissue, and systemic responses.

Metabolism

The whole body metabolism is complex and well integrated. Some of the compounds that are toxic *in vitro* are detoxified by the liver. On the other hand, certain non-toxic or less toxic compounds may be converted to more toxic ones in the liver. For these reasons, it is necessary that the *in vitro* cells are exposed to the same type of compound that is formed in liver after detoxification.

Pharmacokinetics

When the body is exposed to a drug, there occurs tissue penetration, metabolism, clearance and excretion, constituting an organized pharmacokinetics. It is not possible to stimulate these parameters in the isolated cells. However, use of multicellular tumor spheroids has certain advantages with regard to understanding of drug penetration.

ASSAYS FOR CELL VIABILITY AND CYTOTOXICITY

Despite the limitation stated above, there are several assays developed in the laboratory for measuring the cell viability and cytotoxicity. They are broadly categorized into the following types.

- Cytotoxicity and viability assays.
- Survival assays.
- Metabolic assays.
- Transformation assays.
- Inflammation assays.

CYTOTOXICITY AND VIABILITY ASSAYS

A majority of the cytotoxicity and viability assays are based on the measurement of membrane integrity, cellular respiration, radioisotope incorporation, colorimetric assays and luminescence-based tests.

Based on membrane integrity

The most common measurements of cell viability are based on membrane integrity. The damage to membrane may occur due to cell disaggregation, cell separation or freezing and thawing. Membrane integrity can be determined by uptake of dyes to which viable cells are impermeable (e.g. naphthalene black, trypan blue, erythrosin) or release of dyes normally taken up and retained by viable cells (e.g. neutral red, diacetyl fluorescein).

The other assays for membrane integrity are release of labeled chromium (^{51}Cr), enzymes and use of fluorescent probes. Cell viability measurements, based on membrane integrity are immediate that can be detected within a few hours. However, these measurements cannot predict the ultimate survival of cells.

Dye exclusion assay : The principle of this assay is based on the fact that viable cells are impermeable to several dyes such as naphthalene black, trypan blue, eosin Y, nigrosin green and erythrocin B. The technique basically consists of mixing the cells in suspension with the dye and examining them under the microscopy. The stained cells and the total number of cells are counted. The ***percentage of unstained cells represents the viable cells.***

Dye exclusion assay is convenient and suitable to suspension cultures than to monolayers. This is due to the fact that as the dead cells detatch from the monolayers they are lost from the assay.

The major limitation of this assay is that reproductively dead cells do not take up the dye, and will be counted as though they are viable.

Dye uptake assay : The viable cells can take up the dye diacetyl fluorescein and hydrolyse it to fluorescein. The latter is held up by the viable cells, as it is impermeable to membrane. The ***viable cells*** therefore ***emit fluorescin green*** while the dead cells do not. Thus, the viable cells can be identified.

Labeled chromium uptake assay : Labeled chromium (^{51}Cr) binds to the intracellular proteins through basic amino acids. When the cell membrane is damaged, the labeled proteins leak out of the cell, and the degree of leakage is proportional to the amount of damage. Labeled ^{51}Cr uptake method is used in the immunological studies to determine the cytotoxic activity of T-lymphocytes against target cells.

Enzyme release assays : The membrane integrity of cells can also be assessed by estimating the enzymes released. ***Lactate dehydrogenase (LDH)*** has been the most widely used enzyme for this purpose.

Based on cellular respiration

Respiration of the cells measured by oxygen utilization or carbon dioxide production can be used to assess cell viability. This is usually done by using Warburg manometer.

Based on radioisotope incorporation

By using radiolabeled substrates or metabolites, the radiolabel in the products formed can be detected. This method is particularly useful for the cytotoxicity assays of drugs. Some of the important

radioisotope incorporation methods are briefly given.

Labeled nucleotides : Incorporation of (^{3}H) thymidine into DNA, and (^{3}H) uridine into RNA are widely used for the measurement of drug toxicity.

Labeled phosphate : The cells are prelabeled with ^{32}P. When the damage occurs to cells, they release labeled phosphate which can be measured. The efficacy of drugs can be evaluated by this approach.

Based on colorimetric assays

The recent developments in the colorimetric assays by using sophisticated microplate readers are fruitfully utilized for quantitation of cells. A good correlation between the cell number and colorimetric assay are observed. Some highlights of this approach are given below.

- ***Protein*** content can be estimated by methylene blue, amido black, sulforhodamine. In the Lowry method for protein estimation, Folin-Ciocalteau reagent is used.
- ***DNA*** can be quantitated by staining with fluorescence dyes e.g. 2-diaminodino-phenylindone.
- ***Lysosomal and Golgi body activity*** by using neutral red.
- ***Enzyme activity*** assays e.g. hexosaminidase, mitochondrial succinate dehydrogenase.

Based on luminescence test

The viability of cells can be measured with good sensitivity by estimating ATP levels by luminescence based test. The principle is based on the following reaction.

$$ATP + \text{D-Luciferin} + O_2 \xrightarrow{\text{Luciferase}} AMP + 2Pi + CO_2 + \textit{Light}$$

A good sensitivity of this test is reported for the cells in the range of 20 to 2×10^7 cells/ml.

Based on apoptosis

Most of the anticancer drugs kill cells by apoptosis which can be measured for the assessment of cytotoxicity. Apoptosis can be detected by the following ways.

- Changes in the morphology.
- Detection of phosphatidyl serine in the membrane by using annexin V conjugated to fluorescein isothiocyanate (FITC) or biotin.
- DNA laddering.

SURVIVAL ASSAYS

The tests described above for measurement of cell viability and cytotoxicity are short-term, and they identify the dead/live cells at the time of assay. Many times, when the cells are subjected to toxicity (i.e. exposed to drugs, irradiated), the effects are not immediate, but may be observed after several hours or sometimes even days. The assays based on the survival of cells (i.e. retention of regenerative capacity or reproductive integrity) are preferred.

Clonogenic assay

In clonogenic assay, the survival of the cells is ***measured by plating efficiency*** (i.e. the percentage of cells seeded at subculture that give rise to colonies). The plating efficiency measures the proliferative capacity for several cell generations. Clonogenic assay broadly consists of the following stages.

1. Treatment of the cells with varying concentrations of experimental agent for about 24 hours.
2. Trypsinization followed by seeding of cells at low density.
3. Incubation of the cells for 1-3 weeks.
4. Staining and counting of the colonies.

A survival curve (semilog plot) representing the survival fraction of the cells against drug concentration is depicted in ***Fig 38.1.*** The ***inhibitory concentration*** (***IC***) refers to the drug concentration required to inhibit the viability of cells. Thus, IC_{50} and IC_{90} represent the concentrations of a compound that respectively inhibit 50% and 90% of colony formation.

As is evident from the graph, the curve has a knee wherein IC_{50} lies, while IC_{90} falls in the linear range. Therefore, the differences will be more significant in the linear range.

The clonogenic assay is influenced by several factors, the important ones are listed.

- Concentration of the toxic agent.
- Duration of exposure.

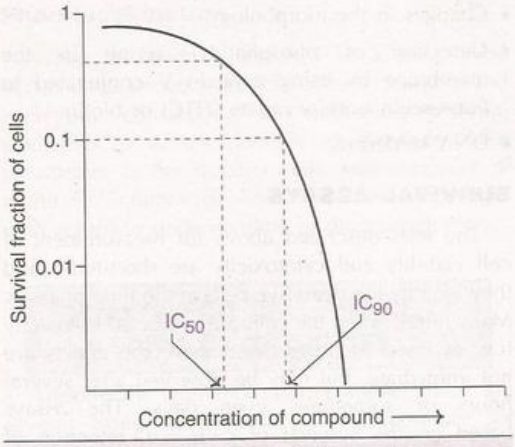


Fig. 38.1 : Survival curve in clonogenic assay.

- Cell density during exposure.
- Cell density during cloning.
- Size of colony.

MTT-based cytotoxicity assay

The tetrazolium salt 3, (4.5-dimethyl-thiazol-2-yl)-2, 5-diphenyl tetrazolium bromide) is commonly known as MTT. It is dye, and is widely used in cytotoxicity assays.

The growing cells in the log phase are exposed to cytotoxic drug. The drug is then removed and the cells are allowed to proliferate for 2-3 population doubling times (PDTs). The number of surviving cells can be detected by MTT dye reduction. The concentration of MTT-formazan formed can be determined spectrophotometrically.

MTT-based cytotoxicity assay is carried out in the following stages.

1. Incubation of monolayer cultures with varying drug concentrations in microtitration plates.

2. Removal of drug and feeding of plates to achieve 2-3 PDTs.

3. Treatment of plates with MTT, and removal of medium and MTT.

4. Measurement of MTT-formazan in an ELISA plate reader.

When the absorbance of test wells/control wells of the microplate is plotted against the concentration of the cytotoxic drug, a sigmoid curve is obtained.

METABOLIC ASSAYS

The metabolic assays are based on the measurements of metabolic responses of the cells. These test are carried out after exposure of the cells to cytotoxic drugs (either immediately or after 2-3 population doublings).

The most commonly used metabolic measurements are DNA, RNA or protein synthesis (by estimating their concentration), besides the assay of certain dehydrogenase enzymes. Some details on the estimation of DNA and protein are already given (See p. 493).

Limitations of metabolic assays : The estimation of the total content of DNA protein may or may not be indicative of increase in cell number. This is because these assays cannot discriminate between the proliferative and metabolic activity of cells. Some workers, therefore prefer to confirm the metabolic measurements by colonogenic survival assay.

TRANSFORMATION ASSAYS

The following are the commonly used assays for measurement of *in vitro* transformation.

- Evidence of mutagenesis.
- Anchorage independence.
- Reduced density limitation of cell proliferation.

Mutagenesis

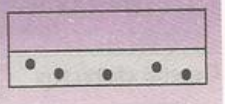
Mutagenesis can be assayed by ***sister chromatid exchange*** (***SCE***). SCE basically involves the reciprocal exchange of DNA segments between sister chromatids at identical loci in the S-phase of cell cycle. Sister chromatid exchanges are more sensitive to mutagenesis than chromosomal breaks. For this reason, SCEs are preferred in mutagenesis research and transformation assay.

The SCE technique basically involves the incorporation of radioactive nucleotides into replicating DNA and detection of SCEs by fluorescence plus Giemsa (FPG) technique.

INFLAMMATION ASSAYS

Inflammation assays are required for testing the various forms of allergy induced by cosmotics, pharmaceuticals and other xenobiotics. These assays are at the early stages of development in the culture cells.

Cell Transformation and Cell Cloning



Cell transformation due to changes in the genetic material, and cell cloning involving the production of a population single cell are described in this chapter.

TRANSFORMATION OF CELLS

Transformation broadly refers to the ***change in phenotype of a cell due to a new genetic material***. As regards the cultured cells, transformation involves spontaneous or induced permanent phenotypic alterations as a result of heritable changes in DNA, and consequently gene expression.

Transformation of cells may occur due to any one of the following causes that ultimately result in a changed genetic material.

- Spontaneous
- Infection with transforming virus.
- From gene transfection.
- Exposure to chemical carcinogens.
- Exposure to ionizing radiations.

Characteristics of transformed cells

The general characters of transformed cells are given in ***Table 39.1***. They are grouped as ***genetic***, ***structural***, ***growth*** and ***neoplastic***, and listed.

Transformation is associated with ***genetic instability***, ***immortalization***, ***aberrant growth control*** and ***malignancy***. These aspects are briefly described.

GENETIC INSTABILITY

In general, the cell lines in culture are prone to genetic instability. A majority of normal finite cell lines are usually genetically stable while cell lines from other species (e.g. mouse) are genetically unstable, and can get easily transformed. The continuous cell lines derived from tumors of all species are unstable.

The normally occurring genetic variations in the cultured cells are due to the following causes.

1. High rate of spontaneous mutations in the *in vitro* conditions, possibly due to high rate of cell proliferation.

2. The continued presence of mutant cells in the culture, as they are not normally eliminated.

IMMORTALIZATION

The ***acquisition of an infinite life span by a cell*** is referred to as immortalization.

Most of the normal cells (from different species) have a finite life span of 20-100 generations. But some cells from mouse, most of the tumor cells have infinite life span, as they go on producing continuous cell lines.

Control of finite life span of cells

The finite life span of cultured cells is regulated by about 10 ***senescence genes***. These dominantly

Table 39.1 General characteristics of transformed cells

Genetic characters	
	Aneuploid
	Heteroploid
	High spontaneous mutation rate
	Overexpressed oncogenes
	Mutated or deleted suppressor genes
Structural characters	
	Altered cytoskeleton
	Changed extracellular matrix
	Modified expression of cell adhesion molecules
	Disrupted cell polarity
Growth characters	
	Immortalized cells
	Loss of contact inhibition
	Anchorage independent
	Density limitation of growth reduced
	Growth factor independent
	Low serum requirement
	Shorter population doubling time
Neoplastic characters	
	Tumorigenic
	Invasive
	Increased protease secretion

acting genes synthesize products which inhibit the cell cycle progression.

It is strongly believed that immortalization occurs due to inactivation of some of the cell cycle regulatory genes e.g. Rb, p^{53} genes.

Immortalization of cells by viral genes

Several viral genes can be used to immortalize cells. Some of these genes are listed below.

SV40LT
HPV16E6/E7
hTRT
Ad5E1a
EBV.

Among the above viral genes, ***SV4OLT is most commonly used to induce immortalization***. The product of this gene (T antigen) binds to senescence genes such as Rb and p^{53}. This binding restricts surveillance activity of senescence genes. The result is an increased genomic instability and activity, leading to further mutations favouring immortalization.

For the process of immortalization, the cells are infected with retroviruses containing immortalizing gene before they enter senescence. By this way, the life span of the cells can be extended by 20-30 population doublings. Thereafter, the cells cease to proliferate, and enter a ***crisis phase*** that may last for several months. At the end of the crisis phase, a small portion of cells can grow, and eventually become immortalized.

Immortalization of human fibroblasts : The human fibroblasts are most successfully immortalized by the viral gene namely SV40LT. The process of fibroblast immortalization is complex and indirect with a very low probability i.e. about 1 in 10^7 cells.

Immortalization of cells by telomerase-induction

The most important cause of finite life span of cells (i.e. senescence) is due to telomeric shortening, followed by cell death (apoptosis).

If the cells are transfected with telomerase gene ***htrt***, the life span of the cells can be extended. And a small proportion of these cells become immortal.

ABERRANT GROWTH CONTROL

The transformed cells and the cells from tumors, grown in culture show many aberrations with respect to growth and its control. The growth characteristics of these cell are listed in ***Table 39.1***, and some of them briefly described hereunder.

Anchorage independence

There occur several changes on the cell surfaces of transformed cells. These include alterations in the cell surface glycoproteins and integrins, and loss of fibronectin. Some of the transformed cells ***may*** totally ***lack cell adhesion molecules (CAMs)***.

The modifications on the surface of transformed cells leads to a decrease in cell — cell, and cell-substrate adhesion. The net result is that there is a

reduced requirement for attachment and spreading of the cells to proliferate. This phenomenon is referred to as anchorage independence.

Anchorage independent cells grow in a disorganized fashion. These cells may be comparable with the tumor cells detached from the native tissue which can grow in foreign tissues i.e. formation of metastases.

Contact inhibition

The ***transformed cells*** are characterized by ***loss of contact inhibition***. This can be observed by the morphological changes in the disoriented and disorganized monolayer cells. This results in a reduced density limitation of growth, consequently leading to higher saturation density compared to normal cells (Refer Chapter 35 and ***Fig. 35.6***).

Low serum requirement

In general, transformed cells or tumor cells have lower serum dependence than the normal cells. This is mostly due to the secrection of autocrine growth factors by the transformed cells (**Note :** The normal cells in culture are dependent on the serum for the supply of growth factors). Some of the growth factors produced by tumor cells are given.

- Colony stimulating factor (CSF).
- Transforming growth factor (TGFa).
- Interleukins 1,2 and 3.
- Vasoactive intestinal peptide (VIP).
- Gastrin releasing peptide.

It may be noted that many normal cells (fibroblasts, endothelial cells) also produce autocrine factors during active stage of cell proliferating. Hence, these factors will not be of much use to serve as markers of cell transformation.

TUMORIGENICITY

Cell transformation is a complex process that often results in the formation of neoplastic cells. The cell lines obtained from malignant tumors are already transformed. Such cells may undergo further transformation in the *in vitro* culture due to

- Increased growth rate.
- Immortalization.
- Reduced anchorage dependence.

For the malignant transformation of cells, several steps may be required. The following two approaches are in use to understand malignant-associated properties of cultured cells.

1. The cells can be cultured from malignant tumors and characterized.

2. Viral genes or chemical carcinogens can be used to transform the untransformed cells.

CELL CLONING

In the traditional culture techniques, the cells are heterogenous in nature. Isolation of pure cell strains is often required for various purposes. ***Cell cloning broadly involves*** the processes connected with the ***production of a population of cells derived from a single cell***. Cloning of continuous cell lines is much easier compared to that of the primary cultures, and finite cell lines.

There are certain limitations for cloning of culture cells derived from normal tissues. These cells survive for a limited number of generations, and therefore cloning may not result in any significant number of cells. On the other hand, cloning of continuous cell lines due to their transformed status is much easier. Thus, the transformed cells have higher cloning efficiency compared to normal cells.

Cloning may be carried out by two approaches-monolayer and suspension cultures.

1. **Monolayer culture :** Petri dishes, multiwell plates or flasks can be used for cloning by monolayer culture. It is relatively easy to remove the individual colonies of cells from the surfaces where they are attached.

2. **Suspension culture :** Cloning can be carried out in suspension by seeding cells into viscous solutions (Methocel) or gels (agar). As the daughter cells are formed in suspension, they remain intact and form colonies in suspension.

DILUTION CLONING

Dilution cloning is the most commonly used technique for cloning of monolayer cells, and involves the following stages (***Fig 39.1***).

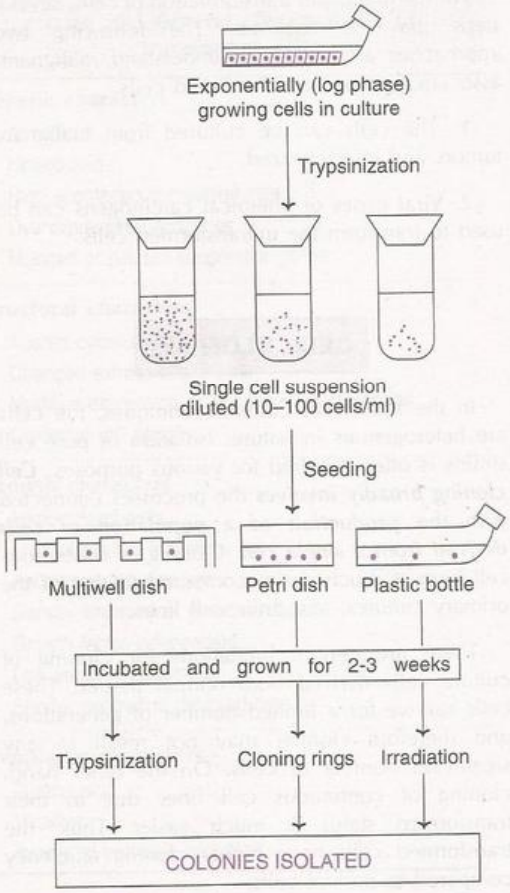


Fig. 39.1 : Dilution cell cloning technique for monolayer cultures.

1. Trypsinization of cells (at log phase) to produce single cell suspensions.

2. Dilution of the cells to about 10-100 cells/ml.

3. Seed the cells in multiwell dishes, petri dishes or plastic bottles.

4. Incubate under appropriate conditions for 1-3 weeks.

5. Isolate individual colonies.

The clones can be isolated directly from the multiwell dishes (by trypsinization), by cloning ring technique from petridishes, and by irradiating the plastic bottle.

STIMULATION OF PLATING EFFICIENCY

Plating efficiency represents the ***percentage of cells seeded at subculture that gives rise to colonies.*** The plating efficiency and ***cloning efficiency*** are said to be identical, if each colony is derived from a single cell.

The plating efficiency is around 10% for continuous cell lines, while for primary cultures and finite cell lines, it is quite low — 0.5 to 5% or sometimes even zero. Several attempts are made to improve the plating efficiency in the culture laboratories. These approaches are based on the assumption that the cells at low densities require more nutrients and/or growth factors.

The stimulation of plating efficiency with regard to ***culture factors***, ***conditioned medium*** and feeder layers is briefly described

Culture factors

Medium : A medium rich in various nutrients is better suited for improving the plating efficiency.

Serum : Fetal bovine serum is better than horse or calf serum, if the addition of serum is required.

Addition of metabolites : Supplementing the medium with intermediary metabolites (e.g. pyruvate, α-ketoglutarate) and nucleosides stimulates plating efficiency.

Addition of hormones : Dexamethasone, a synthetic analogue of hydrocortisone, improves plating efficiency e.g. fibroblasts, melanoma cells, chick myoblasts. Insulin also stimulates plating efficiency of several cell types.

Carbon dioxide : CO_2 significantly influences plating efficiency. Most of the cells require 5% CO_2 while for some cells, a lower CO_2 concentration (around 2% CO_2) is better e.g. human fibroblasts. HEPES is used to protect the medium from pH fluctuations.

Pretreatment of substrate : The plates pretreated with fibronectin or polylysine show an improved plating efficiency.

Use of purified trypsin : Some workers prefer to use purified trypsin instead of crude trypsin for trypsinization so that the plating efficiency is better.

Conditioned medium

A medium that has already been used for the growth of other cells contains certain metabolites, growth factors and other products that stimulate growth. Such a medium, referred to as conditioned medium, when added to cell cloning medium ***improves plating efficiency***.

Feeder layers

A layer growth-arrested living cells, referred to as feeder layer, promotes plating efficiency. This is because the feeder cells provide nutrients, growth factors and matrix constituents that support survival, growth and proliferation of cells.

SUSPENSION CLONING

Certain cells, particularly the transformed fibroblasts and hematopoietic stem cells, can be more conveniently cloned in suspension rather than monolayers.

Suspension cloning can be carried out by using agar or Methocel, which can hold the cells of a given colony together, and prevent mixing of colonies.

The technique of cloning in agar suspension, is carried out in the following stages (***Fig 39.2***).

1. Cells from suspension culture or monolayers can be used. The monolayer cells have to trypsinized while the suspension cells can be directly used.
2. Count the cells and dilute serially so that 10-200 cells/ml are finally present.
3. Freshly prepared agar medium with appropriate dilution is used.
4. The agar medium is inoculated with the diluted cells.
5. Incubation for 1-3 weeks given clones in culture.

ISOLATION OF CLONES

After the cloning is complete, selection and isolation of specific cell strains is the next important step. This is essentially required for the propagation of cells.

If the monolayer cells are cloned directly in the multiwell plates, the colonies can be isolated by trypsinization of the individual wells. However, when the cloning is carried out in petri dishes, the colonies can be separated from the medium by placing stainless steel or ceramic rings around the colonies.

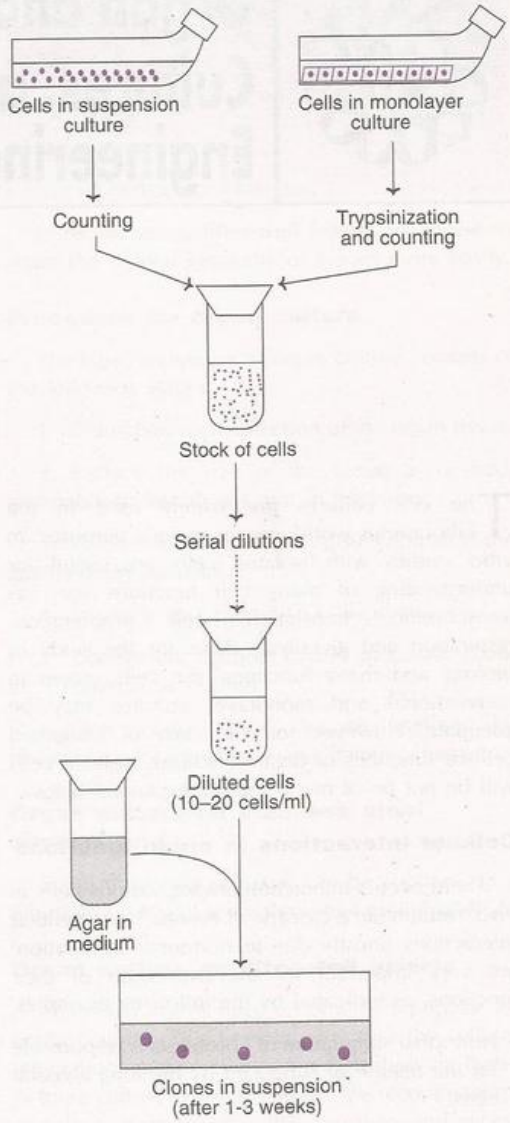
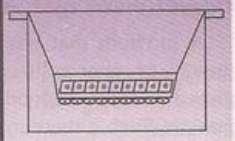


Fig. 39.2 : Suspension cloning technique in agar.

Micromanipulation

The genuineness of a colony, that is the colony is derived from a single cell can be done by micromanipulation. This can be achieved by ***monitoring the colony formation at the early stages of cell cloning***. Micromanipulation, if properly done, is believed to be the conclusive method for determining the genuine clonality of a clone.

Organ and Histotypic Cultures, and Tissue Engineering



The cell cultures are widely used in the laboratories worldover for various purposes. *In vitro* studies with isolated cells are useful for understanding of many cell functions such as transcription, translation, cell proliferation, respiration and glycolysis. Thus for the study of biology and many functions, the cells grown in conventional and monolayer cultures may be adequate. However, for the study of integrated cellular functions or organ functions, isolated cells will be not be of much use, as explained below.

Cellular interactions in organ functions

There occurs interaction among various cells *in vivo*, resulting in a cascade of events. These cellular interactions (mostly due to hormonal stimulation) are very important for the expression of their functions, as indicated by the following examples.

- Hormonal stimulation of fibroblasts is responsible for the release of surfactant by the lung alveolar cells.
- Androgen binding to stromal cells stimulates prostate epithelium.

Besides hormones, nutritional factors and xenobiotics also exert stimulatory effects on the cells to function in a coordinated fashion.

ORGANOTYPIC MODELS

As explained above, the cellular interactions that occur in the *in vivo* system are not possible with isolated cells. The ***recent developments in the organ and histotypic cultures focus to create in vitro models comparable*** (as far as possible in biology and functions) ***to the in vivo systems***. The purpose of this organotypic models is to retain the original structural and functional interactive relationships of the organ. There are three broad approaches in this direction.

1. **Organ cultures :** The whole organs or small fragments of the organs that retain the special and intrinsic properties are used in culture.

2. **Histotypic cultures :** The cell lines grown in three dimensional matrix to high density represent histotypic cultures.

3. **Organotypic cultures :** In this case, the cells from different lineages are put together in the desired ratio and spatial relationships to create a component of an organ in the laboratory.

ORGAN CULTURES

The use of organ cultures (organs or their representative fragments) with reference to structural integrity, nutrient and gas exchange, growth and differentiation, along with the advantages and limitations is briefly described.

Structural integrity

As already stated, the isolated cells are individual, while in the organ culture, the cells are integrated as a single unit. The cell to cell

association, and interactions found in the native tissues or organs are retained to a large extent.

As the structural integrity of the original tissue is preserved, the associated cells can exchange signals through cell adhesion or communications.

Nutrient and gas exchange

There is ***no vascular system*** in the organ culture. This ***limits the nutrient supply and gas exchanges*** of the cells. This happens despite the adequate care taken in the laboratory for the rapid diffusion of nutrients and gases by placing the organ cultures at the interface between the liquid and gaseous phases. As a consequence, some degree of necrosis at the central part of the organ may occur.

Some workers prefer to use high O_2 concentration (sometimes even pure O_2) in the organ cultures. Exposure of cells to high O_2 content is associated with the risk of O_2 induced toxicity e.g. nutrient metabolite exchange is severely affected.

Growth and differentiation

In general, the organ cultures do not grow except some amount of proliferation that may occur on the outer cell layers.

Advantages of organ cultures

- Provide a direct means of studying the behaviour of an integrated tissue in the laboratory.
- Understanding of biochemical and molecular functions of an organ/tissue becomes easy.

Limitations of organ cultures

- Organ cultures cannot be propagated, hence for each experiment there is a need for a fresh organ from a donor.
- Variations are high and reproducibility is low.
- Difficult to prepare, besides being expensive.

TECHNIQUES OF ORGAN CULTURE

The most important requirement of organ or tissue culture is to place them at such a location so that optimal nutrient and gas exchanges occur. This is mostly achieved by keeping the tissue at gas-limited interface of the following supports.

- Semisolid gel of agar.
- Clotted plasma.
- Microporous filter.
- Lens paper.
- Strip of Perspex or Plexiglas.

In recent years, ***filter-well inserts*** are in use to attain the natural geometry of tissues more easily.

Procedure for organ culture

The basic technique of organ culture consists of the following stages.

1. Dissection and collection of the organ tissue.
2. Reduce the size of the tissue as desired, preferably to less than 1 mm in thickness.
3. Place tissue on a support (listed above) at the gas medium interface.
4. Incubate in a humid CO_2 incubator.
5. Change the medium (M199 or CMRL 1066) as frequently as desired.
6. The organ culture can be analysed by histology, autoradiography and immunochemistry.

Organ culture on stainless steel support grid

Small fragments of tissue can be cultured on a filter laid on top of a stainless steel grid (***Fig. 40.1***).

Organ culture on filter-well inserts

Filter-well inserts have become very popular for organ cultures. This is mainly because the cellular interaction, stratification and polarization are better in these culture systems. Further, the recombination of cells to form tissue — like densities, and access to medium and gas-exchange are better.

The four different types of filter wells for growing tissues in the form of cell layers are depicted in ***Fig. 40.2***.

- Growth of cell layer on top of filter (***Fig. 40.2A***).
- Growth of cell layers on matrix (collagen or matrigel) on top of filter (***Fig. 40.2B***).
- Cell layers grown on the interactive cell layers placed on the underside of filter (***Fig. 40.2C***).

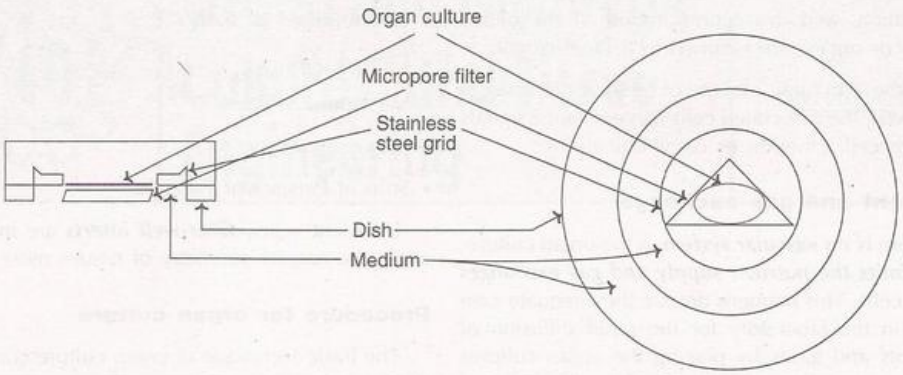


Fig. 40.1 : Organ culture on stainless steel support grid.

- Cell layer grown on the matrix with interactive cell layer on the underside of the filter (***Fig. 40.2D***).

Filter well-inserts with different materials (ceramic, collagen, nitrocellulose) are now commercially available for use in culture laboratories.

Filter-well inserts have been successfully used to develop functionally integrated thyroid epithelium, stratified epidermis, intestinal epithelium and renal (kidney) epithelium.

HISTOTYPIC CULTURES

Growth and propagation of cell lines in ***three-dimensional matrix to high cell density*** represent histotypic cultures. The advantage with this culture system is that dispersed monolayer cultures can be used to regenerate tissue-like structures.

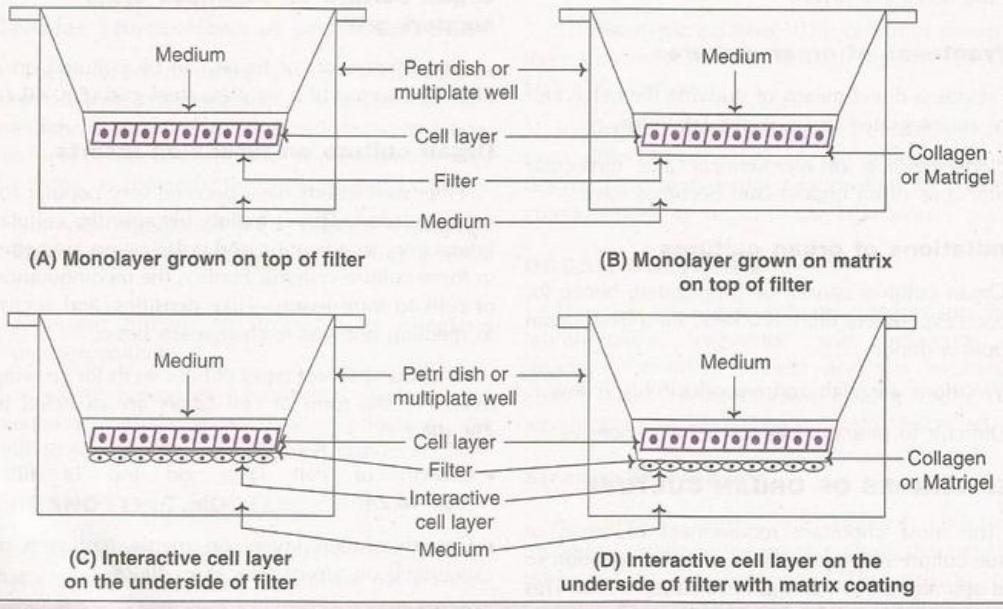


Fig. 40.2 : Diagrammatic representation of organ culture in filter–well inserts.

The commonly used techniques in histotypic cultures use gel and sponge hollow fibers and spheroids.

Gel and sponge technique

The cells (normal or tumor) in culture can penetrate gels (collagen) or sponges (gelatin) which provides a matrix for morphogenesis of primitive cells. This approach has been used for the development of mammary epithelium, and some tubular and glandular structures.

Hollow fibers technique

In recent years, perfusion chambers with a bed of plastic capillary fibers have been developed. The advantage of using hollow fibers in histotypic cultures is that nutrient and gas exchange is more efficient. As the cells attached to capillary fibers grow, there occurs an increase in cell density to form tissue-like structures. Many workers claim that the behaviour of high-density cells formed on hollow fibers is comparable to their *in vivo* behaviour. For instance, choriocarcinoma cells grown in hollow fiber cultures release more chorionic gonadotrophin than in a conventional monolayer.

Hollow fiber culture techniques are regarded as ideal systems for the industrial production of several biologically important compounds. Work is progressing in this direction.

THREE DIMENSIONAL CULTURES

Spheroids in histotypic culture

Spheroids represent the ***clusters of cells usually formed by the reassociation of dissociated cultured cells***. It is known for some years that the dissociated embryonic cells reassemble to form a specialized structure.

The basic principle of using spheroids in histotypic culture is that the cells in heterotypic or homotypic aggregates are capable of sorting out themselves into groups to form tissue-like architecture.

The major drawback of spheroids is the limitation in the diffusion and exchange of nutrients and gases.

Multicellular tumor spheroids (MCTS)

Multicellular tumor spheroids provide an *in vitro* proliferating model for studies on tumor cells. The three dimensional structure of MCTS allows the experimental studies related to drug therapy, penetration of drugs, resistance to radiation etc. Further, MCTS have also been used to study several biological processes.

- Regulation of cell proliferation and differentiation.
- Immune responses.
- Cell death.
- Cell invasion.
- Gene therapy.

The main advantage of three dimensional cell culture (in the form of MCTS) is that they ***provide a well-defined geometry of cells planar or spherical*** which is directly related to the structure and function. It is now well accepted that the MCTS behave like the initial avascular stages of solid tumors *in vivo*. However, beyond a critical size (≥ 500 mm), most of the MCTS develop necrosis (death of cells) at the centre surrounded by viable cells. A diagrammatic representation of MCTS in comparison with tumor is depicted in ***Fig. 40.3***.

Technique of MCTS production

Single-cell suspension obtained from trypsinized monolayer cells or disaggregated tumor is inoculated into the medium in magnetic stirrer flasks or roller tubes. As the incubation is carried out for about 3-5 days, aggregates of cells representing spheroids are formed. It is observed that spheroid formation is more efficient under static conditions on stationary and non-adhesive surfaces. For this reason, agar/agarose-coated culture dishes to which cells do not adhere are frequently used to initiate spheroid formation.

Once the spheroids are formed, they are transferred to 24 well plates for analysis. Spheroid growth is quantified by measuring their diameters regularly. This can be done by using a microscope eyepiece micrometer or an image analysis scanner. Good growth of spheroids is observed when grown in wells.

Transfectant mosaic spheroids : It is now possible to produce spheroids from cells that have been transfected with different genes. Mosaic spheroids are formed by mixing transfected and non-transfected spheroids in the desired proportion.

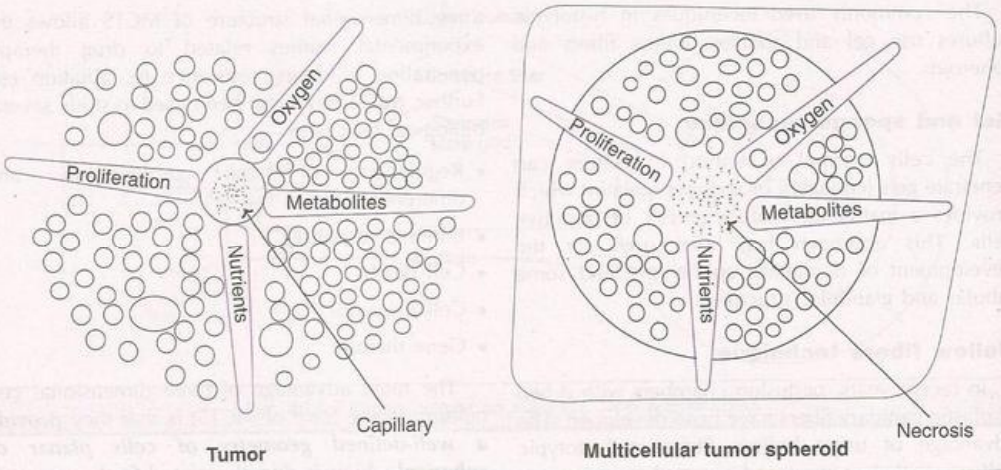


Fig. 40.3 : A diagrammatic representation of a tumor in comparison with a multicellular tumor spheroid.

MCTS co-cultures : MCTS can be produced from heterogenous cells also, forming MCTS co-cultures. This is comparable to heterologous spheroids (in short heterospheroids) consisting of tumor cells in combination with host cells. Some of the MCTS co-cultures are listed.

- MCTS and immune cells.
- MCTS and fibroblasts.
- MCTS and endothelial cells.

Heterospheroids with heterotypic cell interaction serve as good models for studying several *in vivo* processes e.g. inflammation. MCTS co-cultures are very useful in tissue modelling and tissue engineering, the details of which are given later.

Applications of spheroids or MCTS

Spheroids have a wide range of applications. Some of the important ones are listed.

- Serve ***as models for a vascular tumor growth.***
- For the ***study of gene expression*** in a three dimensional configuration of cells.
- To determine the effect cytotoxic drugs, antibodies, radionucleotides used for therapeutic purposes.
- ***To study certain disease processes*** e.g. rheumatoid arthritis.
- For the development of ***gene therapies*** for several diseases e.g. cancer.
- To evaluate radiation effects on target tissues.
- For the development of tissues and tissue models.

ORGANOTYPIC CULTURES

Organotypic culture basically involves the combination of cells from different lineages in a determined ratio to create a component of an organ. With the advances in the organotypic culture techniques, it is now possible to develop certain tissues or tissue models.

- Skin equivalents have been created by co-culturing dermis with epidermis with interviewing layers of collagen.
- Models for prostate and breast.
- Models for control of growth and differentiation of lung.

TISSUE ENGINEERING

Tissue engineering (TE) refers to the ***application of the principles of engineering to cell culture for the construction of functional anatomical units*** (tissues/organs)**.** The ultimate purpose of TE is to supply various body parts for the repair or replacement of damaged tissues or organs. Tissue engineering may be regarded as the ***backbone of reconstructive surgery***. It is possible to supply almost all surgical implants (skin, blood vessels,

ligaments, heart valves, joint surfaces, nerves) through the developments in tissue engineering.

There are two schools of thought while dealing with tissue engineering techniques.

1. Some workers believe that the ***living cells possess an innate potential of biological regeneration***. This implies that when suitable cells are allowed to grow on an appropriate support matrix, the cells proliferate, and ultimately result in an organized and functional tissue. This tissue resembles the original tissue in structure and function. This approach is very simple, and economical, although the success is limited.

2. According the second school of thought, there are several control processes to produce a new and functional tissue. Thus, tissue regeneration *in vivo* or tissue production *in vitro* are very complex. Therefore, tissue engineering is not a simple regeneration of cells, and it requires a comprehensive approach with a thorough understanding of cellular configuration, spacial arrangement and control process.

Tissue engineering is a complicated process. Some fundamental and basic aspects of TE with special reference to the following aspects are briefly described.

- Cell sources and culture
- Cell orientation
- Cell support materials
- Design and engineering of tissues.

CELL SOURCES AND CULTURE

Adequate quantities of cells are required for tissue engineering. There are three types of cell sources-autologous, allogeneic and xenogeneic.

Autologous cell sources

The cell source is said to be autologous when the ***patient's own cells are used in TE***. This is a straight forward approach. A piece of desired tissue is taken by biopsy. It may be enzymatically digested or explant cultured, and the cells are grown to the required number.

The main ***advantages*** of autologous cells in TE are :

- Avoidance of immune complications
- Reduction in the possible transfer of inherent infections.

There are certain ***disadvantages*** associated with autologous cells.

- It is not always possible to obtain sufficient biopsy material from the patient.
- Disease state and age of the patient will be limiting factors.

Allogeneic cell sources

If the ***cells*** are ***taken from a person other than the patient***, the source is said to be allogeneic. The ***advantages*** of allogeneic cell source are listed.

- Obtained in good quantity from a healthy donor.
- Cells can be cultured in a large scale.
- Cost-effective with consistent quality.
- Available as and when required by a patient.

The major ***problem*** of allogeneic cell source is the ***immunological complications*** that may ultimately lead to graft rejection. The immune responses however, are variable depending on the type of cells used. For instance, endothelial cells are more immunogenic while fibroblasts and smooth muscle cells are less immunogenic. The age of the donor is another important factor that contributes to immunological complications. Thus, cells from adult donors are highly immunogenic while fetal or neonatal cells elicit little or no immune response.

Xenogeneic cell sources

When the ***cells are taken from different species*** (e.g. pig source for humans) the source is said to be xenogeneic. This approach is not in common use due to immulogical complications.

CULTURE OF CELLS

The methods adapted for culturing of cells required for tissue engineering depend on the type and functions of cells. For most of the cells, the conventional monolayer cultures serve the purpose. The major drawback of monolayer cultures is that cells may lose their morphology, functions and proliferative capacity after several generations.

Some workers prefer three dimensional cultures for the cells to be used in tissue engineering. For details on these techniques, See p. 471. The nutrient and gaseous exchanges are the limiting factors in three dimensional cultures.

Genetic alterations of cultured cells for use in TE

Gene therapy can be successfully employed in tissue engineering. This can be achieved by transferring the desired genes to cells in culture. The new genes may increase the production of an existing protein or may synthesize a new protein. Some success has been achieved in this direction.

- Genetically altered fibroblasts can produce transferrin, clotting factor VIII and clotting factor IX.
- Modified endothelial cells can synthesize tissue plasminogen activator.
- Genetically engineered keratinocytes can produce transglutaminase-I (This enzyme is lacking in patients suffering from a dermal disorder, lamellar icthyosis). The altered keratinocytes proved successful when transplanted in animal (rat) models of this disease.

CELL ORIENTATION

The orientation of cells with regard to specific shape and spatial arrangement is influenced by the following environmental factors.

1. Substrate or contact guidance.
2. Chemical gradients.
3. Mechanical cues.

Substrate guidance

The topographical features of the substrate determine the contact guidance. These features may be in the form of ridges, aligned fibers etc. It is possible to use differential attachment to substrates as a means of producing different alignment of cells. In recent years, synthetic polymer substrate collagen fibrils and fibronectin are used as bioresorbable templates for tissue engineering.

Chemical gradients

Development of chemical gradients is required for cellular orientation and ***for the stimulation of cellular functions***. Certain growth factors and extracellular macromolecules are capable of creating chemical gradients e.g. vascular endothelial growth factor (VEGF), oligosaccharide fragments of hyaluronan, fibronectin, collagen.

There are certain practical difficulties in maintaining effective chemical gradients for the cells in three dimensional cultures. This is particularly the limiting factor when the cells become dense.

Mechanical cues

The response of the cells to mechanical signals is complex and this may result in any one or more of the following.

- Changes in the cell alignment.
- Deformation of cytoskeleton.
- Altered matrix formation.
- Synthesis of regulatory molecules (e.g growth factors, hormones).

There are mainly three mechanical cues governing cell populations.

1. Tensional forces.
2. Compressional forces.
3. Shear forces.

CELL SUPPORT MATERIALS

The support materials of cells largely determine the nature of adherent cells or cell types, and consequently tissue engineering. There are a large number of support materials which may be broadly categorized as follows.

- Traditional abiotic materials.
- Bioprosthesis materials.
- Synthetic material.
- Natural polymers.
- Semi-natural materials.

Traditional abiotic materials

The traditional abiotic support materials include plastics, ceramics and metals. These materials cannot be resorbed or become biologically integrated into the tissues. Therefore, it is preferable to avoid the traditional materials in tissue engineering.

Bioprosthesis materials

The ***natural materials modified to become biologically inert represent bioprosthesis materials.*** They are formed by extensive chemical cross linking of natural tissues. For instance, the natural collagen-based connective tissue (e.g. porcine heart values) can be stabilized by treatment with glutaraldehyde. The product formed is non-immunogenic that remains unchanged at the site of transplantation for several years. However, growth of some cells or even connective tissue can occur on bioprosthesis materials. The design and fabrication of these materials is done in such a way that their functions are not affected by the surrounding host tissues.

Synthetic materials

A wide range of synthetic bioresorbable polymers are available as support materials. The most commonly used ***polymers in tissue engineering*** are poly (glycolic acid) (PGA), poly (lactic acid) (PLA), and copolymer PLGA (poly (lactic-co-glycolic acid). The composition and dimensions of these polymers can be so adjusted to make them stable *in vivo*, besides supporting *in vivo* cell growth.

There are certain ***advantages*** in using synthetic polymers.

- Production is easy and relatively cheap.
- Composition of polymers is reproducible even in large scale production.

There are however, some ***disadvantages*** also.

- Compatibility with cells is not as good as natural polymers.
- On degradation, they may form some products which cause undesirable cellular effects.

Natural polymers

The most widely used natural polymer materials are collagen-chondroitin sulfate aggregates. These materials are commercially available with varying composition under the trade name ***Integra***. The other natural polymers for cell support are usually obtained by their aggregation in culture as it occurs *in vivo* e.g. collagen gels, fibrin glue, Matrigel and some polysaccharides. Among the polysaccharides, chitosan and hyaluronan are used as hydrated gels.

The natural polymers mainly act on the principle of intermolecular interaction within the polymers to promote intimate molecular packing. The so formed molecules can effectively serve as support materials.

Semi-natural materials

Semi-natural materials are derived from the natural macromolecular polymers or whole tissues. They are the modified materials to achieve aggregation or stabilization. Some examples of semi-natural materials are listed below.

- Chemically cross-linked hyaluronan, stabilized by benzyl esterification.
- Collagen cross-linked with agents such as tannic acid or carbodiimide.

DESIGN AND ENGINEERING OF TISSUES

The following surgical criteria are taken into consideration while dealing with tissue engineering.

- Rapid restoration of the desired function.
- Ease of fixing the tissue.
- Minimal patient discomfort.

For designing tissue engineering, the ***source of donor cells is very critical*** (See cell sources p. 473). Use of patients own cells (autologous cells) is favoured to avoid immunological complications. Allogeneic cells are also used, particularly when the TE construct is designed for temporary repair. It is observed that when the cells are cultured and/or preserved (i.e. cryopreservation), the antigenicity of allogeneic cells is reduced.

Another important criteria in TE is the ***support material***, its degradation products, cell adhesion characteristics and mechanical cues.

The design and tissue engineering with respect to skin, urothelium and peripheral nerve are briefly described hereunder.

Tissue engineered skin

It was first demonstrated in 1975 that ***human keratinocytes could be grown in the laboratory in a form suitable for grafting***. Many improvements have been made since then. It is now possible to grow epithelial cells to produce a continuous sheet which progresses to form cornified layers. The major difficulty with TE skin is the dermal layer

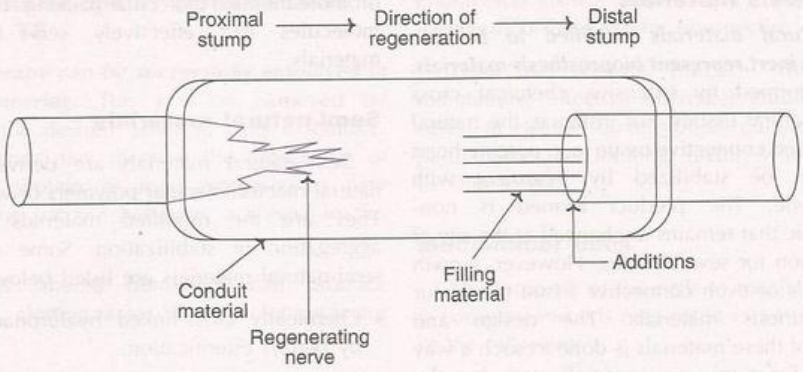


Fig. 40.4 : A diagrammatic representation of the basic design for peripheral nerve implant.

possessing blood capillaries, nerves, sweat glands and other accessory organs.

Some developments have occurred in recent years to produce implantable skin substitutes which may be regarded as tissue engineering skin constructs.

Integra™ : This is a bioartificial material composed of collagen-glycosaminoglycan. Integra™ is not a true TE construct. It is mainly used to carry the seeded cells.

Dermagraft™ : This is composed of poly (glycolic acid) polymer mesh seeded with human dermal fibroblasts from neonatal foreskins.

Apligraf™ : This has human dermal fibroblasts seeded into collagen gel. A layer of human keratinocytes is then placed on the upper surface.

The tissue constructs described above have limited shelf-life (about 5 days). However, they can integrate into the surrounding normal tissue and form a good skin cover. Further, there is no evidence of immunological complications with TE constructs.

Tissue engineered urothelium

It is now possible to culture urothelial cells and bladder smooth muscle cells. This raises the hope that the construction of TE urothelium is possible. In fact, some success has been reported in the development of a functional bladder in dogs. For this purpose, poly (glycolic acid) polymer base was shaped into a bladder and muscle cells were coated on the outer surface. The lumenal surface (i.e. inner surface) coated with pre-cultured urothelial cells. The bladder constructed in this way functioned almost like a normal one, and was maintained for about one year.

Tissue engineered peripheral nerve implants

Peripheral nerve injury is a common occurrence of trauma and tumor resection surgery, often leading to irreversible muscle atrophy. Therefore, the repair of injured peripheral nerves assumes significance.

A diagrammatic representation of the basic design of a peripheral nerve implant is depicted in *Fig 40.4*.

The regeneration of the injured nerve occurs from the proximal stump to rejoin at distal stump. The regeneration is guided by three types of substances.

Conduct material : This is the outer layer and is the primary source of guidance. Conduct material is composed of collagen-glycosaminoglycans, PLGA (poly lactic-co-glycolic acid), hyaluronan and fibronectin. All these are bioresorbable materials.

Filling material : This supports the neural cells for regeneration, besides guiding the process of regeneration. Filling material contains collagen, fibrin, fibronectin and agarose.

Additives : Additives include a large number of growth factors; neurotrophic factors (in different

forms, combinations, ratios) e.g. fibroblast growth factor (FGF), nerve growth factor (NGF). Additions of Schwann cells or transfected fibroblasts promote nerve generation process.

TISSUE MODELLING

Research is in progress to create tissue models in the form of artificial organs. Some of the recent development on experimental tissue modelling are briefly outlined.

Artificial liver

Hepatocytes, cultured as spheroids or hepatocytes and fibroblasts cultured as hetero-spheroids can be used. They are held in the artificial support systems such as porous gelatin sponges, agarose or collagen. Addition of exogenous molecules is useful for the long - term culture of liver cells. Some progress has been reported in creating artificial liver as is evident from the hepatocytes three-dimensional structure and metabolic functions.

Artificial pancreas

Spheroids of insulin secreting cells have been developed from mouse insulinoma beta cells. Some workers could implant fetal islet-like cell clusters under the kidneys of mice, although the functions were not encouraging due to limitation of oxygen supply.

Other tissue models

Pituitary gland : Multicellular spheroids could be created to study certain hormonal release e.g. luteinizing hormone (LH), following stimulation by luteinizing hormone releasing hormone (LHRH). Some success has also been achieved to create spheroids for the production of melatonin.

Thyroid gland : Thyroid cell spheroids can be used for the study of cell adhesion, motility, and thyroid follicle biogenesis.

Brain cell cultures : Three dimensional brain cell cultures have been used for the study of neural myelination and demyelination, neuronal regeneration, and neurotoxicity of lead. Aggregated brain cells are also used for the study of Alzheimer's disease and Parkinson's disease.

Heart cell cultures : Aggregated heart cells have been used for the study of cardiac development and physiology.

EMBRYONIC STEM CELL ENGINEERING

The cells of the body, when grown in culture generally maintain their original character. Thus, liver cells behave like liver cells, and keratinocytes as keratinocytes. This is possible since each type of specialized cell has a memory of its developmental history.

Stem cells are ***undifferentiated cells*** that can divide continuously (indefinitely) to produce daughter cells. The daughter cells undergo differentiation into particular cell types.

Embryonic stem (ES) cells are an extraordinary class of stem cells that can ***proliferate indefinitely*** in culture, as they possess very high developmental potential. The cultured ES cells when put back into the animal can develop into different cell types and tissues. This depends on the site at which they are introduced. For instance, ES cells in liver adopt the character and behaviour of normal liver cells.

ES CELL CULTURES TO PRODUCE DIFFERENTIATED CELLS

The ES cells obtained from an early mouse embryo can be cultured indefinitely. The cultures may be continued as monolayers or allowed to form aggregates called embryoid bodies. The cells of the embryoid bodies can specialize and differentiate under suitable conditions (by using various combinations of signal proteins) into different cells (***Fig. 40.5***). For instance, when treated with certain growth factors, ES cells can form astrocytes and oligodendrocytes. These are the main types of glial cells of the central nervous system.

Cultured ES cells, appropriately treated with growth factors, when injected into the brain can serve as progenitors of glial cells. Thus, it is possible to correct a mouse dificient in myelin-forming oligodendrocytes. The grafted cells are capable of forming myelin sheaths around axons lacking them.

HUMAN EMBRYONIC STEM CELL RESEARCH

It was in 1998, researchers for the first time reported the establishment of human embryonic stem cell lines. These cells, existing indefinitely in culture, are capable of giving rise to any human cell type.

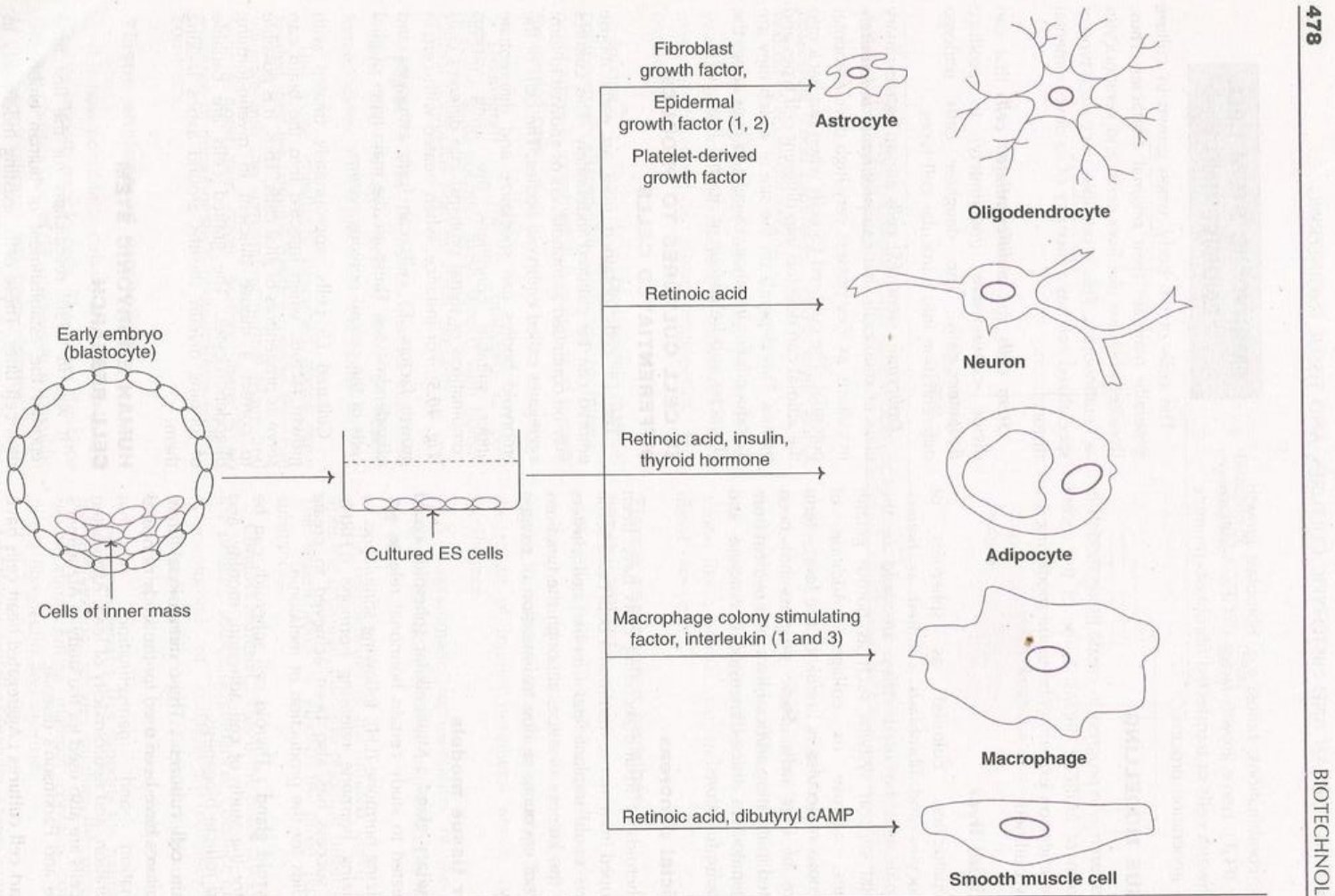


Fig. 40.5 : Culturing of embryonic stem (ES) cells and their differentiation into various cells.

Applications of human ES cells

The applications of human embryonic stem cell research are extraordinary. Some of them are listed below

- Corrections of disorders with loss of normal cells-diabetes, Alzheimer's disease, Parkinson's disease. This can be achieved by appropriate cell therapy.
- Engineering and replacement of various tissues.
- For the discovery of new drugs, and testing safety of drugs.
- To study the development of humans.

Limitations of human ES cells

There are several practical and technical difficulties associated with ES cell cultures and their utility. It is recently (2004) found that contamination with medium (the most commonly used being animal-derived media) created immunological complications when such cells are utilized for therapeutic purposes in humans. This contamination is identified to be due to a non-human molecule namely N-glycolylneuraminic acid. It is suggested that for human ES cell cultures, products of animal origin of any kind should not be used. An ideal alternative is to use human embryo-derived connective tissue cells as the feeder layer in ES cell cultures.

Ethical issues of ES cell research

Many people object to growing of human embryos in the laboratories, even if it is limited to the early stages of development. Some countries have, in fact, banned the use of Government funds for human embryo research. However, some private companies are conducting/financing research on human embryonic cells.

There are many serious ethical, legal and political issues surrounding human ES cell research. This is in addition to the enormous technical difficulties. If may take some more years before the appropriate applications of ES cell research to human welfare becomes a reality.

Transgenic Animals

The dependence of man on animals such as cattle, sheep, poultry, pig and fish for various purposes (milk, meat, eggs, wool etc.) is well known. Improvement in the genetic characteristics of livestock and other domestic animals (e.g., high milk yield, weight gain, etc.), in the early days, was carried out by ***selective breeding methods***. This technique primarily involves a combination of mating and selection of animals with improved genetic triats. Although selective breeding is very time consuming and costly, it was the only method available, till some years ago, to enhance the genetic characteristics of animals. For larger animals with long gestation period, it might take several decades to create a desired character by conventional breeding. With the advent of modern biotechnology, it is now possible to carry out manipulations at the genetic level to get the desired characteristics in animals. ***Transgenesis refers to the phenomenon of introduction of exogeneous DNA into the genome to create and maintain a stable heritable character***. The foreign DNA that is introduced is called ***transgene***. And the animal whose genome is altered by adding one or more transgenes is said to be transgenic. The transgenes behave like other genes present in the animals' genome and are passed on to the offsprings. Thus, transgenic animals are ***genetically engineered*** or ***genetically modified organisms (GMOs)*** with a new heritable character. It was in 1980s, the genetic manipulation of animals by introducing genes into fertilized eggs became a reality.

IMPORTANCE OF TRANSGENIC ANIMALS-GENERAL

Transgenesis has now become a powerful tool for studying the gene expression and developmental processes in higher organisms, besides the improvement in their genetic characteristics. Transgenic animals serve as good models for understanding the human diseases. Further, several proteins produced by transgenic animals are important for medical and pharmaceutical applications. Thus, the transgenic farm animals are a part of the lucrative world-wide biotechnology industry, with geat benefits to mankind.

Transgenesis is important ***for improving the quality and quantity of milk, meat, eggs and wool production***, besides creating drug resistant animals.

Milk as the medium of protein production

Milk is the secretion of mammary glands that can be collected frequently without causing any harm to the animal. Thus, milk from the transgenic animals can serve as a good and authenticated source of human proteins for a wide range of applications. Another advantage with milk is that it contains only a few proteins (casein, lactalbumin, immunoglobuline etc.) in the native state, therefore isolation and purification of a new protein from milk is easy.

Commonly used animals for transgenesis

The first animals used for transgenesis was a mouse. The '***Super Mouse***', was created by inserting a rat gene for growth hormone into the mouse genome. The offspring was much larger than the parents. Super Mouse attracted a lot of public attention, since it was a product of genetic manipulation rather than the normal route of sexual reproduction.

Mouse continues to be an ***animal of choice for*** most ***transgenic experiments***. The other animals used for transgenesis include rat, rabbit, pig, cow, goat, sheep and fish.

TRANSGENIC MICE

As already stated, mouse is the animal of choice for transgenic experiments. Being a small animal, it can be easily handled, and mouse is regarded as researcher-friendly by biotechnologists. Another improtant reason of choosing mice for transgenic experiments is it produces more eggs (normal mouse 5–10 eggs, superovulated mouse can yield up to 40 eggs) unlike the large domestic animals. In the last two decades, hundreds of different genes have been introduced into the various mouse strains. And as such, the methodology of transgenesis is well developed with laboratory mouse.

Transgenic mice have significantly contributed to the understanding of molecular biology, genetics, immunology and cancer, besides creating animal models for several human genetic diseases.

There are three ***methods for introducing a foreign gene*** (transgene) into mice, and in fact the same methods are applicable to other animals as well.

1. ***Retroviral vector method.***
2. ***Microinjection method.***
3. ***Embryonic stem cell method.***

It may be noted here that the term ***transfection*** is used for introducing a foreign gene into animals. This is quite comparable to transformation that is carried out in lower organisms (Refer Chapter 6).

RETROVIRAL VECTOR METHOD

The transfer of small pieces (8 kb) of DNA can be effectively carried out by retroviruses. This method, however, is unsuitable for transfer of larger genes. Further, even for small genes, there is a risk of losing some regulatory sequences. Above all, the biggest drawback is the ***risk of retroviral contamination*** in the products (particularly in foods for human consumption), obtained from transgenic animals. Because of these limitations, the retroviral vector method is not in regular use for transgenesis.

MICROINJECTION METHOD

The introduction of DNA by microinjection method involves the following steps (***Fig. 41.1***).

1. The young virgin female mice (4–5 weeks age) are subjected to superovulation. This is achieved by administration of follicle-stimulating hormone (pregnant mare's serum), followed by (2 days later) human chorionic gonadotropin. The superovulated mouse produces 30–35 eggs (instead of normal 5–10 eggs).

2. The above female mice are mated with males and sacrificed on the following day. The fertilized eggs are removed from the fallopian tubes.

3. By micromanipulation using a micro-injection needle and a holding pipette, the ***DNA is injected into the male pronucleus of the fertilized egg***. Adequate care must be taken to ensure that while the elastic nuclear membrane is punctured, the needle does not touch the nucleoli. A dissection microscope can be used for identifying the male pronucleus (larger in size) and for microinjection.

4. The eggs with the transgenes are kept overnight in an incubator to develop to a 2-cell stage. These eggs are then implanted microsurgically into a foster mother i.e., pseudo-mouse pregnant female mouse which has been mated the previous might with vasectimized (or infertile) male. The foster mother can deliver pups after 3 weeks of implantation.

The presence of transgene in the pups can be identified by polymerase chain reaction or Southern blot hybridization.

The mouse carrying the foreign gene is the transgenic founder from which pure transgenic lines can be established.

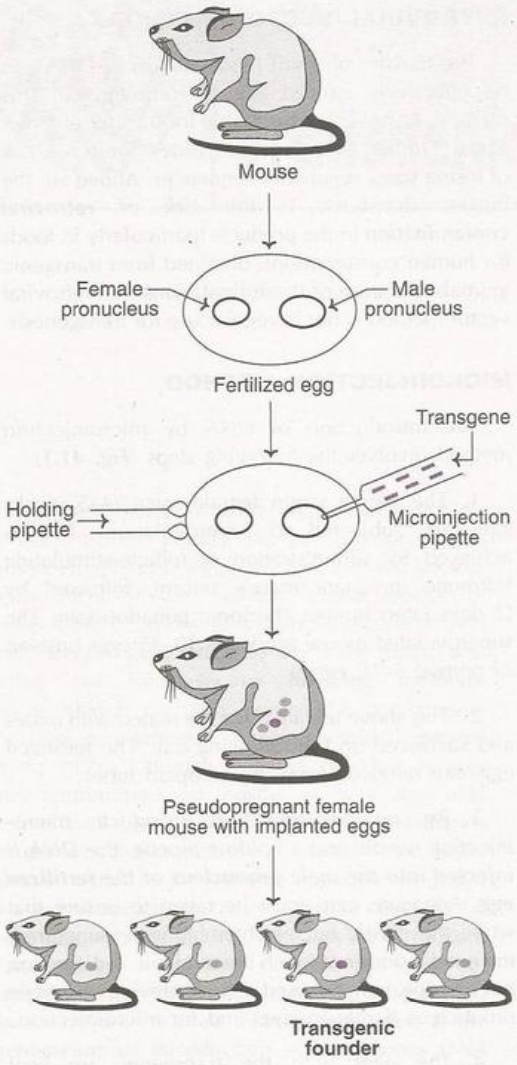


Fig. 41.1 : DNA microinjection method to produce transgenic mice.

The microinjection method involves several steps and none of them is 100% efficient for any animal to develop into transgenic animal. In case of mouse, it was found that about 65% of the fertilized eggs survive microinjection procedure, about 25% of the implanted eggs develop into pups, and only 25% of them are transgenic. Thus, if one starts with 1000 fertilized eggs, only 30 to 50 transgenic pups may be produced i.e., 3–5% of the inoculated eggs develop into transgenic animals.

Limitations of microinjection method

Besides the ***low efficiency*** (discussed above), there are some other disadvantages in this technique. The foreign DNA randomly integrates into the host genome. Sometimes, even many pieces of DNA get incorporated at a single site. Further, ***transgenes may not be expressed*** at all or sometimes underexpressed, or even overexpressed. All these processes will disturb the normal physiology of the transgenic animal. In addition, microinjection procedures are time consuming, costly and labour intensive. Despite all these limitations, this technique is routinely used for producing transgenic animals.

EMBRYONIC STEM CELL METHOD

Cells from the inner cell mass of the blastocyst stage of a developing mouse embryo can proliferate in cell culture. These cells, referred to as pluripotent ***embryonic stem*** (ES) cells, are capable of differentiating into other types of cells (including germ line cells) when transferred to another blastocyst embryo.

The embryonic stem cell technology basically involves the ***introduction of a foreign DNA into ES cell*** (***Fig. 41.2***). Embryonic stem cells in culture can be subjected to genetic manipulations without changing their plutipotency. ***Foreign DNA can be introduced into ES cells by electroporation or microinjection***. The desired genetically engineered cells with transgene can be identified by a selection procedure using a marker gene or PCR analysis (described below). The transfected cells can be cultured, introduced (by microinjection) into blastocyst and then implanted into foster mother (i.e., pseudopregnant female mouse). By this way, transgenic founder mice are produced. Transgenic lines can be established by suitable breeding strategies of the founder mice.

Selection of transgene containing cells

Several strategies have been developed for the selection of transgene containing cells. The important ones are briefly described.

Selection by use of marker gene coding for thymidine kinase : It is worthwhile to know the role of thymidine kinase to understand its utility as a marker gene. There are two pathways for the synthesis of deoxyribonucleotides (dATP, dGTP, dCTP and dTTP), the basic units of DNA structure.

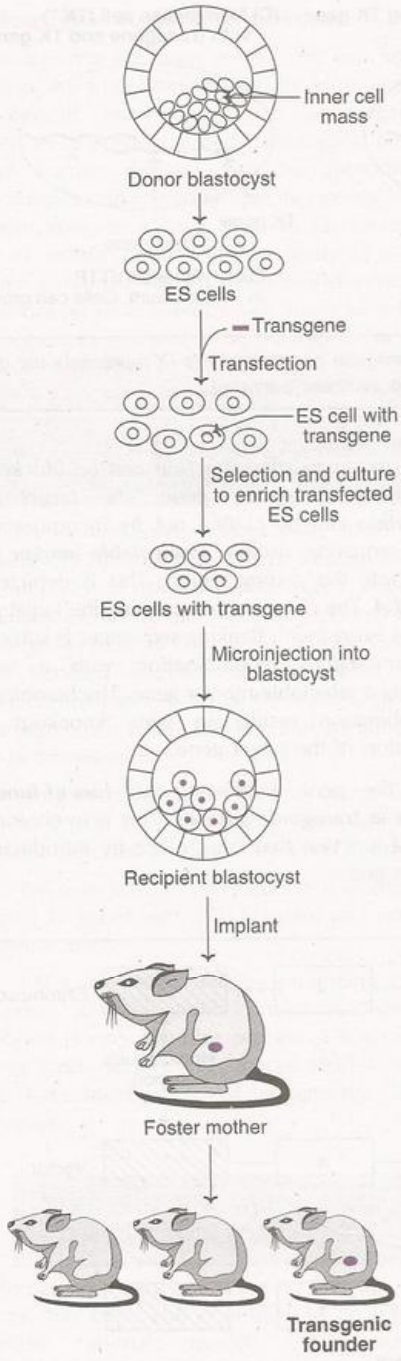


Fig. 41.2 : Embryonic stem cell (ES) method to produce transgenic mice.

One is the ***salvage pathway*** that recycles the degraded nitrogenous bases formed from DNA. The other alternate pathway is an ***endogenous synthetic pathway*** from different precursors (glycine, aspartate, glutamine, methyl tetrahydrofolate etc.).

The enzyme thymidine kinase (TK) is involved in the salvage pathway. TK phosphorylates thymidine to produce thymidine monophosphate (dTMP) which is finally converted to thymidine triphosphate (dTTP).

The gene that encodes the enzyme thymidine kinase can be used as a marker to determine whether the transgene has been inserted. This is illustrated in ***Fig. 41.3***. The mammalian cells are capable of synthesizing dTTP by salvage pathway and endogenous synthetic pathway. The cells lacking TK gene cannot produce dTTP. If such cells are cultured in a ***HAT*** medium containing ***h***ypoxanthine (H), ***a***minopterine (A) and ***t***hymidine (T), they cannot grow and therefore die. This is because thymidine cannot be utilized in the salvage pathway due to lack the enzyme thymidine kinase. Further, aminopterine blocks the endogenous pathway (by inhibiting the enzyme dihydrofolate reductase, required for one carbon metabolism).

If a transgene is joined to a TK gene, inserted into a mammalian cell (TK^-) and then the cells can grow in HAT medium. This is possible only if the TK gene is incorporated into the mammalian cells. And logically, the cells that survive in HAT medium carry the transgene. In this fashion, thymidine kinase can be effectively used as a marker gene.

There are other enzymes that serve as markers for identifying transgene insertion. These include dihydrofolate reductase (resistant to methotrexate) and neomycin phosphotransferase (resistant to antibiotic G418) and ***PCR analysis for selecting transgene containing cells***. The last one is a more direct and recent method, and is successfully used for detecting transgene containing cells.

Promoter sequence to facilitate transgenesis

In the early experiments on transgenesis, the ***mouse metallothionein (MMT)*** gene promoter was used. MMT gene encodes for a metal binding protein that is involved in metal homeostasis. The foreign gene (i.e., a rat growth hormone gene) can be linked to a promoter sequence of MMT. By

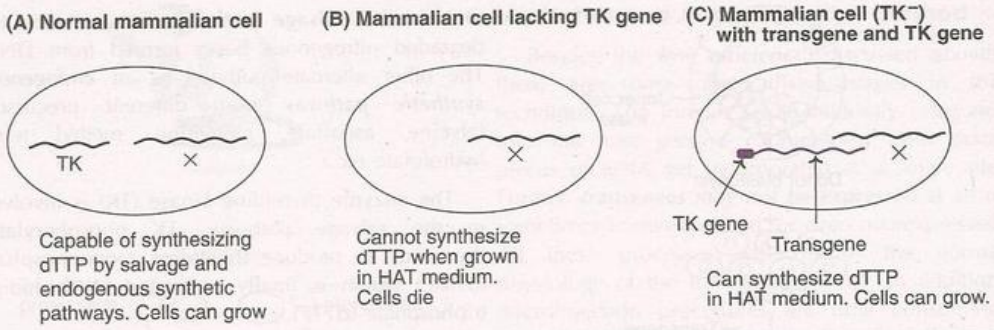


Fig. 41.3 : Thymidine kinase (TK) marker gene for selecting transgene containing cells (X-represents the genes for synthesizing dTTP by endogenous synthetic pathway).

doing so, the promoter switches on the growth hormone gene when the MMT is activated by a metal in the environment (e.g., cadmium). Thus, the metal inducer (Cd) can stimulate the promoter (MMT promoter) to facilitate the transgene (growth hormone gene) to express. Therefore, addition of cadmium triggers the growth hormone production.

GENE KNOCKOUT

By inserting a transgene (foreign gene) into a chromosome, a new function is introduced while producing transgenic animals (described above). On the other hand, in a process referred to as gene knockout an ***existing function can be blocked by destroying a specific gene.*** The ***target gene disruption*** can be carried out ***by*** incorporating a DNA sequence, usually ***a selectable marker gene*** (***smg***) into the coding region. This is depicted in ***Fig. 41.4***. The chromosome carrying the target gene (with 4 exons) with flanking sequences is subjected to homologous recombination with a vector carrying a selectable marker gene. The homologous recombination results in gene knockout i.e., disruption of the target gene.

In the gene knockout, the ***loss-of-function occurs in transgenic animals***. This is in contrast to gain of function that takes place by introducing a foreign gene.

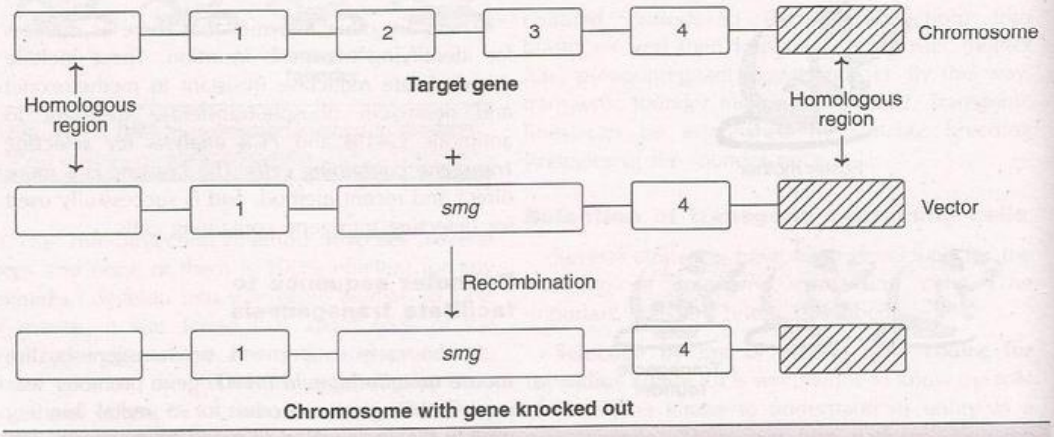


Fig. 41.4 : Gene knockout by homologous recombination (1-4 represent exons of target gene; smg-Selectable marker gene).

Applications of gene knockout

Disruption of the target genes (gene knockout) is important for understanding the development and physiological consequences in an organism. Further, the biochemical and pathological basis of several human diseases can be appropriately understood by inactivating specific genes. About 200 knockout mice have been so far created to serve as animal models for the study of a large number of human abnormalities and disorders. In fact, knockout mouse which lack genes for a single organ or organ system can be produced (For more details, See p. 486).

YEAST ARTIFICIAL CHROMOSOME IN TRANSGENESIS

In the conventional transgenesis, small sized transgenes (≤ 20 kb) are transferred through vectors. This involves a risk of losing some important sequences including the regulatory ones. Further, several genes exist as complexes and are too large to be handled by conventional vectors. Yeast artificial chromosomes (***YACs***) can carry large-sized genes (size range 100–1,000 kb), and are effectively used in transgenesis.

By using microinjection technique or transfection of ES cells with YACs, transgenic mice have been produced. Selected examples are cited below.

1. YACs are used to carry human β-globin gene complex (5 genes with 250 kb size) and produce transgenic mice.

2. YACs are employed for producing human antibodies in transgenic mice. Synthesis of antibodies is very complex process. YACs carrying varying sizes of DNA sequences (800–1,000 kb YACs) have been employed to generate different antibodies.

TRANSGENIC MICE AND THEIR APPLICATIONS

Mouse, although not close to humans in its biology, has been and continues to be the most exploited animal model in transgenesis experiments. The common feature between man and mouse is that both are mammals. Transgenic mice are extensively used as animal models for understanding human diseases and for the production of therapeutic agents. Adequate care, however, must be exercised before extrapolating data of transgenic mice to humans.

Mouse models for several human diseases (cancers, muscular dystrophy, arthritis, Alzheimer's disease, hypertension, allergy, coronary heart disease, endocrine diseases, neurodegenerative disorders etc.) have been developed. A selected few of them are briefly discussed.

THE HUMAN MOUSE

The transgenic mice with ***human immune system*** were produced, and they are commonly referred to as human mice (***Fig. 41.5***). For this purpose, mice with severe combined immunodeficiency (SCID-a condition characterized by total lack of immune system cells) were chosen. Human thymus tissue from an aborted fetus was transplanted under the capsule membrane of the kidney of the mouse. Human lymph node was placed under the opposite kidney. After about a week, immature immune cells (T-lymphocytes) from a human fetus were injected into the mouse tail vein. These lymphocytes enter the thymus tissue under the kidney and mature to T-lymphocytes. The so produced T-lymphocytes enter the circulation and in the lymph node (present under the second kidney), they multiply to form a full-pledged functional immune system. It takes about two weeks after the transplant for the ***mice*** to ***display*** the ***human immune system*** (characteristics of both T-lymphocytes and B-lymphocytes).

The human mouse, being a close ***animal model for human immune system*** is a boon for immunologists, particularly working on ***AIDS***. This is because the various immunological aspects of AIDS including the possible development of an AIDS vaccine can be explored by using human mouse (**Note :** There is no good animal model for AIDS research except chimpanzee).

THE ALZHEIMER'S MOUSE

Alzheimer's disease affects about 1% of the population between 60–65 years old, and about 30% of the population over 80 years old. This disease is characterized by ***progressive loss of memory and personality changes*** (decline in thinking, judgement etc.). The postmortem analysis of brains of Alzheimer's patients revealed plaques of ***dead nerve cells entangled in a protein called amyloid***. This protein was purified and its amino

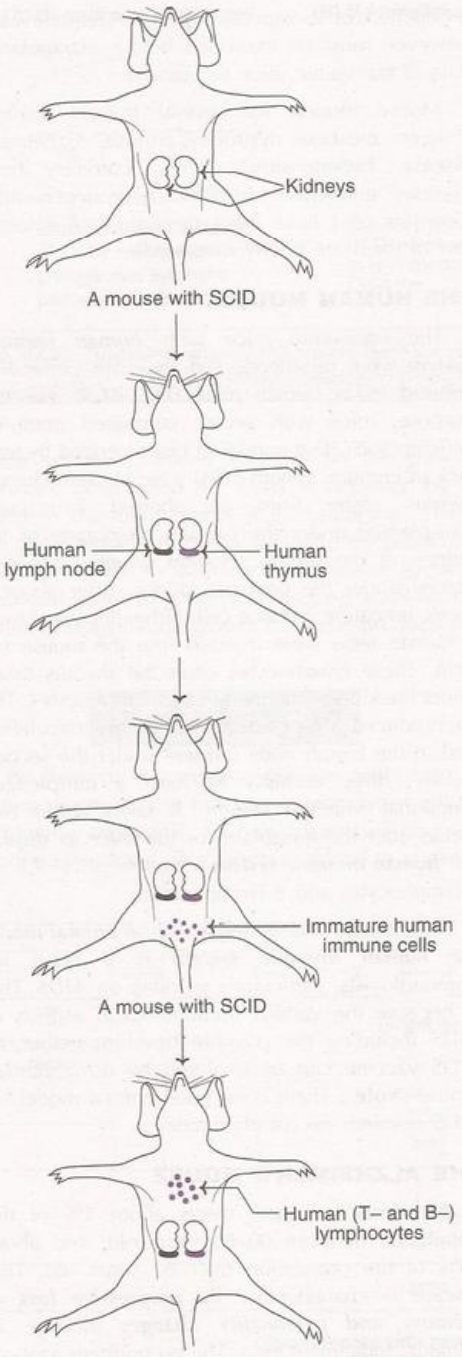


Fig. 2.5 : The human mouse (SCID-Severe combined immunodeficiency).

acid sequence determined. Later, ***amyloid precursor protein*** (***APP***) and its gene sequence were also identified. Alzheimer's disease is more prevalent in some families, clearly indicating a genetic basis for this disease.

Transgenic mice were developed by introducing amyloid precursor gene into fertilized egg cells of mice. The synthesis of human amyloid protein and its accumulation as typical plaques in the mice brain were observed. The Alzheimer's mouse is very useful in understanding the pathological basis of the disease. Certain mutations in APP gene, and the involvement of some other genes are believed to be responsible for Alzheimer's disease.

THE ONCOMOUSE

The ***animal model for cancer*** is the oncomouse (onco refers to cancer). First developed for breast cancer, the oncomouse is useful for understanding of cancer and evolving modalities for cancer therapy.

The ***oncogene c-myc*** in association with mouse mammary tumor (MMT) virus was found to be responsible for breast cancer in animals. Transgenic mice were produced by introducing chimeric DNA consisting of *c-myc* gene and sections MMT virus into fertilized mouse egg cells. Breast cancer developed in adult female mice and the trait was passed on to the offspring.

Oncomouse (the mouse that is genetically altered and is susceptible for cancer) was patented by U.S. Patent office ***in 1988***. In fact, it was the ***very first animal to be patented***.

THE PROSTATE MOUSE

The prostate gland, surrounding the urethra of males is responsible for secreting semen fluid. In the older men, particularly above 60 years of age, prostate gland gets elarged and may become cancerous. The oncogene for prostate cancer was identified (***int-2***). A chimeric DNA by joining ***int-2*** with viral promoter was prepared and introduced into fertilized mouse eggs. In the transgenic mice so developed, enlargement of prostate gland was observed. There prostate mice were also patented in 1991.

THE KNOCKOUT MOUSE

The basic principles underlying gene knockout have been described (See p. 484). Several knockout

mice have been developed. It is not an exaggeration that the knockout mice have become as common as a test tube in the laboratory for a biotechnologist. It must be noted that not all the knockout mice are immediately useful for human health and welfare. A selected few of the knockout mice are described.

SCID mouse

Severe combined immunodeficiency (SCID) is a condition with a total lack of immune system. SCID mice were developed ***by eliminating a single gene and the resultant mice lost the ability to produce B-lymphocytes and T-lymphocytes***. The human mouse was later developed from the SCID mouse (See p. 485).

Knockout mouse for allergy

The receptor sites on certain body cells for ***IgE antibodies*** are believed to be responsible for triggering allergy reactions. Knockout mice were developed for allergy by ***removing the gene encoding for receptor protein***. The result is that antibodies cannot bind to cells due to lack of receptors and the mice are unaffected by allergic reactions. It is expected that some breakthrough may occur in the near future to benefit the millions of sufferers of allergic reactions, spread throughout the world.

Knockout mouse for transplantation

By using a ***suicide gene***, the liver cells were destroyed in a mouse that already lacks immune system. Sample human liver cells were transplanted in this mouse. Encouraged by lack of immune system, the human liver cells can develop. In this fashion, organ replacement in animals is possible.

Knockout mouse with memory loss

The memory processes in brain are believed to be carried out by a specialized area called hippocampus. By a gene knockout technique, researchers have developed ***mice that lack hippocampus***. These knockout mice lacked the ability to remember.

Knockout mouse with retinitis pigmentosa

By inactivating the mouse rhodopsin gene, rods cells of the retinal deteriorate in the transgenic mice. This retinal degeneration is comparable to human retinitis pigmentosa. The ***rhodopsin knockout mouse*** is useful for understanding the retinal degeneration, besides evolving therapeutic strategies.

TRANSGENIC MICE FOR HUMAN DISEASES

Transgenic mice are important for the development of therapeutic drugs and possible gene therapies, besides the understanding of the human disease. Many transgenic mice with human disease equivalents have been developed and a selected few of them are given in ***Table 41.1***.

TRANSGENESIS IN LARGE ANIMALS

In general, transgenesis in large animals is more difficult than with mice. There are several factors for the lower efficiency of transgenesis in large animals. These include less number of eggs they

TABLE 41.1 Transgenic mice with human disease equivalents (selected examples)

Human disease equivalent	*Genetic change in mice*
Sickle-cell anemia	Insertion of β-globin of sickle-cell hemoglobin.
Lesch-Nyhan syndrome	Inactivation of mouse gene encoding hypoxanthine-guanine phosphoribosyltransferase
Kaposis sarcoma	Introduction of HIV *tat* gene
Development of malignancy	Insertion of human *ras* and *c-myc* oncogenes
X-Linked muscular dystrophy	Mutation at a locus on X-chromosome
Hypertension*	Introduction of mouse renin gene

* *This was done in transgenic rats.*

produce and technical difficulties in handling, besides long gestational periods to get the offspring (It takes about 2 years to produce a calf from a fertilized egg).

Some of the early experiments to produce transgenic large animals were far from satisfactory. For instance, transgenic sheep overproducing growth hormone grow leaner with increased feed efficiency. But they are more susceptible to infection, become infertile and tend to die at young age. All this might be due to ineffective control of gene regulation. Several improvements have been made to produce transgenic animals with desirable characters.

Biotechnologists are particularly interested to improve the quality of animals, with improved resistance to diseases, besides enhancing their ability produce foods. '***Building a better animal***', ***being the motto***. Further, production of commercial and pharmaceutical compounds by transgenic animals is also gaining importance in recent years.

The protocol adopted for producing other transgenic animals is comparable with that already described for transgenic mice, with certain modifications.

TRANSGENIC CATTLE

The ***mammary gland*** of the dairy cattle is an ***ideal bioreactor*** for producing several new proteins (of pharmaceutical importance), besides improving the quality and quantity of the existing ones. For instance, a transgenic cow, with an over-expressed casein transgene, can give milk with higher content of casein. If lactase transgene is introduced and expressed in the mammary gland, milk free from lactose will be secreted. Such a milk will be a boon for lactose intolerant people who experience indigestion and other complications, after consuming normal milk and milk products.

Some success has been achieved in creating transgenic cattle with improved resistance to viral, bacterial and parasitic diseases. However, this is not an easy job due to the complexicity of genetic control to combat the disease-producing organisms.

Attempts have been made in recent years to produce cattle with inherited immunological protection by transgenesis. Introduction of genes that code for heavy and light chains of monoclonal antibodies has met with some success in this direction.

In vivo immunization of an animal although not yet fully successful, is ideal for disease protection. *In vivo* immunization primarily involves the insertion of a transgene for an antibody that specifically binds to an antigen.

TRANSGENIC SHEEP AND GOATS

Transgenesis experiments in sheep and goats mostly involve the development of mammary glands as bioreactors for the production of proteins for pharmaceutical use. This is possible despite the fact that quantity of milk produced by sheep and goats is less than that of dairy cattle (cow, buffalo). Some proteins produced by sheep and goats have good pharmaceutical use (***Table 41.2***).

Transgenic sheep with increased wool production

Keratin is the wool protein with highly cross-linked disulfide bridges. For good production of quality wool, the amino acid cysteine (or its precursor methionine) is required in large quantities. However, cysteine supply to sheep is always inadequate, since the microbes harboring the rumen utilize it and release in the form of sulfide. This problem can be overcome by producing transgenic sheep containing bacterial genes for the synthesis of cysteine. The two enzymes, synthesized by the transgenes, are capable of trapping the hydrogen sulfide liberated in the intestine to produce cysteine. Thus, ***good supply of cysteine to the sheep improves the quality and quantity of wool***.

TRANSGENIC PIGS

Transgenic pigs that can ***produce human hemoglobin*** have been successfully developed. This human hemoglobin can be separated from pig hemoglobin by simple analytical techniques. Hemoglobin, the oxygen carrying protein of RBC, can be used as a substitute in blood transfusion experiments. In fact, hemoglobin can be stored for longer period (a few months) than whole blood (weeks only). Further, there is no problem of contamination (like HIV) as is the case with whole blood. However, the free hemoglobin (naked hemoglobin) cannot transport oxygen as effectively as the hemoglobin of RBC. In addition, naked hemoglobin is easily degraded and the breakdown products cause damage to kidney. There also exists a risk of contamination by pig viruses and other

Table 41.2 Transgenic animals as bioreactors for the production of therapeutically important proteins

Transgenic animal	*Protein product*	*Biological importance*
Cow	Lactoferrin	Promotes intestinal iron absorption and hence can be used to overcome iron-deficiency anemias. Possesses antibacterial activity
Cow	Interferon	Provides resistance against viral infections
Sheep	α_1-Antitrypsin	Used in the treatment of emphysema (promotes the exchange of gases in lungs)
Goat	Cystic fibrosis transmembrane regulator (CFTR)	For the treatment of patients suffering from cystic fibrosis (promotes transport of ions)
Goat	Tissue plasminogen activator (tPA)	Used in treating the patients of myocardial infarction (dissolves blood clots)
Goat	Antithrombin III	Regulates blood clotting
Rabbits	α-Glucosidase	Treatment of Pompe's disease (a genetic disorder characterized by block in glycogen degradation)
Mouse	Urokinase	For dissolving blood clots
Mouse	Immunoglobulins (antibodies)	Administration enhances immunity
Pig	Hemoglobin	Blood transfusion
Goat and other animals	Vaccines (?)	To immunize against various diseases

compounds to cause allergic reactions. With these limitations, the initial enthusiasm for substituting blood transfusion with free hemoglobin has remained short-lived.

It is now advised not to use naked hemoglobin for transfusion, when there is a heavy blood loss. However, it can be used during major surgeries for supplementing the whole blood transfusion.

Pig in organ farms

The human organs such as heart, liver, pancreas, kidney and lungs are in great demand for transplantation surgery. The shortage of these transplantable organs can be overcome by developing them in animals. ***Pig is a favourite animal for harvesting human organs***. This is because the physiology of pigs is close to that of humans. Further, pigs do not carry any major infectious diseases transmissible to humans. The use of pigs in organ farming is still at the experimental stages.

In the preliminary experiments, organ transplantation from transgenic pigs into primates showed some promising results. The day may not be very far for utilizing transgenic pigs as donors of human organs (For more details, See p. 492).

TRANSGENIC CHICKENS

The production of transgenic chickens (or other birds) is rather complicated. This is mainly because during fertilization in chickens, several sperms enter the ovum instead of one. This is in contrast to mammals where usually only one sperm enters the egg. The identification of male pronuclei that will fuse with female pronuclei is quite difficult. Further, embryonic stem (ES) cells have not been identified in chicken. Despite all these limitations, transgenic chickens have been developed.

The blastoderm cells (from an egg) can be removed from a donor chicken. They are transfected with transgenes (usually by lipofection with liposomes). The so modified blastoderm cells are reintroduced into the subgerminal space of irradiated blastoderm of freshly laid eggs. Some of the resulting chickens may carry the transgene. Transgenic lines of chickens can be established.

Transgenesis in chicken can be used to develop low fat and cholesterol, and high protein containing eggs. Transgenic chickens that are ***resistant to viral and bacterial diseases have also been developed***. Some attempts have also been made to develop pharmaceutical proteins in the eggs of transgenic chickens.

TRANSGENIC FISH

Several transgenic fish (catfish, salmon, trout etc.) have been developed ***with increase in their growth and size***. This was carried out by introducing ***growth hormone transgene*** (by microinjection or electroporation). The fertilized eggs with inserted transgene are incubated in temperature-regulated holding tanks. (**Note :** The fish egg development is external in contrast to the mammalian embryogenesis). The efficiency of fish transgenesis is as high as 70%.

It was found that the transgenic salmon fish (with growth hormone transgene) were 10 times heavier than the normal ones, at the end of one year.

POSITION EFFECTS

Position effect is the phenomenon of ***different levels of gene expression that is observed after insertion of a new gene at different position in the eukaryotic genome***. This is commonly observed in transgenic animals as well as plants. These transgenic organisms show variable levels and patterns of transgene expression. In a majority of cases, position effects are dependent on the site of transgene integration.

In general, the defective expression is due to the insertion of transgene into a region of highly packed chromatin. The transgene will be more active if inserted into an area of open chromatin.

The positional effects are overcome by a group of DNA sequences called ***insulators***. The sequences referred to as ***specialized chromatin structure (SCS)*** are known to perform the functions of insulators. It has been demonstrated that the expression of the gene is appropriate if the transgene is flanked by insulators.

ANIMAL BIOREACTORS

Transgenesis is wonderfully utilized for production proteins of pharmaceutical and medical use. In fact, any protein synthesized in the human body can be made in the transgenic animals, provided that the genes are correctly programmed. The advantage with transgenic animals is to produce scarce human proteins in huge quantities. Thus, the ***animals serving as factories for production of biologically important products*** are referred to as ***animal bioreactors or*** sometimes ***pharm animals***. Frankly speaking, transgenic animals as bioreactors can be commercially exploited for the benefit of mankind.

Once developed, animal bioreactors are cost-effective for the production of large quantities of human proteins. Routine breeding and healthful living conditions are enough to maintain transgenic animals.

The importance of transgenic animals has already been discussed under the respective individual animals. A selected list of the therapeutically important proteins produced by animal bioreactors is given ***Table 41.2***.

DOLLY-THE TRANSGENIC CLONE

Dolly, the first ever mammal clone was developed by Wilmut and Campbell in 1997. It is a ***sheep*** (female lamb) ***with a mother and no father.***

The ***technique primarily involves nuclear transfer and the phenomenon of totipotency***. The character of a cell to develop to different cells, tissues, organs, and finally an organism is referred to as ***totipotency*** or ***pluropotency***. Totipotency is the basic character of embryonic cells. As the embryo develops, the cells specialize to finally give the whole organism. As such, the cells of an adult lack totipotency. Totipotency was induced into the adult cells for developing Dolly.

The cloning of sheep for producing Dolly, illustrated in ***Fig. 41.6***, is briefly described here. The mammary gland cells from a donor ewe were isolated. They were subjected to total nutrient deprivation (starvation) for five days. By this process, the mammary cells abandon their normal growth cycle, enter a dormant stage and regain totipotency character. An ovum (egg cell) was taken from another ewe, and its nucleus was removed to form an enucleated ovum. The dormant mammary gland cell and the enucleated ovum were fused by pulse electricity. The mammary cell outer

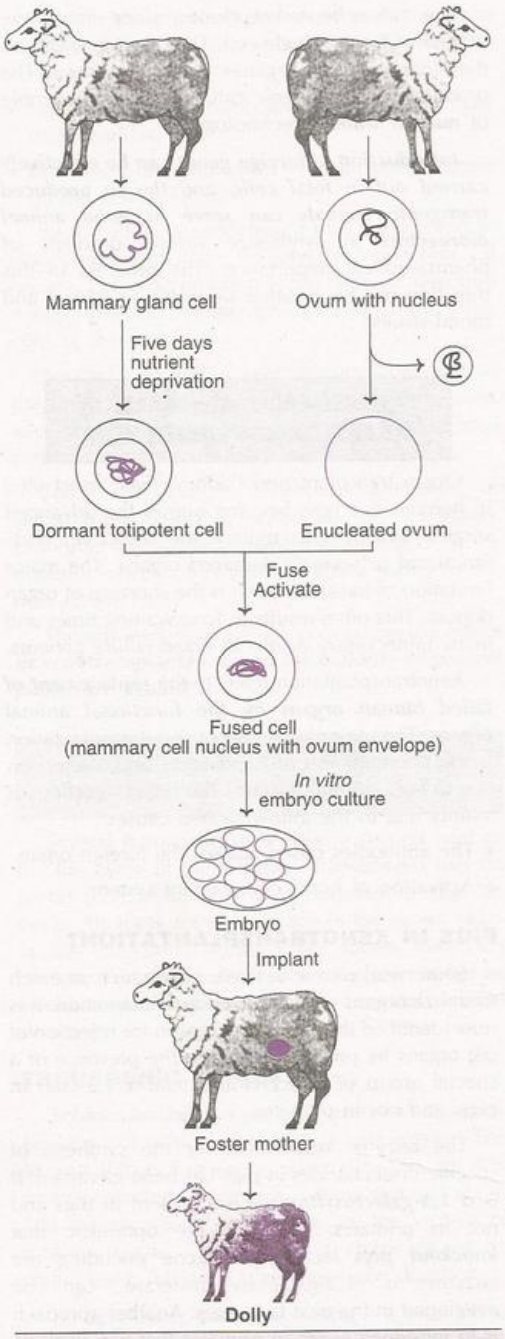


Fig. 41.6 : The cloning of sheep for developing Dolly.

membrane was broken, allowing the ovum to envelope the nucleus. The fused cell, as it had gained totipotency, can multiply and develop into an embryo. This embryo was then implanted into another ewe which serves as a surrogate/foster mother. Five months later, Dolly was born.

As reported by Wilmut and Campbell, they fused 277 ovum cells, achieved 13 pregnancies, and of these only one pregnancy resulted in live birth of the offspring-Dolly.

CLONING OF PET ANIMALS

Some of the companies involved in transgenic experiments have started cloning pet animals like cats and dogs. Little Nicky was the first pet cat that was cloned at a cost of $50,00 by an American company (in Dec. 2004). More cloned cats and dogs will be made available to interested parties (who can afford) in due course.

Some people who own pet animals are interested to continue the same pets which is possible through cloning. There is some opposition to this approach as the cloned animals are less healthy, and have shorter life span, besides the high cost factor.

TRANSGENIC ANIMALS PRODUCED BY CLONING FETAL CELLS

Fetal cells such as fibroblasts are totipotent. Fetal cell cloning was successfully carried out by some workers to produce ***transgenic sheep (Polly), transgenic bull calf (Gene)*** and other animals. Transgenes (foreign genes) can be inserted into the fetal cell genomes to produce desired products by transgenic animals.

As an example, the development of transgenic calves produced by cloning with fetal cells is shown in ***Fig. 41.7***. Fibroblasts were collected from a fifty five day old bovine fetus. These fibroblasts with totipotent character are cultivated in a nutritious medium. The desired foreign genes (transgenes) can be introduced into fibroblasts. The nucleus with the genetically altered DNA (fibroblast nucleus) is removed from the cell. It is then inserted into a bovine enucleated ovum. This ovum cell multiplies to form embryos. They are implanted in a surrogate (foster) mother cow to give birth to transgenic calves.

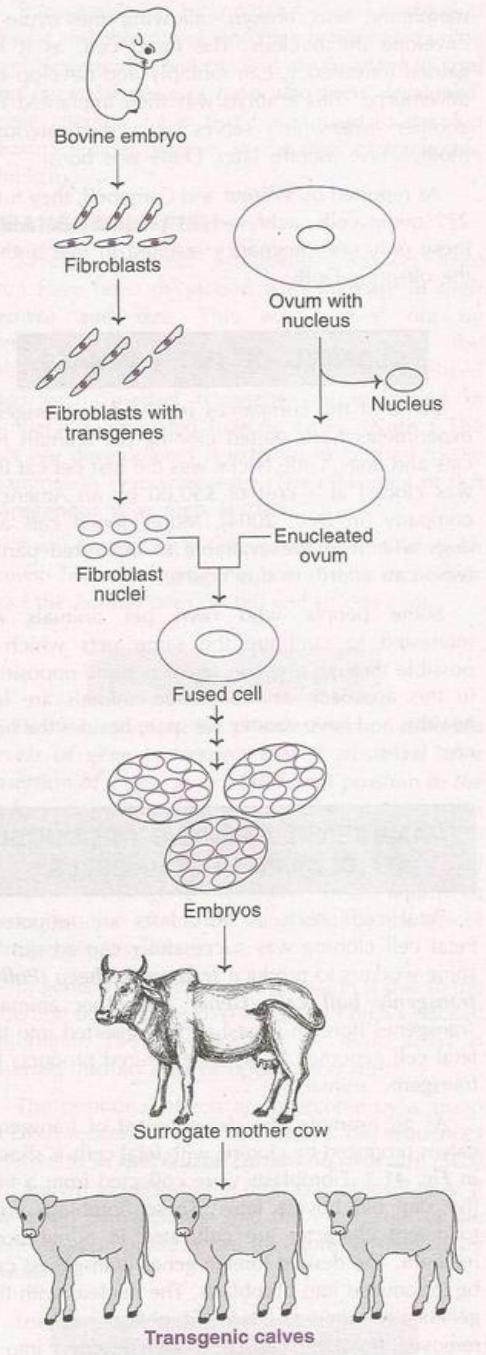


Fig. 41.7 : Transgenic calves produced by cloning fetal fibroblasts.

The calves represent clones, since they have originated from a single cell. They are transgenic as they carry the transgenes (foreign genes). The production of transgenic calves is a good example of ***nuclear transfer*** technology.

Introduction of foreign genes can be effectively carried out in fetal cells, and the so produced transgenic animals can serve as good animal bioreactors to synthesize several products of pharmaceutical importance. The progress in this direction has been rather slow due to ethical and moral issues.

TRANSGENIC ANIMALS IN XENOTRANSPLANTATION

Organ transplantation (kidney, liver, heart etc.) in humans has now become one of the advanced surgical practices to replace the defective, non–functional or severaly damaged organs. The major limitation of transplantation is the shortage of organ donors. This often results in long waiting times and many unnecessary deaths of organ failure patients.

Xenotransplantation refers to the ***replacement of failed human organs by the functional animal organs***. The major limitation of xenotransplantation is the phenomenon of hyperacute organ rejection due to host immune system. The organ rejections is mainly due to the following two causes

- The antibodies raised against the foreign organ.
- Activation of host's complement system.

PIGS IN XENOTRANSPLANTATION?

Some workers are actively conducting research to utilize organs of pigs in xenotransplantation. It is now identified that the major reason for rejection of pig organs by primates is due to the presence of a special group of disaccharides (Gal-α 1,3-Gal) in pigs, and not in primates.

The enzyme responsible for the synthesis of specific disaccharides in pigs has been identified. It is **α 1,3-*galactosyltransferase***, present in pigs and not in primates. Scientists are optimistic that ***knockout pigs*** lacking the gene encoding the enzyme α 1,3-galactosyltransferase can be developed in the next few years. Another approach is to introduce genes in primates that can degrade or modify Gal-α 1,3-Gal disaccharide groups (of pigs). This will reduce immunogenecity.

Besides the above, there are other strategies to avoid hyperactive organ rejection by the hosts in xenotransplantation.

- Expression of antibodies against the pig disaccharides.
- Expression of complement — inactivating protein on the cell surfaces.

By the above approaches, it may be possible to overcome immediate hyperactive rejection of organs. The next problem is the delayed rejection which involves the macrophages and natural killer cells of the host.

Another concern of xenotransplantation is that the endogenous pig retroviruses could get activated after organ transplantation. This may lead to new genetic changes with unknown consequences.

The use of transgenic animals in xenotransplantation is only at the laboratory experimental stages, involving animals. It is doubtful whether this will become a reality in the near future.

There is a vigorous debate concerning the ***ethics of xenotransplantation and the majority of general public are against it.***

TRANSGENIC ORGANISMS TO INTERRUPT DISEASE CYCLES

Several human diseases have a second host in the life cycle of the parasite. Transgenesis will be useful for disease control by interrupting the life cycle. Attempts are being made in the recent years in this direction, although the sucess has been limited. ***Transgenic organisms may soon serve as environmental replacements to control several diseases.***

TRANSGENIC SNAILS

Schistosomiasis is a parasitic diseases caused by *Schistosoma* which develops in the snails. The parasite infects humans by penetrating through skin when contacted in water. It is estimated that about 100 million people worldwide are the victims of schistosomiasis. This disease is characterized by chills, fever, intestinal ulcerations and diarrhea.

By developing transgenic snails, it is possible to interrupt the life cycle of *Schistosoma*. Some workers have attempted to ***create transgenic snails that will resist the invasion of the parasite***. If such snails are released into the environment, they will break the life cycle of schistosoma.

TRANSGENIC MOSQUITOES

Mosquitoes are responsible for the transmission of malaria. Some workers have identified certain critical ***genes in mosquitoes*** (*Anopheles* genus that transmits *Plasmodium* species malarial parasite) that are ***responsible for harbor and transmission of the parasite***. Theoretically, it is possible to alter these genes and produce transgenic mosquito. If such mosquitoes are released into the environment in large numbers, they will dilute the population of native mosquitoes, and halt the transmission of malarial parasite.

Another approach to control malaria is to understand ***vector incompetence*** i.e., the genetic basis of the vector to sustain a parasite and how to alter it so as to make the vector inhospitable for the parasite. Some success has been achieved in developing transgenic mosquitoes with vector incompetence. If such mosquitoes are released into the environment, the life cycle of the malarial parasite will be interrupted.

TRANSGENIC TSETSE FLIES

A protozoal disease African sleeping sickness is transmitted by tsetse fly. This disease affects the nervous system and is often accompanied by coma. Transgenic tsetse flies were developed with a novel approach. Researchers found a ***protein that can kill disease-producing protozoa***. They identified the protein encoding gene and inserted it into the bacteria of tsetse fly gut. These bacteria produce antiprotozoal protein that kills tsetse flies. In this manner the disease transmission can be prevented.

TRANSGENIC BOLLWORMS

Bollworms are the caterpillars that damage the cotton crop. Researches have developed ***transgenic bollworms with a suicide gene***. When released into the environment, the transgenic bollworms will mate with wild bollworms and produce caterpillars that die. The cotton crop can be saved from the damage.

TRANSGENIC MEDFLIES

Medflies (Mediterranean fruit flies) destroy fruit and coffee crops throughout the world. Attempts are being made to develop transgenic medflies to replace wild medflies and save the crops.

SECTION VII

PLANT/ AGRICULTURAL BIOTECHNOLOGY

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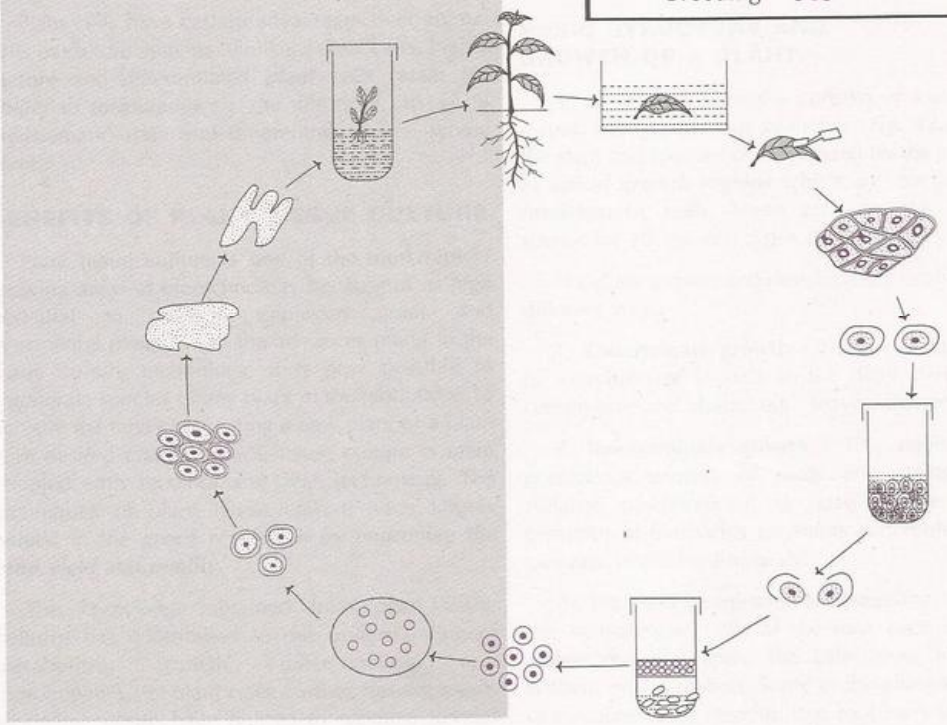


Plant Tissue Culture – General

PLANT TISSUE CULTURE
- Callus culture
- Organ culture (embryo, seed, anther, ovary, meristem, nucellus)
- Cell culture
- Protoplast culture

Plant tissue culture broadly refers to the ***in vitro cultivation of plants, seeds and various parts of the plants*** (organs, embryos, tissues, single cells, protoplasts). The cultivation process is invariably carried out in a nutrient culture medium under aseptic conditions.

Plant cells have certain advantages over animal cells in culture systems. Unlike animal cells, highly ***mature*** and differentiated ***plant cells retain the ability of totipotency*** i.e. the ability of change to meristematic state and differentiate into a whole plant.

BENEFITS OF PLANT TISSUE CULTURE

Plant tissue culture is one of the most rapidly growing areas of biotechnology because of its high potential to develop improved crops and ornamental plants. With the advances made in the tissue culture technology, it is now possible to regenerate species of any plant in the laboratory. To achieve the target of creating a new plant or a plant with desired characteristics, tissue culture is often coupled with recombinant DNA technology. The techniques of plant tissue culture have largely helped in the ***green revolution by improving the crop yield and quality***.

The knowledge obtained from plant tissue cultures has contributed to our understanding of metabolism, growth, differentiation and morphogenesis of plant cells. Further, developments in tissue culture have helped to produce several ***pathogen-free plants***, besides the ***synthesis of many biologically important compounds***, including pharmaceuticals. Because of the wide range of applications, plant tissue culture attracts the attention of molecular biologists, plant breeders and industrialists.

BASIC STRUCTURE AND GROWTH OF A PLANT

An adult plant basically consists of a stem and a root, each with many branches (***Fig. 42.1***). Both the stem and root are characterized by the presence of ***apical growth regions*** which are composed of ***meristematic cells***. These cells are the primary source for all the cell types of a plant.

The plant growth and development occur in two different ways.

1. **Determinate growth :** This is characterized by ceasation of growth as the plant parts attain certain size and shape. e.g., leaves, flowers, fruits.

2. **Indeterminate growth :** This refers to the continuous growth of roots and stems under suitable conditions. It is possible due to the presence of meristems (in stems and roots) which can proliferate continuously.

As the seed germinates and seedling emerges, the meristematic cells of the root apex multiply. Above the root apex, the cells grow in length without multiplication. Some of the elongated cells of the outer layer develop into root hairs to absorb

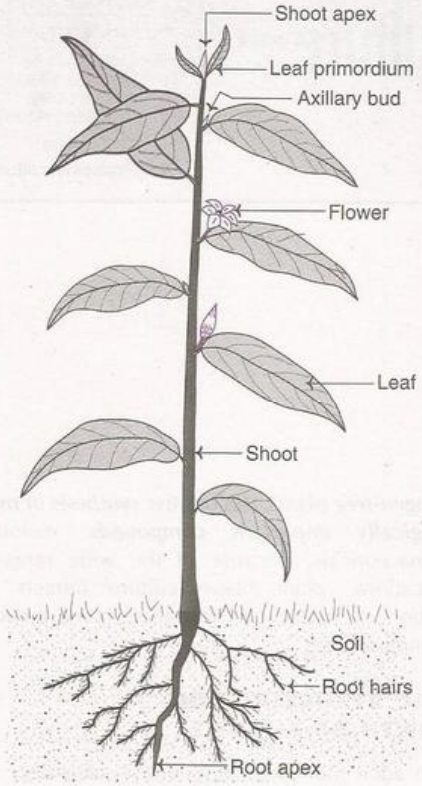


Fig. 42.1 : A diagrammatic view of a plant structure.

water and nutrients from the soil. As the plant grows, root cells differentiate into ***phloem*** and ***xylem***. Phloem is responsible for the absorption of nutrients while xylem absorbs water.

The meristematic cells of the shoot apex divide leading to the growth of stem. Some of the stem cells differentiate and develop into ***leaf primordia***, and then leaves. Axillary buds present between the leaf primordia and elongated stem also possess meristems which can multiply and give rise to branches and flowers.

A diagrammatic view of a plant and a flower are respectively depicted in ***Fig. 42.1*** and ***Fig. 42.2***.

CONVENTIONAL PLANT BREEDING AND PLANT TISSUE CULTURE

Since the time immemorial, man has been closely involved in the improvement of plants to meet his basic needs. The ***conventional*** methods employed for ***crop improvement are very tedious and long time processes*** (sometimes decades). Further, in the conventional breeding methods, it is not possible to introduce desired genes to generate new characters or products.

With the developments in plant tissue culture, it is now possible to reduce the time for the creation of new plants with desired characteristics, transfer of new genes into plant cells and large scale production of commercially important products.

TERMS USED IN TISSUE CULTURE

A selected list of the most commonly used terms in tissue culture are briefly explained

Explant : An excised piece of differentiated tissue or organ is regarded as an explant. The explant may be taken from any part of the plant body e.g., leaf, stem, root.

Callus : The unorganized and undifferentiated mass of plant cells is referred to as callus. Generally, when plant cells are cultured in a suitable medium, they divide to form callus i.e., a mass of parenchymatous cells.

Dedifferentiation : The phenomenon of mature cells reverting to meristematic state to produce callus is dedifferentiation.

Dedifferentiation is possible since the non-dividing quiescent cells of the explant, when grown

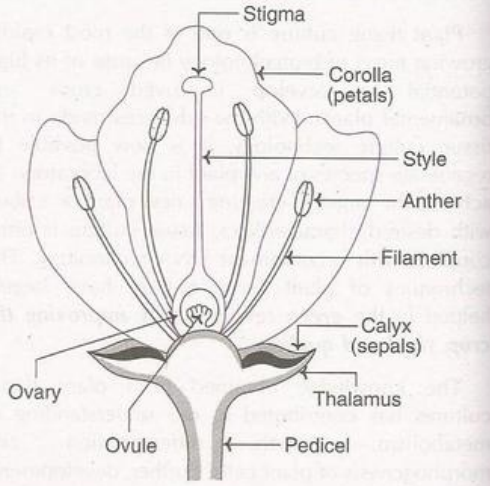


Fig. 42.2 : A diagrammatic representation of important parts in a flower.

in a suitable culture medium revert to meristematic state.

Redifferentiation : The ability of the callus cells to differentiate into a plant organ or a whole plant is regarded as redifferentiation.

Totipotency : The ability of an individual cell to develop into a whole plant is referred to as cellular totipotency. The inherent characteristic features of plant cells namely dedifferentiation and redifferentiation are responsible for the phenomenon of totipotency.

The other terms used in plant tissue culture are explained at appropriate places.

BRIEF HISTORY OF PLANT TISSUE CULTURE

About 250 years ago (1756), Henri-Louis Duhamel du Monceau' demonstrated callus formation on the decorticated regions of elm plants. Many botanists regard this work as the forward for the discovery of plant tissue culture. In 1853, Trecul published pictures of callus formation in plants.

German botanist ***Gottlieb Haberlandt*** (1902), regarded as ***the father of plant tissue culture***, first developed the concept of *in vitro* cell culture. He was the first to culture isolated and fully differentiated plant cells in a nutrient medium.

During 1934–1940, three scientists namely Gautheret, White and Nobecourt largely contributed to the developments made in plant tissue culture.

Good progress and rapid developments occurred after 1940 in plant tissue culture techniques. Steward and Reinert (1959) first discovered somatic embryo production *in vitro*. Maheswari and Guha (1964) from India were the first to develop anther culture and poller culture for the production of haploid plants.

TYPES OF CULTURE

There are different types of plant tissue culture techniques, mainly based on the explant used (***Fig. 42.3***).

Callus culture

This involves the culture of differentiated tissue from explant which dedifferentiates *in vitro* to form callus.

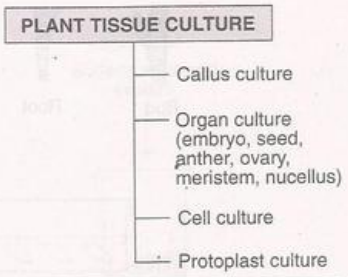


Fig. 42.3 : Major types of plant tissue cultures.

Organ culture

Culture of isolated plant organs is referred to as organ culture. The organ used may be embryo, seed, root, endosperm, anther, ovary, ovule, meristem (shoot tip) or nucellus.

The organ culture may be organized or un-organized.

Organized organ culture : When a well organized structure of a plant (seed, embryo) is used in culture, it is referred to as organized culture. In this type of culture, the characteristic individual organ structure is maintained and the progeny formed is similar in structure as that of the original organ.

Unorganized organ culture : This involves the isolation of cells or tissues of a part of the organ, and their culture *in vitro*. Unorganized culture results in the formation of callus. The callus can be dispersed into aggregates of cells and/or single cells to give a suspension culture.

Cell culture

The culture of isolated individual cells, obtained from an explant tissue or callus is regarded as cell culture. These cultures are carried out in dispension medium and are referred to as ***cell suspension cultures***.

Protoplast culture

Plant protoplasts (i.e., cells devoid of cell walls) are also used in the laboratory for culture.

BASIC TECHNIQUE OF PLANT TISSUE CULTURE

The general procedure adopted for isolation and culture of plant tissues is depicted in ***Fig. 42.4*** and briefly described in the next page.

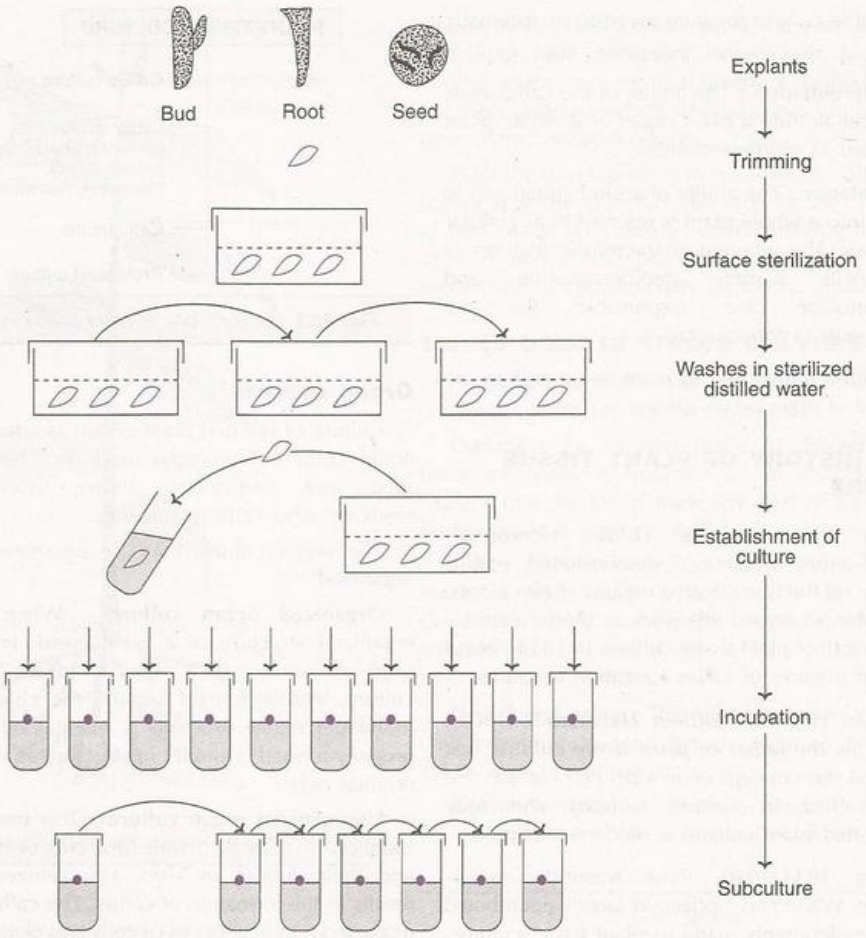


Fig. 42.4 : A diagrammatic outline of the basic procedure in plant tissue culture.

The requisite explants (buds, stem, seeds) are trimmed and then subjected to sterilization in a detergent solution. After washing in sterile distilled water, the explants are placed in a suitable culture medium (liquid or semisolid form) and incubated. This results in the establishment of culture. The mother cultures can be subdivided, as frequently as needed, to give daughter cultures.

The most important aspect of *in vitro* culture technique is to carry out all the operations under aseptic conditions. Bacteria and fungi are the most common contaminants in plant tissue culture. They grow much faster in culture and often kill the plant tissue. Further, the contaminants also produce certain compounds which are toxic to the plant tissue. Therefore, it is absolutely essential that ***aseptic conditions*** are maintained ***throughout the tissue culture operations***.

Some of the culture techniques are described here while a few others are discussed at appropriate places.

CALLUS CULTURE

Callus is the undifferentiated and unorganized mass of plant cells. It is basically a tumor tissue which usually forms on wounds of differentiated

tissues or organs. Callus cells are parenchymatous in nature although not truely homogenous. On careful examination, callus is found to contain some quantity of differentiated tissue, besides the bulk of non-differentiated tissue.

Callus formation *in vivo* is frequently observed as a result of wounds at the cut edges of stems or roots. Invasion of microorganisms or damage by insect feeding usually occurs through callus.

An outline of technique used for callus culture, and initiation of suspension culture is depicted in ***Fig. 42.5***.

Explants for callus culture

The starting materials (explants) for callus culture may be the differentiated tissue from any part of the plant (root, stem, leaf, anther, flower etc.). The selected explant tissues may be at different stages of cell division, cell proliferation and organization into different distinct specialized structures. If the explant used possesses meristematic cells, then the cell division and multiplication will be rapid.

Factors affecting callus culture

Many factors are known to influence callus formation in *in vitro* culture. These include the source of the explant and its genotype, composition of the medium (MS medium most commonly used), physical factors (temperature, light etc.) and growth factors. Other important factors affecting callus culture are — age of the plant, location of explant, physiology and growth conditions of the plant.

Physical factors : A temperature in the range of 22–28°C is suitable for adequate callus formation. As regards the effect of light on callus, it is largely dependent on the plant species-light may be essential for some plants while darkness is required by others.

Growth regulators : The growth regulators to the medium strongly influence callus formation. Based on the nature of the explant and its genotype, and the endogenous content of the hormone, the requirements of growth regulators may be categorized into 3 groups

1. Auxin alone
2. Cytokinin alone
3. Both auxin and cytokinin.

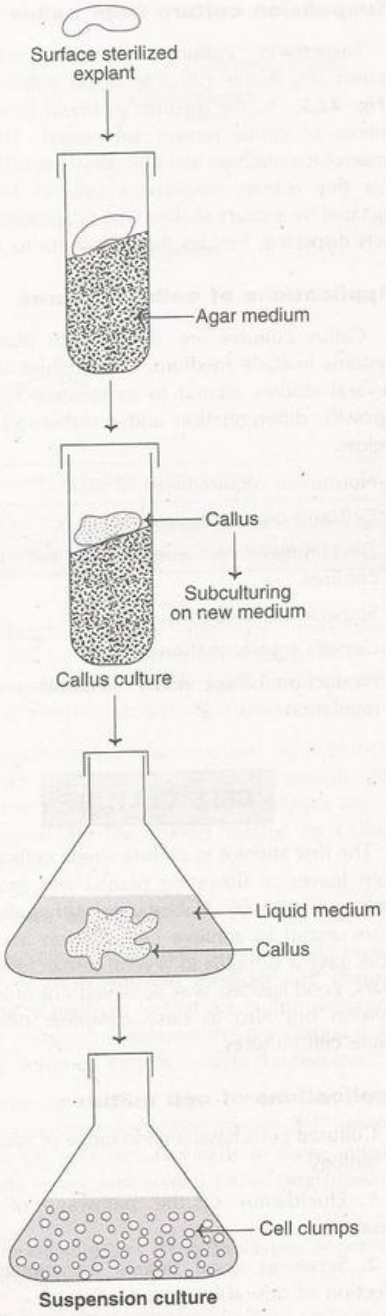


Fig. 42.5 : A diagrammatic representation of callus culture and initiation of suspension culture.

Suspension culture from callus

Suspension cultures can be initiated by transferring friable callus to liquid nutrient medium (***Fig. 42.5***). As the medium is liquid in nature, the pieces of callus remain submerged. This creates anaerobic condition and ultimately the cells may die. For this reason, suspension cultures have to be agitated by a rotary shaker. Due to agitation, the cells gets dispersed, besides their exposure to aeration.

Applications of callus cultures

Callus cultures are slow-growth plant culture systems in static medium. This enables to conduct several studies related to many aspects of plants (growth, differentiation and metabolism) as listed below.

- Nutritional requirements of plants.
- Cell and organ differentiation.
- Development of suspension and protoplast cultures.
- Somaclonal variations.
- Genetic transformations.
- Production of secondary metabolites and their regulation.

CELL CULTURE

The first attempt to culture single cells (obtained from leaves of flowering plants) was made in as early as 1902 by Haberlandt. Although he was unsuccessful to achieve cell division *in vitro*, his work gave a stimulus to several researchers. In later years, good success was achieved not only for cell division but also to raise complete plants from single cell cultures.

Applications of cell cultures

Cultured cells have a wide range of applications in biology.

1. Elucidation of the pathways of cellular metabolism.

2. Serve as good targets for mutation and selection of desirable mutants.

3. Production of secondary metabolites of commercial interest.

4. Good potential for crop improvement.

For more detailed information on the application of cell cultures, See p. 506.

Cell culture technique

The *in vitro* cell culture technique broadly involves the following aspects :

1. Isolation of single cells.
2. Suspension cultures growth and subculturing.
3. Types of suspension cultures.
4. Synchronization of suspension cultures.
5. Measurement of growth of cultures.
6. Measurement of viability of cultured cells.

The salient features of the above steps are briefly described.

ISOLATION OF SINGLE CELLS

The cells employed for *in vitro* culture may be obtained from plant organs, and from cultured tissues.

From plant organs : Plant leaves with homogenous population of cells are the ideal sources for cell culture. Single cells can be isolated from leaves by mechanical or enzymatic methods.

Mechanical method : Surface sterilized leaves are cut into small pieces ($< 1\ cm^2$), suspended in a medium and subjected to grinding in a glass homogeniser tube. The homogenate is filtered through filters and then centrifuged at a low speed to remove the cellular debris. The supernatant is removed and diluted to achieve the required cell density.

Enzymatic method : The enzyme ***macerozyme*** (under suitable osmotic pressure) can release the individual cells from the leaf tissues. Macerozyme degrades middle lamella and cell walls of parenchymatous tissues.

From cultured tissues : Single cells can be isolated from callus cultures (grown from cut pieces of surface sterilized plant parts). Repeated subculturing of callus on agar medium improves the friability of callus so that fine cell suspensions are obtained.

SUSPENSION CULTURES — GROWTH AND SUBCULTURE

The isolated cells are grown in suspension cultures. Cell suspensions are maintained by routine subculturing in a fresh medium. For this

purpose, the cells are picked up in the early stationary phase and transferred.

As the cells are incubated in suspension cultures, the cells divide and enlarge. The incubation period is dependent on :

- Initial cell density
- Duration of lag phase
- Growth rate of cells.

Among these, cell density is very crucial. The initial cell density used in the subcultures is very critical, and largely depends on the type of suspension culture being maintained. With low initial cell densities, the lag phase and log phases of growth get prolonged. Whenever a new suspension culture is started, it is necessary to determine the optical cell density in relation the volume of culture medium, so that maximum cell growth can be achieved. With low cell densities, the culture will not grow well, and requires additional supplementation of metabolites to the medium.

The normal incubation time for the suspension cultures is in the range of 21–28 days.

TYPES OF SUSPENSION CULTURES

There are mainly two types of suspension cultures — batch cultures and continuous cultures.

Batch cultures

A batch culture is a cell suspension culture ***grown in a fixed volume of nutrient culture medium***. In batch culture, cell division and cell growth coupled with increase in biomass occur until one of the factors in the culture environment (nutrient, O_2 supply) becomes limiting. The cells exhibit the following five phases of growth when the cell number in suspension cultures is plotted against the time of incubation (***Fig. 42.6***).

1. ***Lag phase*** characterized by preparation of cells to divide.
2. ***Log phase*** (exponential phase) where the rate of cell multiplication is highest.
3. ***Linear phase*** represented by slowness in cell division and increase in cell size expansion.
4. ***Deceleration phase*** characterized by decrease in cell division and cell expansion.
5. ***Stationary phase*** represented by a constant number of cells and their size.

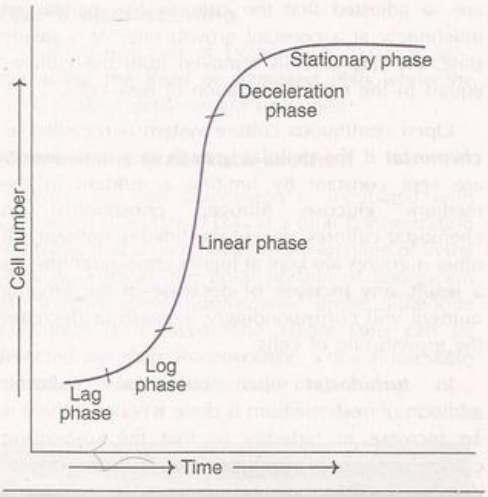


Fig. 42.6 : A graphic representation of different phases of batch suspension culture.

The batch cultures can be maintained continuously by transferring small amounts of the suspension medium (with inoculum) to fresh medium at regular intervals (2-3 days).

Batch cultures are characterized by a constant change in the pattern of cell growth and metabolism. For this reason, these cultures are not ideally suited for the studies related to cellular behaviour.

Continuous cultures

In continuous cultures, there is a regular ***addition of fresh nutrient medium and draining out the used medium*** so that the culture volume is normally constant. These cultures are carried out in specially designed culture vessels (bioreactors).

Continuous cultures are carried out under defined and controlled conditions—cell density, nutrients, O_2, pH etc. The cells in these cultures are mostly at an exponential phase (log phase) of growth.

Continuous cultures are of two types—open and closed.

Open continuous cultures : In these cultures, the inflow of fresh medium is balanced with the outflow of the volume of spent medium along with the cells. The addition of fresh medium and culture harvest

are so adjusted that the cultures are maintained indefinitely at a constant growth rate. At a steady state, the rate of cells removed from the cultures equals to the rate of formation of new cells.

Open continuous culture system is regarded as ***chemostat*** if the cellular growth rate and density are kept constant by limiting a nutrient in the medium (glucose, nitrogen, phosphorus). In chemostat cultures, except the limiting nutrient, all other nutrients are kept at higher concentrations. As a result, any increase or decrease in the limiting nutrient will correspondingly increase or decrease the growth rate of cells.

In ***turbidostat*** open continuous cultures, addition of fresh medium is done whenever there is an increase in turbidity so that the suspension culture system is maintained at a fixed optical density. Thus, in these culture systems, turbidity is preselected on the basis of biomass density in cultures, and they are maintained by intermittent addition of medium and washout of cells.

Closed continuous cultures : In these cultures, the cells are retained while the inflow of fresh medium is balanced with the outflow of corresponding spent medium. The cells present in the outflowing medium are separated (mechanically) and added back to the culture system. As a result, there is a continuous increase in the biomass in closed continuous cultures. These cultures are useful for studies related to cytodifferentiation, and for the production of certain secondary metabolites e.g., polysaccharides, coumarins.

SYNCHRONIZATION OF SUSPENSION CULTURES

In the normal circumstances, the cultured plant cells vary greatly in size, shape, cell cycle etc., and are said to be ***asynchronous***. Due to variations in the cells, they are not suitable for genetic, biochemical and physiological studies. For these reasons, synchronization of cells assumes significance.

Synchronization of cultured cells broadly refers to the ***organized existence of majority of cells in the same cell cycle phase simultaneously***.

A synchronous culture may be regarded as a culture in which the cell cycles or specific phase of cycles for majority of cultured cells occur simultaneously.

Several methods are in use to bring out synchronization of suspension cultures. They may be broadly divided into physical and chemical methods.

Physical methods

The environmental culture growth influencing physical parameters (light, temperature) and the physical properties of the cell (size) can be carefully monitored to achieve reasonably good degree of synchronization. A couple of them are described

Cold treatment : When the suspension cultures are subjected to low temperature (around 4°C) shock synchronization occurs. Cold treatment in combination with nutrient starvation gives better results.

Selection by volume : The cells in suspension culture can be selected based on the size of the aggregates, and by this approach, cell synchronization can be achieved.

Chemical methods

The chemical methods for synchronization of suspension cultures include the use of chemical inhibitors, and deprivation of an essential growth factor (nutrient starvation). By this approach, the ***cell cycle can be arrested at a particular stage***, and then allowed to occur simultaneously so that synchronization is achieved.

Chemical inhibition : Inhibitors of DNA synthesis (5-aminouracil, hydroxyurea, 5-fluorodeoxypurine), when added to the cultures results in the accumulation of cells at G_1 phase. And on removal of the inhibitor, synchronization of cell division occurs.

Colchicine is a strong inhibitor to arrest the growth of cells at metaphase. It inhibits spindle formation during the metaphase stage of cell division. Exposure to colchicine must be done for a short period (during the exponential growth phase), as long duration exposure may lead to mitoses.

Starvation : When an essential nutrient or growth promoting compound is deprived in suspension cultures, this results in stationary growth phase. On supplementation of the missing nutrient compound, cell growth resumption occurs synchronously. Some workers have reported that deprivation and subsequent addition of growth hormone also induces synchronization of cell cultures.

MEASUREMENT OF GROWTH OF CULTURES

It is necessary to assess the growth of cells in cultures. The parameters selected for the measuring growth of suspension cultures include cell counting, packed cell volume and weight increase.

Cell counting

Although cell counting to assess culture growth is reasonably accurate, it is tedious and time consuming. This is because cells in suspension culture mostly exist as colonies in varying sizes. These cells have to be first disrupted (by treating with pectinase or chromic acid), separated, and then ***counted using a haemocytometer***.

Packed cell volume

Packed cell volume (PCV) is expressed as ***ml of pellet per ml of culture***. To determine PCV, a measured volume of suspension culture is centrifuged (usually at 2000 xg for 5 minutes) and the volume of the pellet or packed cell volume is recorded.

After centrifugation the supernatant can be discarded, the pellet washed, dried overnight and weighed. This gives ***cell dry weight***.

Cell fresh weight

The wet cells are collected on a preweighed nylon fabric filter (supported in funnel). They are washed to remove the medium, drained under vacuum and weighed. This gives the fresh weight of cells. However, ***large samples have to be used*** for accurate weights.

MEASUREMENT OF VIABILITY OF CULTURED CELLS

The viability of cells is the most important factor for the growth of cells. Viability of cultured cells can be measured by microscopic examination of cells directly or after staining them.

Phase contrast microscopy

The viable cells can be detected by the presence of healthy nuclei. Phase contrast microscope is used for this purpose.

Evan's blue staining

A dilute solution of Evan's blue (0.025% w/v) dye ***stains the dead or damaged cells*** while the living (viable) cells remain unstained.

Fluorescein diacetate method

When the cell suspension is incubated with fluorescein diacetate (***FDA***) at a final concentration of 0.01%, it is cleaved by esterase enzyme of living cells. As a result, the polar portion of fluorescein which emits green fluoroscence under ultraviolet (UV) light is released. The ***viable cells can be detected by their fluoroscence***, since fluorescein accumulates in the living cells only.

CULTURE OF ISOLATED SINGLE CELLS (SINGLE CELL CLONES)

A ***clone is a mass of cells***, all of them ***derived*** through mitosis ***from a single cell***. The cells of the clone are expected to be identical with regard to genotype and karyotype. However, changes in these cells may occur after cloning. Single cells separated from plant tissues under suitable conditions can form clones.

The various methods for the isolation of single cells have already been described (See p. 502). Single cells can be cultured by the following methods :

1. Filter paper raft-nurse tissue technique
2. Microchamber technique
3. Microdrop method
4. Bergman's plating technique.

Filter paper raft-nurse tissue technique

Small pieces of sterile filter papers are placed on established callus cultures several days before the start of single cell culture. Single cell is now placed on the filter paper (***Fig. 42.7A***). This filter paper, wetted by the exudates from callus tissue (by diffusion) supplies the nutrients to the single cell. The ***cell divides and forms clonies on the filter paper***. These colonies can be isolated and cultured.

Microchamber technique

A microscopic ***slide or a coverslip can be used to create a microchamber***. Sometimes, a cavity

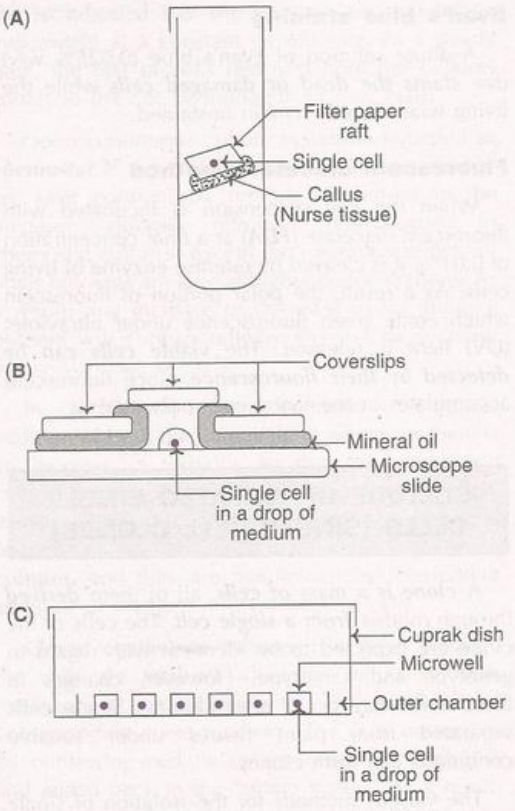


Fig. 42.7 : Techniques for the culture of single cells (A) Filter paper raft-nurse tissue technique (B) Microchamber technique (side view), (C) Microdrop method (Refer Fig. 42.8 for Bergmann's plating technique).

slide can be directly used. A drop of the medium containing a single cell is placed in the microchamber. A drop of mineral oil is placed on either side of the culture drop which is covered with a coverslip (***Fig. 42.7B***). On incubation, single cell colonies are formed.

Microdrop method

For the culture of single cells by microdrop method, a specially designed dish (cuprak dish) is used. It has a small outer chamber (to be filled with sterile distilled water) and a large inner chamber with a number of microwells (***Fig. 42.7C***). The cell density of the medium is adjusted in such a way that it contains one cell per droplet.

Bergmann's plating technique

Bergmann (1960) developed a technique for cloning of single cells. Now a days, Bergmann's plating technique is the ***most widely used method*** for culture of isolated single cells. This method is depicted in ***Fig. 42.8*** and briefly described hereunder.

The cell suspension is filtered through a sieve to obtain single cells in the filtrate. The free cells are suspended in a liquid medium, at a density twice than the required density for cell plating. Now, equal volumes of melted agar (30–35°C) and medium containing cells are mixed. The agar medium with single cells is poured and spread out in a petridish so that the cells are evenly distributed on a thin layer (of agar after it solidifies). The petridishes (culture dishes) are sealed with a parafilm and incubated at 25°C in dark or diffused light. The single cells divide and develop into clones.

The viability of cells in single clones can be measured by the same techniques that have been described for suspension cultures.

APPLICATIONS OF PLANT TISSUE CULTURES

Plant tissue cultures are associated with a wide range of applications—the most important being the ***production of pharmaceutical, medicinal and other industrially important compounds***. In addition, tissue cultures are useful for several other purposes listed below.

1. To study the respiration and metabolism in plants.
2. For the evaluation of organ functions in plants.
3. To study the various plant diseases and work out methods for their elimination.
4. Single cell clones are useful for genetic, morphological and pathological studies.
5. Embryonic cell suspensions can be used for ***large scale clonal propagation***.
6. Somatic embryos from cell suspensions can be stored for long term in germplasm banks.
7. In the production of variant clones with new characteristics, a phenomenon referred to as ***somaclonal variations***.

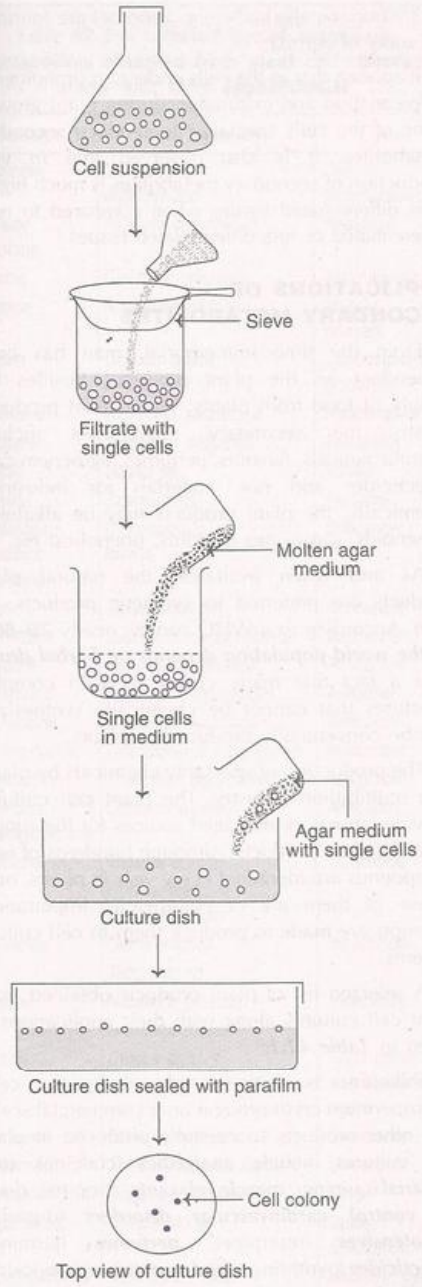


Fig. 42.8 : A diagrammatic representation of Bergmann's cell plating technique.

8. ***Production of haploids*** (with a single set of chromosomes) for improving crops.

9. Mutant cells can be selected from cultures and used for crop improvement.

10. Immature embryos can be cultured *in vitro* to produce hybrids, a process referred to ***as embryo rescue***.

SECONDARY METABOLITES IN PLANT CULTURES

The chemical compounds produced by plants are collectively referred to as ***phytochemicals***. Biotechnologists have special interest in plant tissue culture for the large scale production of commercially important compounds. These include pharmaceuticals, flavours, fragrances, cosmetics, food additives, feedstocks and antimicrobials. Most of these products are ***secondary metabolites—chemical compounds that do not participate in metabolism of plants***. Thus, secondary metabolites are not directly needed by plants as they do not perform any physiological function (as is the case with primary metabolites such as amino acids, nucleic acids etc.).

Although the native plants are capable of producing the secondary metabolites of commercial interest, tissue culture systems are preferred. The advantages and limitations are listed.

Major advantages

1. Compounds can be produced under controlled conditions as per market demands.

2. Culture systems are ***independent of environmental factors***, seasonal variations, pest and microbial diseases and geographical constraints.

3. Cell growth can be controlled to facilitate improved product formation.

4. The quality of the product will be consistent as it is produced by a specific cell line.

5. Recovery of the product will be easy.

6. Plant cultures are particularly useful in case of plants which are difficult or expensive to be grown in the fields.

7. Mutant cell lines can be developed for the ***production of novel compounds*** of commercial importance, which are not normally found in plants.

8. Biotransformation reactions (converting specific substrates to valuable products) can be carried out with certain cultured cells.

9. The production control is not at the mercy of political interference.

10. The production time is less and labour costs are minimal.

Considering the advantages listed above, ***about 25–30% of medicines for human use***, and the various chemical materials for industrial purposes are ***obtained from plant tissue cultures***. In general, tissue culture production of natural materials is cheaper compared to synthetic production. However, there are certain limitations associated with tissue cultures.

Limitations/disadvantages

1. In general, ***in vitro production of secondary metabolites is lower*** when compared to intact plants.

2. Many a times, secondary metabolites are formed in differentiated tissues/organs. In such a case, culture cells which are non-differentiated can produce little.

3. ***Cultured cells are genetically unstable and may undergo mutation***. The production of secondary metabolite may be drastically reduced, as the culture ages.

4. Vigorous stirring is necessary to prevent aggregation of cultured cells. This may often damage the cells.

5. Strict aseptic conditions have to be maintained during culture technique. Any infection to the culture adversely affects product formation.

Why do plants produce secondary metabolites?

Based on the existing evidence, it is believed that the production of some secondary metabolites is linked to the induction of morphological differentiation.

Consider the following examples

1. Cardiac glycosides are found in the leaves of *Digitalis*.

2. Quinine and quinidine are present in the bark of *Cinchona*.

3. Tropane alkaloids (e.g. atropine) are found in the roots of *Atropa*.

It appears that as the cells undergo morphological differentiation and maturation during plant growth, some of the cells specialise to produce secondary metabolites. It is also observed that *in vitro* production of secondary metabolites is much higher from differentiated tissues when compared to non-differentiated or less differentiated tissues.

APPLICATIONS OF SECONDARY METABOLITES

From the time immemorial, man has been dependent on the plant products, besides the supply of food from plants. These plant products, mostly the secondary metabolites include pharmaceuticals, flavours, perfumes, agrochemicals, insecticides and raw materials for industries. Chemically, the plant products may be alkaloids, terpenoids, glycosides (steroids, phenolics) etc.

As and when available, the natural plant products are preferred to synthetic products, by man. According to a WHO survey, nearly ***70–80% of the world population depends on herbal drugs***. It is a fact that many chemicals with complex structures that cannot be chemically synthesized can be conveniently produced in plants.

The production of speciality chemicals by plants is a multibillion industry. The plant cell cultures provide laboratory managed sources for the supply of useful plant products. Although hundreds of new compounds are identified every year in plants, only a few of them are of commercial importance. Attempts are made to produce them in cell culture systems.

A selected list of plant products obtained from plant cell cultures along with their applications is given in ***Table 42.1***.

Shikonine is a dye produced by the cells *Lithospermum erythrorhizon* on a commercial scale. The other products successfully produced in plant cell cultures include ***analgistics*** (codeine) ***anti-malarial*** (quinine), ***muscle relaxants*** (atropine), ***drugs to control cardiovascular disorders*** (digoxin), ***hypotensives*** (reserpine), ***perfumes*** (jasmine), ***insecticides*** (pyrithrins), ***food sweeteners*** (stevioside) and ***anticancer agents*** (vincristine). Sometimes, the cost of the plant products is unimaginably high. For instance, one kg of vincristine and vinblastine respectively cost $ 3,500,00 and $ 1,000,000!

TABLE 42.1 A selected list of secondary metabolites obtained from plant cell cultures along with their application(s)

Product	*Plant species*	*Uses*
Shikonine	*Lithospermum erythrorhizon*	Dye, pharmaceutical
Codeine, morphine	*Papaver somniferum*	Analgistic
Quinine	*Cinchona officinalis*	Antimalarial
Atropine	*Atropa belladonna*	Muscle relaxant
Digoxin	*Digitalis lanata*	Cardiovascular disorders
Reserpine	*Rauwolfia serpentina*	Hypotensive
Diosgenin	*Dioscorea deltoidea*	Antifertility
Vanillin	*Vanilla* sp	Vanilla
Jasmine	*Jasmium* sp	Perfume
Vinblastine, ajmalicine, vincristine	*Catharanthus roseus*	Anticancer
Taxol	*Taxus brevifolia*	Anticancer
Baccharine	*Baccharis megapotanica*	Anticancer
Cesaline	*Caesalpinia gillisesii*	Anticancer
Fagaronine	*Fagara zanthoxyloides*	Anticaner
Maytansine	*Maytenus bucchananii*	Anticancer
Harring tonine	*Cephalotaxus harringtonia*	Anticancer
Thalicarpine	*Thalictrum dasycarpum*	Anticancer
Ellipticine, 3-deoxycolchine	*Ochrosia mocrei*	Anticancer
Pyrithrins	*Tagetus erecta* *Chrysanthemum cinerariefolium*	Insecticide
Rotenoids	*Derris elliptica* *Tephrosia* sp	Insecticide
Nicotine	*Nicotiana tabacum* *Nicotiana rustica*	Insecticide
Saffron	*Crocus sativus*	Food colour and flavouring agent
Stevioside	*Stevia rabaudiana*	Sweetener
Thaumatin	*Thaumatococcus damielli*	Sweetener
Capsaicin	*Capsicum frutesus*	Chilli
Rosamarinic acid	*Coleus blunei*	Spice, antioxidant
Anthraquinones	*Morinda citrifolia*	Laxative, dye
Berberine	*Coptis japonica*	Antibacterial
Sarcoplasmine (hyoscine)	*Datura stramonium*	Treatment of nausea

PRODUCTION OF SECONDARY METABOLITES

The process of *in vitro* culture of cells for the large scale production of secondary metabolites is complex, and involves the following aspects

1. Selection of cell lines for high yield of secondary metabolites.
2. Large scale cultivation of plant cells.
3. Medium composition and effect of nutrients.
4. Elicitor-induced production of secondary metabolites.
5. Effect of environmental factors.
6. Biotransformation using plant cell cultures.
7. Secondary metabolite release and analysis.

SELECTION OF CELL LINES FOR HIGH YIELD OF SECONDARY METABOLITES

The very purpose of tissue culture is to produce high amounts of secondary metabolites. However, in general, majority of callus and suspension cultures produce less quantities of secondary metabolites. This is mainly due to the lack of fully differentiated cells in the cultures. Some special techniques have been devised to select cell lines that can produce higher amounts of desired metabolites. These methods are ultimately useful for the ***separation of producer cells from the non-producer cells***. The techniques commonly employed for cell line selection are cell cloning, visual or chemical analysis and selection for resistance.

Cell cloning

This is a simple procedure and involves the growth of single cells (taken from a suspension cultures) in a suitable medium. Each cell population is then screened for the secondary metabolite formation. And only those cells with high-yielding ability are selected and maintained by subcloning.

Single cell cloning : There are certain practical difficulties in the isolation and culture of single cells.

Cell aggregate cloning : Compared to single cell cloning, cell aggregate cloning is much easier, hence preferred by many workers.

A schematic representation of cell aggregate cloning for the selection of cells yielding high

quantities of secondary metabolites is given in ***Fig. 42.9***. A high yielding plant of the desired metabolite is selected and its explants are first cultured on a solid medium. After establishing the callus cultures, high metabolite producing calluses are identified, and they are grown in suspension cultures. Cell aggregates from these cultures are grown on solid medium. The freshly developed cell aggregates (calluses) are divided into two parts. One half is grown further, while the other half is used for the quantitative analysis of the desired metabolite produced. The cell lines with high yield of secondary metabolites are selected and used for scale-up in suspension cultures. This is followed by large scale tissue culture in a bioreactor.

Visual or chemical analysis

A direct measurement of some of the secondary metabolites produced by cell lines can be done either by visual or chemical analysis.

Visual identification of cell lines producing ***coloured secondary metabolites*** (pigments e.g., β-carotene, shikonin) will help ***in the selection*** of high-yielding cells. This method is quite simple and non-destructive. The major limitation is that the desired metabolite should be coloured.

Certain secondary metabolites emit fluorescence under UV light, and the corresponding clones can be identified.

Some workers use simple, sensitive and inexpensive chemical analytical methods for quantitative estimation of desired metabolites. Analysis is carried out in some colonies derived from single cell cultures. Radioimmunoassay is the most commonly used analytical method. Microspectrophotometry and fluorescent antibody techniques are also in use.

Selection for resistance

Certain cells resistant to toxic compounds may lead to the ***formation of mutant cells*** which can overproduce a primary metabolite, and then a secondary metabolite. Such mutants can be selected and used to produce the desired metabolite in large quantities. One example is described.

Cell lines selected for resistance of 5-methyl-tryptophan (analogue of tryptophan) produce strains which can overproduce tryptophan. These

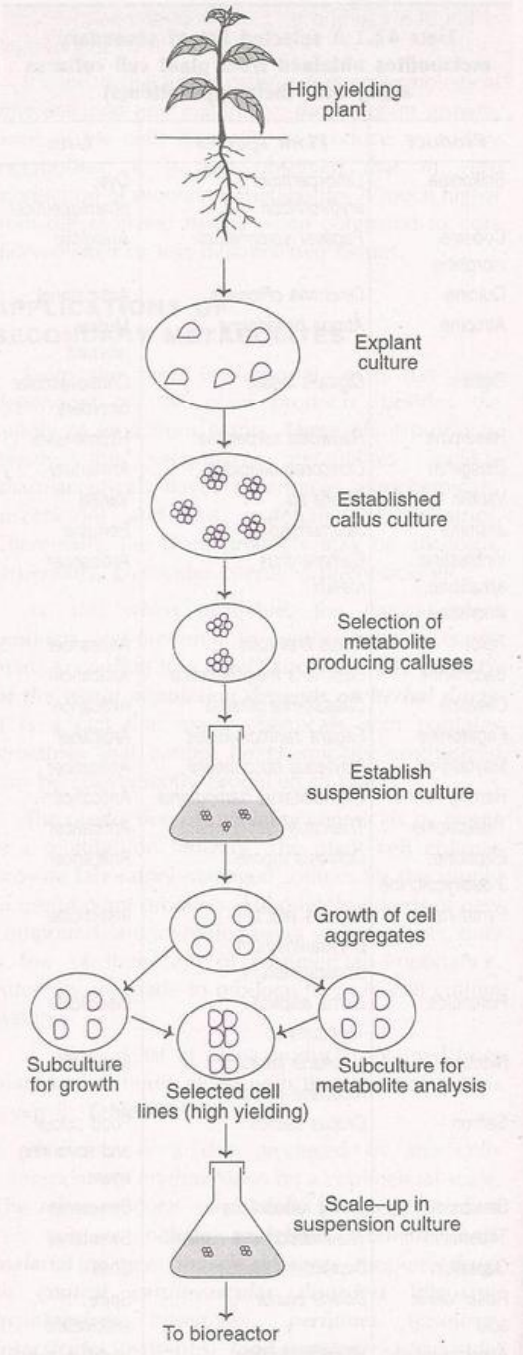


Fig. 42.9 : A schematic representation of cell aggregate cloning for the selection of high yielding cell lines.

tryptophan overproducing strains can synthesize 10–50 times higher levels of the natural auxin namely indole acetic acid (**Note :** The secondary metabolite indole acetic acid is derived from the primary metabolite tryptophan).

LARGE SCALE (MASS) CULTIVATION OF PLANT CELLS

In order to achieve industrial production of the desired metabolite, large scale cultivation of plant cells is required. Plant cells (20–150 µm in diameter) are generally 10–100 times larger than bacterial or fungal cell. When cultured, plant cells exhibit changes in volumes and thus variable shapes and sizes. Further, cultured cells have low growth rate and genetic instability. All these aspects have to be considered for mass cultivation of cells. The following four different culture systems are widely used

1. Free-cell suspension culture
2. Immobilized cell culture
3. Two-phase system culture
4. Hairy root culture.

Free-cell suspension culture

Mass cultivation of plant cells is most frequently carried out by cell suspension cultures. Care should be taken to achieve good growth rate of cells and efficient formation of the desired secondary metabolite. Many specially designed bioreactors are in use for free-cell suspension cultures. Some of these are listed below :

- Batch bioreactors
- Continuous bioreactors
- Multistage bioreactors
- Airlift bioreactors
- Stirred tank bioreactors.

Two important aspects have to be considered for good success of suspension cultures.

1. Adequate and continuous oxygen supply.
2. Minimal generation of hydrodynamic stresses due to aeration agitation.

Immobilized cell cultures

Plant cells can be made immobile or immovable and used in culture systems. The cells are physically immobilized by entrapment. Besides individual cells, it is also possible to immobilize aggregate cells or even calluses. Homogenous suspensions of cells are most suitable for immobilization.

Surface immobilized plant cell (SIPC) technique efficiently retains the cells and allows them to ***grow at a higher rate***. Further, through immobilization, there is better cell-to-cell contact, and the cells are protected from high liquid shear stresses. All this helps in the maximal production the secondary metabolite.

The common methods adopted for entrapment of cells are briefly described.

1. **Entrapment of cells in gels :** The cells or the protoplasts can be entrapped in several gels e.g., alginate, agar, agarose, carrageenin. The gels may be used either individually or in combination. The techniques employed for the immobilization of plant cells are comparable to those used for immobilization of microorganisms or other cells (Refer Chapter 21).

2. **Entrapment of cells in nets or foams :** Polyurethane foams or nets with various pore sizes are used. The actively growing plant cells in suspension can be immobilized on these foams. The cells divide within the compartments of foam and form aggregates.

3. **Entrapment of cells in hollow-fibre membranes :** Tubular hollow fibres composed of cellulose acetate silicone polycarbonate and organized into parallel bundles are used for immobilization of cells. It is possible to entrap cells within and between the fibres.

Membrane entrapment is mechanically stable. However, it is more expensive than gel or foam immobilization.

Bioreactors for use of immobilized cells

Fluidized bed or fixed bed bioreactors are employed to use immobilized cells for large scale cultivation.

In the fluidized-bed reactors, the immobilized cells are agitated by a flow of air or by pumping the medium. In contrast, in the fixed-bed bioreactor, the immobilized cells are held stationary (not agitated) and perfused at a slow rate with an aerated culture medium.

Table 42.2 A selected list of plant species with immobilized cells employed for the production secondary metabolite(s)

Plant culture species	*Immobilization method*	*Substrate*	*Product*
Catharanthus roseus	Entrapment in agarose	Cathenamine	Ajmalicine
Digitalis lanata	Entrapment in alginate	Digitoxin	Digoxin
Capsicum frutescens	Entrapment in polyurethane foam	Sucrose	Capsaicin
Catheranthus roseus	Entrapment in alginate, agarose, carrageenin	Sucrose	Ajmalicine
Petunia hybrida	Entrapment in hollow fibres	Sucrose	Phenolics
Morinda citrifolia	Entrapment in alginate	Sucrose	Anthraquinone
Solanum aviculare	Attachment polyphenylene beads	Sucrose	Steroid glycosides
Glycine max	Entrapment in hollow fibre	Sucrose	Phenolics

Biochemicals produced by using immobilized cells

A selected list of the immobilized cells from selected plants and their utility to produce important biochemicals is given in ***Table 42.2***.

Two-phase system culture

Plant cells can be cultivated in an aqueous two phase system for the production of secondary metabolites. In this technique, the cells are kept apart from the product by separation in the bioreactor. This is advantageous since the product can be removed continuously. Certain polymers (e.g., dextran and polyethylene glycol for the separation of phenolic compounds) are used for the separation of phases.

Hairy root culture

Hairy root cultures are used ***for the production of root-associated metabolites***. In general, these cultures have high growth rate and genetic stability.

For the production of hairy root cultures, the explant material (plant tissue) is inoculated with the cells of the pathogenic bacterium, *Agrobacterium rhizogenes*. This organism contains root-inducing (Ri) plasmid that causes genetic transformation of plant tissues, which finally results in hairy root cultures. Hairy roots produced by plant tissues have metabolite features similar to that of normal roots.

Hairy root cultures are most recent organ culture systems and are successfully used for the commercial production of secondary metabolites. A selected list of the plants employed in hairy root cultures and the secondary metabolites produced is given in ***Table 42.3***.

MEDIUM COMPOSITION AND EFFECT OF NUTRIENTS

The *in vitro* growth of the plant cells occurs in a suitable medium containing all the requisite elements. The ingradients of the medium effect the growth and metabolism of cells.

For optimal production of secondary metabolites, a two-medium approach is desirable. The first medium is required for good growth of cells (biomass growth) while the second medium, referred to as ***production medium promotes secondary metabolite*** formation. The effect of

Table 42.3 A selected list of plant species used in hairy root cultures for the production of secondary metabolite(s)

Plant species	*Secondary metabolite(s)*
Nicotiana tabacum	Nicotine, anatabine
Atropa belladonna	Atropine
Datura stramonium	Hyoscyamine
Lithospermum erythrorhizon	Shikonin
Catharanthus roseus	Ajmalicine, serpentine
Cinchona ledgeriana	Quinine alkaloids
Mentha vulgaris	Monoterpenes
Solanum laciniatum	Steroid alkaloids

nutrients (carbon and nitrogen sources, phosphate, growth regulators, precursors, vitamins, metal ions) on different species in relation to metabolite formation are variable, some of them are briefly described.

Effect of carbon source

Carbohydrates influence the production of phytochemicals. Some examples are given below.

1. Increase in sucrose concentration (in the range 4–10%) increases allkaloid production in *Catharanthus roseus* cultures.

2. ***Sucrose is a better carbon source*** than fructose or galactose for diosgenin production by *Dioscorea deltoidea* or *Dalanites aegyptiaca* cultures.

3. Low concentration of sucrose increases the production of ubiquinone-10 in tobacco cell cultures.

Effect of nitrogen source

The standard culture media usually contain a mixture of nitrate and ammonia as nitrogen source. Majority of plant cells can tolerate high levels of ammonia. The cultured cells utilize nitrogen for the biosynthesis of amino acids, proteins (including enzymes) and nucleic acids. The ***nitrogen containing primary metabolites directly influence the secondary metabolites.***

In general, high ammonium ion concentrations inhibit secondary metabolite formation while lowering of ammonium nitrogen increases. It is reported that addition of KNO_3 and NH_4NO_3 inhibited anthocyanin (by 90%) and alkaloid (by 80%) production.

Effect of phosphate

Inorganic phosphate is essential for photosynthesis and respiration (glycolysis). In addition, many secondary metabolites are produced through phosphorylated intermediates, which subsequently release the phosphate e.g., phenylpropanoids, terpenes, terpenoids. In general, high phosphate levels promote cell growth and primary metabolism while low phosphate concentrations are beneficial for secondary product formation. However, this is not always correct.

Increase in phosphate concentration in the medium may increase, decrease or may not affect product formation e.g.

1. Increased phosphate concentration increases alkaloid (in *Catharanthus roseus*), anthraquinone (in *Morinda citrifolia*) and diosgenin (in *Dioscorea deltoidea*) production.

2. Decreased phosphate level in the medium increases the formation of anthocyanins and phenolics (in *Catharanthus roseus*), alkaloids (in *Peganum harmala*) and solasodine (in *Solanum lanciatum*).

3. Phosphate concentration (increase or decrease) has no effect on protoberberine (an alkaloid) production by *Berberis* sp.

Effect of plant growth regulators

Plant growth regulators (auxins, cytokinins) ***influence growth, metabolism and differentiation of cultured cells.*** (For more details on the nature and different types of auxins and cytokinins, Refer Chapter 43). There are a large number of reports on the influence of growth regulators for the production of secondary metabolites in cultured cells. A few examples are given.

1. Addition of ***auxins*** (indole acetic acid, indole pyruvic acid, naphthalene acetic acid) enhanced the production of diosgenin in the cultures of *Balanites aegyptiaca.*

2. Auxins may inhibit the production of certain secondary metabolites e.g., naphthalene acetic acid and indole acetic acid inhibited the synthesis of anthocyanin in carrot cultures.

3. Another auxin, 2, 4-dichlorophenoxy acetate (2, 4-D) inhibits the production of alkaloids in the cultures of tobacco, and shikonin formation in the cultures of *Lithospermum erythrorhizon.*

4. ***Cytokinins*** promote the production of secondary metabolites in many tissue cultures e.g., ajmalicine in *Catharanthus roseus*; scopolin and scopoletin in tobacco; carotene in *Ricinus* sp.

5. In some tissue cultures, cytokinins inhibit product formation e.g., anthroquinones in *Morinda citrifolia*; shikonin in *Lithospermum erythroshizon*; nicotine in tobacco.

In the above examples, auxins and cytokinins are separately discussed. In actual practice, a combination of auxins and cytokinins is used to achieve maximum production of secondary metabolites in culture systems.

Effect of precursors

The substrate molecules that are incorporated into the secondary metabolites are referred to as precursors. ***In general, addition of precursors to the medium enhances product formation***, although they usually inhibit the growth of the culture e.g., alkaloid synthesis in *Datura* cultures in increased while growth is inhibited by the addition of ornithine, phenylalanine, tyrosine and sodium phenylpyruvate; precursors tryptamine and secologanin increase ajmalicine production in *C. roseus* cultures.

ELICITOR-INDUCED PRODUCTION OF SECONDARY METABOLITES

The production of secondary metabolites in plant cultures is generally low and does not meet the commercial demands. There are continuous efforts to understand the mechanism of product formation at the molecular level, and exploit for increased production. The synthesis of majority of secondary metabolites involves multistep reactions and many enzymes. It is possible to stimulate any step to increase product formation.

Elicitors are the compounds of biological origin which stimulate the production of secondary metabolites, and the phenomenon of such stimulation is referred to as ***elicitation***.

Elicitors produced within the plant cells are ***endogenous elicitors*** e.g., pectin, pectic acid, cellulose, other polysaccharides. When the elicitors are produced by the microorganisms, they are referred to as ***exogenous elicitors*** e.g., chitin, chitosan, glucans. All the elicitors of biological origin are ***biotic elicitors***. The term ***abiotic elicitors*** is used to represent the physical (cold, heat, UV light, osmotic pressure) and chemical agents (ethylene, fungicides, antibiotics, salts of heavy metals) that can also increase the product formation. However, the term abiotic stress is used for abiotic elicitors, while elicitors exclusively represent biological compounds.

Phytoalexins

Plants are capable of defending themselves when attacked by microorganisms, by producing ***antimicrobial compounds*** collectively referred to as phytoalexins. Phytoalexins are the chemical weapons of defense against pathogenic microorganisms. Some of the phytoalexins that induce the production of secondary metabolites are regarded as elicitors. Some chemicals can also act as elicitors e.g., actinomycin-D, sodium salt of arachidonic acid, ribonuclease-A, chitosan, poly-L-lysine, nigeran. These compounds are ***regarded as chemically defined elicitors***.

Interactions for elicitor formation

Elicitors are compounds involved in plant-microbe interaction. Three different types of interactions between plants and microorganisms are known that lead to the formation of elicitors.

1. Direct release of elicitor by the microorganisms.

2. Microbial enzymes that can act as elicitors. e.g. endopolygalacturonic acid lyase from *Erwinia carotovara*.

3. Release of phytoalexins by the action of plant enzymes on cell walls of microorganisms which in turn stimulate formation elicitors from plant cell walls e.g., chitosan from *Fusarium* cell walls; α-1, 3-endoglucanase from *Phytophthora* cell walls.

Methodology of elicitation

Selection of microorganisms : A wide range of microorganisms (viruses, bacteria, algae and fungi) that need not be pathogens have been tried in cultures for elicitor induced production of secondary metabolites. Based on the favourable elicitor response, an ideal microorganism is selected. The quantity of the microbial inoculum is important for the formation elicitor.

Co-culture : Plant cultures (frequently suspension cultures) are inoculated with the selected microorganism to form co-cultures. The cultures are transferred to a fresh medium prior to the inoculation with microorganism. This helps to stimulate the secondary metabolism.

Co-cultures of plant cells with microorganisms may sometimes have inhibitory effect on the plant cells. In such a case, elicitor preparations can be obtained by culturing the selected microorganism on a tissue culture medium, followed by homogenisation and autoclaving of the entire culture. This process releases elicitors. In case of heat labile elicitors, the culture homogenate has to be filter sterilized (instead of autoclaving).

In some co-culture systems, direct contact of plant cells and microorganisms can be prevented by immobilization (entrapment) of one of them. In these cultures, plant microbial interaction occurs by diffusion of the elicitor compounds through the medium.

Mechanism of action of elicitors

Elicitors are found to activate genes and increase the synthesis of mRNAs encoding enzymes responsible for the ultimate biosynthesis secondary metabolites.

There are some recent reports suggesting the involvement of elicitor mediated calcium-based signal transduction systems that promotes the product formation. When the cells are pretreated with a calcium chelate (EDTA) prior to the addition of elicitor, there occurs a decrease in the production of secondary metabolite.

Elicitor-induced products in cultures

In ***Table 42.4***, a selected list of elicitor-induced secondary metabolites produced in culture systems are given.

Table 42.4 A selected list of elicitor-induced secondary metabolite production in plant cell cultures

Elicitor microorganism	*Plant cell culture(s)*	*Secondary metabolite(s)*
Aspergillus niger	*Cinchona ledgeriana, Rubia tinctoria*	Anthraquinones
Pythium aphanidermatum	*Catharanthus roseus*	Ajmalicine, Strictosidine Catharanthine
Botrytis sp	*Papaver somniferum*	Sanguinarine
Phytophthora megasperma	*Glycine max*	Isoflavonoids Gluceollin
Dendryphion sp	*Papaver somniferum*	Sanguinarine
Alternaria sp	*Phaseolus vulgaris*	Phaseollin
Fusarium sp	*Apium graveolens*	Furanocoumarins
Phythium aphanidermatum	*Daucus carota*	Anthocynins
Penicillium expansum	*Sanguinaria canadensis*	Benzophenan-thridine Alkaloids

EFFECT OF ENVIRONMENTAL FACTORS

The physical factors namely light, incubation temperature, pH of the medium and aeration of cultures influence the production of secondary metabolites in cultures.

Effect of light

Light is absolutely essential for the carbon fixation (photosynthesis) of field-grown plants. Since the carbon fixation is almost absent or very low in plant tissue cultures, light has no effect on the primary metabolism. However, the ***light-mediated enzymatic reactions indirectly influence the secondary metabolite formation***. The quality of light is also important. Some examples of light-stimulated product formations are given

1. Blue light enhances anthocyanin production in *Haplopappus gracilis* cell suspensions.

2. White light increases the formation of anthocyanin in the cultures of *Catharanthus roseus*, *Daucus carota* and *Helianthus tuberosus*.

3. White or blue light inhibits naphthoquinone biosynthesis in callus cultures of *Lithospermum erythrorhizon*.

Effect of incubation temperature

The growth of cultured cells is increased with increase in temperature up to an optimal temperature (25–30°C). However, at least for the production some secondary metabolites lower temperature is advantageous. For instance, in *C. roseus* cultures, indole alkaloid production is increased by two fold when incubated at 16°C instead of 27°C. Increased temperature was also found to reduce the production of caffeine (by *C. sineneis*) and nicotine (by *N. tabacum*).

Effect of pH of the medium

For good growth of cultures, the pH of the medium is in the range of 5 to 6. There are reports

indicating that pH of the medium influences the formation of secondary metabolites. e.g., production of anthocyanin by cultures of *Daucus carota* was much less when incubated at pH 5.5 than at pH 4.5. This is attributed to the increased degradation of anthocyanin at higher pH.

Aeration of cultures

Continuous aeration is needed for good growth of cultures, and also for the efficient production of secondary metabolites.

BIOTRANSFORMATION USING PLANT CELL CULTURES

The ***conversion of one chemical into another*** (i.e., a substrate into a final product) ***by using biological systems*** (i.e. cell suspensions) as biocatalysts is regarded as biotransformation or bioconversion. The biocatalyst may be free or immobilized, and the process of biotransformation may involve one or more enzymes. Biotransformation involving microorganisms and animal cells are described elsewhere (Chapter 22).

The biotechnological application of plant cell cultures in biotransformation reactions involves the conversion of some less important substances to valuable medicinal or commercially important products. In biotransformation, it is necessary to select such cell lines that possess the enzymes for catalysing the desired reactions. Bioconversions may involve many types of reactions e.g., hydroxylation, reduction, glycosylation.

A good example of biotransformation by plant cell cultures is the large scale ***production of cardiovascular drug digoxin*** from digitoxin by *Digitali lanata*. Digoxin production is carried out by immobilized cells of *D. lanata* in airlift bioreactors. Cell cultures of *Digitalis purpurea* or *Stevia rebaudiana* can convert steviol into steviobiocide and steviocide which are 100 times sweeter than cane sugar.

A selected list of biotransformations carried out in plant cell cultures is given in ***Table 42.5***.

Table 42.5 Selected examples of biotransformations by plant cell cultures

Plant cell culture	*Substrate*	*Product*
Digitalis lanata	Digitoxin	Digoxin
Papaver somniferum	Codeinone	Codeine
Nicotiana tobacum	Carvoxine	Cavaxone
Daucus carota	Digitoxigenin	Periplogenin
Mucuna pruriens	L-Tyrosine	L-Dihydroxy-phenylalanine (L-DOPA)
Mentha sp	(–)-Menthone	(+)-Neomenthol
Coffea arabica	Vanillin	Vanillin-D-glucoside
Solanum tuberosum	Solavetivone	Hydroxylated derivatives
Galium mollugo	2-Succinyl benzoate	Anthra quinones
Datura sp	Hydroquinone	Arbutin
Citrus sp	Valencene	Nootkatone
Choisya ternata	Ellipticine	5-Formyl-ellipticine
Digitalis purpurea / *Stvia rebandiana*	Steviol	Steviocide, Steviobiocide

SECONDARY METABOLITE RELEASE AND ANALYSIS

The methods employed for the separation and purification of secondary metabolites from cell cultures are the same as that used for plants.

Sometimes, the products formed within the cells are released into the medium, making the isolation and analysis easy. For the secondary metabolites stored within the vacuoles of cells, two membranes (plasma membrane and tonoplast) have to be disrupted. Permeabilizing agents such as dimethyl sulfoxide (DMSO) can be used for the release of products.

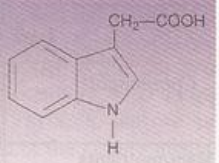
In general, separation and purification of products from plant cell cultures are expensive, therefore every effort is made to make them cost-effective. Two approaches are made in this direction :

1. Production of secondary metabolite should be as high as possible.

2. Formation of side product(s) which interfere with separation must be made minimal.

Once a good quantity of the product is released into the medium, separation and purification techniques (e.g. extraction) can be used for its recovery. These techniques largely depend on the nature of the secondary metabolite.

43 Plant Tissue Culture Media



Culture media are largely responsible for the *in vitro* growth and morphogenesis of plant tissues. The success of the plant tissue culture depends on the choice of the nutrient medium. In fact, the cells of most plant cells can be grown in culture media.

Basically, the plant tissue culture media should contain the same nutrients as required by the whole plant. It may be noted that plants in nature can synthesize their own food material. However, ***plants growing in vitro are mainly heterotrophic i.e. they cannot synthesize their own food.***

Composition of media

The composition of the culture media is primarily dependent on two parameters.

1. The particular species of the plant.
2. The type of material used for culture i.e. cells, tissues, organs, protoplasts.

Thus, the composition of a medium is formulated considering the specific requirements of a given culture system. The media used may be solid (***solid medium***) or liquid (***liquid medium***) in nature. The selection of solid or liquid medium is dependent on the better response of a culture.

MAJOR TYPES OF MEDIA

The composition of the most commonly used tissue culutre media is given in ***Table 43.1***, and briefly discribed below.

White's medium : This is one of the earliest plant tissue culture media developed for root culture.

MS medium : Murashige and Skoog (MS) originally formulated a medium to induce organogenesis, and regeneration of plants in cultured tissues. These days, MS medium is widely used for many types of culture systems.

B5 medium : Developed by Gamborg, B5 medium was originally designed for cell suspension and callus cultures. At present with certain modifications, this medium is used for protoplast culture.

N6 medium : Chu formulated this medium and it is used for cereal anther culture, besides other tissue cultures.

Nitsch's medium : This medium was developed by Nitsch and Nitsch and frequently used for anther cultures.

Among the media referred above, MS medium is most frequently used in plant tissue culture work due to its success with several plant species and culture systems.

Synthetic and natural media : When a medium is composed of chemically defined components, it is referred to as a synthetic medium. On the other hand, if a medium contains chemically undefined compounds (e.g., vegetable extract, fruit juice, plant extract), it is regarded as a natural medium.

TABLE 43.1 Composition of commonly used plant tissue culture media

Components	*Amount (mg l^{-1})*				
	White's	*Murashige and Skoog (MS)*	*Gamborg (B5)*	*Chu(N6)*	*Nitsch's*
Macronutrients					
$MgSO_4.7H_2O$	750	370	250	185	185
KH_2PO_4	—	170	—	400	68
$NaH_2PO_4.H_2O$	19	—	150	—	—
KNO_3	80	1900	2500	2830	950
NH_4NO_3	—	1650	—	—	720
$CaCl_2.2H_2O$	—	440	150	166	—
$(NH_4)_2.SO_4$	—	—	134	463	—
Micronutrients					
H_3BO_3	1.5	6.2	3	1.6	—
$MnSO_4.4H_2O$	5	22.3	—	4.4	25
$MnSO_4.H_2O$	—	—	10	3.3	—
$ZnSO_4.7H_2O$	3	8.6	2	1.5	10
$Na_2MoO_4.2H_2O$	—	0.25	0.25	—	0.25
$CuSO_4.5H_2O$	0.01	0.025	0.025	—	0.025
$CoCl_2.6H_2O$	—	0.025	0.025	—	0.025
KI	0.75	0.83	0.75	0.8	—
$FeSO_4.7H_2O$	—	27.8	—	27.8	27.8
$Na_2EDTA.2H_2O$	—	37.3	—	37.3	37.3
Sucrose (g)	20	30	20	50	20
Organic supplements					
Vitamins					
Thiamine HCl	0.01	0.5	10	1	0.5
Pyridoxine (HCl)	0.01	0.5	1	0.5	0.5
Nicotinic acid	0.05	0.5	1	0.5	5
Myoinositol	—	100	100	—	100
Others					
Glycine	3	2	—	—	2
Folic acid	—	—	—	—	0.5
Biotin	—	—	—	—	0.05
pH	5.8	5.8	5.5	5.8	5.8

Synthetic media have almost replaced the natural media for tissue culture.

Expression of concentrations in media : The concentrations of inorganic and organic constituents in culture media are usually expressed as mass values (mg/l or ppm or mg l^{-1}). However, as per the recommendations of the International Association of Plant Physiology, the concentrations of macronutrients should be expressed as mmol/l^{-} and micronutrients as µmol/l^{-}.

CONSTITUENTS OF MEDIA

Many elements are needed for plant nutrition and their physiological functions. Thus, these elements have to be supplied in the culture medium to support adequate growth of cultures *in vitro*. A selected list of the elements and their functions in plants is given in ***Table 43.2***.

The culture media usually contain the following constituents

1. Inorganic nutrients
2. Carbon and energy sources
3. Organic supplements
4. Growth regulators
5. Solidifying agents
6. pH of medium

Inorganic nutrients

The inorganic nutrients consist of ***macronutrients*** (concentration >0.5 mmol/l^{-}) and ***micronutrients*** (concentration <0.5 mmol/l^{-}). A wide range of mineral salts (elements) supply the macro- and micronutrients. The inorganic salts in water undergo dissociation and ionization. Consequently, one type of ion may be contributed by more than one salt. For instance, in MS medium, K^+ ions are contributed by KNO_3 and KH_2PO_4 while NO_3^- ions come from KNO_3 and NH_4NO_3.

Macronutrient elements : The six elements namely ***nitrogen, phosphorus, potassium, calcium, magnesium and sulfur*** are the essential macronutrients for tissue culture. The ideal concentration of nitrogen, and potassium is around 25 mmol l^{-1} while for calcium, phosphorus, sulfur and magnesium, it is in the range of 1-3 mmol l^{-}. For the supply of nitrogen in the medium, nitrates and ammonium salts are together used.

Micronutrients : Although their requirement is in minute quantities, micronutrients are essential for plant cells and tissues. These include iron, manganese, zinc, boron, copper and molybdenum. Among the microelements, ***iron requirement is very critical***. Chelated forms of iron and copper are commonly used in culture media.

Carbon and energy sources

Plant cells and tissues in the culture medium are heterotrophic and therefore, are dependent on the external carbon for energy. Among the energy sources, ***sucrose is the most preferred***. During the course of sterilization (by autoclaving) of the medium, sucrose gets hydrolysed to glucose and fructose. The plant cells in culture first utilize glucose and then fructose. In fact, glucose or fructose can be directly used in the culture media. It may be noted that for energy supply, glucose is as efficient as sucrose while fructose is less efficient.

TABLE 43.2 A selected list of elements and their functions in plants

Element	*Function(s)*
Nitrogen	Essential component of proteins, nucleic acids and some coenzymes. (Required in most abundant quantity)
Calcium	Synthesis of cell wall, membrane function, cell signalling.
Magnesium	Component of chlorophyll, cofactor for some enzymes.
Potassium	Major inorganic cation, regulates osmotic potential.
Phosphorus	Component of nucleic acids and various intermediates in respiration and photosynthesis, involved in energy transfer.
Sulfur	Component of certain amino acids (methionine, cysteine and cystine, and some cofactors).
Manganese	Cofactor for certain enzymes.
Iron	Component of cytochromes, involved in electron transfer.
Chlorine	Participates in photosynthesis.
Copper	Involved in electron transfer reactions, Cofactor for some enzymes.
Cobalt	Component of vitamin B_{12}.
Molybdenum	Component of certain enzymes (e.g., nitrate reductase), cofactor for some enzymes.
Zinc	Required for chlorophyll biosynthesis, cofactor for certain enzymes.

It is a common observation that ***cultures grow better on*** a medium with ***autoclaved sucrose*** than on a medium with filter-sterilized sucrose. This clearly indicates that the hydrolysed products of sucrose (particularly glucose) are efficient sources of energy. Direct use of fructose in the medium subjected to autoclaving, is found to be detrimental to the growth of plant cells.

Besides sucrose and glucose, other carbohydrates such as lactose, maltose, galactose, raffinose, trehalose and cellobiose have been used in culture media but with a very limited success.

Organic supplements

The organic supplements include vitamins, amino acids, organic acids, organic extracts, activated charcoal and antibiotics.

Vitamins : Plant cells and tissues in culture (like the natural plants) are capable of synthesizing vitamins but in suboptimal quantities, inadequate to support growth. Therefore the medium should be supplemented with ***vitamins to achieve good growth of cells***. The vitamins added to the media include thiamine, riboflavin, niacin, pyridoxine, folic acid, pantothenic acid, biotin, ascorbic acid, myo-inositol, para-amino benzoic acid and vitamin E.

Amino acids : Although the cultured plant cells can synthesize amino acids to a certain extent, media supplemented with ***amino acids stimulate cell growth and help in establishment of cells lines***. Further, organic nitrogen (in the form of amino acids such as L-glutamine, L-asparagine, L-arginine, L-cysteine) is more readily taken up than inorganic nitrogen by the plant cells.

Organic acids : Addition of Krebs cycle intermediates such as citrate, malate, succinate or fumarate allow the growth of plant cells. Pyruvate also enhances the growth of cultured cells.

Organic extracts : It has been a practice to supplement culture media with organic extracts such as ***yeast***, ***casein hydrolysate***, coconut milk, orange juice, tomato juice and potato extract.

It is however, preferable to avoid the use of natural extracts due to high variations in the quality and quantity of growth promoting factors in them. In recent years, natural extracts have been replaced by specific organic compounds e.g., replacement of yeast extract by L-asparagine; replacement of fruit extracts by L-glutamine.

Activated charcoal : Supplementation of the medium with activated charcoal stimulates the growth and differentiation of certain plant cells (carrot, tomato, orchids). Some ***toxic/inhibitory compounds*** (e.g. phenols) produced by cultured plants are ***removed*** (by adsorption) ***by activated charcoal***, and this facilitatis efficient cell growth in cultures.

Addition of activated charcoal to certain cultures (tobacco, soybean) is found to be inhibitory, probably due to adsorption of growth stimulants such as phytohormones.

Antibiotics : It is sometimes necessary to add antibiotics to the medium ***to prevent the growth of microorganisms***. For this purpose, low concentrations of streptomycin or kanamycin are used. As far as possible, addition of antibiotics to the medium is avoided as they have an inhibitory influence on the cell growth.

Growth regulators

Plant hormones or ***phytohormones*** are a group of natural organic compounds that promote growth, development and differentiation of plants. Four broad classes of growth regulators or hormones are used for culture of plant cells-***auxins, cytokinins, gibberellins*** (***Fig. 43.1***) and ***abscisic acid***. They promote growth, differentiation and organogenesis of plant tissues in cultures.

Auxins : Auxins ***induce cell division***, cell elongation, and formation of callus in cultures. At a low concentration, auxins promote root formation while at a high concentration callus formation occurs. A selected list of auxins used in tissue cultures is given in ***Table 43.3***.

Among the auxins, 2, 4-dichlorophenoxy acetic acid is most effective and is widely used in culture media.

Cytokinins : Chemically, cytokinins are derivatives of a purine namely adenine. These adenine derivatives are involved in cell division, ***shoot differentiation and somatic embryo formation***. Cytokinins promote RNA synthesis and thus stimulate protein and enzyme activities in tissues. The most commonly used cytokinins are given in ***Table 43.3***.

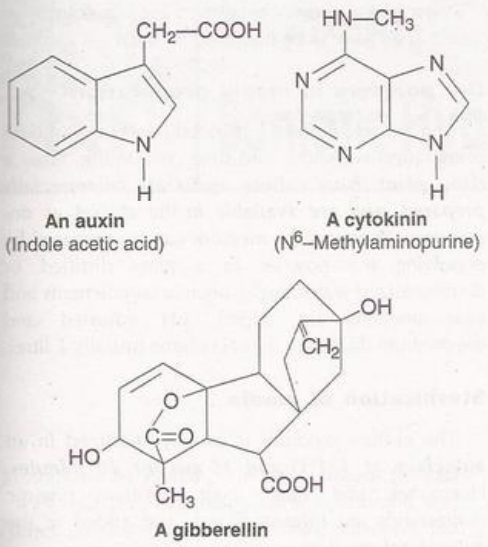


Fig. 43.1 : Structures of selected plant growth regulators.

Among the cytokinins, kinetin and benzyl-aminopurine are frequently used in culture media.

Ratio of auxins and cytokinins : The relative concentrations of the growth factors namely auxins and cytokinins are crucial for the morphogenesis of culture systems. When the ***ratio of auxins to cytokinins is high, embryogenesis, callus initiation and root initiation occur. On the other hand, for axillary and shoot proliferation, the ratio of auxins to cytokinins is low***. For all practical purposes, it is considered that the formation and maintenance of callus cultures require both auxin and cytokinin, while auxin is needed for root culture and cytokinin for shoot culture.

The actual concentrations of the growth regulators in culture media are variable depending on the type of tissue explant and the plant species.

Gibberellins : About 20 different gibberellins have been identified as growth regulators. Of these, gibberellin A_3 (GA_3) is the most commonly used for tisue culture. GA_3 promotes growth of cultured cells, ***enhances callus growth*** and induces dwarf plantlets to elongate.

Gibberellins are capable of promoting or inhibiting tissue cultures, depending on the plant species. They usually inhibit adventitious root and shoot formation.

Abscisic acid (ABA) : The callus growth of cultures may be stimulated or inhibited by ABA. This largely depends on the nature of the plant species. Abscisic acid is an important growth regulation for induction of embryogenesis.

Solidifying agents

For the preparation of semisolid or solid tissue culture media, solidifying or ***gelling agents*** are required. In fact, solidifying agents extend support to tissues growing in the static conditions.

Agar : Agar, a polysaccharide obtained from seaweeds, is most commonly used as a gelling agent for the following reasons (next page).

Table 43.3 A selected list of plant growth regulators used in culture media

Growth regulator (abbreviation/name)	*Chemical name*
Auxins	
IAA	Indole 3-acetic acid
IBA	Indole 3-butyric acid
NAA	1-Naphthyl acetic acid
2, 4-D	2, 4-Dichlorophenoxy acetic acid
2, 4, 5-T	2, 4, 5-Trichlorophenoxy acetic acid
4—CPA	4-Chlorophenoxy acetic acid
NOA	2-Naphthyloxy acetic acid
MCPA	2-Methyl 4-chlorophenoxy acetic acid
Dicamba	2-Methoxy 3, 6-dichlorobenzoic acid
Picloram	4-Amino 2, 5, 6-trichloropicolinic acid
Cytokinins	
BAP	6-Benzyl aminopurine
BA	Benzyl adenine
2 iP (IPA)	N^6-(2-isopentyl) adenine
DPU	Diphenyl urea
Kinetin	6-Furfuryl aminopurine
Zeatin	4-Hydroxy 3-methyltrans 2-butenyl aminopurine
Thidiazuron	1-Phenyl 3-(1, 2, 3-thiadiazol-5 yl) urea

1. It does not react with media constituents.

2. It is not digested by plant enzymes and is stable at culture temperature.

Agar at a concentration of 0.5 to 1% in the medium can form a gel.

Gelatin : It is used at a high concentration (10%) with a limited success. This is mainly because gelatin melts at low temperature (25°C), and consequently the gelling property is lost.

Other gelling agents : Biogel (polyacrylamide pellets), phytagel, gelrite and purified agarose are other solidifying agents, although less frequently used. It is in fact advantageous to use synthetic gelling compounds, since they can form gels at a relatively low concentration (1.0 to 2.5 g l^{-1}).

pH of medium

The optimal pH for most tissue cultures is ***in the range of 5.0-6.0***. The pH generally falls by 0.3-0.5 units after autoclaving. Before sterilization, pH can be adjusted to the required optimal level while preparing the medium. It is usually not necessary to use buffers for the pH maintenance of culture media.

At a pH higher than 7.0 and lower than 4.5, the plant cells stop growing in cultures. If the pH falls during the plant tissue culture, then fresh medium should be prepared. In general, pH above 6.0 gives the medium hard appearance, while pH below 5.0 does not allow gelling of the medium.

PREPARATION OF MEDIA

The general methodology for a medium preparation involves preparation of stock solutions (in the range of 10x to 100x concentrations) using high purity chemicals and demineralized water. The stock solutions can be stored (in glass or plastic containers) frozen and used as and when required.

Most of the growth regulators are not soluble in water. They have to be dissolved in NaOH or alcohol.

Dry powders in media preparation

The conventional procedure for media preparation is tedious and time consuming. Now a days, ***plant tissue culture media are commercially prepared, and are available in*** the market as dry powders. The requisite medium can be prepared by dissolving the powder in a glass distilled or demineralized water. Sugar, organic supplements and agar (melted) are added, pH adjusted and the medium diluted to a final volume (usually 1 litre).

Sterilization of media

The culture medium is usually sterilized in an ***autoclave at 121°C and 15 psi for 20 minutes.*** Hormones and other heat sensitive organic compounds are filter-sterilized, and added to the autoclaved medium.

SELECTION OF A SUITABLE MEDIUM

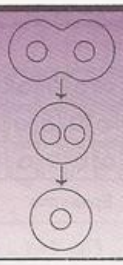
In order to select a suitable medium for a particular plant culture system, it is customary to start with a known medium (e.g. MS medium, B5 medium) and then develop a new medium with the desired characteristics. Among the constituents of a medium, growth regulators (auxins, cytokinins) are highly variable depending on the culture system. In practice, 3-5 different concentrations of growth regulators in different combinations are used and the best among them are selected.

For the selection of appropriate concentrations of minerals and organic constituents in the medium, similar approach referred above, can be employed.

Medium-utmost important for culture

For tissue culture techniques, it is absolutely essential that the medium preparation and composition are carefully followed. Any mistake in the preparation of the medium is likely to do a great harm to the culture system as a whole.

Protoplast Culture and Somatic Hybridization



Protoplasts are ***naked plant cells without the cell wall***, but they possess plasma membrane and all other cellular components. They represent the functional plant cells but for the lack of the barrier, cell wall. ***Protoplasts of different species can be fused*** to generate a hybrid and this process is referred to as ***somatic hybridization*** (or protoplast fusion). Cybridization is the phenomenon of fusion of a normal protoplast with an enucleated (without nucleus) protoplast that results in the formation of a ***cybrid*** or ***cytoplast*** (cytoplasmic hybrids).

Historical developments

The term protoplast was introduced in 1880 by Hanstein. The first isolation of protoplasts was achieved by Klercker (1892) employing a mechanical method. A real beginning in protoplast research was made in 1960 by Cocking who used an enzymatic method for the removal of cell wall. Rakabe and his associates (1971) were successful to achieve the regeneration of whole tobacco plant from protoplasts. Rapid progress occurred after 1980 in protoplast fusion to improve plant genetic material, and the development of transgenic plants.

IMPORTANCE OF PROTOPLASTS AND THEIR CULTURES

The isolation, culture and fusion of protoplasts is a fascinating field in plant research. Protoplast isolation and their cultures provide ***millions of single cells*** (comparable to microbial cells) ***for a viriety of studies***. Protoplasts have a wide range of applications, some of them are listed below.

1. The protoplast in culture can be regenerated into a whole plant.
2. Hybrids can be developed from protoplast fusion.
3. It is easy to perform single cell cloning with protoplasts.
4. Genetic transformations can be achieved through genetic engineering of protoplast DNA.
5. Protoplasts are excellent materials for ultrastructural studies.
6. Isolation of cell organelles and chromosomes is easy from protoplasts.
7. Protoplasts are useful for membrane studies (transport and uptake processes).
8. Isolation of mutants from protoplast cultures is easy.

ISOLATION OF PROTOPLASTS

Protoplasts are isolated by two techniques

1. Mechanical method
2. Enzymatic method

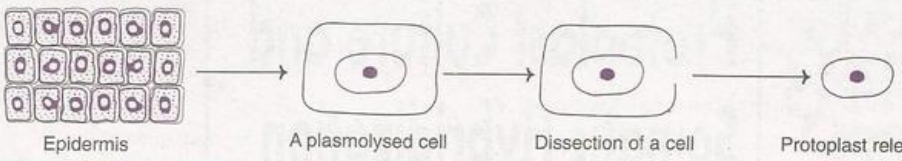


Fig. 44.1 : A diagrammatic representation of mechanical method for the isolation of protoplasts.

MECHANICAL METHOD

Protoplas isolation by mechanical method is a crude and tedious procedure. This results in the isolation of a very small number of protoplasts. The technique involves the following stages (***Fig. 44.1***).

1. A small piece of epidermis from a plant is selected.

2. The cells are subjected to plasmolysis. This causes protoplasts to shrink away from the cell walls.

3. The tissue is dissected to release the protoplasts.

Mechanical method for protoplast isolation is no more in use because of the following limitations

- Yield of protoplasts and their viability is low.
- It is restricted to certain tissues with vacuolated cells.
- The method is laborious and tedious.

However, some workers ***prefer mechanical methods if the cell wall degrading enzymes*** (of enzymatic method) ***cause deleterious effects to protoplasts.***

ENZYMATIC METHOD

Enzymatic method is ***a very widely used technique*** for the isolation of protoplasts. The advantages of enzymatic method include good yield of viable cells, and minimal or no damage to the protoplasts.

Sources of protoplasts

Protoplasts can be isolated from a wide variety of tissues and organs that include leaves, roots, shoot apices, fruits, embryos and microspores. Among these, the mesophyll tissue of fully expanded leaves of young plants or new shoots are most frequently used. In addition, callus and suspension cultures also serve as good sources for protoplast isolation.

Enzymes for protoplast isolation

The enzymes that can digest the cell walls are required for protoplast isolation. Chemically, the plant cell wall is mainly composed of cellulose, hemicellulose and pectin which can be respectively degraded by the enzymes cellulase, hemicellulase and pectinase. The different enzymes for protoplast isolation and the corresponding sources are given in ***Table 44.1***.

In fact, the various enzymes for protoplast isolation are commercially available. The enzymes are usually used at a pH 4.5 to 6.0, temperature 25–30°C with a wide variation in incubation period that may range from half an hour to 20 hours. The enzymatic isolation of protoplasts can be carried out by two approaches.

TABLE 44.1 A selected list of commercially available enzymes for protoplast isolation, and their sources

Enzyme	*Source*
Cellulases	
Cellulase onozuka R-10	*Trichoderma viride*
Cellulase YC	*Trichoderma viride*
Cellulysin	*Trichoderma viride*
Driselase	*Irpex lactus*
Hemocellulases	
Hemicellulase	*Aspergillus niger*
Helicase	*Helix pomatia*
Rhozyme HP-150	*Aspergillus niger*
Hemicellulase H-2125	*Rhizopus* sp
Pectinases	
Macerase	*Rhizopus arrhizus*
Pectolyase	*Aspergillus japonicus*
Macerozyme R-10	*Rhizopus arrhizus*
Zymolyase	*Arthrobacter luteus*

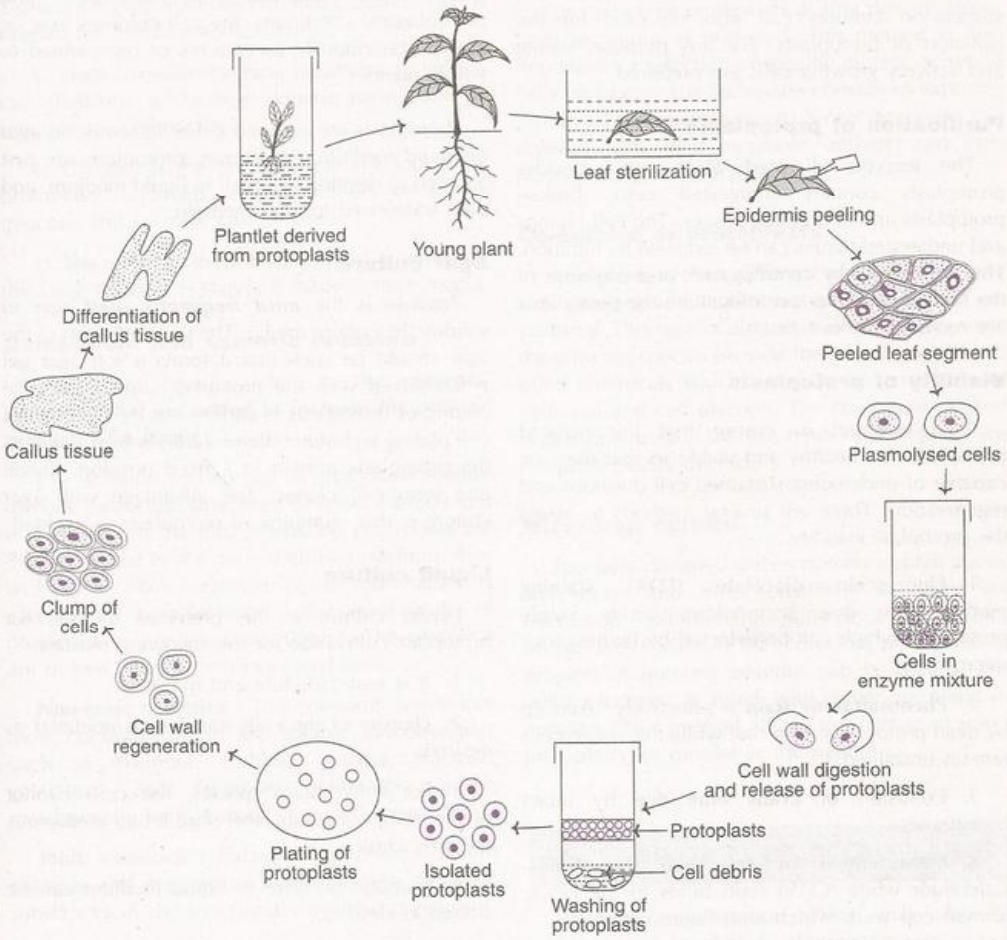


Fig. 44.2 : Major steps involved in protoplast isolation, culture and regeneration of plants.

1. **Two step or sequential method :** The tissue is first treated with pectinase (macerozyme) to separate cells by degrading middle lamella. These free cells are then exposed to cellulase to release protoplasts. Pectinase breaks up the cell aggregates into individual cells while cellulase removes the cell wall proper.

2. **One step or simultaneous method :** This is the preferred method for protoplast isolation. It involves the simultaneous use of both the enzymes — macerozyme and cellulase.

Isolation of protoplasts from leaves

Leaves are ***most commonly used***, for protoplast isolation, since it is possible to isolate uniform cells in large numbers. The procedure broadly involves the following steps (***Fig. 44.2***).

1. Sterilization of leaves.
2. Removal of epidermal cell layer.
3. Treatment with enzymes.
4. Isolation of protoplasts.

Besides leaves, callus cultures and cell suspension cultures can also be used for the isolation of protoplasts. For this purpose, young and actively growing cells are preferred.

Purification of protoplasts

The enzyme digested plant cells, besides protoplasts contain undigested cells, broken protoplasts and undigested tissues. The cell clumps and undigested tissues can be removed by filtration. This is followed by centrifugation and washings of the protoplasts. After centrifugation, the protoplasts are recovered above Percoll.

Viability of protoplasts

It is essential to ensure that the isolated protoplasts are healthy and viable so that they are capable of undergoing sustained cell divisions and regeneration. There are several methods to assess the protoplast viability.

1. **Fluorescein diacetate (FDA) staining** method—The dye accumulates inside viable protoplasts which can be detected by fluorescence microscopy.

2. **Phenosafranine stain** is selectively taken up by dead protoplasts (turn red) while the viable cells remain unstained.

3. Exclusion of **Evans blue dye** by intact membranes.

4. **Measurement of cell wall formation**—Calcofluor white (CFW) stain binds to the newly formed cell walls which emit fluorescence.

5. **Oxygen uptake** by protoplasts can be measured by oxygen electrode.

6. **Photosynthetic activity** of protoplasts.

7. The ability of protoplasts to undergo continuous **mitotic divisions** (this is a direct measure).

CULTURE OF PROTOPLASTS

The very ***first step*** in protoplast culture is the ***development of a cell wall*** around the membrane of the protoplast. This is followed by the cell divisions that give rise to a small colony. With suitable manipulations of nutritional and physiological conditions, the cell colonies may be grown continuously as cultures or regenerated to whole plants.

Protoplasts are cultured either in semisolid agar or liquid medium. Sometimes, protoplasts are first allowed to develop cell wall in liquid medium, and then transferred to agar medium.

Agar culture

Agarose is the ***most frequently used*** agar to solidify the culture media. The concentration of the agar should be such that it forms a soft agar gel when mixed with the protoplast suspension. The plating of protoplasts is carried out by Bergmann's cell plating technique (Refer ***42.8***). In agar cultures, the protoplasts remain in a fixed position, divide and form cell clones. The advantage with agar culture is that clumping of protoplasts is avoided.

Liquid culture

Liquid culture is the preferred method for protoplast cultivation for the following reasons.

1. It is easy to dilute and transfer.

2. Density of the cells can be manipulated as desired.

3. For some plant species, the cells cannot divide in agar medium, therefore liquid medium is the only choice.

4. Osmotic pressure of liquid medium can be altered as desired.

CULTURE MEDIA

The culture media with regard to nutritional components and osmoticum are briefly described.

Nutritional components

In general, the nutritional requirements of protoplasts are ***similar to those of cultured plant cells*** (callus and suspension cultures). Mostly, MS and B5 media (Refer Chapter 43) with suitable modifications are used. Some of the special features of protoplast culture media are listed below.

1. The medium should be devoid of ammonium, and the quantities of iron and zinc should be less.

2. The concentration of calcium should be 2-4-times higher than used for cell cultures. This is needed for membrane stability.

3. High auxin/kinetin ratio is suitable to induce cell divisions while high kinetin/auxin ratio is required for regeneration.

4. Glucose is the preferred carbon source by protoplasts although a combination of sugars (glucose and sucrose) can be used.

5. The vitamins used for protoplast cultures are the same as used in standard tissue culture media.

Osmoticum and osmotic pressure

Osmoticum broadly refers to the ***reagents/chemicals that are added to increase the osmotic pressure of a liquid.***

The isolation and culture of protoplasts require osmotic protection until they develop a strong cell wall. In fact, if the freshly isolated protoplasts are directly added to the normal culture medium, they will burst. Thus, addition of an osmoticum is essential for both isolation and culture media of protoplast to prevent their rupture. The osmotica are of two types — non-ionic and ionic.

Non-ionic osmotica : The non-ionic substances most commonly used are soluble carbohydrates such as mannitol, sorbitol, glucose, fructose, galactose and sucrose. Mannitol, being metabolically inert, is most frequently used.

Ionic osmotica : Potassium chloride, calcium chloride and magnesium phosphate are the ionic substances in use to maintain osmotic pressure.

When the protoplasts are transferred to a culture medium, the use of metabolically active osmotic stabilizers (e.g., glucose, sucrose) along with metabolically inert osmotic stabilizers (mannitol) is advantageous. As the growth of protoplasts and cell wall regeneration occurs, the metabolically active compounds are utilized, and this results in the reduced osmotic pressure so that proper osmolarity is maintained.

CULTURE METHODS

The culture techniques of protoplasts are almost the same that are used for cell culture with suitable modifications. Some important aspects are briefly given.

Feeder layer technique

For culture of protoplasts at low density feeder layer technique is preferred. This method is also important for selection of specific mutant or hybrid cells on plates. The technique consists of exposing protoplast cell suspensions to X-rays (to inhibit cell division with good metabolic activity) and then plating them on agar plates.

Co-culture of protoplasts

Protoplasts of ***two different plant species*** (one slow growing and another fast growing) ***can be co-cultured.*** This type of culture is advantageous since the growing species provide the growth factors and other chemicals which helps in the generation of cell wall and cell division. The co-culture method is generally used if the two types of protoplasts are morphologically distinct.

Microdrop culture

Specially designed dishes namely cuprak dishes with outer and inner chambers are used for microdrop culture. The inner chamber carries several wells wherein the individual protoplasts in droplets of nutrient medium can be added. The outer chamber is filled with water to maintain humidity. This method allows the culture of fewer protoplasts for droplet of the medium.

REGENERATION OF PROTOPLASTS

Protoplast regeneration which may also be regarded as ***protoplast development*** occurs in two stages.

1. Formation of cell wall.
2. Development of callus/whole plant.

Formation of cell wall

The process of cell wall formation in cultured protoplasts starts within a few hours after isolation that ***may take two to several days*** under suitable conditions. As the cell wall development occurs, the protoplasts lose their characteristic spherical shape. The newly developed cell wall by protoplasts can be identified by using calcofluor white fluorescent stain.

The freshly formed cell wall is composed of loosely bound microfibrils which get organized to form a typical cell wall. This process of cell wall development ***requires continuous supply of nutrients***, particularly a readily metabolised carbon source (e.g. sucrose). Cell wall development is found to be improper in the presence of ionic osmotic stabilizers in the medium.

The protoplasts with proper cell wall development undergo normal cell division. On the other hand, protoplasts with poorly regenerated cell wall show budding and fail to undergo normal mitosis.

Development of callus/whole plant

As the cell wall formation around protoplasts is complete, the cells increase in size, and the first division generally occurs within 2-7 days. Subsequent divisions result in small colonies, and by the end of third week, visible colonies (macroscopic colonies) are formed. These colonies are then transferred to an osmotic-free (mannitol or sorbitol-free) medium for further development to form callus. With induction and appropriate manipulations, the callus can undergo organogenic or embryogenic differentiation to finally form the whole plant. A general view of the protoplast isolation, culture and regeneration is represented in ***Fig. 44.2*** (See p. 525).

Plant regeneration can be done from the callus obtained either from protoplasts or from the culture of plant organs. There are however, certain differences in these two calluses. The callus derived from plant organs carries preformed buds or organized structures, while the callus from protoplast culture does not have such structures.

The first success of regeneration of plants from protoplast cultures of *Nicotiana tabacum* was achieved by Takebe *et al* (in 1971). Since then, several species of plants have been regenerated by using protoplasts (***Table 44.2***).

Table 44.2 Selected examples of plant species regenerated from protoplasts

Category	*Plant species*
Cereals	*Oryza sativa*
	Zea mays
	Hordeum vulgare
Vegetables	*Cucumis sativus*
	Brassica oleracea
	Capsicum annuum
Woody trees	*Larix eurolepsis*
	Coffea canephora
	Prunus avium
Ornamentals	*Rosa* sp
	Chrysanthemum sp
	Pelargonium sp
Tubers and roots	*Beta vulgaris*
	Ipomoca batatas
Oil crops	*Helianthus annuces*
	Brassica napus
Legumes	*Glycine max*

SUB-PROTOPLASTS

The ***fragments derived from protoplasts*** that do not contain all the contents of plant cells are referred to as sub-protoplasts. It is possible to experimentally induce fragmentation of protoplasts to form sub-protoplasts. This can be done by application of different centrifugal forces created by discontinuous gradients during centrifugation. Exposure of protoplasts to cytochalasin B in association with centrifugation is a better approach for fragmentation of protoplasts. There are three types of sub-protoplasts (***Fig. 44.3***).

1. **Miniprotoplasts :** These are also called as ***karyoplasts*** and contain the nucleus. Miniprotoplasts can divide and are capable of regeneration into plants.

2. **Cytoplasts :** These are sub-protoplasts containing the original cytoplasmic material (in part or full) but lack nucleus. Thus, cytoplasts are ***nuclear-free sub-protoplasts*** which cannot divide, but they can be used for ***cybridization***.

3. **Microprotoplasts :** This term was suggested for sub-protoplasts that contain not all but a few chromosomes.

SOMATIC HYBRIDIZATION

The conventional method to improve the characteristics of cultivated plants, for years, has been ***sexual hybridization***. The major limitation of sexual hybridization is that it can be performed

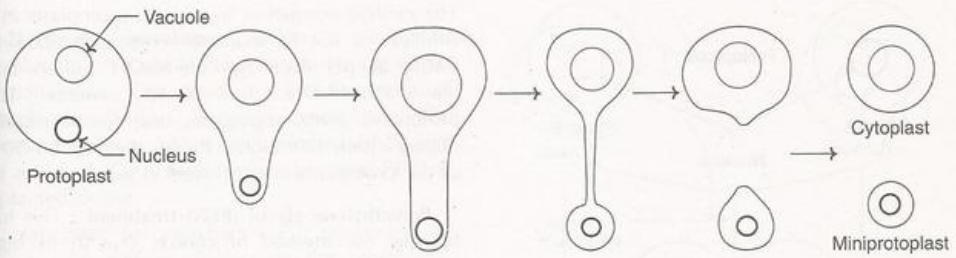


Fig. 44.3 : A diagrammatic representation of fragmentation of protoplast to form sub–protoplasts (cytoplasts and miniprotoplasts)

within a plant species or very closely related species. This restricts the improvements that can be done in plants.

The species barriers for plant improvement encountered in sexual hybridization can be overcome by somatic cell fusion that can form a viable hybrids. Somatic hybridization broadly involves ***in vitro fusion of isolated protoplasts to form a hybrid cell and its subsequent development to form a hybrid plant***. Plant protoplasts are of immense utility in somatic plant cell genetic manipulations and imporovement of crops. Thus, protoplasts provide a novel opportunity to create cells with new genetic constitution. And protoplast fusion is a wonderful approach to overcome sexual incompatibility between different species of plants. More details on the applications of somatic hybridization are given later.

Somatic hubridization involves the following aspects :

1. Fusion of protoplasts
2. Selection of hybrid cells
3. Identification of hybrid plants.

FUSION OF PROTOPLASTS

As the isolated protoplasts are devoid of cell walls, their *in vitro* fusion becomes relatively easy. There are no barriers of incompatibility (at interspecific, intergeneric or even at interkingdom levels) for the protoplast fusion.

Protoplast fusion that involves mixing of protoplasts of two different genomes can be achieved by spontaneous, mechanical, or induced fusion methods.

Spontaneous fusion

Cell fusion is a natural process as is observed in case of egg fertilization. During the course of enzymatic degradation of cell walls, some of the adjoining protoplasts may fuse to form ***homokaryocytes*** (homokaryons). These fused cells may sometimes contain high number of nuclei (2-40). This is mainly bacause of expansion and subsequent coalescence of plasmodermal connections between cells. The frequency of homokaryon formation was found to be high in protoplasts isolated from dividing cultured cells.

Spontaneously fused protoplasts, however, cannot regenerate into whole plants, except undergoing a few cell divisions.

Mechanical fusion

The protoplasts can be pushed together mechanically to fuse. Protoplasts of *Lilium* and *Trillium* in enzyme solutions can be fused by gentle trapping in a depression slide. Mechanical fusion may damage protoplasts by causing injuries.

Induced fusion

Freshly isolated protoplasts can be fused by induction. There are several ***fusion-inducing agents*** which are collectively referred to as ***fusogens*** e.g. $NaNO_3$, high pH/Ca^{2+}, polyethylene glycol, polyvinyl alcohol, lysozyme, concavalin A, dextran, dextran sulfate, fatty acids and esters, electrofusion. Some of the fusogens and their use in induced fusion are described. A diagrammatic representation of protoplast fusion is depicted in ***Fig. 44.4***.

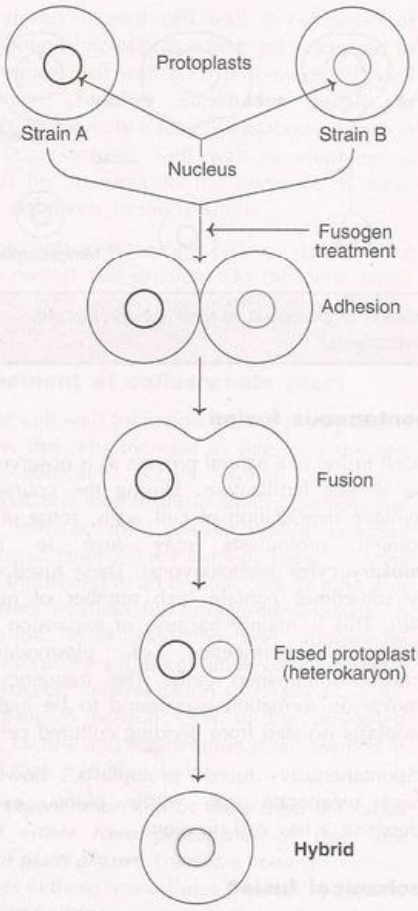


Fig. 44.4 : A diagrammatic representation of protoplast fusion.

Treatment with sodium nitrate : The isolated protoplasts are exposed to a mixture of 5.5% $NaNO_3$ in 10% sucrose solution. Incubation is carried out for 5 minutes at 35°C, followed by centrifugation (200 xg for 5 min). The protoplast pellet is kept in a water bath at 30°C for about 30 minutes, during which period protoplast fusion occurs. $NaNO_3$ treatment results in a low frequency of heterokaryon formation, particularly when mesophyll protoplasts are fused.

High pH and high Ca^{2+} ion treatment : This method was first used for the fusion of tobacco protoplasts, and is now in use for other plants also. The method consists of incubating protoplasts in a solution of 0.4 M mannitol containing 0.05 M $CaCl_2$ at pH 10.5 (glycine-NaOH buffer) and temperature 37°C for 30–40 minutes. The protoplasts form aggregates, and fusion usually occurs within 10 minutes. By this method, 20–50% of the protoplasts are involved in fusion.

Polyethylene glycol (PEG) treatment : This has become the ***method of choice***, due to its high success rate, for the fusion of protoplasts from many plant species. The isolated protoplasts in culture medium (1 ml) are mixed with equal volume (1 ml) of 28–56% PEG (mol. wt. 1500–6000 daltons) in a tube. PEG enhances fusion of protoplasts in several species. This tube is shaken and then allowed to settle. The settled protoplasts are washed several times with culture medium.

PEG treatment method is widely used for protoplast fusion as it has several ***advantages***

- It results in a reproducible high-frequency of heterokaryon formation.
- Low toxicity to cells.
- Reduced formation of binucleate heterokaryons.
- PEG-induced fusion is non-specific and therefore can be used for a wide range of plants.

Electrofusion : In this method, electrical field is used for protoplast fusion. When the protoplasts are placed in a culture vessel fitted with micro-electrodes and an electrical shock is applied, protoplasts are induced to fuse. Electrofusion technique is simple, quick and efficient and hence preferred by many workers. Further, the cells formed due to electrofusion do not show cytotoxic responses as is the case with the use of fusogens (including PEG). The major ***limitation*** of this method is the ***requirement of specialized and costly equipment***.

Mechanism of fusion

The fusion of protoplasts involves three phases agglutination, plasma membrane fusion and formation of heterokaryons.

1. **Agglutination (adhesion) :** When two protoplasts are in close contact with each other, adhesion occurs. Agglutination can be induced by fusogens e.g. PEG, high pH and high Ca^{2+}.

2. **Plasma membrane fusion :** Protoplast membranes get fused at localized sites at the points of adhension. This leads to the formation of cytoplasmic bridges between protoplasts. The plasma membrane fusion can be ***increased by high pH and high Ca^{2+}, high temperature and PEG***, as explained below.

(a) High pH and high Ca^{2+} ions neutralise the surface charges on the protoplasts. This allows closer contact and membrane fusion between agglutinated protoplasts.

(b) High temperature helps in the intermingling of lipid molecules of agglutinated protoplast membranes so that membrane fusion occurs.

(c) PEG causes rapid agglutination and formation of clumps of protoplasts. This results in the formation of tight adhesions of membranes and consequently their fusion.

3. **Formation of heterokaryons :** The fused protoplasts get rounded as a result of cytoplasmic bridges leading to the formation of spherical homokaryon or heterokaryon.

SELECTION OF HYBRID CELLS

About 20–25% of the protoplasts are actually involved in the fusion. After the fusion process, the protoplast population consists of a heterogenous mixture of unfused chloroplasts, ***homokaryons*** and ***heterokaryons*** (***Fig. 44.5***). It is therefore necessary to select the hybrid cells (heterokaryons). The commonly used methods employed for the selection of hybrid cells are ***biochemical***, ***visual*** and ***cytometric methods***.

Biochemical methods

The biochemical methods for selection of hybrid cells are based on the use of biochemical compounds in the medium (selection medium). These compounds help to sort out the hybrid and parental cells based on their differences in the expression of characters. Drug sensitivity and auxotrophic mutant selection methods are described below.

1. **Drug sensitivity :** This method is useful for the selection hybrids of two plant species, if one of them is sensitive to a drug. Protoplasts of *Petunia hybride* (species A) can form macroscopic callus

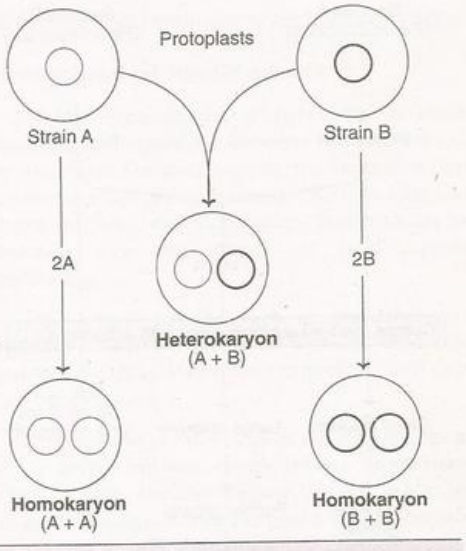


Fig. 44.5 : Fusion products of protoplasts.

on MS medium, but are sensitive to (inhibited by) actinomycin D. *Petunia parodii* protoplasts (species B) form small colonies, but are resistant to actinomycin D. When these two species are fused, the fused protoplasts derive both the characters — formation of macroscopic colonies and resistance to actinomycin D on MS medium. This helps in the selection of hybrids (***Fig. 44.6***). The parental protoplasts of both the species fail to grow. Protoplasts of *P. parodii* form very small colonies while that of *P. hybrida* are inhibited by actinomycin D.

Drug sensitivity technique was originally developed by Power *et al* (1976) for the selection of hybrids of *Petunia* sp (described above). A similar procedure is in use for the selection of other somatic hybrids e.g., hybrids between *Nicotiana sylvestris* and *Nicotiana knightiana*.

2. **Auxotrophic mutants :** Auxotrophs are mutants that cannot grow on a minimal medium and therefore require specific compounds to be added to the medium. Nitrate reductase deficient mutants of tobacco (*N. tabacum*) are known. The parental protoplasts of such species cannot grow with nitrate as the sole source of nitrogen while the hybrids can grow. Two species of nitrate reductase deficiency— one due to lack of apoenzyme (nia-type mutant) and the other due to lack of molybdenum cofactor (cnx-

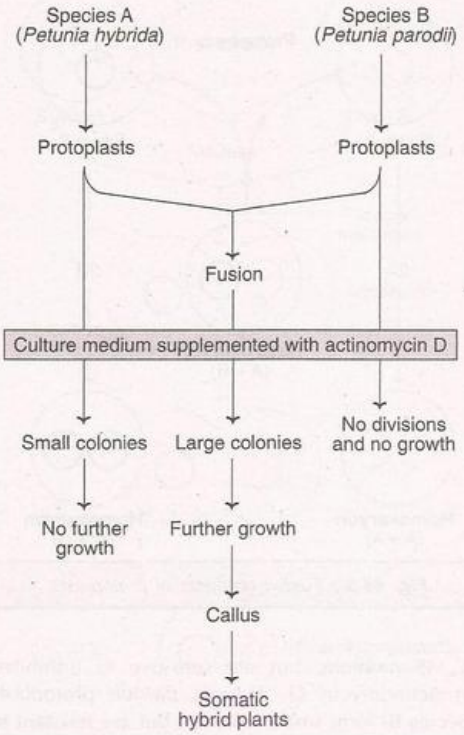


Fig. 44.6 : A diagrammatic representation of drug sensitivity method for the isolation of hybrid cells.

type mutant) are known. The parental protoplasts cannot grow on nitrate medium while the hybrid protoplasts can grow (***Fig. 44.7***).

The selection of auxotrophic mutants is possible only if the hybrid cells can grow on a minimal medium. Another limitation of the technique is the paucity of higher plant auxotrophs.

Visual methods

Visual selection of hybrid cells, although tedious is very efficient. In some of the somatic hybridization experiments, chloroplast deficient (albino or non-green) protoplasts of one parent are fused with green protoplasts of another parent. This facilitates the visual identification of haterokaryons under light microscope. The ***heterokaryons are bigger and green in colour while the parental protoplasts are either small or colourless***. Further identification of these heterokaryons has to be carried out to develop the specific hybrid plant.

There are two approaches in this direction — growth on selection medium, and mechanical isolation.

1. **Visual selection coupled with differential media growth :** There exist certain natural differences in the sensitivity of protoplasts to the nutrients of a given medium. Thus, some media can selectively support the development of hybrids but not the parental protoplasts.

A diagrammatic representation of visual selection coupled with the growth of heterokaryons on a selection medium is given in ***Fig. 44.8***.

2. **Mechanical isolation :** The visually identified heterokaryons under the microscope can be isolated by mechanical means. This involves the use of a special pipette namely Drummond pipette. The so isolated heterokaryons can be cloned to finally produce somatic hybrid plants. The major limitation of this method is that each type of hybrid

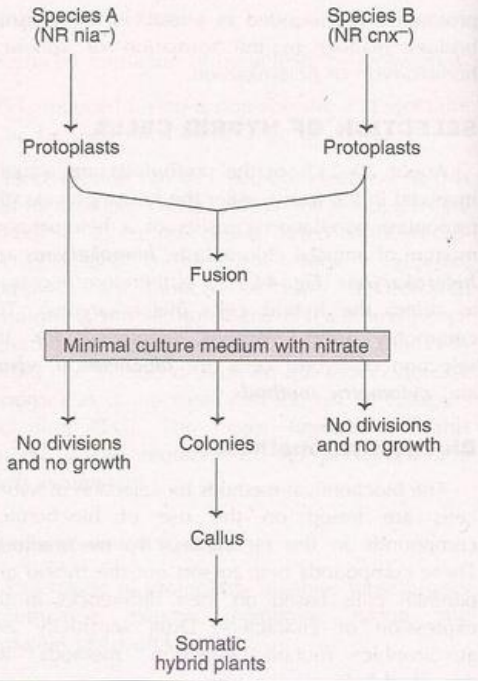


Fig. 44.7 : A diagrammatic representation of hybrid selection based on auxotrophic mutant (NR nia⁻— Nitrate reductase apoenzyme deficient; NR cnx⁻— Nitrate reductase lacking molybdenum cofactor).

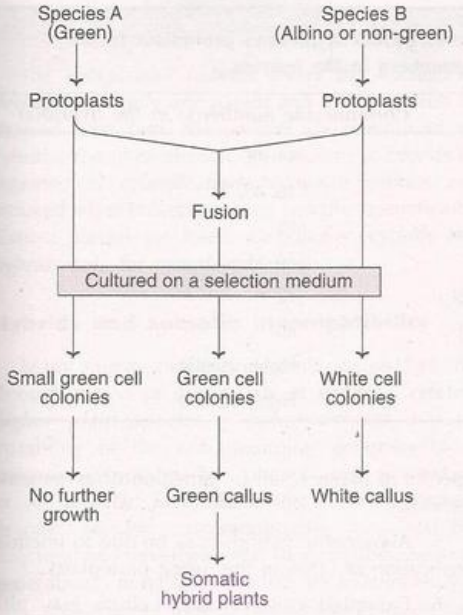


Fig. 44.8 : A diagrammatic representation of visual selection of hybrids coupled with growth on selection medium.

cell requires a special culture medium for its growth. This can be overcome by employing microdrop culture of single cells using feeder layers (Refer p. 506).

Cytometric methods

Some workers use flow cytometry and fluorescent-activated cell sorting techniques for the analysis of plant protoplasts while their viability is maintained. The same techniques can also be applied for sorting and selection of heterokaryons. The hybrid cells derived from such selections have proved useful for the development of certain somatic hybrid plants.

IDENTIFICATION OF HYBRID (CELLS) PLANTS

The development of hybrid cells followed by the generation of hybrid plants requires a clear proof of genetic countribution from both the parental protoplasts. The hybridity must be established only from euploid and not from aneuploid hybrids. Some of the commonly used approaches for the identification of hybrid plants are briefly described.

Morphology of hybrid plants

Morphological features of hybrid plants which ***usually are intermediate between two parents*** can be identified. For this purpose, the vegetative and floral characters are considered. These include leaf shape, leaf area, root morphology, flower shape, its structure, size and colour, and seed capsule morphology.

The somatic hybrids such as ***pomatoes*** and ***topatoes*** which are the fused products of potato and tomato show abnormal morphology, and thus can be identified.

Although the genetic basis of the morphological characters has not been clearly known, intermediate morphological features suggest that the traits are under the control of multiple genes. It is preferrable to support hybrid morphological characters with evidence of genetic data.

Isoenzyme analysis of hybrid plants

The multiple forms of an enzyme catalysing the same reaction are referred to as isoenzymes. Electrophoretic patterns of isoenzymes have been widely used to verify hybridity. ***Somatic hybrids posses specific isoenzymes*** (of certain enzymes) of one or the other parent or both the parents simultaneously.

There are many enzymes possessing unique isoenzymes that can be used for the identification of somatic hybrids e.g. amylase, esterase, aspartate aminotransferase, phosphodiesterase, isoperoxidase, and hydrogenases (of alcohol, lactate, malate). If the enzyme is dimeric (having two subunits), somatic hybrids usually contain an isoenzyme with an intermediate mobility properties.

The isoenzymes are often variable within the same plant. Therefore, it is necessary to use the same enzyme from each plant (parents and somatic hybrids), from a specific tissue with the same age.

Chromosomal constitution

The number of chromosomes present in the hybrid cells can be directly counted. This provides information on the ploidy state of the cells. The somatic hybrids are expected to possess chromosomes that are equal to the total number of

Table 44.3 A selected list of interspecific hybrids produced through protoplast fusion along with the chromosome numbers in the hybrids

Plant species with their chromosome number	*Chromosome number(s) in the hybrid(s)*
Petunia parodii (2n = 48) + *P. hybrida* (2n = 14)	44–48
Datura innoxia (2n = 24) + *D. stramonium* (2n = 24)	46, 48, 72
Nicotiana tabacum (2n = 48) + *N. nesophila* (2n = 48)	96
Nicotiana tabacum (2n = 48) + *N. glutinosa* (2n = 24)	50–58
Lycopersicon esculentum (2n = 24) + *L. peruvianum* (2n = 24)	72
Solanum tuberosum (2n = 24, 48) + *S. chacoense* (2n = 14)	60
Brassica oleracea (2n = 18) + *B. campestris* (2n = 18)	Highly variable
Brassica napus (2n = 38) + *B. juncea* (2n = 36)	Highly variable

chromosomes originally present in the parental protoplasts. Sometimes, the hybrids are found to contain more chromosomes than the total of both the parents. The presence of chromosomal markers is greatly useful for the genetic analysis of hybrid cells.

Molecular techniques

Many recent developments in molecular biology have improved the understanding of genetic constitution of somatic plant hybrids. Some of them are listed below.

1. Differences in the restriction patterns of chloroplast and mitochondrial DNAs.

2. Molecular markers such as RFLP, AFLP, RAPD and microsatellites.

3. PCR technology.

CHROMOSOME NUMBER IN SOMATIC HYRBIDS

The chromosome number in the somatic hybrids is ***generally more than the total number of both of the parental protoplasts.*** However, wide variations are reported which may be due to the following reasons

1. Fusion of more than two protoplasts.

2. Irregularities in mitotic cell divisions.

3. In fusogen or electro-induced fusions, about one third of the fusions occur between more than two protoplasts.

4. Differences in the status of protoplasts (actively dividing or quiescent) from the two species of plants result in formation of asymmetric hybrids.

5. Asymmetric hybrids may be due to unequal replication of DNA in the fusing protoplasts.

6. Protoplast isolation and culture may also lead to somaclonal variations, and thus variations in chromosome number.

A selected list of interspecific hybrids produced through protoplast fusion along with the number of chromosomes in the hybrids is given in ***Table 44.3***.

Symmetric and asymmetric hybrids

If the ***chromosome number in the hybrid is the sum of the chromosomes*** of the two parental protoplasts, the hybrid is said to be ***symmetric***. Symmetric hybrids between incompatible species are usually sterile. This may be due to production of 3n hybrids by fusing 2n of one species with n of another species.

Asymmetric hybrids have abnormal or wide variations in the chromosome number than the exact total of two species. These hybrids are usually formated with full somatic complement of one parental species while all or nearly all of the chromosomes of other parental species are lost during mitotic divisions. Asymmetric hybrids may be regarded as cybrids but for the introgressed genes.

As given in ***Table 44.3***, protoplast fusion between *N. tabacum* (2n = 48) and *N. nesophila* (2n = 24) results in a symmetric hybrids, while asymmetric hybrids are formed when *B. napus* and *B. junea* are fused.

CYBRIDS

The ***cytoplasmic hybrids where the nucleus is derived from only one parent and the cytoplasm is derived from both the parents*** are referred to as cybrids. The phenomenon of formation of cybrids is regarded as ***cybridization***. Normally, cybrids are prduced when protoplasts from two phytogenetically distinct species are fused. Genetically, ***cybrids are hybrids only for cytoplasmic traits.***

Hybrids and somatic incompatibility

Many a times, production of full-pledged hybrids through fusion of ***protoplasts of distantly related higher plant species*** is rather difficult due to instability of the two dissimilar genomes in a common cytoplasm. This phenomenon is referred to as ***somatic incompatibility***. Hybrids formed despite somatic incompatibility may exhibit structural and developmental abnormalities. Several generations may be required to eliminate the undesirable genes. Due to this limitation in somatic hybridization, cybridization involving protoplast fusion for partial genome transfer is gaining importance in recent years.

Methodology of cybridization

A diagrammatic representation of the formation of hybrids and cybrids is given in ***Fig. 44.9***.

As the formation of heterokaryon occurs during hybridization, the nuclei can be stimulated to segregate so that one protoplast contributes to the cytoplasm while the other contributes nucleus alone (or both nucleus and cytoplasm). In this way cybridization can be achieved. Some of the approaches of cybridization are given hereunder.

1. The protoplasts of cytoplasm donor species are ***irradiated with X-rays*** or ***γ-rays***. This treatment renders the protoplasts inactive and non-dividing, but they are efficient donors of cytoplasmic constituents when fused with recipient protoplasts.

2. Normal ***protoplasts can be*** directly ***fused with enucleated protoplasts***. Enucleated protoplasts can be isolated by high-speed centrifugation.

3. Protoplasts are ***inactivated by metabolic inhibitors*** such as iodoacetate. In practice, iodoacetate treated protoplasts are fused with X-rays irradiated protoplasts for more efficient formation of cybrids.

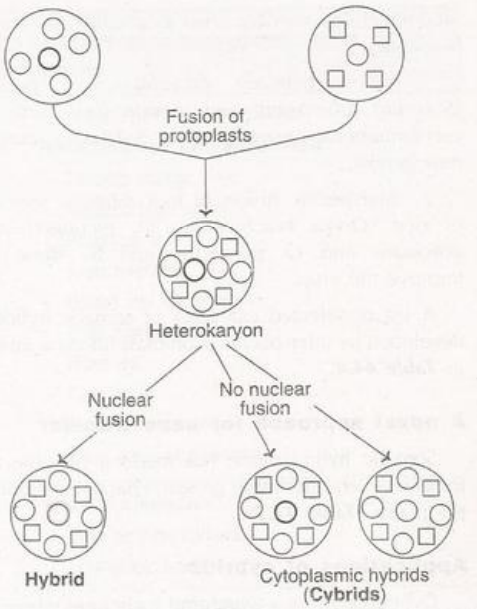


*Fig. 44.9 : A diagrammatic representation of hybridization and cybridization (**Note :** The plastids are shown as □ and ○)*

4. It is possible to suppress nuclear division in some protoplasts and fuse them with normal protoplasts.

Genetic recombination in asexual or sterile plants

There are many plants that cannot reproduce sexually. Somatic hybridization is a novel approach through which two parental genomes of a sexual or sterile plants can be brought together. Thus, by fusing parental protoplasts, fertile diploids and polyploids can be produced.

Overcoming barriers of sexual incompatibility

Sexual crossing between two different species (interspecific) and two different genus (intergeneric) is impossible by conventional breeding methods. Somatic hybridization overcomes the sexual

incompatibility barriers. Two examples are given hereunder.

1. Fusion between protoplasts of ***po***tato (*Solanum tuberosum*) and to***mato*** (*Lycopersicon esculentum*) has created ***pomato*** (*Solanopersicon*, a new genus).

2. Interspecific fusion of four different species of rice (*Oryza brachyantha, O. eichngeri, O. officinalis* and *O. perrieri*) could be done to improve the crop.

A list of selected examples of somatic hybrids developed by interspecific protoplast fusion is given in ***Table 44.4***.

Table 44.4 Selected examples of somatic hybrids developed by interspecific protoplast fusion

Common name (botanical name)	*Common name (botanical name)*
Tomato (*Lycopersicon esculentum*)	Potato (*Solanum tuberosum*)
Petunia (*Petunia parodii*)	Petunia (*Petunia parviflora*)
Thorn apple (*Datura innoxia*)	Deadly nightshade (*Atropa belladonna*)
Turnip (*Brassica campestis*)	Thale cress (*Arabidopsis thaliana*)

A novel approach for gene transfer

Somatic hybridization has made it possible to transfer several desirable genetic characters among the plants (***Table 44.5***).

Applications of cybrids

Cybridization is a wonderful technique wherein the ***desired cytoplasm can be transferred in a single step***. Cybrids are important for the transfer of cytoplasmic male sterility (CMS), antibiotic and herbicide resistance in agriculturally useful plants.

Some of the genetic triats in certain plants are cytoplasmically controlled. This includes some types of ***male sterility, resistance to certain antibiotics and herbicides***. A selected list of agronomic characters transferred through cybrids is given in ***Table 44.5*** (along with somatic hybrids).

Cybridization has been successfully used to transfer CMS in rice. Cybrids of *Brassica raphanus* that contain nucleus of *B. napus,* chloroplasts of atrazinc resistant *B. campestris* and male sterility from *Raphanus sativas* have been developed.

APPLICATIONS OF SOMATIC HYBRIDIZATION

Somatic hybridization has opened new possibilites for the *in vitro* ***genetic manipulation of plants to improve the crops***. Some of the practical application are briefly given

1. **Disease resistance :** Several interspecific and intergeneric hybrids with disease resistance have been created. Many disease resistance genes (e.g., tobacco mosaic virus, potato virus X, club rot disease) could be successfully transferred from one species to another. For example, resistance has been introduced in tomato against diseases such as TMV, spotted wilt virus and insect pests.

2. **Environmental tolerance :** The genes responsible for the tolerance of cold, frost and salt could be successfully introduced through somatic hybridization. e.g., introduction of cold tolerance gene in tomato.

3. **Quality characters :** Somatic hybrids for the production of high nicotine content, and low erucic acid have been developed (***Table 44.5***).

4. **Cytoplasmic male sterility :** A modification of hybridization in the form of cybridization has made it possible to transfer cytoplasmic male sterility. Details given in the preceeding section.

Other application of somatic hybridization

1. Somatic hybridization has helped to study the cytoplasmic genes and their functions. In fact, the information is successfully used in plant breeding programmes.

2. Protoplast fusion will help in the combination of mitochondria and chloroplasts to result in a unique nuclear-cytoplasmic genetic combination.

3. Somatic hybridization can be done in plants that are still in juvenile phase.

4. Protoplast transformation (with traits like nitrogen fixation by incorporating exogenous DNA) followed by somatic hybridization will yield innovative plants.

TABLE 44.5 A selected list of genetic traits transferred through protoplast fusion in crop plant species

Somatic hybrids	*Genetic trait in new plant/ resistance (or tolerance) trait*
Nicotiana tobaccum + N. nesophila	Tobacco mosaic virus
	Tobacco horn worm
Solanum tuberosum + S. chacoense	Potato virus X
Solanum tuberosum + S. commersounii	Frost tolerance
Solanum tuberosum + S. brevidens	Potato leaf roll virus
Solanum ochranthum + Lycopersicon esculentum	Cold tolerance
Brassica oleracea + B. napus	Black rot
Citrullus lanatus + Cucumis melo	Club rot
Raphanus sativus + Brassica napus	Beet cyst nematode
Hordeum vulgare + Daucus carota	Frost and salt tolerance
Somatic hybrids	***Quality character***
Nicotiana tabacum + N. rustica	High nicotine content
Brassica napus + Eruca sativa	Low erucic acid
Cytoplasmic hybrids (cybrids)	***Agronomic character***
Nicotiana tabacum + N. sylvestris	Streptomycin resistance
Nicotiana nigrum + Solanum tuberosum	Triazine resistance
Brassica nigra + B. napus	Hygromycin resistance
Solanum nigrum + S. tuberosum	Triazine resistance
Nicotiana tabacum + N. sylvestris	Cytoplasmic male sterility
Brassica napus + B. tournefortii	Cytoplasmic male sterility
Brassica campestris + B. napus	Cytoplasmic male sterility
Lycopersicon esculentum + Solanum acaule	Cytoplasmic male sterility
Brasica napus + B. campestris + Raphanus sativa	Cytoplasmic male sterility and triazine resistance

LIMITATIONS OF SOMATIC HYBRIDIZATION

Although somatic hybridization is a novel approach in plant biotechnology, there are several problems and limitations. The success of the technique largely depends on overcoming these limitations, some of which are listed below.

1. Somatic hybridization does ***not always produce plants that give fertile and visible seeds.***

2. Regenerated plants obtained from somatic hybridization are often ***variable due to somaclonal variations***, chromosomal elimination, organelle segregation etc.

3. Protoplast culture is frequently associated with genetic instability.

4. Protoplast fusion between different species/ genus is easy, but the production of viable somatic hybrids is not possible in all instances.

5. Some of the somatic hybrids, particularly when produced by the fusion of taxonomically different partners, are unbalanced and not viable.

6. There are limitations in the selection methods of hybrids, as many of them are not efficient.

7. There is no certainity as regards the expression of any specific character in somatic hybridization.

8. Somatic hybridization between two diploids results in the formation of an amphidiploid which is not favourable. For this reason, haploid protoplasts are recommended in somatic hybridization.

Production of Haploid Plants

Haploid plants are characterized by ***possessing only a single set of chromosomes*** (gametophytic number of chromosomes i.e. n) in the sporophyte. This is in contrast to diploids which contain two sets (2n) of chromosomes. Haploid plants are of great significance for the production of ***homozygous lines*** (homozygous plants) and for the improvement of plants in plant breeding programmes.

Brief history

The existence of haploids was discovered (as early as 1921) by Bergner in *Datura stramonium*. Plant breeders have been conducting extensive research to develop haploids. The Indian scientists ***Guha and Maheswari (1964) reported the direct development of haploid embryos and plantlets from microspores of Datura innoxia*** by the cultures of excised anthers. Subsequently, Bourgin and Hitsch (1967) obtained the first full-pledged haploid plants from *Nicotiana tabacum*. Thereafter, much progress has been made in the anther cultures of wheat, rice, maize, pepper and a wide range of economically important species.

Grouping of haploids

Haploids may be divided into two broad categories.

1. **Monoploids (monohaploids) :** These are the haploids that possess half the number of chromosomes from a diploid species e.g. maize, barley.

2. **Polyhaploids :** The haploids possessing half the number of chromosomes from a polyploid species are regarded as polyhaploids e.g. wheat, potato.

It may be noted that when the term haploid is generally used it applies to any plant originating from a sporophyte (2n) and containing half the number (n) of chromosomes.

IN VIVO AND *IN VITRO* APPROACHES

The importance of haploids in the field of plant breeding and genetics was realised long ago. Their practical application, however, has been restricted due to very a low frequency (< 0.001%) of their formation in nature. The process of apomixis or ***parthenogenesis*** (***development of embryo from an unfertilized egg***) is responsible for the spontaneous ***natural production of haploids***. Many attempts were made, both by *in vivo* and *in vitro* methods to develop haploids. The success was much higher by *in vitro* techniques.

In vivo techniques for haploid production

There are several methods to induce haploid production *in vivo*. Some of them are listed below.

1. **Androgenesis :** Development of an egg cell containing ***male nucleus*** to a haploid is referred to as androgenesis. For a successful *in vivo* androgenesis, the egg nucleus has to be inactivated or eliminated before fertilization.

2. **Gynogenesis :** An ***unfertilized egg*** can be manipulated (by delayed pollination) to develop into a haploid plant.

3. **Distant hybridization :** Hybrids can be produced by elimination of one of the parental genomes as a result of distant (interspecific or intergeneric crosses) hybridization.

4. **Irradiation effects :** Ultra violet rays or X-rays may be used to induce chromosomal breakage and their subsequent elimination to produce haploids.

5. **Chemical treatment :** Certain chemicals (e.g., chloramphenicol, colchicine, nitrous oxide, maleic hydrazide) can induce chromosomal elimination in somatic cells which may result in haploids.

In vitro techniques for haploid production

In the plant biotechnology programmes, haploid production is achieved by two methods.

1. **Androgenesis :** Haploid production occurs through anther or pollen culture, and they are referred to as ***androgenic haploids***.

2. **Gynogenesis :** Ovary or ovule culture that results in the production of haploids, known as ***gynogenic haploids***.

ANDROGENESIS

In androgenesis, the male gametophyte (microspore or immature pollen) produces haploid plant. The basic principle is to ***stop the development of pollen cell into a gamete*** (sex cell) ***and force it to develop into a haploid plant***.

There are two approaches in androgenesis—anther culture and pollen (microspore) culture. Young plants, grown under optimal conditions of light, temperature and humidity, are suitable for androgenesis.

ANTHER CULTURE

The selected flower buds of young plants are surface-sterilized and anthers removed along with their filaments. The anthers are excised under aseptic conditions, and crushed in 1% acetocarmine to test the stage of pollen development. If they are at the correct stage, each anther is gently separated (from the filament) and the intact anthers are inoculated on a nutrient medium. Injured anthers should not be used in cultures as they result in callusing of anther wall tissue.

The anther cultures are maintained in alternating periods of light (12–18 hr) and darkness (6–12 hrs) at 28°C. As the anthers proliferate, they produce callus which later forms an embryo and then a haploid plant (***Fig. 45.1***).

POLLEN (MICROSPORE) CULTURE

Haploid plants can be produced from immature pollen or microspores (male gametophytic cells). The pollen can be extracted by pressing and squeezing the anthers with a glass rod against the sides of a beaker. The pollen suspension is filtered to remove anther tissue debris. Viable and large pollen (smaller pollen do not regenerate) are concentrated by filtration, washed and collected. These pollen are cultured on a solid or liquid medium. The callus/embryo formed is transferred to a suitable medium to finally produce a haploid plant (***Fig. 45.1***), and then a diploid plant (on colchicine treatment).

Comparison between anther and pollen cultures

Anther culture is easy, quick and practicable. Anther walls act as conditioning factors and promote culture growth. Thus, anther cultures are reasonably efficient for haploid production. The major limitation is that the plants not only originate from pollen but also from other parts of anther. This results in the population of plants at different ploidy levels (diploids, aneuploids). The disadvantages associated with anther culture can be overcome by pollen culture.

Many workers prefer pollen culture, even though the degree of success is low, as it offers the following ***advantages***

- Undesirable effects of anther wall and associated tissues can be avoided.
- Androgenesis, starting from a single cell, can be better regulated.
- Isolated microspores (pollen) are ideal for various genetic manipulations (transformation, mutagenesis).
- The yield of haploid plants is relatively higher.

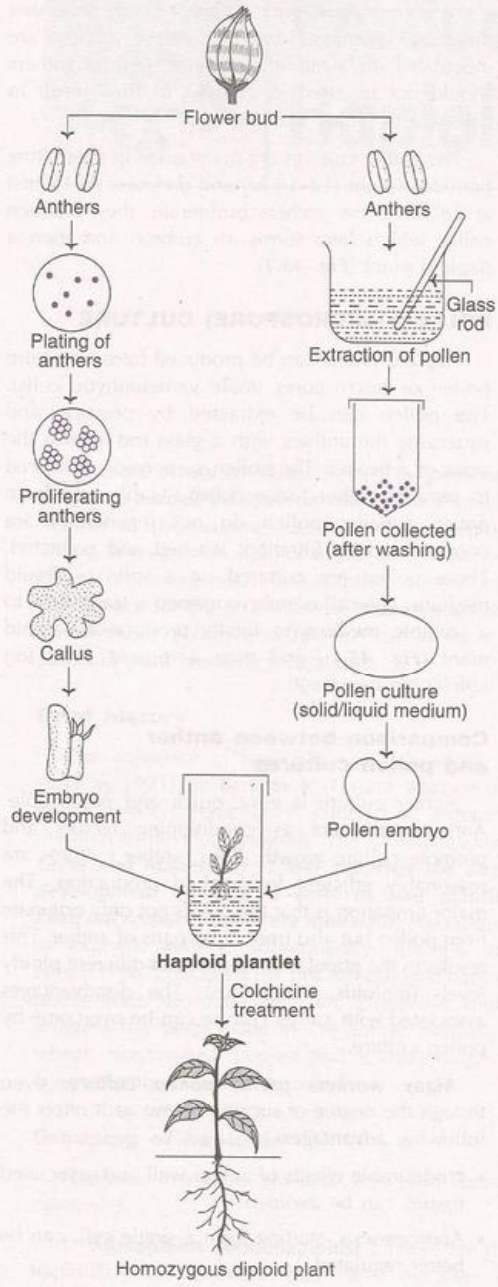


Fig. 45.1 : Diagrammatic representation of anther and pollen cultures for the production of haploid and diploid plants.

DEVELOPMENT OF ANDROGENIC HAPLOIDS

The process of *in vitro* androgenesis for the ultimate production of haploid plants is depicted in ***Fig. 45.2***.

The cultured microspores mainly follow four distinct pathways during he initial stages of *in vitro* androgenesis.

Pathway I : The uninucleate microspore undergoes equal division to form ***two daughter cells*** of equal size e.g. *Datura innoxia.*

Pathway II : In certain plants, the microspore divides unequally to give bigger vegetative cell and a smaller generative cell. It is the ***vegetative cell*** that undergoes further divisions to ***form callus*** or embryo. The generative cell, on the other hand, degenerates after one or two divisions—e.g., *Nicotiana tabacum, Capsicum annuum.*

Pathway III : In this case, the microspore undergoes unequal division. The ***embryos are formed from the generative cell*** while the vegetative cell does not divide at all or undergoes limited number of divisions e.g. *Hyoscyamus niger.*

Pathway IV : The microspore divides unequally as in pathways I and II. However, in this case, both ***vegetative and generative cells*** can further divide and ***contribute to the development of haploid*** plant e.g. *Datura metel, Atropa belladonna.*

At the initial stages, the microspore may follow any one of the four pathways described above. As the cells divide, the pollen grain becomes multicellular and burst open. This multicelluar mass may form a callus which later differentiates into a plant (through ***callus phase***). Alternately, the multicellular mass may produce the plant through ***direct embryogenesis*** (***Fig. 45.1***).

FACTORS AFFECTING ANDROGENESIS

A good knowledge of the various factors that influence androgenesis will help to improve the production of androgenic haploids. Some of these factors are briefly described.

Genotype of donar plants

The success of anther or pollen culture largely depends on the genotype of the donor plant. It is therefore important to ***select*** only ***highly responsive genotypes***. Some workers choose a breeding approach for improvement of genotype before they are used in androgenesis.

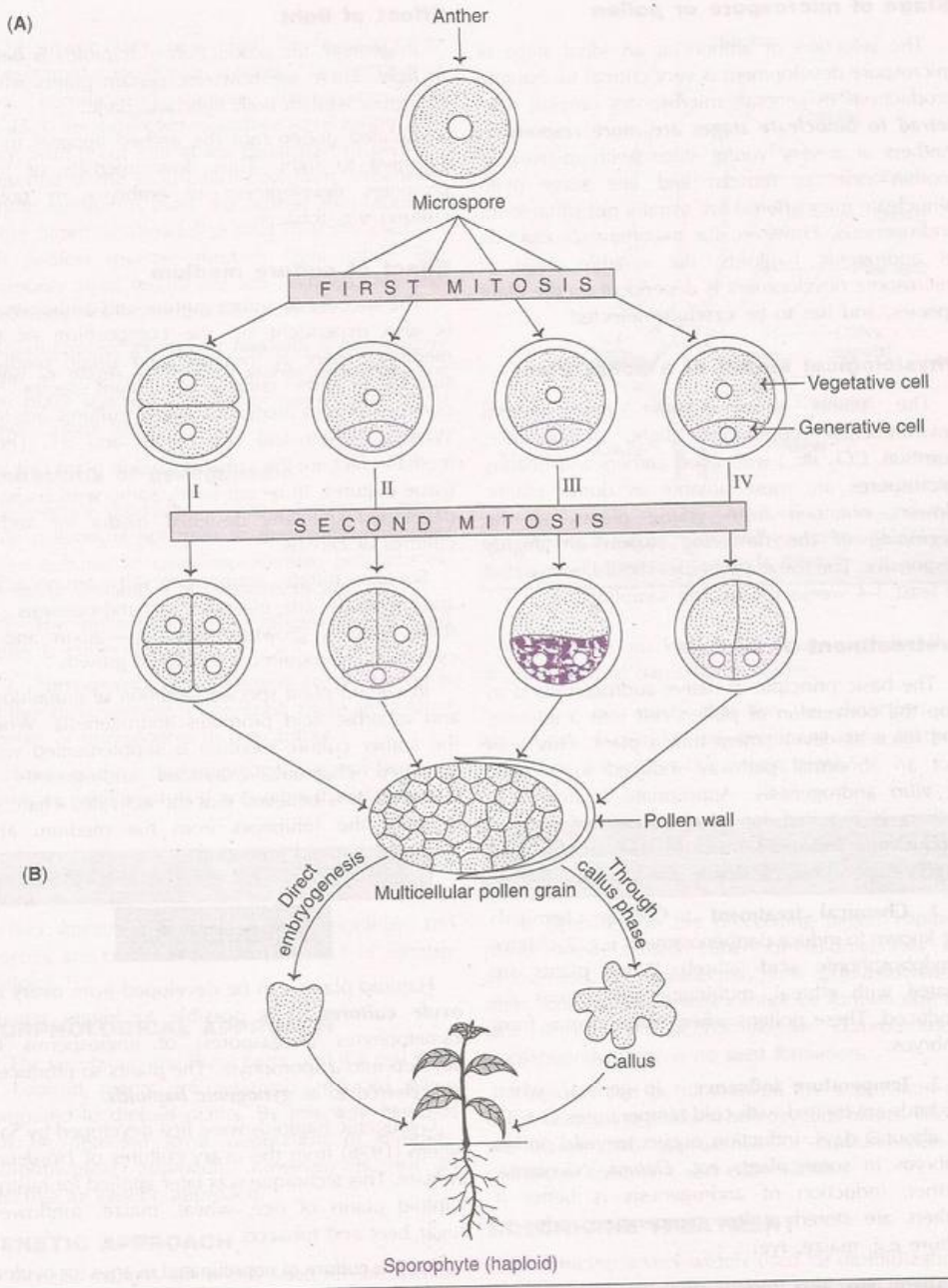


Fig. 45.2 : *Diagrammatic representation of microscope divisions leading to the formation of a multicelluar pollen grain (**A**), followed by the formation of haploid sporophyte (**B**) (**Note :** I, II, III and IV indicate respective pathways).*

Stage of microspore or pollen

The selection of anthers at an ideal stage of microspore development is very critical for haploid production. In general, microspores ranging from ***tetrad to binucleate stages are more responsive***. Anthers at a very young stage (with microspore mother cells or tetrads) and late stage (with binucleate microspores) are usually not suitable for androgenesis. However, for maximum production of androgenic haploids, the suitable stage of microspore development is dependent on the plant species, and has to be carefully selected.

Physiological status of a donar plant

The plants grown under best natural environmental conditions (light, temperature, nutrition, CO_2 etc.) with good anthers and healthy microspores are most suitable as donor plants. Flowers obtained from young plants, at the ***beginning of the flowering season are highly responsive***. The use of pesticides should be avoided at least 3-4 weeks preceeding sampling.

Pretreatment of anthers

The basic principle of native androgenesis is to stop the conversion of pollen cell into a gamete, and force its development into a plant. This is in fact an abnormal pathway induced to achieve *in vitro* androgenesis. Appropriate treatment of anthers is required for good success of haploid production. Treatment methods are variable and largely depend on the donor plant species.

1. **Chemical treatment :** Certain chemicals are known to induce parthenogenesis e.g. 2-chloro-ethylphosphonic acid (ethrel). When plants are treated with ethreal, multinucleated pollens are produced. These pollens when cultured may form embryos.

2. **Temperature influence :** In general, when the buds are treated with cold temperatures (3-6°C) for about 3 days, induction occurs to yield pollen embryos in some plants e.g. *Datura, Nicotiana*. Further, induction of androgenesis is better if anthers are stored at low temperature, prior to culture e.g. maize, rye.

There are also reports that pretreatment of anthers of certain plants at higher temperatures (35°C) stimulates androgenesis e.g. some species of *Brassica* and *Capsicum*.

Effect of light

In general, the production of haploids ***is better in light***. There are however, certain plants which can grow well in both light and dark.

Isolated pollen (not the anther) appears to be sensitive to light. Thus, low intensity of light promotes development of embryos in pollen cultures e.g. tobacco.

Effect of culture medium

The success of anther culture and androgenesis is also dependent on the composition of the medium. There is, however, no single medium suitable for anther cultures of all plant species. The commonly used media for anther cultures are MS, White's, Nitsch and Nitsch, N6 and B5. These media in fact are the same as used in plant cell and tissue cultures. In recent years, some workers have developed specially designed media for anther cultures of cereals.

Sucrose, nitrate, ammonium salts, amino acids and minerals are essential for androgenesis. In some species, growth regulators — auxin and/or cytokinin are required for optimal growth.

In certain plant species, addition of glutathione and ascorbic acid promotes androgenesis. When the anther culture medium is supplemented with activated charcoal, enhanced androgenesis is observed. It is believed that the activated charcoal removes the inhibitors from the medium and facilitates haploid formation.

GYNOGENESIS

Haploid plants can be developed from ***ovary*** or ***ovule cultures***. It is possible to trigger female gametophytes (megaspores) of angiosperms to develop into a sporophyte. The plants so produced are referred to as ***gynogenic haploids***.

Gynogenic haploids were first developed by San Noem (1976) from the ovary cultures of *Hordeum vulgare*. This technique was later applied for raising haploid plants of rice, wheat, maize, sunflower, sugar beet and tobacco.

In vitro culture of unpollinated ovaries (or ovules) is usually employed when the anther cultures give unsatisfactory results for the production of haploid plants. The procedure for gynogenic haploid production is briefly described.

The flower buds are excised 24–48 hr prior to anthesis from unpollinated ovaries. After removal of calyx, corolla and stamens, the ovaries (see ***Fig. 45.3)*** are subjected to surface sterilization. The ovary, with a cut end at the distal part of pedicel, is inserted in the solid culture medium. Whenever a liquid medium is used, the ovarier are placed on a filter paper or allowed to float over the medium with pedicel inserted through filter paper. The commonly used media are MS, White's, N6 and Nitsch, supplemented growth factors.

Production of ***gynogenic haploids is particularly useful in plants with male sterile genotype***. For such plant species, this technique is superior to anther culture technique.

Limitations of gynogenesis

In practice, production of haploid plants by ovary/ovule cultures is not used as frequently as anther/pollen cultures in crop improvement programmes. The major limitations of gynogenesis are listed.

1. The dissection of unfertilized ovaries and ovules is rather difficult.

2. The presence of only one ovary per flower is another disadvantage. In contrast, there are a large number of microspores in one anther.

However, the future of gynogenesis may be more promising with improved and refined methods.

IDENTIFICATION OF HAPLOIDS

Two approaches based on morphology and genetics are commonly used to detect or identify haploids.

MORPHOLOGICAL APPROACH

The vegetative and floral parts, and the cell sizes of haploid plants are relatively reduced when compared to diploid plants. By this way haploids can be detected in a population of diploids. Morphological approach, however, is not as effective as genetic approach.

GENETIC APPROACH

Genetic markers are widely used for the specific identification of haploids. Several markers are in use.

- 'a_1' marker for brown coloured aleurone.
- 'A' marker for purple colour.
- 'Lg' marker for liguleless character.

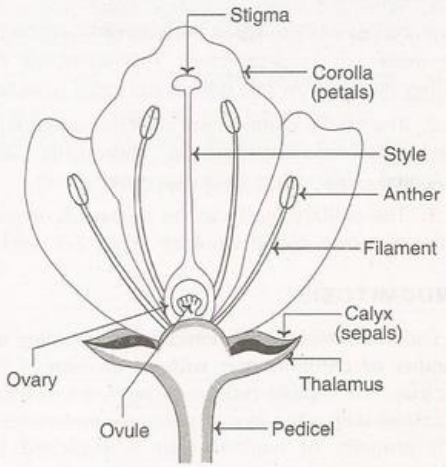


Fig. 45.3 : A diagrammatic representation of important parts in a flower.

The above markers have been used for the development of haploids of maize.

It may be noted that for the detection of androgenic haploids, the dominant gene marker should be present in the female plant.

DIPLOIDIZATION OF HAPLOID PLANTS (PRODUCTION OF HOMOZYGOUS PLANTS)

As described in the preceeding pages, haploid plants are obtained either by androgenesis or gynogenesis. These plants may grow up to a flowering stage, but viable gametes cannot be formed due to lack of one set of homologous chromosomes. Consequently, there is no seed formation.

Haploids can be diploidized (by duplication of chromosomes) to produce homozygous plants. There are mainly two approaches for diploidization—colchicine treatment and endomitosis.

COLCHICINE TREATMENT

Colchicine is very widely used for diploidization of homologous chromosomes. It acts as an inhibitor of spindle formation during mitosis and ***induces chromosome duplication***. There are many ways of colchicine treatment to achieve diploidization for production of homozygous plants.

1. When the plants are mature, colchicine in the form of a paste is applied to the axils of leaves. Now, the main axis is decapitated. This stimulates the axillary buds to grow into diploid and fertile branches.

2. The young plantlets are directly treated with colchicine solution, washed thoroughly and replanted. This results in homozygous plants.

3. The axillary buds can be repeatedly treated with colchicine cotton wool for about 2-3 weeks.

ENDOMITOSIS

Endomitosis is the phenomenon of ***doubling the number of chromosomes without division*** of the nucleus. The haploid cells, in general, are unstable in culture with a tendency to undergo endomitosis. This property of haploid cells is exploited for diploidization to produce homozygous plants.

The procedure involves growing a small segment of haploid plant stem in a suitable medium supplemented with growth regulators (auxin and cytokinin). This induces callus formation followed by differentiation. During the growth of callus, chromosomal doubling occurs by endomitosis. This results in the production of diploid homozygous cells and ultimately plants.

APPLICATIONS OF HAPLOIDS

In vitro production of haploids is of great ***significance in plant breeding programmes***. Some of them are listed.

Development of homozygous lines

It is now possible to develop ***homozygous lines within a span of few months or a year*** by employing anther/pollen culture. This is in contrast to the conventional plant breeding programme that might take several years (6–10 yrs). In this way, production of haploids is highly useful for research related to plant genetics and breeding.

Generation of exclusive male plants

By the process of androgenesis to produce haploids, followed by chromosome doubling, it is possible to develop exclusive male plants. The male plants are particularly useful when their productivity and applications are much more than female plants.

Induction of mutations

In general, majority of induced mutations are recessive and therefore are not expressed in diploid cells (due to the presence of dominant allele). Haploids provide a convenient system for the induction of mutations and selection of mutants with desired traits. In fact, the haploid cells can be cultured and handled in a fashion similar to microorganisms.

Mutants from several plant species that are ***resistant to antibiotics, toxins, herbicides*** etc. have ***been developed***. When the haploid cells of tobacco plant (*Nicotiana tabacum*) were exposed to methionine sulfoximine (a mutagen), mutants which showed lower level of infection to *Pseudomonas tabaci* were produced.

Production of disease resistance plants

Disease resistance genes can be introduced while producing haploids. The so developed haploids are screened for the desired resistance, and then diploidized. Some examples of disease resistance plants are listed.

1. Hwansambye, a rice variety resistant to leaf blast, bacterial leaf blight and rice stripe tenui virus.

2. Barley accession Q21681 resistant to stem rust, leaf rust and powdery mildew.

More examples of disease resistance crops are given in ***Table 45.1***.

Production of insect resistance plants

Some varieties of rice resistant to insects have been developed e.g. Hwacheongbyeo resistant to brown plant hopper. Other varieties of rice that are resistant to pests have also been produced.

Production of salt tolerance plants

The plant species with salt tolerance are needed for their cultivation in some areas. Anther cultures have resulted in some varieties of rice and wheat with good salt tolerance e.g. wheat Hua Bain 124–4.

Cytogenetic research

Haploids are useful in several areas of cytogenetic research. These include

- Production of aneuploids

Table 45.1 A selected list of improved varieties of crops developed by using anther culture

Crop	*Varieties*	*Improvements made*
Wheat (*Triticum aestivum*)	Lunghua 1, Zing Hua 1, Zing Hua 2 Huapei 1, Florin, Ambitus, Jingdan 2288	High yield, rust resistance, cold resistance, large spikes, more tillers.
Rice (*Oryza sativa*)	Tangfong 1, Xin Xiu, Zhog Hua 8 Zhong Hua 9, Hua yu 1, Hua Yu 2, Huapei Shanyou 63, Zhe keng 66, Ta Be 78, Nonhua 5, Hirohikari, Hirohonami.	High yield, good quality, disease resistance
Tobacco (*Nicotiana tabacum*)	Tanyu 1, Tanyu 2, Tanyu 3, F 211 Hai Hua 19, Hai Hua 30	Mild smoking, disease resistance
Brassica napus	Jai kisan	Low erucic acid

- Determination of the nature of ploidy
- Determination of basic chromosome number
- Evaluation of origin of chromosomes.

Induction of genetic variability

Besides the development of haploid mutants, it is also possible to produce plants with various ploidy levels through androgenesis.

Doubled haploids in genome mapping

Genome mapping, a recent development in molecular biology, can be more conveniently achieved by using doubled haploid plant species.

Evolutionary studies

A comparison of dihaploids (doubled haploids) with diploid wild plant species will be useful to trace the evolutionary origin of various plants. The close evolutionary relationship between tomato and potato has been evaluated by this approach.

CROPS DEVELOPED THROUGH ANTHER CULTURES

Anther culture has made it possible to develop many new varieties of plants, a selected list is given in ***Table 45.1***.

In general, the new crops are high-yielding varieties with disease resistance. Some of them are resistance to cold, salt etc. Many plant breeders these days use anther cultures in their regular breeding programmes.

LIMITATIONS OF HAPLOIDS

It has been possible to raise several haploid plants through anther cultures. However, haploid breeding programmes have not yielded the expected and desired results, despite heavy investments made during the past 3 decades. There are many problems encountered with haploid production. Some of them are listed below.

1. The ***frequency of haploid production is very low***, hence selection is often difficult.

2. The operation of tissue culture to develop haploids requires high level expertise and management.

3. Besides haploids, ***different ploidy levels are also produced***.

4. Haploids with deleterious traits frequently develop in cultures.

5. The callus derived from anther culture is usually detrimental to haploid production.

6. Isolation of haploids from cultures is often difficult since ***polyploids outgrow haploids***.

7. The embryos derived from haploids often get aborted.

8. Haploid production is associated with high incidence of albino (colourless) plants.

9. The doubling of haploids (***diploidization***) may ***not always lead to the formation of homozygous plant***.

10. *In vitro* haploid production is not economically viable due to very low success rate.

Somaclonal Variations

The ***genetic variations found in the in vitro cultured cells*** are collectively referred to as somaclonal variations. The plants derived from such cells are referred to ***somaclones***.

Some authors use the terms ***calliclones*** and ***protoclones*** to represent cultures obtained from callus and protoplasts respectively.

The growth of plant cells *in vitro* is an asexual process involving only mitotic division of cells. Thus, culturing of cells is the method to clone a particular genotype. It is therefore expected that plants arising from a given tissue culture should be the exact copies of the parental plant. The occurrence of phenotypic variants among the regenerated plants (from tissue cultures) has been known for several years. These variations were earlier dismissed as tissue culture artefacts.

The term somaclonal variations was first used by Larkin and Scowcraft (1981) for variations arising due to culture of cells. i.e., ***variability generated by a tissue culture***. This term is now universally accepted.

As described elsewhere (Chapter 42), the explant used in tissue culture may come from any part of the plant organs or cells. These include leaves, roots, protoplasts, microspores and embryos. Somaclonal variations are reported in all types of plant tissue cultures.

In recent years, the term ***gametoclonal variations*** is used for the variations observed in the regenerated plants from gametic cells (e.g., anther cultures). For the plants obtained from protoplast cultures, ***protoclonal variations*** is used.

BASIS OF SOMACLONAL VARIATIONS

Somaclonal variations occur as a result of ***genetic heterogeneity*** (change in chromosome number and/or structure) in plant tissue cultures. This may be due to

- Expression of chromosomal mosaicism or genetic disorders.
- Spontaneous mutations due to culture conditions.

The genetic changes associated with somaclonal variations include polyploidy, aneuploidy, chromosomal breakage, deletion, translocation, and gene amplifications, besides several mutations. In fact, the presence of several chromosomal aberrations—reciprocal translocation, deletions, inversions, chromosomal breakage, multicentric, acentric fragments have been found among the somaclones of barley, garlic and oat.

The occurrence of mutations in cultures is relatively low. Mutations may be due varied nutrients, culture conditions and mutagenic effects of metabolic products that accumulate in the medium.

Somaclonal variations due to transposable elements, mitotic crossing over and changes in the cytoplasmic genome have also been reported.

Nomenclature of somaclones

The somaclones that are regenerated from tissue cultures directly are regarded as R_0 or R plants. The self-fertilized progeny of R_0 plants represent R_1 plants. R_2, R_3, R_4 etc. plants are the subsequent generations. Some workers use other nomenclature — somaclones ($SC_1 = R_0$), SC_2, SC_3, SC_4 etc. for subsequent generations.

ISOLATION OF SOMACLONAL VARIANTS

There are two procedures commonly used for obtaining the crop plants with somaclonal variations

1. Without *in vitro* selection
2. With *in vitro* selection.

WITHOUT *IN VITRO* SELECTION

An explant (leaf, stem, root etc.) is cultured on a suitable medium, supplemented with growth regulators. The unorganized callus and cells do not contain any selective agent (toxic or inhibitory substance). These cultures are normally subcultured, and transferred to shoot induction medium for regeneration of plants. The so produced plants are grown in pots, transferred to field, and analyzed for somaclonal variants (***Fig. 46.1***).

Somaclonal variants of several crops have been successfully obtained by this approach e.g., sugarcane, potato, tomato, cereals etc. A selected list of disease resistant crop plants obtained from somaclonal variations without *in vitro* selection along with the pathogenic organisms is given in ***Table 46.1***.

Limitations of without *in vitro* selection approach

There is no directed and specific approach for the isolation of somaclones without *in vitro* selection. Consequently, the appearance of a desired trait is purely by chance. Further, this procedure is time consuming and requires screening of many plants.

WITH *IN VITRO* SELECTION

Isolation of somaclones with *in vitro* selection method basically involves handling of plant cells in cultures (protoplast, callus) like microorganisms and selection of biochemical mutants. The cell lines are screened from plant cultures for their ability to survive in the presence of a toxic/inhibitory substance in the medium or under conditions of environmental stress.

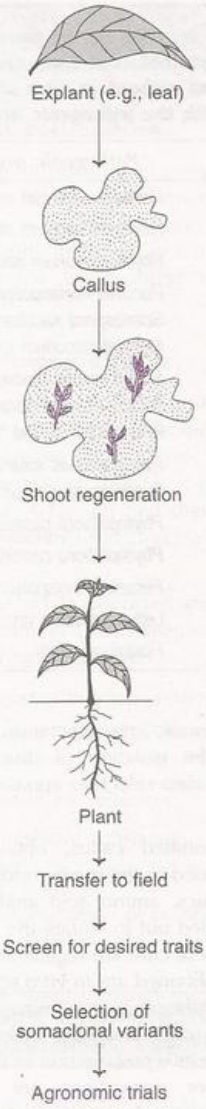


***Fig. 46.1** : A diagrammatic representation of isolation of somaclones without* in vitro *selection.*

Table 46.1 A selected list of disease resistant crop plants obtained from somaclonal variations without *in vitro* selection, along with the pathogenic organisms

Crop	Pathogenic organism(s)
Rice	*Helminthosporium oryzae*
Maize	*Helminthosporium maydis*
Barley	*Rhynchosporium secalis*
Sugarcane	*Puccinia melanocephala*, *Sclerospora sacchari* *Helminthosporium sacchari*, Fiji virus
Potato	*Streptomyces scabie*, *Alternaria solani*, *Phytophthora infestans*, Potato virus X and Y.
Tomato	*Pseudomonas solanacearum*, *Fusarium oxysporum*
Tobacco	*Phytophthora parasitica*
Apple	*Phytophthora cactorum*
Banana	*Fusarium oxysporum*
Lettuce	Lettuce mosaic virus
Alfalfa	*Fusarium solani*

A diagrammatic representation of *in vitro* protocol for the isolation of disease resistance plants with *in vitro* selection approach is given in ***Fig. 46.2***.

The differentiated callus, obtained from an explant is exposed in the medium to inhibitors like toxins, antibiotics, amino acid analogs. Selection cycles are carried out to isolate the tolerant callus cultures and these calli are regenerated into plants. The plants so obtained are *in vitro* screened against the toxin (or pathogen or any other inhibitor). The plants resistant to the toxin are selected and grown further by vegetative propagation or self pollination. The subsequent generations are analysed for ***disease resistant plants against the specific pathogenic organism***.

A selected list of disease resistant crop plants obtained from somaclonal variations by *in vitro* selection along with pathogenic organisms and selection agents is given in ***Table 46.2***.

Besides the disease resistant plants (described above), plants with herbicide resistance and antibiotic resistance have also been developed with *in vitro* selection approach.

Environmental stress tolerance

High salt concentration in the soil is the major constraint limiting the crop development and yield. Many plants with salt tolerance (salinity) have been developed e.g. rice, tobacco.

Several attempts are being made by plant biotechnologists to develop cold tolerant crops, although the success rate has been very limited.

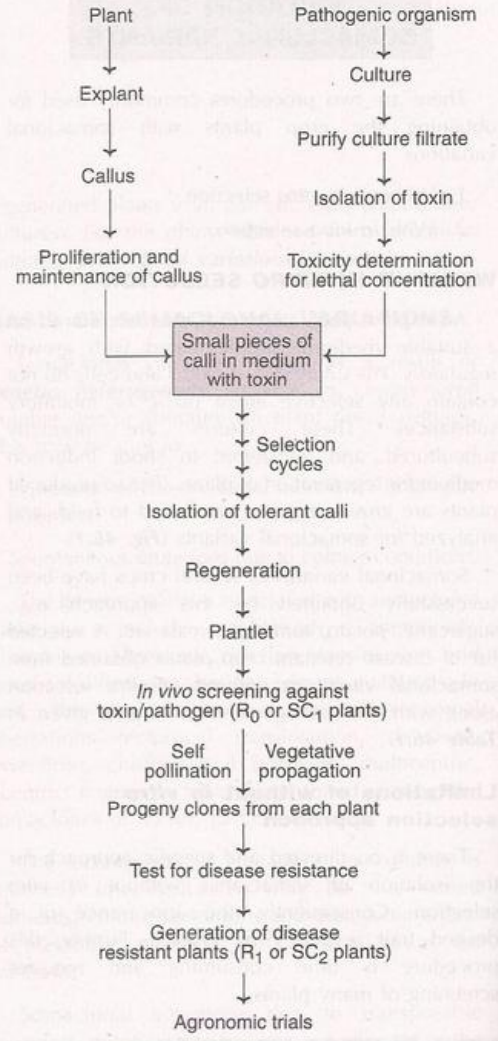


Fig. 46.2 : A diagrammatic representation of isolation of disease resistant plants with in vitro selection.

TABLE 46.2 A selected list of disease resistant crop plants obtained from somaclonal variations with *in vitro* selection along with pathogenic organisms and selection agents

Crop	*Pathogenic organism(s)*	*Selection agent*
Rice	*Xanthomonas oryzae*	Bacterial cells
Wheat	*Helminthosporium sativum*	Crude toxin
	Pseudomonas syringae	Syringomycin
Barley	*Fusarium* sp	Fusaric acid
	Helminthosporium sativum	Crude toxin
Maize	*Helminthosporium maydis*	Hm T toxin
Sugarcane	*Helminthosporium sacchari*	Toxin
Potato	*Fusarium oxysporum*	Culture filtrate
	Ercoinia carotovora	Pathogen
Tomato	*Pseudomonas solanacearum*	Crude filtrate
	Tobacco mosaic virus	Virus
Alfalfa	*Foxysporum* sp	Crude filtrate
Tobacco	Tobacco mosaic virus	Virus
	Pseudomonas syringae	Methionine sulfoximine

Advantages of with *in vitro* selection appraoch

The major advantage of with *in vitro* selection method is the specific selection of the desired trait rather than a general variation found at the plant level. This procedure is less time consuming when compared to without *in vitro* selection approach.

FACTORS AFFECTING PRODUCTION OF SOMACLONAL VARIANTS

Some of the important factors that influence development of somaclonal variants by without *in vitro* selection and with *in vitro* selection are briefly described.

Genotype and explant source

The nature of genotype of the plants ***influences the frequency of regeneration and frequency of production of somaclones.***

Explants can be taken from any part of plant — leaves, roots, internodes, ovaries etc. The source of explant is very critical for somaclonal variations. For instance, potato plants regenerated from callus of rachis and petiole are much higher (~50%) compared to those regenerated from callus of leaves (~12%).

Duration of cell culture

In general, for many plant cultures, somaclonal variations are higher with increased duration of cultures. For example, it was reported that genetic variability increased in tobacco protoplasts from 1.5 to 6% by doubling the duration of cultures.

Growth hormone effects

The plant growth regulators in the medium will influence the karyotypic alterations in qultured cells, and therefore development of somaclones. Growth hormones such as 2, 4-dichlorophenoxy acetic acid (2, 4-D) and naphthalene acetic acid (NAA) are frequently used to achieve chromosomal variability.

Besides the factors discussed above, selection of somaclones with *in vitro* selection are dependent on the following parameters.

- Selection propagule (cells, protoplasts, calli).
- Selection agent (toxin, herbicide, amino acid analogue).
- Technique used for selection.
- Stability of resistant substance.
- *In vivo* testing procedure.
- Ability for regeneration of plants.

APPLICATIONS OF SOMACLONAL VARIATIONS

Somaclonal variations (and also gametoclonal variations, described later) are highly useful in plant breeding programmes. The genetic variations with desirable (or improved characters), besides the existing favourable characters can be introduced into the plants. In general, the ***methodology*** adopted for induction of somaclonal variations are ***simpler and easier compared to recombinant DNA technology***. Hence, they are ***preferred by some workers***. The important applications of somaclonal variations are briefly described.

TABLE 46.3 A selected list of somaclonal variants obtained from different crop species with their morphological characters

Crop	*Character(s)*
Rice	Flowering period, panicle size; tiller number; plant height; leaf length, shape and colour; frequency of fertile seed; sterility mutants.
Wheat	Plant height; tiller number; grain colour; seed storage protein.
Maize	Reduced pollen fertility and male sterility; twin stalks from a single node.
Sugarcane	High sugar yield; increased stalk length, diameter, weight and density.
Barley	Increased grain yield; leaf shape; heading date; ash content.
Oats	Plant height; heading date; morphology and fertility.
Soybean	Variable height; maturity; seed protein and oil content.
Potato	Higher yield; growth habit, maturity and morphology.
Tomato	Dwarf habit; early flowering; orange fruit colour.
Carrot	Higher carotene content.
Pineapple	Foliage density; leaf colour, width and spine formation.
Brassica	Multiple branching stem; altered leaf; slow growth; failure to flower or delay in flowering; large pollen grains.
Tobacco	Increased yield; plant height; leaf number, shape, width and yield; type of inflorescence and yield.

Production of agronomically useful plants

As a result of somaclonal variations, several novel variants of existing crops have been developed. e.g., pure thornless blackberries.

In ***Table 46.3***, somaclonal variations in a selected list of crops with useful and improved morphological characters are given. The crops include rice, wheat, maize, sugarcane, potato, carrot etc.

Resistance to diseases

Somaclonal variations have largely contributed towards the development of disease resistance in many crops e.g. rice, wheat, maize, sugarcane, tobacco, apple, tomato.

Selected crops somaclonal variants, with increased disease resistance developed, without *in vitro* selection and with *in vitro* selection are respectively given in ***Tables 46.1 and 46.2***.

Resistance to abiotic stresses

It has been possible to develop biochemical mutants with abiotic stress resistance.

- Freezing tolerance e.g. wheat.
- Salt tolerance e.g., rice, maize, tobacco.
- Aluminium tolerance e.g., carrot, sorghum, tomato.

Resistance to herbicides

Certain somaclonal variants with herbicide resistance have been developed. Selected examples are given

- Tobacco resistant to glyphosate, sulfonylurea and picloram.
- Carrot resistant to glyphosate.
- Lotus resistant to 2, 4-dichlorophenoxy acetic acid (2, 2-D).

Improved seed quality

A new variety of *Lathyrus sativa* seeds (Lathyrus Bio L 212) with a ***low content of neurotoxin*** has been developed through somaclonal variations. (**Note :** Lathyrism is a crippling disease caused by consumption of Lathyrus seeds i.e. kesari dal, in many parts of central India).

LIMITATIONS OF SOMACLONAL VARIATIONS

Despite several applications of somaclonal variations, there are certain limitations/ disadvantages also.

- Most of the somaclonal variations ***may not be useful***.
- The variations occur in an unpredictable and uncontrolled manner.

- Many a times the genetic traits obtained by somaclonal variations are ***not stable and heritable***.
- Somaclonal variations are cultivar-dependent which is frequently a time consuming process.
- Somaclones can be produced in only those species which regenerate to complete plants.
- Many cell lines (calli) may not exhibit regeneration potential.

GAMETOCLONAL VARIATIONS

The ***variations*** observed while ***culturing*** the ***gametic cells*** are regarded as gametoclonal variations. This is in contrast to the somaclonal variations detected in the cultures of somatic tissues.

The term ***gametoclones*** (in place of somaclones) is used for the products of gametoclonal variations.

As the somatic cells divide by mitosis, the genetic material is equally distributed to the daughter cells. In contrast, the gametes, being the products of meiosis, possess only half of the parent cell genetic material.

The gametoclonal variations differ from somaclonal variations by three distinct features.

1. Mutants obtained from gametoclonal variations give rise to haploid plants since a single set of chromosomes are present.

2. Meiotic crossing over is the recombination process observed in gametoclonal variations.

3. The gametoclones can be stabilized by doubling the chromosome number.

TABLE 46.4 A selected list of plants regenerated from gametoclonal variations

Crop	*Characteristics*
Oryza sativa	Plant height; time of flowering; seed size and protein content; level of tillering; waxy mutant; chloroplast content.
Nicotiana tabacum	Plant size; leaf shape; number of leaves; alkaloid content; virus resistance; time of flowering.
Brassica napus	Leaf shape and colour; time to flower; type of flower; glucinolate content; pod size and shape.
Hordeum vulgare	Plant height; days to maturity; grain yield; fertility.

Production of gametoclones

Gametoclones can be developed by culturing male or female gametic cells. The cultures of anthers or isolated microspores are widely used. (For details, Refer Chapter 43).

A selected list of plants regenerated from gametoclonal variations is given in ***Table 46.4***. Improvements have been made in several plant species through development of gametoclones e.g., rice, wheat, tobacco.

Source of gametoclonal variations

There are three sources of genetic variations in the gametoclones.

1. Cell culture technique may induce genetic variations.

2. Variations may be induced while doubling the haploid chromosomes.

3. Genetic variations may occur due to heterozygosity of the diploids.

Clonal Propagation (Micropropagation)

Plants can be propagated by sexual (through generation of seeds) or asexual (through multiplication of vegetative parts) means. Clonal propagation refers to the process of ***asexual reproduction by multiplication of genetically identical copies of individual plants***. The term ***clone*** is used to represent a plant population derived from a single individual by asexual reproduction.

Asexual reproduction through ***multiplication of vegetative parts*** is the only method for the *in vivo* propagation of certain plants, as they do not produce viable seeds e.g. banana, grape, fig, chrysanthemum. Clonal propagation has been successfully applied for the propagation of apple, potato, tuberous and several ornamental plants.

Advantages of vegetative propagation

Asexual (vegetative) propagation of plants has certain advantages over sexual propagation.

- Faster multiplication — large number of plants can be produced from a single individual in a short period.
- Possible to produce genetically identical plants.
- Sexually — derived sterile hybrids can be propagated.
- Seed — raised plants pass through an undesirable juvenile phase which is avoided in asexual propagation.
- Gene banks can be more easily established by clonally propagated plants.

In vitro clonal propagation

The *in vivo* clonal propagation of plants is tedious, expensive and frequently unsuccessful. ***In vitro clonal propagation through tissue culture is referred to as micropropagation***. Use of tissue culture technique for micropropagation was first started by Morel (1960) for propagation of orchids, and is now applied to several plants. Micropropagation is a handy technique for rapid multiplication of plants.

TECHNIQUE OF MICROPROPAGATION

Micropropagation is a complicated process and mainly involves 3 stages (I, II and III). Some authors add two more stages (stage 0 and IV) for more comprehensive representation of micropropagation. All these stages are represented in ***Fig. 47.1***, and briefly described hereunder.

Stage 0 : This is the initial step in micropropagation, and involves the selection and growth of stock plants for about 3 months under controlled conditions.

Stage I : In this stage, the initiation and establishment of culture in a suitable medium is achieved. Selection of appropriate explants is important. The most commonly used explants are organs, shoot tips and axillary buds. The chosen explant is surface sterilized and washed before use.

Stage	Methodology involved
Stage 0	Selection of mother plant and its maintenance
↓	
Stage I	Initiation and establishment of culture
↓	
Stage II	Multiplication of shoots or rapid somatic embryo formation
↓	
Stage III	*In vitro* germination of somatic embryos and/or rooting of shoots
↓	
Stage IV	Transfer of plantlets to sterilized soil for hardening under greenhouse environment

Fig. 47.1 : Major stages involved in micropropagation.

Stage II : It is in this stage, the major activity of micropropagation occurs in a defined culture medium. Stage II mainly involves multiplication of shoots or rapid embryo formation from the explant.

Stage III : This stage involves the transfer of shoots to a medium for rapid development into shoots. Sometimes, the shoots are directly planted in soil to develop roots. *In vitro* rooting of shoots is preferred while simultaneously handling a large number of species.

Stage IV : This stage involves the establishment of plantlets in soil. This is done by transferring the plantlets of stage III from the laboratory to the environment of greenhouse. For some plant species, stage III is skipped, and unrooted stage II shoots are planted in pots or in suitable compost mixture.

The different stages described above for micropropagation are particularly useful for comparison between two or more plant systems, besides better understanding. It may however, be noted that not all plant species need to be propagated *in vitro* through all the five stages referred above.

Micropropagation mostly involves *in vitro* clonal propagation by two approaches.

1. Multiplication by axillary buds/apical shoots.
2. Multiplication by adventitious shoots.

Besides the above two approaches, the ***plant regeneration*** processes namely ***organogenesis*** and ***somatic embryogenesis*** may also be treated as micropropagation, hence they are also described in this chapter.

3. **Organogenesis** : The formation of individual organs such as shoots, roots, directly from an explant (lacking preformed meristem) or from the callus and cell culture induced from the explant.

4. **Somatic embryogenesis** : The regeneration of embryos from somatic cells, tissues or organs.

MULTIPLICATION BY AXILLARY BUDS AND APICAL SHOOTS

Quiescent or actively dividing meristems are present at the axillary and apical shoots (shoot tips). The axillary buds located in the axils of leaves are ***capable of developing into shoots***. In the *in vivo* state, however only a limited number of axillary meristems can form shoots. By means of induced *in vitro* multiplication in micropropagation, it is possible to develop plants from meristem and shoot tip cultures and from bud cultures.

Meristem and shoot tip cultures

Apical meristem is a dome of tissue located at the extreme tip of a shoot. The apical meristem along with the young leaf primordia constitutes the shoot apex. For the development of disease-free plants, meristem tips should be cultured.

Meristem or shoot tip is isolated from a stem by a V-shaped cut. The size (frequently 0.2 to 0.5 mm) of the tip is critical for culture. In general, the larger the explant (shoot tip), the better are the chances for culture survival. For good results of micropropagation, explants should be taken from the actively growing shoot tips, and the ideal timing is at the end of the plants dormancy period.

The most widely used media for meristem culture are MS medium and White's medium. A diagrammatic representation of shoot tip (or meristem) culture in micropropagation is given in ***Fig 47.2***, and briefly described hereunder.

In stage I, the culture of meristem is established. Addition of growth regulators namely cytokinins (kinetin, BA) and auxins (NAA or IBA) will support the growth and development.

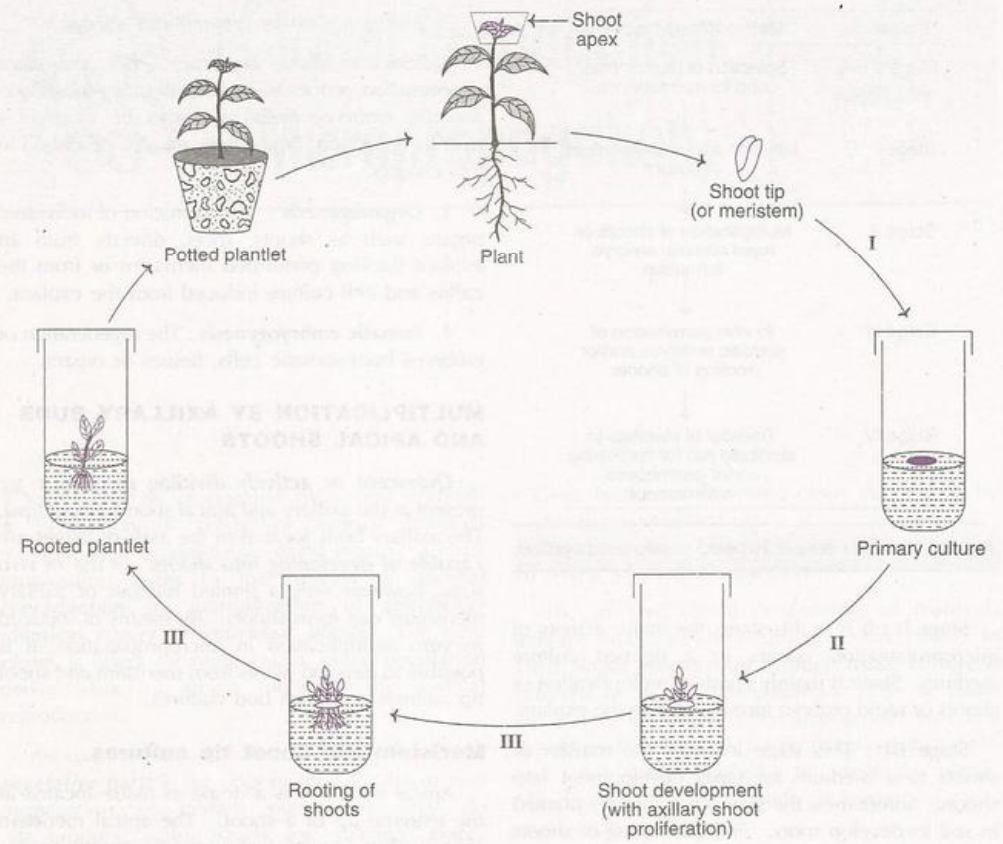


*Fig. 47.2 : A diagrammatic representation of shoot tip (or meristem) culture in micropropagation (**Note : I, II and III** represent stages in micropropagation)*

In stage II, shoot development along with axillary shoot proliferation occurs. High levels of cytokinins are required for this purpose.

Stage III is associated with rooting of shoots and further growth of plantlet. The root formation is facilitated by low cytokinin and high auxin concentration. This is opposite to shoot formation since high level of cytokinins is required (in stage II). Consequently, ***stage II medium and stage III medium should be different in composition.***

The optimal temperature for culture is in the range of 20–28°C (for majority 24-26°C). Lower light intensity is more appropriate for good micropropagation.

Bud cultures

The plant buds possess quiescent or active meristems depending on the physiological state of the plant. Two types of bud cultures are used—single node culture and axillary bud culture.

Single node culture : This is a natural method for vegetative propagation of plants both in *in vivo* and *in vitro* conditions. The bud found in the axil of leaf is comparable to the stem tip, for its ability in micropropagation. A bud along with a piece of stem is isolated and cultured to develop into a plantlet. Closed buds are used to reduce the chances of infections.

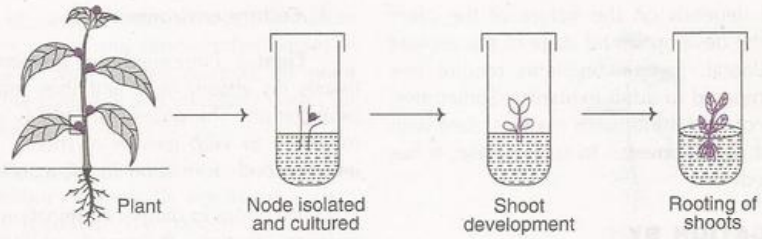


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Fig. 47.3 : A diagrammatic representation of micropropagation by single node technique.

A diagrammatic representation of single node culture is depicted in ***Fig 47.3***. In single node culture, no cytokinin is added.

Axillary bud culture : In this method, a shoot tip along with axillary bud is isolated. The cultures are carried out with high cytokinin concentration. As a result of this, apical dominance stops and axillary buds develop. A schematic representation of axillary bud culture for a rosette plant and an elongate plant is given in ***Fig 47.4***.

For a ***good axillary bud culture, the cytokinin/ auxin ratio is around 10 : 1***. This is however,

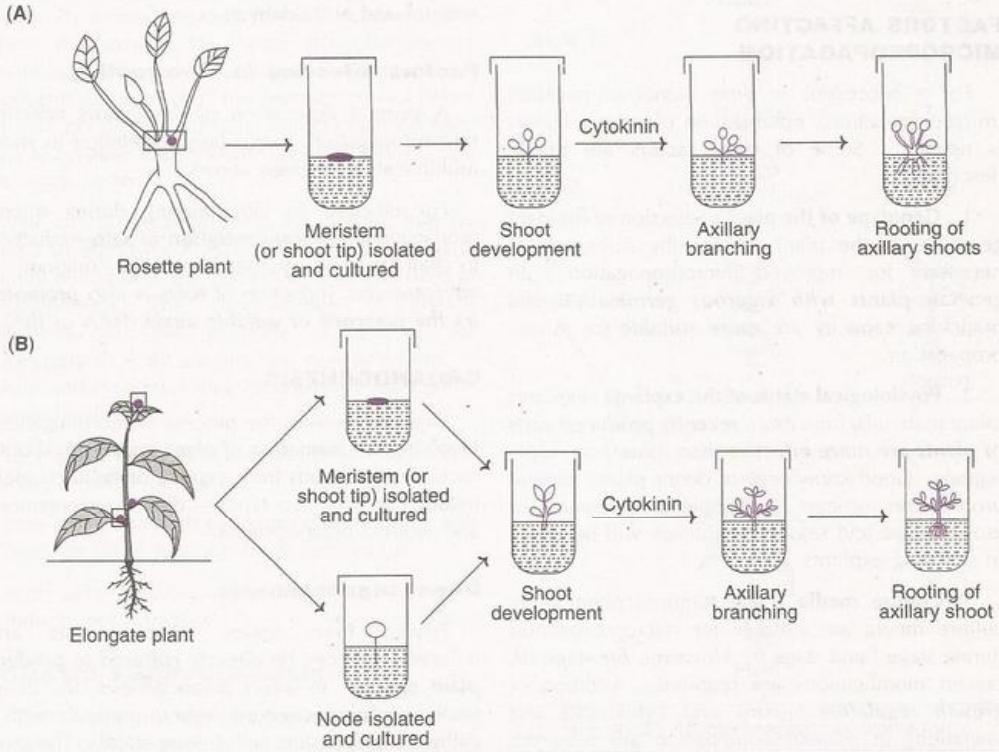


*Fig. 47.4 : A diagrammatic representation of micropropagation of plants by axillary bud (or shoot tip) method (**A**) Rosette plant (**B**) Elongate plant.*

variable and depends on the nature of the plant species and the developmental stage of the explant used. In general, juvenile explants require less cytokinin compared to adult explants. Sometimes, the presence of apical meristem may interfere with axillary shoot development. In such a case, it has to be removed.

MULTIPLICATION BY ADVENTITIOUS SHOOTS

The stem and leaf structures that are naturally formed on plant tissues located in sites other than the normal leaf axil regions are regarded as adventitious shoots. There are many adventitious shoots which include stems, bulbs, tubers and rhizomes. The adventitious shoots are useful for *in vivo* and *in vitro* clonal propagation.

The meristematic regions of adventitious shoots can be induced in a suitable medium to regenerate to plants.

FACTORS AFFECTING MICROPROPAGATION

For a successful *in vitro* clonal propagation (micropropagation), optimization of several factors is needed. Some of these factors are briefly described.

1. **Genotype of the plant :** Selection of the right genotype of the plant species (by screening) is necessary for improved micropropagation. In general, ***plants with vigorous germination*** and branching capacity are ***more suitable*** for micropropagation.

2. **Physiological status of the explants :** Explants (plant materials) from more ***recently produced parts of plants are more effective*** than those from older regions. Good knowledge of donor plants' natural propagation process with special reference to growth stage and seasonal influence will be useful in selecting explants.

3. **Culture media :** The standard plant tissue culture media are suitable for micropropagation during stage I and stage II. However, ***for stage III***, certain modifications are required. Addition of ***growth regulators*** (auxins and cytokinins) and alterations in mineral composition are required. This is largely dependent on the type of culture (meristem, bud etc).

4. **Culture environment**

Light : Photosynthetic pigment in cultured tissues do absorb light and thus influence micropropagation. The quality of light is also known to influence *in vitro* growth of shoots. e.g blue light induced bud formation in tobacco shoots.

Variations in diurnal illumination also influence micropropagation. In general, an illumination of 16 hours day and 8 hours night is satisfactory for shoot proliferation.

Temperature : Majority of the culture for micropropagation requires an ***optimal*** temperature ***around 25°C***. There are however, some exceptions e.g. *Begonia* X *Cheimantha* hybrid tissue grow at a low temperature (around 18°C).

Composition of gas phase : The constitution of the gas phase in the culture vessels also influences micropropagation. Unorganized growth of cells is generally promoted by ethylene, O_2, CO_2 ethanol and acetaldehyde.

Factors affecting *in vitro* rooting

A general description of the factors affecting micropropagation, particularly in relation to shoot multiplication is given above.

For efficient *in vitro* rooting during micropropagation, ***low concentration of salts*** (reduction to half to one quarter from the original) is advantageous. Induction of roots is also ***promoted by the presence of suitable auxin*** (NAA or IBA).

ORGANOGENESIS

Organogenesis is the process of morphogenesis involving the ***formation of plant organs*** i.e. shoots, roots, flowers, buds from explant or cultured plant tissues. It is of two types — direct organogenesis and indirect organogenesis.

Direct organogenesis

Tissues from leaves, stems, roots and inflorescences can be ***directly cultured to produce plant organs***. In direct organogenesis, the tissue undergoes morphogenesis without going through a callus or suspension cell culture stage. The term ***direct adventitious organ formation*** is also used for direct organogenesis.

Induction of adventitious shoot formation directly on roots, leaves and various other organs of intact plants is a widely used method for plant propagation. This approach is particularly useful for herbaceous species.

For appropriate organogenesis in culture system, exogenous addition of growth regulators—auxin and cytokinin is required. The concentration of the growth promoting substance depends on the age and nature of the explant, besides the growth conditions.

Indirect organogenesis

When the organogenesis occurs ***through callus or suspension cell culture formation***, it is regarded as indirect organogenesis (***Fig 47.5 B and C***). Callus growth can be established from many explants (leaves, roots, cotyledons, stems, flower petals etc.) for subsequent organogenesis.

The explants for good organogenesis should be mitotically active immature tissues. In general, the bigger the explant the better the chances for obtaining viable callus/cell suspension cultures. It is advantageous to select meristematic tissues (shoot tip, leaf, petiole) for efficient indirect organogenesis. This is because their growth rate and survival rate are much better.

For indirect organogenesis, the cultures may be grown in liquid medium or solid medium. Many culture media (MS, B5, White's etc.) can be used in organogenesis. The concentration of growth regulators in the medium is critical for organogenesis. By varying the concentrations of auxins and cytokinins, *in vitro* organogenesis can be manipulated.

- Low auxin and low cytokinin concentration will induce callus formation.
- Low auxin and high cytokinin concentration will promote shoot organogenesis from callus.
- High auxin and low cytokinin concentration will induce root formation.

SOMATIC EMBRYOGENESIS

The process of ***regeneration of embryos from somatic cells***, tissues or organs is regarded as somatic (or asexual) embryogenesis. Somatic embryogenesis may result in non-zygotic embryos or somatic embryos (directly formed from somatic

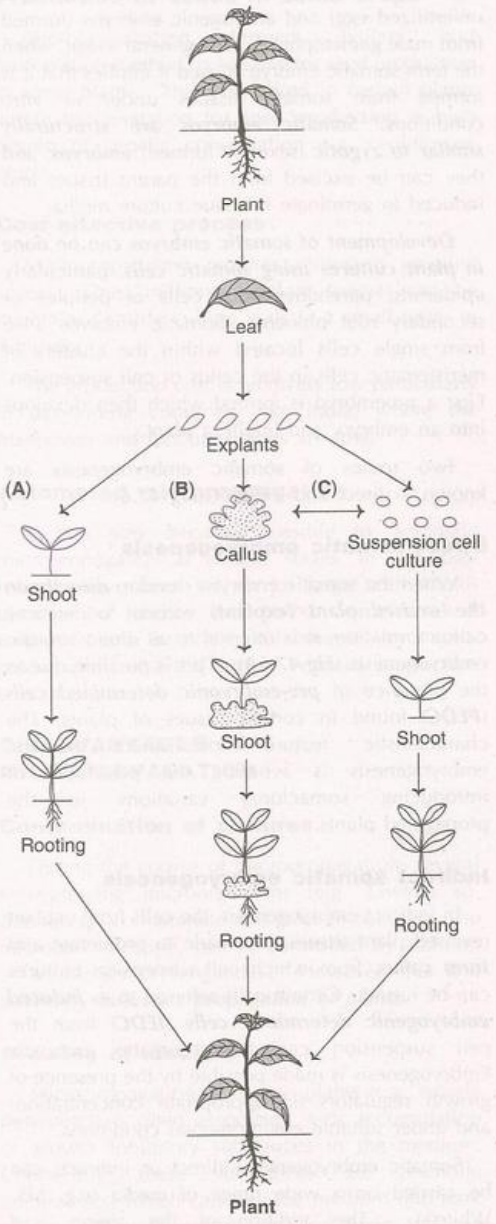


Fig. 47.5 : *Micropropagation of plants by organogenesis (**A**) Direct organogenesis (**B**) Indirect organogenesis through callus (**C**) Indirect organogenesis through suspension cell culture.*

organs), parthogenetic embryos (formed from unfertilized egg) and androgenic embryos (formed from male gametophyte). In a general usage, when the term somatic embryo is used it implies that it is formed from somatic tissues under *in vitro* conditions. ***Somatic embryos are structurally similar to zygotic*** (sexually formed) ***embryos***, and they can be excised from the parent tissues and induced to germinate in tissue culture media.

Development of somatic embryos can be done in plant cultures using somatic cells, particularly epidermis, parenchymatous cells of petioles or secondary root phloem. Somatic embryos arise from single cells located within the clusters of meristematic cells in the callus or cell suspension. First a proembryo is formed which then develops into an embryo, and finally a plant.

Two routes of somatic embryogenesis are known — direct and indirect (***Fig 47.6***).

Direct somatic embryogenesis

When the somatic embryos develop ***directly on the excised plant (explant)*** without undergoing callus formation, it is referred to as direct somatic embryogenesis (***Fig 47.6A***). This is possible due to the presence of ***pre-embryonic determined cells*** (***PEDC***) found in certain tissues of plants. The characteristic features of direct somatic embryogenesis is avoiding the possibility of introducing somaclonal variations in the propagated plants.

Indirect somatic embryogenesis

In indirect embryogenesis, the cells from explant (excised plant tissues) are made to proliferate and ***form callus***, from which cell suspension cultures can be raised. Certain cells referred to as ***induced embryogenic determined cells*** (***IEDC***) from the cell suspension can form somatic embryos. Embryogenesis is made possible by the presence of growth regulators (in appropriate concentration) and under suitable environmental conditions.

Somatic embryogenesis (direct or indirect) can be carried on a wide range of media (e.g. MS, White's). The addition of the amino acid ***L-glutamine promotes embryogenesis***. The presence of auxin such as 2, 4-dichloro-phenoxy acetic acid is essential for embryo initiation. On a low auxin or no auxin medium, the embryogenic clumps develop into mature embryos.

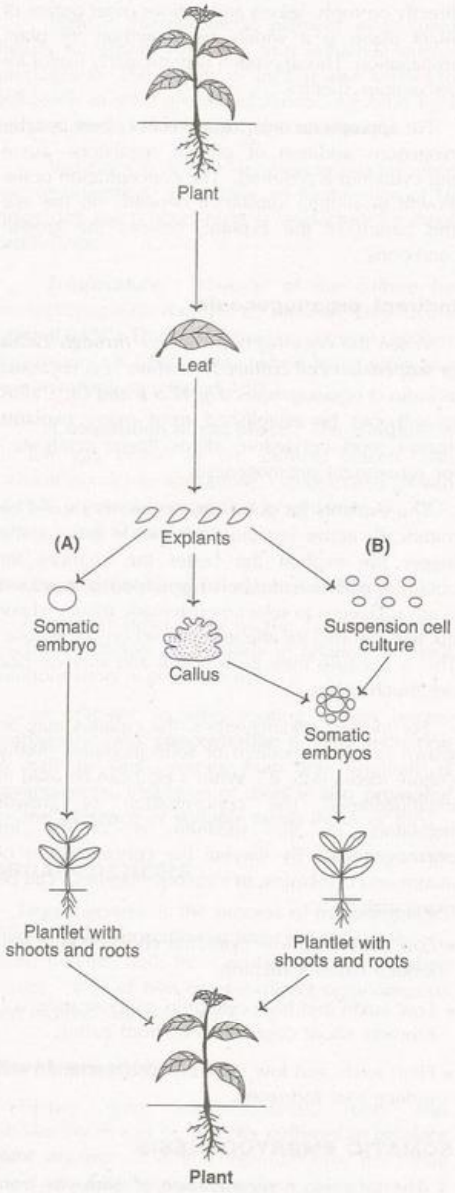


*Fig. 47.6 : Micropropagation of plants by embryogenesis (**A**) Direct embryogenesis (**B**) Indirect embryogenesis*

Indirect somatic embryogenesis is commercially very attractive since a large number of embryos can be generated in a small volume of culture medium. The somatic embryos so formed are synchronous and with good regeneration capability.

Artificial seeds from somatic embryos

Artificial seeds can be made by ***encapsulation of somatic embryos***. The embryos, coated with sodium alginate and nutrient solution, are dipped in calcium chloride solution. The calcium ions induce rapid cross-linking of sodium alginate to produce small gel beads, each containing an encapsulated embryo. These artificial seeds (encapsulated embryos) can be maintained in a viable state till they are planted.

APPLICATIONS OF MICROPROPAGATION

Micropropagation has become a suitable alternative to conventional methods of vegetative propagation of plants. There are ***several advantages*** of micropropagation.

High rate of plant propagation

Through micropropagation, a ***large number of plants can be grown from a piece of plant tissue within a short period***. Another advantage is that micropropagation can be carried out throughout the year, irrespective of the seasonal variations. Further, for many plants that are highly resistant to conventional propagation, micropropagation is the suitable alternative.

The small sized propagules obtained in micropropagation can be easily stored for many years (germplasm storage), and transported across international boundaries.

Production of disease-free plants

It is possible to produce disease-free plants through micropropagation. ***Meristem tip cultures*** are generally employed to ***develop pathogen-free plants*** (details given later in this chapter). In fact, micropropagation is successfully used for the production of virus-free plants of sweet potato (*Ipomea batatus*), cassava (*Manihot esculenta*) and yam (*Discorea rotundata*).

Production of seeds in some crops

Micropropagation, through axillary bud proliferation method, is suitable for seed production in some plants. This is required in certain plants where the limitation for seed production is high degree of genetic conservation e.g. cauliflower, onion.

Cost-effective process

Micropropagation requires minimum growing space. Thus, millions of plant species can be maintained inside culture vials in a small room in a nursery.

The production cost is relatively low particularly in developing countries (like India) where the manpower and labour charges are low.

Automated micropropagation

It has now become possible to automate micropropagation at various stages. In fact, bio-reactors have been set up for large scale multiplication of shoots and bulbs. Some workers employ robots (in place of labourers) for micropropagation, and this further reduces production cost of plants.

DISADVANTAGES OF MICROPROPAGATION

Contamination of cultures

During the course of micropropagation, several slow-growing microorganisms (e.g. *Eswinia* sp, *Bacillus* sp) contaminate and grow in cultures. The microbial infection, can be controlled by addition of antibiotics or fungicides. However, this will ***adversely influence propagation of plants***.

Brewing of medium

Micropropagation of certain plants (e.g. woody perennials) is often associated with accumulation of growth inhibitory substances in the medium. Chemically, these substances are phenolic compounds, which can turn the medium into dark colour. ***Phenolic compounds are toxic and can inhibit the growth of tissues***.

Brewing of the medium can be prevented by the addition of ascorbic acid or citric acid or polyvinyl pyrrolidone to the medium.

Genetic variability

When micropropagation is carried out through shoot tip cultures, genetic variability is very low. However, use of adventitious shoots is often associated with pronounced genetic variability.

Vitrification

During the course of repeated *in vitro* shoot multiplication, the cultures exhibit water soaked or almost translucent leaves. Such ***shoots cannot grow and even may die***. This phenomenon is referred to as vitrification. Vitrification may be prevented by increasing the agar concentration (from 0.6 to1%) in the medium. However, increased agar concentration reduces the growth rate of tissues.

Cost factor

For some micropropagation techniques, expensive equipment, sophisticated facilities and trained manpower are needed. This limits its use.

PRODUCTION OF DISEASE-FREE PLANTS

Many plant species are infected with pathogens — viruses, bacteria, fungi, mycoplasma and nematodes that cause systemic diseases. Although these diseases do not always result in the death of plants, they reduce the quality and yield of plants. The plants infected with bacteria and fungi frequently respond to chemical treatment by bactericides and fungicides. However, it is very difficult to cure the virus-infected plants. Further, viral disease are easily transferred in seed-propagated as well as vegetatively propagated plant species. Plant breeders are always interested to develop disease-free plants, particularly viral disease-free plants. This have become ***a reality through tissue cultures.***

Apical meristems with low concentration of viruses

In general, the apical meristems of the pathogen infected and disease harbouring plants are either free or carry a low concentration of viruses, for the following reasons.

- Absence of vascular tissue in the meristems through which viruses readily move in the plant body.
- Rapidly dividing meristematic cells with high metabolic activity do not allow viruses to multiply.
- Virus replication is inhibited by a high concentration of endogenous auxin in shoot apices. Tissue culture techniques employing ***meristem-tips*** are successfully ***used for the production of disease-free plants,*** caused by several pathogens — viruses, bacteria, fungi, mycoplasmas.

METHODS TO ELIMINATE VIRUSES IN PLANTS

In general, plants are infected with many viruses; the nature of some of them may be unknown. The usage virus-free plant implies that the given plant is free from all the viruses, although this may not be always true. The commonly used methods for virus elimination in plants are listed below, and briefly described next.

- Heat treatment of plant
- Meristem-tip culture
- Chemical treatment of media
- Other *in vitro* methods

Heat treatment (thermotherapy) of plants

In the early days, before the advent of meristem cultures, *in vivo* eradication of viruses from plants was achieved by heat treatment of whole plants. The underlying principle is that many viruses in plant tissues are either partially or completely inactivated at higher temperatures with minimal injury to the host plant. Thermotherapy (at temperatures 35–40°C) was carried out by using hot water or hot air for elimination viruses from growing shoots and buds.

There are two ***limitations*** of viral elimination by heat treatment.

1. Most of the viruses are not sensitive to heat treatment.

2. Many plant species do not survive after thermotherapy.

With the above disadvantages, heat treatment has not become popular for virus elimination.

Meristem-tip culture

A general description of the methodology adopted for meristem and shoot tip cultures has been described (see ***Fig 47.2***).

Table 47.1 A selected list of the plants with virus elimitation by meristem cultures

Plant species	*Virus eliminated*
Solanum tuberosum (potato)	Leaf roll, potato viruses — A, X, Y, S
Nicotiana tabacum (tobacco)	Tobacco mosaic virus
Saccharum officinarum (sugar cane)	Mosaic virus
Allium sativum (garlic)	Mosaic virus
Anenas sativus (pineapple)	Mosaic virus
Brassica oleracea (cauliflower)	Cauliflower/mosaic virus turnip mosaic virus
Ipomoea batata (sweet potato)	Fealthery mottle virus
Ribes grassularia	Vein banding virus
Humulus lupulus	Hop latent virus
Armoracia rusticena	Turnip mosaic virus
Musa sp (Banana)	Cucumber mosaic virus
Hycinthus sp	Hycinth mosaic virus
Dahlia sp	Dahlia mosaic virus
Chrysanthemum sp	Virus B
Petunia sp	Tobacco mosaic virus
Iris sp	Iris mosaic virus
Cymbidium sp	Cymbidium mosaic virus
Fragaria sp	Pallidosis virus, yellow virus complex
Freesia sp	Freesia mosaic virus

For viral elimination, the size of the meristem used in cultures is very critical. This is due to the fact that most of the viruses exist by establishing a gradient in plant tissues. ***In general, the regeneration of virus-free plants through cultures is inversely proportional to the size of the meristem*** used. The meristem-tip explant used for viral elimination cultures are too small. A stereoscopic microscope is usually employed for this purpose.

Meristem-tip cultures are influenced by the following factors.

- Physiological condition of the explant — actively growing buds are more effective.
- Thermotherapy prior to meristem-tip culture — for certain plants (possessing viruses in the meristematic regions), heat treatment is first given and then the meristem-tips are isolated and cultured.
- Culture medium — MS medium with low concentrations of auxins and cytokinins is ideal.

A selected list of the plants from which viruses have been eliminated by meristem cultures is given in ***Table 47.1.***

Chemical treatment of media

Some workers have attempted to eradicate viruses from infected plants by chemical treatment of the tissue culture media. The commonly used chemicals are growth substances (e.g. cytokinins) and antimetabolites (e.g thiouracil, acetyl salicylic acid). There are however, conflicting reports on the elimination viruses by chemical treatment of the media. For instance, addition of cytokinin suppressed the multiplication of certain viruses while for some other viruses, it actually stimulated.

Other *in vitro* methods

Besides meristem-tip culture, other *in vitro* methods are also used for raising virus-free plants. In this regard ***callus cultures*** have been successful to some extent. The callus derived from the infected tissue does not carry the pathogens throughout the cells. In fact, the uneven distribution of tobacco mosaic virus in tobacco leaves was exploited to develop virus-free plants of tobacco.

Somatic cell hybridization, gene transformation and ***somaclonal variations*** also useful to raise disease-free plants.

ELIMINATION OF PATHOGENS OTHER THAN VIRUSES

Besides the elimination of viruses, meristem-tip cultures and callus cultures are also useful for eradication bacteria, fungi and mycoplasmas. Some examples are given

- The fungus *Fusarium roseum* has been successfully eliminated through meristem cultures from carnation plants.
- Certain bacteria (*Pseudomonas carophylli, Pectobacterium parthenii*) are eradicated from carnation plants by using meristem cultures.

MERITS AND DEMERITS OF DISEASE-FREE PLANT PRODUCTION

Among the culture techniques, ***meristem-tip culture is the most reliable method*** for virus and other pathogen elimination. This, however, requires good knowledge of plant pathology and tissue culture.

Virus-free plants exhibit increased growth and vigour of plants, higher yield (e.g. potato), increased flower size (e.g. *Chrysanthemum*), improved rooting of stem cuttings (e.g. *Pelargonium*)

Virus-free plants are more susceptible to the same virus when exposed again. This is the major limitation. Reinfection of disease-free plants can be minimized with good knowledge of greenhouse maintenance.

EMBRYO CULTURE

Embryo culture deals with the sterile isolation and *in vitro* growth of a mature or an immature embryo with an ultimate objective of obtaining a viable plant. Conventionally, the term embryo culture refers to the sexually produced zygotic embryo culture. This is different from the somatic embryogenesis, already described in this chapter.

There are two types of embryo culture — mature embryo culture and immature embryo culture (embryo rescue).

MATURE EMBRYO CULTURE

Mature embryos are isolated from ripe seeds and cultured *in vitro*. Mature embryo cultures are carried out in the following conditions.

- When the embryos remain dormant for long periods.
- Low survival of embryos *in vivo*.
- To avoid inhibition in the seed for germination.
- For converting sterile seeds to viable seedlings.

Seed dormancy in plant species is a common occurrence. This may be due to chemical inhibitors or mechanical resistance exerted by the structures covering the embryo. Seed dormancy can be successfully bypassed by culturing the embryos *in vitro*.

Embryo culture is relatively easy as they can be grown on a simple inorganic medium supplemented with energy source (usually sucrose). This is possible since the mature embryos excised from the developing seeds are autotrophic in nature.

EMBRYO RESCUE

Embryo rescue involves the ***culture of immature embryos*** to rescue them from unripe or hybrid seeds ***which fail to germinate***. This approach is very useful to avoid embryo abortion and produce a viable plant.

Wild hybridization involving crossing of two different species of plants from the same genus or different genera often results in failure. This is mainly because the normal development of zygote and seed is hindered due to genetic barriers. Consequently, hybrid endosperm fails to develop leading the abortion of hybrid embryo. The endosperm may also produce toxins that ultimately kill the embryo.

In the normal circumstances, endosperm first develops and supports embryo development nutritionally. Thus, majority of embryo abortions are due to failure in endosperm development. Embryo abortion can be avoided by isolating and culturing the hybrid embryos prior to abortion.

The most important ***application*** of embryo rescue is the ***production of interspecific and intergeneric hybrids from wild plant species***.

Culture technique for embryo rescue

The isolation of immature embryos often poses some difficulty. The aseptically isolated embryos can be grown in a suitable medium under optimal conditions. In general, a complex nutrient medium is required for culture methods involving embryo rescue.

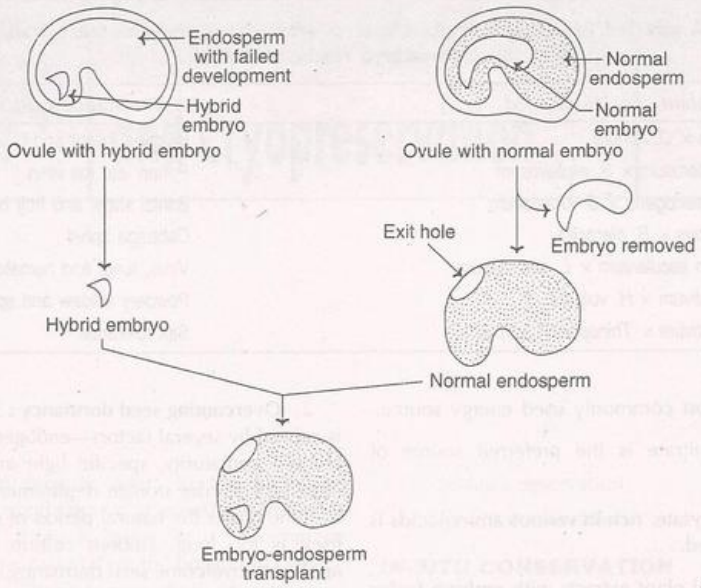


Fig. 47.7 : Embryo–endosperm transplant technique used in embryo rescue (or immature embryo culture).

For adequate nutritional support of immature embryos, embryo-endosperm transplant is used.

Embryo-endosperm transplant : The endosperm transplant technique used for culturing immature embryos is given in ***Fig. 47.7***, and briefly described below.

The hybrid embryo from the ovule in which endosperm development has failed is taken out by excision. Another normally developed ovule with endosperm enclosing an embryo is chosen. This ovule is dissected and the normal embryo is pressed out. This leaves a normal endosperm with an exit hole. Now, the hybrid embryo can be inserted into the normal endosperm through exit hole. This results in embryo-endosperm transplant which can be cultured in a suitable medium.

By using embryo-endosperm transplant, many interspecific and intergeneric plants have been raised e.g., hybrid plants of legumes.

NUTRITIONAL REQUIREMENTS OF EMBRYO CULTURES

There are two phases in the embryo development and the nutritional requirement varies accordingly

1. **Heterotrophic phase :** This is an early phase and the embryo is mostly dependent on the endosperm and maternal tissues for nutrient supply.

2. **Autotrophic phase :** This phase is characterized by the metabolic capability of the embryo to synthesize substances required for its growth which slowly makes it independent.

The critical stage is the intervening phase between the heterotrophic and autotrophic phases. The nutrient supply is highly variable at this phase which mostly depends on the plant species.

In general, the composition of the medium for culturing immature embryos is more complex than that required by mature embryos which can grow on a simple inorganic medium. Further, the transfer of embryos from one medium to another is frequently needed in order to achieve full development of embryos.

Composition of the medium : Some salient features of medium and culture conditions are listed below

- Inorganic constituents of MS, B5 or White's media are adequate.

TABLE 47.2 A selected list of distant plant species crossed and the resistance traits developed through embryo rescue technique

Distant plant species crossed	*Resistance trait(s)*
Oryza sativa × *O. minuta*	Bacterial blight and blast
Solanum tuberosum × *S. etuberosum*	Potato leaf roll virus
Solanum melanogena × *S. khasianum*	Brinjal shoot and fruit borer
Brassica napus × *B. oleracea*	Cabbage aphid
Lycopersicon esculentum × *L. peruvianum*	Virus, fungi and nematodes
Hordeum sativum × *H. vulgare*	Powdery mildew and spot blotch
Triticum aestivum × *Thinopyrum scirpeum*	Salt tolerance

- Sucrose is most commonly used energy source.
- Ammonium nitrate is the preferred source of nitrogen.
- Casein hydrolysate, rich in various amino acids is frequently used.
- Certain natural plant extracts with ***embryo factor*** promote embryo cultures e.g. liquid endosperm of coconut milk. The embryo factor is believed to supply certain amino acids, sugars, growth regulators etc.
- In general, growth regulators are not required, as they induce callus formation.
- Embryos grow well in the pH range of 5-7.5.
- An incubation temperature of 24–26°C is ideal.
- Better growth of embryos is observed in darkness which are then transferred to light for germination.

During the culture conditions, the embryos are grown into plantlets, and then transferred to sterile soil for full-pledged growth to maturity.

APPLICATIONS OF EMBRYO CULTURE

1. **Prevention of embryo abortion :** Incompatibility barriers in interspecific and intergeneric hybridization programmes leading to embryo abortion can be successfully overcome by embryo rescue. In fact, many distant hybrids have been obtained through embryo rescue techniques. A selected list of distant plant species crossed and the resistance traits developed by employing embryo rescue technique is given in ***Table 47.2***.

2. **Overcoming seed dormancy :** Seed dormancy is caused by several factors—endogenous inhibitors, embryo immaturity, specific light and temperature requirements, dry storage requirements etc. Further, in some plants the natural period of seed dormancy itself is too long. Embryo culture is successfully applied to overcome seed dormancy, and to produce viable seedlings in these plant species.

3. **Shortening of breeding cycle :** Some of the plants in their natural state have long breeding cycles. This is mostly due to seed dormancy attributed to seed coat and/or endosperm. The embryos can be excised and cultured *in vitro* to develop into plants within a short period. For instance, Hollies, a Christmas decoration plant can be ***grown in 2-3 weeks*** time through embryo cultures ***in contrast to 3 years*** period required through seed germination.

4. **Production of haploids :** Embryo culture has been successfully used for the production of haploid (or monoploid) plants e.g. barley.

5. **Overcoming seed sterility :** Certain plant species produce sterile seeds that do not germinate e.g. early ripening varieties of cherry, apricot, plum. Seed sterility is mostly associated with incomplete embryo development which leads to the death of the germinating embryo. Using embryo cultures, it is possible to raise seedlings from sterile seeds of early ripening fruits e.g. apricot, plum.

6. **Clonal propagation :** Embryos are ideally suited for *in vitro* clonal propagation. This is due to the fact that embryos are juvenile in nature with high regenerative potential. Further, it is possible to induce organogenesis and somatic embryogenesis from embryonic tissues.

Germplasm Conservation and Cryopreservation

Germplasm broadly refers to the ***hereditary material*** (total content of genes) transmitted to the offspring through germ cells. Germplasm provides the raw material for the breeder to develop various crops. Thus, conservation of germplasm assumes significance in all breeding programmes.

As the primitive man learnt about the utility of plants for food and shelter, he cultivated the habit of saving selected seeds or vegetative propagules from one season to the next one. In other words, this may be regarded as ***primitive*** but ***conventional germplasm preservation*** and management, which is highly valuable in breeding programmes.

The very ***objective*** of germplasm conservation (or storage) ***is to preserve the genetic diversity of a particular plant or genetic stock for its use at any time in future***. In recent years, many new plant species with desired and improved characteristics have started replacing the primitive and conventionally used agricultural plants. It is important to conserve the endangered plants or else some of the valuable genetic traits present in the primitive plants may be lost.

A global body namely ***International Board of Plant Genetic Resources (IBPGR)*** has been established for germplasm conservation. Its main objective is to provide necessary support for collection, conservation and utilization of plant genetic resources throughout the world.

There are two approaches for germplasm conservation of plant genetic materials.

1. *In-situ* conservation
2. *Ex-situ* conservation

IN-SITU CONSERVATION

The ***conservation*** of germplasm ***in their natural environment*** by establishing biosphere reserves (or national parks/gene sanctuaries) is regarded as *in-situ* conservation. This approach is particularly useful for preservation of land plants in a near natural habitat along with several wild relatives with genetic diversity. The *in-situ* conservation is considered as a high priority germplasm preservation programme.

The major ***limitations*** of *in-situ* conservation are listed below.

- The risk of losing germplasm due to environmental hazards
- The cost of maintenance of a large number of genotypes is very high.

EX-SITU CONSERVATION

Ex-situ conservation is the ***chief method*** for the preservation of germplasm obtained from cultivated and wild plant materials. The ***genetic materials*** in the form of seeds or from *in vitro* cultures (plant cells, tissues or organs) can be ***preserved as gene banks for long term storage*** under suitable conditions. For successful establishment of gene banks, adequate knowledge of genetic structure of plant populations, and the techniques involved in sampling, regeneration, maintenance of gene pools etc. are essential.

Germplasm conservation in the form of seeds

Usually, seeds are the most ***common and convenient materials*** to conserve plant germplasm. This is because many plants are propagated through seeds, and seeds occupy relatively ***small space***. Further, seeds can be ***easily transported*** to various places.

There are however, certain ***limitations*** in the conservation of seeds

- Viability of seeds is reduced or lost with passage of time.
- Seeds are susceptible to insect or pathogen attack, often leading to their destruction.
- This approach is exclusively confined to seed propagating plants, and therefore it is of no use for vegetatively propagated plants e.g. potato, *Ipomoea, Dioscorea*.
- It is difficult to maintain clones through seed conservation.

Certain seeds are heterogeneous and therefore, are not suitable for true genotype maintenance.

In vitro methods for germplasm conservation

In vitro methods employing shoots, meristems and embryos are ideally suited for the conservation of germplasm of vegetatively propagated plants. The plants with recalcitrant seeds and genetically engineered materials can also be preserved by this *in vitro* approach.

There are several advantages associated with *in vitro* germplasm conservation

- Large quantities of materials can be preserved in small space.
- The germplasm preserved can be maintained in an environment, free from pathogens.
- It can be protected against the nature's hazards.
- From the germplasm stock, large number of plants can be obtained whenever needed.
- Obstacles for their transport through national and international borders are minimal (since the germplasm is maintained under aspectic conditions).

There are mainly three approaches for the *in vitro* conservation of germplasm

1. Cryopreservation (freeze-preservation)
2. Cold storage
3. Low-pressure and low-oxygen storage

CRYOPRESERVATION

Cryopreservation (Greek, krayos-frost) literally means ***preservation in the frozen state***. The principle involved in cryopreservation is to bring the ***plant cell and tissue cultures to a zero metabolism or non-dividing state*** by reducing the temperature in the presence of cryoprotectants.

Cryopreservation broadly means the storage of germplasm at very low temperatures.

- Over solid carbon dioxide (at –79°C)
- Low temperature deep freezers (at –80°C)
- In vapour phase nitrogen (at –150°C)
- In liquid nitrogen (at –196°C)

Among these, the most commonly used cryopreservation is by employing ***liquid nitrogen***. At the temperature of liquid nitrogen (–196°C), the cells stay in a completely inactive state and thus can be conserved for long periods. In fact, cryopreservation has been successfully applied for germplasm conservation of a wide range of plant species e.g. rice, wheat, peanut, cassava, sugarcane, straberry, coconut. Several plants can be regenerated from cells, meristems and embryos stored in cryopreservation.

Mechanism of cryopreservation

The technique of freeze preservation is based on the transfer of water present in the cells from a liquid to a solid state. Due to the presence of salts and organic molecules in the cells, the cell water requires much more lower temperature to freeze (even up to –68°C) compared to the freezing point of pure water (around 0°C). When stored at low temperature, the metabolic processes and biological deteriorations in the cells/tissues almost come to a standstill.

Precautions/limitations for successful cryopreservation

Good technical and theoretical knowledge of living plant cells and as well as cryopreservation

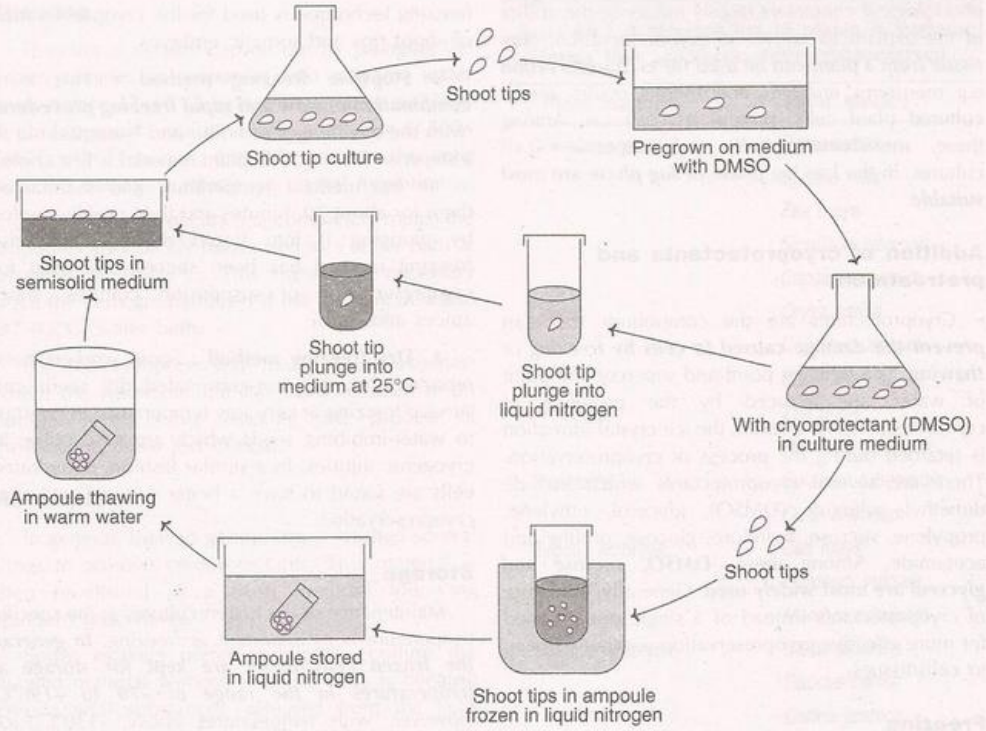


Fig. 48.1 : An outline of the protocol for cryopreservation of shoot tip (DMSO–Dimethyl sulfoxide).

technique are essential. Other ***precautions*** (the limitations that should be overcome) for successful cryopreservation are listed below

- Formation ice crystals inside the cells should be prevented as they cause injury to the organelles and the cell.
- High intracellular concentration of solutes may also damage cells.
- Sometimes, certain solutes from the cell may leak out during freezing.
- Cryoprotectants also affect the viability of cells.
- The physiological status of the plant material is also important.

TECHNIQUE OF CRYOPRESERVATION

An outline of the protocol for cryopreservation of shoot tip is depicted in ***Fig. 48.1***. The cryopreservation of plant cell culture followed by the regeneration of plants broadly involves the following stages

1. Development of sterile tissue cultures
2. Addition of cryoprotectants and pretreatment
3. Freezing
4. Storage
5. Thawing
6. Reculture
7. Measurement of survival/viability
8. Plant regeneration.

The salient features of the above stages are briefly described.

Development of sterile tissue culture

The selection of plant species and the tissues with particular reference to the morphological and

physiological characters largely influence the ability of the explant to survive in cryopreservation. ***Any tissue from a plant can be used for cryopreservation*** e.g. meristems, embryos, endosperms, ovules, seeds, cultured plant cells, protoplasts, calluses. Among these, ***meristematic cells*** and suspension cell cultures, in the late lag phase or ***log phase are most suitable***.

Addition of cryoprotectants and pretreatment

Cryoprotectants are the compounds that can ***prevent the damage caused to cells by freezing*** or ***thawing***. The freezing point and supercooling point of water are reduced by the presence of cryoprotectants. As a result, the ice crystal formation is retarded during the process of cryopreservation. There are several cryoprotectants which include dimethyl sulfoxide (DMSO), glycerol, ethylene, propylene, sucrose, mannose, glucose, proline and acetamide. Among these, ***DMSO, sucrose and glycerol are most widely used***. Generally, a mixture of cryoprotectants instead of a single one, is used for more effective cryopreservation without damage to cells/tissues.

Freezing

The sensitivity of the cells to low temperature is variable and largely depends on the plant species. Four different types of freezing methods are used.

1. **Slow-freezing method :** The tissue or the requisite plant material is slowly frozen at a slow ***cooling rates of 0.5-5°C/min*** from 0°C to –100°C, and then transferred to liquid nitrogen. The advantage of slow-freezing method is that some amount of water flows from the cells to the outside. This promotes extracellular ice fromation rather than intracellular freezing. As a result of this, the plant cells are partially dehydrated and survive better. The slow-freezing procedure is successfully used for the ***cryopreservation of suspension cultures***.

2. **Rapid freezing method :** This technique is quite simple and involves plunging of the vial containing plant material into liquid nitrogen. During rapid freezing, a ***decrease in temperature –300° to –1000°C/min occurs***. The freezing process is carried out so quickly that small ice crystals are formed within the cells. Further, the growth of intracellular ice crystals is also minimal. Rapid freezing technique is used for the cryopreservation of shoot tips and somatic embryos.

3. **Stepwise freezing method :** This is a ***combination of slow and rapid freezing procedures*** (with the advantages of both), and is carried out in a stepwise manner. The plant material is first cooled to an intermediate temperature and maintained there for about 30 minutes and then rapidly cooled by plunging it into liquid nitrogen. Stepwise freezing method has been successfully used for cryopreservation of suspension cultures, shoot apices and buds.

4. **Dry freezing method :** Some workers have reported that the non-germinated dry seeds can survive freezing at very low temperature in contrast to water-imbibing seeds which are susceptible to cryogenic injuries. In a similar fashion, dehydrated cells are found to have a better survival rate after cryopreservation.

Storage

Maintenance of the frozen cultures at the specific temperature is as important as freezing. ***In general, the frozen cells/tissues are kept for storage at temperatures in the range of –70 to –196°C.*** However, with temperatures above –130°C, ice crystal growth may occur inside the cells which reduces viability of cells. Storage is ideally done in liquid nitrogen refrigerator — at 150°C in the vapour phase, or at –196°C in the liquid phase.

The ultimate objective of storage is to stop all the cellular metabolic activities and maintain their viability. ***For long term storge***, temperature at ***–196°C in liquid nitrogen is ideal***. A regular and constant supply of liquid nitrogen to the liquid nitrogen refrigerator is essential. It is necessary to check the viability of the germplasm periodically in some samples.

Proper documentation of the germplasm storage has to be done. The documented information must be comprehensive with the following particulars.

- Taxonomic classification of the material
- History of culture
- Morphogenic potential
- Genetic manipulations done
- Somaclonal variations
- Culture medium
- Growth kinetics

Thawing

Thawing is usually carried out by ***plunging*** the frozen samples in ***ampoules into a warm water*** (temperature 37–45°C) bath with vigorous swirling. By this approach, rapid thawing (at the rate of 500–750°C min^{-1}) occurs, and this protects the cells from the damaging effects ice crystal formation.

As the thawing occurs (ice completely melts) the ampoules are quickly transferred to a water bath at temperature 20–25°C. This transfer is necessary since the cells get damaged if left for long in warm (37–45°C) water bath.

For the cryopreserved material (cells/tissues) where the water content has been reduced to an optimal level before freezing, the process of thowing becomes less critical.

Reculture

In general, thawed germplasm is washed several times to remove cryoprotectants. This material is then recultured in a fresh medium following standard procedures.

Some workers prefer to directly culture the thawed material without washing. This is because certain vital substances, released from the cells during freezing, are believed to promote *in vitro* cultures.

Measurement of survival/viability

The viability/survival of the frozen cells can be measured at any stage of cryopreservation or after thawing or reculture.

The techniques employed to determine viability of cryopreserved cells are the same as used for cell cultures (Refer Chapter 42). ***Staining techniques*** using triphenyl tetrazolium chloride (TTC), Evan's blue and fluorescein diacetate (FDA) are ***commonly used***.

The best indicator to measure the viability of cryopreserved cells is their entry into cell division and regrowth in culture. This can be evaluated by the following expression.

$$\frac{\text{No. of cells/organs growing}}{\text{No. of cells/organs thawed}} \times 100$$

Plant regeneration

The ultimate purpose of cryopreservation of germplasm is to regenerate the desired plant. For appropriate plant growth and regeneration, the cryopreserved cells/tissues have to be carefully nursed, and grown. Addition of certain growth promoting substances, besides maintenance of appropriate environmental conditions is often necessary for successful plant regeneration.

A selected list of plants (in various forms) that have been successfully used for cryopreservation is given in ***Table 48.1***.

TABLE 48.1 A selected list of plants in various forms that are successfully cryopreserved

Plant material	*Plant species*
Cell suspensions	*Oryza sativa*
	Glycine max
	Zea mays
	Nicotiana tabacum
	Capsicum annum
Callus	*Oryza sativa*
	Capsicum annum
	Saccharum sp
Protoplast	*Zea mays*
	Nicotiana tabacum
Meristems	*Solanum tuberosum*
	Cicer arietinum
Zygotic embryos	*Zea mays*
	Hordeum vulgare
	Manihot esculenta
Somatic embryos	*Citrus sinensis*
	Daucus carota
	Coffea arabica
Pollen embryos	*Nicotiana tabacum*
	Citrus sp
	Atropa belladona

COLD STORAGE

Cold storage basically involves germplasm conservation at a ***low and non-freezing temperatures (1-9°C)***. The growth of the plant material is slowed down in cold storage in contrast to complete stoppage in cryopreservation. Hence,

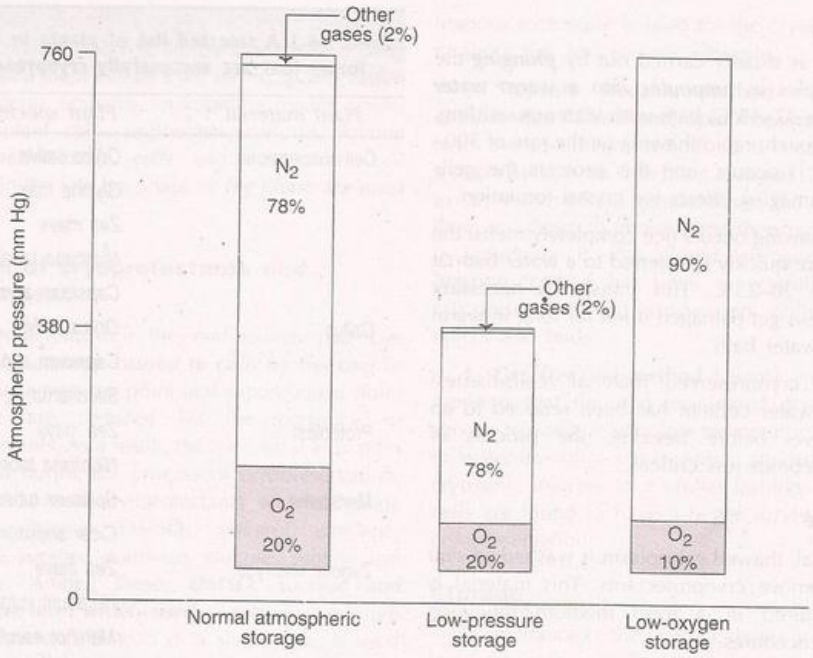


Fig. 48.2 : A graphic representation of tissue culture storage under normal atmospheric pressure, low-pressure, and low-oxygen.

cold storage is regarded as a ***slow growth germplasm conservation*** method. The major advantage of this approach is that the plant material (cells/tissues) is not subjected to cryogenic injuries.

Long-term cold storage is simple, cost-effective and yields germplasm with good survival rate. Many *in vitro* developed shoots/plants of fruit tree species have been successfully stored by this approach e.g. grape plants, strawberry plants. Virus-free strawberry plants could be preserved at 10°C for about 6 years, with the addition of a few drops of medium periodically (once in 2-3 months). Several grape plants have been stored for over 15 years by cold storage (at around 9°C) by transferring them yearly to a fresh medium.

LOW-PRESSURE AND LOW-OXYGEN STORAGE

As alternatives to cryopreservation and cold storage, low-pressure storage (LPS) and low-oxygen storage (LOS) have been developed for germplasm conservation. A graphic representation of tissue culture storage under normal atmospheric pressure, low-pressure and low-oxygen is depicted in ***Fig. 48.2***.

Low-pressure storage (LPS)

In low-pressure storage, the atmospheric pressure surrounding the plant material is reduced. This results in a partial decrease of the pressure exerted by the gases around the germplasm. The ***lowered partial pressure reduces the in vitro growth of plants*** (of organized or unorganized tissues).

Low-pressure storage systems are useful for short-term and long-term storage of plant materials. The short-term storage is particularly useful to increase the shelf life of many plant materials e.g. fruits, vegetables, cut flowers, plant cuttings. The germplasm grown in cultures can be stored for long term under low pressure.

Besides germplasm preservation, LPS reduces the activity of pathogenic organisms and inhibits spore germination in the plant culture systems.

Low-oxygen storage (LOS)

In the low-oxygen storage, the oxygen concentration is reduced, but the atmospheric pressure (260 mm Hg) is maintained by the addition of inert gases (particularly nitrogen).

The partial pressure of oxygen below 50 mm Hg reduces plant tissue growth (organized or unorganized tissue). This is due to the fact that with reduced availability of O_2, the production of CO_2 is low. As a consequence, the ***photosynthetic activity is reduced, thereby inhibiting the plant tissue growth and dimension.***

Limitations of LOS : The long-term conservation of plant materials by low-oxygen storage is likely to inhibit the plant growth after certain dimensions.

APPLICATIONS OF GERMPLASM STORAGE

The germplasm storage has become a boon to plant breeders and biotechnologists. Some of the applications are briefly described.

1. **Maintenance of stock cultures :** Plant materials (cell/tissue cultures) of several species can be cryopreserved and maintained for several years, and used as and when needed. This is in contrast to an *in vitro* cell line maintenance which has to be subcultured and transferred periodically to extend viability. Thus, germplasm storage is an ideal method to avoid subculturing, and maintain cells/tissues in a viable state for many years.

2. Cryopreservation is an ideal method ***for long term conservation of cell cultures*** which produce secondary metabolites (e.g. medicines).

3. Disease (pathogen)-free plant materials can be frozen, and propagated whenever required.

4. Recalcitrant seeds can be maintained for long.

5. Conservation of somaclonal and gameto̲clonal variations in cultures.

6. Plant materials from ***endangered species can be conserved.***

7. Conservation of pollen for enhancing longevity.

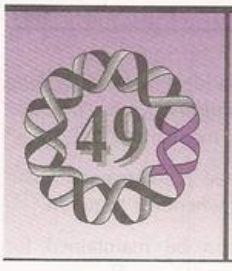
8. Rare germplasms developed through somatic hybridization and other genetic manipulations can be stored.

9. Cryopreservation is a good method for the selection of cold resistant mutant cell lines which could develop into frost resistant plants.

10. Establishment of ***germplasm banks for exchange of information at the international level.***

LIMITATIONS OF GERMPLASM STORAGE

The major limitations of germplasm storage are the ***expensive equipment and the trained personnel.*** It may, however, be possible in the near future to develop low cost technology for cryopreservation of plant materials.



Genetic Engineering of Plants – Methodology

By the ***conventional plant breeding techniques***, significant achievements have been made in the improvement of several food crops. These age-old classical methods, involving gene transfer through sexual and vegetative propagation, ***take very long time***. For instance, about 6-8 years may be required to develop a new rice or a wheat by sexual propagation. Rapid advances in gene structure and function, coupled with the recent developments made in the genetic engineering techniques have dramatically improved the plant breeding methods to yield the desired results in a short period.

Plant genetic transformation technology basically deals with ***the transfer of desirable gene(s) from one plant species to another*** (or insertion of totally new genes) with subsequent integration and expression of the foreign gene(s) in the host genome. The term ***transgene*** is used to ***represent the transferred gene***, and the genetic transformation in plants is broadly referred to as ***plant transgenesis***. The ***genetically transformed*** new plants are regarded as ***transgenic plants***.

The development of transgenic plants is the outcome of an integrated application of recombinant DNA (rDNA) technology, gene transfer-methods and tissue culture techniques. (**Note :** The reader must ***refer Chapters 6 and 7*** for basic information on genetic engineering, and for common techniques employed in rDNA technology).

Why transgenic plants?

It is only through the recombinant approach (genetic engineering) of biotechnology, new genes with desired characters (that may or may not be present in other plants) can be introduced into the plants. Further, it is possible to manipulate the existing genes to make the proteins with suitable alterations e.g. increase in the content of an essential amino acid. The most important ***reasons for developing transgenic plants*** are listed

- To improve agricultural, horticultural or ornamental value of plants.
- To develop ***plant bioreactors*** for inexpensive manufacture of commercially important products e.g. proteins, medicines, pharmaceutical compounds.
- To study the action of genes in plants during development and various biological processes.

Genetic traits introduced into transgenic plants

Since plant cells are ***totipotent*** (i.e. a single plant cell can regenerate into a whole plant), the genetically engineered cells with new gene(s) can produce a transgenic plant. This plant carrying the desired trait will give raise to successive generations. ***Many genetic traits have been introduced*** into plants through genetic engineering

- Resistance to herbicides

Table 49.1 Gene transfer (DNA delivery) methods in plants

Method	*Salient features*
I. Vector–mediated gene transfer	
Agrobacterium (Ti plasmid)-mediated gene transfer	Very efficient, but limited to a selected group of plants
Plant viral vectors	Ineffective method, hence not widely used
II. Direct or vectorless DNA transfer	
(A) Physical methods	
Electroporation	Mostly confined to protoplasts that can be regenerated to viable plants. Many cereal crops developed.
Microprojectile (particle bombardment)	Very successful method used for a wide range of plants/ tissues. Risk of gene rearrangement high.
Microinjection	Limited use since only one cell can be microinjected at a time. Technical personnel should be highly skilled.
Liposome fusion	Confined to protoplasts that can be regenerated into viable whole plants.
Silicon carbide fibres	Requires regenerable cell suspensions. The fibres, however, require careful handling.
(B) Chemical methods	
Polyethylene glycol (PEG)-mediated	Confined to protoplasts. Regeneration of fertile plants is frequently problematical.
Diethylaminoethyl (DEAE) dextran-mediated	Does not result in stable transformants.

- Protection against viral infections
- Insecticidal activity
- Improved nutritional quality
- Altered flower pigmentation
- Tolerance to environmental stresses
- Self incompatibility

Criteria for commercial use of genetically transformed plants

For large-scale commercial application of genetically engineered plants, the following requirements have to be satisfied

- Introduction of desirable gene(s) to all plant cells.
- Expression of cloned genes in the appropriate cells at the right time.
- Stable maintenance of new gene(s) inserted.
- Transmission of new genetic information to subsequent generations.

GENE TRANSFER METHODS

The gene transfer techniques in plant genetic transformation are broadly grouped into two categories

I. Vector-mediated gene transfer

II. Direct or vectorless DNA transfer

The salient features of the commonly used gene (DNA) transfer methods are given in ***Table 49.1***.

VECTOR-MEDIATED GENE TRANSFER

Vector-mediated gene transfer is carried out either by *Agrobacterium*-mediated transformation or by use of plant viruses as vectors.

AGROBACTERIUM-MEDIATED GENE TRANSFER

Agrobacterium tumefaciens is a soil-borne, Gram-negative bacterium. It is rod shaped and motile, and belongs to the bacterial family of

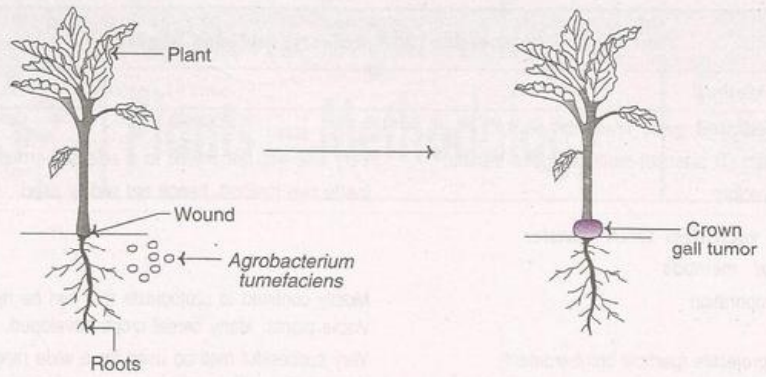


Fig. 49.1 : Formation of a crown gall tumor in a plant infected with Agrobacterium tumefaciens.

Rhizobiaceae. ***A. tumefaciens*** is a phytopathogen, and is treated as the ***nature's most effective plant genetic engineer***. Some workers consider this bacterium as the ***natural expert of interkingdom gene transfer***. In fact, the major credit for the development of plant transformation techniques goes to the natural unique capability of *A. tumefaciens*. Thus, this bacterium is the most beloved by plant biotechnologists.

There are mainly two species of *Agrobacterium* :

- *A. tumefaciens* that induces crown gall disease.
- *A. rhizogenes* that induces hairy root disease.

CROWN GALL DISEASE AND Ti PLASMID

Almost 100 years ago (1907), Smith and Townsend postulated that a bacterium was the causative agent of crown gall tumors, although its importance was recognized much later.

As ***A. tumefaciens*** infects wounded or damaged plant tissues, in induces the formation of a ***plant tumor called crown gall*** (***Fig. 49.1***). The entry of the bacterium into the plant tissues is facilitated by the release of certain phenolic compounds (acetosyringone, hydroxyacetosyringone) by the wounded sites. Crown gall formation occurs when the bacterium releases its ***Ti plasmid*** (tumor-inducing plasmid) into the plant cell cytoplasm. A ***fragment*** (segment) ***of Ti plasmid***, referred to as ***T-DNA***, is actually transferred from the bacterium into the host where it gets integrated into the plant cell chromosome (i.e. host genome). Thus, ***crown gall disease is a naturally evolved genetic engineering process.***

The T-DNA carries genes that code for proteins involved in the biosynthesis of growth hormones (auxin and cytokinin) and novel plant metabolites namely ***opines*** — amino acid derivatives and ***agropines*** — sugar derivatives (***Fig. 49.2***). The growth hormones cause plant cells to proliferate and form the gall while opines and agropines are utilized by *A. tumefaciens* as sources of carbon and energy. As such, opines and agropines are not normally part of the plant metabolism (neither produced nor metabolised). Thus, *A. tumefaciens* genetically transforms plant cells and creates a biosynthetic machinery to produce nutrients for its own use.

As the bacteria multiply and continue infection, grown gall develops which is a visible mass of the accumulated bacteria and plant material. Crown gall formation is the consequence of the transfer, integration and expression of genes of T-DNA (or Ti plasmid) of *A. tumefaciens* in the infected plant. The genetic transformation leads to the formation of crown gall tumors, which interfere with the normal growth of the plant. Several ***dicotyledonous plants (dicots)*** are affected by crown gall disease e.g. grapes, roses, stone-fruit trees.

Organization of Ti plasmid

The Ti plasmids (approximate size 200 kb each) exist as independent replicating circular DNA molecules within the *Agrobacterium* cells. The T-DNA (transferred DNA) is variable in length in

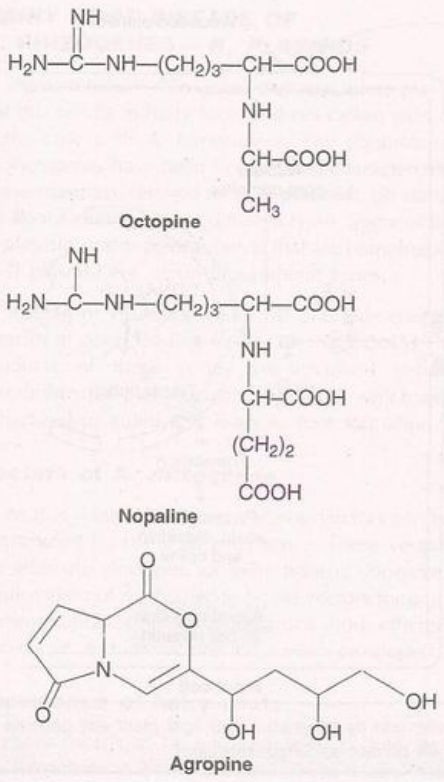


Fig. 49.2 : Structures of three opines–octopine, nopaline, and agropine.

the range of 12 to 24 kb, which depends on the bacterial strain from which Ti plasmids come. Nopaline strains of Ti plasmid have one T-DNA with length of 20 kb while octopine strains have two T-DNA regions referred to as T_L and T_R that are respectively 14 kb and 7 kb in length.

A diagrammatic representation of a Ti plasmid is depicted in ***Fig. 49.3***. The Ti plasmid has three important regions.

1. **T-DNA region :** This region has the genes for the biosynthesis of ***auxin (aux)***, ***cytokinin (cyt)*** and ***opine (ocs)***, and is flanked by left and right borders. These three genes-*aux, cyto* and *ocs* are referred to as ***oncogenes***, as they are the determinants of the tumor phenotype.

T-DNA borders — A set of 24 kb sequences present on either side (right and left) of T-DNA are also transferred to the plant cells. It is now clearly established that the right border is more critical for T-DNA transfer and tumorigenesis.

2. **Virulence region :** The genes responsible for the transfer of T-DNA into the host plant are located outside T-DNA and the rigion is referred to as *vir* or virulence region. *Vir* region codes for proteins involved in T-DNA transfer. At least nine *vir*-gene operons have been identified. These include *vir A, vir G, vir* B_1, *vir* C_1, *vir* D_1, D_2 and D_4, and *vir* E_1 and E_2.

3. **Opine catabolism region :** This region codes for proteins involved in the uptake and metabolisms of opines.

Besides the above three, there is ***ori region*** that is responsible for the origin of DNA replication which permits the Ti plasmid to be stably maintained in *A. tumefaciens*.

T-DNA transfer and integration

The process of T-DNA transfer and it integration into the host plant genome is depicted in ***Fig. 49.4***, and is briefly described.

1. **Signal induction to *Agrobacterium* :** The wounded plant cells release certain chemicals-phenolic compounds and sugars which are recognized as signals by *Agrobacterium*. The signals induced result in a sequence of biochemical events in *Agrobacterium* that ultimately helps in the transfer of T-DNA of T-plasmid.

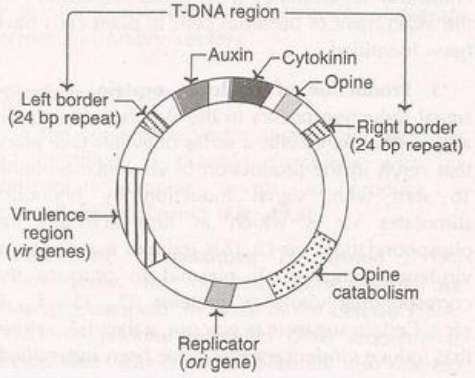


Fig. 49.3 : A dagrammatic representation of a Ti plasmid.

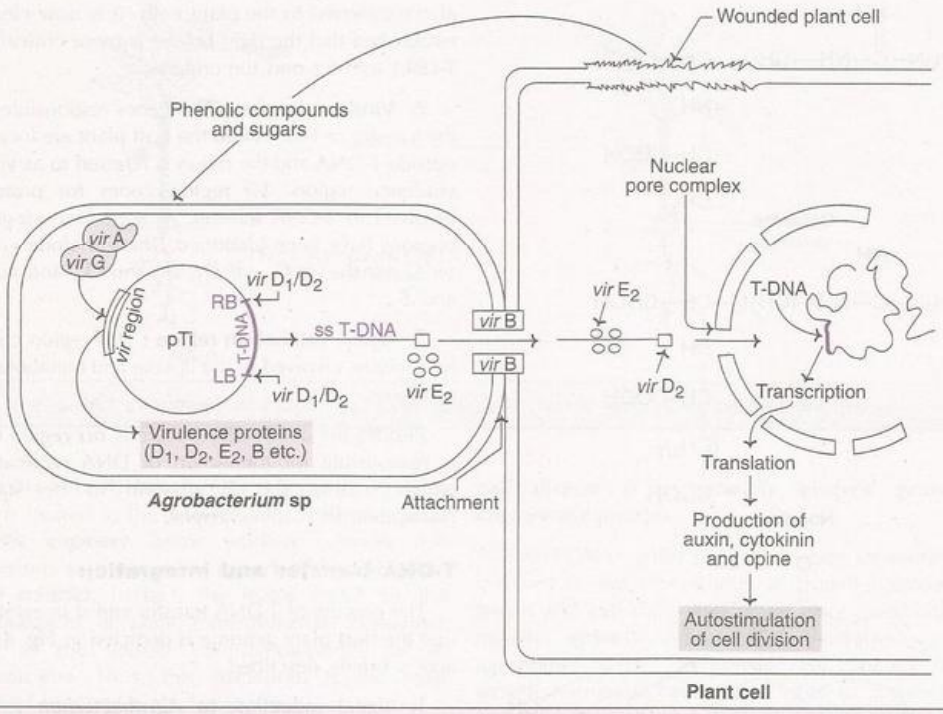


Fig. 49.4 : A diagrammatic representation of T-DNA transfer and its integration into host plant cell genome (pTi-Ti plasmid; RB-Right border; LB-Left border; ss-Single-stranded).

2. **Attachment of *Agrobacterium* to plant cells :** The *Agrobacterium* attaches to plant cells through polysaccharides, particularly cellulose fibres produced by the bacterium. Several chromosomal virulence (*chv*) genes responsible for the attachment of bacterial cells to plant cells have been identified.

3. **Production of virulence proteins :** As the signal induction occurs in the *Agrobacterium* cells attached to plant cells, a series of events take place that result in the production of virulence proteins. To start with, signal induction by phenolics stimulates *vir A* which in turn activates (by phosphorylation) *vir G*. This induces expression of virulence genes of Ti plasmid to produce the corresponding virulence proteins (D_1, D_2, E_2, B etc.). Certain sugars (e.g. glucose, galactose, xylose) that induce virulence genes have been indentified.

4. **Production of T-DNA strand :** The right and left borders of T-DNA are recognized by vir D_1/vir D_2 proteins. These proteins are involved in the production single-stranded T-DNA (ss DNA), its protection and export to plant cells. The ss T-DNA gets attached to vir D_2.

5. **Transfer of T-DNA out of *Agrobacterium* :** The ss T-DNA — vir D_2 complex in association with vir G is exported from the bacterial cell. *Vir B* products form the transport apparatus.

6. **Transfer of T-DNA into plant cells and integration :** The T-DNA-vir D_2 complex crosses the plant plasma membrane. In the plant cells, T-DNA gets covered with vir E_2. This covering protects the T-DNA from degradation by nucleases. vir D_2 and vir E_2 interact with a variety of plant proteins which influences T-DNA transport and integration. The T-DNA-vir D_2-vir E_2 — plant protein complex enters the nucleus through nuclear pore complex. Within the nucleus, the T-DNA gets integrated into the plant chromosome through a process referred to ***illegitimate recombination***. This is different from the homologous recombination, as it does not depend on the sequence similarity.

HAIRY ROOT DISEASE OF A. RHIZOGENES — R_1 PLASMIDS

Agrobacterium rhizogenes can also infect plants. But this results in hairy roots and not crown galls as is the case with *A. tumefaciens*. The plasmids, of *A. rhizogenes* have been isolated and characterized. These plasmids, referred to as ***Ri plasmids***, (Ri stands for ***R***oot ***i***nducing) are of different types. Some of the Ri plasmid strains possess genes that are homologous to Ti plasmid e.g. auxin biosynthetic genes.

Instead of virulence genes, Ri plasmids contain a series of open reading frames on the T-DNA. The products of these genes are involved in the metabolism of plant growth regulators which gets sensetized to auxin and leads to root formation.

Vectors of A. *rhizogenes*

As it is done with *A tumefaciens*, vectors can be constructed by using *A. rhizogenes*. These vectors are alternate strategies for gene transfer. However, employment of *A. rhizogene*-based vectors for plant transformation is not common since more efficient systems of *A. tumefaciens* have been developed.

Importance of hairy roots

Hairy roots can be cultured *in vitro*, and thus are important in plant biotechnology. Hairy root systems are useful for the production of secondary metabolites, particularly pharmaceutical proteins.

Ti PLASMID-DERIVED VECTOR SYSTEMS

The ability of Ti plasmid of *Agrobacterium* to genetically transform plants has been described. It is possible to insert a desired DNA sequence (gene) into the T-DNA region (of Ti plasmid), and then use *A. tumefaciens* to deliver this gene(s) into the genome of plant cell. In this process, Ti plasmids serve as natural vectors. However, there are several ***limitations to use Ti plasmids directly as cloning vectors.***

- Ti plasmids are large in size (200–800 kb). Smaller vectors are preferred for recombinant experiments. For this reason, large segments of DNA of Ti plasmid, not essential for cloning, must be removed.
- Absence of unique restriction enzyme sites on Ti plasmids.
- The phytohormones (auxin, cytokinin) produced prevent the plant cells being regenerated into plants. Therefore auxin and cytokinin genes must be removed.
- Opine production in transformed plant cells lowers the plant yield. Therefore opine synthesizing genes which are of no use to plants should be removed.
- Ti plasmids cannot replicate in *E. coli*. This limits their utility as *E. coli* is widely used in recombinant experiments. An alternate arrangement is to add an origin of replication to Ti plasmid that allows the plasmid to replicate in *E. coli*.

Considering the above limitations, ***Ti plasmid-based vectors with suitable modifications have been constructed***. These vectors are mainly composed of the following components.

1. The right border sequence of T-DNA which is absolutely required for T-DNA integration into plant cell DNA.

2. A multiple cloning site (polylinker DNA) that promotes the insertion of cloned gene into the region between T-DNA borders.

3. An origin of DNA replication that allows the plasmids to multiply in *E. coli*.

4. A selectable marker gene (e.g. neomycin phosphotransferase) for appropriate selection of the transformed cells.

Two types of Ti plasmid-derived vectors are used for genetic transformation of plants— ***cointegrate vectors*** and ***binary vectors***.

Cointegrate vector

In the cointegrate vector system, the disarmed and modified Ti plasmid combines with an intermediate cloning vector to produce a recombinant Ti plasmid (***Fig. 49.5***).

Production of disarmed Ti plasmid : The T-DNA genes for hormone biosynthesis are removed (disarmed). In place of the deleted DNA, a bacterial plasmid (pBR322) DNA sequence is incorporated. This disarmed plasmid, also referred to as ***receptor plasmid***, has the basic structure of T-DNA (right and left borders, virulence genes etc.) necessary to transfer the plant cells.

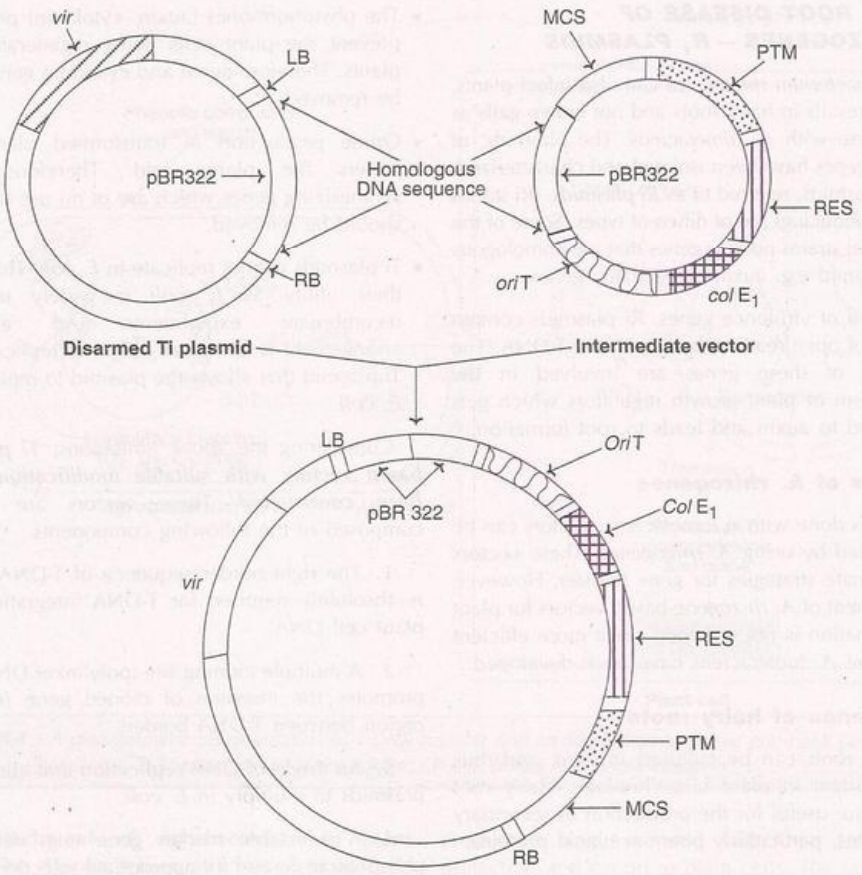


Fig. 49.5 : Cointegrate vector system (vir-Ti plasmid virulence region; pBR322-Bacterial plasmid 322; LB-Left border; RB-Right border; MCS-Multiple cloning site; PTM-Plant transformation marker; RES-Bacterial resistance marker; col E_1-Origin of a replication from col E_1 plasmid; ori T-Origin of transfer site for conjugative plasmid mobilization).

Construction of intermediate vector : The intermediate vector is constructed with the following components.

- A pBR322 sequence ***DNA homologous*** to that found in the receptor Ti plasmid.
- A ***plant transformation marker*** (***PTM***) e.g. a gene coding for neomycin phosphotransferase II (*npt* II). This gene confers resistance to kanamycin in the plant cells and thus permits their isolation.
- A ***bacterial resistance marker*** e.g. a gene coding for spectinomycin resistance. This gene confers spectinomycin resistance to recipient bacterial cells and thus permits their selective isolation.
- A ***multiple cloning site*** (***MCS***) where ***foreign genes*** can be ***inserted***.
- A *Col*E_1 origin of replication which allows the replication of plasmid in *E. coli* but not in *Agrobacterium*.
- An *ori*T sequence with basis of mobilization (*bom*) site for the transfer of intermediate vector from *E. coli* to *Agrobacterium*.

Production and use of cointegrate vectors : The desired foreign gene (target-gene) is first cloned in the multiple cloning site of the intermediate vector.

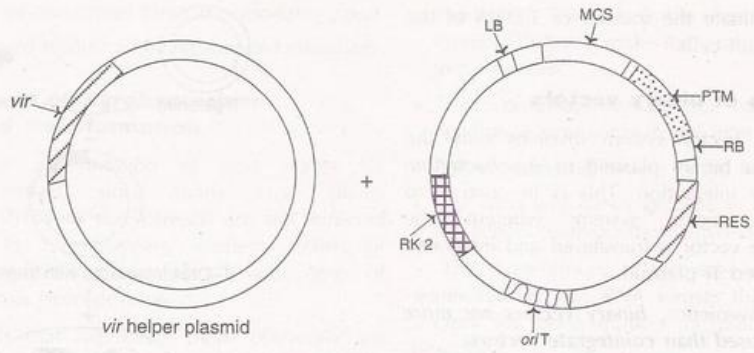


Fig. 49.6 : Binary vector system (vir-Ti plasmid virulence region; LB-Left border; RB-Right border; MCS-Multiple cloning site; PTM-Plant transformation marker; RES-Bacterial resistance marker; oriT-Origin of transfer site for conjugative plasmid mobilization; RK_2-Origin of replication from plasmid).

The cloning process is carried out in *E. coli*, the bacterium where the cloning is most efficient. The intermediate vector is mated with *Agrobacterium* so that the foreign gene is mobilised into the latter. The transformed *Agrobacterium* cells with receptor Ti plasmid and intermediate vector are selectively isolated when grown on a minimal medium containing spectinomycin. The selection process becomes easy since *E. coli* does not grow on a minimal medium in which *Agrobacterium* grows. Within the *Agrobacterium* cells, intermediate plasmid gets integrated into the receptor Ti plasmid to produce cointegrate plasmid. This plasmid containing plant transformation marker (e.g. *npt II*) gene and cloned target gene between T-DNA borders is transferred to plant cells. The transformed plant cells can be selected on a medium containing kanamycin when the plant and *Agrobacterium* cells are incubated together.

Advantages of cointegrate vector

- Target genes can be easily cloned
- The plasmid is relatively small with a number of restriction sites.
- Intermediate plasmid is conveniently cloned in *E. coli* and transferred to *Agrobacterium*.

Binary vector

The binary vector system ***consists of an Agrobacterium strain along with a disarmed Ti plasmid called vir helper*** plasmid (the entire T-DNA region including borders deleted while *vir* gene is retained). It may be noted that both of them are not physically linked (or integrated). A binary vector with T-DNA can replicate in *E. coli* and *Agrobacterium*.

A diagrammatic representation of a typical binary vector system is depicted in ***Fig. 49.6***. The binary vector has the following components.

1. Left and right borders that delimit the T-DNA region.

2. A plant transformation marker (PTM) e.g. *npt II* that confers kanamycin resistance in plant transformed cells.

3. A multiple cloning site (MCS) for introducing target/foreign genes.

4. A bacterial resistance marker e.g. tetracycline resistance gene for selecting binary vector colonies in *E. coli* and *Agrobacterium*.

5. *oriT* sequence for conjugal mobilization of the binary vector from *E. coli* to *Agrobacterium*.

6. A broad host-range origin of replication such as RK_2 that allows the replication of binary vector in *Agrobacterium*.

Production and use of binary vector : The target (foreign) gene of interest is inserted into the multiple cloning site of the binary vector. In this way, the target gene is placed between the right and left border repeats and cloned in *E. coli*. By a mating process, the binary vector is mobilised from *E. coli* to *Agrobacterium*. Now, the virulence gene proteins

of T-DNA facilitate the transfer of T-DNA of the vector into plant cells.

Advantages of binary vectors

- The binary vector system involves only the transfer of a binary plasmid to *Agrobacterium* without any integration. This is in contrast to cointegrate vector system wherein the intermediate vector is transferred and integrated with disarmed Ti plasmid.
- Due to convenience, ***binary vectors are more frequently used than cointegrate vectors.***

PLANT TRANSFORMATION TECHNIQUE USING *AGROBACTERIUM*

Agrobacterium-mediated technique is the ***most widely used*** for the transformation of plants and generation of transgenic plants. The important requirements for gene transfer in higher plants through *Agrobacterium* mediation are listed.

- The explants of the plant must produce phenolic compounds (e.g. autosyringone) for activation of virulence genes.
- Transformed cells/tissues should be capable to regenerate into whole plants.

In general, most of the *Agrobacterium*-mediated plant transformations have the following basic protocol (***Fig. 49.7***)

1. Development of *Agrobacterium* carrying the cointegrate or binary vector with the desired gene.

2. Identification of a suitable explant e.g. cells, protoplasts, tissues, calluses, organs.

3. Co-culture of explants with *Agrobacterium*.

4. Killing of *Agrobacterium* with a suitable antibiotic without harming the plant tissue.

5. Selection of transformed plant cells.

6. Regeneration of whole plants.

Advantages of *Agrobacterium*-mediated transformation

- This is a natural method of gene transfer.
- *Agrobacterium* can conveniently infect any explant (cells/tissues/organs).
- Even large fragments of DNA can be efficiently transferred.

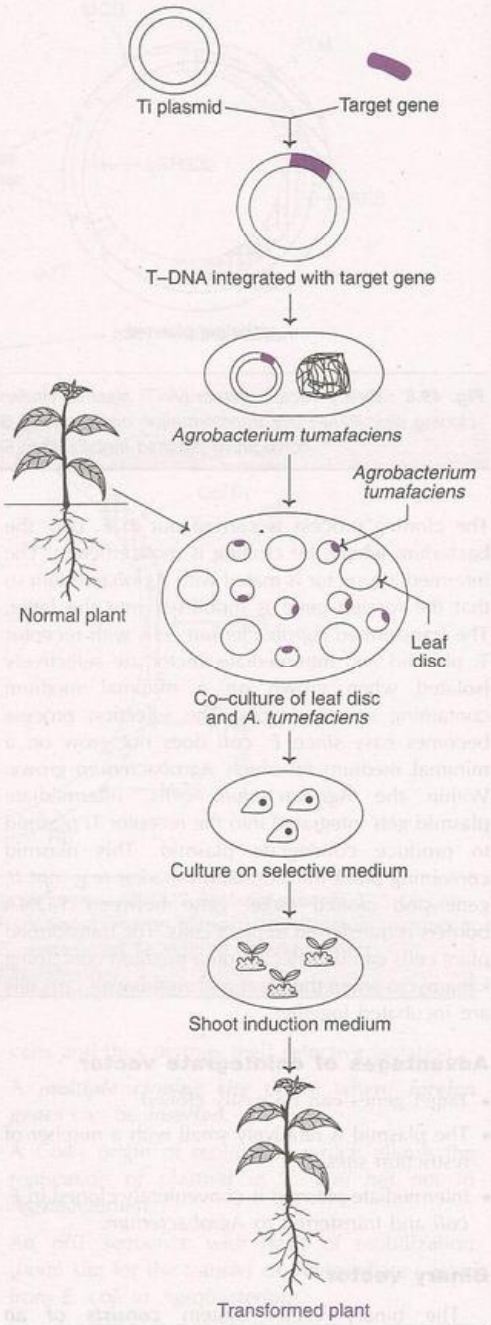


Fig. 49.7 : *Transformation technique using* Agrobacterium*-mediated gene transfer.*

- Stability of transferred DNA is reasonably good.
- Transformed plants can be regenerated effectively.

Limitations of *Agrobacterium*-mediated transformation

- There is a limitation of host plants for *Agrobacterium*, since many crop plants (***monocotyledons*** e.g. cereals) are ***not infected*** by it. In recent years, virulent strains of *Agrobacterium* that can infect a wide range of plants have been developed.
- The cells that regenerate more efficiently are often difficult to transform. e.g. embryonic cells lie in deep layers which are not easy targets for *Agrobacterium*.

VIRUS-MEDIATED GENE TRANSFER (PLANT VIRUSES AS VECTORS)

Plant viruses are considered as ***efficient gene transfer agents*** as they can infect the intact plants and amplify the transferred genes through viral genome replication. Viruses are natural vectors for genetic engineering. They can introduce the desirable gene(s) into almost all the plant cells since the viral infections are mostly systemic.

Plant viruses are nonintegrative vectors

The plant viruses do not integrate into the host genome in contrast to the vectors based on T-DNA of *A. tumefaciens* which are integrative. The viral genomes are suitably modified by introducing desired foreign genes.

These recombinant viruses are transferred, multiplied and expressed in plant cells. They spread systemically within the host plant where the new genetic material is expressed.

Criteria for a plant virus vector

An ideal plant virus for its effective use in gene transfer is expected to posses the following characteristics.

- The virus must be capable of spreading from cell to cell through plasmodesmata.
- The viral genome shoule be able to replicate in the absence of viral coat protein and spread from cell to cell. This is desirable since the insertion of foreign DNA will make the viral genome too big to be packed.
- The recombinant viral genome must elicit little or no disease symptoms in the infected plants.
- The virus should have a broad host range.
- The virus with DNA genome is preferred since the genetic manipulations involve plant DNA.

The three groups of viruses — ***caulimoviruses***, ***geminiviruses*** and ***RNA viruses*** that are used as vectors for gene transfer in plants are briefly described.

CAULIMOVIRUSES AS VECTORS

The caulimoviruses contain circular double-stranded DNA, and are spherical in shape. Caulimoviruses are widely distributed and are responsible for a number of economically important diseases in various crops. The caulimovirus group has around 15 viruses and among these ***cauliflower mosaic virus*** (CaMV) is the most ***important for gene transfer***. The other caulimoviruses include carnation etched virus, dahlia mosaic virus, mirabilis mosaic virus and strawberry vein banding virus.

Cauliflower mosaic virus (CaMV)

CaMV infects many plants (e.g. members of Cruciferae, Datura) and can be easily transmitted, even mechanically. Another attractive feature of CaMV is that the ***infection is systemic***, and large quantities of viruses are found in infected cells.

A diagrammatic view of the CaMV genetic map is depicted in ***Fig. 49.8***. The genome of CaMV consists of a 8 kb (8024 bp) relaxed but tightly packed circular DNA with six major and two minor coding regions. The genes II and VII are not essential for viral infection.

Use of CaMV in gene transfer

For appropriate transmission of CaMV, the foreign DNA must be encapsulated in viral protein. Further, the newly inserted foreign DNA must not interfere with the native assembly of the virus. CaMV genome does not contain any non-coding regions wherein foreign DNA can be inserted. It is fortunate that two genes namely ***gene II and gene VII have no essential functions*** for the virus. It is therefore possible to ***replace one of them and insert the desired foreign gene***.

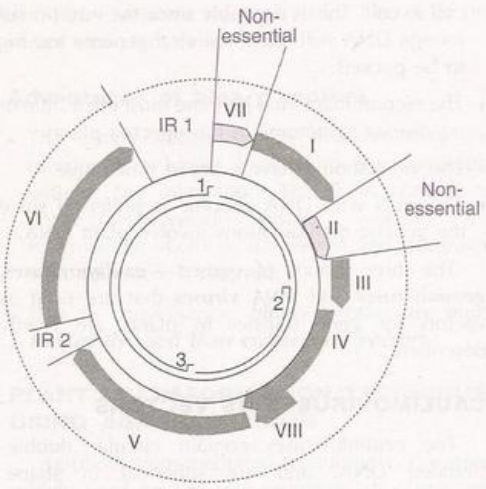


Fig. 49.8 : A diagrammatic representation of the genetic map of cauliflower mosaic virus genome (I...VIII represent coding regions; IR 1and IR 2 are intergeneric regions; The outside dotted circle represents 30S transcript; The two circular lines at the centre indicate viral DNA strands).

Gene II of CaMV has been successfully replaced with a bacterial gene encoding dihydrofolate reductase that provides resistance to methotrexate. When the chimeric CaMV was transmitted to turnip plants, they were systemically infected and the plants developed resistance to methotrexate.

Limitations of CaMV as a vector

- CaMV vector has a limited capacity for insertion of foreign genes.
- Infective capacity of CaMV is lost if more than a few hundred nucleotides are introduced.
- Helper viruses cannot be used since the foreign DNA gets expelled and wild-type viruses are produced.

GEMINIVIRUSES AS VECTORS

The geminiviruses are so named because they have geminate (Gemini literally means heavenly twins) morphological particles i.e. twin and paired capsid structures. These viruses are characterized by possessing one or two single-stranded circular DNAs (ss DNA). On replications, ss DNA forms an intermediate double-stranded DNA.

The geminiviruses can infect a wide range of crop plants (monocotyledons and dicotyledons) which attract plant biotechnologists to employ these viruses for gene transfer. Curly top virus (CTV) and maize streak virus (MSV) and bean golden mosaic virus (BGMV) are among the important geminiviruses.

It has been observed that a large number of replicative forms of a geminivirus genome accumulate inside the nuclei of infected cells. The single-stranded genomic DNA replicates in the nucleus to form a double-stranded intermediate.

Geminivirus vectors can be used to deliver, amplify and express foreign genes in several plants/explants (protoplasts, cultured cells). However, the serious drawback in employing geminiviruses as vectors is that it is very difficult to introduce purified viral DNA into the plants. An alternate arrangement is to take the help of *Agrobacterium* and carry out gene transfer.

RNA PLANT VIRUSES AS VECTORS

There are mainly two types single-stranded RNA viruses.

1. **Monopartite viruses :** These viruses are usually large and contain undivided genomes for all the genetic information e.g. tobacco mosaic virus (TMV).

2. **Multipartite viruses :** The genome in these viruses is divided into small RNAs which may be in the same particle or different particles. e.g. brome mosaic virus (BMV). HMV contains four RNAs divided between three particles.

Plant RNA viruses, in general, are characterized by high level of gene expression, good efficiency to infect cells and spread to different tissues. But the major limitation to use them as vectors is the difficulty of joining RNA molecules *in vitro*.

Use of cDNA for gene transfer

Complementary DNA (cDNA) copies of RNA viruses are prepared *in vitro*. The cDNA so generated can be used as a vector for gene transfer in plants. This approach is tedious and cumbersome. However, some success has been reported. A gene sequence encoding chloramphenicol resistance (enzyme-chloramphenicol acetyltransferase) has been inserted into brome mosaic virus genome. This gene expression, however, has been confined to protoplasts.

LIMITATIONS OF VIRAL VECTORS IN GENE TRANSFER

The ultimate objective of gene transfer is to ***transmit the desired genes to subsequent generations***. With virus vectors, this is ***not possible*** unless the virus is seed-transmitted. However, in case of vegetatively propagated plants, transmission of desired traits can be done e.g. potatoes. Even in these plants, there is always a risk for the transferred gene to be lost anytime.

For the reasons referred above, plant biotechnologists prefer to insert the desired genes of interest into a plant chromosome.

DIRECT OR VECTORLESS DNA TRANSFER

The term direct or vectorless transfer of DNA is used when the ***foreign DNA is directly introduced into the plant genome***. Direct DNA transfer methods rely on the delivery of naked DNA into the plant cells. This is in contrast to the *Agrobacterium* or vector-mediated DNA transfer which may be regarded as indirect methods.

Majority of the direct DNA transfer methods are simple and effective. And in fact, several transgenic plants have been developed by this approach.

Limitations of direct DNA transfer

The major disadvantage of direct gene transfer is that the frequency of ***transgene rearrangements is high***. This results in higher transgene copy number, and ***high frequencies of gene silencing*** (described on p. 593).

Types of direct DNA transfer

The direct DNA transfer can be broadly divided into three categories.

1. ***Physical gene transfer methods***—electroporation, particle bombardment, microinjection, liposome fusion, silicon carbide fibres.

2. ***Chemical gene transfer methods***—Polyethylene glycol (PEG)-mediated, diethyl aminoethyl (DEAE) dextran-mediated, calcium phosphate precipitation.

3. DNA imbibition by cells/tissues/organs.

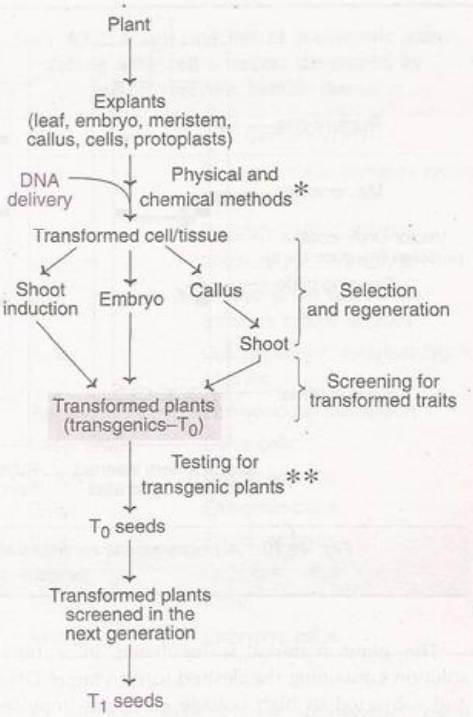


*Fig. 49.9 : An overview of the protocol for the production of transgenic plants using direct DNA delivery methods (*Electroporation, microinjection, macroinjection, bombardment, etc.; **Polymerase chain reaction, Southern hybridization)*

The salient features of the different methods for direct DNA transfer are given in ***Table 49.1*** (See p. 573). The commonly used methods are briefly described in the following pages.

PHYSICAL GENE TRANSFER METHODS

An overview of the general scheme for the production of transgenic plants by employing physical transfer methods is depicted in ***Fig. 49.9***. Some details of the different techniques are described.

ELECTROPORATION

Electroporation basically involves the use of high field strength ***electrical impulses to reversibly permeabilize the cell membranes for the uptake of DNA***. This technique can be used for the delivery of DNA into intact plant cells and protoplasts.

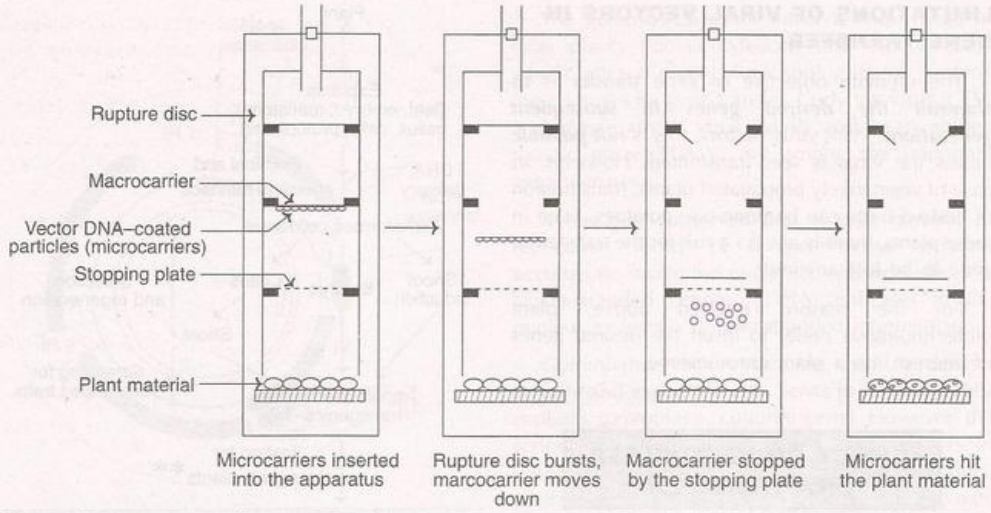


Fig. 49.10 : A diagrammatic representation of particle bombardment (biolistics) system for gene transfer in plants.

The plant material is incubated in a buffer solution containing the desired foreign/target DNA, and subjected to high voltage electrical impulses. This results in the formation of pores in the plasma membrane through which DNA enters and gets integrated into the host cell genome. In the early years, only protoplasts were used for gene transfer by electroporation. Now a days, intact cells, callus cultures and immature embryos can be used with suitable pre- and post-electroporation treatments. Electroporation has been successfully used for the production of transgenic plants of many cereals e.g. rice, wheat, maize.

Advantages of electroporation

- This technique is simple, convenient and rapid, besides being cost-effective.
- The transformed cells are at the same physiological state after electroporation.
- Efficiency of transformation can be improved by optimising the electrical field strength, and addition of spermidine.

Limitations of electroporation

- Under normal conditions, the amount of DNA delivered into plant cells is very low.
- Efficiency of electroporation is highly variable depending on the plant material and the treatment conditions.
- Regeneration of plants is not very easy, particularly when protoplasts are used.

PARTICLE BOMBARDMENT (BIOLISTICS)

Particle (or ***microprojectile*** ***bombardment*** is the most effective method for gene transfer, and creation of transgenic plants. This method is versatile due to the fact that it can be successfully used for the DNA transfer in mammalian cells and microorganisms.

The microprojectile bombardment method was initially named as biolistics by its inventor Sanford (1988). ***Biolistics*** is a combination of ***bio***logical and bal***listics***. There are other names for this technique- ***particle gun***, ***gene gun***, ***bioblaster***.

A diagrammatic representation of micro-projectile bombardment system for the transfer of genes in plants is depicted in ***Fig. 49.10***, and briefly described below.

Microcarriers (microprojectiles), the tungsten or gold particles coated with DNA, are carried by macrocarriers (macroprojectiles). These macro-

carriers are inserted into the apparatus and pushed downward by rupturing the disc. The stopping plate does not permit the movement of macrocarrier while the microcarriers (with DNA) are propelled at a high speed into the plant material. Here the DNA segments are released which enter the plant cells and integrate with the genome.

Plant material used in bombardment

Two types of plant tissue are commonly used for particle bombardment.

1. ***Primary explants*** which can be subjected to bombardment that are subsequently induced to become embryogenic and regenerate.

2. ***Proliferating embryonic tissues*** that can be bombarded in cultures and then allowed to proliferate and regenerate.

In order to protect plant tissues from being damaged by bombardment, cultures are maintained on high osmoticum media or subjected to limited plasmolysis.

Transgene integration in bombardment

It is believed (based on the gene transfer in rice by biolistics) that the gene transfer in particle bombardment is a two stage process.

1. In the ***preintegration phase***, the vector DNA molecules are spliced together. This results in fragments carrying multiple gene copies.

2. ***Integrative phase*** is characterized by the insertion of gene copies into the host plant genome.

The integrative phase facilitates further transgene integration which may occur at the same point or a point close to it. The net result is that particle bombardment is frequently associated with high copy number at a single locus. This type of single locus may be beneficial for regeneration of plants.

The success of bombardment

The particle bombardment technique was first introduced in 1987. It has been successfully used for the transformation of many cereals. e.g. rice, wheat, maize. In fact, the ***first commercial genetically modified (GM) crops such as maize*** containing ***Bt-toxin gene were developed*** by this approach.

Table 49.2 A selected list of transgenic plants (along with cell sources) developed by microprojectile bombardment

Plant	*Cell source(s)*
Rice	Embryonic callus, immature zygotic embryos
Wheat	Immature zygotic embryos
Sorghum	Immature zygotic embryos
Corn	Embryonic cell suspension, immature zygotic embryos
Barley	Cell suspension, immature zygotic embryos
Banana	Embryonic cell suspension
Sweet potato	Callus cells
Cotton	Zygotic embryos
Grape	Embryonic callus
Peas	Zygotic embryos
Peanut	Embryonic callus
Tobacco	Pollen
Alfalfa	Embryonic callus

A selected list of the ***transgenic plants*** (developed by biolistics) along with the sources of the plant materials used is given in ***Table 49.2***.

Factors affecting bombardment

Several attempts are made to study the various factors, and optimize the system of particle bombardment for its most efficient use. Some of the important parameters are described.

Nature of microparticles : Inert metals such as tungsten, gold and platinum are used as microparticles to carry DNA. These ***particles with relatively higher mass will have a better chance*** to move fast when bombarded and ***penetrate the tissues***.

Nature of tissues/cells : The target cells that are capable of undergoing division are suitable for transformation. Some more details on the choice of plant material used in bombardment are already given.

Amount of DNA : The transformation may be low when too little DNA is used. On the other hand, too much DNA may result is high copy number and rearrangement of transgenes.

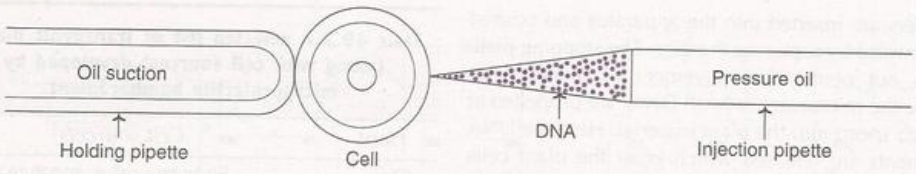


Fig. 49.11 : Microinjection of DNA by holding pipette method.

Therefore, the quantity of ***DNA used should be balanced***. Recently, some workers have started using the chemical aminosiloxane to coat the microparticles with low quantities of DNA adequate enough to achieve high efficiency of transformation.

Environmental parameters : Many environmental variables are known to influence particle bombardment. These factors (temperature, humidity, photoperiod etc.) influence the physiology of the plant material, and consequently the gene transfer. It is also observed that some explants, after bombardment may require special regimes of light, humidity, temperature etc.

The technology of particle bombardment has been improved in recent years, particularly with regard to the use of equipment. A ***commercially produced particle bombardment apparatus*** namely ***PDS-1000/HC*** is widely used these days.

Advantages of particle bombardment

- Gene transfer can be efficiently done in organized tissues.
- Different species of plants can be used to develop transgenic plants.

Limitations of particle bombardment

- The major complication is the production of high transgene copy number. This may result in instability of transgene expression due to ***gene silencing***.
- The target tissue may often get damaged due to lack of control of bombardment velocity.
- Sometimes, undesirable chimeric plants may be regenerated.

MICROINJECTION

Microinjection is a ***direct physical method involving the mechanical insertion of the desirable DNA into a target cell***. The target cell may be the one identified from intact cells, protoplasts, callus, embryos, meristems etc. Microinjection is used for the transfer of cellular organelles and for the manipulation of chromosomes.

The technique of microinjection involves the transfer of the gene through a micropipette (0.5–10.0 µm tip) into the cytoplasm/nucleus of a plant cell or protoplast. While the gene transfer is done, the recipient cells are kept immobilized in agarose embedding, and held by a suction holding pipette (***Fig. 49.11***).

As the process of microinjection is complete, the transformed cell is cultured and grown to develop into a transgenic plant. In fact, transgenic tobacco and *Brassica napus* have been developed by this approach.

The major limitations of microinjection are that it is slow, expensive, and has to be performed by trained and skilled personnel.

LIPOSOME-MEDIATED TRANSFORMATION

Liposomes are artificially created lipid vesicles containing a phospholipid membrane. They are successfully used in mammalian cells for the delivery of proteins, drugs etc. Liposomes carrying genes can be employed to fuse with protoplasts and transfer the genes. The efficiency of transformation increases when the process is carried out in conjunction with polyethylene glycol (PEG).

Liposome-mediated transformation involves ***adhesion of liposomes to the protoplast surface, its fusion at the site of attachment and release of plasmids inside the cell*** (***Fig. 49.12***).

Advantages of liposome fusion

- Being present in an encapsulated form of liposomes, DNA is protected from environmental insults and damage.

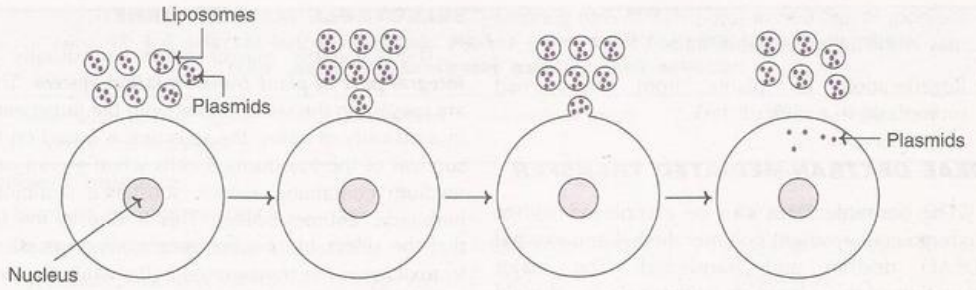


Fig. 49.12 : A diagrammatic representation of fusion of plasmid–filled liposomes with protoplasts.

- DNA is stable and can be stored for sometime in liposomes prior to transfer.
- Applicable to a wide range of plant cells.
- There is good reproducibility in the technique.

Limitations of liposome fusion

The major problem with liposome-mediated transformation is the difficulty associated with the regeneration of plants from transformed protoplasts.

SILICON CARBIDE FIBRE-MEDIATED TRANSFORMATION

The silicon carbide fibres (***SCF***) are about 0.3-0.6 µm in diameter and 10–100 µm in length. These fibres are capable of penetrating the cell wall and plasma membrane, and thus can deliver DNA into the cells.

The DNA coated silicon carbide fibres are vortexed with plant material (suspension culture, calluses). During the mixing, DNA adhering to the fibres enters the cells and gets stably integrated with the host genome. The silicon carbide fibres with the trade name ***Whiskers*** are available in the market.

Advantages of SCF-mediated transformation

- Direct delivery of DNA into intact walled cells. This avoids the protoplast isolation.
- Procedure is simple and does not involve costly equipment.

Disadvantages of SCF-mediated transformation

- Silicon carbide fibres are cancinogenic and therefore have to be carefully handled.
- The embryonic plant cells are hard and compact and are resistant to SCF penetration.

In recent years, some improvements have been made in SCF-mediated transformation. This has helped in the transformation of rice, wheat, maize and barley by using this technique.

CHEMICAL GENE TRANSFER METHODS

POLYETHYLENE GLYCOL-MEDIATED TRANSFORMATION

Polyethylene glycol (PEG), in the presence of divalent cations (using Ca^{2+}), ***destabilizes the plasma membrane of protoplasts and renders it permeable to naked DNA***. In this way, the DNA enters nucleus of the protoplasts and gets integrated with the genome.

The procedure involves the isolation of protoplasts and their suspension, addition of plasmid DNA, followed by a slow addition of 40% PEG-4000 (w/v) dissolved in mannitol and calcium nitrate solution. As this mixture is incubated, protoplasts get transformed.

Advantages of PEG-mediated transformation

- A large number of protoplasts can be simultaneously transformed.
- This technique can be successfully used for a wide range of plant species.

Limitations of PEG-mediated transformation

- The DNA is susceptible for degradation and rearrangement.

- Random integration of foreign DNA into genome may result in undesirable traits.
- Regeneration of plants from transformed protoplasts is a difficult task.

DEAE DEXTRAN-MEDIATED TRANSFER

The desirable DNA can be complexed with a high molecular weight polymer diethyl amino ethyl (DEAE) dextran and transferred. The major limitation of this approach is that it does not yield stable transformants.

CALCIUM PHOSPHATE CO-PRECIPITATION-MEDIATED TRANSFER

The DNA is allowed to mix with calcium chloride solution and isotonic phosphate buffer to form DNA-calcium phosphate precipitate. When the actively dividing cells in culture are exposed to this precipitate for several hours, the cells get transformed. The success of this method is dependent on the high concentration of DNA and the protection of the complex precipitate. Addition of ***dimethyl sulfoxide*** (DMSO) ***increases the efficiency of transformation***.

DNA IMBIBITION BY CELLS/TISSUES

Some workers have seriously tried to transform cells by incubating cell suspensions, tissues, embryos and even seeds with DNA. The belief is that the DNA gets imbibed, and the cells get transformed. DNA imbibition approach has met with little or no success.

MARKER GENES FOR PLANT TRANSFORMATION

In general, genetic transformation of plants is a low-frequency event. Some ***methods for selecting the transformed plant materials*** (cells/tissues) have been devised ***by using a set of genes*** referred to as marker genes. These marker genes are introduced into the plant material along with the target gene. The marker genes are of two types.

I. Selectable marker genes

II. Reporter genes

SELECTABLE MARKER GENES

The selectable marker genes are usually an ***integral part of plant transformation system***. They are present in the vector along with the target gene. In a majority of cases, the selection is based on the survival of the transformed cells when grown on a medium containing a toxic substance (antibiotic, herbicide, antimetabolite). This is due to the fact that the selectable marker gene confers resistance to toxicity in the transformed cells, while the non-transformed cells get killed.

A large number of selectable marker genes are available and they are grouped into three categories—antibiotic resistance genes, antimetabolite marker genes, and herbicide resistance genes (***Table 49.3***).

ANTIBIOTIC RESISTANCE GENES

In many plant transformation systems, antibiotic resistance genes (particularly of *E. coli*) are used as selectable markers. Despite the plants being eukaryotic in nature, antibiotics can effectively inhibit the protein biosynthesis in the cellular organelles, particularly in chloroplasts. Some of the antibiotic resistance selectable marker genes are briefly described.

Neomycin phosphotransferase II (*npt II* gene)

The ***most widely used selectable marker*** is *npt II* gene encoding the enzyme neomycin phosphotransferase II (NPT II). This marker gene confers resistance to the antibiotic kanamycin. The transformants and the plants derived from them can be checked by applying kanamycin solution and the resistant progeny can be selected.

Hygromycin phosphotransferase (*hpt* gene)

The antibiotic hygromycin is more toxic than neomycin and therefore can kill non-transformed plant cells much faster. Hygromycin phosphotransferase (*hpt*) gene thus provides resistance to transformed cells.

Aminoglycoside adenyltransferase (*aadA* gene)

Aminoglycoside 3′-adenyltransferase (*aadA*) gene confers resistance to transformed plant cells against the antibiotics streptomycin and spectionomycin.

TABLE 49.3 A selected list of selectable marker genes used for gene transfer in plants, their source and substrates used for their selection

Selectable marker gene (encoded enzyme)	*Abbreviation*	*Source of gene*	*Substrate(s) used for selection*
Antibiotic resistance			
Neomycin phosphotransferase II	*nptII*	*E. coli*	Kanamycin, geneticin (G418)
Neomycin phosphotransferase IiI	*nptIII*	*Streptococcus faecalis*	Kanamycin, geneticin (G418)
Hygromycin phosphotransferase	*hpt/hyg*	*E. coli*	Hygromycin
Bleomycin resistance	*ble*	*E. coli*	Bleomycin
Aminoglycoside adenyltransferase	*aadA*	*Shigella flexneri*	Streptomycin, spectinomycin
Antimetabolite markers			
Dihydrofolate reductase	*dhfr*	Mouse	Methotrexate
Dihydropteroate synthase	*dhps/sul*	*E. coli*	Sulfonamides
Herbicide resistance			
Phosphinothricin acetyltransferase	*bar/pat*	*Streptomyces hygroscopicus/ S. viridochromogenes*	Glufosinate, L-phosphinothricin, Bialophos
Enolpyruvyl shikimate phosphate synthase	*epsps/aroA*	*Agrobacterium* sp/ *Petunia hybrida*	Glyphosate
Acetolactase synthase	*als*	*Arabidopsis* sp/maize/tobacco	Sulfonylureas
Glyphosate oxidoreductase	*gox*	*Achromobacter LBAA*	Glyphosate
Bromoxynil nitrilase	*bxn*	*Klebsiella pneumoniae*	Bromoxynil
Others			
β-Glucuronidase	*gus/uidA*	*E. coli*	Cytokinin glucuronide
Xylose isomerase	*xylA*	*Thermoanaerobcterium thermosulfurogenes*	Xylose
Mannose 6-phosphate isomerase	*pmi/manA*	*E. coli*	Mannose
Betaine aldehyde dehydrogenase	*badh*	Spinach	Betaine aldehyde

ANTIMETABOLITE MARKER GENES

Dihydrofolate reductase (*dhfr* gene)

The enzyme dihydrofolate reductase, produced by *dhfr* gene is inhibited by the antimetabolite methotrexate. A mutant *dhfr* gene in mouse that codes for this enzyme which has a low affinity to methotrexate has been identified. This *dhfr* gene fused with CaMV promoter results in a methotrexate resistant marker which can be used for the selection of transformed plants.

HERBICIDE RESISTANCE MARKERS

Genes that confer resistance to herbicides are in use as markers for the selection of transgenic plants.

Phosphinothricin acetytransferase (*pat/bar* gene)

Bialophos, phosphinothricin and glufosinate are commonly used herbicides. The *pat/bar* genes code for phosphinothricin acetyltransferase which converts these herbicides into acetylated forms that are non-herbicidal. Thus, *pat/bar* genes confer resistance to the transformed plants.

Enolpyruvylshikimate phosphate synthase (*epsps/aroA* genes)

The herbicide glyphosate inhibits photosynthesis. It blocks the activity of enolpyruvylshikimate phosphate (EPSP) synthase, a key enzyme involved in the biosynthesis of phenylalanine, tyrosine and

tryptophan. Mutant strains of *Agrobacterium* and *Petunia hybrida* that are resistant to glyphosate have been identified. The genes *epsps/aroA* confer resistance to transgenic plants which can be selected.

Bromoxynil nitrilase (*bxn* gene)

The herbicide bromoxynil inhibits photosynthesis (photosystem II). Bromoxynil nitrilase enzyme coded by the gene *bxn* inactivates this herbicide. The gene *bxn* can be successfully used as a selectable marker for the selection of transformed plants.

PRODUCTION OF MARKER-FREE TRANSGENIC PLANTS

There is a growing ***concern among the public*** regarding the use of ***antibiotic or herbicide resistance genes as selectable markers*** of plant transformation

- The products of some marker genes may be toxic or allergic.
- The antibiotic resistance might be transferred to pathogenic microorganisms in the soil.
- There is a possibility of creation of superweeds that are resistant to normally used herbicides.
- A transgenic plant with selectable marker genes cannot be transformed again by using the same selectable markers.

In light of the apprehensions listed above, the public is concerned about the safety of transgenic technology, particularly related to the selectable marker genes (antibiotic/herbicide resistance genes). There are fears about the safety of consumption of foodstuffs derived from genetically engineered plants. This is despite the fact that ***so far none of the marker genes have been shown to adversely affect human, animal or environmental safety***.

CLEAN GENE TECHNOLOGY

The process of ***developing transgenic plants without the presence of selectable marker genes*** or by use of more acceptable marker genes is regarded as clean gene technology. And this will result in the production of many marker-free transgenic plants that will be readily acceptable by the public. Some of the approaches for clean gene technology are given.

Avoiding selectable marker genes

Theoretically, it is possible to totally avoid marker genes and introduce only the transgene of interest. The transformed palnts can then be screened by an advanced technique like polymerase chain reaction and the desirable plants selected. This approach is not practicable due to cost factor.

Cotransformation with two DNAs

The transgenic plants can be produced by employing two separate DNAs — one carrying the desired target gene and the other the marker gene. The transformed plants contain both the genes, but at different sites on the chromosomal DNA. Traditional breeding techniques (a few rounds) can be used to get rid of the transgenic plants with selectable markers.

Removal of selectable markers

It is possible to selectively remove the selectable marker genes from the plant genome. For this purpose, ***site-specific recombinase systems*** are utilized. Several recombinase systems are in fact available which can be used to selectively excise the marker genes from the plant genome.

Cloning of selectable markers between transposable elements

A selectable marker gene can be cloned between plant transposable elements (Ds elements) and then inserted. The selectable marker is planked by the sequences that increase the intrachromosomal recombination. This results in the excision of the marker gene.

REPORTER GENES

A reporter gene may be regarded as the ***test gene whose expression can be quantified***. The plant transformation can be assessed by the expression of reporter genes (also called as screenable or scoreable genes). In general, an assay for the reporter gene is carried out by estimating the quantity of the protein it produces or the final products formed.

A selected list of the reporter genes along with the detection assays is given in ***Table 49.4***, some of the important ones are discussed below.

TABLE 49.4 A selected list of reporter genes used for gene transfer in plants, their sources, and detection assays

Reporter gene (enzyme/protein encoded)	*Abbreviation*	*Source of gene*	*Detection assay*
Octopine synthase	*ocs*	*Agrobacterium tumefaciens*	Electrophoresis, chromatography
Nopaline synthase	*nos*	*Agrobacterium tumefaciens*	Electrophoresis, chromatography
β-Glucuronidase	*gus/uidA*	*E. coli*	Fluorometric or histochemical or colorimetric
Green fluorescent protein	*gfp*	*Aequorea victoria* (jelly fish)	Fluorescence
Luciferase (bacterial)	*luxA/luxB*	*Vibrio harveyi*	Bioluminescence
Luciferase (firefly)	*luc*	*Photonus pyralis*	Bioluminescence
Chloramphenicol acetyltransferase	*cat*	*E. coli*	Autoradiography

Opine synthase (*ocs, nos* genes)

The common opines present in T-DNA of Ti or Ri plasmids of *Agrobacterium* are octopine and nopaline, respectively produced by the synthase genes *ocs* and *nos*. The transformed status of the plant cells can be easily detected by the presence of these opines. ***Opines*** can be ***separated*** by electrophoresis ***and identified***. Alternately, the enzyme activities responsible for the production of opines can also be assayed.

β-Glucuronidase (*gus/uidA* gene)

β-Glucuronidase producing gene (*gus/uidA*) is the ***most commonly used reporter gene*** in assessing plant transformation for the following reasons.

- β-Glucuronidase assays are very sensitive.
- Quantitative estimation of the enzyme can be done by fluorometric method (using substrate 4-methylumbelliferryl β-D-glucuronide which is hydrolysed to 4-methylumbelliferone).
- Qualitative data on the enzyme can be obtained by histochemical means (enzyme localization can be detected by chromogenic substance such as substrate X-gluc).
- No need to extract and identify DNA.

Green fluorescent protein (*gfp* gene)

Green fluorescent protein (GFP), coded by *gfp* gene, is being widely used in recent years. In fact, in many instances, GFP has replaced GUS since ***assays of GFP are easier and non-destructive***. Thus, screening of even the primary transplants can be done by GFP which is not possible with other reporter genes.

Gene for GFP has been isolated from jelly fish *Aequorea victoria* which is a luminescent organism. The original *gfp* gene has been significantly modified to make it more useful as a reporter gene. GFP emits fluorescence which can be detected under a fluorescent microscope.

Bacterial luciferase (*luxA/luxB* genes)

The bacterial luciferase genes (*luxA* and *luxB*) have originated from *Vibrio harveyi*. They can be detected in some plant transformation vectors. The ***detection assay*** of the enzyme is based on the principle of ***bioluminescence***. Bacterial luciferase catalyses the oxidation of long-chain fatty aldehydes that results in the emission of light which can be measured.

Firefly luciferase (*luc gene*)

The enzyme firefly luciferase, encoded by the gene *luc*, catalyses the oxidation of D-luciferin (ATP dependent) which results in the ***emission of light*** that can be detected by sensitive luminometers.

The firefly luciferase gene, however, is not widely used as a marker gene since the assay of the enzyme is rather cumbersome.

Chloramphenicol acetyl transferase (*cat* gene)

The *cat* gene producing chloramphenicol acetyl transferase (CAT) is a widely used reporter gene in mammalian cells. Due to the availability of GUS and GFP reporter systems for plant transformants, CAT is not commonly used. However, some workers continue to use CAT by a sensitive radioactive assay, for the detection of the reporter gene *cat*.

PROMOTERS AND TERMINATORS

The ultimate objective of gene transfer is its correct expression to provide the desired character/trait. The ***appropriate expression of genes is made possible by the presence of promoters and terminators***. The DNA sequence upstream the coding region is the promoter while the terminator is the sequence at the 3′ terminus. The promoters are responsible for the commencement of transcription whereas the terminators ensure the ceasation of transcription at the correct position.

Promoters possess certain inherent characters such as promoter strength, tissue specificity and developmental regulation which determine the efficiency of promoter function in gene expression.

Agrobacterium-derived promoters and terminators

The genes coding for ***napaline synthase*** (*nos*) in Ti plasmid of *Agrobacterium* are frequently used as promoters and terminators in plant transformation vectors. Originally derived from bacteria, the genes coding for opine synthesis are well adapted to function in plants. In fact, *nos* promoter is regarded as constitutive by many plant biotechnologists.

The 30S promoter of CaMV

The cauliflower mosaic virus (CaMV) 30S promoter is the ***most widely used as a promoter in plants***. This promoter, (a 30S RNA gene) is expressed in almost all the tissues of the transgenic plants. The driving strength of 30S promoter is much higher in dicots compared to monocots.

The 30S promoter is preferred for the appropriate expression of sclectable marker genes and reporter genes. In recent years, the efficiency of 30S promoter has been further increased by suitable modifications in the enhancer region.

Promoters for monocots : Since 30S promoter is not very efficient for monocots, alternates are used. ***Ubiquitin I promoter*** or the ***rice actin promoter*** are frequently used for high expression of transgenes in monocots.

Inducible promoters

In recent years, certain inducible promoter systems for transgene expression have been developed. These promoters differ from 30S promoter which is constitutive in nature. There are mainly three inducible expression systems :

1. **Non-plant-derived systems :** These inducible promoters are independent of the normal plant processes. A specific exogenous chemical will induce their expression. e.g. tetracycline, ethanol, steroid, copper. Non-plant-derived inducible promoters are useful, but not economically viable.

2. **Plant-derived systems based on response to environmental signals :** These systems are not independent, since they are plant-derived and actually form a part of normal plant processes. Plant-derived systems respond to a variety of environmental signals e.g. wound, heat shock.

3. **Plant-derived systems based on developmental control of gene expression :** Certain genes that express at a particular stage of plant development have been identified. The promoters of some of these developmental genes are useful for transgene expression. e.g.

- Senescence-specific gene expression
- Abscisic acid inducible gene expression
- Auxin inducible gene expression

Although plant-derived systems (2 and 3) are not independent as non-plant derived one (1), they are still useful in plant biotechnology since no exogenous inducer is required.

Tissue-specific promoters

Efforts are on in recent years to isolate promoters that can drive gene expression in a tissue specific manner. The advantage with tissue-specific promoters is that the expression (which may produce a harmful compound) is confined to selected tissues that are not consumed by humans or animals. In fact, some tissue-specific promoters have been isolated, and are in use.

TRANSGENE STABILITY, EXPRESSION AND GENE SILENCING

The genetically transformed plant cells are grown *in vitro* to finally regenerate to plants. At the initial stages of plant cell growth, some evidence for the success of plant transformation can be obtained e.g. resistance to antibiotics, herbicides etc.

The ***integration of transgenes with the host plant genome can be confirmed by some molecular techniques*** — Southern hybridization, polymerase chain reaction. This is usually carried out by analysing the seeds at T_1 generation.

What is actually required for the success of plant transformation is the efficient and stable expression of transgene, and not just its presence in plant genome. It is just useless to have a desired gene without appropriate expression.

SCAFFOLD ATTACHMENT REGIONS AND GENE STABILITY

Scaffold attachment regions (***SARs***; also known as ***matrix attachment/associated regions***, ***MARs***) are the regions of DNA isolated based on their ability to bind to the nuclear scaffold (protein depleted chromosome forms a central scaffold surrounded by DNA). SARs are known to stabilize gene expression in transgenic plants. During the course of transformation, SARs are ligated to the flanking regions of foreign gene. It is observed that the transformation efficiency by particle bombardment is increased by use of SARs.

The ***presence of SARs along with transgene stabilizes or normalizes gene expression***. e.g. SARs from yeast and tobacco have helped to stabilize gene expression in plants. The degree of stabilization of genes by SARs is dependent on the affinity of SARs towards the nuclear matrix of the target plant cells. SARs are particularly useful for stabilizing the gene expression when the copy number of DNA introduced is high. In the absence of SARs, high DNA copy number would result in an instable and highly variable gene expression.

INTRONS AND GENE EXPRESSION

The presence of introns between the promoter and coding regions of a gene will significantly influence the gene expression. The gene expression is enhanced when introns are used in monocotyledonous plant species. For instance, introduction of introns between cauliflower mosaic virus 35S promoter and β-glucuronidase significantly improved gene expression in maize. It is believed that introns may stabilize mRNAs and increase the protein biosynthesis.

There are conflicting reports on the affect of introns on the gene expression in dicotyledonous plants. Some workers have reported stimulation while others contradict this claim.

It is now accepted that the ***intron-containing constructs will enhance gene expression*** in ***transgenic monocots***, although the picture is less clear as regards dicots.

GENE SILENCING

There are in fact many instances of gene transfer that are not properly expressed. The ***instability of gene expression*** (or inadequate gene expression) in ***transgenic plants is*** referred to as ***gene silencing***. The mechanism of gene silencing is not well understood although it is predominantly due to the phenomenon of ***homology dependent gene silencing*** (***HDGS***).

Two types of gene silencing are known in plant biotechnology—transcriptional gene silencing and post-transcriptional gene silencing. The two differ at the level of gene expression where silencing occurs.

Transcriptional gene silencing (TGS)

When the transgenes share homology in their promoter regions, TGS occurs. It is due to altered methylation patterns and altered chromatin conformation. Gene silencing occurs by repression of transcription.

Post-transcriptional gene silencing (PTGS)

Sometimes, the expression of a homologous transgene may inhibit the expression of the transgene and endogenous gene. This primarily occurs by decreasing the stability of RNA, and thus a reduced expression. This is often observed when strong promoters are used to drive the transgene expression which may result in co-suppresion or PTGS, and thus a lower level of gene expression. For this reason, some workers prefer to use weak promoters rather than strong promoters.

Post-transcriptional gene silencing is due to the production of double-stranded RNA either in the nucleus or cytoplasm. The double-stranded RNA is formed when antisense RNA is produced due to the activity of RNA dependent RNA polymerase.

Strategies to avoid gene silencing

The control of gene silencing is not an easy job, since it occurs in an unpredictable fashion. Some general recommendations, are, however, made to minimize the impact of gene silencing in transgenic plants

- Reduction in the number of transgenes inserted (i.e. reduced copy number).
- Avoiding the use of promoters and transgenes with high degree of homology.
- Minimizing/avoiding the use of multiple copies of the same promoter or terminator.

Gene silencing is not observed in chloroplast (plastid) transformation, hence it is preferred in recent years.

CHLOROPLAST TRANSFORMATION

The chloroplasts (plastids) and mitochondria are believed to have evolved from prokaryotes during the course of evolution. Both these organelles have their own genome, although it is much simpler when compared to nuclear genome. Further, many of the proteins that function in chloroplasts and mitochondria are encoded by nuclear genes and then transported to the organelle.

Chloroplast genome

Most of the higher plants have about 100 chloroplasts per leaf cell. Each chloroplast contains approximately 100 copies of chloroplast DNA genome. The chloroplast genome (the ***plastome***) is a ***circular double-stranded DNA*** molecule (or chromosome) located in the stroma. Majority of chloroplast genomes are in the size of 120–160 kbp and contain about 120–140 genes. About 100 chloroplast genes are known to code for proteins. The protein synthesis in chloroplasts resembles that of prokaryotes.

CHLOROPLAST ENGINEERING

Genetic engineering of chloroplast that leads to chloroplast (plastid) transformation is an important and exciting field in modern biotechnology as it offers the following advantages.

1. ***Chloroplasts are maternally inherited***, hence there is no danger of gene transfer through pollen to related weeds. This is because pollen does not contain transgenes.

2. Multigene transfer can be conveniently carried out in chloroplasts which is rather difficult with nuclear genome.

3. Chloroplasts genome is functionally comparable to prokaryotic genome. A single promoter can control the expression of group of genes (transgenes). It is therefore possible to introduce desirable multiple genes which can be expressed under the control of a single promoter.

4. High level of transgene expression is possible with chloroplasts. There are about 100 chloroplasts per cell, each containing about 100 copies of genome. Thus, there is possibility of 10,000 copies of transgenes per cell! This is a tremendous number of transgenes carried by transformed chloroplasts. There is a tremendous ***potential for a very high level of gene expression*** and large scale production of active proteins.

5. Chloroplast transformation is ***not associated with gene silencing*** which is a major problem with nuclear genome transformation.

6. Antibiotic resistance genes need not be used as selectable markers. Even if used, they can be easily excised.

7. Toxicity associated with foreign protein production in chloroplasts is much less when compared to nuclear-controlled foreign proteins.

Design of vectors for chloroplast transformation

A diagrammatic representation of two vector constructs for chloroplast transformation is depicted in ***Fig. 49.13***.

1. **A construct for expression of a single gene :** The vector for chloroplast transformation is based on the selectable marker gene *aadA* that provides resistance to antibiotic spectinomycin. The single foreign (desirable) gene is fused to regulatory sequences (promoter and terminator) which in turn is flanked on either side by chloroplast DNA (Cp DNA) (***Fig. 49.13A***).

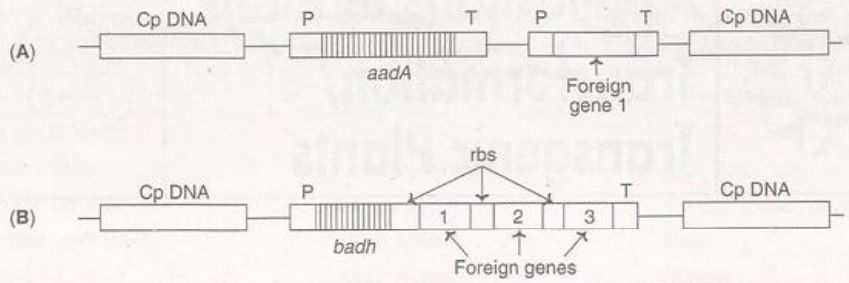


*Fig. 49.13 : A diagrammatic representation of vectors for chloroplast transformation (**A**) A construct designed for a single foreign gene (**B**) A construct designed for multiple foreign genes (Cp DNA-Chloroplast DNA; P-Promoter;* aadA-*A selectable marker gene that confers resistance to antibiotic spectinomycin; T-Terminator;* badh-*A selectable marker gene encoding betaine aldehyde dehydrogenase; rbs-Ribosome binding site).*

2. **A construct for expression of multiple genes :** In this case, the selectable marker is the betaine-aldehyde dehydrogenase (*badh*) gene.

It is flanked by a promoter and the multiple transgenes are flanked by a terminator. At both ends chloroplast DNA sequences are present. In between the transgenes, these are ribosome-binding sites (one between two transgenes) to ensure efficient translation (***Fig. 49.13B***).

Introduction of foreign genes into chloroplast genome

Most of the methods used for introducing the foreign genes into nuclear genome are not useful for chloroplast transformation. The most successful method for inserting foreign genes into chloroplasts is ***particle gun bombardment***.

After the bombardment, homologous recombination occurs between the chloroplast DNA sequences on the vector and those of on the genome. This is a site-specific integration and thus avoids the frequent problems associated with randon insertion of foreign genes into nuclear genome.

The regenerated plants derived from the modified plastome (chloroplast genome) are regarded as ***transplastomic plants***.

The future of chloroplast transformation

The technology of chloroplast transformation is ***in the developing stages***. In fact, it has not become as routine as transformation of nuclear genomes of plants.

Chloroplast engineering, however, ***holds a great promise in plant biotechnology*** being an efficient, clean and environmental-friendly approach for the production of transgenic plants.

50 Applications of Plant Transformation/ Transgenic Plants

The genetic manipulations carried out in plants for the production of transgenic plants have been described (Chapter 49). The ultimate goal of transgenics (involving introduction, integration, and expression of foreign genes) is ***to improve the crops, with the desired traits***. Some of the important ones are listed.

- Resistance to biotic stresses i.e. resistance to diseases caused by insects, viruses, fungi and bacteria.
- Resistance to abiotic stresses-herbicides, temperature (heat, chilling, freezing), drought, salinity, ozone, intense light.
- Improvement of crop yield, and quality e.g. storage, longer shelf life of fruits and flowers.
- Transgenic plants with improved nutrition.
- Transgenic plants as bioreactors for the manufacture of commercial products e.g. proteins, vaccines, and biodegradable plastics.

Environmental stresses to plants

The different types of external stresses that influence the plant growth and development are depicted in ***Fig. 50.1***, These stresses are grouped based on their characters—***biotic and abiotic stresses.*** The biotic stresses are caused by insects, pathogens (viruses, fungi, bacteria), and wounds. The abiotic stresses are due to herbicides, water deficiency (caused by drought, temperature, salinity), ozone and intense light.

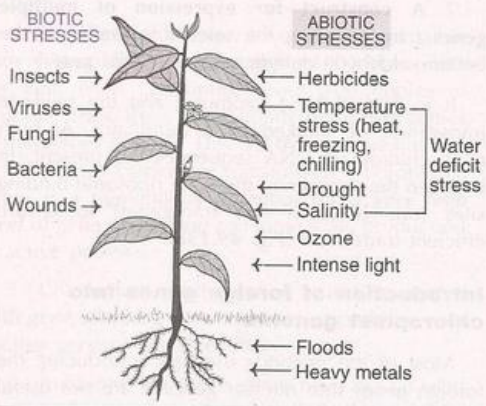


Fig. 50.1 : Biotic and abiotic stresses that affect plant growth, development and yield.

Almost all the stresses, either directly or indirectly, lead to the production of reactive oxygen species (ROS) that create oxidative stress to plants. This damages the cellular constitutents of plants which is associated with a reduction in plant yield.

The major ***objective of plant biotechnology is to develop plants that are resistant to biotic and abiotic stresses.***

RESISTANCE TO BIOTIC STRESSES

Genetic engineering of plants has led to the development of crops with increased resistance to

Table 50.1 A selected list of common insect pests along with the major crops damaged by them

Common name of pest	*Botanical name*	*Crop(s) damaged*
Cotton bollworm	*Helicoverpa zea*	Cotton
Cotton leafworm	*Spodoptera littiralis*	Rice, maize, cotton, tobacco
Tobacco hornworm	*Manduca sexta*	Tobacco, tomato, potato
European corn borer	*Ostrinia nubilalis*	Maize
Locust	*Locusta migratoria*	Grasses
Tobacco budworm	*Heliothis virescens*	Tobacco, cotton
Tomato fruitworm	*Heliothis armigera*	Tomato, cotton
Cowpea seed beetle	*Callosobruchus maculatus*	Cowpea, soybean
Colorado beetle	*Leptinotarsa decemlineata*	Potato
Brown plant hopper	*Nilaparvata lugens*	Rice

biotic stresses which is described in three major categories

1. Insect resistance.
2. Virus resistance.
3. Fungal and bacterial disease resistance.

INSECT (PEST) RESISTANCE

It is estimated that about 15% of the world's crop yield is lost to insects or pests. A selected list of the common insects and the crops damaged is given in ***Table 50.1.***

The damage to crops is mainly caused by insect larvae and to some extent adult insects. The majority of the insects that damage crops belong to the following orders (with examples).

- Lipidoptera (bollworms).
- Coleoptera (beetles).
- Orthoptera (grasshoppers).
- Homoptera (aphids).

Till sometime ago, chemical pesticides are the only means of pest control. Scientists have been looking for alternate methods of pest control for the following reasons (i.e. limitations of pesticide use).

- About 95% of the pesticide sprayed is washed away from the plant surface and accumulates in the soil.
- It is difficult to deliver pesticides to vulnerable parts of plants such as roots, stems and fruits.
- Chemical pesticides are not efficiently degraded in the soil, causing environmental pollution.
- Pesticides, in general, are toxic to non-target organisms, particularly humans and animals.

It is fortunate that scientists have been able to discover new biotechnological alternatives to chemical pesticides thereby providing insect resistance to crop plants. Transgenic plants with insect resistance transgenes have been developed. About 40 genes obtained from microorganisms of higher plants and animals have been used to provide insect resistance in crop plants. Some of the approaches for the ***biocontrol of insects*** are briefly described.

RESISTANCE GENES FROM MICROORGANISMS

Bacillus thuringiensis (Bt) toxin

Bacillus thuringiensis was first discovered by Ishiwaki in 1901, although its commercial importance was ignored until 1951. *B. thuringiensis* is a Gram negative, soil bacterium. This bacterium produces a parasporal crystalline proteinous toxin with insecticidal activity. The protein produced by *B.thuringiensis* is referred to as ***insecticidal crystalline protein (ICP).*** ICPs are among the endotoxins produced by sporulating bacteria, and were originally classified as ***δ-endotoxins*** (to distinguish them from other classes of α-, β- and γ-endotoxins).

Bt toxin genes

Several strains of *B. thuringiensis* producing a wide range of ***crystal (cry) proteins*** have been identified. Further, the structure of *cry* genes and their corresponding toxin (δ-endotoxin) products have been characterized. The *cry* genes are classified (latest in 1998) into a large number of distinct families (about 40) designated as *cry 1.....cry 40,* based on their size and sequence similarities. And within each family, there may be sub-families. Thus, the total number of genes producing *Bt* toxins (Cry proteins) is more than 100.

There are differences in the structure of different Cry proteins, besides certain sequence similarities. The molecular weights of Cry proteins may be either large (~130 KDa) or small (~70KDa). Despite the differences in the Cry proteins, they share a common active core of three domains.

Mode of action of Cry proteins

Most of the *Bt* toxins (Cry proteins) are active against Lepidopteran larvae, while some of them are specific against Dipteran and Coleopteran insects. The protoxin of Cry I toxin group has a molecular mass of 130 kilodaltons (130 KDa). When this parasporal crystal is ingested by the target insect, the protoxin gets activated within its gut by a combination of alkaline pH (7.5 to 8.5) and proteolytic enzymes. This results in the conversion of protoxin into an active toxin with a molecular weight of 68 KDa (***Fig. 50.2***).

The active form of toxin protein gets itself inserted into the membrane of the gut epithelial cells of the insect. This result in the formation of ion channels through which there occurs an excessive loss of cellular ATP. As a consequence, cellular metabolism ceases, insect stops feeding, and becomes dehydrated and finally dies. Some workers in the recent years suggest that the *Bt* toxin opens cation-selective pores in the membranes, leading to the inflow of cations into the cells that causes osmotic lysis and destruction of epithelial cells (and finally the death of insect larvae).

The *Bt* toxin is not toxic to humans and animals since the conversion of protoxin to toxin requires alkaline pH and specific proteases (These are absent humans and animals).

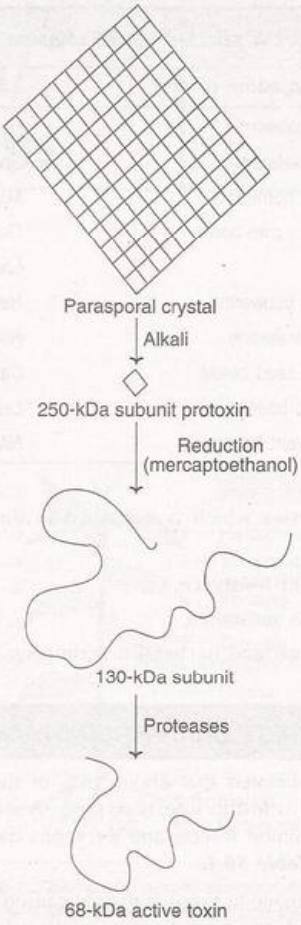


Fig. 50.2 : *A diagrammatic representation of the formation of active toxin from the parasporal crystal of* Bacillus thuringiensis.

Bt toxin as biopesticide

Preparations of *Bt* spores or isolated crystals have been used as organic biopesticide for about 50 years. This approach has not met with much success for the following reasons.

- Low persistence and stability (sunlight degrades toxin) of the toxin on the surface of plants.
- The *Bt* toxin cannot effectively penetrate into various parts of plants, particularly roots.
- Cost of production is high.

TABLE 50.2 *Bt*-based genetically modified crop plants developed for commercial use

Crop	*Trade name*	*Bt protein*	*Resistance to insect(s)*
Cotton	Bollgard	Cry 1Ac	Cotton bollworm, tobacco budworm
Maize	YieldGard knockout	Cry 1Ab	European corn borer
Maize	Starlink	Cry 9C	European corn borer
Maize	Herculex I	Cry 1F	European corn borer
Maize	Bt-Xtra	Cry 1Ac	European corn borer
Potato	New-leaf	Cry 3A	Colarado beetle

Bt-based genetic transformation of plants

It has been possible to genetically modify (***GM*** ***plants*** by inserting *Bt* genes and provide pest resistance to these transformed plants. For an effective pest resistance, the bacterial gene in transgenic plants must possess high level expression. This obviously means that the transgene transcription should be under the effective control of promoter and terminator sequences.

The early attempts to express cry 1A and cry 3A proteins under the control of CaMV 35S or *Agrobacterium* T-DNA promoters resulted in a very low expression in tobacco, tomato and potato plants.

Modification of *Bt* cry 1A gene : The wild type transgene *Bt* cry 1A(b) was found to express at a very low levels in transgenic plants. The nucleotide sequence of this gene was modified (G + C content altered, several polyadenylation signals removed, ATTTA sequence deleted etc). With appropriate sequence changes, an enormous increase (about 100 fold) in the *Bt* toxin product formation was observed.

The transgenic *Bt* crops that were found to provide effective protection against insect damage were given approval for commercial planting by USA in the mid-1990s (***Table 50.2***). Some biotechnological companies with their own trade names introduced several transgenic crops into the fields. Among these, only maize and cotton *Bt* crops are currently in use in USA. The other genetically modified plants met with failure for various reasons.

Advantages of transgenic plants with *Bt* genes

- *Bt* genes could be expressed in all parts of the plants, including the roots and internal regions of stems and fruits. This is not possible by any chemical pesticide.
- Toxic proteins are produced within the plants; hence they are environmental-friendly.
- *Bt* toxins are rapidly degraded in the environment.

The problem of insect resistance to *Bt* crops

The major limitation of *Bt*-gene possessing transgenic plants is the ***development of Bt-resistant insects.*** The *Bt* toxin is a protein, and the membrane receptor (of the gut) through which the toxin mediates its action is also a protein. It is possible that the appropriate mutations in the insect gene coding for receptor protein may reduce the toxin binding and render it ineffective. This may happen within a few generations by repeated growing of *Bt* crops.

Several approaches are made to avoid the development of resistance in insects.

- Introduction of two different *Bt* toxin genes for the same target insect.
- Development of transgenic plants with two types of insect resistance genes e.g. *Bt* gene and proteinase inhibitor gene.
- Rotating *Bt* crops with non-*Bt* crops may also prevent the build-up of resistance in insect population.

The environmental impact of *Bt* crops

The most serious impact of *Bt* crops on environment is the ***build-up of resistance in the pest population.***

In 1999, another issue was brought to light about *Bt* crops. It was reported that the pollen from

Table 50.3 A selected list of plant insecticidal (non-*Bt*) genes used for developing transgenic plants with insect resistance

Plant gene	*Transgenic plant(s)*	*Encoded protein*	*Resistance to insect(s)*
Protease inhibitors			
CpTi	Potato, apple, rice, sunflower, wheat, tomato	Trypsin	Coleoptera, Lepidoptera
CII	Tobacco, potato	Serine protease	Coleoptera, Lepidoptera
PI-IV	Potato, tobacco	Serine protease	Lebidoptera
OC-1	Tobacco, oilseed rape	Cysteine protease	Coleoptera, Homoptera
CMe	Tobacco	Trypsin	Lebidoptera
α-Amylase inhibitors			
α-A1-Pv	Pea, tobacco	α-Amylase	Coleoptera
WMAI-1	Tobacco	α-Amylase	Lepidoptera
Lectins			
GNA	Potato, rice, sugarcane sweet potato, tobacco	Lectin	Homoptera, Lepidoptera
WGA	Maize	Agglutin	Lepidoptera, Coleoptera
Others			
BCH	Potato	Chitinase	Homoptera, Lepidoptera
TDC	Tobacco	Tryptophan decarboxylase	Homoptera

Bt maize might be toxic to the larvae of Monarch butterfly. This generated considerable opposition to *Bt* crops by the public, since Monarch butterfly is one of the most colorful natives in USA. It was later proved that the fears about the impact of *Bt* crops on the monarch butterfly were without the required scientific evidence.

The lesson learnt from the monarch butterfly episode is that the risks of GM crops should be thoroughly assessed before they are reported.

Usage of *Bt* : The usage *Bt* is commonly used for a transgenic crop with a *cry* gene e.g. *Bt* cotton. In the same way, Cry proteins are also referred to as *Bt* proteins. It may also be stated here that the authors use four different names for the same group of proteins-δ-endotoxin, insecticidal crystal protein (ICP), Cry and now *Bt*.

Resistance genes from other microorganisms

The *Bt* toxin genes from *B.thuringiensis* have been described in the preceeding pages. There are certain other insect resistant genes from other microorganisms. Some of the important ones are listed.

1. ***Cholesterol oxidase*** of *Strepotmyces* culture filtrate was found to be toxic to boll weevil larvae. Cholesterol oxidase gene has been introduced into tobacco to develop a transgenic plant.

2. ***Isopentenyl transferase gene*** from *Agrobacterium tumefaciens* has been introduced into tobacco and tomato. This gene codes for an important enzyme in the synthesis of cytokinin. The transgenic plants with this transgene were found to reduce the leaf consumption by tobacco hornworm and decrease the survival of peach potato aphid.

RESISTANCE GENES FROM HIGHER PLANTS

Certain genes from higher plants were also found to result in the synthesis of products possessing insecticidal activity. Some authors regard them as ***non-Bt insecticidal proteins***. A selected list of plant insecticidal (non-*Bt*) genes used for developing transgenic plants with insect resistance is given ***Table 50.3***. Some of them are briefly described.

PROTEINASE (PROTEASE) INHIBITORS

Proteinase inhibitors are the proteins that inhibit the activity of proteinase enzymes. Certain plants naturally produce proteinase inhibitors to provide defence against herbivorous insects. This is possible since the inhibitors when ingested by insects interfere with the digestive enzymes of the insect. This results in the nutrient deprivation causing death of the insects.

It is possible to control insects by introducing proteinase inhibitor genes into crop plants that normally do not produce these proteins.

Cowpea trypsin inhibitor gene

It was observed that the wild species of cowpea plants growing in Africa were resistant to attack by a wide range of insects. Research findings revealed that insecticidal protein was a trypsin inhibitor that was capable of ***destroying insects*** belonging to the orders Lepidoptera (*e.g Heliothis virescans*), Orthaptera (e.g.*Locusta migratoria*) and Coleoptera (e.g.*Anthonous grandis*). Cowpea trypsin inhibitor (*CpTi*) has no affect on mammalian trypsins; hence it is ***non-toxic to mammals***.

CpTi gene was introduced into tobacco, potato and oilseed rape for developing transgenic plants. Survival of insects and damage to plants were much lower in plants possessing *CpTi* gene.

Advantages of proteinase inhibitors

- Many insects, not controlled by *Bt*, can be effectively controlled.
- Use of proteinase gene along with *Bt* gene will help to overcome *Bt* resistance development in plants.

Limitations of proteinase inhibitors

- Unlike *Bt* toxin, high levels of proteinase inhibitors are required to kill insects.
- It is necessary that the expression of proteinase inhibitors should be very low in the plant parts consumed by humans, while the expression should be high in the parts of plants utilized by insects.

α-Amylase inhibitors

The insect larvae secrete a gut enzyme α- amylase to digest starch. By blocking the activity of this enzyme by α-amylase inhibitor, the larvae can be starved and killed. α-Amylase inhibitor gene (α-AI-Pv) isolated from bean has been successfully transferred and expressed in tobacco. It provides ***resistance against Coleoptera*** (e.g. *Zabrotes subfasciatus*).

LECTINS

Lectins are plant glycoproteins and they provide resistance to insects by acting as toxins. The lectin gene (*GNA*) from snowdrop (*Galanthus nivalis*) has been transferred and expressed in potato and tomato. The major limitations of lectin are that it acts only against piercing and sucking insect, and high doses are required.

RESISTANCE GENES FROM ANIMALS

Proteinase inhibitor genes from mammals have also been transferred and expressed in plants to provide resistance against insects, although the success in this direction is very limited. Bovine pancreatic trypsin inhibitor (BPTI) and α_1-antitrypsin genes appear to be promising to offer insect resistance to transgenic plants.

INSECT RESISTANCE THROUGH COPY NATURE STRATEGY

Some of the limitations experienced by transferring the insecticidal genes (particularly *Bt*) and developing transgenic plants have prompted scientists to look for better alternatives. The copy nature strategy was introduced in 1993 (by Boulter) with the objective of ***insect pest control which is relatively sustainable and environmentally friendly.*** The copy nature strategy for the development of insect -resistant transgenic plants has the following stages.

1. **Identification of leads**: The first step in copy nature strategy is to identify the plants (from worldover) that are naturally resistant to insect damage.

2. **Isolation and purification of protein**: The protein with insecticidal properties (from the resistant plants) is isolated and purified. The sequence of the protein is determined, and the gene responsible for its production identified.

3. **Bioassay of isolated protein**: The activity of the protein against the target insects is determined by performing a bioassay in the laboratory.

4. **Testing for toxicity in mammals:** It is absolutely necessary to test the toxicity of the protein against mammals, particularly humans. If the protein is found to have any adverse affect, the copy nature strategy should be discontinued.

5. **Gene transfer :** By the conventional techniques of genetic engineering, the isolated gene corresponding to the protein toxin is introduced into the crop plants.

6. **Selection of transgenic plants** : After the gene transfer, the transgenic plants developed should be tested for the inheritance and appropriate expression of transgene. The efficiency of insecticidal protein to destroy insects is also evaluated.

7. **Evaluation for biosafety** : Field trails have to be conducted to evaluate the crop yield, damage to insects, influence on the environment with respect to the transgenic plant.

The copy nature strategy, though time consuming and many a times unsuccessful, takes into account the complex interplay in biological communities (between plants animals, microorganisms) and physical environment.

VIRUS RESISTANCE

Virus infections of crops may result in retarded cell division (***hypoplasia***), excessive cell division (***hyperplasia***), and cell death (***necrosis***). The overall effects of virus infections are growth retardation, lowered product yield and sometimes complete crop failure.

The ***chemical methods*** used to control various plant pathogens ***will be ineffective*** with respect to plant viruses since the viruses are intracellular obligate parasites.

There are however, certain ***safe agricultural practices*** to control/reduce viral infections to plants.

- Use of seeds that are virus - free.
- Control insects that spread plant viruses.
- Control weeds that serve as alternate hosts for viruses.
- Use cultivars that possess virus resistance.

It is possible to immunize plants against viral damages by expressing viral proteins in the plant cells. With the advances made in genetic engineering, it has become a reality to develop ***transgenic plants with virus resistance***. This is mostly done by employing ***virus-encoded genes*** - virus coat proteins, movement proteins, transmission proteins, satellite RNA, antisense RNAs and ribozymes. In recent years, some attempts are also made to provide virus resistance to plants by using ***animal genes***. Some of the developments are briefly described.

Virus coat proteins

The virus coat protein-mediated approach is the most successful one to provide virus resistance to plants. It was in 1986, transgenic tobacco plants expressing tobacco mosaic virus (TMV) coat protein gene were first developed. These plants exhibited high levels of resistance to TMV. Excited by this remarkable success, scientists have worked with many more viruses (around 30 or so) and developed crops with virus coat protein-mediated protection.

A selected list of the virus resistant transgenic plants with sources of virus coat protein genes is given in ***Table 50.4***. The transgenic plant providing coat protein-mediated resistance to virus are rice, potato, wheat, tobacco, peanut, sugar beet, alfalfa etc. The viruses that have been used include alfalfa mosaic virus (AlMV), cucumber mosaic virus (CMV), potato virus X (PVX), potato virus Y (PVY), citrus tristeza virus (CTV) and R rice stripe virus (RSV).

Advantages of virus coat proteins : The coat protein gene from one virus sometimes provides resistance (cross protection) to some other viruses, which may be unrelated e.g. TMV of tobacco plant provides resistance to potato virus X, alfalfa mosaic virus and cucumber mosaic virus.

Limitation of virus coat proteins

The virus coat protein-mediated protection is successful for viruses with single-stranded RNA genomes. However, this approach is of not much use for viruses with genomes containing double-stranded RNA and single-stranded DNA.

Mechanism of action of virus coat proteins : As the transgenic plant expresses the gene for coat

protein of a given virus, the ability of the same virus to infect the plants again is drastically reduced. Despite a remarkable success in the virus coat protein-mediated protection, the molecular mechanism of the protection is not clearly known.

Movement proteins

As the virus infects the plant cells, its rapid spread through intercellular junctions (plasmadesmata) of vascular tissue occurs through the participation of movement proteins produced by the viruses. Good examples of movement proteins are 30KDa protein of tobacco mosaic virus (TMV) and 32KDa protein of brome mosaic virus (BMV).

Transgenic tobacco plants that express a mutated 30KDa movement protein have been developed. The TMV infection to these plants is much less. It is believed that the mutated movement protein competes with the wild-type TMV-coded protein thereby reducing the spread of the virus (TMV). In recent years, a recombinant movement protein having the components of golden mosaic virus and African cassava mosaic virus have been developed. This protein effectively interferes with the spread of both the viruses.

The ***advantage*** with movement protein strategy is that it is ***applicable to single-stranded DNA viruses*** (Geminiviruses) also.

TABLE 50.4 A selected list of the virus-resistant transgenic plants with sources of virus protein coat genes

Plant	*Source(s) of virus coat protein gene*
Tobacco	*TMV, CMV, AIMV*
Rice	*RTSV, RSV, RYMV*
Wheat	*SBWMV, BYDV*
Potato	*PVX, PVY, PLRV*
Squash	*CMV, ZYMV*
Sugar beet	*BNYVV*
Peanut	*TSWV*
Papaya	*PRSV*
Citrus	*CTV*
Alfalfa	*AIMV*

TMV-Tobacco mosaic virus; CMV-Cucumber mosaic virus; AIMV-Alfalfa mosaic virus; RTSV-Rice tungro spherical virus; RSV-Rice stripe virus; RYMV-Rice yellow mottle virus; SBWMV-Soil-borne wheat mosaic virus; BYDV-Barley yellow dwarf virus; PVX-Potato virus X; PVY-Potato virus Y; PLRV-Potato leaf froll virus; ZYMV-Zucchini yellow mosaic virus; BNYVV-Beet necrotic yellow vein virus; TSWV-Tomato spotted wilt virus; PRSV-Papaya ringspot virus; CTV-Citrus tristeza virus.

Transmission proteins

There is a good coordination and interaction between plant viruses and insect vectors for the spread of viruses from one plant to another. Certain viral-encoded transmission proteins do this job effectively. It is possible to produce mutated transmission proteins and block the spread of viruses. Thus, the ***spread of insect-transmitted viruses*** can be ***prevented by*** engineering crops to express a ***defective virus-transmission protein***.

Satellite RNAs

Plant viral satellite RNAs are small RNA molecules that multiply in the host cells with the help of specific helper viruses. These satellite RNAs are encapsulated together with the respective helper viruses. In general, the presence of satellite RNAs reduces the severity of the viral disease and the symptoms, and thus reduces the effect of the virus. Transgenic plants containing satellite sequences have been developed to provide resistance to virus diseases. One example is given here. When cucumber mosaic virus (CMV) infects pepper plants, severe symptoms appear. These symptoms can be minimized with higher plant yield when CMV is coinoculated with a satellite RNA.

Satellite RNA approach is ***not widely used*** due to several ***limitations***.

- Some of the satellite RNA may increase the severity of disease symptoms in some plants.
- Satellite RNAs mutate very rapidly which may sometimes result in a highly virulent agent.
- Recombinations between satellite RNAs have been detected. This may lead to serious consequences.

Antisense RNAs

The antisense RNA approach is designed to specifically interfere with virus replication.

By use of genetic engineering, a complementary DNA strand of a gene (DNA sequence) can be inserted in reverse orientation (3′ → 5′ as opposed

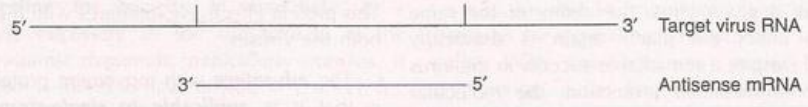


Fig. 50.3 : Hybridization of antisense mRNA with virus RNA to block replication.

to 5′ → 3′) and this is referred to as antisense gene. The mRNA produced by antisense gene is complementary to the mRNA synthesized by normal gene. As a result, both these mRNAs hybridize and thus the normal translation of mRNA is blocked. The net effect of employing an ***antisense gene*** into a cell is that it ***blocks a specific gene expression.***

It is possible to introduce viral antisense genes into plants and produce mRNAs complementary to viral sequences involved in virus replication. The antisense mRNAs can block the replication of viruses (***Fig 50.3***). Initially, antisense RNA approach was carried out in single-stranded RNA viruses. The success of this approach however, was limited probably due to the following reasons.

- High concentration of antisense mRNA may be required.
- Protein association with mRNA interferes with hybridization (between sense mRNA and antisense mRNA).

Antisense RNA approach may be more useful for DNA viruses. In fact, tomato golden mosaic virus (TGMV) replicase coding sequence was cloned in antisense orientation and introduced into tobacco plants. The transgenic tobacco plants expressed antisense RNA of TGMV replicase. These plants were resistant to TGMV infection.

Ribozymes

Ribozymes are small RNA molecules which promote the catalytic cleavage of RNA. For providing virus resistance, ribozymes in the form of antisense RNAs capable of cleaving the target viral (sense) RNAs have been developed. The ribozyme (antisense RNA) binds to a small sequence of viral RNA and splits (***Fig. 50.4).***

In this way, it is possible to block the replication of viral RNA. However, the ribozymes approach has not been very successful in plants.

FUNGAL AND BACTERIAL DISEASES

The plants do posses general defence systems against invading pathogens. This is however, not truely comparable with the immune system of the animals.

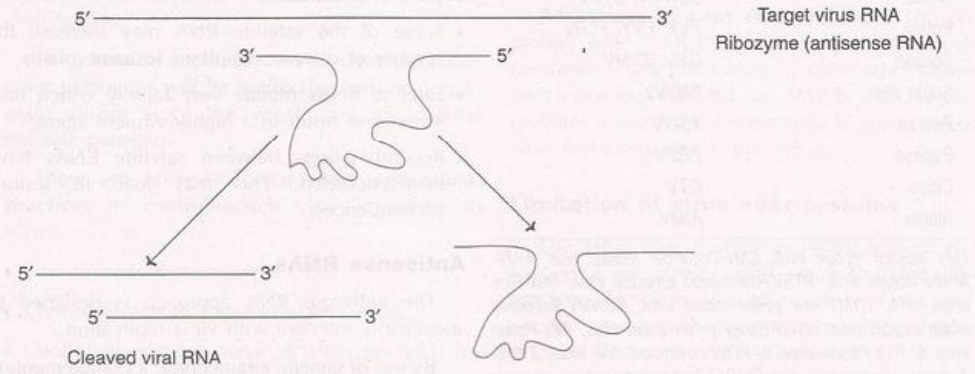


Fig. 50.4 : Action of ribozymes as antisense RNAs to block virus replication.

Whenever there is a cellular damage caused by pathogens (fungi, bacteria) and plant pests, the general defence system of plants get geared up to provide some amount of protection to the plant. This natural disease resistance of plants is inadequate. However, knowledge on the natural systems of plant resistance is useful for the biotechnological approaches to develop disease resistance.

Some of the defenses of plants and the biotechnological approaches are briefly described.

PATHOGENESIS-RELATED (PR) PROTEINS

To defend themselves against the invading pathogens (fungi and bacteria), plants accumulate ***low molecular weight proteins*** which are collectively regarded as pathogenesis-related (PR) proteins. The different types of PR proteins and their properties are given in ***Table 50.5***. Some of the most important ones are described.

Table 50.5 Different types of pathogenesis-related (PR) proteins produced in plants

Type	*Properties*
PR-1	Antifungal
PR-2	Endo β-1, 3-glucanases
PR-3	Endochitinases
PR-4	Antifungal, endochitinase
PR-5	Antifungal, thaumatin-like proteins, osmotins
PR-6	Protease inhibitors
PR-7	Endoprotease
PR-8	Chitinase/lysozyme
PR-9	Peroxidases
PR-10	Ribonucleases
PR-11	Endochitinase acitivty
PR-12	Defensins
PR-13	Thionins
PR-14	Lipid transfer proteins (non-specific)

Chitinase

Chitin is a constituent of fungal cell walls which can be hydrolysed by the enzyme chitinase. Certain chitinase genes from plants have been isolated and characterized. A bacterial chitinase gene obtained from a soil bacterium (*Serratia marcescens*) was introduced and expressed in tobacco leaves. Some other workers isolated a chitinase gene from bean (*Phaseolus vulgaris*) and developed transgenic plants of tobacco and *Brassica napus* with this gene.

The transformed tobacco plants were found to be resistant to infection of the pathogen *Rhizoctonia solani*. In case of *B.napus*, the protection however, was comparatively less.

Glucanase

Glucanase is another enzyme that degrades the cell wall of many fungi. The most widely used glucanase is β-1, 4-glucanase. The gene encoding for β-1, 4-glucanase has been isolated from barley, introduced, and expressed in transgenic tobacco plants. This gene provided good protection against soil-borne fungal pathogen *Rhizoctonia solani*.

The resistance to fungal pathogens is much higher if both chitinase and glucanase producing genes are present in transgenic plants. By this approach, fungal resistant tobacco, tomato and carrot have been developed.

RIBOSOME-INACTIVATING PROTEINS (RIPs)

Ribosome-inactivating proteins offer protection against fungal infections. They act on the large rRNA of eukaryote and prokaryote ribosomes (remove an adenine residue from a specific site), and thus inhibit protein biosynthesis. Certain RIPs that do not inhibit plant ribosomes were identified and the corresponding genes have been used to develop transgenic plants e.g. Type-I barley RIP is used to provide resistance to fungal infections.

Some authors use the term ***antimicrobial proteins*** to RIPs. The other examples of antimicrobial proteins are lectins, defensins, lysozyme, thionins etc.

Lysozyme

Lysozyme degrades chitin and peptidoglycan of cell wall, and in this way fungal infection can be reduced. Transgenic potato plants with lysozyme gene providing resistance to *Eswinia carotovora* have been developed.

Table 50.6 A selected list of transgenic plants developed along with the genes transferred and the resistance provided against the pathogens (fungus/bacterium)

Crop	*Gene(s) transferred*	*Resistance against pathogen*
Pathogenesis-related (PR) proteins		
Tobacco	Chitinase from bacterium (*Serratia marcescens*)	*Alternaria longipes*
	Bean chitinase	*Rhizoctonia solani, Phytophthora parasitica*
	Chitinase and 1, 3-β glucanase	*Cercospora nicotinae*
Rice	Chitinase	*Rhizoctonia solani*
Carrot	Chitinase and 1, 3-β glucanase	*Alternaria dauci, A. radicina*
Tomato	Chitinase and 1, 3-β glucanase	*Fusarium oxysporum*
Brassica napus	Chitinase	*Rhizoctonia solani*
Antimicrobial proteins		
Tobacco	Barley ribosome inactivating protein	*Rhizoctonia solani*
Tobacco	Defensin from radish	*Alternaria longipes*
Tobacco	α-Thionin gene from barley	*Pseudomonas syringae*
Potato	Bacteriophage T-4 lysozyme	*Erwinia carotovora*
Phytoalexins		
Rice	Stilbene synthase	*Pyricularia oryzae*
Tobacco	Stilbene synthase	*Botrytis cinerea*

Defensins

Defensins are antimicrobial peptides (26–50 amino acid residues) found in all the plant cells. They attack the microbial plasma membrane, however this is not adequate to provide resistance to pathogens.

In recent years, an artificial defensin gene has been developed and introduced into potatoes. These potatoes developed resistance to the bacterium *Eswinia carotovora*.

Thionins

Thionin proteins also offer protection against bacteria. Thionin coding genes have been introduced into tobacco and the transgenic plant so developed showed resistance to *Pseudomonas syringae*.

PHYTOALEXINS

Phytoalexins are secondary metabolites produced in plants in response to infection. They are low-molecular weight and antimicrobial in nature. The phytoalexins usually present in specialized cells or organelles are mobilized when infection occurs. Further, during infection there occurs induction of genes for increased production of phytoalexins. Stilbene synthase is a key enzyme for the synthesis of a common phytoalexin. The gene coding stilbene synthase has been isolated from peanut and introduced into tobacco, rice and *Brassica napus*. The transgenic plants carrying stilbene synthase gene were resistant against some fungi.

A selected list of transgenic plants developed, along with the genes transferred and the controlled pathogens is given in ***Table 50.6.***

NEMATODE RESISTANCE

Nematodes are simple worms found in the soil. They possess a complete digestive tract. The annual crop loss of the world due to nematode (roundworm) infestation is very high.

Some workers have identified and cloned a nematode resistance gene from wild beet plants. It

is proposed that this gene encodes a protein that detects the pests (nematodes) and triggers a defensive reactions in the plant. It is believed that some chemical compounds that destroy the gut of the nematode are produced.

Attempts were also made to transfer the nematode resistance gene to sugar beet. Not much success was reported, the major limitation being the difficulty in cultivating the gene-altered cells of sugar beet.

RESISTANCE TO ABIOTIC STRESSES

Plants are constantly being subjected to environmental stresses (See ***Fig 50.1***) that may result in deterioration of crop plants, and a very low or even no yield. Plants are dependent on the subtle internal mechanisms for tolerance of various stresses. The *in situ* tolerance of crop plants, whenever present, is inadequate and therefore cannot give protection against the stresses. A wide range of strategies are required to engineer plants against a particular type of stress tolerance.

Some of the abiotic stresses and the recombinant strategies developed to overcome them are described.

HERBICIDE RESISTANCE

Weeds (wild herbs) are unwanted and useless plants that grow along with the crop plants. Weeds compete with crops for light and nutrients, besides harbouring various pathogens. It is estimated that the world's crop yield is reduced by 10-15% due to the presence of weeds. To tackle the problem of weeds, modern agriculture has developed a wide range of ***weedkillers*** which are collectively referred to as ***herbicides***. In general, majority of the herbicides are broad-spectrum as they can kill a wide range of weeds. A good or an ***ideal herbicide*** is expected to possess the ***following characteristics***.

- Capable of killing weeds without affecting crop plants.
- Not toxic to animals and microorganisms.
- Rapidly translocated within the target plant.
- Rapidly degraded in the soil.

None of the commercially available herbicides fulfils all the above criteria. The major limitation of the herbicides is that they cannot descriminate weeds from crop plants. For this reason, the ***crops are*** also ***affected by herbicides, hence the need to develop herbicide-resistant plants.*** Thus, these plants provide an opportunity to effectively kill the weeds (by herbicides) without damaging the crop plants.

Strategies for engineering herbicide resistance

A number of biological manipulations particularly involving genetic engineering are in use to develop herbicide-resistant plants.

1. **Overexpression of the target protein** : The target protein, being acted by the herbicide can be produced in large quantities so that the affect of the herbicide becomes insignificant. Overexpression can be achieved by integrating multiple copies of the genes and/or by using a strong promoter.

2. **Improved plant detoxification** : The plants do possess natural defense systems against toxic compounds (herbicides). Detoxification involves the conversion of toxic herbicide to non-toxic or less toxic compound. By enhancing the plant detoxification system, the impact of the herbicide can be reduced.

3. **Detoxification of herbicide by using a foreign gene** : By introducing a foreign gene into the crop plant, the herbicide can be effectively detoxified.

4. **Mutation of the target protein**: The target protein which is being affected by the herbicide can be suitably modified. The changed protein should be capable of discharging the functions of the native protein but is resistant to inhibition by the herbicide. Once the resistant target protein gene is identified, it can be introduced into the plant genomes, and thus herbicide-resistant plants can be developed.

For success in the development of herbicide resistant plants, good knowledge of the target protein and the action of herbicides is required. Some of the developments made in the herbicide resistance of plant are briefly described.

GLYPHOSATE RESISTANCE

Glyphosate, is a glycine derivative. It acts as a broad-spectrum herbicide and is effective against 76 of the world's worst 78 weeds. Glyphosate is less toxic to animals and is rapidly degraded by microorganisms. In addition, it has a short half-life.

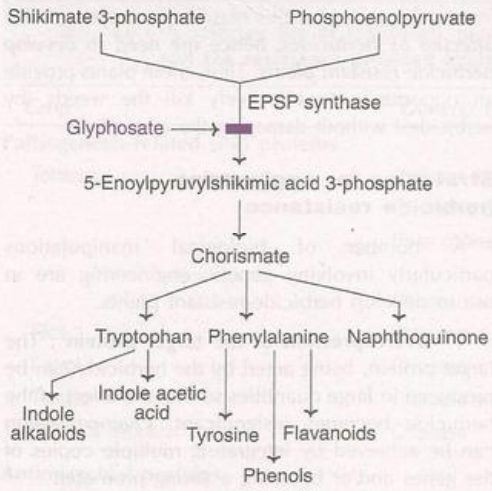


Fig. 50.5 : Shikimate pathway indicating the action of the herbicide glyphosate (EPSP synthase-5-Enolylpyruvylshikimate 3-phosphate synthase)

The American chemical company Monsanto markets glyphosate as ***Round up***.

Mechanism of action of glyphosate

Glyphosate is rapidly transported to the growing points of plants. It is capable of killing the plants even at a low concentration. Glyphosate acts as a ***competitive inhibitor*** of the enzyme 5-***enoyl-pyruvylshikimate 3-phosphate synthase (EPSPS).*** This is a key enzyme in ***shikimic acid pathway*** that results in the formation of aromatic amino acids (tryptophan, phenylalanine and tyrosine), phenols and certain secondary metabolites (***Fig. 50.5***).

The enzyme EPSPS catalyses the synthesis of 5-enoylpyruvylshikimate 3-phosphate from shikimate 3-phosphate and phosphoenoylpyruvate. Glyphosate has some structural similarily with the substrate phosphoenol pyruvate (***Fig 50.6***). Consequently, glyphosate binds more tightly with EPSPS and blocks the normal shikimic acid pathway. Thus, the herbicide glyphosate inhibits the biosynthesis of aromatic amino acids and other important products. This results in inhibition of protein biosynthesis (due to lack of aromatic amino acids). As a consequence, cell division and plant growth are blocked. Further, the plant growth regulator indole acetic acid (an auxin) is also produced from tryptophan. The net result of glyphosate is the death of the plants. Glyphosate is toxic to microorganisms as they also possess shikimate pathway.

Glyphosate is non-toxic to animals (including humans), since they do not possess shikimate pathway. Of the three aromatic amino acids (synthesized in this pathway), tryptophan and phenylalanine are essential and they have to be supplied in the diet, while tyrosine can be formed from phenylalanine.

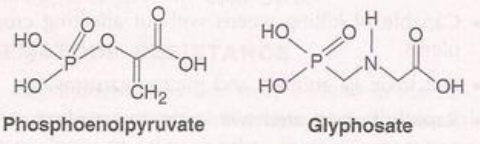
Strategies for glyphosate resistance

There are three distinct strategies to provide glyphosphate resistance to plants.

1. **Overexpression of crop plant EPSPS gene** : An overexpressing gene of EPSPS was detected in *Petunia*. This expression was found to be due to gene amplification rather than an increased expression of the gene. EPSPS gene from *Petunia* was isolated and introduced into other plants. The increased synthesis of EPSPS (by about 40 fold) in transgenic plants provides resistance to glyphosate. These plants can tolerate glyphosate at a dose of 2-4 times higher than that required to kill wild-type plants.

2. **Use of mutant EPSPS genes :** An EPSPS mutant gene that conferred resistance to glyphosate was first detected in the bacterium *Salmonella typhimurium*. It was found that ***a single base substitution*** (***C*** to ***T***) resulted in the change of an amino acid from proline to serine in EPSPS. This ***modified enzyme cannot bind to glyphosate***, and thus provides resistance.

The mutant EPSPS gene was introduced into tobacco plants using *Agrobacterium* Ti plasmid vectors. The transgene produced high quantities of the enzyme EPSPS. However, the transformed tobacco plants provided only marginal resistance to



Phosphoenolpyruvate **Glyphosate**

Fig. 50.6 : Structures of phosphoenolpyruvate (the substrate) and the herbicide glyphosate (the competitive inhibitor).

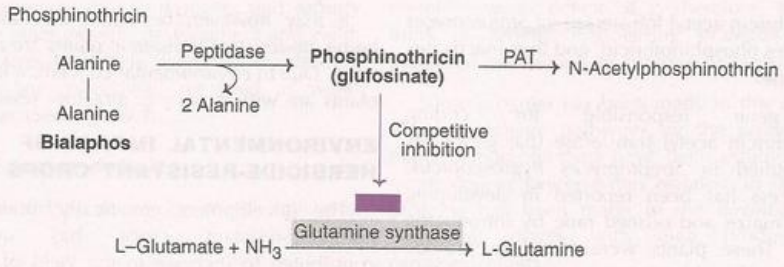


Fig. 50.7 : The formation, mode of action and detoxification of phosphinothricin (PAT-Phosphinothricin acetyl transferase).

glyphosate. The reason for this was not immediately identified. It was later known that the shikimate pathway occurs in the chloroplasts while the glyphosate resistant EPSPS was produced only in the cytoplasm. This enzyme was not transported to the chloroplasts, hence the problem to provide resistance. This episode made scientists to realize the importance of chloroplasts in genetic engineering.

In later years, the mutant EPSPS gene was tagged with a chloroplast-specific transit peptide sequence. By this approach, the glyphosate-resistant EPSPS enzyme was directed to freely ***enter chloroplast and confer resistance against the herbicide***.

3. **Detoxification of glyphosate** : The soil microorganisms possess the enzyme glyphosate oxidase that converts glyphosate to glyoxylate and aminomethylphosponic acid. The gene encoding glyphosate oxidase has been isolated from a soil organism *Ochrobactrum anthropi*. With suitable modifications, this gene was introduced into crop plants e.g. oilseed rape. The transgenic plants were found to exhibit very good glyphosate resistance in the field.

Use of a combined strategy : More efficient resistance of plants against glyphosate can be provided by employing a combined strategy. Thus, resistant (i.e. mutant) EPSPS gene in combination with glyphosate oxidase gene are used. By this approach, there occurs glyphosate resistance (due to mutant EPSPS gene) as well as its detoxification (due to glyphosate oxidase gene).

PHOSPHINOTHRICIN RESISTANCE

Phosphinothricin (or ***glufosinate***) is also a broad spectrum herbicide like glyphosate. Phosphinothricin is more effective against broad-leafed weeds but least effective against perennials. (**Note** : Phosphinothricin and glufosinate are two names for the same herbicide. However, to avoid confusion between glyphosate and glufosinate, phosphinothricin is more commonly used. ***Basta Aventis*** and ***Liberty*** are the ***trade names*** for phosphinothricin).

Phosphinothricin-a natural herbicide

Phosphinothricin is an unusual herbicide, being a derivative of a natural product namely ***bialaphos***. Certain species of *Streptomyces* produce bialaphos which is a combination of phosphinothricin bound to two alanine residues, forming a tripeptide. By the action of a peptidase, bialaphos is converted to active phosphinothricin (***Fig 50.7***).

Mechanism of action of phosphinothricin

Phosphinothricin acts as a competitive inhibitor of the enzyme glutamine synthase (***Fig 50.7***). This is possible since phosphinothricin has some structural similarity with the substrate glutamate. As a consequence of the inhibition of glutamine synthase, ammonia accumulates and kills the plant cells. Further, disturbance in glutamine synthesis also inhibits photosynthesis. Thus, the herbicidal activity of phosphinothricin is due to the combined effects of ammonia toxicity and inhibition of photosynthesis.

Strategy for phosphinothricin resistance

The natural detoxifying mechanism of phosphinothricin observed in *Streptomyces* sp has prompted scientists to develop resistant plants against this herbicide. The enzyme

phosphinothricin acetyl transferase (of *Streptomyces* sp) acetylates phosphinothricin, and thus inactivates the herbicide.

The gene responsible for coding phosphinothricin acetyl transferase (*bar* gene) has been identified in *Streptomyces hygroscopicus*. Some success has been reported in developing transgenic maize and oilseed rape by introducing *bar* gene. These plants were found to provide resistance to phosphinothricin.

SULFONYLUREAS AND IMIDAZOLINONES RESISTANCE

The herbicides namely sulfonylureas and imidazolinones inhibit the enzyme acetolactate synthase (ALS), a key enzyme in the synthesis of branched chain amino acids namely isoleucine, leucine and valine. Mutant forms of this enzyme and the corresponding genes have been isolated, identified and characterized. Transgenic plants with the mutant genes of ALS were found to be resistant to sulfonylureas and imidazolinones e.g. maize, tomato, sugarbeet.

Resistance to other herbicides

Besides the above, some other herbicide resistant plants have also been developed e.g. bromoxynil, atrazine, phenocarboxylic acids, cyanamide. A list of selected examples of gene transferred herbicide resistant plants is given in ***Table 50.7***.

It may however, be noted that some of the herbicide-resistant transgenic plants are at field-trial stage. Due to environmental concern, a few of these plants are withdrawn e.g. atrazine- resistant crops.

ENVIRONMENTAL IMPACT OF HERBICIDE-RESISTANT CROPS

The development genetically modified (GM) herbicide-resistant crops has undoubtedly contributed to increase in the yield of crops. For this reason, farmers particularly in the developed countries (e.g. USA) have started using these GM crops. Thus, the proportion of herbicide resistant soybean plants grown in USA increased from 17% in 1997 to 68% in 2001. The farmer is immensely benefited as there is a reduction in the cost of herbicide usage.

It is believed that the impact of herbicide-resistant plants on the environment is much lower than the direct use of the herbicides in huge quantities. There are however, other environmental concerns.

- Disturbance in biodiversity due to elimination of weeds.
- Rapid development of herbicide–resistance weeds that may finally lead to the production of ***super weeds***.

TOLERANCE TO WATER DEFICIT STRESSES

The environmental conditions such as ***temperature*** (heat, freezing, chilling), ***water***

TABLE 50.7 Selected examples of gene transferred herbicide resistant plants

Herbicide	*Gene transfer/mechanism of resistance*	*Transgenic crop(s)*
Glyphosate	Inhibition of EPSPS	Soybean, tomato
Glyphosate	Detoxification by glyphosate oxidase	Maize, soybean
Phosphinothricin	*bar* gene coding phosphinothricin acetyltransferase	Maize, rice, wheat, cotton, potato, tomato, sugarbeet
Sulfonylureas/imidazolinones	Mutant plant with acetolactate synthase	Rice, tomato, maize, sugarbeet
Bromoxynil	Nitrilase detoxification	Cotton, potato, tomato
Atrazine	Mutant plant with chloroplast *psb A* gene	Soybean
Phenocarboxylic acids	Monooxygenase detoxification (e.g. 2,4-D and 2,4,5-T)	Maize cotton
Cyanamide	Cyanamide hydratase gene	Tobacco

(EPSPS-5-Enoylpyruvylshikimate 3-phosphate synthase; 2,4-D-2,4-Dichlorophenoxy acetic acid; 2,4,5-T-2,4,5-Trichlorophenoxy acetic acid)

availability (shortage due to drought), and ***salinity*** influence the plant growth, development and yield. The abiotic stresses due to temperature, drought and salinity are collectively regarded as water deficit stresses (See ***Fig 50.1***).

Causes of water deficit

Water deficit may occur due to the following causes.

- Reduced soil water potential.
- Increased water evaporation (in dry, hot and windy conditions).
- High salt concentration in the soil (decrease soil water potential).
- Low temperature resulting in the formation of ice crystals.

Effects of water deficit

- Results in osmotic stress.
- Inhibits photosynthesis.
- Increases the concentration of toxic ions (reactive oxygen species) within the cells.
- Loss of water from the cell causing plasmolysis and finally cell death.

Tolerance to osmotic stress

The plant cells are subjected to severe osmotic stress due to water deficit. They however, produce certain compounds, collectively referred to as ***osmoprotectants*** or ***osmolytes***, to overcome the osmotic stress. Osmoprotectants are non-toxic compatable solutes and are divided into two groups.

1. **Sugar and sugar alcohols** e.g. mannitol, sorbitol, pinitol, ononitol, trehalose, fructans.

2. **Zwitterionic compounds :** These osmoprotectants carry positive and negative charges e.g. proline, glycine betaine.

The production of a given osmoprotectant is species dependent. The formation of mannitol, proline and glycine betaine are more closely linked to osmotic tolerance.

Strategies to develop water deficit tolerance plants

As explained above osmoprotectants offer good protection to plants against osmotic stress and therefore water deficit. It is therefore, logical to think of genetic engineering strategies for the increased production of osmoprotectants.

Some progress has been made in this direction. The biosynthetic pathways for the production of many osmoprotectants have been established and genes coding key enzymes isolated. In fact, some progress has been made in the development of transgenic plants with high production of osmoprotectants.

Transgenic plants with glycine betaine production

Glycine betaine is a quaternary ammonium compound and is electrically neutral. Besides functioning as a cellular osmolytes, glycine betaine stabilizes proteins and memebrane structures.

Some of the key enzymes for the production of glycine betaine have been identified e.g. choline monooxygenase, choline dehydrogenase, betaine aldehyde dehydrogenase. The genes coding these enzymes were transferred to develop transgenic plants.

By using choline oxidase gene from *Arthrobacter* sp, transgenic rice that produces higher glycine betaine (which offers tolerance against water deficit stress) has been developed.

RESISTANCE AGAINST ICE-NUCLEATING BACTERIA

Formation of ice on the plant cells (outer membrane) is a complex chemical process. The importance of ice-nucleating bacteria is recognized in recent years. The occurrence of these bacteria has been reported in most of the plants — cereals, fruits and vegetable crops. The ice-nucleating bacteria synthesize proteins, which coalesce with water molecules to form ice crystals at temperature around 32°F. As the ice crystals grow, they can pierce the plant cells and severely damage the plants.

Chemical treatment of plants to protect from ice formation

Plants can be treated with copper containing compounds to kill the bacteria. Another approach is to use urea solution so that the ice formation is minimized.

Ice-minus bacteria to resist plants from cold temperatures

The bacterium *Pseudomonas syringae* is one of the highly prevalent ice-forming organisms in nature. With genetic manipulations, the gene that directs the synthesis of ice-related bacterial proteins in *P. syringae* was removed. These newly developed bacteria are referred to as ice-minus bacteria.

The researchers proposed to spray the transgenic ice-minus bacteria on to young plants. The intention was that these bacteria would ***give frost tolerance to the plants***; and thus increase the crop yield. The opponents of DNA technology were against this approach—the main fear being that the bacterial mutants may create some health complication in humans. The researchers argued and justified that no new genetic information is introduced into *P. syringae* and it is closely related in all aspect to the parent one which is already in the environment. After prolonged court proceedings in USA, clearance was given for spraying the ice-minus bacteria in the fields.

It was in 1987, ice-minus bacteria were sprayed on to the field of potato plants and strawberry plants. Another strain of *P. syringae* commercially labeled, as ***Frostban*** was later developed and used in crop fieds.

It may be noted here that ***ice-minus bacteria of P. syringae were the first transgenic bacteria*** that were ***used outside the laboratory***. Fortunately, the experiments yielded encouraging results, since crop damage due to frost formation was found to be reduced.

Arabidopsis with cold-tolerant genes

Scientists were successful in developing cold-tolerant genes (around 20) in *Arabidopsis* when this plant was gradually exposed to slowly declining temperatures. They also identified a ***coordinating gene*** that encodes a protein, which acts as a transcription factor for regulating the expression of cold-tolerant genes. By introducing the coordinating gene, expression of cold-tolerant genes was triggered, and this protected the plants against cold temperatures. More work is in progress in this direction.

IMPROVEMENT OF CROP YIELD AND QUALITY

With the advances made in plant genetic engineering, improvement in crop yield and quality have become a reality. The crop yield is primarily dependent on the photosynthetic efficiency and the harvest index (the fraction of the dry matter allocated to the harvested part of the crop). The quality of the crop is dependent on a wide range of desirable characters-nutritional composition of edible parts, flavour, processing quality, shelf-life etc.

GREEN REVOLUTION

The 'Green Revolution' led by Borlaug, Swaminathan and Khus enabled the ***world's food supply to be tripled during the last three decades of 20th century***. This was made possible by adopting genetically improved varieties of crops, coupled with advances in crop management.

The development of high-yielding varieties of wheat and rice has enabled several developing countries (a good example being India) to move from a position of food scarcity to become net exporter of these cereals. The Green Revolution became a reality as the farmers adopted to new cereal seeds, besides employing high-input methods of agriculture — use of nitrogen fertilizers, herbicides, pesticides, modern equipment of agriculture etc.

Selected examples of crops for quality and yield

There are a wide range of crops that have been manipulated by scientists for improved yield and quality. Only selected examples are briefly described.

GENETIC ENGINEERING FOR EXTENDED SHELF-LIFE OF FRUITS

The genetic manipulation of fruit ripening has become an important commercial aspect in plant genetic engineering. Delay in fruit ripening has many advantages.

- It extends the shelf-life, keeping the quality of the fruit intact.
- Long distance transport becomes easy without damage to fruit.
- Slow ripening improves the flavour.

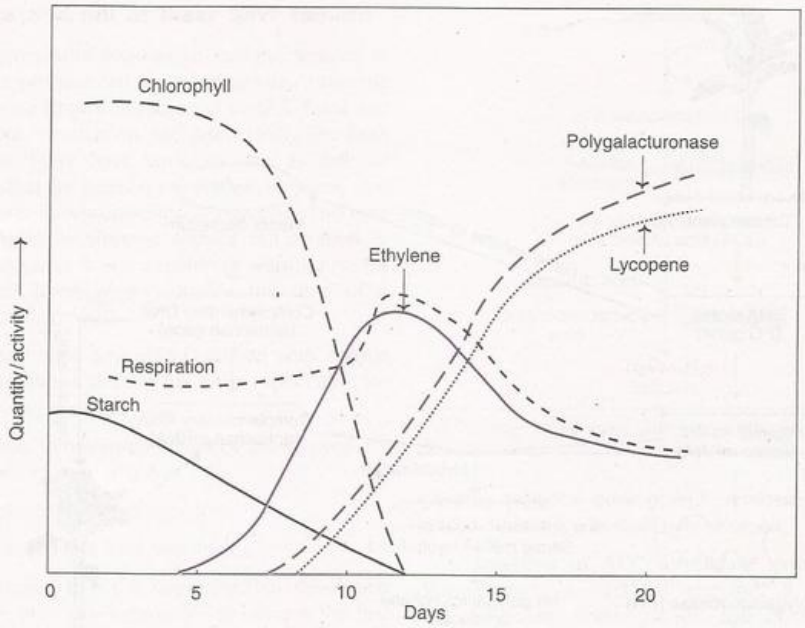


Fig. 50.8 : Biochemical changes during the process of tomato ripening.

Genetic engineering work has been extensively carried out in tomatoes, and some of the development are described.

BIOCHEMICAL CHANGES DURING TOMATO RIPENING

Fruit ripening is an active process. It is characterized by increased respiration accompanied by a rapid increase in ethylene synthesis. As the chlorophyll gets degraded, the green colour of the fruit disappears, and a red pigment, lycopene is synthesized (***Fig. 50.8***). The fruit gets softened as a result of the activity of cell wall degrading enzymes namely ***polygalacturonase*** (PG) and ***pectin methyl esterase***. The phytohormone ethylene production is intimately linked to fruit ripening as it triggers the ripening process of fruit. Addition of exogenous ethylene promotes fruit ripening, while inhibition of ethylene biosynthesis drastically reduces ripening.

The breakdown of starch to sugars, and accumulation of a large number of secondary products improves the flavour, taste and smell of the fruit.

Three distinct genes involved in tomato ripening have been isolated and cloned. The enzymes encoded by these genes and their respective role in fruit ripening are given in ***Table 50.8***.

TABLE 50.8 Clones of tomato ripening genes and their functions

Gene clone	*Enzyme synthesized*	*Function in ripening*
pTOM 5	Phytoene synthase	Lycopene synthesis that gives red colouration
pTOM 6	Polygalacturonase	Degradation of cell wall, resulting in fruit softening
pTOM 13	ACC oxidase	Ethylene formation that triggers fruit ripening

(ACC-1-Amino-cyclopropane-1-carboxylic acid)

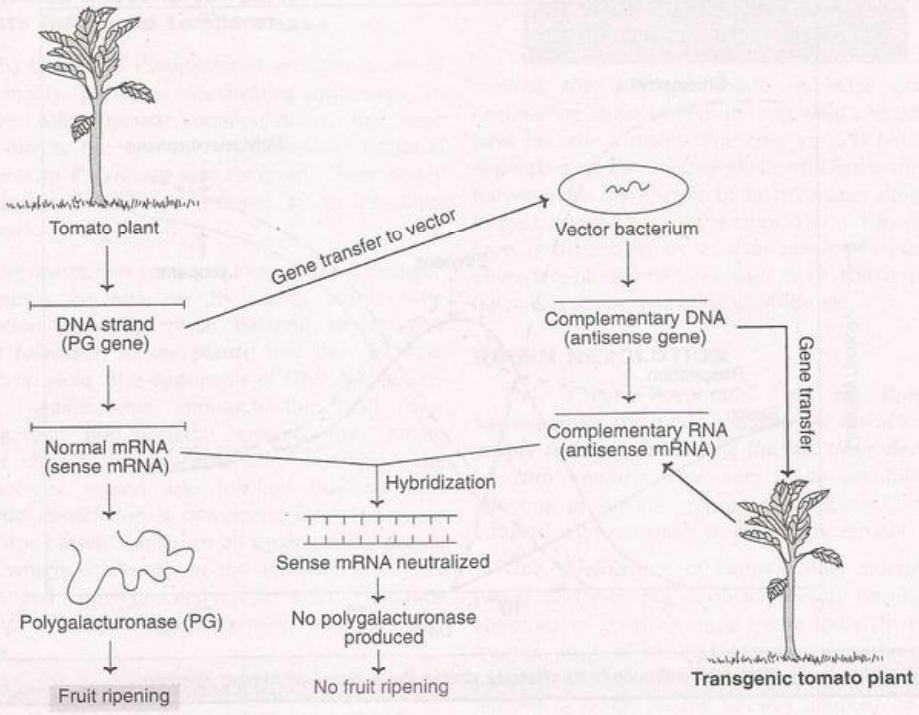


*Fig. 50.9 : Genetic manipulation of the enzyme polygalacturonase (PG) by antisense RNA approach (Production of **Flavr Savr** tomato plant)*

GENETIC MANIPULATIONS OF FRUIT RIPENING

Scientists have been trying to genetically manipulate and delay the fruit ripening process. Almost all the attempts involve ***antisense RNA approach***.

Manipulation of the enzyme polygalacturonase (development of Flavr Savr tomato)

As already stated, softening of the fruit is largely due to degradation of the cell wall (pectin) by the enzyme polygalacturonase (PG). The gene responsible for PG, the rotting enzyme, has been cloned (pTOM 6). The genetic manipulation of polygalacturonase by antisense RNA approach for the development of ***Flavr Savr tomato*** (by Calgene company in USA) is depicted in ***Fig. 50.9***, and mainly involves the following stages.

1. Isolation of the DNA from tomato plant that encodes the enzyme polygalacturonase (PG).

2. Transfer of PG gene to a vector bacteria and production of complementary DNA molecules.

3. Introduction of complementary DNA into a fresh tomato plant to produce a transgenic plant.

Mechanism of PG antisense RNA approach : In the normal tomato plant, PG gene encodes a normal (sense) mRNA that produces the enzyme polygalacturonase that is actively involved in fruit ripening. The complementary DNA of PG encodes for antisense mRNA, which is complementary to normal (sense) mRNA. The hybridization between the sense and antisense mRNAs renders the sense mRNA ineffective. Consequently, no polygalacturonase is produced, hence fruit ripening is delayed.

The rise and fall of Flavr Savr tomato

The genetically engineered tomato, known as Flavr Savr (pronounced flavour saver) by employing PG antisense RNA was approved by U.S. Food and Drug Administration on 18th May 1994. The FDA ruled that Flavr Savr tomatoes are as safe as tomatoes that are bred by conventional means, and therefore no special labeling is required. The new tomato could be shipped without refrigeration to far off places, as it was capable of resisting rot for more than three weeks (double the time of a conventional tomato).

Although Flavr Savr was launched with a great fanfare in 1995, it did not fulfill the expectation for the following reasons.

- Transgenic tomatoes could not be grown properly in different parts of U.S.A.
- The yield of tomatoes was low.
- The cost of Flavr Savr was high.

It is argued that the company that developed Flavr Savr, in its overenthusiasm to become the first Biotech Company to market a bioengineered food, had not taken adequate care in developing the transgenic plant. And unfortunately, within a year after its entry, Flavr Savr was withdrawn, and it is now almost forgotten!

Manipulation of ethylene biosynthesis

It has been clearly established that ethylene plays a key role in the ripening of fruits. The biosynthetic pathway of ethylene is depicted in ***Fig 50.10***. Ethylene is synthesized from S-adenosyl methonine via the formation of an intermediate, namely ***1-aminocyclopropane-1 carboxylic acid (ACC)***, catalysed by the enzyme ACC synthase. The next step is the conversion of ACC to ethylene by ACC oxidase.

Three different strategies have been developed to block ethylene biosynthesis, and thereby reduce fruit ripening.

1. **Antisense gene of ACC oxidase :** Transgenic plants with antisense gene of ACC oxidase have been developed. In these plants, production of ethylene was reduced by about 97% with a significant delay in fruit ripening.

2. **Antisense gene of ACC synthase :** Ethylene biosynthesis was inhibited to an extent of 99.5% by inserting antisense gene of ACC synthase, and the tomato ripening was markedly delayed.

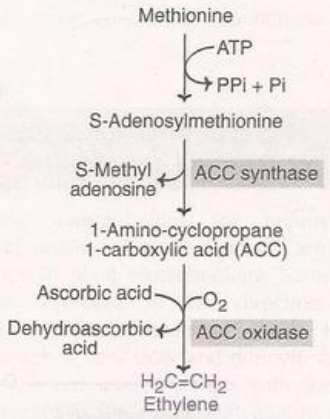


Fig. 50.10 : Biosynthesis of ethylene.

3. **Insertion of ACC deaminase gene :** ACC deaminase is a bacterial enzyme. It acts on ACC (removes amino group), and consequently the substrate availability for ethylene biosynthesis is reduced. The bacterial gene encoding ACC deaminase has been transferred and expressed in tomato plants. These transgenic plants inhibited about 90% of ethylene biosynthesis. The fruit ripening was delayed by about six weeks.

The strategies ***1 and 2*** may be referred to as ***antisense ethylene technology***.

LONGER SHELF-LIFE OF FRUITS AND VEGETABLES

The spoilage of fruits, vegetables and senescence of picked flowers, collectively referred to as ***post-harvest spoilage*** is major concern in agriculture. This hampers the distribution system particularly when the transport is done to far off places.

The successful manipulations to delay ripening, senescence and spoilage of various foods will significantly contribute to the appropriate food distribution and thus good economic practices in agriculture.

Suppressing the biosynthesis of ethylene appears to be a promising area to reduce the spoilage of fruits, vegetables and senescence of flowers. The three different strategies to block ethylene synthesis

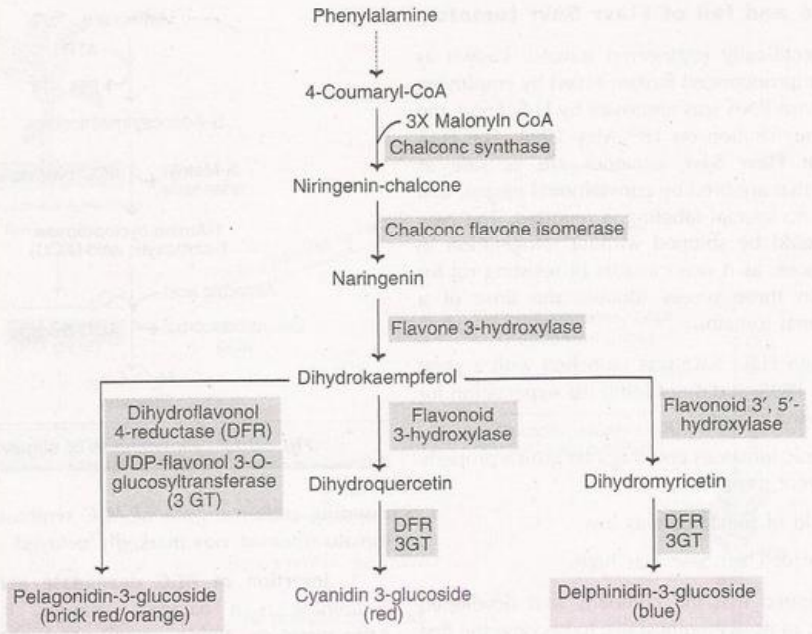


Fig. 50.11 : Biosynthesis of anthocyanins.

in tomato have been described. The same approaches in fact can be successfully used for other fruits, vegetables etc., to ***achieve longer shelf-life***

GENETIC ENGINEERING FOR PREVENTING DISCOLORATION

Discoloration of fruits and vegetables is a major postharvest problem encountered in food industry. Certain food additives are added to prevent discoloration. However, these additives may cause health complications in humans.

Biochemically, discoloration of fruits and vegetables is mainly due to the oxidation of phenols (mono- and diphenols) to quinones, catalysed by a group of enzymes namely ***polyphenol oxidases***. These enzymes are localized in the membranes of mitochondria and chloroplasts.

Genetic manipulations using antisense approach to inhibit the synthesis of polyphenol oxidase has been carried. Some success has been reported in preventing the discoloration of potatoes by this strategy.

GENETIC ENGINEERING FOR FLOWER PIGMENTATION

There are continuous attempts in flower industry to make the ornamental flowers more attractive (by improving or creating new colours), besides prolonging post-harvest lifetime. The cut flower industry is mostly (about 70%) dominated by four plants—roses, tulips, chrysanthemums and carnations.

The most common type of flower pigments are anthocyanins, a group of flavonoids. They are synthesized by a series of reactions, starting from the amino acid phenylalanine (***Fig. 50.11***). The colour of the flower is dependent on the chemical nature of the anthocyanin produced.

- Pelagonidin 3-glucoside — brick red/orange.
- Cyanidin 3-glucoside — red.
- Delphinidin 3-glucoside — blue to purple.

Manipulation of anthocyanin pathway enzymes

The enzymes responsible for different reactions, in the anthocyanin pathway have been identified.

By genetic manipulations and mutations, it is possible to develop flowers with the desired colours.

Most of the flowers (roses, carnations chrysanthemums) lack blue colour due to the absense of the key enzyme flavonoids 3', 5'-hydroxylase (F 3'5' H) that produces delphinidine 3 glucoside. One company, by the name Florigene, has genetically manipulated and introduced the gene encoding the enzyme F 3' 5' H (from *Petunia hybrida*) into the following plants.

The ***world's first genetically modified (GM) flower*** was introduced in 1996. It was a mauve (bluish) coloured carnation with a trade name ***Moondust***™. Subsequently, many other flowers have been produced and marketed.

Can GM-flowers be eaten?

In majority of countries, flowers are used for ornamental purposes and not usually eaten. However, in some countries like Japan, flower petals are used for decoration of foods, and frequently eaten also. This raises an important question about the safety of GM flowers, since they are not thoroughly screened for human consumption. But the present belief is that the anthocyanins (the colouring chemical molecules) are natural plant materials, and their consumption may be in fact beneficial to health.

GENETIC ENGINEERING FOR MALE STERILITY

The plants may inherit male sterility either from the nucleus or cytoplasm. ***cytoplasmic male sterility (cms)*** is due to the defects in the mitochondiral genome. It is possible to introduce male sterility through genetic manipulations while the female plants maintain fertility.

In tobacco plants, male sterility was introduced by using a mitochondrial mutated gene encoding the enzyme ribonuclease. The gene encoding ribonuclease namely *barnase* gene from *Bacillus amyloliquefaciens* was transferred to tobacco plants. The ribonuclease is toxic to tapetal cells, and thus prevents the development of pollen, ultimately leading to male sterility. By this approach, transgenic plants of tobacco, cauliflower, cotton, tomato, corn, lettuce etc., with male sterility have been developed.

It is possible to restore male sterility in the above plants by crossing them with a second set of transgenic plants containing ribonuclease inhibitor gene.

TRANSGENIC PLANTS WITH IMPROVED NUTRITION

Genetic manipulations for improving the nutritional quality of plant products are of great importance in plant biotechnology. Some success has been achieved in this direction through conventional cross-breeding of plants. However, this approach is very slow and difficult, and many a times will not give the traits with the desired improvements in the nutritional quality. Selected examples of genetic engineering with improved nutritional contents are described.

AMINO ACIDS OF SEED STORAGE PROTEINS

Of the 20 amino acids present in the humans, 10 are essential while the other 10 can be synthesized by the body. The 10 essential amino acids (EAAs) have to be supplied through the diet. Cereals (rice, wheat, maize, corn) are the predominant suppliers of EAAs. However, cereals do not contain adequate quantity of the essential amino acid lysine. On the other hand, pulses (Bengal gram, red gram, soybean) are rich in lysine and limited in sulfur-containing amino acids (the essential one being methionine).

Transgenic routes have been developed to improve the essential amino acid contents in the seed storage proteins of various crop plants.

Overproduction of lysine by deregulation

The four essential amino acids namely lysine, methionine, threonine and isoleucine are produced from a non-essential amino acid aspartic acid (***Fig. 50.12***). The formation of lysine is regulated by feedback inhibition of the enzymes aspartokinase (AK) and dihydrodipicolinate synthase (DHDPS). Theoretically, it is possible to overproduce lysine by abolishing the feedback regulation. This is what has been accomplished.

The lysine feedback-insensitive genes encoding the enzymes AK and DHPDS have been respectively isolated from *E.coli* and *Cornynebacterium* with appropriate genetic

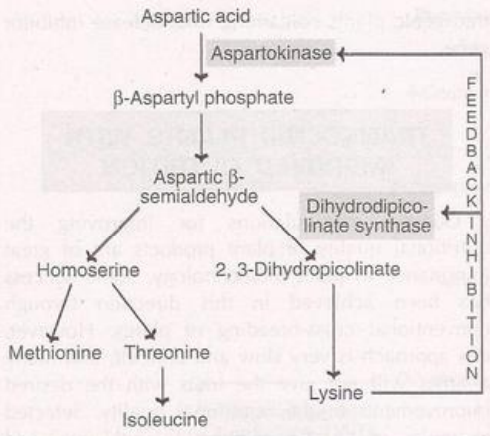


*Fig. 50.12 : An overview of the biosynthetic pathway of some essential amino acids derived from aspartic acid (**Note** : Lysine regulates its own synthesis by feedback inhibition).*

manipulations, these genes were introduced into soybean and canola plants. The transgenic plants so developed produced high quantities of lysine.

Transfer of genes encoding methionine-rich proteins

Several genes encoding methonine-rich proteins have been identified.

- In maize, 21 KDa zein with 28% methionine.
- In rice, 10 KDa prolamin with 20% methionine.
- In sunflower, seed albumin with 16% methionine.

These genes have been introduced into some crops such as soybean, maize and canola.

The transgenic plants produced proteins with high contents of sulfur-containing amino acids.

Production of lysine-rich glycinin in rice

Glycinin is a lysine-rich protein of soybean. The gene encoding glycinin has been introduced into rice and successfully expressed. The transgenic rice plants produced glycinin with high contents of lysine. Another added advantage of ***glycinin*** is that its consumption in humans is associated with a ***reduction in serum cholesterol*** (hypo-cholesterolaemic effect).

Construction of artificial genes to produce proteins rich in EAAs

Attempts are being made to construct artificial genes that code for proteins containing the essential amino acids in the desired proportion. Some success has been reported in the production of one synthetic protein containing 13% methionine residues.

GENETIC ENGINEERING FOR IMPROVING PALATABILITY OF FOODS

More than the nutritive value, taste of the food is important for attracting humans. It is customary to make food palatable by adding salt, sugar, flavors and many other ingradients. It would be nice if a food has an intrinsically appetizing character.

A protein ***monellin*** isolated from an African plant (*Dioscorephyllum cumminsii*) is about ***100,000 sweeter than sucrose*** on molar basis. Monellin gene has been introduced into tomato and lettuce plants. Some success has been reported in the production of monellin in these plants, improving the palatability.

GOLDEN RICE — THE PROVITAMIN A ENRICHED RICE

About one-third of the world's population is dependent on rice as staple food. The milled rice that is usually consumed is almost deficit in β-carotene, the provitamin A. As such, vitamin A deficiency (causing night blindness) is major nutritional disorder worldover, particularly in people subsisting on rice.

To overcome vitamin A deficiency, it was proposed to genetically manipulate rice to produce β-carotene, in the rice endosperm. The presence of ***β-carotene in the rice gives*** a characteristic ***yellow/orange colour***, hence the provitamin A-enriched rice are appropriately considered as Golden Rice.

In ***Fig. 50.13*** an outline of the biosynthetic pathway for the formation of β-carotene is given. The genetic manipulation to produce Golden Rice required the introduction of three genes encoding the enzymes phytoene synthase, carotene desaturase and lycopene β-cyclase. It took about 7 years to insert three genes for developing Golden Rice.

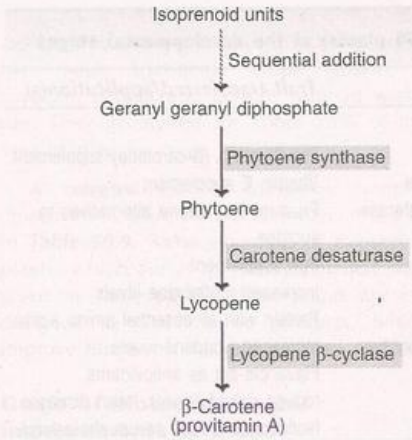


Fig. 50.13 : An outline of pathway for the biosynthesis of provitamin A (β-carotene).

Golden Rice has met almost all the objections raised by the opponents of GM foods. However, many people are still against the large scale production of Golden Rice, as this will open door to the entry of many other GM foods. Another argument put forth against the consumption of Golden Rice is that it can supply only about 20% of daily requirement of vitamin A. But the proponents justify that since rice is a part of a mixed diet consumed (along with many other foods), the contribution of provitamin A through Golden Rice is quite substantial.

Recently (in 2004), a group of British scientists have developed an improved version of Golden Rice. The new strain, Golden Rice 2 contains more than 20 times the amount of provitamin A than its predecessor. It is claimed that a daily consumption of 70 g rice can meet the recommended dietary allowance for vitamin A.

TABLE 50.9 A selected list of transgenic crop plants (GM crops approved in USA) for commercial use

Crop plant	*Genetically altered trait*	*Product name*
Cotton	Insect resistance	Bollguard
	Glyphosate resistance	Roundup ready
	Bromoxynil resistance	BXN
	Sulfonylurea resistance	—
Maize	Insect resistance	Yield Guard
	Insect resistance	Maximizer
	Glyphosate resistance	Roundup Ready
	Glufosinate resistance	Liberty Link
Rice	Vitamin A enrichment	Golden Rice
Tomato	Delayed ripening	Flavr Savr
	Delayed ripening	Endless Summer
	Virus resistance	—
Soybean	Glyphosate resistance	Roundup Ready
Potato	Insect resistance	Newleaf
	Modified starch	—
Oilseed rape (canola)	Glufosinate resistance	Innovator
	Glyphosate resistance	Roundup Ready
	High lauric acid	Laurical
	Male sterility hybrid	—
Squash	Virus resistance	Freedom II
Tobacco	Virus resistance	—
Capsicum	Virus resistance	—
Carnation	Modified flower colour	—

TABLE 50.10 Some examples of transgenic crop plants (GM plants) at the developmental stages

Plant	*Gene transfer*	*Trait transferred/application(s)*
For improving human health		
Tomato	Phytoene desaturase	Provitamin A (β-carotene) supplement
Canola	γ-Tocopherol methyl transferase	Vitamin E supplement
Sugar beet	Sucrose-sucrose fructosyl transferase	Fructans-low calorie alternatives to sucrose
Rice	Ferritin	Iron supplement
Potato	Antisense threonine synthase	Increased methionine levels
Potato	Seed albumin	Protein with all essential amino acids
Tomato	S-Adenosylmethionine decarboxylase	Increased lycopene levels
Tomato	Chalcone isomerase	Flavanols-act as antioxidants, reduce risk of cancer, heart diseases
Arabidopsis	Isoflavone synthase	Isoflavones-reduce serum cholesterol, and reduce osteoporosis
Canola	Modified acyl-acyl carrier protein thioesterase	*cis*-Stearates-lower the risk of heart diseases
For increased crop yield		
Rice	Phosphoenol pyruvate carboxylase	Increased efficiency of photosynthesis
Tobacco	Phytochrome A	Avoids shades
Lettuce	Gibberellic acid (GA) oxidase	Inhibits GA accumulation and stem growth (dwarfing)
Potato	Phytochrome B	Increased photosynthesis and longer life span
Others		
Tobacco and soybean	Cytochrome P_{450}	Synthesis of epoxy fatty acids for manufacture of adhesives and paints
Rice	Nicotianamine aminotransferase	Tolerance to low iron availability
Tobacco	Nitroreductase	Reduces land contamination by trinitrotoluene

GENETIC ENGINEERING TO INCREASE VITAMINS AND MINERALS

The transgenic rice (Golden Rice) developed with high provitamin A content is described above. Transgenic crop plants are also being developed for increased production of other vitamins and minerals.

A transgenic *Arabidopsis thaliana* that can produce ten-fold higher ***vitamin E*** (α-tocopherol) than the native plant has been developed. This was done by a novel approach. *A. thaliana* possesses the biochemical machinery to produce a compound close in structure to α-tocopherol. A gene that can finally produce α-tocopherol is also present, but is not expressed. This dormant gene was activated by inserting a regulatory gene from a bacterium. This resulted in an efficient production of vitamin E.

Some workers are trying to increase the mineral contents of edible plants by enhancing their ability to absorb from the soil. Some success has been reported with regard to increased concentration of iron.

COMMERCIAL TRANSGENIC CROP PLANTS

The very purpose of production of transgenic plants is for their commercial importance with high productivity. It was in 1995-96, transgenic plants

(potato and cotton) were, for the first time, made available to farmers in USA. By the year 1998-99, five major transgenic crops (cotton, maize, soybean, canola and potato) were in widespread use. They accounted for about 75% of the total area planted by crops in USA.

A selected list of transgenic crop plants (approved in USA) for the commercial use is given in ***Table 50.9.*** Some examples of transgenic crop plants, which are at the developmental stages are given in ***Table 50.10***. These plants are carefully designed to give rise to products, which will improve human health and increase of crop yield.

Goals of biotechnological improvements in crops

There are about 30–40 crops that have been genetically modified, and many more are being added. However, very few of them have got the clearance for commercial use. A selected list is already given in ***Table 50.9.***

The ultimate goals of genetically modified (GM) crop plants are listed below.

- Resistance to diseases (insect, microorganisms).
- Improved nitrogen fixing ability.
- Higher yielding capacity.
- Resistance to drought and soil salinity.
- Better nutritional properties.
- Improved storage qualities.
- Production of pharmaceutically important compounds.
- Absence of allergens.
- Modified sensory attributes e.g. increased sweetness as in thaumatin.

Concerns about transgenic plants

The fears about the harmful environmental and hazardous health affects of transgenic plants still exists, despite the fact that there have been no reports so far in this regard.

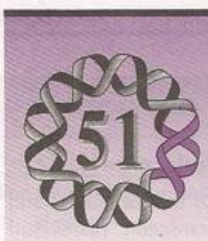
The transfer of almost all the transgenic plants from the laboratory to the crop fields is invariably associated with legal and regulatory hurdles, besides the social and economic concerns.

The major concern expressed by public (also acknowledged by biotechnologists) is the development of resistance genes in insects, generation of super weeds etc. Several remedial measures are advocated to overcome these problems.

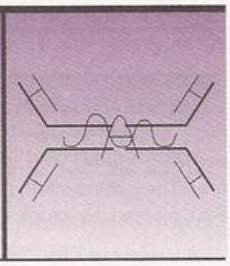
The farmers in developing countries are much worried about the ***seed terminator technology*** which forces them to buy seeds for every new crop. These farmers are traditionally habituated to use the seeds from the previous crop which is now not possible due to seed termintor technology.

TRANSGENIC PLANTS AS BIOREACTORS

Another important application of genetically transformed plants is their utility as bioreactors to produce a wide range of metabolic and industrial products. These aspects are described in the next chapter.



Transgenic Plants as Bioreactors



Genetically modified plants can be manipulated to act as bioreactors to produce a wide range of biologically important compounds. These include carbohydrates, lipids and proteins, besides the secondary products. The commercial products of plants are useful for industries or for improving the human or animal health.

The terms ***molecular farming/pharming or metabolic engineering of plants*** are also used to collectively represent the molecular biological techniques for the synthesis of commercial products in plants. Metabolic engineering of transgenic plants is gaining importance in recent years as an attractive alternative to animals and micro-organisms for the manufacture of biologically important products.

METABOLIC ENGINEERING OF CARBOHYDRATES

An outline of the biosynthetic pathways for the production of different carbohydrates is given in ***Fig 51.1***. As is evident, the important carbohydrates are synthesized and stored in different cellular compartments.

- Starch and its derivatives are synthesized in the plastids.
- Sugars and sugar derivatives are produced in the cytosol and accumulate throughout the cell.
- Fructans synthesized and stored in the vacuoles.

STARCH

Starch is a polymer of glucose, and is composed of amylose and amylopectin. Starch is used as a food, feed and for industrial purposes. Genetic engineering techniques can be used to manipulate the quantity and quality of starch.

Increasing the production of starch

Biosynthesis of starch in plants is a highly regulated process involving the key enzyme ***ADP-glucose pyrophosphorylase***. This enzyme is allosterically controlled (feedback inhibition) by metabolites such as phosphate.

ADP-glucose pyrophosphorylase in *E.coli* was mutated to alter its allosteric properties. This mutant gene was transferred and expressed in potato plants. The ***transgenic potato plants produced high quantities of starch***.

Synthesis of amylopectin starch

Starch normally contains 20–30% amylose and 70–80% amylopectin. The relative proportion of amylose and amylopectin determine the physico-chemical properties of starch. Amylopectin gels are more stable and therefore are preferred in food processing industries.

The enzyme namely ***granule-bound starch synthase (GBSS)*** is responsible for the synthesis of amylose while ***starch branching enzyme (SBE)*** produces amylopectin (***Fig 51.1***). By using an

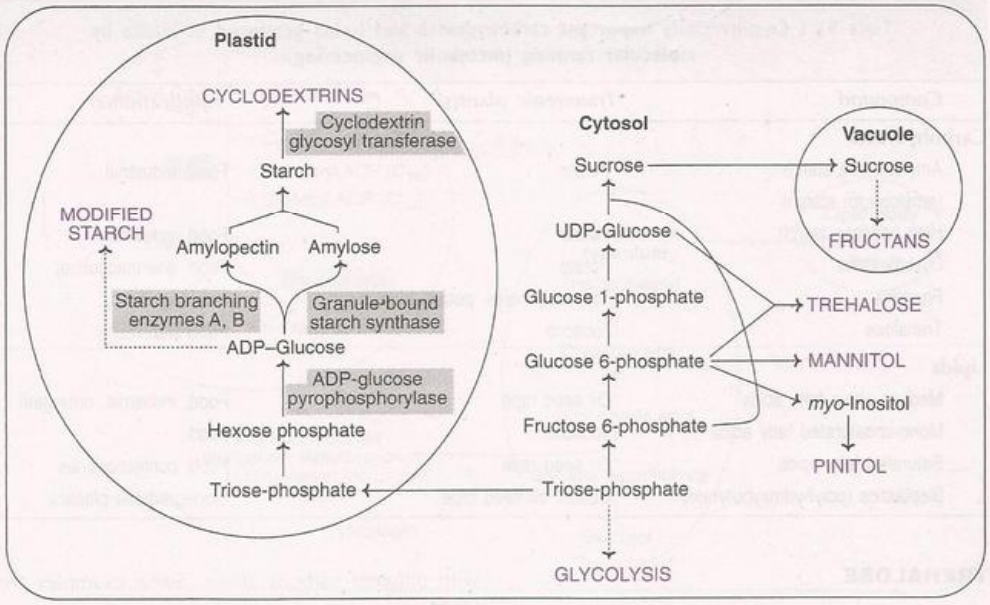


Fig. 51.1 : *An outline of the metabolic pathway for the biosynthesis of carbohydrates and their important derivatives in a plant cell.*

antisense approach, the enzyme GBSS was inhibited in potato plants. Consequently, amylose-free starch i.e. starch containing only amylopectin was produced.

Synthesis of high-amylose starch

High-amylose starch with limited branching is useful as a food, feed and for certain industrial purposes. It is possible to produce high-amylose starch in potatoes by antisense inhibition of starch branching enzyme A and B (SBE A and SBE B).

CYCLODEXTRINS

Cyclodextrins are cone-shaped rings formed by 6-8 glucopyranose subunits. They are hydrophilic in nature and can pocket hydrophobic compounds. Due to this property, cyclodextrins are used as therapeutic agents to solubilise hydrophobic pharmaceuticals such as steroids.

A key enzyme responsible for the synthesis of cyclodextrins from starch namely ***cyclodextrin glycosyl transferase*** has been identified. A bacterial gene coding for this enzyme was transferred to potato plants. However, the transgenic potato plants did not show any enhancement in the synthesis of cyclodextrins due to various reasons.

FRUCTANS

Fructans, also referred to as polyfructans, are soluble polymers of fructose. They are synthesized and stored in cellular vacuoles. Short-chain fructans (oligofructans) are almost as sweet as sucrose. Thus, they can be used as sugar substitutes in food and soft drink industries. Another advantage is that fructans are not digested in the gut and therefore, they serve as low-calorie sweeteners.

Transgenic potato plant with ***fructosyl transferase gene*** (from *Streptomyces*) has been developed. Fructosyl transferase, being a key enzyme enhanced the synthesis of fructans.

In recent years, oligofructans of sugarbeet (fructan beet) with suitable genetic manipulation have been produced. They serve as low calorie sweeteners.

Table 51.1 Commercially important carbohydrates and lipids produced in plants by molecular farming (metabolic engineering)

Compound	*Transgenic plant(s)*	*Application(s)*
Carbohydrates		
Amylose-free starch (amylopectin starch)	Potato	Food, industrial
High amylose starch	Potato	Food, industrial
Cyclodextrins	Potato	Food, pharmaceutical
Fructans	Tobacco, maize, potato, sugarbeet	Industrial, food
Trehalose	Tobacco	Food stabilizer
Lipids		
Medium chain fatty acids	Oil seed rape	Food, industrial, detergent
Mono-unsaturated fatty acids	Tobacco	Food
Saturated fatty acids	Oil seed rape	Food, confectioneries
Bioplastics (polyhydroxybutyrate)	Cotton, oil seed rape	Biodegradable plastics

TREHALOSE

Trehalose is a disaccharide produced in plants and microorganisms in response to osmotic stress. Genetic manipulations have been successfully carried out for enhanced synthesis of trehalose to withstand abiotic stress (that cause water deficit).

Trehalose is useful as a food preservative. Some workers are attempting to produce trehalose in bulk quantities for its utility in food industries.

METABOLIC ENGINEERING OF LIPIDS

Plant oils are useful as foods (about two thirds), and for industrial purposes (about one third). The industrial applications of plant oils include their use in the manufacture of soaps, detergents, lubricants and biofuels.

An outline of the synthesis of oils (chemically triacylglycerols) is depicted in ***Fig 51.2***. There is an involvement of plastids, cytoplasm and endoplasmic reticulum for the production of oils. After their synthesis, oils are stored in lipid bodies.

Acetate from the cytoplasm is taken up by the plastids and converted to acetyl CoA and malonyl CoA. These two molecules undergo a series of reactions, catalysed by the enzyme fatty acid synthase (FAS), to produce fatty acyl carrier proteins with different carbons atoms. Some examples are listed.

- Laurate — C_{12}
- Myristate — C_{14}
- Palmitate — C_{16}
- Stearate — C_{18}
- Oleate — $C_{18:1}$

By the action of acyl-ACP thioesterases, the fatty acids are released and exported to the cytoplasm. Here, they may undergo certain modifications — elongation, desaturation (insertion of double bonds), hydroxylation etc. These modified fatty acids react with glycerol 3-phosphate (in the endoplasmic reticulum) to finally form triacylglycerols (TG). TG are transported to lipid oil bodies and stored. The oil bodies of seeds also contain proteins called ***oleosins***, in the lipid monolayers (for discussion on oleosins, See p. 631).

Some of the approaches of genetic engineering for the manipulation of plant oil biosynthesis are briefly described.

PRODUCTION OF SHORTER CHAIN FATTY ACIDS

Majority of the plant oils contain more than 16-carbons e.g. palmitic, stearic, oleic acids. Oils with shorter chains (C_8—C_{14}) are more useful in

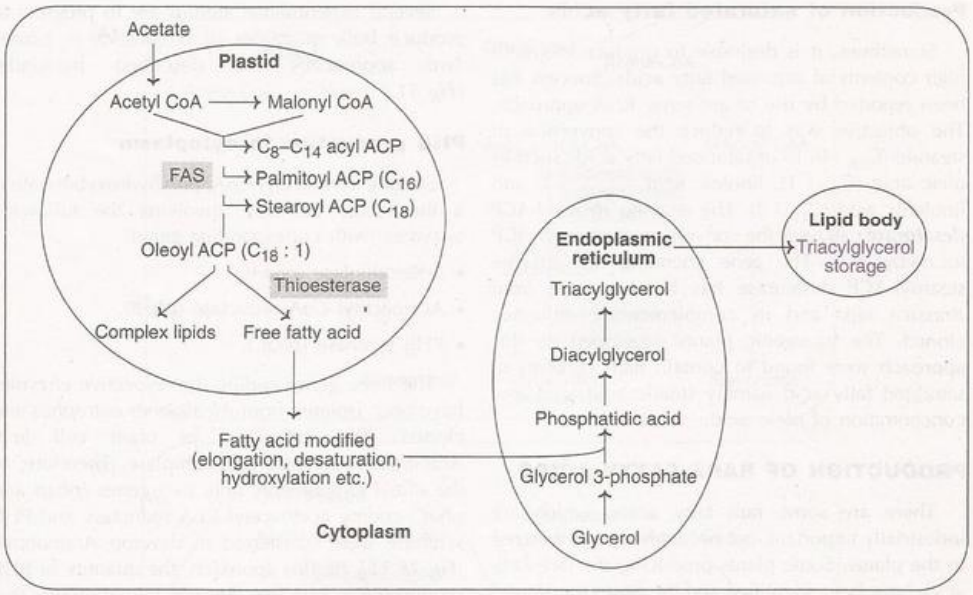


Fig. 51.2 : An outline of the biosynthesis of triacylglycerols in plants (FAS-Fatty acid synthase; ACP-Acyl carrier protein).

many industries — for the production of soaps, detergents, cosmetics etc.

It is possible to terminate the hydrolysis of acyl-ACP by specific thioesterases and produce high proportion of selected fatty acids. One acyl-ACP thioesterase that specifically hydrolyses lauroyl-ACP has been isolated from California bay tree (*Umbellularia californica*). The gene encoding this enzyme has been cloned and transferred to oilseed rape. The transgenic plants were found to produce oils with high proportion of lauric acid (12-carbon fatty acid).

PRODUCTION OF LONGER CHAIN FATTY ACIDS

For the application as industrial oils, triacylglycerols with longer chain fatty acids (> C_{18}) are preferred. Thus, erucic acid (22C) containing oils are useful in industries. However, erucic acid is unsuitable for human consumption.

By conventional plant breeding two distinct crops of oilseed rape have been developed — ***high-erucic acid rape (HEAR)*** and ***low-erucic acid rape (LEAR)***.

Some attempts are being made to genetically manipulate crops to yield oils with high contents of erucic acid. The approaches include to overexpress genes encoding the enzymes elongases and transfer of genes to produce enzymes that can preferentially incorporate erucic acid into triacylglycerols.

PRODUCTION OF FATTY ACIDS WITH MODIFIED DEGREE OF SATURATION

A fatty acid is said to be saturated if it has no double bonds. Saturated and unsaturated (with on or more double bonds) fatty acids occur in nature. It is possible to genetically manipulate the degree of saturation in fatty acids.

Production of unsaturated fatty acids

Oleic acid (C_{18} : 1) rich oils are useful as food, feed and in some industries. There has been some success in the transfer of antisense gene encoding the enzyme ***desaturase*** in oilseed rape and soybean plants. These transgenic plants were found to produce oils with very high proportion of oleic acid.

Production of saturated fatty acids

Sometimes, it is desirable to produce oils with high contents of saturated fatty acids. Success has been reported by use of antisense RNA approach. The objective was to reduce the conversion of stearate (C_{18} : 0) to unsaturated fatty acids such as oleic acid (C_{18} : 1), linoleic acid (C_{1818} : 2) and linolenic acid (C_{18} : 3). The enzyme ***stearoyl-ACP desaturase*** catalyses the conversion of stearoyl-ACP to oleoyl-ACP. The gene encoding the enzyme stearoyl-ACP desaturase has been isolated from *Brassica rapa* and its complementary sequence cloned. The transgenic plants developed by this approach were found to contain high contents of saturated fatty acid namely stearic acid, and low concentration of oleic acid.

PRODUCTION OF RARE FATTY ACIDS

There are some rare fatty acids, which are industrially important, but not normally synthesized in the plants. Some plants producing the rare fatty acids have been identified and the genes transferred for their overproduction.

One good example is the production of ***petroselenic acid*** that is found in coriander. Tobacco plants were transformed with coriander ***acyl-ACP desaturase*** that led to a high production of petroselenic acid in tobacco plants. Petroselenic acid is commercially important. On oxidation by ozone, it forms lauric acid (for use in soap and detergent production) and adipic acid (for the manufacture of nylon).

Some success has also been reported in the increased production of several unsaturated fatty acids through genetic manipulation by incorporating genes coding special enzymes namely ***front-end desaturases*** e.g. γ-linolenic acid, arachidonic acid, ricolenic acid.

BIODEGRADABLE PLASTICS (BIOPLASTICS)

Biodegradable plastics (or bioplastics) are chemically ***polyhydroxy alkanoates (PHAs)***. They are currently being produced in large quantities by microbial fermentation. The structures and the biosynthetic pathways for the production of PHAs are given in Chapter 30. Among the PHAs, ***polyhydroxy butyrate*** (PHB) is the most important one.

Several experimental studies are in progress to produce bulk quantities of bioplastics in plants. Two approaches are described hereunder ***(Fig 51.3).***

PHB production in cytoplasm

Starting from acetyl CoA, polyhydroxybutyrate is a three-stage pathway, involving the following enzymes (with corresponding gene).

- 3-Ketothiolase (*phaA*).
- Acetoacetyl CoA reductase (*phaB).*
- PHB synthase (*phaC).*

The three genes coding the respective enzymes have been isolated from *Alcaligenes eutrophus* and cloned. The cytoplasm of plant cell (e.g. *Arabidopsis*) contains 3-ketothiolase. Therefore, in the initial experiments only two genes (*phaB* and *phaC)* coding acetoacetyl CoA reductase and PHB synthase were transferred to develop *Arabidopsis* ***(Fig 51.3A).*** By this approach, the quantity of BHP produced was very low. Another limitation was that the plants had stunted growth.

PHB production in plastids

The problems encountered in the above approach were overcome by transferring all three genes (*phaA, phaB, phaC*) of PHB synthesis and targeting them to chloroplast (***Fig 51.3B***). To achieve this, each gene was separately fused with a coding sequence of transit peptid bound to N-terminal fragment of ***Rubisco*** (ribulose 1,5-bisphosphate carboxylase) subunit protein. The genes expression was carried out by CaMV 35S promoter. Transgenic *Arabidopsis* plants with each gene construct were first developed. Then a series of sexual crossings were carried out between the individual transformants. The transgenic plants developed by this approach yielded good quantity of bioplastics (about 14% dry weight of the plant), and in addition there was no observable adverse affect on the growth or fertility of these plants.

Production of bioplastics in cotton fibres

Cotton fibres contains the enzyme β-ketothiolase. Therefore, the genes for the other two enzymes of PHB pathway (*phaB* and *phaC*) from *Alcaligenes eutrophus* were transferred into meristems of cotton

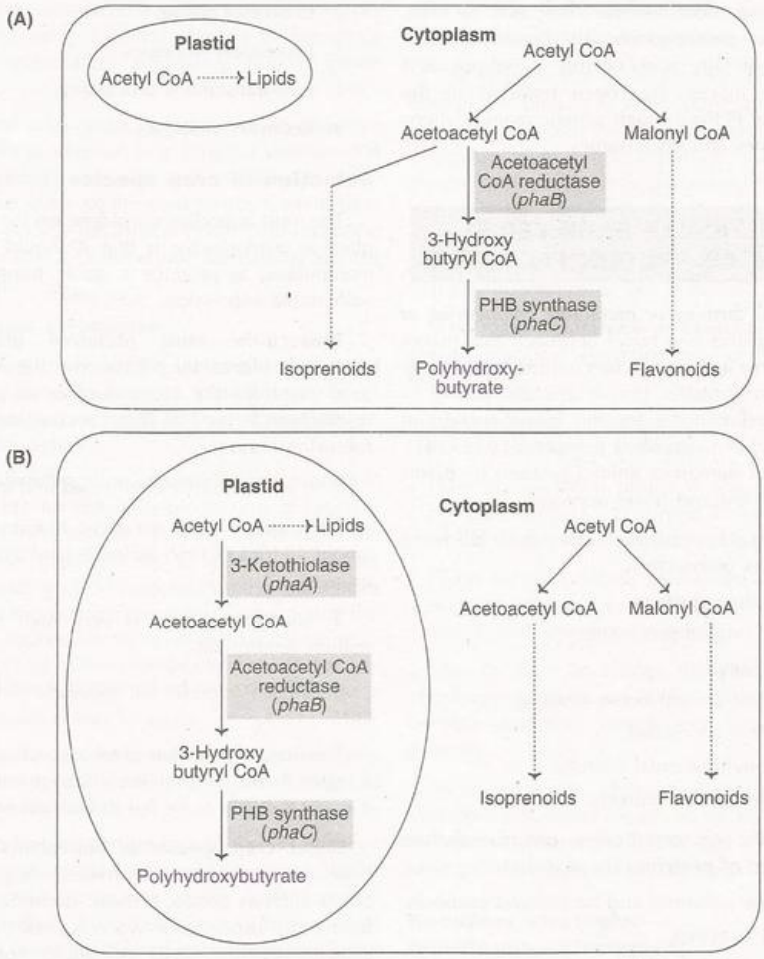


Fig. 51.3 : *Development of transgenic* Arabidopsis *for the biosynthesis of polyhydroxybutyrate (PHB) (**A**) Cytoplasmic synthesis of PHB (**B**) Chloroplast (plastid) synthesis of PHB (**Note :** phaA, phaB and phaC represent genes for the corresponding enzymes).*

plant by particle bombardment. Adequate synthesis of PHB was reported in the fibres of transgenic cotton plants.

Production of polyhydroxyalkanoate co-polymers

The most important bioplastic of commercial importance is polyhydroxybutyrate. The genetic engineering approaches for its manufacture in plants are described above. But the disadvantage with PHB production is that it is found as stiff and bristle plastics, since it forms highly crystalline polymers.

The other bioplastic, composed of polylydroxyalkanoate (PHA) co-polymer is a polymer made up of longer monomers. It is less

crystalline and more flexible compared to PHB. The PHAs are produced from the intermediates of β-oxidation of fatty acids-notably 3-hydroxy acyl CoA. Some success has been reported in the production of PHAs through genetic manipulations of peroxisomes and glyoxisomes.

GENETICALLY ENGINEERED PLANTS AS PROTEIN FACTORIES

Molecular farming or metabolic engineering of proteins in plants is a novel approach that makes ***plants to serve as living factors*** (bioreactors) for the production of proteins. Use of animals, animal or microbial cell cultures for the bioproduction of proteins has been described (Chapters 15, 33, 41). Production of human or animal proteins by plants is a more recent and novel approach.

The transgenic ***plants*** have the following ***advantages as bioreactors.***

- Low cost of production.
- Eukaryotic protein processing.
- Unlimited supply.
- No spread of animal borne diseases.
- Reduced time to market.
- Safe and environmental friendly.
- No need for skilled workers.

Genetically engineered plants ***can manufacture a wide range of proteins***. These include :

- Enzymes for industrial and agricultural purposes.
- Lysosomal enzymes.
- Antibodies (plantibodies).
- Subunit vaccines.
- Other biopharmaceuticals or medically related proteins.

The general approaches for the production of desired proteins in plants are first described, followed by the different products of commercial importance.

APPROACHES FOR PROTEIN PRODUCTION IN PLANTS

The general methodology for the production of proteins in plants involves the following aspects

1. Selection of crop species.
2. Choice of tissue.
3. Expression strategies.
4. Post-translations processing.
5. Recovery strategies.

Selection of crop species

The most important consideration for choosing a plant as a bioreactor is that it should be readily manipulated to produce a stable transgenic line, with stable expression.

Tobacco-the most preferred plant as a transgenic bioreactor : Tobacco is the most widely used plant for the bioproduction of proteins by researchers as well as biotech companies, for the following reasons.

1. It can be easily engineered and transformed.
2. Tobacco is an excellent biomass producer (about 40 tons of fresh leaf/acre land against 4 tons by rice).
3. Seed production is very high (about one million seeds/plant).
4. Tobacco can be harvested several times in a year.

The major ***limitations*** of tobacco plant are — it is a regional crop and relatively labour intensive, and it is not suitable to be fed to humans or animals.

Other crop species as bioreactors : For the development of edible products (e.g. vaccines) plants such as potato, tomato, banana, maize and lettuce are used. Some workers prefer to produce proteins in the seeds of plants such canola, soybean, corn, rice and barley. This is mainly because seeds can accumulate and store good quantities of proteins.

Choice of tissue

The preferred tissues for protein production by plants include leaves, storage organs (e.g. tubers) and seeds. This choice of tissue mainly depends on the following factors.

- Compatibility with the desired protein.
- Correct processing.
- Stable accumulation.
- Efficiency of recovery.

Many proteins have been produced in leaves (particularly using tobacco plant) e.g. lysosomal enzymes, mammalian antibodies, human serum albumin.

In recent years, seeds have become an attractive high protein production (e.g. hirudin) vehicles due to good storage capacity and stability. It is no surprise that some of the seed proteins retain their biological activity for several years. One limitation of seeds is that it is sometimes difficult to recover the protein.

Expression strategies

There are two major approaches for the transgene expression in plants to produce proteins-stable integration approach and use of plant viruses as transient vectors.

1. **Stable integration approach** : This strategy is most suitable for the bulk production of proteins particularly in leaves. In this approach, the transgene is regulated by a strong constitutive promoter such as 35S promoter. Another alternative is to use tissue-specific promoter. By restricting the transgene expression to a particular tissue (e.g. seed), the yield of the protein can be substantially increased. In addition, the protein is stable for a long time from weeks to years.

A diagrammatic representation of stable gene expression approach for protein production is depicted in ***Fig 51.4.***

2. **Transient expression by using plant viruses :** Bioproduction strategies involving viruses are useful for the protein production at the desired period (discrete period). By genetic manipulations, it is possible to insert the desired gene (for protein) into the coat protein gene of the virus genome. The so developed infectious RNA viruses are introduced into the plants for the production of proteins.

Transient expression system by viruses has been successfully used in tobacco plants by employing tobacco mosaic virus (TMV). Tobacco plants at an appropriate age were inoculated with genetically modified TMV. Recombinant protein could be extracted within 2-3 weeks of harvesting.

Post-translational processing

The ultimate purpose of the use of plants as bioreactors is to produce proteins with native conformation, good biological activity and biocompatibility. The specific folding of the protein with disulfide bonds and suitably modified amino acids is important for this purpose.

The changes that occur after (or sometimes even during) the protein is synthesized are collectively referred to as post-translational modifications e.g. phosphorylation, hydroxylation, carboxylation, glycosylation, proteolytic processing etc. Variations in post-translational changes are often limiting the successful production and use of recombinant proteins either in plants or even in other systems (transgenic animals or cell cultures).

The major ***limitations of plants*** with regard to post-translational modifications of proteins are listed.

- Glycosylation (incorporation of carbohydrate moiety) is different in plants.
- Plant glycosylation results in the production of complex glycans containing fucose or xylose, which do not occur in humans.
- Plants cannot perform specialized carboxylation necessary for certain clotting factors (II, VII, IX, and X) and enzymes (protein C).

Due to above limitations, many a times proteins of transgenic plants are inefficient or less efficient in their function, besides exhibiting immuno-genecity.

In recent years, cloning of genes responsible for post-translational modifications of proteins (e.g. glycation, carboxylation) have met with success to produce more biologically active proteins.

Recovery strategies (purification strategies)

A rational approach for a cost-effective purification strategy is desirable for the recovery of the proteins (downstream processing) produced in transgenic plants. The degree of purity is dependent on the purpose for which the protein is produced. For instance, industrial enzymes in general, do not require high degree of purity. Sometimes, purification may not be necessary at all. A good example is the production of the enzyme phytase in seeds. Phytase acts on phytate and increases the availability of phosphate to animals. For this purpose, milled transgenic seeds containing phytase can be directly added to the feed. This totally avoids the purification/downstream processing costs.

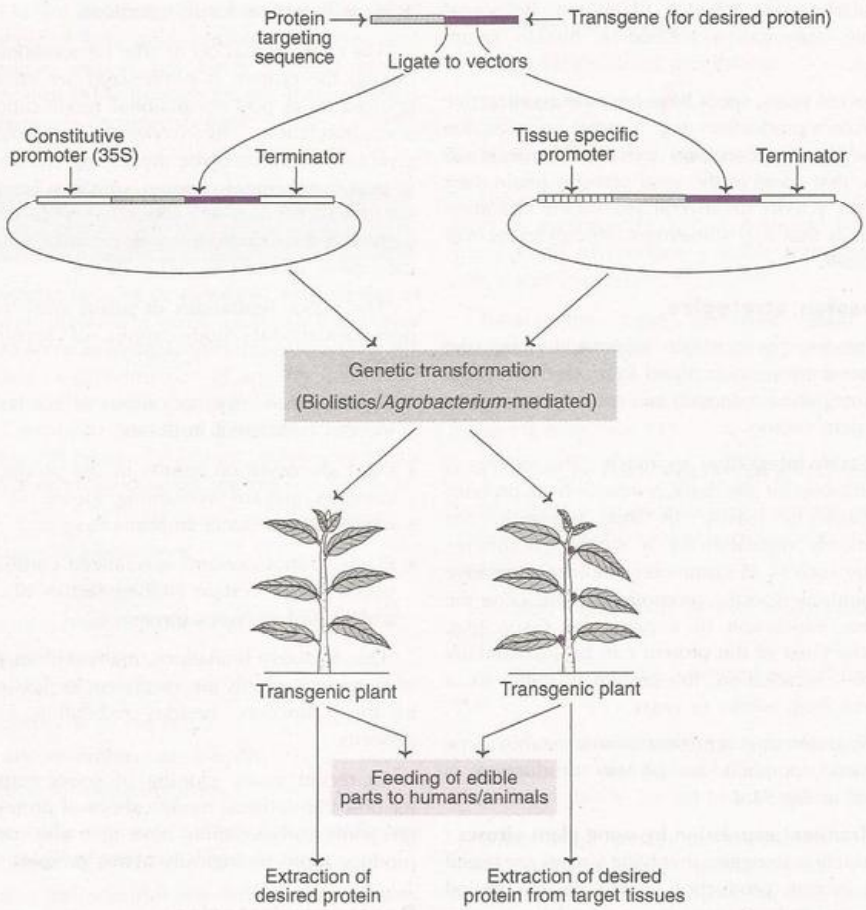


***Fig. 51.4** : Stable gene expression strategy for the production of desired proteins in plants.*

For a great majority of proteins, particularly pharmaceutical proteins, purification strategy is absolutely required. It is therefore necessary to develop efficient methods for downstream processing. Some of the recovery strategies are briefly described.

Affinity tag-based purification : The use of affinity tags for the purification of recombinant proteins has been described elsewhere (Chapter 11), and a similar strategy works in plant systems as well. The technique involves the fusion of desired protein with another protein or peptide (tag), which can bind to ligand, and thus the protein can be isolated. Following purification of the desired protein, the affinity tag can be removed. Some workers have been successful in purifying the enzyme ***glucocerebrosidase*** by this approach.

Purification through compartmentalization : It is possible to direct the accumulation of proteins in a specific sub-cellular organelle. This is achieved by using signal peptides or fusion peptides. Once the protein is localized, the technique of sub-cellular fractionation can be used for preliminary isolation of protein followed by its purification.

The human neuropeptide leu-eukephalin has been compartmentalized in the endoplasmic reticulum and vacuoles by this strategy. Then by appropriate purification techniques, it is isolated.

Protein purification by production in oil bodies : Oil bodies are natural sub-cellular organelles that store triacylglycerols (TG). Oil bodies are composed of triacylglycerols enveloped by a phospholipid monolayer. A group of proteins, referred to as ***oleosins*** are associated with the surface of the oil bodies.

Oil bodies are found in the oil/seeds that represent the storage form of energy (as TG). Oil seeds contain oleosin at a concentration in the range of 2–10% of the total seed proteins.

Oil bodies can be used for the production of foreign proteins in plants. The development of transgenic plants and the purification of the desired protein by employing oil bodies are shown in ***Fig. 51.5***.

A transgene with oleosin and desired protein can be constructed. By introducing this gene, transgenic plants are developed. Oleosins tolerate foreign proteins; hence it is possible to produce oleosin-fusion proteins that accumulate in the oil bodies.

The plant materials (mainly seeds oil bodies) are crushed and centrifuged to separate into three fractions.

- A pellet of insoluble plant material at the bottom of the tube.
- An aqueous phase containing soluble components of the seeds at the middle.
- A floating layer at the top with oil bodies and their associated oleosin fusion proteins.

The top layer of oil bodies is isolated, resuspended in a buffer and re-centrifuged. The concentrated oil bodies are treated with the proteolytic enzyme protease that cleaves the desired protein from oleosin. On further centrifugation, the pharmaceutically important protein (aqueous phase) can be separated from the oil bodies.

OLEOSIN PARTITION TECHNOLOGY

The ***purification of foreign proteins through their production in oil bodies*** (described above) is referred to as oleosin partition technology (***Fig 51.5***). The oil bodies along with the oleosin-fusion proteins are quite stable within the seed and during the course of processing. The pharmaceutical protein of interest remains stable retaining its biological activity within the seeds for years. Further, the purification oleosin-fusion proteins is relatively easier. Hence oleosin partition technology is preferred for the purification of many pharmaceutically important proteins. Two functional proteins namely ***hirudin*** and ***β-glucuronidase*** have been recovered by oleosin partition technology.

Production of hirudin in *Brassica napus*

Hirudin is a natural anticoagulant protein produced in the salivary glands of leeches (*Hirudo medicinalis*). It is an effective therapeutic agent as it specifically inhibits thrombin and blocks blood coagulation.

Till sometime ago, hirudin was produced by employing bacteria and yeasts. The annual requirement of hirudin is very high, and it is a good candidate for producing in transgenic plants.

A synthetic gene encoding hirudin was fused with *Arabidopsis* oleosin gene. An endopeptidase factor Xa gene was inserted between hirudin and oleosin genes. (The endopeptidase gene serves as a recognition site for the protease action). The transgene construct was introduced into *Brassica napus*. The so developed transgenic plants produce oil bodies containing ***oleosin-hirudin fusion protein***. Hirudin can be purified by the procedure described already and depicted in ***Fig 51.5.***

Oleosin partition technology holds a great promise for the successful commercial production of several pharmaceutically important proteins.

PRODUCTION OF INDUSTRIAL ENZYMES IN PLANTS

There is an ever-growing demand for industrial enzymes. Existing sources may not be able to meet the requirements. Large-scale production of plant-derived transgenic enzymes for commercial and industrial purposes are preferred. Besides several advantages of plants as bioreactors (See p. 628), there is no fear of spread of animal pathogens (like BSE) as is the case with animal production.

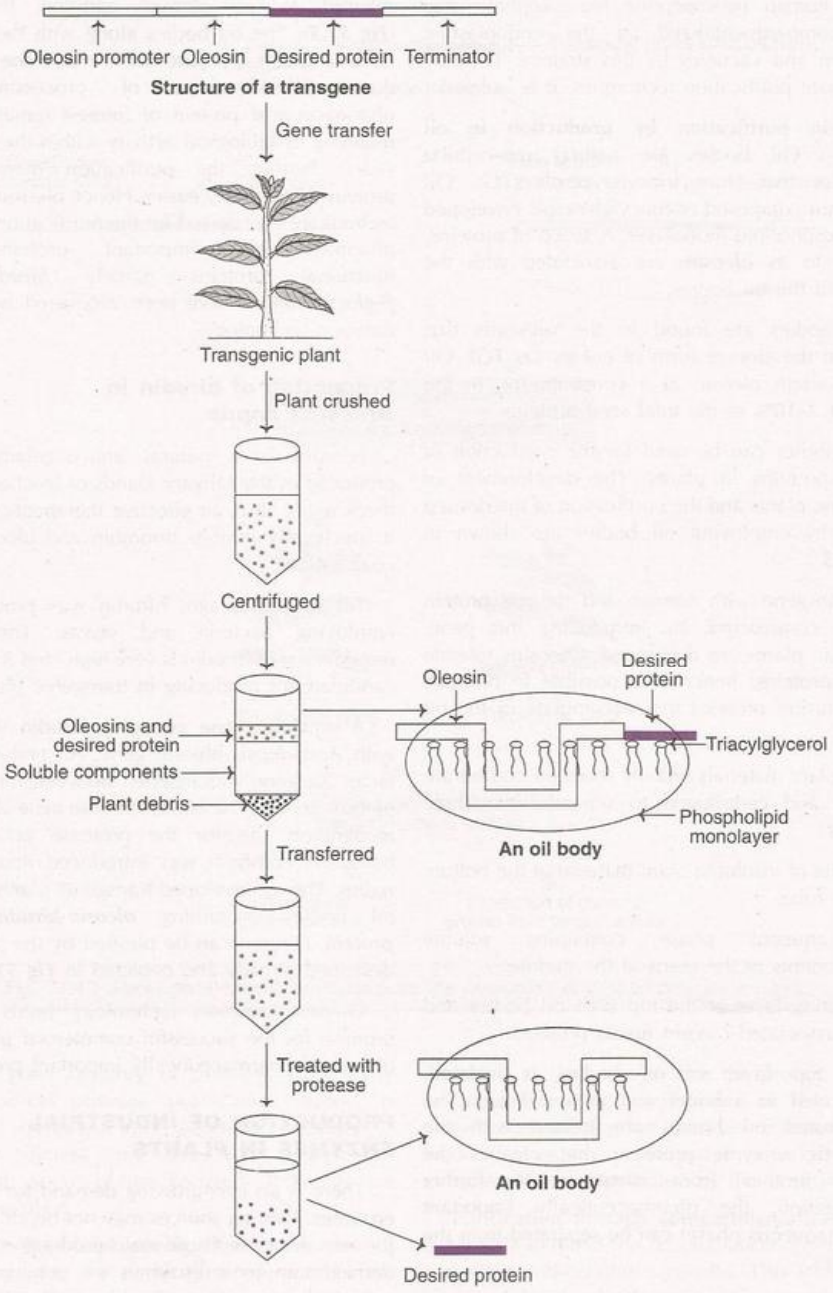


*Fig. 51.5 : The purification strategy of a desired (pharmaceutical) protein by production in oil bodies (**oleosin partition technology**).*

Table 51.2 Selected examples of industrial enzymes produced in plants and their application(s)

Enzyme	*Application(s)*
α-Amylase	Food processing
Avidin	In diagnostic kits
Cellulase	Production of alcohol from wastes (cellulose)
β-Glucanase	In brewing industry
β-Glucuronidase	In diagnostic kits
Lignin peroxidase	In paper manufacture
Phytase	Improved phosphate utilization (by phytate breakdown)
Trypsin	Pharmaceutical
Xylanase	Biomass processing, paper and textile industries

A selected list of industrial enzymes produced in transgenic plants and their important applications is given in ***Table 51.2.*** Some of the important enzymes are described.

Avidin and β-glucuronidase

The first proteins/enzymes that were produced in transgenic plants (maize) are avidin and β-glucuronidase. They are used in diagnostic kits.

Trypsin

Trypsin is an important proteolytic enzyme and its production by conventional recombinant approaches is rather difficult. Trypsin is now produced in plants. It has a wide range of applications-for the production of insulin, vaccines, wound care etc.

Phytase

Phytase is a hydrolytic enzyme that catalyses the hydrolysis of phytate (inositol hexaphosphate) to inositol and inorganic phosphate.

Phytate is present in high quantities in many plant seeds used as a feed to pigs and poultry (monogastric animals). These animals do not possess the enzyme phytase; hence they cannot derive the nutrient phosphate from phytate. The undigested phytate gets excreted and accumulates in the soil and water, leading to eutrophication.

Transgenic plants capable of synthesizing phytase in their seeds have been developed. These seeds (no need to isolate the enzyme) can be mixed with the feed of animals, and fed. The phytase enzyme has successfully solved nutritional (phosphate) and environmental (eutrophication) problems.

Cellulase and xylanase

The two enzymes namely cellulase and xylanase are used in several industries- bioethanol, textile, paper and pulp. In all these processes, they are basically involved in the degradation of plant materials (predominantly cellulose).

By a novel approach, it is possible to produce cellulase and xylanase in the plants for which the target itself is the digestion of plant cell. The risk of autodigestion of the plant cells (by these enzymes) is overcome by genetic manipulation to produce thermostable enzymes. Thus, cellulase and xylanase, produced by transgenic plants, are inactive at the temperatures at which plants normally grow. The activity of these enzymes is restored on heating the plant extracts. And these extracts are very successfully used in many industries.

While the recovery of enzymes from transgenic plants is often required, there are some instances where the crude extracts can be directly used. The industrial applications of cellulase, xylanase and phytase (described above) are some good examples.

PRODUCTION OF LYSOSOMAL ENZYMES IN PLANTS

Lysosomes are the cellular organelles responsible for the degradation of macromolecules. The degradative enzymes present in lysosomes include proteases, nucleases, lipases, glycosidases, phosphatases, phospolipases and sulfatases. Deficiency of a specific lysosomal enzyme results in the accumulation of undegraded substrate causing clinical manifestations. Some examples of lysosomal disorders and the corresponding enzyme deficiencies are listed below.

- Tay-Sachs disease — α-hexosaminidase.
- Gaucher's disease — glucocerebrosidase.
- Hurler syndrome — iduronidase.

- Mucopolysaccharidoses (a group of diseases) — Several enzyme of glycosaminoglycan degradation.

A couple of lysosomal enzymes produced in transgenic tobacco plants are described below.

Glucocerebrosidase

Glucocerebrosidase is a lysosomal enzyme that degrades glucocerebroside (predominantly produced when erythrocytes are destroyed due to aging or damage) to glucose and ceramide. The deficiency of this enzyme causes Gaucher's disease, characterized by swelling of spleen, liver, and bone damage causing extreme pain.

Gaucher's disease can be effectively ***treated by*** administering ***the enzyme glucocerebrosidase***. At present, this enzyme is obtained from human placenta or from mammalian cell cultures, both being very costly. It is estimated that a single patient requires 10–12 tons of human placenta annually costing $100,000 to 400,000!

Glucocerebroside was the first lysosomal enzyme ***produced*** in plants (***Nicotiana tobacum***). The production process of this enzyme has been patented, and FDA approved it for use in humans. It is hoped that through the plant-based production of glucocerebrosidase (marketed as ***Cerezyme***), the patients of Gaucher's disease can be treated in a cost-effective manner.

α-L-Iduronidase

The deficiency of the enzyme iduronidase causes Hurler's syndrome, the most common disorder of defective mucopolysaccharide degradation.

Tobacco plants have been genetically engineered to produce α-L-iduronidase.

Limitations for the production of lysosomal enzymes in plants

Both the enzymes referred above are glycoproteins (glucocerebrosidase and iduronidase). As described elsewhere (See p. 629), appropriate glycation of animal proteins is not feasible in plants. This poses a major problem for the production of lysosomal enzymes in plants. Through a series of genetic manipulations, biotechnologists were successful in achieving the desired glycation for the production glucocerebrosidase and iduronidase.

PRODUCTION OF ANTIBODIES (PLANTIBODIES) IN PLANTS

Antibodies or immunoglobulins are the defense proteins produced in mammals. For details on the structure and functions of different immunoglobulins, refer Chapter 68.

The use of plants for commercial production of antibodies, referred to as plantibodies, is a novel approach in biotechnology. The first successful production of a functional antibody, namely a mouse immunoglobulin IgGI in plants, was reported in 1989. This was achieved by developing two transgenic tobacco plants-one synthesizing heavy γ-chain and the other light κ-chain, and crossing them to generate progeny that can produce an assembled functional antibody.

Production of secretory IgA (sIgA)

Secretory IgA is an immunoglobulin that protects against dental caries produced by *Streptococcus mutans*. For the production of sIgA, transgenic plants were developed to synthesize different subunits and then crossed to produce the functional antibody (***Fig 51.6***).

Four separate transgenic plants synthesizing four distinct pieces of antibody (H-, L-, J-chains and secretory component) were developed. The plants expressing H- and L-chains were crossed (cross 1) giving plants that produce IgA. When these plants were crossed (cross 2) with plants expressing J-chain, the progeny produced dimeric IgA. The dimeric IgA synthesizing plants were then crossed (cross 3) with the plants expressing secretory component, the functional sIgA could be produced.

Secretory antibodies have many advantages. They are resistant to proteolytic degradation; hence their yield in plants is substantially higher. sIgA are the predominant antibodies that protect against mucosal infections of microorganisms. They bind to antigens more effectively and offer good protection.

The secretory antibodies produced in transgenic plants have been tested on humans. When sIgA was topically applied to teeth, it could prevent the colonization by *Streptococcus mutans* up to 4 months. This is comparable to the protection offered by immunoglobulins produced through hybridoma technology. This was possible despite some structural differences between plantibodies and monoclonal antibodies. It appears that there is no difference in the binding properties between the two types of antibodies.

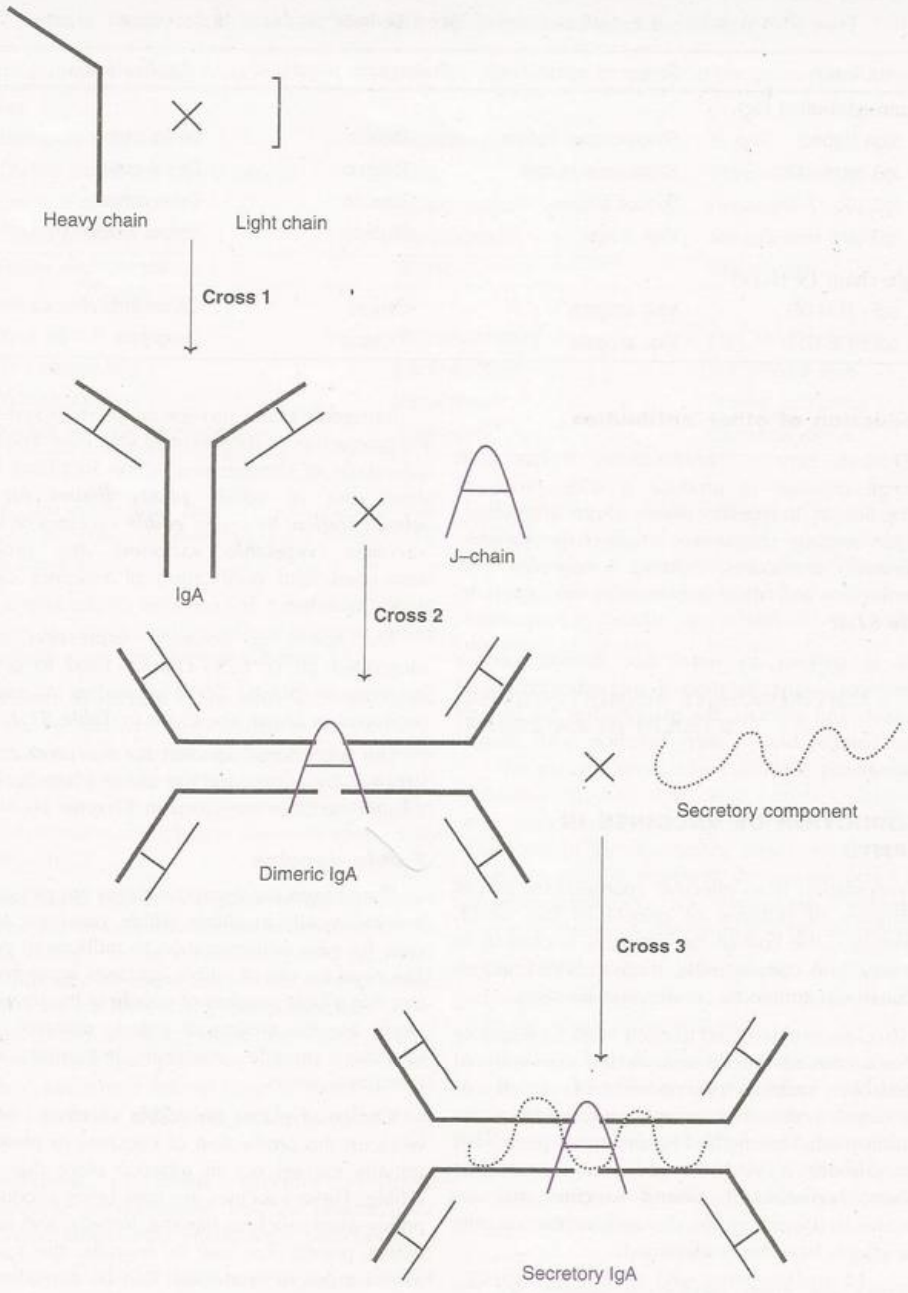


Fig. 51.6 : Production of secretory IgA (sIgA) molecule in transgenic plants.

Table 51.3 A selected list of antibodies (plantibodies) produced in transgenic plants

Antibody	*Targeted against*	*Transgenic plant*	*Application*
Immunoglobulins (Ig)			
sIgA (hybrid)	*Streptococcus mutans*	Tobacco	Dental caries
IgG (guy's–13)	*Streptoccus mutans*	Tobacco	Dental caries
IgG (Co 17–1A)	Surface antigen	Tobacco	Colon cancer
IgG (anti HSV–2)	Viral antigen	Soybean	Herpes simplex virus
Single-chain Fv (scFv)			
scFv (T84.66)	Viral antigens	Cereals	Carcinoembryonic cancer
scFv (38 C13)	Viral antigens	Tobacco	Lymphoma

Production of other antibodies

Through genetic manipulations, it has now become possible to produce a wide range of antibodies in transgenic plants-whole antibodies, antigen-binding fragments, single-chain variable fragment antibodies. Some examples of plantibodies and their applications are given in ***Table 51.3.***

It is striking to note that despite certain differences in glycosylation of antibodies produced in plants and animals, the plantibodies are by and large comparable in their function with animal antibodies.

PRODUCTION OF VACCINES IN PLANTS

Vaccination is an effective approach for proper healthcare of millions of people in the world. Sometimes, the cost of the vaccines is coming in the way, and consequently, thousands of children become susceptible to preventable diseases.

Vaccines are designed to elicit immune response without causing the disease. In the conventional approach, vaccines composed of killed or attenuated disease-causing organisms are administered. This method is sometimes associated with side effects. With the advances in molecular biology, ***recombinant subunit vaccines*** that are effective in preventing the disease, without causing side effects have been identified.

There are many production systems for the synthesis of recombinant vaccines. These include mammalian cell cultures, yeast, bacteria and insect cell cultures, and transgenic animals.

Transgenic plants provide an alternate system for the production of recombinant vaccines. The major advantage of vaccine production in plants is the direct use of ***edible plants tissues for oral administration***. By use of ***edible vaccines*** or ***veggie vaccines*** (***vegetable vaccines***), the problems associated with purification of vaccines can be totally avoided.

The stable or transient expression system (described on p. 629) can be used to produce vaccines in plants. Some examples of vaccines produced in plants are given in ***Table 51.4***.

The procedures adopted for the production of vaccines by plants, and the use of plants as edible subunit vaccines are given in Chapter 16.

Edible vaccine

Plant-based production systems are designed to provide locally available edible vaccines, at low-cost, for easy administration to millions of people. The need for use of edible vaccines arose from the fact that a large number of people in the developing world are the victims of enteric diseases. Edible vaccines provide mucosal immunity against infectious agents.

Choice of plants for edible vaccines : Most of work on the production of vaccines in plants was initially carried out in tobacco plant that is not edible. These vaccines are now being produced in edible plants such as banana, tomato, and to some extent potato. For use in animals, the common fodder crops or food crops can be considered.

Some workers consider banana as an ideal system for the production of edible vaccines. This is mainly because it is grown in most parts of the

Table 51.4 Some examples of vaccines produced in transgenic plants

Recombinant protein (vaccine)	*Transgenic plant*	*Application (protection against)*
Human		
Heat-labile enterotoxin B	Tobacco	*E. coli*
Cholera Ctox A and Ctox B subunits	Potato	*Vibrio cholerae*
Envelope surface protein	Tobacco/potato	Hepatitis B
Capsid protein	Tobacco/potato	Morwalk virus
Rabies virus glycoprotein	Tomato	Rabies virus
Animal		
Virus epitope VP_1	Alfalfa/*Arabidopsis*	Foot and mouth virus
Viral epitope VP_2	Blackeyed bean	Milk enteritis virus
Viral glycoprotein	Maize/tobacco	Porcine coronavirus
Peptide from VP_2 capsid protein	*Arabidopsis*	Canine parvovirus

world, and eaten raw. Another advantage is that children (to whom vaccines are most needed) like eating banana.

The procedures adopted for the production of vaccines by plants, and the use of plants as edible subunit vaccines are given in Chapter 16.

Delivery of vaccine to the gut : Vaccines, being proteins, are likely to be degraded in the stomach. This is true to some extent. However, it is now known that the orally administered plant materials (edible vaccine) can induce immune response. The particulate materials of plants are more effective in this regard.

Food-based tablets to replace edible vaccines : It is true that the edible vaccines may work well in the laboratory or under controlled conditions. There is a difficulty of dose adjustment when edible vaccines are consumed as a part of foodstuff. Some workers prefer to discontinue the direct use of plant materials, and prefer to opt for food-based tablets containing a known dose of vaccine. This approach is being applied to vaccines produced in tomatoes.

Limitations of edible vaccines : Direct consumption of transgenic fruit or vegetable or food-based tablets may create some problems.

- The risk of loss of vaccines by the action of enzymes in stomach and intestine.
- The possibility of allergic reactions as they enter circulation.

The future of edible vaccines : Despite several limitations, biotechnologists see a great hope for the new wave of ***transgenic veggie vaccines.*** But it may take several years before the banana or tomato replaces the needle as a vehicle for vaccine delivery!

PRODUCTION OF THERAPEUTIC PROTEINS IN PLANTS

The general approaches, and the production of industrial enzymes, lysosomal enzymes, antibodies and vaccines in transgenic plants have been described in the proceeding pages. Many of these products actually represent therapeutic proteins or biopharmaceuticals e.g. lysosomal antibodies, vaccines. Therefore, they must also be referred again.

In ***Table 51.5,*** a selected list of therapeutic proteins, and the plants producing them, along with the applications is given.

The major ***limitations*** of protein production by transgenic plants are the ***low product yield and*** the difficulties associated with ***recovery.*** With improvements in genetic engineering and recovery processes, it is hoped that the production of therapeutic proteins will substantially increase in the coming years.

Chloroplasts in the production of therapeutic proteins

Chloroplasts are capable of folding the foreign proteins, besides bringing out most of the post-

Table 51.5 Selected examples of therapeutic proteins produced in transgenic plants

Therapeutic protein	*Transgenic plant*	*Application(s)*
Hemoglobin	Tobacco	Blood substitute
Serum albumin	Tobacco	Burns/fluid replacement, blood extender
Protein C	Tobacco	Anticoagulant
Hirudin	Canola	Anticoagulant
α_1-Interferon	Rice	Viral protection, anticancer
β-Interferon	Rice/tobacco/turnip	Treatment for hepatitis B + C
γ-Interferon	Tobacco	Phagocyte activator
Collagen	Tobacco	As collagen
Somatotrophin	Tobacco	As growth hormome
Trypsin inhibitor	Maize	Transplant surgery
α_1-Antitrypsin	Rice	Cystic fibrosis
Erythropoietin	Tobacco	As mitogen
Lactoferrin	Potato	Antimicrobial
Tissue growth factor	Tobacco	As mitogen
Enkephalin	Canola/*Arabidopsis*	Opiate
Angiotension converting enzyme	Tobacco/tomato	Hypertension
Glucocerebrosidase	Tobacco	Gaucher's disease
α-Galactosidase	Tobacco	Fabry disease
Trichosanthin-α	Tobacco	HIV therapy, cancer

translational modification. As the chloroplasts are inherited maternally, they will not spread via pollen to non-transgenic systems. This type of biotechnological approach will help to get clearance from the regulatory bodies of biotechnology. Some of the therapeutic proteins produced in chloroplasts are listed.

- Interferons
- Serum albumin
- Hemoglobin
- Proinsulin
- Monoclonal antibodies
- Protective antigen of *Bacillus anthracis*.

52 Growth-Promoting Bacteria in Plants

Certain beneficial microorganisms, present in the soil, are known to influence the plant growth, development and yield. These ***bacteria and fungi*** may ***provide growth-promoting products to plants or inhibit the growth of soil pathogenic microorganisms*** (phytopathogens), which hinder the plant growth. The former is the direct effect while the latter is the indirect effect of growth-promoting bacteria in plants.

The growth-promoting activity of micro-organisms and the biotechnological approaches are described briefly with respect to the following aspects.

1. Biological nitrogen fixation.
2. Biocontrol of phytopathogens.
3. Biofertilizers.

BIOLOGICAL NITROGEN FIXATION

Nitrogen is an essential element of many biomolecules, the most important being nucleic acids and amino acids. Although nitrogen is the most abundant gas (about 80%) in the atmosphere, neither animals nor plants can use this nitrogen to synthesize biological compounds. However, there are certain microorganisms on which the living plants (and animals) are dependent to bring nitrogen into their biological systems.

The phenomenon of ***fixation of atmospheric nitrogen by microorganisms*** is known as ***diazotrophy*** and these organisms are collectively referred to as ***diazotrophs***. Diazotrophs are biological nitrogen fixers, and are prokaryotic in nature.

NITROGEN CYCLE

An outline of the nitrogen cycle is depicted in ***Fig. 52.1***. Nitrogen enters the soil with the deposits of dead animals and plants, and urea of urine. These waste materials (proteins, urea) are decomposed by soil bacteria into ammonia and other products. The ammonia is converted to nitrite (NO_2) and then nitrate (NO_3) by certain bacteria belonging to the genera *Nitrosomonas* and *Nitrobacter*. The nitrate is degraded by various microorganisms to release nitrogen that enters atmosphere. This atmospheric nitrogen is taken up by the nitrogen fixing bacteria (present on the roots of leguminous plants) and used for the synthesis of biomolecules (e.g. amino acids). As the animals consume the leguminous plants as food, the nitrogen cycle is complete.

NITROGEN FIXING BACTERIA

It is estimated that about 50% of the nitrogen needed by the plant comes from nitrogen fixing bacteria. These are two types of nitrogen fixing microorganisms-asymbiotic and symbiotic.

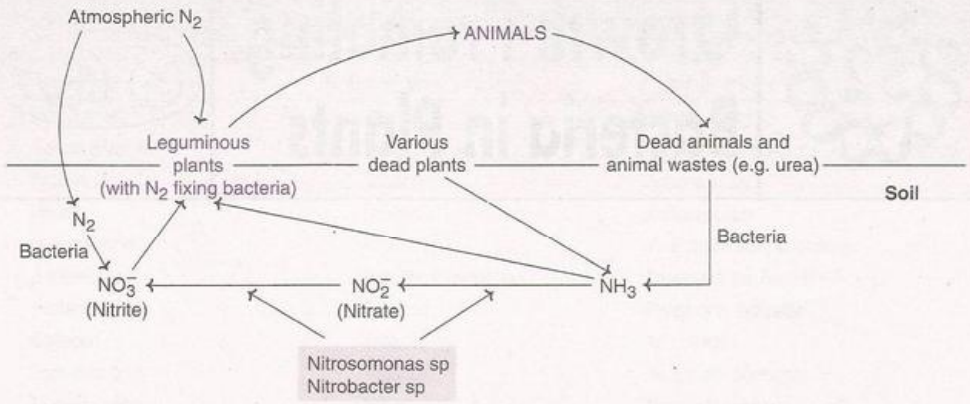


Fig. 52.1 : An outline of the nitrogen cycle.

Asymbiotic nitrogen fixing microorganisms

The gaseous nitrogen of the atmosphere is directly and independently utilized to produce nitrogen-rich compounds. When these non-symbiotic organisms die, they enrich the soil with nitrogenous compounds. Several species of bacteria and fungi can do this job e.g. *Clostridium pasturianum, Azatobacter chrooccum.*

The mechanism of nitrogen fixation by asymbiotic bacteria is not clearly understood. It is believed that nitrogen is first converted to hydroxylamine or ammonium nitrate, and then incorporated into biomolecules.

Symbiotic nitrogen fixing microorganisms

These microorganisms live together with the plants in a mutually beneficial relationship, phenomenon referred to as ***symbiosis***. The most important microorganisms involved in symbiosis belong to two related genera namely *Rhizobium* and *Bradyrhizobium*. These symbiotic bacteria also referred to as nodule bacteria are Gram negative, flagellated and rod-shaped. The host plants harbouring these bacteria are known as ***legumes*** e.g. soybean, peas, beans, alfalfa, peanuts, and clover.

Each one of the species of *Rhizobium* and *Bradyrizobium* are specific for a limited number of plants, which survive as the natural hosts (***Table 52.1***). It is now clearly known that these bacteria do not interact with plants other than the natural hosts.

The relationship between the symbiotic bacteria and the legumes is well recognized. On the roots of legumes, there are a number of nodules (swellings) in which *Rhizobium* sp thrive. These bacteria trap atmospheric nitrogen and synthesize nitrogen-rich compounds (amino acids, proteins etc) used by the legumes. At the same time, the legumes supply important nitrogen compounds for the metabolism of Rhizobia.

The growth of legumes has been known to enrich the soil fertility. This is due to the fact that the concentration of nitrogen compounds in the soil increases as a result of the presence of symbiotic bacteria. For this reason normally,

Table 52.1 Symbiotic nitrogen fixing bacteria along with the host plant(s)

Bacterial species	*Host plant(s)*
Rhizobium leguminosarum	Pea, broad bean
Rhizobium meliloti	Alfalfa
Rhizobium fredii	Soybean
Rhizobium loti	Lotus
Rhizobium elti	Mung bean, kidney bean
Rhizobium ciceri	Chickpea
Bradyrhizobium japonicum	Soybean
Bradyrhizobium elkanii	Soybean

nitrogen fertilizers are not needed in the fields cultivated legumes.

MECHANISM OF NITROGEN FIXATION

Inside the root nodules of leguminous plants, the bacteria proliferate. These bacteria exist in a form that has no cell wall.

The bacteria of the nodules are capable of fixing nitrogen by means of the specific enzyme namely ***nitrogenase.***

Nitrogenase

Nitrogenase is a complex enzyme containing two oxygen sensitive components. Component I has two α-protein subunits and two β-protein subunits, 24 molecules of iron, two molecules of molybdenum and an ***iron molybdenum cofactor (FeMoCo).*** Component II possesses two α-protein subunits (different from that of component I) and a large number of iron molecules.

Component I of nitrogenase catalyses the actual conversion of N_2 to ammonia while component II donates electrons to component I (***Fig. 52.2***).

Leghemoglobin

A protein comparable of hemoglobin in animals has been identified ***in the nodules of leguminous plants***. Leghemoglobin (***LHb***) contains iron and is red in colour. It is an oxygen binding protein. The heme part of leghemoglobin is synthesized by the bacterium while the protein (globin) portion is produced by the host plant. Leghemoglobin is absolutely necessary for nitrogen fixation. The nodules that lack LHb are not capable of fixing nitrogen.

It is LHb that facilitates the appropriate transfer of oxygen (by forming oxyLHb) to the bacteria for respiration to produce ATP. And energy in the form of ATP is absolutely required for nitrogen fixation.

Another important function of LHb is that it prevents the damaging effects of direct exposure of O_2 on nitrogenase.

In ***Fig. 52.2***, the fixation of nitrogen by symbiotic bacteria is depicted. As the oxyLHb supplies O_2, bacterial respiration occurs. The ATP generated is used for fixing nitrogen to produce ammonia. The complex reaction is summarized below.

$$N_2 + 8H^+ + 8e^- + 16\ ATP \downarrow 2NH_3 + H_2\uparrow + 16\ ADP + 16\ Pi$$

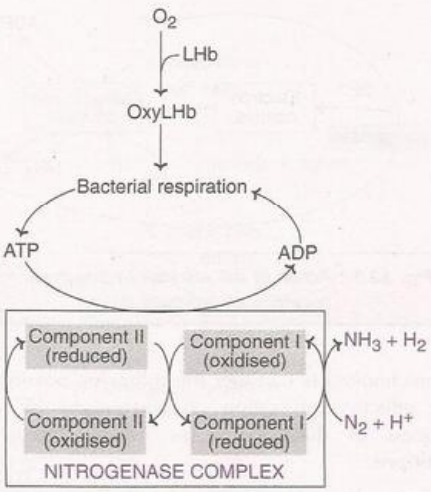


Fig. 52.2 : Nitrogen fixation in symbiotic bacteria.

Hydrogenase

During the course of nitrogen fixation by nitrogenase, an undesirable reaction also occurs. That is reduction of H^+ to H_2 (hydrogen gas). For the production of hydrogen, ATP is utilized, rather wasted. Consequently the efficiency of nitrogen fixation is drastically lowered. It is possible theoretically to reduce the energy wastage by recycling H_2 to form H^+.

In fact, some strains of ***Bradyrhizobium japonicum*** in soybean plants were found to use hydrogen as the energy source. These strains were found to ***possess*** an enzyme namely ***hydrogenase***. Recycling of the hydrogen gas that is formed as a byproduct in nitrogen fixation is shown in ***Fig. 52.3***.

It is advantageous for nitrogen fixation if the symbiotic bacteria possess the enzyme hydrogenase. However, the naturally occurring strains of *Rhizobium* and *Bradhyrhizobium* do not normally possess the gene encoding hydrogenase.

GENETIC MANIPULATIONS FOR NITROGEN FIXATION

The nitrogen-fixing bacteria that are closely associated with the world's food supply are among the favoured organisms for genetic manipulations.

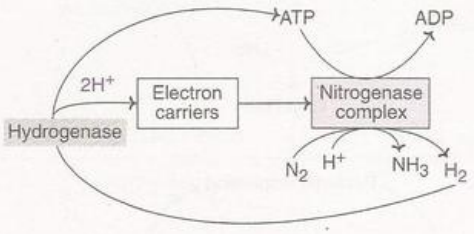


Fig. 52.3 : Action of the enzyme hydrogenase for recycling of hydrogen gas.

Biotechnologists consider the following possibities for effective utilization of diazotrophs (or the process of diazotrophy) as natural biological fertilizers.

- Gene alterations in *Rhizobium* sp to improve nitrogen fixing efficiency, and bacteria-host plant interactions
- Genetic engineering of *Rhizobium* sp so that it can form a symbiotic relationship with non-leguminous plants such as wheat, rice and corn.
- Transfer of genes for nitrogen fixation from *Rhizobium* sp to other bacteria such as *Agrobacterium tumefaciens*. The so developed *A. tumefaciens* can infect several important crops (e.g. tomato, tobacco, petunia) and fix nitrogen.

Some other considerations include the insertion of nitrogen-fixing genes into plants or even animals including humans. This approach, which enables the plants, animals and humans to directly trap and utilize nitrogen from the atmosphere is rather theoretical. Life would be certainly different if introduction of nitrogen-fixing genes in man can dispense with the habit of eating daily!

Despite several possibilities and optimistic predictions, the ***success in the genetic engineering in relation to nitrogen-fixation is very limited***, and briefly described below.

GENETIC ENGINEERING OF NITROGENASE GENE

It is absolutely necessary to identify, isolate and characterize the ***ni***trogen-***fi***xing genes, ***nif*** genes to undertake any kind of genetic manipulations.

Genetic complementation is the common technique used for the isolation *nif* genes. The procedure basically involves the identification and characterization of clones from a wild-type library to restore nitrogen fixation in the various mutants of the original organism.

The *nif* genes were first isolated from the clone banks of the diazotroph *Klebsiella pneumonia*. This organism is found in soil, water and human intestine, and its molecular biology is well known. The isolation of *nif* genes by genetic complementation from *K. pneumonia* is depicted in ***Fig. 52.4***, and involves the following steps.

1. The wild-type DNA of *K. pneumonia* (Nif^+) was cut with restriction endonucleases.

A clone bank was constructed with a vector and maintained in *E.coli*.

2. *K. pneumonia* cells are exposed to a mutagenic agent. This may result in the mutation of *nif* genes to form Nif^- cells.

3. The Nif^- *K. pneumonia* cells are then conjugated with *E.coli* cells, carrying the clone bank on a vector.

4. The transformants of *K. pneumonia* possessing Nif^+ phenotype can be selected by growing the cells on a minimal medium that does not contain fixed nitrogen.

5. The DNA fragment in the plasmid that contains *nif* genes, which complements Nif^- mutation in *K. pneumonia* can be isolated.

The isolation of *nif* genes will be more effective if a series of independently derived Nif^- mutants are employed in genetic complementation experiments. This approach increases the likelihood of different *nif* genes isolation. Further, by using DNA hybridization probes of *nif* genes, DNA clone bank of *K. pneumonia* can be screened. This allows the identification of additional genes that are involved in nitrogen fixation.

Nitrogenase gene cluster

The entire set of nitrogenase genes from *K. pneumonia* has been identified. Some highlights are listed.

- The nitrogenaes (*nif*) genes are located as a single cluster, occupying approximately 24kb of the bacterial genome.
- There are seven distinct operons that encode 20 different proteins.

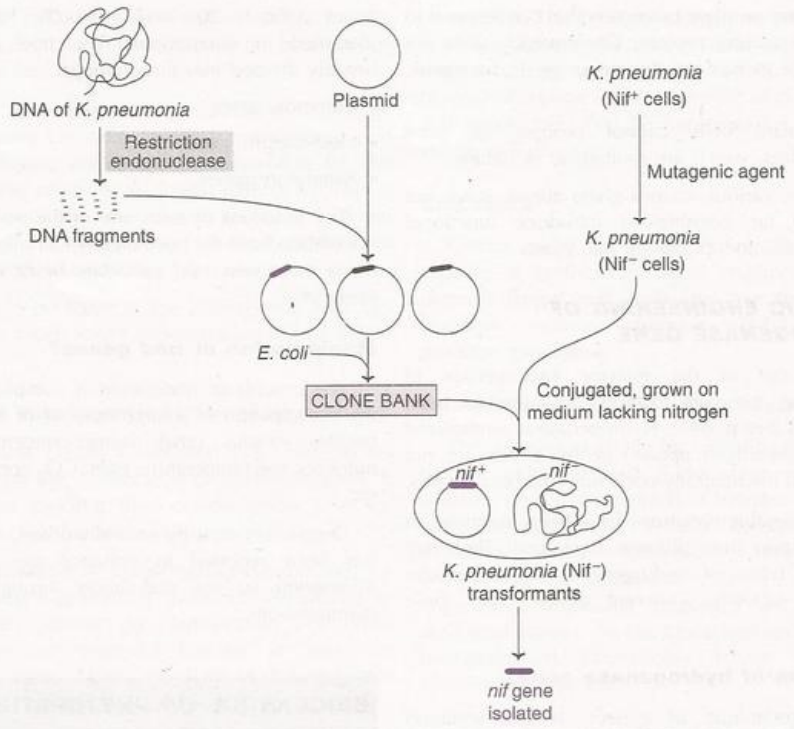


Fig. 52.4 : *An outline of the method for the isolation of* nif *genes from* K. pneumonia *by genetic complementation.*

- All the *nif* genes transcribe and translate in a well-coordinated fashion.
- The *nif* genes are under the regulatory control of two genes namely *nifA and nifL.*

Although the *nif* genes have been characterized from *K. pneumonia,* for technical reasons, the contribution of this organism for biological nitrogen fixation is almost insignificant. However, the *nif* genes of *K. pneumonia* have been used as hybridization probes to identify *nif* genes from the DNA clone banks of diazotrophic organisms (particularly *Rhizobium* sp). It is now known that almost all the nitrogen-fixing bacteria possess similar type of *nif* gene clusters.

Manipulation of *nif* genes

Some workers have been successful in introducing extra copies of nitrogenase regulatory genes namely *nifA* and *nifL* into diazotrophs. For instance, insertion of more *nifA* genes into *Rhizobium meliloti* resulted in an increased biomass production in alfalfa plants. This is due to the fact that nitrogen fixation in these plants is much increased. But the major limitation of this approach is that the genetically engineered bacteria have a diminished growth rate. As a result, these newly developed bacteria loss their efficiency as plant-growth promoting agents.

Despite continuous efforts by several groups of workers, no significant success has been reported in the transfer of *nif* genes into plants. The major limitations are listed.

- Transfer of 24 kb *nif* gene cluster has not been effective since the normal cellular concentration of oxygen would inactivate nitrogenase enzyme. Any reduction in O_2 results in cell death.
- Transfer of one or two genes of *nif* gene cluster is useless.

- There are no plant promoters that can respond to *nifA* regulatory protein. Consequently, it is not possible to turn on the *nif* genes in transgenic plants.
- The plant cells cannot process *nif* gene transcripts, which are multigenic in nature.

For the various reasons given above, it has not been so far possible to introduce functional nitrogen fixation capability into plants.

GENETIC ENGINEERING OF HYDROGENASE GENE

The role of the enzyme hydrogenase in promoting nitrogen fixation has already been described (See p. 641). Hydrogenase is synthesized by ***hup*** (***h***ydrogen ***up***take) genes, which are not present in the naturally occurring Rhizobial strains.

Considerable variations have been identified in hydrogenases from different organisms. There are different types of hydrogenases, which usually contain subunits. Different genes code these subunits.

Isolation of hydrogenase genes

The technique of genetic complementation (similar to that described for *nif* genes on p. 642) can be successfully employed for this isolation of hydrogenase (*hup*) genes. In fact, *hup* genes were first isolated from *E.coli* by this approach. Later, *hup* genes from *B.japonicum* have also been isolated.

Manipulation of *hup* genes

The hydrogenase (*hup*) genes have been introduced into *R.leguminosarum*. These ***Hup***$^+$ strains of bacteria, when inoculated into legumes, resulted in higher nitrogen fixation.

GENETIC ENGINEERING OF NODULATION GENES

Establishment of nodules on the roots of leguminous plants is a prerequisite for nitrogen fixation. Certain genes involved in nodulation namely ***nod genes*** have been identified in *Rhizobium melitoti*. The technique of genetic complementation (similar to that described for *nif* genes on p. 642) has been used to isolate *nod* genes from *R. melitoti*. A large number of *nod* genes (about 20 *nodA* — *nodX*) have been identified in diazotrophs. The *nod* genes are broadly divided into three groups.

- Common genes
- Host-specific genes
- Regulatory genes.

The functions of each one of the *nod* genes in nodulation have not been clearly identified. Further, many more new *nod* genes are being discovered every year.

Manipulation of *nod* genes?

The process of nodulation is complex through the participation of a large number of *nod* genes, besides various other factors-concentration of nutrients, soil temperature, light, CO_2 concentration etc.

Despite attempts by several workers, no success has been reported to enhance the ability of *Rhizobium* sp for nodulation through genetic manipulations.

BIOCONTROL OF PHYTOPATHOGENS

Phytopathogens can drastically reduce the crop yield, which may be in the range of 25–100%. Chemical agents are commonly used to control them. This is associated with ill-health affects on humans, besides environmental pollution.

There are certain bacteria that can lessen or prevent the deleterious effects of plant pathogens-fungi or bacteria. And thus, they promote plant growth by indirect means.

Plant growth-promoting bacteria are capable of producing a wide range of substances that can restrict the damage caused by phytopathogens to plants. The important plant growth-promoting substances are ***siderophores, antibiotics*** and certain ***enzymes.***

SIDEROPHORES

There are some bacteria in the soil that can synthesize a low molecular weight (400–1,000 Daltons) ***iron-binding peptides***, collectively referred to as siderophores. Siderophores have high affinity to bind to iron in the soil and transport it to the

microbial cell. This is required since iron is essential, and cannot directly enter the bacteria due to a very low solubility.

The growth-promoting bacteria, through siderophores can take up large quantities of iron from the soil, and make it unavailable for the growth and existence of fungal pathogens. This is possible since the siderophores produced by fungi have a very low affinity when compared to siderophores of bacteria.

There is no harm to the plants since they can grow at a much lower concentration of iron in the soil.

Genetic manipulations for siderophores

It is now clearly established that siderophores can prevent the proliferation of phytopathogens. It is therefore logical to think of siderophore genes for more effective biocontrol of plant pathogens.

Pseudobactin is a siderophore synthesized by plant growth-promoting bacteria *Pseudomonas putida*. By genetic complementation and other techniques of molecular biology, at least five separate gene clusters involved in pseudobactin production have been identified.

Genetic manipulation for improved synthesis of siderophore is not an easy job, since it is a complex process involving a large number of genes. Some success, however, has been reported in cloning the genes for iron siderophore receptors from certain plant growth-promoting bacteria, and introducing them into other bacteria.

ANTIBIOTICS

Certain antibiotics produced by plant growth-promoting bacteria, can prevent the growth and proliferation of plant pathogens. Some of the antibiotics synthesized by pseudomonads are agrocin 84, agrocin 434, herbicolin, phenazine, oomycin, pyrrolinitrin.

Genetic manipulations for antibiotics

It is possible to enhance the growth-promoting activity of the bacteria by inserting genes encoding the synthesis of antibiotics.

In fact, a genetically engineered biocontrol bacterium of *Agrobacterium radiobacter* has been developed, and is being marked commercially since 1989. This transgenic organism produces the antibiotic agrocin 84 that is toxic to crown gall disease-causing *Agrobacterium tumefaciens*. By this biocontrol approach, the crop yield of almond trees and peach trees can be improved by preventing crown gall disease.

ENZYMES

Certain plant growth-promoting bacteria are capable of synthesizing some enzymes that can ***degrade fungal cell walls and lyse them***. These enzymes include ***chitinase, β-1,3-glucanase, protease*** and ***lipase***.

Genetic manipulation for enzymes

The genes responsible for encoding the enzyme namely chitinase and β-glucanase have been isolated and characterized. Chitinase gene has been isolated from *Serralia mercescens* and introduced into *Rhizobium meliloti* and *Trichoderma harzianum*. The genetically engineered organisms displayed an increased antifungal activity. By this biocontrol approach, the phytopathogen *Rhizoctonia solani* has been effectively controlled.

BIOFERTILIZERS

It is estimated that more than 100 million tons of fixed nitrogen are needed for global food production. The use of chemical/synthetic fertilizers is the common practice to increase crop yields. Besides the cost factor, the use of fertilizers is associated with environmental pollution.

Scientists are on a constant look for alternate, cheap and environmental-friendly sources of nitrogen and other nutrients for plants. The term biofertilizers is used to refer to the ***nutrient inputs of biological origin to support plant growth***. This can be achieved by the addition of ***microbial inoculants*** as a source of biofertilizers.

Biofertilizers broadly includes the following categories.

- Symbiotic nitrogen fixers
- Asymbiotic nitrogen fixers
- Phosphate solubilising bacteria
- Organic fertilizers.

The most important microorganisms used as biofertilizers along with the crops are listed in *Table 52.2*.

Some of the important features of these biofertilizers are briefly described.

SYMBIOTIC NITROGEN FIXERS

The ***diazotrophic microorganisms*** are the symbiotic nitrogen fixers that serve as biofertilizers. e.g. *Rhizobium* sp and *Bradyrhizobium* sp. The details on these bacteria with special reference to nitrogen fixation must be referred now. Many attempts are being made (although the success has been limited) to genetically modify the symbiotic bacteria for improving nitrogen fixation.

Green manuring

It is a farming practice wherein the leguminous plants which are benefited by the symbiotic nitrogen fixing bacteria are ploughed into the soil and a non-leguminous crop is grown to take benefits from the already fixed nitrogen. Green manuring has been in practice in India for several centuries. It is a ***natural way of enriching the soil with nitrogen***, and minimizing the use of chemical fertilizers. *Rhizobium* sp can fix about 50-150 kg nitrogen/hectare/annum.

ASYMBIOTIC NITROGEN FIXERS

The asymbiotic nitrogen-fixing bacteria can directly convert the gaseous nitrogen to nitrogen-rich compounds. When these asymbiotic nitrogen fixers die, they enrich the soil with nitrogenous compounds, and thus serve as biofertilizers e.g. *Azobacter* sp, *Azospirillum* sp.

Blue-green algae (cynobacteria)

Blue-green algae multiply in water logging conditions. They can fix nitrogen in the form of organic compounds (proteins, amino acids). The term ***algalization*** is used to the process of cultivation of blue-green algae in the field as a source of biofertilizer.

Blue-green algae, besides fixing nitrogen, accumulate biomass, which improves the physical properties of the soil. This is useful for reclamation of alkaline soils besides providing partial tolerance to pesticides. Cynobacteria are particularly useful for paddy fields. The most common blue-green algae are *Azobacter* sp and *Azospirillum* sp.

TABLE 52.2 A selected list of microorganisms used as biofertilizers along with the crops

Category/microorganism	*Useful for crops*
Symbiotic nitrogen fixers	Legumes (pulses, oilseeds)
Rhizobium leguminosarum, R. meliloti, R. ciceri, Bradyrhizobium japonicum	
Asymbiotic nitrogen fixers	Wheat, rice, sugarcane, jower, vegetables
Azobacter, Azospirillum	
Blue green algae (Anabaena, Nostic, Plectonemia)	Rice
Phosphate solubilizing bacteria	Pulses
Thiobacillus, Bacillus	
Mycorrhiza (Glomus)	Pulses

Azolla

Azolla is an aquatic fern, which contains an endophytic cynobacterium *Anabaena azollae* in the leaf cavities providing a symbiotic relationship. *Azolla* with *Anaebaena* is useful as biofertilizer. But due to certain limitation (listed below), the use of *Azolla* has not become popular.

- Azolla plant requires adequate supply of water.
- It can be easily damaged by pest diseases.
- Azolla cultivation is labour intensive.

PHOSPHATE SOLUBILIZING BACTERIA

Certain bacteria (e.g. *Thiobacillus, Bacillus*) are capable of converting non-available inorganic phosphorus present in the soil to utilizable (organic or inorganic) form of phosphate. These bacteria can also produce siderophores, which chelates with iron, and makes it unavailable to pathogenic bacteria. Thus, besides making phosphate available, the plants are protected from disease - causing microorganisms.

Mycorrhizas

Mycorrhizas are the ***fungus roots*** (e.g. *Glomus* sp) with distinct morphological structure. They are developed as a result of mutual symbiosis between certain root-inhabiting fungi and plant roots. Mycorrhizas are formed in plants, which are limited with nutrient supply. These plants may be herbs, shrubs and trees.

For the development of mycorrhizas, the fungus may be located on the root surface (***ectomycorrhizas***) or inside the root (***endomycorrhizas***).

In recent years, an artificially produced inoculum of mycorrhizal fungi is used in crop fields. This practice improves plant growth and yield, besides providing ***resistance against biotic*** (pathogens) and ***abiotic*** (climatic changes) ***stress.*** Mycorrhizas also produce plant growth-promoting substances.

ORGANIC FERTILIZERS

There are several organic wastes, which are useful as fertilizers. These include animal dung, urine, urban garbage, sewage, crop residues and oil cakes. A majority of these wastes remain unutilized as organic fertilizers. There exists a good potential for the development of organic manures from these wastes.

BENEFITS OF BIOFERTILIZERS

- Low cost and easy to produce. Small farmers are immensely benefited.
- Fertility of the soil is increased year after year.
- Free from environmental pollution.
- Besides nutrient supply, some other compounds, which promote plant growth, are also produced e.g. plant growth hormones, antibiotics.
- Biofertilizers increase physico-chemical properties of the soil, soil texture, and water holding capacity.
- Reclaimation of saline or alkaline soil is possible by using biofertilizers.
- Biofertilizers improve the tolerance of plants against toxic heavy metals.
- Plants can better withstand biotic and abiotic stresses and improve in product yield.

LIMITATIONS OF BIOFERTILIZERS

- Biofertilizers cannot meet the total needs of the plants for nutrient supply.
- They cannot produce spectacular results, as is the case with synthetic fertilizers.

Considering the advantages and disadvantages of biofertilizers, a realistic and pragmatic approach is to use combination of biofertilizers and synthetic fertilizers for optimum crop yield.

Molecular Marker-Aided Plant Breeding

Plant breaders are always interested to improve the agricultural production. This is possible to produce new/improved plant varieties with the desired characters. An important task of plant breeders is the correct identification, and continued maintenance of the plants with improved characters. There are different types of markers used for this purpose. The basic requirement of a ***marker*** is that it ***must be polymorphic*** i.e. it must exist in different forms. The polymorphism can be detected by three different types of markers

1. Morphological markers
2. Biochemical markers
3. Molecular markers.

Morphological markers

The morphological or phenotype markers represent the quantitative traits that can be visually detected e.g. altered leaf morphology, albinism, dwarfism etc. Morphological markers may be dominant or recessive. These markers are not widely used to detect polymorphism these days due to the following ***limitations***.

- Morphological markers influenced by the environment, and thus may not represent the desired genetic variation.
- Majority of the visible markers may not be required in the plant improvement programmes.

Biochemical markers

The proteins produced by gene expression are also used as markers in plant breeding programmes. The most commonly used are ***isoenzymes***, the different molecular forms of the same enzyme. The isoenzymes can be separated by electrophoresis and identified. However, the polymorphism of isoenzymes is rather poor. Hence, they are not widely used as markers.

MOLECULAR MARKERS

A molecular marker is a ***DNA sequence in the genome which can be located and identified***. As a result of genetic alterations (mutations, insertions, deletions), the base composition at a particular location of the genome may be different in different plants. These differences, collectively called as polymorphisms can be mapped and identified. Plant breeders always prefer to detect the gene as the molecular marker, although this is not always possible. The alternative is to have markers which are closely associated with genes and inherited together.

The molecular markers are highly reliable and ***advantageous*** in plant breeding programmes

- Molecular markers provide a true representations of the genetic make up at the DNA level.

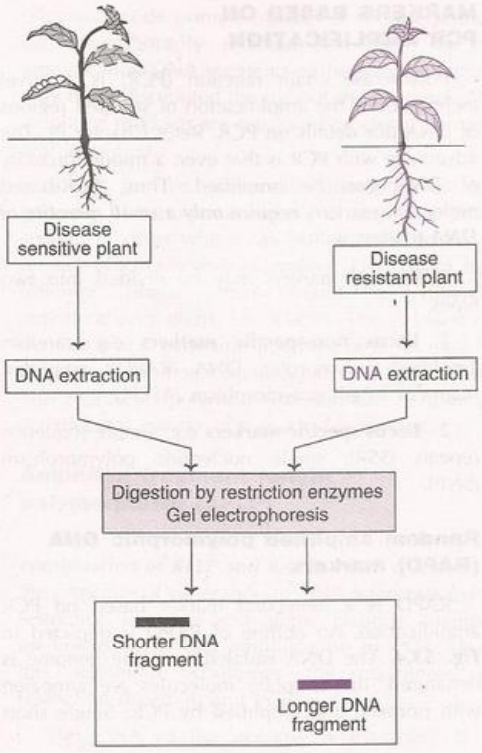


Fig. 53.1 : Basic principle of molecular marker detection (screening of genotypes for the identification of DNA markers).

- They are consistent and not affected by environmental factors.
- Molecular markers can be detected much before development of plants occur.
- A large number of markers can be generated as per the needs.

Basic principle of molecular marker detection

Let us assume that there are two plants of the same species—one with disease sensitivity and the other with disease resistance. If there is DNA marker that can identify these two alleles, then the genome can be extracted, digested by restriction enzymes, and separated by gel electrophoresis. The DNA fragments can be detected by their separation. For instance, the disease resistant plant may have a shorter DNA fragment while the disease — sensitive plant may have a longer DNA fragment (***Fig. 53.1***). Molecular markers are of two types.

1. Based on nucleic acid (DNA) hybridization (non-PCR based approaches).

2. Based on PCR amplification (PCR-based approaches).

MARKERS BASED ON DNA HYBRIDIZATION

The DNA piece can be cloned, and allowed to hybridize with the genomic DNA which can be detected. Marker-based DNA hybridization is widely used. The major ***limitation*** of this approach is that it requires ***large quantities of DNA*** and the use of radioactivity (labeled probes).

Restriction fragment length polymorphism (RFLP)

RFLP was the very first technology employed for the detection of polymorphism, based on the DNA sequence differences. RFLP is mainly ***based on the altered restriction enzyme sites***, as a result of mutations and recombinations of genomic DNA.

An outline of the RFLP analysis is given in ***Fig. 53.2***, and schematically depicted in ***Fig. 53.3***.

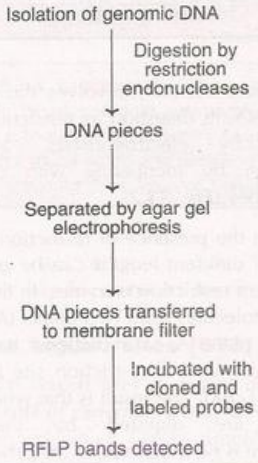


Fig. 53.2 : An outline of restriction fragment length polymorphism (RFLP) analysis as a molecular marker in plant breeding.

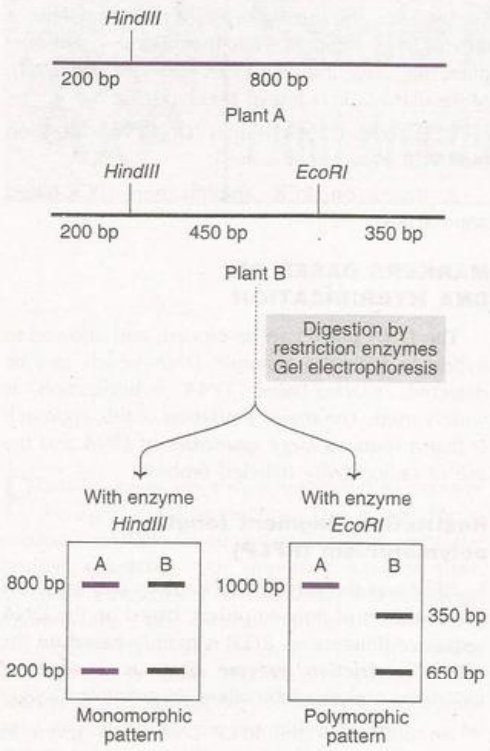


Fig. 53.3 : A schematic representation of restriction fragment length polymorphism (RFLP) analysis as molecular marker.

The procedure basically involves the isolation of genomic DNA, its digestion by restriction enzymes, separation by electrophoresis, and finally hybridization by incubating with cloned and labeled probes (***Fig. 53.2***).

Based on the presence of restriction sites, DNA fragments of different lengths can be generated by using different restriction enzymes. In the ***Fig. 53.3***, two DNA molecules from two plants (A and B) are shown. In plant A, a mutations has occurred leading to the loss of restriction site that can be digested by *EcoRI*. The result is that when the DNA molecules are digested by the enzyme *HindIII*, there is no difference in the DNA fragments separated. However, with the enzyme *EcoRI*, plant A DNA molecules is not digested while plant B DNA molecule is digested. This results in a polymorphic pattern of separation.

MARKERS BASED ON PCR AMPLIFICATION

Polymerase chain reaction (PCR) is a novel technique for the amplification of selected regions of DNA (for details on PCR, Refer Chapter 8). The advantage with PCR is that even a minute quantity of DNA can be amplified. Thus, PCR-based molecular markers ***require only a small quantity of DNA*** to start with.

PCR-based markers may be divided into two types.

1. **Locus non-specific markers** e.g. random amplified polymorphic DNA (RAPD); amplified fragment length polymorphism (AFLP).

2. **Locus specific markers** e.g. simple sequence repeats (SSR); single nucleotide polymorphism (SNP).

Random amplified polymorphic DNA (RAPD) markers

RAPD is a molecular marker based on PCR amplification. An outline of RAPD is depicted in ***Fig. 53.4***. The DNA isolated from the genome is denatured, the template molecules are annealed with primers, and amplified by PCR. Single short

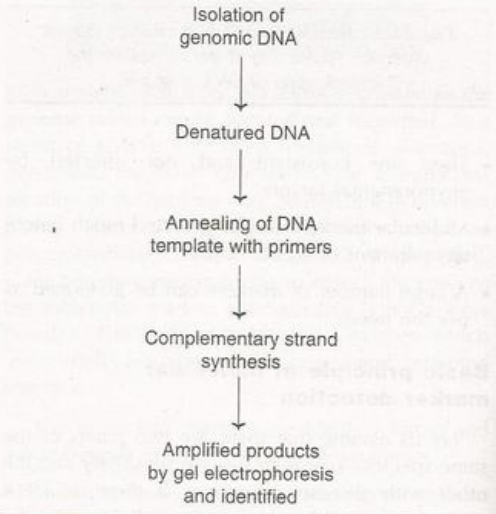


Fig. 53.4 : An outline of random amplified polymorphic DNA (RAPD) analysis as a molecular marker in plant breeding.

oligonucleotide primers (usually a 10-base primer) can be arbitrarily selected and used for the amplification DNA segments of the genome (which may be in distributed throughout the genome). The amplified products are separated on electrophoresis and identified.

Based on the nucleotide alterations in the genome, the polymorphisms of amplified DNA sequences differ which can be identified as bends on gel electrophoresis. Genomic DNA from two different plants often results in, different amplification patterns i.e. RAPDs. This is based on the fact that a particular fragment of DNA may be generated from one individual, and not from others. This represents polymorphism and can be used as a molecular marker of a particular species.

Amplified fragment length polymorphism (AFLP)

AFLP is a novel technique involving a ***combination of RFLP and RAPD***. AFLP is based on the principle of generation of DNA fragments using restriction enzymes and oligonucleotide adaptors (or linkers), and their amplification by PCR. Thus, this technique combines the usefulness of restriction digestion and PCR.

The DNA of the genome is extracted. It is subjected to restriction digestion by two enzymes (a rare cutter e.g. *MseI;* a frequent cutter e.g. *EcoRI*). The cut ends on both sides are then ligated to known sequences of oligonucleotides (***Fig. 53.5***).

PCR is now performed for the preselection of a fragment of DNA which has a single specific nucleotide. By this approach of preselective amplification, the pool of fragments can be reduced from the original mixture. In the second round of amplification by PCR, three nucleotide sequences are amplified. This further reduces the pool of DNA fragments to a manageable level (< 100). Autoradiography can be performed for the detection of DNA fragments. Use of radiolabeled primers, and fluorescently labeled fragments quickens AFLP.

AFLP analysis is tedious and requires the involvement of skilled technical personnel. Hence some people are not in favour of this technique. In recent years, commercial kits are made available for AFLP analysis.

AFLP is very sensitive and reproducible. It does not require prior knowledge of sequence information: By AFLP, a large number of polymorphic bands can be produced and detected.

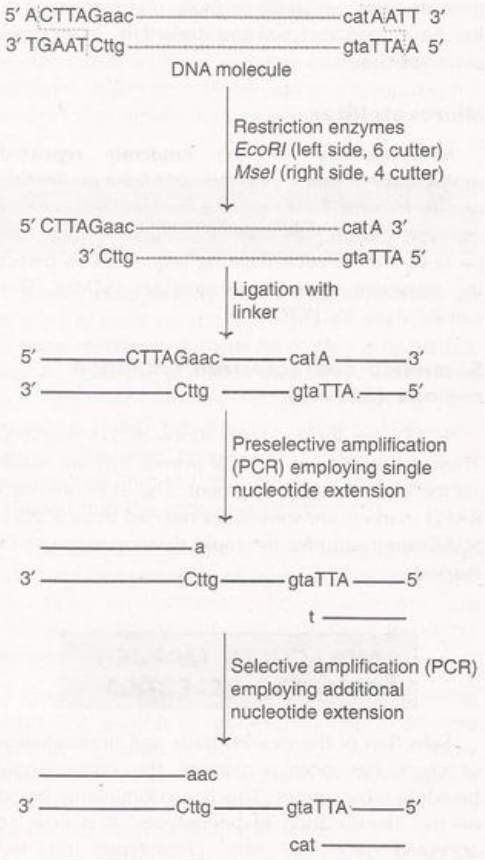


***Fig. 53.5** : A diagrammatic representation of the amplified fragment length polymorphism (AFLP) (**Note** : The lower case letters represent the sequences found within the amplified region; the coloured lines indicate linkers).*

Sequence tagged sites (STS)

Sequence tagged sites represent unique ***simple copy segments of genomes***, whose DNA sequences are known, and which can be amplified by using PCR. STS markers are based on the polymorphism of simple nucleotide repeats e.g. $(GA)_n$, $(GT)_n$, $(CAA)_n$ etc. on the genome. STS have been recently developed in plants. When the STS loci contain ***simple sequence length polymorphisms (SSLPs)***,

they are highly valuable as molecular markers. STS loci have been analysed and studied in a number of plant species.

Microsatellites

Microsatellites are the ***tandemly repeated multicopies of mono-, di-, tri- and tetra nucleotide motifs***. In some instances, the flanking sequence of the repeat sequences may be unique. Primers can be designed for such flanking sequences to detect the sequence tagged microsatellites (STMS). This can be done by PCR.

Sequence characterized amplified regions (SCARs)

SCARs are the modified forms of STS markers. They are developed by PCR primer that are made for the ends of RAPD fragment. The STS-converted RAPD markers are sometimes referred to as SCARs. SCARs are useful for the rapid development of STS markers.

MOLECULAR MARKER ASSISTED SELECTION

Selection of the desired traits and improvement of crops has been a part of the conventional breeding programmes. This is predominantly based on the identification of phenotypes. It is now an accepted fact that the phenotypes do not necessarily represent the genotypes. Many a times the environment may mark the genotype. Thus, the plant's genetic potential is not truely reflected in the phenotypic expression for various reasons.

The molecular marker assisted selection is based on the identification of DNA markers that link/ represent the plant traits. These traits include resistance to pathogens and insects, tolerance to abiotic stresses, and various other qualitative and quantitative traits. The advantage with a molecular marker is that a plant breeder can select a ***suitable marker*** for the desired trait which ***can be detected well in advance***. Accordingly, breeding programmes can be planned.

The following are the ***major requirements*** for the molecular marked selection in plant breeding

- The marker should be closely linked with the desired trait.
- The marker screening methods must be efficient, reproducible and easy to carry out.
- The analysis should be economical.

MOLECULAR BREEDING

With rapid progress in molecular biology and genetic engineering, there is now a possibility of improving the crop plants with respect to yield and quality. The term molecular breeding is frequently used to represent the ***breeding methods that are coupled with genetic engineering techniques***.

Improved agriculture to meet the food demands of the world is a high priority area. For several years, the conventional plant breeding programmes (although time consuming) have certainly helped to improve grain yield and cereal production. The development of dwarf and semi-dwarf varieties of rice and wheat have been responsible for the ***'Green Revolution'***, which has helped to feed millions of poverty-stricken people around the world.

Many development on the agriculture front are expected in the coming years as a result of molecular breeding.

Linkage analysis

Linkage analysis basically deals with studies to ***correlate the link between the molecular marker and a desired trait***. This is an important aspect of molecular breeding programmes. Linkage analysis has to be carried out among the populations of several generations to establish the appropriate linkage.

In the earlier years, linkage analysis was carried out by use of isoenzymes and the associated polymorphisms. Molecular markers are now being used. The techniques employed for this purpose have already been described.

QUANTITATIVE TRAIT LOCI

These are many characteristics controlled by several genes in a complex manner. Some good examples are growth habit, yield, adaptability to environment, and disease resistance. These are referred to as ***quantitative traits***. The ***locations on the chromosomes for these genes*** are regarded as quantitative trait loci (QTL).

The major problem, the plant breeder faces is how to improve the a complex character controlled

by many genes. It is not an easy job to manipulate multiple genes in genetic engineering. Therefore, it is a very difficult and time consuming process. For instance, development of Golden Rice (with enriched provitamin A) involving the insertion of just three genes took about seven years.

ARID AND SEMI-ARID PLANT BIOTECHNOLOGY

The terms ***arid zone*** is used to refer to ***harsh environmental conditions with extreme heat and cold***. The fileds have limited water and minerals. It is different task to grow plants and achieve good crop yield in arid zones.

Semi-arid regions are characterized by ***unpredictable weather***, inconsistent rainfall, long dry seasons, and poor nutrients in the soil. Most parts of India and many other developing countries (Africa, Latin America, Southeast Asia) have semi-arid regions.

Crops like sorghum, millet, groundnut and cowpea are mostly grown in semi-arid tropics. Besides unpredictable weather, biotic and abiotic stresses contribute to crop loss in these areas.

The biotechnological approaches for the breeding programmes in the semi-arid regions should cover the following areas.

- Development of crops that are tolerant to drought and salinity.
- Improvements to withstand various biotic and abiotic stresses.
- Micropropagation techniques to spread economically important plants which can withstand harsh environmental conditions.

Some success has been achieved in improving sorghum, millet and legume crops that are grown in semi-arid regions. Genetic transformation in sorghum was possible by using microprojectile method (For details, Refer Chapter 49).

GREENHOUSE AND GREENHOME TECHNOLOGY

Greenhouse literally means a building made up of glass to grow plants. Green houses are required to ***grow regenerated plants for further propagation, and for growing plants to maturity***. Greenhouses are the intermediary stages involving the transitional step between the plant cultures and plant fields. The purpose of greenhouses is to acclimatize and test the plants before they are released into the natural environment.

The plants are grown in greenhouse to develop adequate root systems and leaves so as to withstand the field environment. The greenhouses are normally equipped with cooling systems to control temperature.

Greenhouses have chambers fitted with artificial lights. It is possible to subject the plants to different lighting profiles. In recent years many improvements have been made in the development of more suitable greenhouses. These include the parameters such as soil, and humidity.

The major limitation of greenhouse technology is an increase in CO_2 production that in turn increases temperature. Some approaches are available to control temperature.

Greenhome technology is a recent development. In this case, ***temperature is controlled by using minimum energy***.

SECTION VIII

ENVIRONMENTAL BIOTECHNOLOGY

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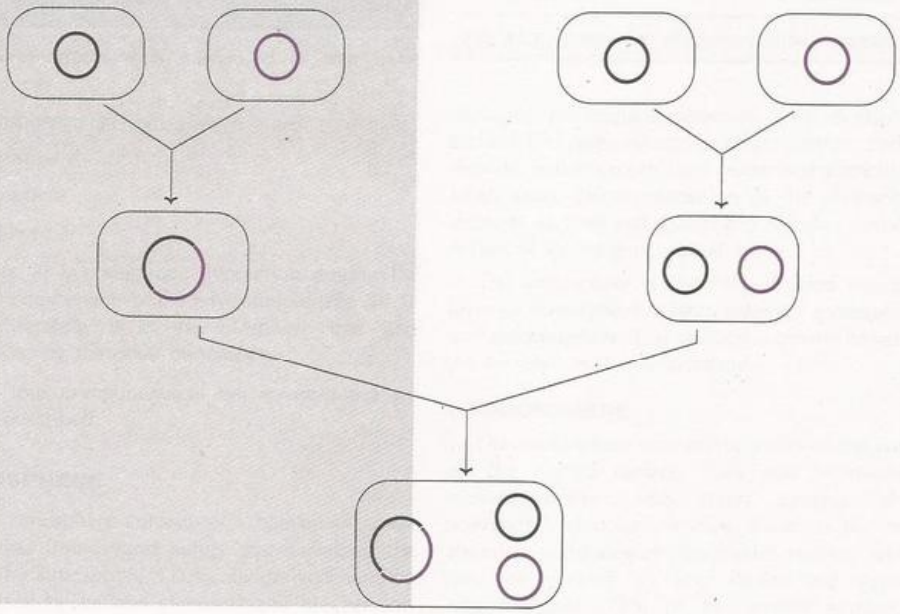
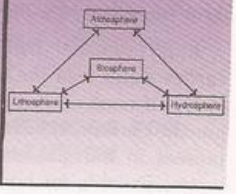


54 Environmental Pollution — General



Environment may be considered as our surroundings which includes everything around us, i.e. the ***non-living*** (***abiotic***) and ***living*** (***biotic***) ***environment***. The abiotic environment consists of air, water and soil, while the biotic environment includes all the living organisms (plants, animals, microorganisms) that we regularly come in contact.

ENVIRONMENT — BASIC CONCEPTS

The environment is composed of four basic components.

- Atmosphere
- Hydrosphere
- Lithosphere
- Biosphere.

There is a continuous interaction among the various components of the environment (***Fig. 54.1***). And ultimately, it is the biosphere that gets influenced by the other components.

The four components of the environment are briefly described.

ATMOSPHERE

The atmosphere consists of a blanket of gases, suspended liquids and solids that envelope the earth. The atmosphere is basically derived from the earth itself by various chemical and biochemical reactions. The major components of the atmosphere include the gases nitrogen, oxygen, argon, carbon dioxide, water vapour, and suspended particulates (dust, soot). The composition of the atmosphere depends on time and space, and is highly variable. A liter of air weights around 1.3 g.

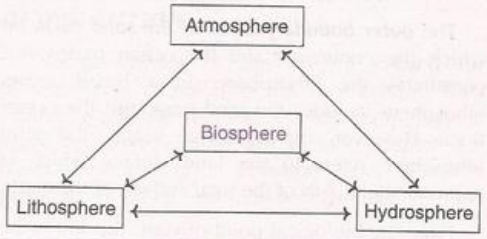


Fig. 54.1 : Interaction of environmental components.

The atmosphere is vertically divided into four layes — ***troposphere***, ***stratosphere***, ***mesosphere***, and ***thermosphere***. This division is mainly based on the increase in the temperature.

HYDROSPHERE

The hydrosphere primarily consists of the ***water on the earth's surface***. Thus, the hydrosphere includes oceans, seas, rivers, streams, lakes, reservoirs and polar ice caps. Water is the most abundant substance on the earth's surface, which may be present as ice, liquid and vapour. Approximately, 71% of the earth's surface is

covered with water, mainly in the form of oceans.

It is estimated that ***about 97% of the total earth's water is in the oceans and inland seas*** with high salt content. And this water is not useful for human consumption. Around 2% of the water is present in the glaciers and ice caps. The actual water available for human consumption is around 1% of the total earth's water. This includes the ground water, water from lakes and rivers and soil moisture.

Humans uses water in the homes, industries, agriculture and recreation. There is a continuous decrease in the consumable global water. Therefore, there is a need for precious use of water, and its conservation.

LITHOSPHERE

The ***outer boundary layer of the solid earth*** on which the continents and the ocean basins rest constitutes the lithosphere. In a broad sense, lithosphere includes the land mass and the ocean floor. However, in a general usage, the term lithosphere refers to the land surface which is approximately $\frac{3}{10}$th of the total surface of the earth.

From the biological point of view, the soil is the most important part of the lithosphere because it contains the organic matter and supports growth of plants and microorganisms. Lithosphere is involved in the production of food for humans and animals, besides the decomposition of organic wastes.

BIOSPHERE

The biosphere comprises of all the ***zones on earth in which life is present***. Biosphere is spread over the lower part of the atmosphere, the top of the lithosphere and the entire hydrosphere. In other words, the broad spectrum of ***bioresources of the earth***, supporting life constitutes the biosphere. It is estimated that the biosphere contains more than 3.5 lakh species of plants (including algae, fungi, mosses and higher plants) and more than 110 lakh species of animals (unicellular, multicellular and higher animals). The biosphere provides the essential requisites (water, light, heat, air, food, space etc.) for the existence of life.

The biosphere is very vast, and for the sake of understanding, it is divided into smaller units namely ***ecosystems***. An ecosystem may be considered as the smallest unit of biosphere that possesses the requisite characteristics to sustain life e.g., ponds, seas, deserts, cities.

ENVIRONMENTAL POLLUTION — SOURCES AND NATURE

Man lives in two worlds—a ***natural world*** of the native environment and a ***built-world*** created by himself. The built-world, an outcome of the advances made in the science and technology, is associated with environmental pollution. Environmental pollution is a global phenomenon, and therefore a matter of concern for everyone.

Pollution broadly refers to the presence of undesirable substances in the environment which are harmful to man and other organisms. The presence of unwanted substances in the environment may occur due to human activity discharging byproducts, a wide spectrum of waste products and several harmful secondary products (**Note :** It must be remembered that the natural world of environment is not totally free from impurities or undesirable substances. However, such impurities are usually not harmful).

SOURCES OF POLLUTION

As already stated, environmental pollution is mostly due to direct or indirect human activities, arising out of the built-world created by him. There are six ***major sources of environmental pollution***.

1. Industrial sources
2. Agricultural sources
3. Biogenic sources
4. Anthropogenic sources
5. Unnatural sources
6. Extra-terrestrial sources.

The relative importance of each one of these sources depends on the site-specific situation. For instance in cities, anthropogenic sources are the major contributors while in rural areas, agricultural sources significantly add to pollution.

Types of pollution

The environmental pollution may be categorized into six major groups.

1. Air/atmosphere pollution
2. Water pollution
3. Land/soil pollution
4. Noise pollution
5. Thermal pollution
6. Radioactive pollution.

The details of air pollution and water pollution are described elsewhere (Chapters 55 and 56).

NATURE OF POLLUTANTS

The pollutants that occur in the environment may be chemical, biological and physical in their nature.

Chemical pollutants : Gaseous pollutants (sulfur dioxide, nitrogen dioxide), toxic metals, pesticides, herbicides, hydrocarbons, toxins, acidic substances, carcinogens.

Biological pollutants : Pathogenic organisms, products of biological origin.

Physical pollutants : Heat (thermal), sound, odours, radiation and radioactive substances.

POLLUTION MONITORING/ MEASUREMENT

Environmental pollution poses a big threat to the healthy existence of humankind. The Governments worldover pay serious attention to continuously monitor and minimize pollution. The public and non-governmental organizations (NGOs) are also actively involved in this venture. Broadly, there are ***four levels of pollution monitoring agences or environmental protection agencies*** (***EPAs***).

Primary level : This is at the district or block level. The non-governmental organizations and the rural development agencies are involved in pollution monitoring.

Secondary level : Existing at the state level, the monitoring is done by the respective state pollution control boards.

Tertiary level : This is at the national/country level. Each country has its own environmental protection agency to monitor pollution.

Quaternary level : International/inter-Governmental bodies are closely associated with monitoring of pollution which is a global phenomenon. World Health Organization and United Nation Environmental Programme are actively involved.

BIOTECHNOLOGICAL METHODS FOR MEASUREMENT OF POLLUTION

In recent years, environmental pollution detection and monitoring is being done by approaches involving ***biosystems***. For this, purpose, several groups of plants, animals and microorganisms are utilized. The environmental protection agencis (EPAs) consider ***biomonitoring*** of pollution as a useful device to monitor environmental pollution from the point of diagnostic, preventive and remedial measures.

CRITERIA FOR BIOMONITORING OF POLLUTION

The parameters or the criteria chosen for biomonitoring of pollution are very crucial. They should be reliable, reproducible and cost-effective. Three types of criteria are mostly adopted for biomonitoring of pollution-visual rating, genotoxicity rating and metabolic rating.

Visual rating

In the visual rating, the ***growth rate and productivity*** are considered. When microorganisms are used in the test assay, the growth can be measured by turbidometric analysis. In case of higher plants, growth rate of different parts, visual damage to leaves, seed viability and germination frequency are taken into account.

As regards animals (fishes are commonly used), the concept of LD_{50} is used i.e. the dose at which 50% of the test organism is affected.

Sometimes, the presence or absence of a particular species of an organism serves as an indicator for the environmental pollution.

Genotoxicity rating

Genotoxicity tests ***measure the extent of damage caused to an organism by environmental pollution at the cellular and sub-cellular levels***. The genotoxic lesions may be detected on the cellular organelles (membranes commonly used), genomes, immune systems, biomolecules, etc.

Cytotoxic tests such as measurement of chromosomal damage (including breakage), sister chromatid exchange (SCE) and micronuclei counting are also useful for pollution detection. The cell viability can be measured by detecting *in vitro* lysosomal viability. In recent yeas, DNA probes are in use for the identification of disease-causing organisms in water.

Metabolic rating

The ***biochemical changes*** with environmental pollution can be measured (qualitatively and quantitatively) in selected organisms. In fact, certain metabolic parameters can be used as ***biomarkers to assess the pollution stress***. The biomarkers used in metabolic rating include chlorophyll, proteins, nucleic acids (DNA and RNA) and changes in enzyme activities.

The biotechnological methods adopted for pollution measurement are briefly described in the following order.

1. General bioassays
2. Cell biological assays
3. Molecular biological assays
4. Biosensors.

BIOASSAYS IN ENVIRONMENTAL MONITORING

In the early years, the conventional physical and chemical methods were used for the detection of environmental pollution. Bioassays are preferred these days, since the biological responses that reflect the damages to the living organisms are very crucial for the actual assessment of pollution.

The organisms employed in the bioassays for pollution detection are expected to satisfy the following criteria.

- It should readily take up the pollutant (absorption or adsorption).
- The organism should be sensitive to the pollutant.
- It should possess measurable features to detect pollution.
- The organism should have wide occurrence, and available round the year.
- The bioassay should be simple, reproducible and cost-effective.

The most commonly used plants and animals in the bioassays are briefly described.

Plant test systems in bioassays

Certain algae, bacteria, lichens, mosses and vascular macrophytes are commonly used in bioassays.

Algal bioassays : Among the plant systems, algal bioassays are the most commonly used. Algae are considered to be reliable indicators of pollution due to their high sensitivity and easy availability, besides simple culturing techniques. The criteria adopted for algal bioassays are the growth rate, biomass accumulation and photosynthetic efficiency.

The algae used in the test assays include *Chlorella, Microcystis, Spirulina, Navicula, Scenedesmus, Anabaena, Ulva, Codium, Fucus* and *Laminaria*. In water, organic pollution can be detected by using the blue green algae, *Microcystis*, while metal pollution can be measured by *Navicula*.

Bacterial bioassays : These are commonly used for the detection of fecal pollution in potable water, the most widely employed test being ***coliform test***. ***Ames test*** that detects mutagenic pollutants is carried out by the bacterium *Salmonella*.

Bacterial bioluminescence is a recent technique used for the measurement of gaseous pollutants and other compounds e.g. sulfur dioxide, formaldehyde, ethylacetate. *Photobacterium phosphoreum* is the organism of choice for bacterial bioluminescence.

Lichens in bioassays : Lichens are widely used for the detection of atmospheric gas pollution, particularly in cities. Lichens are very sensitive for the measurement of sulfur dioxide.

Mosses in bioassays : Environmental metal pollution can be detected by using certain forest and aquatic mosses e.g. *Stereophyllum, Sphagnum, Brynus*.

Vascular macrophytes in bioassays : Water hyacinth (*Eichormia crassipes*) and duck weed (*Lemna minor*) are in use to detect aquatic metal pollution. In fact, certain biochemical parameters of macrophytes are used to serve as ***biomarkers*** of pollution e.g., peroxidase activity increases due to metal pollution; inhibition of nitrate reductase activity by mercury. The other commonly used

bioassay parameters are the estimation of soluble proteins, nucleic acids, chlorophyll, and assay of enzyme (e.g. catalase, peroxidase) activities.

Pollution-induced peptides in bioassays : Very recently, some workers have identified the presence of small peptides within the plant cells which are pollution-induced. These peptides, referred to as ***phytochelatins***, are formed as a result of metal pollution. They are reasonably reliable ***for the detection of metal pollution.***

Animal test systems in bioassays

Among the animals, certain fishes, protozoa and helminths are employed in bioassays.

Fishes in bioassays : Toxic effects of environmental pollutants on fishes have been in use for quite some time as a measure of bioassays. In fact, the concept of LD_{50} (i.e. the dose of the pollutant at which 50% of the test organisms are affected) has originated from the studies on fishes. The criteria or parameters used for assessment of fish bioassays include changes in the morphology and organs, behavioural pattern and modifications in metabolisms. The alterations in the ***enzyme acetylcholine esterase serves as a reliable marker for pesticide pollution.***

The most commonly used fishes in bioassays are *Catla, Teleost, Labeo* and *Channa.*

Protozoa in bioassays : The ciliated protozoa serve as good bioassay systems for the detection of environmental pollution. The toxic effects of the pollutants can be measured by the ***changes in the behavioural patterns*** of protozoa, recorded on an ***ethogram.***

Helminths in bioassays : Rotifers are a group of helminths that grow on aquatic vegetation. They are used for the detection of organic matter in water (given by BOD). Rotifers, with round the year availability, easy cultivation, slow growth rate and easy recognition are used for biomonitoring of water.

Pollution-induced peptides in bioassays : As already described in case of plant bioassays (above), pollution-induced small peptides are found in animal cells also. They are collectively referred to as ***metallothioneins*** (comparable to phytochelatins in plants). Metallothioneins are useful ***for the detection of metal pollution.***

Biomonitoring of pollution with multiple species

Most often, bioassays using a single organism are not adequate to detect pollution. In such a case, multiple species of organisms are used.

CELL BIOLOGY IN ENVIRONMENTAL MONITORING

Cell biology deals with the study of the structural and functional aspects of cells and the cellular organelles. It is successfully exploited for environmental pollution detection, particularly with reference to mutagens and carcinogens.

The cell biological methods primarily aim to trace the harmful effects of pollutants on different cellular components — membranes, chloroplasts, mitochondria, chromosomes. In addition, the macromolecules namely nucleic acids (particularly DNA) and proteins are also used. Further, cell biological methods help in understanding the mechanisms of toxicity of pollutants.

Some important cell biological methods used in environmental pollution monitoring are described.

Membrane damage in bioassay

The plasma membrane, an envelope surrounding the cell, protects the cell from hostile environment. It is the first cellular component to be directly exposed to pollutants. Many toxic substances that cause damage to cell structure and its functions are known.

For the purpose of bioassay, the physical damages caused by pollutants or their deposition on the membranes can be detected by light, phase contrast and electron microscopy. This approach may not be always practicable. The alterations in the semipermeable properties of the membranes due to pollutants can be detected by leakage of enzymes (e.g., lactate dehydrogenase), efflux of electrolytes or uptake of trypan blue. ***Lysosomes*** are also useful as ***biomarkers for measurement of cell viability.*** This can be done by ***neutral red retention test.*** The damaged lysosomes cannot retain this dye.

In recent years, animal and plant tissue culture techniques are also used for pollution monitoring. This is made possible by measuring cellular damages observed in cell cycles. A good example

is the use of human lymphocyte culture to monitor the persons exposed to toxic pollutants.

Cytogenetic bioassays

The genetic damages of the cells, as reflected by changes in the chromosomes, can be effectively used in biomonitoring of pollution. For this purpose, animals (e.g. insect *Drosophila*) and plants (e.g. *Arabidiopsis*) with shorter life cycles are preferred. Other plants such as pea, maize and soy bean are also used in cytogenetic bioassays.

Chromosomal damage : The pollutants may cause several types of chromosomal damages-fragmentation, bridge formation, disruption in cell division. The ***chromosomal alterations can be effectively used for pollution detection***. It has been clearly established that the severity of chromosomal damage depends on the chemical nature of the pollutant.

Micronucleus test : Severe damage to chromosomes by pollutants may result in large scale fragmentation of chromosomes, followed by micronuclei formation. The degree of micronuclei development is directly related to the severity of the damage. Micronucleus test (MNT) is used for ***screening of mutagenic compounds***.

Sister-chromatid exchange : The damages caused by pollutants results in misexchange of chromosomal segments (chromatids) during cell division. The sister chromated exchange (SCE) can be detected by using a fluorescent dye technique.

Ames test in bioassays

Ames test can be used for the ***detection of chemical mutagens and their carcinogenecity***. This is very widely used bioassay for screening of various pollutants, drugs, cosmotics, food additives and metals.

Ames test employs the use of a special mutant strain of bacterium namely *Salmonella typhimurium* (His^-). This organism cannot synthesize histidine, hence the same should be supplied in the medium for its growth. Addition of chemical carcinogens causes mutations (reverse mutation) restoring the ability of this bacterium to synthesize histidine (His^+). By detecting the strain of *Salmonella* (His^+) in the colony of agar plates, the chemical mutagens can be identified. The ***Ames assay can detect about 90% of the chemical carcinogens***.

Recently, the yeast cells (*Saccharomyces cerevisae*) are also used for the detection of chemical carcinogens.

MOLECULAR BIOLOGY IN ENVIRONMENTAL MONITORING

The use of ***molecular probes*** and ***immunoassays*** in monitoring of environmental pollution is gaining importance in recent years. Molecular biological bioassays are particularly useful for the detection of bacteria, viruses and other pathogenic organisms that cause diseases.

DNA probes

DNA probes and ***polymerase chain reaction (PCR)*** can be effectively used for water quality monitoring, particularly potable water. However, these techniques are expensive and not practicable at all places. For more details on DNA probes and PCR, Refer Chapters 9 and 8 respectively.

Immunoassays

Immunological techniques are useful for the detection of pollutants (pesticides, herbicides) and identification of pathogens that exhibit immunological properties. Immunoassays are in use for the measurement of several pesticides e.g. aldrin, triazines DDT, glyphosate. Metabolic products of certain bacteria can also be detected by immunoassays. For instance, assay systems have been developed for the detection of toxins of *cholera* and *Salmonella*.

In recent years, use of ***monoclonal antibodies (MAbs)*** in the detection and biomonitoring of environmental pollution is gaining importance. In fact, assay techniques are available for detection of pestiçide and herbicide contamination in water.

Bioluminescent bioassays using Lux reporter genes

Certain genes, referred to as Lux reporter genes, on the plasmids produce assayable signals. Whenever these genes are expressed in luminescent bacteria like *Photobacterium* and *Vibrio*. Some bacterial strains have been developed through gene cloning (employing Lux reporter genes) for the detection of pollutants and their degradation. For instance, genetically altered *Pseudomonas* can be used for detecting naphthalene, xylene, toluene and salicylate.

BIOSENSORS IN ENVIRONMENTAL MONITORING

A biosensor is an ***analytical device containing an immobilized biological material*** (enzyme, organelle, cell) which can specifically interact with an analyte (a compound whose concentration is to be determined) and produce physical, chemical or electrical signals that can be measured. Biosensors are highly specific and accurate in their function.

The details on biosensors—principles of working, types, various applications are described elsewhere (Refer Chapter 21). Some of the important biosensors used in environmental pollution monitoring are briefly described.

BOD biosensor

Biological oxygen demand (BOD_5) is a widely used test for the detection of organic pollution. This test requires five days of incubation. A BOD biosensor using the yeast *Trichosporon cutaneum* with oxygen probe takes just 15 minutes for detecting organic pollution.

Gas biosensors

Microbial biosensors for the detection of gases such as sulfur dioxide (SO_2), methane and carbon dioxide have been developed. *Thiobacillus*-based biosensor can detect the pollutant SO_2, while methane (CH_4) can be detected by immobilized *Methalomonas*. For carbon dioxide monitoring, a particular strain of *Pseudomonas* is used.

Immunoassay biosensors

Immunoelectrodes as biosensors are useful for the detection of low concentrations of pollutants. Pesticide specific antibodies can detect the presence of low concentrations of triazines, malathion and carbamates, by employing immunoassays.

Other biosensors

Biosensors employing ***acetylcholine esterase*** (obtained from bovine RBC) can be used for the detection of organophosphorus compounds in water. In fact, portable pesticide monitors are commercially available in some developed countries.

TABLE 54.1 A selected list of pollutants measured by employing biosensors

Pollutant measured	*Biosensor–biological component*
BOD	*Trichosporon cutaneum*
SO_2	*Thiobacillus* sp
CH_4	*Methanomonas flagellae*
CO_2	*Pseudomonas* sp
Nitrate	*Azobacter vinelandi*
NH_3 and NO_2	Mixed nitrifying bacteria
Nerve gas	Acetylcholine esterase
Ethanol	NADH and dehydrogenase
Phenol	Phenol oxidase
Parathion	Antibody to parathion
Herbicide–induced electron transport chain inhibition	*Synechococcus*

Biosensors for the detection of polychlorinated biphenyls (PCBs) and chlorinated hydrocarbons and certain other organic compounds have been developed.

Phenol oxidase enzyme (obtained from potatoes and mushrooms) containing biosensor is used for the detection of phenol.

A graphite electrode with *Cynobacterium* and *Synechococcus* has been developed to measure the degree of electron transport inhibition during photosynthesis due to certain pollutants e.g. herbicides.

A selected list of environmental pollutants measured by employing biosensors is given in ***Table 54.1***.

BIOTECHNOLOGICAL METHODS FOR MANAGEMENT OF POLLUTION

The different approaches for the management of environmental pollution are described in the respective chapters for air pollution, refer Chapter 55, and for water pollution Chapter 56.

The general concepts of the biotechnological approaches for the abatement of pollution with special reference to the following aspects are described in this chapter.

1. Atmospheric CO_2 reduction.
2. Sewage treatment by bacteria and algae.
3. Eutrophication and phosphorus pollution.
4. Management of metal pollution.
5. Immobilized cells in the management of pollution.

ATMOSPHERIC CO_2 REDUCTION

The lower atmosphere contains CO_2 at a concentration of 0.0314% by volume. There is a continuous addition of CO_2, particularly coming from industrial processes. Any increase in the content CO_2 has to be viewed very seriously, since it is the principal gas that causes ***green house effect*** and rises atmospheric temperature. It is suggested that the global temperature will increase by around 2-5°C on doubling the atmospheric concentration of CO_2. The present belief is that during the past 100–150 years, the CO_2 level increased by about 25% with an increase in the atmospheric temperature by about 0.5%. Thus, CO_2 is closely linked with ***global warming***.

For the reasons stated above, the reduction in atmospheric CO_2 concentration assumes significance. There are mainly two approaches for the biotechnological reduction of CO_2 in the atmosphere.

1. Photosynthesis
2. Biological calcification.

PHOTOSYNTHESIS TO REDUCE ATMOSPHERIC CO_2

Utilization of CO_2 for photosynthesis by plants is the most significant process to reduce CO_2 content in the atmosphere. Photosynthesis may be represented by the following equation.

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

Higher plant photosynthesis

The fast growing trees are more efficient in utilizing CO_2 for photosynthesis, hence their propagation is advocated. Further, ***micro-propagation*** and ***synthetic seed*** production through plant tissue culture techniques are also important. The major problem is the continuous deforestation that is associated with a possible increase in atmospheric CO_2. The conservation of forests and plantations is the need of the hour.

Certain biotechnological approaches have also been made to improve the CO_2 utilization by enhancing photosynthesis. The enzyme ***ribulose-bisphosphate carboxylase*** (RUBP-case) is closely linked with CO_2 fixation. There are some attempts to genetically manipulate this enzyme so that the photosynthetic efficiency is increased.

Microalgal photosynthesis

Certain microalgae are more efficient than higher plants in utilizing atmospheric CO_2 for photosynthesis e.g. *Chlorella pyrenodiosa, Spirulina maxima*. These organisms are capable of generating more O_2 than the amount of CO_2 consumed.

It is advocated that growing microalgae in the vicinity of industries and power plants where the production CO_2 is very high will help to minimise the polluting effects of CO_2. In addition, these microalgae, appropriately regarded as ***photobioreactors***, accumulate as ***biomass*** which can be utilized for the extraction of protein for use as food or feed.

Genetic manipulations are in progress to develop new strains of microalgae that can tolerate high concentrations of CO_2, and yield larger biomass. Some success has been reported in the mutants of *Anacystis nidulans* and *Oocystis* sp.

BIOLOGICAL CALCIFICATION TO REDUCE ATMOSPHERIC CO_2

Certain organisms present in the deep sea (corals, green and red algae) are capable of storing CO_2 through a process of biological calcification. The overall process of calcification may be represented as follows.

$$H_2O + CO_2 \longrightarrow H_2CO_3$$

$$H_2CO_3 + Ca^{2+} \longrightarrow CaCO_3 + CO_2 + H_2O$$

As the $CaCO_3$ gets precipitated, more and more atmospheric CO_2 can be utilized for its formation.

SEWAGE TREATMENT BY BACTERIA AND ALGAE

The ***waste water*** resulting from various human activities (domestic), agricultural and industrial is technically referred to as sewage. The sewage is mostly composed of organic and inorganic compounds, toxic substances, heavy metals and pathogenic organisms.

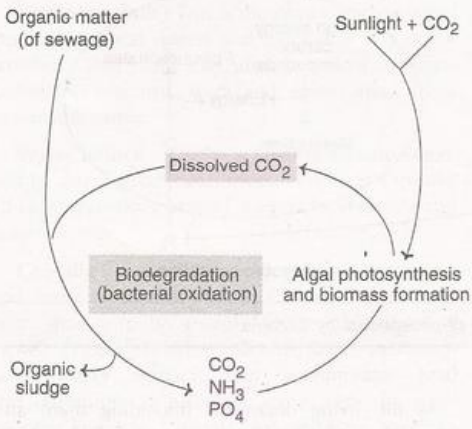


Fig. 54.2 : Sewage treatment by algae.

In the biological treatment of sewage, the organic matter is subjected to ***biodegradation*** or bacterial oxidation. In this way, the organic matter is degraded to smaller molecules (CO_2, NH_3, PO_4 etc.), Biodegradation requires a constant supply of O_2. This can be done by continuous bubbling of atmospheric O_2 through a specially designed equipment. This is an expensive process, besides the involvement of manpower.

A continuous supply of O_2 can be achieved by growing microalgae in the ponds where sewage treatment is being carried out. The algae are very efficient in photosynthesis and they release O_2 in to the ponds. This is the form of dissolved O_2 (***Fig. 54.2***). Thus, ***algal-bacterial symbiosis is responsible for photosynthetic oxygenation and the biodegradation of sewage organic matter***. This is a natural and inexpensive process.

In addition to the supply of CO_2, the algae used in sewage treatment can adsorb certain toxic heavy metals also. This is possible due to the negative charges on the algal cell surface which can take up the positively charged metals. The algal treatment of sewage is useful in several other ways.

- Support fish growth, as the algal cells are good food for fishes, besides supplying dissolved O_2.
- The algal cells, rich in protein, serve as good food and feed.
- Certain metals (that are adsorbed from the sewage) can be recovered.
- The load of pathogenic microorganisms is very much reduced (they either die or settle at the bottom of the pond).

The algal species of different genera ideally suited for sewage treatment are *Chlorella, Euglene, Chlamydomonas, Scenedesmus, Ulothrix* and *Thribonima.*

EUTROPHICATION AND PHOSPHORUS POLLUTION

Sewage, and waste waters from industries and agriculture are rich in organic and inorganic nutrients, containing phosphorus and to a lesser extent nitrogen. These nutrients favour ***excessive growth of algae*** which results in ***oxygen depletion (deoxygenation)*** and this phenomenon is referred to as ***eutrophication***. Eutrophication results in the death of non-resistant organisms (e.g. fishes), and foul smell.

Water blooms : Certain species of algae (particularly blue green algae e.g. *Lynhbia, Microcystis, Oscillatoria, Anabaena*) that can tolerate eutrophication, grow well and form water blooms in waste/sewage water. From the extensive formation of water blooms, one can predict that pond has accumulation of very high quantities of phosphorus. The algae are capable of absorbing phosphorus, and store them as polyphosphates. Phosphorus is a strong promoter of algal growth and water bloom formation. Water blooms are associated with the production of certain toxins (e.g. lipopolysaccharides) that are harmful to fishes and birds. These toxins may also cause certain diseases in humans e.g. diarrhea, gastroenteritis, nausea.

Control of eutrophication : There are broadly two ways of controlling algal growth resulting in eutrophication. In the ***chemical approach***, algicides such are copper sulfate, sodium arsenate and 2, 3-dichloronaphthoquinone are used. But the chemical treatment is associated with increase in sludge volume. In the ***biological approach, cyanophages*** (i.e. the viruses that can kill algal cells) can be used to contain eutrophication.

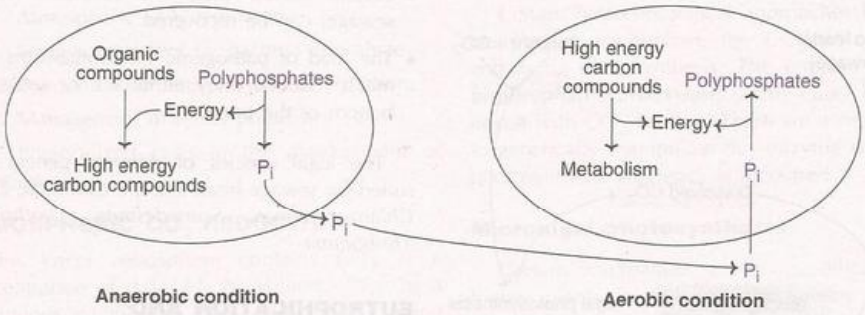


Fig. 54.3 : Biological removal of phosphorus by bacteria.

BIOLOGICAL REMOVAL OF PHOSPHORUS

As stated above, phosphorus significantly contributes to algal growth in waste water. Removal of phosphorus therefore can protect from eutrophication. It is possible to precipitate phosphorus with salts of calcium, magnesium and aluminium. This however, is not practicable on large scale. In addition, ***chemical methods*** add to the sludge volume. Hence, biological methods are preferred.

Biological removal of phosphorus by bacteria is carried out through aerobic and anaerobic processes, as depicted in ***Fig. 54.3***. Under anaerobic conditions, certain organic compounds (e.g. acetate) obtain their energy from polyphosphates to store as carbon reserves. In this process, free inorganic phosphorus is released. In the subsequent aerobic process, the organic carbon reserve compounds are oxidized to generate energy. A small portion of this energy is utilized for the conversion of phosphates to polyphosphates. The net effect is that phosphorus accumulates in the bacterial cell in the form of polyphosphates.

MANAGEMENT OF METAL POLLUTION

Environmental pollution with heavy metals (e.g., lead, cadmium, mercury) causes several toxic manifestations in living organisms, including cancer. Bioaccumulation, biomagnification and biomethylation are some of the characteristic features of metal pollution.

As the living organisms (including man) are constantly exposed to metals, they accumulate by a process referred to as ***bioaccumulation***. Continuous exposure and accumulation of a given metal in the organisms results in its increased concentration, a phenomenon referred to as ***biomagnification***. Biomagnification usually occurs ***through food chain*** and man is the ultimate victim.

Biomethylation is the process of transfer of methyl groups from organic compounds to metals. This is carried out by microorganisms in the soil and water. Although it is true that some metals get detoxified by methylation (e.g. arsenic), some of them may even become more toxic (mercury).

BIOSCAVENGERS OF METALS

Certain aquatic plants namely ***phytoplanktons*** (plants that freely float on water surface) or ***benthics*** (plants attached to some substratum at the bottom of aquatic reservoirs) and microorganisms can take up the metals from the bathing media (waste water, ponds) and act as ***natural bioscavengers***. Besides combating the pollution effects of metals, this process is also useful for the recovery of industrially important metals from the bioscavengers.

In the recent years, environmental protection agencies (EPAs) worldover are advocating the use of aquatic plants and microorganisms as low cost bioabsorbants for the removal of toxic metals from the environmental pollution.

Some of the plants and microorganisms useful for the management of metal pollution are briefly described.

Plants in metal pollution management

Water hyacinth : This is the most common weed found in tropical waters and its control is a major problem. This plant can absorb and accumulate cadmium, mercury, lead and even silver to a significant extent.

Water lettuce : This is a floating aquatic weed, and is also a good metal bioscavenger. It can take up large quantities of lead, cadmium, mercury and arsenic.

Cat-tails : These plants possess long erect leaves and grow in marshy areas. Cat-tail leaves have been shown to be useful for the absorption of nickel, copper, besides calcium and magnesium salts. Water spinach that accumulates lead and chromium is eaten in some areas by poor people. The toxic metals are certainly toxic to humans.

Ferns : The fern namely *Azolla filicoides* accumulates cadmium, copper and uranium. *Salvinia* sp can absorb several metals including lead, nickel and chromium.

Microorganisms in metal pollution management

Algae : Several metals from the fresh water can be absorbed by ***algal blooms***. For instance, *Chlorella vulgaris* can take up copper, mercury and uranium. The marine microalgae (e.g., *Ulva, Laminaria*) are also useful for the abatement of metal pollutants.

Fungi : Certain fungal species are good absorbers of heavy metals (Pb, Hg). e.g. *Rhizopus, Aspergillus, Penicillum, Neurospora.*

Bacteria : A few bacterial species can accumulate metals on their cell walls. For instance, *E. coli* can take up mercury while *Bacillus circulans* can accumulate copper.

Attempts are being made to improve the strains of bacteria (through genetic manipulations) for more efficient uptake and accumulation of toxic metals. Further, such species will also be useful for the recovery of metals from the bacterial biomass.

MECHANISMS OF METAL SCAVENGING

The aquatic plants and microorganisms may take up the metal pollutants through absorption and/or adsorption. The mechanism of accumulation of metals by the bioscavengers is very complex and may involve one or more of the following processes.

Binding of metals to cell walls

Bioaccumulation of metals on the cell walls is commonly observed in lower plants-algae, fungi and bacteria. Even some higher plants can concentrate metals on the cell walls. There is some evidence to show that metals are held to certain organic biopolymers on the cell walls, till the binding sites get saturated.

Accumulation of metals in cellular compartments

Plants are capable of transporting metals to intracellular and intercellular free spaces, and the cellular organelles (lysosomes, vacuoles). Certain metals occur as immobilized metal containing crystals. For instance, heavy metal complexes of calcium oxalate crystals have been detected in algae.

Synthesis of metal binding proteins

Several algae and certain fungal species are known to synthesize metal binding proteins or peptides. ***Phytochelatin*** is an ubiquitious protein present in all the plants. It is a metal chelating protein and is regared as a common buffering molecule for the homeostasis of metals. Phytochelatin is rich in cysteine and can form salt metal complexes through sulfhydryl (SH) groups. It is comparable in its functions to ***metallothionein*** found in animals.

Certain metals are known to induce the synthesis of phytochelatin. Some workers, in recent years, have suggested ***phytochelatin*** can be used as a ***biomarker*** for metal pollution detection.

IMMOBILIZED CELLS IN THE MANAGEMENT OF POLLUTION

The use of immobilized cells, paticularly microbial whole cells, for the abatement of environmental pollution is a recent development. The general aspects of cell immobilization, techniques involved, and their applications are described elsewhere (Chapter 21).

The immobilized cells are useful for the waste water treatment, and for the recovery of metals

TABLE 54.2 A selected list of pollutants and the corresponding immobilized microorganisms used for their abatement

Pollutant	*Immobilized microorganism*
Phenol	*Pseudomonas putida*
	Cryptococcus elinovii
Triethyl lead	*Arthrobacter* sp
4-Chlorophenol	*E. coli*
	Enterobacter sp
	Alcaligenes sp
Ammonia	*Thiosphaera* sp
Copper	*Rhizopus arrhizus*

from industrial effluents. A selected list of the pollutants and the immobilized microorganisms used for their abatement is given in ***Table 54.2***. For instance, phenol and triethyl lead can be removed respectively by immobilized *Pseudomonas putida* and *Arthrobacter* sp.

For more efficient management of a group of environmental pollutants simultaneously, immobilized systems with more than one type of microorganisms are used. In the recent past, ***genetically engineered microorganisms*** (***GEMs***) are being developed for the treatment of polluted water and soils. Some details on this aspect are described in Chapter 59.

55 Air Pollution and Control

The earth's atmosphere contains various gases, water vapor and suspended particles. The ***dry air*** of the atmosphere is composed of four major gases ***nitrogen***, ***oxygen***, ***argon*** and ***carbon dioxide***, that account for more than 99.5% (***Table 55.1***). The other gases found in traces in the air include helium, methane, krypton, hydrogen, carbon monoxide, nitrous oxide (N_2O), nitrogen dioxide (NO_2), ammonia, ozone, sulfur dioxide and hydrogen sulfide. The lower part of the atmosphere (up to about 12 km) also contains water vapor at a concentration, ranging from 0.01 to 5.0%. This water is mostly contributed by evaporation from the hydrosphere.

Table 55.1 Composition of the dry air in the lower atmosphere

Gas	*Composition (% by volume)*
Nitrogen	78.08
Oxygen	20.95
Argon	0.93
Carbon dioxide	0.03
Trace gases (He, CH_4, Kr, H_2, CO, N_2O, NO_2, NH_3, O_3, SO_2, H_2S)	> 0.02

The air is getting contaminated by pollution due the natural and unnatural activities of man. ***Air pollution*** is basically the ***presence of foreign substances in the air*** at a concentration that will ***adversely affect the health and property*** of the individual.

As per the World Health Organization (WHO) criteria, air pollution refers to 'the substances put into the air by the activity of mankind into concentration sufficient to cause harmful effects to his health, vegetables, property or to interfere with the enjoyment of his property.'

According to Indian Standards Institution 'air pollution is the presence in ambient atmosphere of substances generally resulting from the activity of man, in sufficient concentration, present for sufficient time and under circumstances which interfere significantly with the comfort, health or welfare of a person or with the full use or enjoyment of his property.'

The term ***pollutant*** refers to a substance which increases in quantity in the air and adversely affects the environment e.g. carbon monoxide, sulfur dioxide, lead. On the other hand, a ***contaminant*** is a substance which is ***not present in nature***, but released due to human activity e.g. methyl isocyanate. DDT, malathion. However, this distinction is not very rigid, and most authors use the term pollutant to represent both (pollutant as well as contaminant).

SOURCES OF AIR POLLUTANTS

The sources that contribute to air pollution may be broadly categorized into two types.

TABLE 55.2 Major industrial sources of air pollution

Industry	*Major air pollutants*
Thermal power plants	NO_2, N_2O, SO_2, particulates
Steel industries	Smoke, particulates, CO, fluoride
Petroleum refineries	SO_2, smoke, particulates
Metal smelters	SO_2, NO_2, N_2O, smoke, particulates
Fertilizer plants	SO_2, NO_2, N_2O, NH_3, fluoride
Acid plants	SO_2, NO_2, N_2O
Cement plants	SO_2, smoke, particulates
Soap and detergent plants	Particulates, odour
Paper mills	SO_2, particulates, odour

1. **Natural sources :** These include volcanic eruptions, sand storms, decomposition of organic matter, forest fire, pollen grains and cormic dust. The problem of pollution due to natural sources in general, is considered to be minimal.

2. **Anthropogenic (man-made) sources :** Air pollution due to human-induced activities is very high. The sources of pollution include industries, burning of fossil fuels, emissions from vehicles, agricultural activities and warfares.

The sources of air pollution may also considered as ***stationary*** (industries, open combustion) or ***mobile*** (motor vehicles, trains, aircraft) in nature.

Industrial pollutants

The major problem of air pollution is due to industrial activities. Among the several industries, nine types of industries are considered to be the major air pollutant generating industries (***Table 55.2***). Among these, thermal power plants, steel industries, petroleum refineries and metal smelters are the most dangerous polluting industries.

CLASSIFICATION OF AIR POLLUTANTS

Air pollutants are classified based on their origin, chemical composition and the state of the matter.

Classification based on origin

Air pollutants are divided into two categories, based on their origin-primary and secondary.

Primary air pollutants : These pollutants are ***directly emitted into the atmosphere*** and present there as such (i.e. in the form they are originally emitted). Primary air pollutants contribute to as much as ***90% global air pollution***. Particulates, carbon monoxide, (CO), oxides of sulfur (SO_x), oxides of nitrogen (NO_x), hydrocarbons (HCs), radioactive compounds, pollen and bacteria are the major primary air pollutants.

Secondary air pollutants : These are ***produced*** in the air ***as a result of interaction among the primary pollutants***, or by a reaction that occurs between a primary pollutant and a normal constituent of the atmosphere. Good examples of secondary air pollutants are ozone (O_3), peroxyacetyl nitrate (PAN), formaldehyde and smog.

Classification based on chemical composition

According to chemical composition, air pollutants are categorized as organic and inorganic.

Organic air pollutants : These pollutants are mainly composed of carbon and hydrogen. In addition, oxygen, nitrogen, sulfur and phosphorus may also be present e.g. hydrocarbons, organic sulfur compounds, aldehydes, ketones, carboxylic acids.

Inorganic air pollutants : These are purely inorganic in nature e.g. carbon dioxide, carbon monoxide, oxides of sulfur, oxides of nitrogen, ozone.

Classification based on the state of matter

Air pollutants may be divided into two types, based on the state in which they exist-particulate and gaseous.

Particulate air pollutants : The solids and liquids dispersed in the atmosphere constitute the particulate air pollutants. ***Solid particulates*** e.g. smoke, dust, fly ash. ***Liquid particulates*** e.g. mist, fog, spray.

Gaseous air pollutants : These are the organic and inorganic gases that are present in the air as pollutants. ***Organic gases*** e.g. methane, butane, aldehydes. ***Inorganic gases*** e.g. CO_2, SO_2, NO_2, NH_3, H_2S.

EFFECTS OF AIR POLLUTION

Majority of the air pollutants are present normally in the atmosphere, although at low concentrations. Such pollutants are unlikely to cause any significant harmful effects. When the concentrations of the air pollutants go beyond the acceptable limits (variable for each pollutant) they are dangerous and cause several harmful effects. The WHO has set guidelines and fixed ***threshold limit values*** (***TLV***) for each air pollutant. The general effects of air pollutants on humans, plants, animals, materials and global climate are briefly described.

Effects of air pollutants on humans

On an average, man breathes around 22,000 times and inhales about 16 kg of air each day. Primarily, the air we breath is life sustaining. However, pollutants present in the air ***often cause harmful effects***. The nature of the pollutant, its concentration and duration of exposure are among the factors that affect the health of humans. In general, ***infants, elderly people and those with respiratory diseases are more susceptible to air pollution***. Further, adverse health effects of air pollutants are maximum during winter compared to other seasons.

Some of the important air pollutants and their effects on human health are given in ***Table 55.3***. ***Sulfur dioxide is the most dangerous pollutant gas to man***. It is produced in many industries and causes respiratory disorders. Oxides of nitrogen and carbon monoxide reduce O_2 supply to the tissues. Lead pollution is associated with tissue damage and abnormal behavioural pattern. Suspended particulate matter (SPM) mainly causes chronic pulmonary diseases. Prolonged exposure to radioactive isotopes results in anemia, besides a heavy risk of cancer and genetic defects.

Effects of air pollutants on plants

At low concentration, the air pollutants may not cause any visible damage to the plants. However, even at this level, the pollutants may be stored and introduced into the food chain. This in turn, may affect animals as well as humans.

TABLE 55.3 Health effects of some of the air pollutants

Pollutant	*Effects*
Sulfur dioxide (SO_2)	Nose and throat irritation, respiratory illness. Prolonged exposure may lead to chronic bronchitis.
Nitrogen dioxide (NO_2) and nitric oxide (NO)	Irritation of eyes, nose and throat. NO combines with O_2 and reduces supply of O_2 to tissues.
Carbon monoxide (CO)	Binds to hemoglobin and drastically reduces O_2 supply to tissues. May result in cardiovascular and pulmonary diseases.
Heavy metals (Pb)	Damage to liver, kidney and brain; causes anemia, neurobehavioural changes, abnormalities in fertility and pregnancy.
Aliphatic hydrocarbons and polycyclic organic compounds	Cancer
Suspended particulate matter (SPM)	Chronic pulmonary diseases–bronchitis, asthma.
Radioactive isotopes (I^{131}, P^{32}, Co^{60})	Anemia, cancer and genetic defects.

The entry of the gas air pollutants (SO_2, NO_2) predominantly occurs through the stomata, openings on the leaves. Stomata are located at the bottom of the leaves through which CO_2 enters for photosynthesis. In the same way as CO_2 enters the leaves, the gaseous pollutants also enter and cause various effects.

The solid particles are adsorbed on the surfaces of leaves. This may result in clogging of the stomata and a reduced intake of CO_2. Further, the suspended particles deposited on the leaves, may adversely affect the leaf functions (reduced exposure to sun light, decrease in chlorophyll content, interruption of gaseous exchange).

The effects of common air pollutants on leaves are given in ***Table 55.4***. The sensitivity of plants to pollutants depends on many factors such as climatic conditions (light, temperature, humidity),

Table 55.4 Effects of air pollutants on plants

Pollutant	*Effects*
Sulfur dioxide	Bleaching of leaves and necrosis (killing of leaves)
Nitrogen dioxide	Bleaching and suppressed growth
Ozone	Bleaching and necrosis
Fluorides	Marginal necrosis
Ammonia	Leaves become dull green
Ethylene and propylene	Leaf curling and leaf dropping
Peroxyacetylnitrate (PAN)	Suppressed growth and silvering of lower leaf surfaces
Particulates (with toxic metals)	Bleaching and necrosis

soil and water, and the individual plant's response to a particular pollutant. The major ***air pollulants affecting plants are SO_2, NO_2, O_3, fluorides, ammonia and ethylene***, besides the ***particulates***. They may damage the plants to varying degrees as outlined in ***Table 55.4***.

The sensitivity of certain selected plants to air pollutants can be used for biomonitoring of air pollution (details given later).

Effects of air pollutants on animals

The effect of air pollution on animals is mostly indirect, as it occurs after they eat polluted plants or foliage. Fluorine, lead and arsenic are three main air pollutants that cause harmful effects to livestock.

Fluorine : Fluorine causes loss of weight, muscular weakness, diarrhea, wearing of teeth, and even death. Fluorosis mainly affects the ruminants, particularly dairy cows.

Lead : Lead poisoning is commonly observed in animals grazing near lead mines. It is associated with loss of appetite, difficulty in breathing, diarrhea and paralysis.

Arsenic : Arsenic toxicity in animals is associated with increased salivation, thirst, irregular pulses, abnormal body temperature and paralysis. Chronic poisoning may result in anemia, diarrhea, paralysis and even death.

Effects of air pollutants on materials

Air pollutants cause immense damage to various materials — stone, metal, paint work, fibre materials, glass, ceramic, textiles, rubber, architecture etc. The adverse effects of air pollution on materials and properties are associated with ***severe economic losses throughout the world***.

Effects of air pollutants on global climatic changes

Air pollution is associated with significant changes in global climate and the related processes e.g., ozone depletion, green house effects, acid rain (Refer Chapter 60).

BIOMONITORING OF AIR POLLUTION

Plants are used to ***monitor*** (biomonitoring) air pollution and such plants are referred to as ***indicator plants***. This is based on the principle of sensitivity and response of the plants to air pollutants. A list of the plants used for biomonitoring and the corresponding air pollutants is given in ***Table 55.5***. Among these, lichens and mosses are most commonly used to check the quality of air. The pattern of occurrence of patches on lichens serves as an index for biomonitoring of air pollution.

Bioluminescence in air pollution monitoring

Bioluminescence is the phenomenon of emission of visible light by an organism. It occurs as an enzymatically catalysed light emitting

Table 55.5 Air pollutants and the plants for their biomonitoring

Air pollutant	*Plant(s)*
Sulfur dioxide	Lichens, moss, white pine
Ozone	Tobacco, garden bean
Peroxyacetylnitrate (PAN)	Lettuce, bean
Ethylene	Orchids, marigold, cucumber
Fluoride	Apricot, peach
Heavy metals	Moss, lichens

reaction in living cells. Bioluminescence is an indicator for the analysis of atmospheric gases such as SO_2, formaldehyde and ethyl acetate. The microorganism, *Photobacterium phosphoreum* can be used as a special photodetector. The changes in the emission of light due to a pollutant can be detected by a sensor, amplified and recorded.

AIR POLLUTION CONTROL

The air pollution load particularly from industries, can be reduced by several measures-replacement of burning fuel by electricity or solar energy, improvement in fuel burning process, dispersion and dilution of pollutants, and reduction at source by using control equipment. These measures, however, are useful only to a limited extent.

In the nature itself, there are certain devices, commonly referred to as ***atmospheric self-cleansing processes*** for the removal of air pollutants. These natural processes are very slow and limited, and cannot cope up with the present increased demands of pollution control. However, the artificially devised pollution control measures are based on the same principles of atmosphere self-cleansing processes. These principles include ***dispersion, gravitational settling, flocculation, absorption*** and ***rain out***.

There are ***two categories*** of ***air control devices***—devices to control ***particulate pollutants***, and to control ***gaseous pollutants***.

CONTROL DEVICES FOR PARTICULATE POLLUTANTS

In general, greater emphesis is given to control particulate pollutants, may be because they are visible. It may however, be noted that in terms of volume of air pollution, gaseous pollutant contribution is much higher.

The important devices/equipment used to control particulate pollutants are listed.

- Gravity settling chambers
- Cyclone collectors
- Dynamic precipitators
- Spray towers
- Electrostatic precipitators
- Fabric filters.

Gravity settling chambers

Settling chambers are the oldest and very simple type of equipment used for collection of solid particles (***Fig. 55.1A***). As the air is passed through the chambers at a low velocity, the ***dust particles*** (size 40–100 µm diameter) ***settle by gravity*** and the clean air comes out. Although this technique is simple, inexpensive with low cost of maintenance, it is not possible to settle smaller size dust particles.

Cyclone collectors

Cyclone collectors also operate on the ***principle of gravity settlement***. The equipment mainly consist of a vertically placed cylinder with an inverted cone attached to its base (***Fig. 55.1B***). As the air enters the cylinder, it takes a helical path downwards. Due to rapid spiralling movement of the air, the particles are thrown towards the walls by centrifugal force. These dust particles settle down in the hopper at the bottom, while the dust free air passes through a pipe and comes out.

The cyclone collectors are simple, low cost and easy for maintenance. They are ideal for separation of dust particles with sizes 15–50 µm. Cyclone collectors cannot settle particles less than 10 µm size.

Dynamic precipitators

Dynamic precipitators work on the ***principle of centrifugal force, generated by rotating blades*** (***Fig. 55.1C***). The dust particles of the air are concentrated on the rotating blades from where they are collected in a concentrated stream. By employing this equipment, dust particles of 5–20 µm size can be separated. However, dynamic precipitators are not suitable for sticky or fibrous dust particles, as they stick to the blades.

Spray towers

In spray towers, the ***dust particles*** (10 µm in size) are made to ***settle down by spraying water*** (***Fig. 55.1D***). This technique is particularly useful for separating a heavy load of particles.

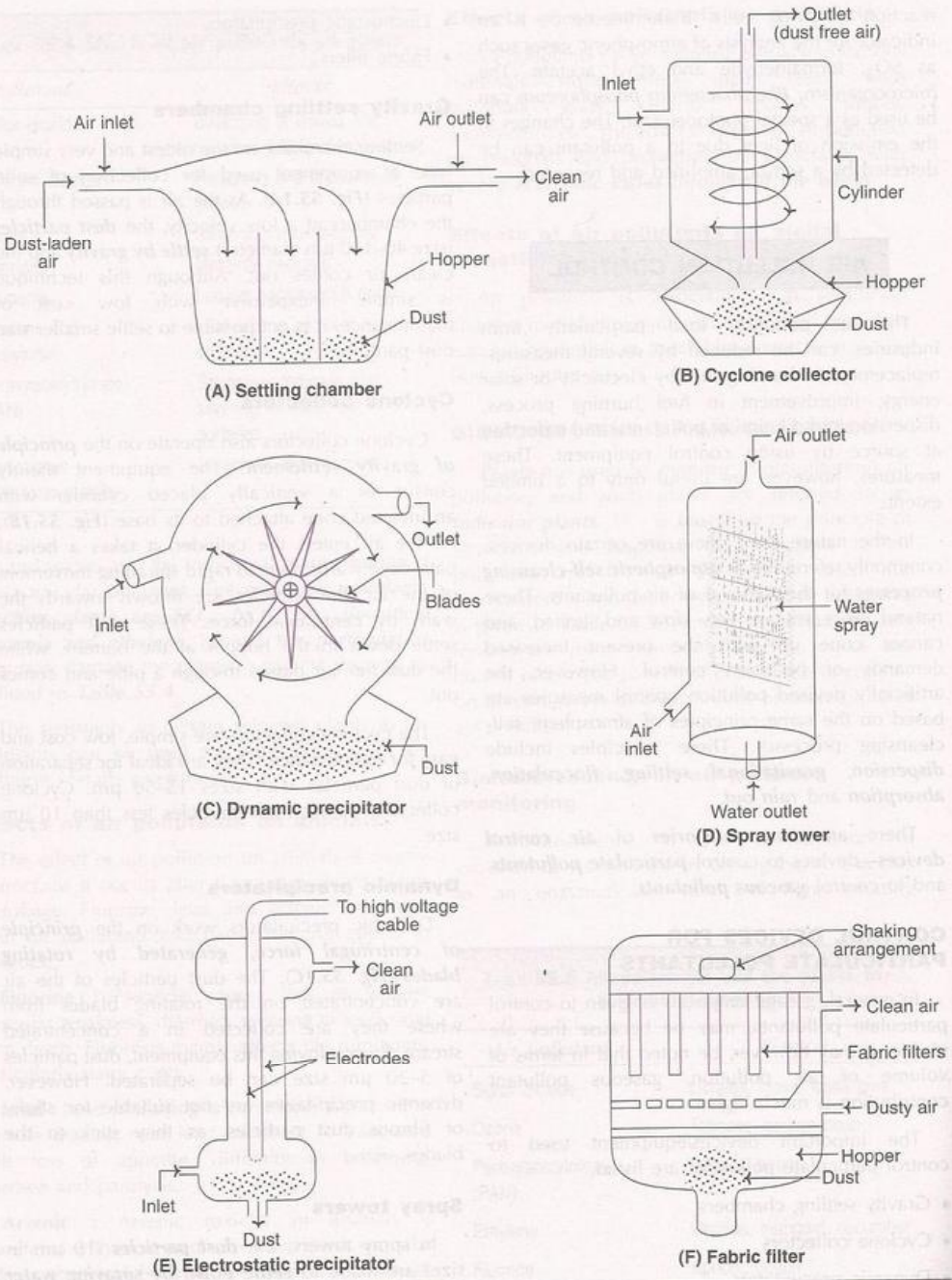


Fig. 55.1 : Control devices for particulate pollutants.

Electrostatic precipitators

Electrostatic precipitators (***ESPs***) are very efficient and versatile. They work on the ***basis of electrostatic attraction***. An ESP consists of a thick cylinder fitted with an inlet at bottom and an outlet at the top (***Fig. 55.1E***). An electrode (more than one electrode are commonly used) fitted at the centre of the cylinder is connected to a high voltage cable. As the dust laden air passes through ESP, larger sized particles settle down due to gravity. The smaller charged particles are attached to the oppositively charged electrodes which gradually fall down to the bottom. The dust free air comes out.

Electrostatic precipitators are said to be ***dry precipitators*** if the dust particles can be removed by scrapping. In case of ***wet electrostatic precipitators***, the particles are removed by using water or any other fluid. Wet precipitators are more efficient, however, dry precipitators are preferably used for practical reasons.

Fabric filters

Fabric filters (***Fig. 55.1F***) are the most efficient and can separate particles with sizes less than 0.5 μm in diameter. As the air or gas is passed over fabric like mats of wool, celluloss etc., ***the dust is trapped while the gas passes out***. Collection of dust on the fabric filter results in the formation of a dust layer (commonly referred to as filter cake). This in turn serves as a more efficient dust collector and helps to capture even fine dust particles. The fabric filters have to be frequently cleaned. Several models of fabric filters have been developed for commercial applications.

CONTROL DEVICES FOR GASEOUS POLLUTANTS

The principle gas pollutants are SO_2, NO_2, N_2O, CO, hydrocarbons, and organic and inorganic acid gases. The control devices used for gaseous pollutants are based on the ***principles of adsorption, absorption, condensation and combustion***. The important aspects of selected devices for controlling gaseous pollutants are briefly described.

Adsorption

When a stream of gas is passed through a porous solid material (absorbent), it can attract and hold the gas (adsorbate) molecules.

TABLE 55.6 A selected list of adsorbents and their major applications

Adsorbent	*Major application(s)*
Silicate or zeolates (as molecular sieves)	Control of SO_2, NO_2 and mercury
Activated alumina	Drying air, gases and liquids
Activated carbon	Removing odours, purification of gases
Silica gel	Drying and purifying gases
Bauxite	Drying of gases and liquids, treatment of petroleum fractions
Fuller's earth	Refining of oils and fats

If the adsorption is purely a physical phenomenon (held by van der Waal's forces), it is referred to as ***physical adsorption***. This process is rapid and easily ***reversible***. In ***chemical adsorption*** (***chemisorption***), the pollutant gas molecules form chemical bonds with the adsorbent. Chemisorption is very slow, usually requires the supply of energy and is irreversible. Pressure and temperature significantly influence chemical adsorption.

In the ***Table 55.6***, a selected list of the commonly used adsorbents along with their major applications is given. The equipment used for adsorption, referred to as ***adsorbers***, may be fixed, moving or fluidized beds.

Multiple fixed bed adsorber : In this, adsorbent is arranged in the form of beds or layers as depicted in ***Fig. 55.2A***. As the adsorption occurs, the adsorbent gets saturated with the adsorbate. For reuse, the gas collected on the adsorbent must be removed. If this is not possible, the adsorbent must be replaced.

Absorption

The pollutant gas (absorbate), when brought in contact with a solvent or liquid (absorbent), gets adsorbed. The process of absorption may occur due to physical or chemical phenomena.

Absorption process is used to control the pollutants SO_2, NO_2, H_2S and hydrocarbons. Ammonia is used as an absorbent for controlling SO_2.

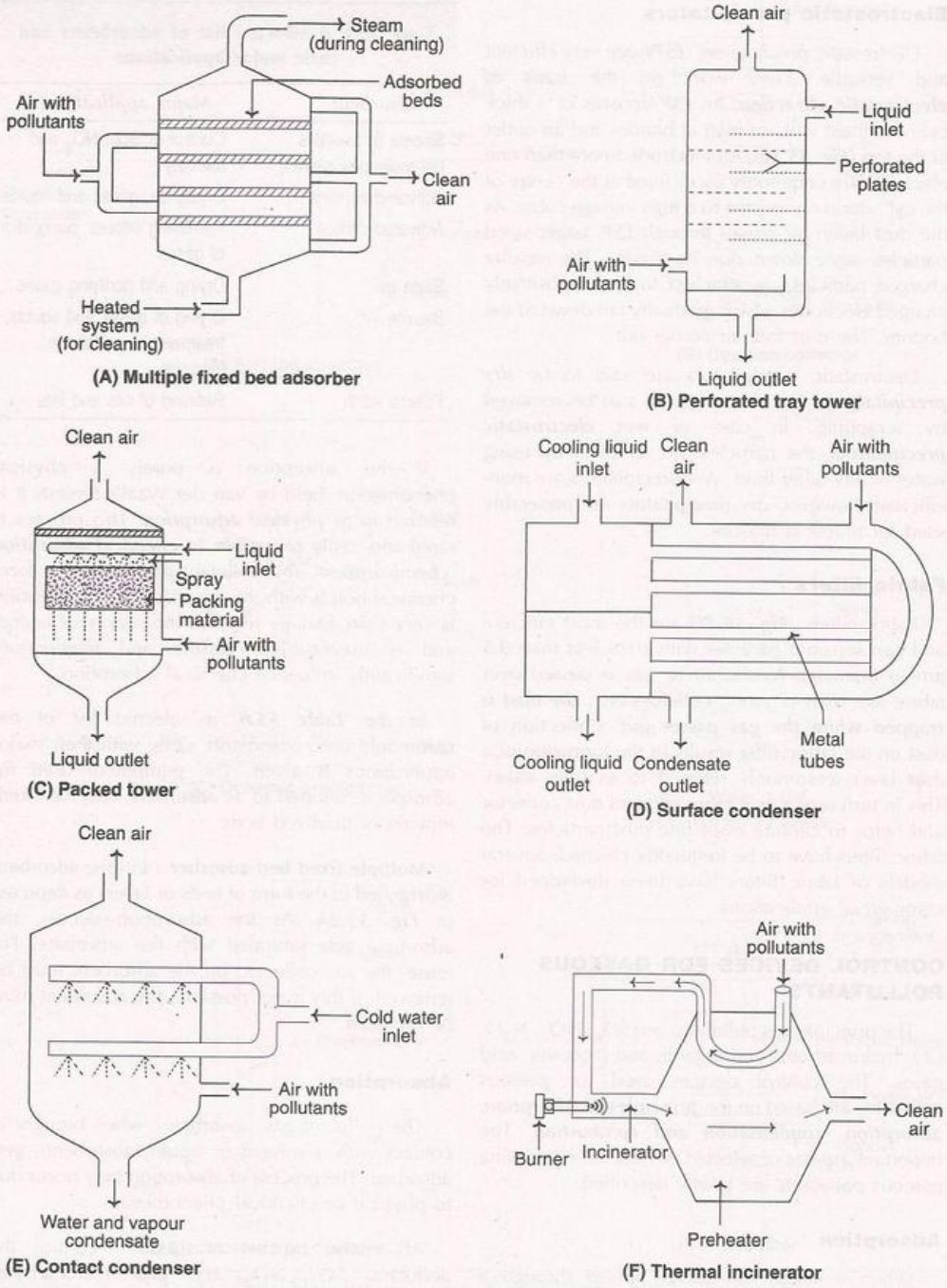


Fig. 55.2 : Control devices for gaseous pollutants.

Different types of ***absorbers*** are used for control of gaseous pollutants. These include spray towers, plate or tray towers and packed towers. The gas absorption units are so designed that there occurs an intimate contact between the gas and the liquid. This will ensure optimal absorption of gas into the liquid.

Spray towers (Refer ***Fig. 55.1D***) **:** They can efficiently remove gaseous pollutants, besides the particulates. These devices are particularly useful for the removal of pollutant gases with high concentrations of particulates.

Plate or tray towers : These absorbers are designed with horizontal trays or plates so as to provide large liquid-gas interfacial surface (***Fig. 55.2B***). In the perforated tray tower, the absorbent enters the column from the top, spills and flows in a zig zag fashion. The polluted air enters the column from an inlet at the side of the bottom. As the air passes through the openings of the trays, it gets in close contact with liquid due to repeated exposure. This enables the removal of gaseous as well as particulate pollutants. The clean gas emerges at the top.

Packed towers : In this case, the liquid (absorbent) is sprayed over a packing material (e.g. all rings, Berl saddles) with large surface to volume ratio (***Fig. 55.2C***). The air with gaseous pollutants enters from the side of the bottom of the tower and the clean air comes out from the top.

Condensation

When the temperature of a gaseous compound is reduced, its condensation occurs. There are two devices commonly used to control gaseous pollutants, based on the principle of condensation.

Surface condenser : In this device, the cooling medium is passed through metal tubes over which the gas with pollutants is passed (***Fig. 55.2D***). The vapour condenses on the surface of the tubes. This gets collected as a film of liquid which can be drained off.

Contact condenser : In contact condenser, the cooling medium and the gaseous pollutants are made to have a direct contact with each other (***Fig. 55.2E***). As the vapors get cooled, they condense and are easily removed along with the water, and the clean gas comes out from the top of the condenser.

Combustion

It is a fact that combustion or incineration is a major source of air pollution. And surprisingly, the same combustion processes effectively serve to control air pollutants. This is carried out by converting certain pollutants (e.g., hydrocarbons, carbon monoxide) to carbon dioxide and water. For efficient oxidation during combustion, supply of oxygen, optimal temperature and time are necessary. There are different types of combustion processes depending on the pollutant to be oxidized. The most important ones are ***direct-flame combustion***, ***thermal combustion*** and ***catalytic combustion***.

Direct-flame combustion : In this device, the waste gases are directly burnt with or without the addition of auxillary fuels. Direct-flame combustions are most often used in petrochemical plants and refineries.

Thermal combustion : This process usually becomes necessary when the gaseous pollutant concentration is too low, and it is difficult to carry out direct-flame combustion. Thermal combustion is carried out by a ***thermal incinerator*** or ***after burner*** (***Fig. 55.2F***). An heat exchanger preheats the air/gas with gaseous pollutants. This preheated gas is passed over to the incineration equipped with a burner. The temperature of the incinerator may be in the range 500–1000°C, and this actually depends on the nature of the pollutant to be oxidized. Combustion with thermal incinerators has to be carefully carried out so that the oxidation process is efficient and complete. Incomplete combustion may produce more pollutants e.g. carbon monoxide.

Catalytic combustion : This is required when the ***gaseous pollutants are too low in concentration***. By employing a catalyst, the combustion can be made faster and complete, and the catalyst can be used again and again.

Catalytic combustion is used for the removal of NO_2 (in the tail gas) from the nitric acid plants. Here platinum catalyst is used. Carbon monoxide can be removed by using copper (Cu^{2+}) catalyst.

The major limitation of catalytic combution is the high cost of the catalyst and maintenance of the catalytic combustion.

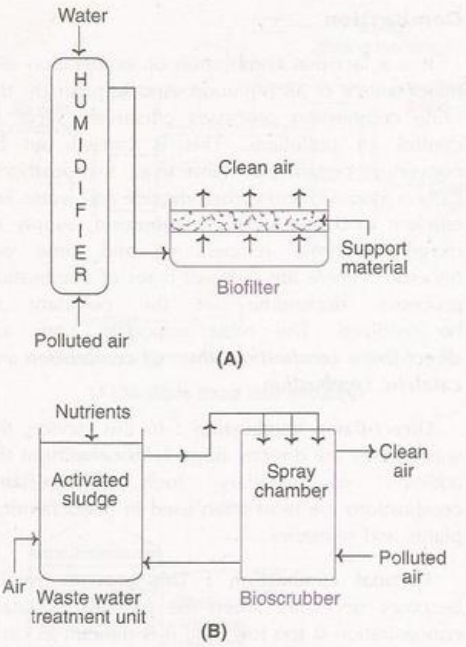


Fig. 55.3 : Treatment processes for the removal of volatile organic compounds (A) Biofilter (B) Bioscrubber.

CONTROL DEVICES FOR VOLATILE ORGANIC COMPOUNDS (VOCs)

There are several volatile organic compounds, arising from industrial and domestic activities, that cause air pollution. Good examples of VOCs are alcohols, ketones, organic acids, phenols and organic solvents. The conventional methods (e.g. combustion, adsorption) of treatment VOCs require large quantities of energy, besides creating secondary pollution.

Biodegradation by microorganisms is a preferred method for controlling air pollution of VOCs. The efficiency of biodegradation depends on the individual compounds and based on this, VOCs are broadly categorized as follows :

- ***Rapidly degraded VOCs*** — alcohols, ketones, aldehydes, organic acids.
- ***Slowly degraded VOCs*** — hydrocarbons, phenols, organic solvents (chloroethene).
- ***Very slowly degraded VOCs*** — polyaromatic hydrocarbons, polyhalogenated hydrocarbons.

There are mainly two types of reactor designs for the biodegradation of VOCs — biofilters and bioscrubbers (***Fig. 55.3***).

Biofilters

A biofilter is composed of a porous medium (of compost, wood chips) that supports the microorganisms forming a film. The polluted air is first passed through a humidifier and then through the biofilter. A great majority of VOCs in the air (approximately 90%) get biodegraded, and clean air comes out of the biofilter (***Fig. 55.3A***).

Advantages of biofilters : They are simple and cost-effective. Certain poorly ***soluble pollutants*** (hydrocarbons) can ***be easily removed*** by this approach. Biofilters can be successfully used for the biodegradation of several xenobiotics e.g. chloromethane.

Disadvantages of biofilters : It is rather difficult to control process conditions (pH, temperature). When compost is used as support material, it generates unpleasant odours. Biofilters require large areas.

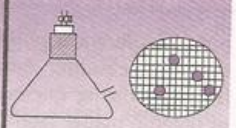
Bioscrubbers

Bioscrubber is an improvement over biofilter (***Fig. 55.3B***). When the polluted air is passed through a liquid stream in a spray chamber, the pollutant (VOC) gets transferred to the liquid. The sprayed liquid contains a suspension of microorganisms which cycles between the spray chamber and waste water treatment unit (activated sludge unit). The microbial biodegradation of VOC occurs in the waste water treatment unit. The clean air comes out, after recycle, from the spray chamber. Besides a continuous nutrient supply, it is essential to maintain the pH in the activated sludge unit.

Biotrickle filters

Biotrickle filter is a recent development for more efficient biodegradation of VOCs. It ***works on the principles of adsorption, followed by biodegradation***. The pollutant gases are adsorbed on a solid surface (e.g. activated carbon) and then subjected to biodegradation by a stream of medium containing the desired microorganisms. As of now, biotrickle filters are at an experimental stage, and soon they may be available for use.

56 Water Pollution and Sewage



Water is absolutely ***essential for the existence of life*** (animal or plant). It provides a wonderful chemical medium in which all biological and biochemical processes occur. Water dissolves various nutrients, distributes them to cells and removes waste products.

About 60% of the human body is composed of water. One can survive without food for some weeks, but cannot survive without water for more than a few days. Besides drinking, man requires water for bathing, washing, livestock raising, industrial, agricultural and various other purposes.

Water-a scarce natural resource

Water, covering more than 70% of the earth's surface, exists in all three states-liquid, solid and gas. Most of the water on earth is salt water, and is unsuitable for drinking and other needs of man. It is estimated that ***only around 0.007% of the earth's water is available for direct human consumption***. Thus, ***consumable water is a scarce natural resource***.

Water is required for development and progress of people, and therefore a nation. In fact historically, the places of availability of water were the major determinants for the settlement of people. This is because besides regular use, water is essential to raise the crops. It is an accepted fact that a community with good water supply has good growth, progress and prosperity.

For the reasons stated above, it is well recognized worldover that there is an ***urgent need for water conservation and management***. The Governmental policies also support this concept.

DISSOLVED OXYGEN

Oxygen is soluble in water, and this dissolved oxygen (DO) is ***essential to support fish and other aquatic life in water***. The solubility of atmospheric oxygen in fresh water is around 7–15 mg/dl, depending on the temperature and pressure. Increase in temperature and salts in water decreases O_2 solubility.

Oxygen gets dissolved in water by one or more of the following ways.

- Direct entry from the atmosphere.
- By aquatic plant (algae) photosynthesis.
- Introduction by artificial means (aerators).

Importance of dissolved oxygen

A minimum level of dissolved oxygen at a concentration around 4 mg/dl is considered to be necessary to support healthy aquatic life. If good aerobic conditions are not maintained, anaerobic microorganisms take over and cause harmful effects. And this is what happens in polluted waters, particularly from the industrial and municipal wastes.

Determination of dissolved oxygen is very important in water pollution. This is mostly carried out by measuring ***biochemical oxygen demand*** (***BOD***), utilizing microorganisms for the oxidation of organic matter (See p. 683).

WATER POLLUTION

Several kinds of natural, and man made activities (domestic, industrial, agricultural etc.) contribute to water pollution. Water pollution is characterized by certain ***observable disturbances in the normal properties and functions of fresh water***. These include offensive odours, decrease in aquatic life (e.g. fishes), bad taste and unchecked growth of aquatic weeds.

Ground water pollution

The ground water is generally considered to be safe and is useful for drinking, agricultural and industrial purposes. In fact, ***ground water is less prone to pollution***. However, in recent years, it is recognized that contamination with fluoride, arsenic and nitrate in the ground water poses a serious threat to human health.

Surface water pollution

The rivers, lakes and reservoirs are highly susceptible for pollution due to natural and man-made activites (industrial, domestic, agricultural etc.). The routes for the entry of pollutants include sewage outfalls, industrial outfalls, and outfalls from nuclear stations. Surface water pollution is a major threat for the survival of life itself. It is therefore necessary that regular monitoring of various routes of water contamination is done and effective protective measures are taken to minimise the pollution.

NATURE OF WATER POLLUTANTS

There are a large number of water pollutants which may be in dissolved, suspended or colloidal state. The pollutants may be broadly categorized as follows.

- Organic pollutants
- Inorganic pollutants
- Microbiological pollutants
- Radioactive pollutants.

The salient features of these pollutants and the sources of their contribution are briefly described.

ORGANIC POLLUTANTS

Organic compounds as such are absolutely essential for the existence of life. However, their entry into water causes pollution resulting in bad odour and unpleasant taste. In addition, some of the organic compounds may be toxic or even carcinogenic to humans.

Sources of organic pollutants

There are several natural sources of organic pollutants. These include the decay of leaves, plants, and dead animals, besides the release of organic materials from aquatic plants and microorganisms.

Several synthetic organic pollutants (e.g. pesticides, herbicides, ethylbenzene) and synthetic volatile organic chemicals (e.g. carbon tetrachloride, tetrachloroethylene) pollute water. Certain organic pollutants such as dioxin and polychlorinated biphenyls (PCBs) are very toxic, besides being carcinogenic to humans.

INORGANIC POLLUTANTS

In the natural fresh water (source may be ground or surface), there occurs varying concentrations of inorganic chemicals. Most of them, in general, are within the acceptable limits.

Sources of inorganic pollutants

The industrial and agricultural outflows pollute water with inorganic compounds e.g. metal complexes, inorganic salts, mineral acids, trace elements. Among these, water pollution due to toxic metals assumes significance e.g. lead, mercury, cadmium, nickel, arsenic and aluminium.

MICROBIOLOGICAL POLLUTANTS

A wide range of microorganisms (viruses, bacteria, protozoa, algae, helminths) contribute to water pollution. A majority of these organisms are harmless to humans. However, there are certain organisms that cause diseases, referred to as ***water borne diseases***. Such organisms causing some form of sickness in humans are referred to as ***pathogens*** or ***pathogenic organisms***. A selected list of pathogenic organisms and the corresponding diseases is given in ***Table 56.1***.

Table 56.1 Common pathogenic organisms found in water (domestic waste water) and the respective diseases

Organism	*Disease*	*Major disease symptom*
Bacteria		
Escherichia coli (enteropathogenic)	Gastroenteritis	Diarrhea
Salmonella typhi	Typhoid	High fever, diarrhea
Salmonella paratyphi	Paratyphoid	Fever
Salmonella sp	Salmonellosis	Food poisoning
Vibrio cholerae	Cholera	Diarrhea, dehydration
Shigella sp	Shigellosis	Bacillary dysentery
Legionella pneumophila	Legionellosis	Acute respiratory illness
Viruses		
Adenovirus	Respiratory diseases	Breathing difficulty
Enteroviruses	Gastroenteritis	Diarrhea
Reovirus	Gastroenteritis	Diarrhea
Rotavirus	Gastroenteritis	Diarrhea
Hepatitis A	Infectious hepatitis	Jaundice
Protozoa		
Entamoeba histolytica	Amoebic dysentery	Prolonged diarrhea
Giardia lamblia	Giardiasis	Mild diarrhea, nausea
Balantidium coli	Balantidiasis	Dysentery
Helminths		
Ascaris lumbricoides	Ascariasis	Roundworm infestation
Schistosoma sp	Schistosomiasis	Hematuria, diarrhea
Fasciola hepatica	Fascioliasis	Jaundice
Taenia saginata	Taeniasis	Intestinal infection
Algae		
Microcystis aeruginosa	Gastroenteritis	Diarrhea
Aphanizomenon flosaquae	Gastroenteritis	Diarrhea
Shizothrix calciola	Gastroenteritis	Diarrhea

Sources of pathogens

The pathogens are mainly present in the waste water and for most of them ***human feces are the prime source of pollution***. The human waste can pollute the water from sewage outfalls or from a flow of waste from ground (mostly from failed septic system).

The best way to control water contamination with pathogenic organisms is to prevent the human wastes (particularly feces) from entering potable water.

RADIOACTIVE POLLUTANTS

Water may get polluted with radioactive materials arising from nuclear plants, radioisotopes used in medical, industrial and research purposes, besides the processing of ores to produce radioisotpes. The radioactive pollutants are carcinogenic e.g. uranium, radium, thorium.

Table 56.2 Typical composition of domestic waste water or sewage (untreated)

Constituent(s)	*Concentration (mg/l)**		
	Weak	*Medium*	*Strong*
Total solids (TS)	350	700	1,250
Dissolved solids (DS)	250	500	900
Suspended solids (SS)	100	200	350
Total nitrogen	20	45	90
Organic N	10	15	40
Inorganic N (as free NH_3)	10	30	50
Total phosphorus	4	8	16
Organic	1	3	6
Inorganic	3	5	9
Chlorides	25	50	100
Alkalinity (as $CaCO_3$)	50	100	200
Grease	50	50	100 150
Volatile organic compounds (µg / l)	<100	100–400	>400
Biochemical oxygen demand (BOD)	100	200	400
Chemical oxygen demand (COD)	250	500	1000
Total coliform (no/dl)	10^6–10^7	10^7–10^8	10^8–10^9

* *The units for volatile organic compounds and total coliform are given along with them.*

WASTE WATER OR SEWAGE

Sewage is the ***liquid waste*** or waste arising mainly ***from domestic*** (residential, institutional, commercial) and ***industrial sources***.

COMPOSITION OF SEWAGE

The actual composition of sewage depends on the source from which it comes. In general, about 97–99% of sewage is composed of water while the rest (1-3%) is solids. The most important organic, inorganic compounds, and living organisms found in waste water are listed below.

Organic—carbohydrates, fats, proteins, amino acids and urea, besides the products of their degradation.

Inorganic—sand, mud, mineral ash, mineral salts, lead, arsenic, mercury and cyanides.

Living organisms — bacteria, viruses, algae, fungi and protozoa. Some details on pathogenic organisms and the corresponding diseases are given in ***Table 56.1***.

TYPES OF SEWAGE

The waste water (sewage) is broadly categorized as ***weak***, ***medium*** and ***strong***, based on the composition (***Table 56.2***). It must however, be noted that the composition of sewage is highly variable which mostly depends on the local conditions. The sewage mainly consists of solids, organic and inorganic compounds of nitrogen and phosphorus, chlorides, grease and volatile organic compounds. In ***Table 56.2***, the data on biochemical oxygen demand (BOD), chemical oxygen demand (COD) and total coliform for the three types of sewage are also given.

MEASUREMENT OF WATER POLLUTION

The measurement of water pollution in sewage waste water with special reference to organic matter and pathogenic organisms is described.

MEASUREMENT OF ORGANIC MATTER OF SEWAGE

The organic matter present in the sewage is regarded as ***biologically active***, if it can be ***oxidized by the bacteria***. On the other hand, the organic matter resistant ***to bacterial degradation*** is considered as ***biologically inactive***. For the purpose of sewage treatment, the biologically active organic matter is important.

The commonly used laboratory methods for the measurement of organic matter in sewage are given below.

- Biochemical oxygen demand (BOD)
- Chemical oxygen demand (COD)
- Total organic carbon (TOC)
- Theoretical oxygen demand (ThOD).

Biochemical oxygen demand (BOD)

Biochemical oxygen demand is the most widely used parameter to ***measure the organic pollution*** in sewage as well as surface water. BOD basically involves ***the measurement of dissolved oxygen (DO) utilized by the microorganisms for the biochemical oxidation of organic matter***. The demand for oxygen and the process of oxidation depends on the type and quantity of organic matter, temperature and type of the organism used.

In general, biochemical oxygen demand is measured for an ***incubation period of five days*** (hence appropriately referred to as BOD_5) at a temperature of 20°C. If the organic content of the sewage is high, it needs to be diluted for the measurement of BOD. Further, for waste water with less population of microorganisms, seeding with bacterial culture is necessary.

BOD indicates the amount of organic matter present in the sewage. Thus, the more is organic content, the higher is the BOD (Refer ***Table 56.2*** for BOD values of different types of sewages). If the available oxygen (dissolved O_2) is less than the BOD, the organic matter decomposes anaerobically, putrifies and produces foul smell. Thus, ***BOD is a measure of nuisance potential of sewage***.

Limitations of BOD

1. BOD measures only biodegradable organic matter.
2. A high concentration of bacterial load is required.
3. For measuring BOD of toxic waste water, pretreatment is necessary.
4. Requires long period of incubation i.e. 5 days.

Despite these drawbacks, BOD is very widely used worldover for practical and economic reasons.

Chemical oxygen demand (COD)

Chemical oxygen demand refers to the ***oxygen equivalents of organic matter that can be oxidized by using strong chemical oxidizing agents***. Usually, potassium dichromate in the presence of a catalyst, in acidic medium is employed for this purpose. The overall reaction is given below.

$$\text{Organic matter} + Cr_2O_7^{2-} + H^+ \longrightarrow Cr^{3+} + CO_2 + H_2O$$

When compared with BOD, COD oxidizes more organic compounds, hence COD values are higher than BOD values. Chemical oxygen demand can be determined in just three hours, in contrast to BOD requiring five days. Some workers determine COD and calculate BOD. This is possible since there exists a reasonably good correlation between COD and BOD.

Total organic carbon (TOC)

Measurement of total organic carbon is required when the concentration of organic matter is very low. TOC can be determined by oxidizing organic carbon, in the presence of a catalyst to CO_2, which can be measured.

Theoretical oxygen demand (ThOD)

The organic matter of sewage is mainly composed of carbohydrates, proteins, fats and products of their decomposition. If the chemical formulae of the organic matter (i.e. individual compounds) are known, the theoretical oxygen demand can be calculated.

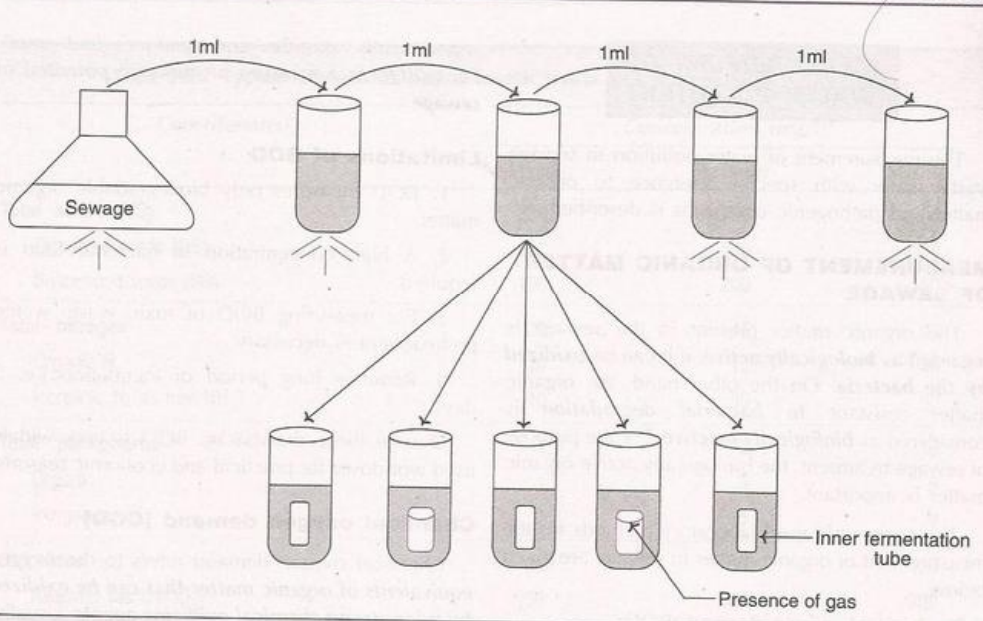


Fig. 56.1 : Multiple-tube fermentation technique for the detection of coliform bacteria.

DETECTION OF PATHOGENIC ORGANISMS OF SEWAGE

Human beings infected with disease-causing microorganisms (i.e. pathogens) are the carriers and they can discharge pathogenic organisms in waste water. Several species of bacteria, viruses, protozoa and helminths are responsible for diseases in humans. A selected list of the pathogenic organisms and the major diseases is given in (***Table 56.1***). Among the diseases (water-borne diseases) originating from sewage, typhoid, paratyphoid, cholera, gastroenteritis, diarrhea, and dysentery are important. All these diseases are highly infectious and are responsible for the death of thousands of people in countries (particularly developing ones) with poor sanitation.

Indicator or index organisms of sewage

It is tedious and time consuming to isolate and identify pathogenic organisms in water waste. Some other non-pathogenic bacteria of sewage polluted water are employed for this purpose e.g. *E. coli, Streptococus faecalis, Clostridium perfingens, Klebisella* sp. These organisms are collectively referred to as ***indicator*** or ***index organisms***.

Among the indicator organisms, the rod-shaped bacteria, commonly known as ***coliform organisms*** (*E. coli, Aerobacter* sp) are the most commonly used. The intestinal tract of man is very rich in coliform bacteria. It is estimated that each person discharges about 100–400 billions of coliform organisms per day. Therefore, the detection of coliform organisms is a clear indication of fecal contamination and thus the presence of pathogenic organisms. On the other hand, the absence of these organisms indicates that the water is free from disease-causing organisms. Detection of coliform organisms is based on the fact that a ***great majority of water-borne diseases have fecal origin***.

LABORATORY METHODS FOR DETECTION OF COLIFORM ORGANISMS

There are three methods commonly employed for the detection of coliform organisms in water samples-multiple tube-fermentation technique, membrane filter technique and colilert technique.

Multiple-tube fermentation (MTF) technique

This is basically a test that involves acid fermentation. Lactose broth fermentation is tested at around 35°C for 1-2 days in a series of tubes,

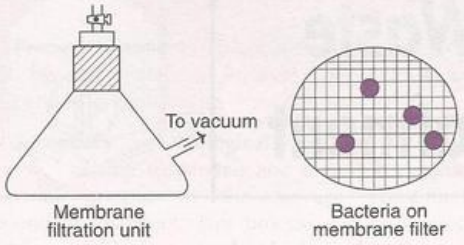


Fig. 56.2 : Membrane filtration technique for the detection of coliform bacteria.

involving sequential tests-presumptive test, confirmed test and completed test.

Presumptive test : The waste water sample is serially diluted. One ml of the sample of each dilution is then transferred to each of the five fermentation tubes. These tubes contain lactose culture medium and inverted gas collection tubes (***Fig. 56.1***). At the end of 24 hour incubation period, if there occurs ***gas collection in the inverted tube, the test is positive indicating the presence of coliform organisms.***

Confirmed test : If no gas formation occurs in the presumptive test, the incubation is continued for another 24 hours. If gas formation is detected now, the test confirms the presence of coliform bacteria. And if no gas is produced even in the confirmed test, it can be assumed that the samples are free from coliform.

Completed test : In this, a sample from positive coliform test is grown on eosin methylene blue (EMD) agar. The incubation is carried out at 35°C for 24 hours. The presence of ***E. coli*** can be ***detected*** by the ***occurrence*** of ***greenish metallic sheen*** on the plates.

Membrane filtration technique (MFT)

The presence of coliform organisms in water can be detected by using cellulose acetate ester with a pore size of 0.3 to 0.5 µm. For this purpose, the membrane is first sterilized for 20 minutes at 80°C. The filtration is carried out in aseptic conditions under vacuum.

As the filtration occurs, the ***bacteria are held on the membrane surface*** (***Fig. 56.2***). The trapped bacteria are dried and stained with a dye such as erythrosine, and detected.

Advantages and disadvantages : The membrane filtration technique is rapid and can be carried out outside the laboratory (fields) also. This is in contrast to multiple tube fermentation technique. The major limitation MFT is that it cannot be used for turbid and/or heavily polluted water. This is because the membrane pores get clogged.

Colilert technique

Although commonly used, both the tests described above (MFT and MF) sometimes give false-positive and false-negative results. Colilert technique is a recent and novel technique designed to specifically detect *E. coli* and other coliform bacteria. This is based on the principle that when ***certain substrates are digested by microorganisms, chromogens*** (coloured substances) ***are produced.*** Thus, when O-nitrophenyl β-D-galactopyranoside (ONPG) is used, it is converted to a yellow compound by coliform bacteria. With the substrate 4-methylumbelliferyl β-D-glucuronide, *E. coli* produces a fluorescent compound.

TECHNIQUES TO DISTINGUISH FECAL FROM NON-FECAL BACTERIA

A positive laboratory test for coliform bacteria need not necessarily be due to pathogenic organisms, originating from intestines. For instance, *Enterobacter (Aerobacter) aerogenes*, found in decaying plant materials can ferment lactose and give a positive coliform test. Thus, the presence of *E. aerogenes* in water does not indicate contamination with coliform bacteria.

A series of laboratory tests are available to differentiate fecal from non-fecal organisms. These techniques are collectively referred to as ***IMViC test***, with the following connotations.

I stands for ***indole test***

M represents ***methy red test***

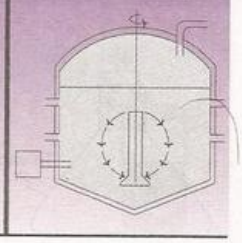
V is for ***Voges-Proskauer test***

C stands for ***citrate test***

(**Note :** ***i*** is used only for phonetic purpose).

These four tests are used to distinguish *E. coli* from *Enterobacter aerogenes*. *E. coli* gives positive indole and methyl red tests, and indicates that the source of pollution is of fecal origin. On the other hand, *Enterobacter aerogenes* gives positive Voges-Proskauer and citrate tests which shows that the water pollution is due to non-fecal origin.

57 Sewage/Waste Water treatment



The water pollutants, composition of sewage/ waste water, and the measurement of pollution in sewage are described in the previous chapter. Discharge of ***sewage or waste water*** to the environment is a matter of concern, as it poses ***serious threat to public health***.

In the rural communities, recycling of wastes (human, animal, vegetable) has been practised for centuries. Such recycling processes provide fertilizers and fuel. However, these waste and the crude recycling processes are very dangerous, as they are the major sources of disease-causing pathogens. And in many developing countries, crude, and unscientific recycling of wastes is still in use ***exposing the people to health hazards***. In some countries, the wastes (of rural or urban origin) are carefully treated and disposed off. Waste water/ sewage treatment was started in 19th century (150 years ago), and certainly it is the most important factor for the well being of present day man, besides increasing the life expectancy.

The process designed for the treatment of sewage should have the following objectives.

- Removal of floatable and suspended particles.
- Treatment of biodegradable organic materials.
- Elimination of disease-causing (pathogenic) organisms.

CLASSIFICATION OF TREATMENT PROCESSES

The sewage treatment methods may be classified in two different ways based on the principle of operation or the stages of treatment process.

Based on the principle of operation

Physical unit operations : These methods involve the applications of physical forces for sewage treatment. The physical processes involved include screening, mixing, sedimentation, flocculation and filtration.

Chemical unit operations : These processes operate on the principles of chemical reactions that may involve precipitation, disinfection, adsorption etc.

Biological unit operations : The removal of pollutants, particularly the biodegradable organic materials, can be carried out through biological processes.

Based on the treatment process

The sewage treatment processes are more conveniently classified based on the stages of treatment as ***preliminary, primary, secondary*** and ***tertiary*** treatments.

Preliminary treatment : This process basically involves the removal of floating materials (leaves, rags, papers, animals etc.) settleable inorganic solids (sand, grit etc.), and grease, fats and oils.

Primary treatment : This process, usually carried out by sedimentation, involves the removal of suspended organic solids from the sewage.

Secondary or biological treatment : The removal of fine suspended and dissolved organic material present in the sewage constitutes secondary treatment. This process can be carried out by ***aerobic*** or ***anaerobic biological units***.

Tertiary or advanced treatment : This is the final treatment process of sewage, and is carried out mostly in the developed countries. In general, sewage is disposed off after the secondary treatment. When needed, the tertiary treatment is carried out by disinfection (i.e. chlorination).

Some details on the various treatment processes are briefly described, in the following pages

PRELIMINARY TREATMENT

As already stated, preliminary treatment involves the removal of floating materials (leaves, papers, rags) and settleable inorganic solids (sand, grit), besides oily substances (fats, oils, greases). The three major types of equipment-screeners, grit chambers, and skimming tanks, employed in preliminary screening are briefly described.

Screeners

A screener is a device with openings (usually uniform in size) to remove the floating materials and suspended particles. The process of screening can be carried out by passing sewage through different types of screeners (with different pore sizes). The screeners are classified as ***coarse***, ***medium*** or ***fine***, depending on the size of the openings. The coarse screen has larger openings (75–150 mm). The openings for medium and fine screens respectively are 20–50 mm and less than 20 mm.

Different types of screens-fixed bar screen (coarse or medium) disc type fine screen, drum type fine screen are in use. A diagrammatic representation of a ***fixed bar screen*** is shown in ***Fig. 57.1A***.

A ***shredder or comminutor*** is a special screen that can cut and retain the floating and suspended materials (***Fig. 57.1B***).

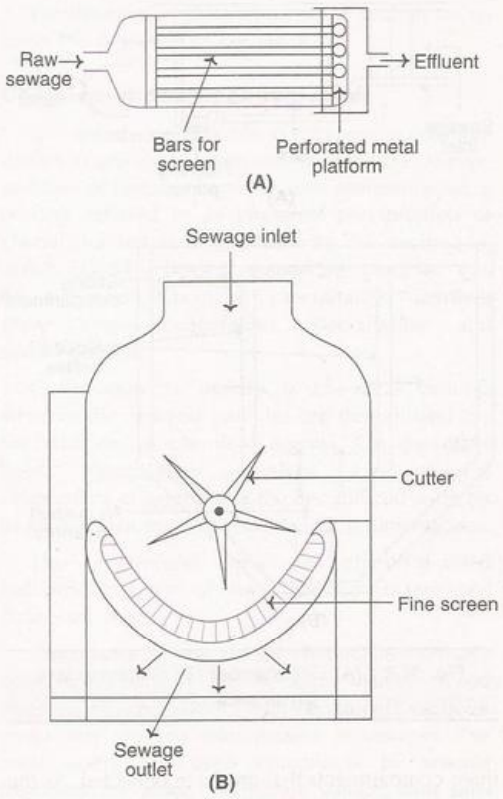


Fig. 57.1 : Diagrammatic representation of screeners (A) Fixed bar screen (coarse or medium) (B) Shredder.

Grit chambers

The heavy ***inorganic materials*** (specific gravity 2.4-2.7) like sand, ash and others ***can be removed*** by using grit chambers. This technique is based on the process of sedimentation due to gravitational forces. Grit chambers may be kept either before or after the screens (described above). A diagrammatic representation of a typical grit chamber is depicted in ***Fig. 57.2A***.

Skimming tanks

Several ***greasy and oily materials*** (fats, oils, waxes, soaps etc.) from the domestic or industrial outlets find their entry into the sewage. They ***can be removed by*** using a skimming tank which is fitted with baffle walls that divide the tank (***Fig. 57.2B***). The skimming tank is divided into

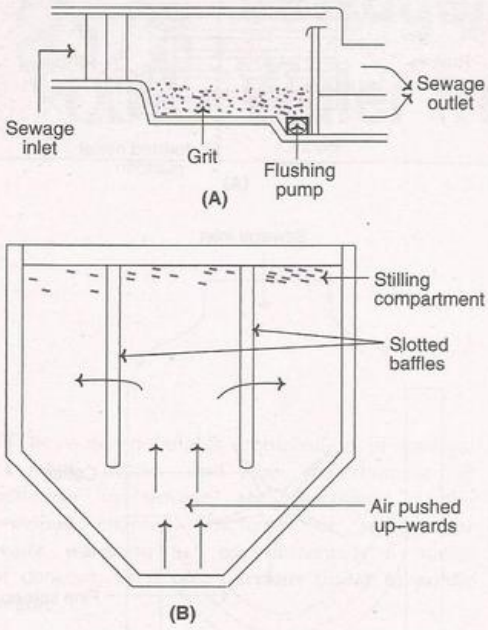


Fig. 57.2 : (A) Grit chamber (B) Skimming tank (cross section).

three compartments that are interconnected. As the compressed air is pushed from the floor of the tank, the raising air bubbles coagulate and solidify the oily and greasy materials present in the sewage. This material is pushed to the side compartment referred to as stilling compartment from where it can be removed manually or mechanically.

PRIMARY TREATMENT

Primary treatment is aimed at the removal of fine suspended organic solids that cannot be removed in the preliminary treatment. Primary treatment basically involves the process of ***sedimentation*** or ***settling***. In the normal process of sewage treatment, sedimentation is usually carried out twice-once before the secondary treatment, referred to as ***primary sedimentation***, and then after the secondary treatment is complete, a process known as ***secondary sedimentation***. It is sometimes necessary to use chemical coagulants to facilitate or aid sedimentation, and this process is referred to as ***chemical precipitation*** or ***coagulation-aided sedimentation***.

Principle of sedimentation

The solid particle of the sewage tend to settle down due to gravity. However, most of the solid particles of organic compounds remain in a suspended state in a flowing sewage. If the flow of the sewage is stopped and if it is stored in a tank referred to as ***sedimentation tank***, the solid particles can settle down at the bottom.

The process of sedimentation is influenced by several factors. These include the size, shape and specific gravity of particles, besides viscosity and flow velocity of sewage.

Types of settling

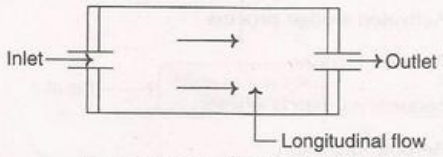
There ar four major types of settling—discrete settling, flocculant settling, hindered or zone settling and compression. This categorization is mainly based on the tendency of the particles to interact and form solids.

Discrete settling : The particles which do not change their size, shape and weight are referred to as discrete particles or ***granular particles***. The use of grit in sewage may be considered as an example of discrete settling.

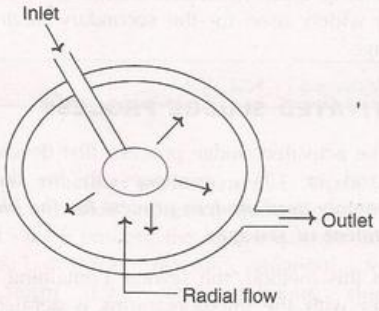
Flocculant settling : The flocculant particles can change their size, shape and weight, and thus lose their identity. These particles actually coalesce during settling. Settling of bioflocs, and chemical flocs in secondary sedimentation tanks are good examples of flocculant settling.

Hindered or zone settling : The particles as such, tend to remain in a fixed position with respect to each other. When flocculated, the whole mass of particles settle as a unit or a zone. In the hindered settling the concentration of particles increases from top to the bottom and this results in the thickening of the sludge. Zone settling is employed in conjuction with biological treatment facilities.

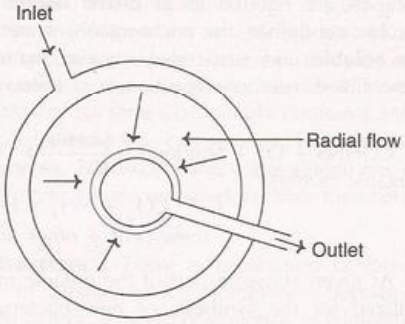
Compression : Settlement of particles in the lower layers can occur by compression of the weight of the particles on the upper layers. This process facilitates sludge thickening at the bottom.



Rectangular sedimentation tank



Center–fed circular sedimentation tank



Peripheral–fed sedimentation tank

Fig. 57.3 : Selected examples of sedimentation tanks.

Types of sedimentation tanks (clarifiers)

There are different ways of classifying sedimentation tanks.

Based on the shape — rectangular, circular and square.

Based on the flow of sewage — longitudinal, vertical, radial and spiral.

Based on the purpose and position — primary, secondary, coagulation-cum-sedimentation tanks, grit chambers, septic and Imhoff tanks.

Based on the operation — batch type and continuous flow type.

For the sake of illustration, some sedimentation tanks are depicted in ***Fig. 57.3***.

Chemical-aided sedimentation

It is not always possible to remove the colloidal wastes in sewage by plain sedimentation. However, addition of certain chemicals aids sedimentation, a process referred to as ***chemical precipitation*** or chemical-aided sedimentation. By this technique, about 60–80% of the suspended particles can be removed. Chemical precipitation ***involves three stages—coagulation***, ***flocculation*** and ***sedimentation***.

Coagulation is mainly a chemical process wherein the charged particles are destabilized (by the addition of chemical agents). On the other hand, flocculation involves the physical phenomena of aggregating the destabilized particles to finally form settleable solids (i.e. sedimentation).

The chemicals used in chemical-aided sedimentation are of two types-coagulants and coagulant aids.

Coagulants : These are the chemicalls (normally positively charged) which form insoluble and gelatinous precipitates with colloidal particles (negatively charged ones present in sewage). The most commonly used coagulants in sewage treatment are ***alum*** (aluminium sulfate) ***iron salts*** (ferric sulfate, ferrous sulfate, ferric chloride), ***lime and soda ash*** (sodium carbonate), sodium silicate and sodium aluminate.

Coagulant-aids : These chemicals aid or facilitate the process of coagulation. This is brought out by enchancing the action of coagulants and reducing the amount of sludge formed. The common coagulant aids are ***activated silica***, ***weighting agents*** (e.g. powdered lime stone or silica) and ***polyelectrolytes***.

SECONDARY OR BIOLOGICAL TREATMENT

Biological treatment of sewage is required for the removal of dissolved and fine colloidal organic matter. This process involves the use of microorganisms (bacteria, algae, fungi, protozoa, rotifers, nematodes) that decompose the unstable organic matter to stable inorganic forms.

TABLE 57.1 A list of major biological (secondary) treatment processes used for sewage treatment

Category	*Common process(s)*
Aerobic processes	
Suspended-growth systems	Activated sludge process
	Plug-flow process
	Sequencing batch reactor
	Contact stabilization
	Oxidation ditch
	Deep shaft
	Suspended-growth nitrification
	Aerobic lagoons
	Aerobic digestion
Attached-growth systems	Trickling filters
	Roughing filters
	Rotating biological contactors (RBC)
	Packed-bed reactors
Suspended-attached growth combined systems	Activated biofilters (e.g. trickling filter solids-contact process)
Anaerobic processes	
Suspended-growth systems	Anaerobic digestion (single-stage/two-stage)
	Anaerobic contact process
Attached-growth systems	Anaerobic filter process
	Expanded-bed process
Pond processes	
	Aerobic ponds
	Anaerobic ponds
	Facultative ponds

The biological treatment processes of sewage are broadly classified as ***aerobic***, ***anaerobic*** and ***pond processes***. Depending on the nature of the use of the microorganisms, the biological processes are categorized as ***suspended growth systems*** and ***attached growth systems***. A list of the major secondary (biological) treatment processes is given in ***Table 57.1***, and the most important ones are briefly described.

AEROBIC SUSPENDED-GROWTH TREATMENT PROCESSES

The most important suspended-growth biological treatment systems used for the removal of organic matter are listed.

- Activated sludge process
- Aerated lagoons
- Sequencing batch reactor
- Aerobic digestion.

Among these, activated sludge process is the most widely used for the secondary treatment of sewage.

ACTIVATED SLUDGE PROCESS

The activated sludge process, first developed in England in 1914, continues to be the ***most commonly used modern process for the biological treatment of sewage***.

In this method, the sewage containing organic matter with the microorganisms is aerated (by a mechanical aerator) in an aeration tank. The reactor contents are referred to as ***mixed liquor***. Under aerobic conditions, the microorganisms metabolise the soluble and suspended organic matter. The generalized metabolic reaction is as follows.

$$\underset{\text{(organic matter)}}{\text{COHNS} + O_2 + \text{Nutrients}} \xrightarrow{\text{Bacteria}} \underset{\text{(new cells + other products)}}{CO_2 + NH_3 + H_2O}$$

As given above, a part of the organic matter is utilized for the synthesis of new bacterial cells while the remaining gets oxidized to CO_2 and H_2O. The newly formed ***microorganisms are agglomerated to form flocs, technically referred to as sludge***. The separated sludge which is not in contact with organic matter becomes activated. It is separated from the settling tank, and returned to the aeration tank, and recycled. The ***activated sludge recycled*** in aeration tank serves as a ***seed*** or ***inoculum***. The excess and waste sludge can be removed.

For efficient operation of activated sludge process, it is necessary to maintain a constant supply of O_2 which can be done by mechanical aeration or through the use of rotating paddles. Growth of protozoa in a sludge is an indication of its healthy condition.

The disposal of a waste sludge is a problem. It may be used as a fertilizer in crop lands or as landfills, after drying.

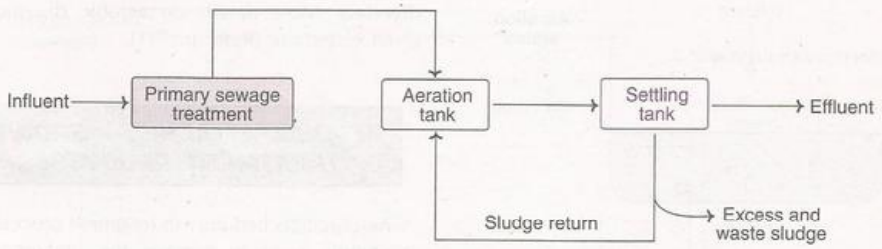


Fig. 57.4 : Flow diagram of activated–sludge process.

Factors affecting performance : There are several factors that influence the efficiency of activated sludge process, the most important being the type of the reactor, aeration, food microorganism (F/M) ratio, nutrients, sludge recirculation rate, besides pH and temperature.

Advantages : The activated sludge process is a very compact, low-cost and an efficient biological treatment system for sewage treatment. It is worked out that under ideal conditions, up to 95% of BOD_5, 98% of bacteria (particularly coliform) and 95% of suspended solids can removed by activated sludge process. The excess and waste sludge has a higher fertilizer value compared to other treatment processes.

Disadvantages : There is production of large volumes of sludge which sometimes becomes difficult to handle. Power consumption is relatively high for operation. Supervision by skilled personnel is necessary.

Conventional activated sludge process

In the normal treatment of sewage, the activated sludge is preceeded by primary sedimentation tank. The conventional activated sludge system consists of a separation tank, settling or sedimentation tank and sludge removal line (***Fig. 57.4***). The sewage after the primary treatment is introduced at the head of the tank. It is desirable to supply O_2 uniformly throughout the tank.

Modified activated sludge processes

For increasing the performance of the activated sludge system, several modifications have been done in the recent years. Most of them are directed to bring out efficient aeration. Aeration can be done by ***step aeration, tapered aeration, high rate aeration by complete mixing and extended aeration***.

AERATED LAGOONS

Aerated lagoons, also called as ***aerated ponds***, are the faculative stabilization ponds wherein surface aerators are installed to overcome the bad adours (due to overload of organic materials). The microbiological treatment of aerated ponds is comparable to the activated sludge process. The major difference is the large surface area in aerated ponds and this is more susceptible for temperature effects.

It is possible to carry out continuous nitrification in aerated lagoons. This however, depends on the design and operating conditions of the pond (particularly the temperature).

SEQUENCING BATCH REACTOR

Sequencing batch reactor (SBR) is a modification of activated sludge treatment system. The processes namely aeration and sedimentation are carried out in both the systems. The major difference is that while in the conventional activated sludge system, aeration and sedimentation occur simultaneously in separate tanks, these two processes are carried out sequentially in the same tank in SBR. Thus, the sequencing batch reactor may be regarded as ***fill-and-draw activated sludge process***.

The operating sequence of a typical SBR is depicted in ***Fig. 57.5***. The process is carried out in a sequence of five steps — ***filling, aeration*** (reacting) ***sedimentation*** (settling), ***decanting*** and ***idle***. Several modifications and improvements have been made in the SBR for more efficient operation.

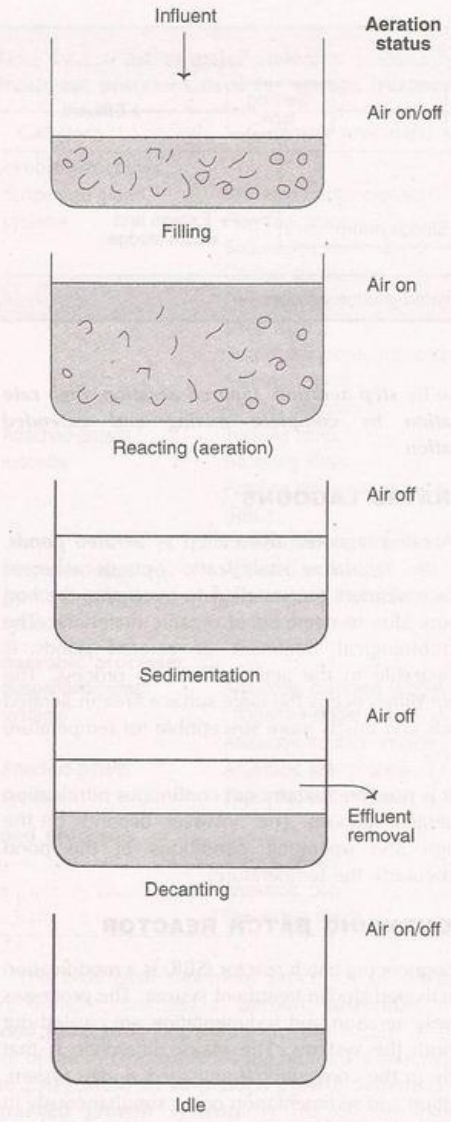


Fig. 57.5 : Operating sequence of a typical sequencing batch reactor (SBR).

AEROBIC DIGESTION

The organic sludges produced from various treatment processes (activated sludge treatment, trickling filter-sludge) are subjected to aerobic digestion in special reactors referred to as ***aerobic digesters***. More details on aerobic digestion are given elsewhere (Refer p. 711).

AEROBIC ATTACHED — GROWTH TREATMENT PROCESSES

Aerobic attached-growth treatment processes are commonly used to remove the organic matter found in the sewage. These processes are also useful for the ***nitrification*** (conversion of ammonia to nitrate). The commonly used attached-growth processes are listed.

- Trickling filters
- Roughing filters
- Rotating biological contractors
- Packed bed reactors.

Among these, trickling filter is most widely used.

TRICKLING FILTERS

Trickling filters, also known as ***percolating*** or ***sprinkling filters***, are commonly used for the biological treatment of domestic sewage and industrial waste water. In a strict sense, trickling filters are not filters, but they are ***oxidation units***.

A diagrammatic representation of trickling filter is depicted in ***Fig. 57.6***. It has a bed of coarse, hard and porous material over which sewage is sprayed. In about two weeks time, the biomass attached to the media surface grows and forms a layer, referred to as ***biological film*** or ***microbial slime***. This film has a thickness of 0.1 to 2.0 mm and is rich in microorganisms. As the liquid (sewage) trickles through the biofilm, the organic matter gets oxidized to CO_2 and NO_2 by the microbial metabolism. This oxidation is carried out by the aerobic organisms (particularly bacteria) that are present on the upper portion of the biological film.

The biological film is rich in the bacteria-*Pseudomonas, Flavobacterium, Alcaligenes*, and algae-*Chlorella, Utothrix*, and *Stigeoclonium*, besides some fungi and yeasts. Biofilms with a thickness in the range of 70–100 µm are efficient for the treatment process.

As the biofilm ages, its thickness increases and it automatically settles to the bottom of the tank.

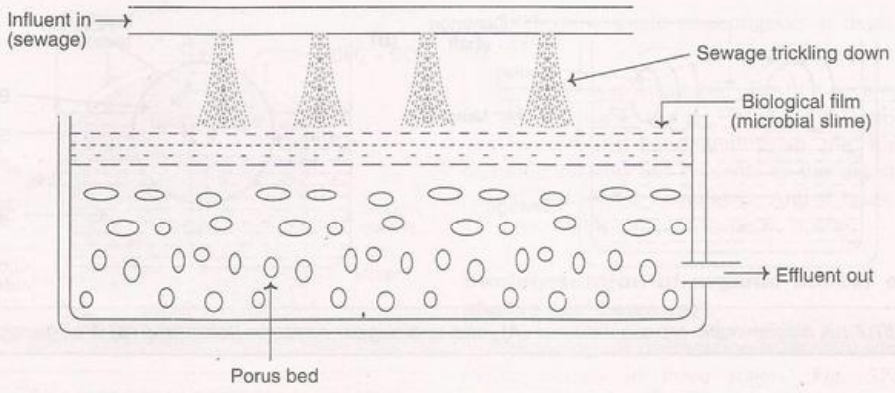


Fig. 57.6 : A diagrammatic representation of trickling filter.

The ***waste waters obtained in milk processing, paper mills*** and ***pharmaceutical industries are treated by trickling filters.***

Types of trickling filters

The trickling filters are classified as ***low rate*** (the conventional one), ***high rate*** and ***super rate***. This categorization is mainly based on the hydraulic and organic loading rates of the sewage. The low rate filters are suitable for the treatment of domestic sewage, while high rate filters and super rate filters are useful for industrial sewage.

Factors affecting performance of trickling filters

The type of the media and its depth, organic and hydraulic loading, filter staging, recirculation rate and flow distribution are the important factors that influence the performance of the trickling filters.

Advantages : Trickling filters are simple, occupy less space and the operating costs are low. They operate efficiently in hot climate and thus are ***suitable for most developing countries*** (like India).

Disadvantages : Removal of BOD is moderate (around 70%), and disposal of excess sludge is necessary. Primary sedimentation is required, since trickling filters cannot handle raw sewage.

ROUGHING FILTERS

Roughing filters are a special type of trickling filters that are designed to operate at high hydraulic loading rates. These filters are mostly used to reduce the organic matter in downstream processing, besides nitrification applications. Due to high hydraulic loading, there is a continuous sloughing of the biological film. In such a case, the unsettled filter effluent can be recycled, and this increases the efficiency of the treatment process. The retention time on the biofilm being less, organic materials that are not readily degradable remain unaffected.

ROTATING BIOLOGICAL CONTACTORS (RBC)

Rotating biological contactor (RBC) is a recent device for the biological treatment of sewage. It operates on the ***principle of aerobic attached-growth system operated on the moving media***. RBC are suitable for the treatment of domestic and industrial sewage in small and medium towns.

A rotating biological contactor is composed of a series of closely spaced and light weight circular discs (***Fig. 57.7***). They are made up of inert materials such as polystyrene or polyvinyl chloride (PVC) or polyethylene. These discs are mounted on a horizontal shaft in a tank through which waste water flows. The shaft is rotated slowly (less than 10 revolutions per minute) by a low speed motor. The discs of the shaft, referred to as ***biodiscs***, are partially (40–60%) submerged in sewage. As the biodiscs are rotated, the biomass attached to them is alternately submerged in sewage. This enables the discs to pick up a thin layer of sewage, and then to ***oxidize the absorbed substrates***. The unoxidized substrates fall back into the sewage.

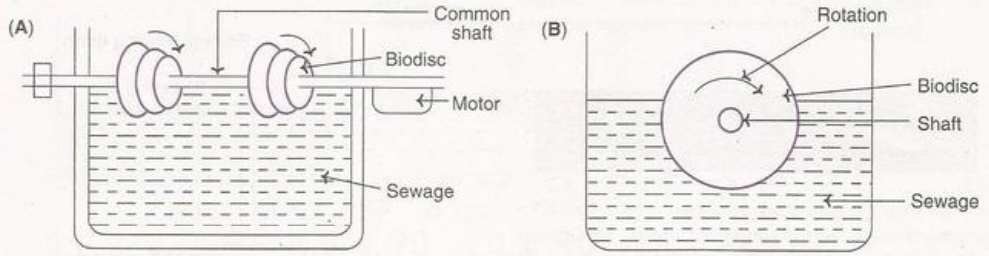


*Fig. 57.7 : A diagrammatic representation of (**A**) rotating biological contactor (RBC) and (**B**) A single biodisc.*

And this process in repeated again and again by the rotating biodiscs.

Rotating biological contractor is basically a ***film flow bioreactor***. The ***microbial biofilm*** is built upon the partly submerged support medium containing biodiscs.

The RBC is very efficient for the removal of organic matter in the sewage (about 90% of BOD). As and when there is an excess growth of biomass, it has to be removed. This process can be carried in a similar fashion, as it is done for the trickling filter (described already).

RBC are commonly used for the ***treatment of municipal waste water***. In addition, they are widely used for the biological processing of ***industrial wastes***, coming from several industries such as ***vegetables***, ***pulp***, ***meat*** and ***textiles***.

Factors affecting performance of RBC : The treatment process in RBC is influenced by rotation speed of shaft, waste water retention time, temperature, disc submergence, organic loading, media density, and number of stages.

Advantages : RBC is compact and requires moderate energy input. It has ***high BOD removal efficiency***.

Disadvantages : Disposal of sludge formed is a major problem with RBC. The treatment process is frequently associated with odour formation. RBC operation requires skilled personnel.

PACKED-BED REACTORS

Packed-bed reactors or ***fluidized bed reactors*** are used for the removal of BOD and nitrification. A reactor is packed with a medium to which the microorganisms get attached and form biofilms. The sewage along with air (or pure oxygen) is introduced from the bottom of the reactor (***Fig. 57.8***). The main advantage with packed bed reactors is that they have high surface area of biofilms for unit of reactor.

ANAEROBIC SUSPENDED — GROWTH TREATMENT PROCESSES

Anaerobic processes basically involve the decomposition of organic and inorgnic matter in the absence of oxygen. Anaerobic processing systems are important for the treatment of sludges,

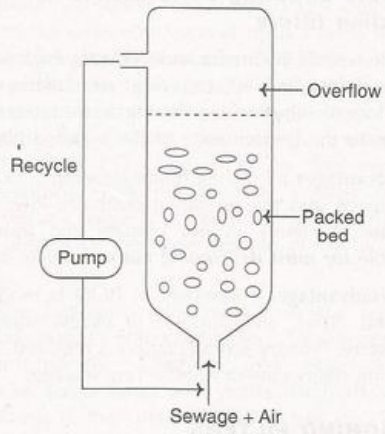


Fig. 57.8 : A diagrammatic representation of a packed-bed reactor.

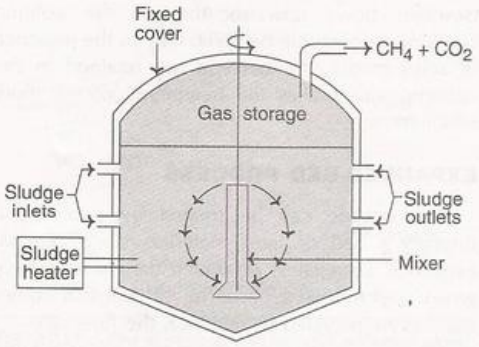


Fig. 57.9 : High-rate complete mix, single-stage anaerobic digester.

and high strength organic sewage. Several reactor systems for anaerobic suspended-growth treatment process have been developed. Among these, the ***complete-mix anaerobic digestion process*** is most widely used for the treatment of sewage.

ANAEROBIC DIGESTION

Anaerobic digestion is mostly useful for the stabilization of concentrated sludges that are produced on the treatment of industrial sewage. A diagrammatic representation of a typical high-rate, complete mix, single stage digester is depicted in ***Fig. 57.9***.

The process of anaerobic digestion is carried out in an air tight reactor. Sludge is introduced continuously or intermittently. In the high-rate digestion system, the contents of the digester are heated and mixed completely. And it takes about 15 days for the process to be complete.

Biodegradation of organic matter of sludge (or sewage)

The biological degradation of organic matter of sludge occurs in three stages (***Fig. 57.10***) — hydrolysis, acidogenesis and methanogenesis.

Hydrolysis : In the enzyme-catalysed reactions, high molecular weight compounds (proteins, polysaccharides, lipids and nucleic acids) are degraded to low molecular weight compounds (amino acids, monosaccharides, fatty acids, purines and pyrimidines). The latter serve as substrates for energy supply and microbial growth.

Acidogenesis : The low molecular weight compounds (referred above) are converted to acidic products (propionate, butyrate, lactate).

Methanogenesis : This is the third and final stage and involves the production of methane and carbon dioxide, from the intermediates formed in

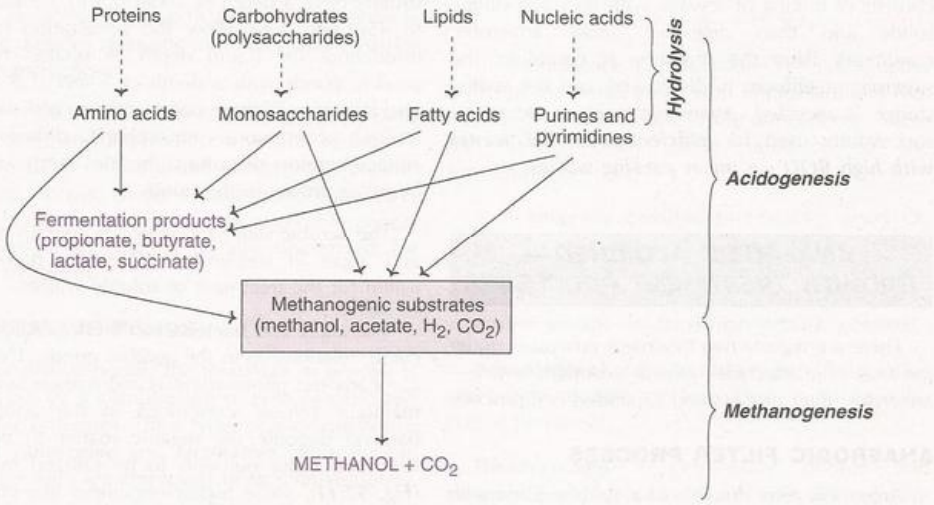


Fig. 57.10 : A diagrammatic view of the degradation of organic materials in anaerobic digestion.

acidogenesis. Methane gas is highly insoluble and its departure from the digester represents the stabilization of sewage or sludge.

Microorganisms to degrade organic matter of sludge (or sewage)

A consortium of anearobic microorganisms work together for degradation of sludge (or sewage) organic matter. They may be categorized into two types.

1. **Acid-forming bacteria :** These are also known as ***acidogens*** or ***non-methanogenic bacteria***. They bring out the hydrolysis of macromolecules (e.g. carbohydrate) to simple substrates (e.g. monosaccharides), and the latter to acids e.g. *Clostridium* sp, *Corynebacterium* sp, *Lactobacillus* sp, *Actinomyces* sp, *Staphylococcus* sp, *Peptococcus anaerobus, Escherichia coli.*

2. **Methanogenic bacteria :** These bacteria, also referred to as ***methanogens*** or ***methane formers*** are responsible for the conversion of acetic acid and hydrogen to methane and carbon dioxide. The most important methanogens belong to the genera *Methanobacterium, Methanobacillus, Methanococcus* and *Methanosarcina.*

ANAEROBIC CONTACT PROCESS

Anaerobic contact process is carried out in a specially designed reactors. The treatment process consists of mixing of sewage with recycled sludge solids and then digestion under anaerobic conditions. After the digestion is complete, the supernatant effluent is discharged and the settled sludge is recycled. Anaerobic contact process is successfully used for ***efficient industrial wastes with high BOD*** e.g. ***meat packing wastes.***

ANAEROBIC ATTACHED—GROWTH TREATMENT PROCESSES

There are mainly two treatment processes under the anaerobic attached—growth treatment system—anaerobic filter process and expanded bed process.

ANAEROBIC FILTER PROCESS

Anaerobic filter consists of a column filled with solid media for the treatment of organic matter in sewage. In this process system, waste water (sewage) flows upwards through the column containing anaerobic bacteria. Due to the presence of solid media, the bacteria are retained in the column. This makes the treatment process more efficient.

EXPANDED-BED PROCESS

The sewage can be treated by pumping it through a bed of inert materials (sand or coal expanded aggregates) on which the bacteria have grown and formed a film. The effluent that comes out can be recycled to maintain the flow rate.

POND TREATMENT PROCESSES

Pond treatment processes for the treatment of sewage (containing biodegradable wastes) are carried out by specially designed and constructed ponds. These ponds, referred to as ***stabilization ponds***, are large, shallow earthen basins. The treatment process is a natural one involving the combined use of bacteria and algae. The stabilization ponds are classified as ***aerobic***, ***anaerobic*** and ***facultative ponds***.

AEROBIC PONDS

The aerobic ponds, as the name indicates, maintain complete aerobic conditions. These ponds usually have a depth of about 0.5 to 1.5 feet (150 to 450 mm) and allow the penetration of light throughout the liquid depth. A second type of aerobic ponds with a depth of 5 feet (1.5 m) are also in use. In all these ponds, oxygen is maintained through continuous atmospheric diffusion (by surface aerators or pumps), besides the production by algae grown in the pond.

The aerobic stabilization ponds contain bacteria and algae in suspension. They are particularly useful for the treatment of soluble wastes.

The algae and bacteria exhibit a symbiotic and cyclic relationship in the aerobic ponds. The algae can carry out photosynthesis and release oxygen to maintain aerobic conditions in the pond. The bacteria degrade the organic matter to produce CO_2 and other nutrients to be utilized by algae (***Fig. 57.11***). Some higher organisms like protozoa and rotifers present in the pond are responsible for the polishing of the effluent.

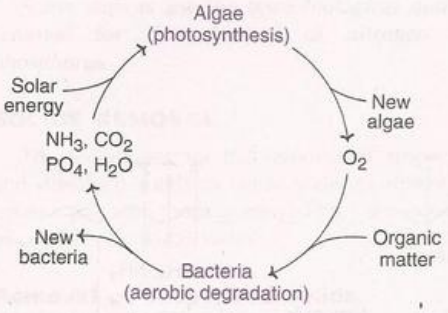


Fig. 57.11 : The symbiotic relationship between algae and bacteria in an aerobic pond.

Factors affecting aerobic ponds

The species of the algae and bacteria present in the pond significantly influence the efficiency of the aerobic ponds. The other factors such as the quality and quantity of organic material, degree of pond mixing, nutrients, sunlight, pH and temperature are also important for the successful operation of these ponds.

ANAEROBIC PONDS

Anaerobic ponds are useful for the ***treatment of high-strength organic matter and solid*** containing sewage/waste water. These ponds are completely devoid of dissolved O_2. They are very deep (up to 30 feet i.e. about 9 m), so that heat conservation is possible, besides requiring minimum land area (for pond construction).

When the sewage is added to the pond, precipitation and anaerobic conversion of organic waste to CO_2, methane and other gases, organic acids etc. occurs. Under suitable conditions, 75% of the BOD can be removed in anaerobic ponds. The clarified effluent is usually discharged for further treatment.

FACULTATIVE PONDS

In facultative ponds, the treatment of sewage is carried out by a combination of both aerobic and anaerobic processes. Three types of microorganisms ***aerobic***, ***anaerobic*** and ***facultative*** (both aerobic and anaerobic) are employed in facultative ponds.

The term ***oxidation pond*** or ***stabilization pond*** is frequently used for facultative ponds.

A diagrammatic view of a facultative pond is depicted in ***Fig. 57.12***. It consists of three zones.

1. **The surface aerobic zone :** It has the aerobic bacteria and algae, existing in a symbiotic relation (as explained on p. 696).

2. **The bottom anaerobic zone :** It contains anaerobic bacteria and solids that undergo decomposition.

3. **Intermediate facultative zone :** This zone is partly aerobic and partly anaerobic and contains both types (aerobic and anaerobic) of bacteria.

Processes that occur in facultative ponds

The sewage organic matter is stabilized by both aerobic and anaerobic processes. The algae present in the aerobic zone carry out photosynthesis and release O_2. This oxygen is utilized by the aerobic and facultative bacteria to oxidize soluble and colloidal organic matter. The organic solids present in the anaerobic zone (bottom sludge) are degraded to dissolved organic compounds (organic acids) and gases such as CO_2, CH_4 and H_2S. The organic acids can be oxidized by the aerobic bacteria while the gases produced (CO_2, CH_4, H_2S, NH_3) may be vented to the atmosphere. In fact, most of CO_2 is utilized by the algae for photosynthesis, while H_2S combines with O_2 to form sulfuric acid.

$$H_2S + 2O_2 \longrightarrow H_2SO_4$$

In the absence of adequate O_2 in the upper layers of the pond, gases with unpleasant and foul odours (H_2S) are vented to the atmosphere. These foul smells often cause nuisance to the surroundings.

It is generally possible to maintain good O_2 supply by the algae and facultative bacteria. Sometimes, surface aerators are used to enhance the efficiency of facultative ponds (particularly when the sewage contains high organic content).

Advantages : For the facultative ponds, the initial and operating costs are low. There is no need for skilled personnel.

Disadvantages : Unpleasant odours and mosquite breeding are frequently seen in facultative ponds. The requirement of land area for construction of these ponds is more.

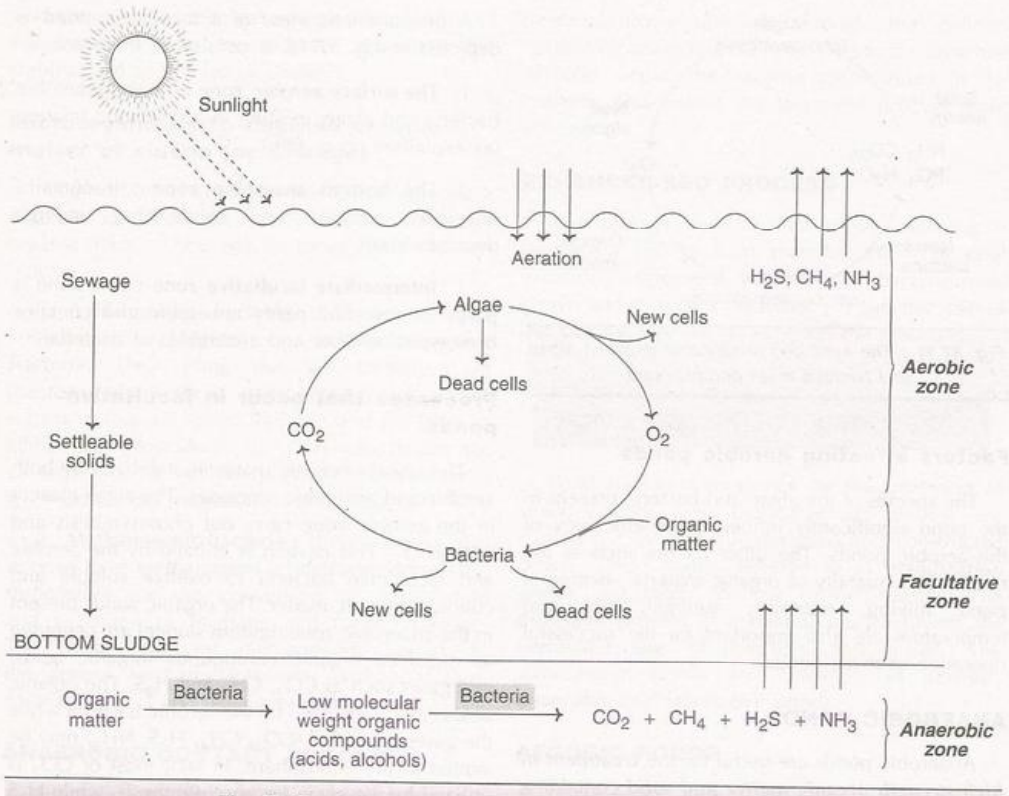


Fig. 57.12 : A diagrammatic representation of a facultative pond.

TERTIARY TREATMENT

Tertiary treatment or ***advanced treatment*** is sometimes needed ***for the removal of suspended and dissolved substances***, after the conventional primary and secondary treatments. In general, the effluent of the sewage obtained after secondary treatment can be conveniently disposed without causing any nuisance. However, tertiary treatment is needed under the following circumstances.

- When the quality of the effluent to be discharged does not meet the standard requirements (particularly in the developed countries).
- When there is a necessary to reuse the sewage/ waste water (***reclamation of water*** is quite expensive, but is required in certain situations of water shortage).
- For the removal of nitrogen and phosphorus compounds.

Tertiary treatment process broadly involves the removal of suspended and dissolved solids, nitrogen, phosphorus and pathogenic organisms. In the conventional heirarchy of sewage treatment, the unit operations are carried out in the order of preliminary, primary, secondary and finally tertiary treatment. However, sometimes advanced (tertiary) treatment process may be directly carried out bypassing the other unit operations. This mainly depends on the composition of waste water and the requirements.

There are four major processes under the tertiary treatment.

1. Solids removal
2. Biological nitrogen removal
3. Biological phosphorus removal
4. Disinfection.

Some authors use the term ***biological nutrient removal for the removal of nitrogen and phosphorus***.

SOLIDS REMOVAL

The techniques for the removal of suspended and dissolved solids in waste water treatment are comparable with those employed for the processing of potable (drinking) water.

Removal of suspended solids

The effluents obtained from secondary treatment may contain suspended solids in the size 0.1 to 100 μm. The concentration of these solids is variable, and is usually 20–40 mg/l. The removal of suspended solids is carried out by ***granular medium (sand) filtration*** and ***microscreening***. Sometimes, diatomaceous earth filters and coagulation-cum sedimentation techniques are also used.

Removal of dissolved solids

The dissolved solids can be removed mainly by two techniques—***adsorption*** and ***ion-exchange***.

Adsorption by activated carbon : Activated carbon is highly porous and provides large surface area for the adsorption of dissolved solids in the advanced treatment. The compounds that can be removed by adsorption include ***organic materials*** (herbicides, pesticides, tannins, lignins, colour and odour producing substances), ***inorganic materials*** (toxic trace metals) and several other pollutants.

Granular activated carbon contactors

Granular activated carbon contactors (***GAC***) are the most widely used for the advanced treatment of sewage. The GAC consists of a tank (usually cylindrical) which is loaded with granular activated carbon (***Fig. 57.13***). The waste water/sewage (influent) is passed from top through the activated carbon bed and the effluent comes out from the bottom. The solids attached to the carbon bed hamper further adsorption process. These particles can be removed by backwashing.

The activated carbon requires periodical regeneration. This can be done by heating in a furnance at about 800°C in the absence of O_2. At this temperature, the adsorbed organic compounds are converted to gases and released. The ***carbon granulates are reactivated for reuse***.

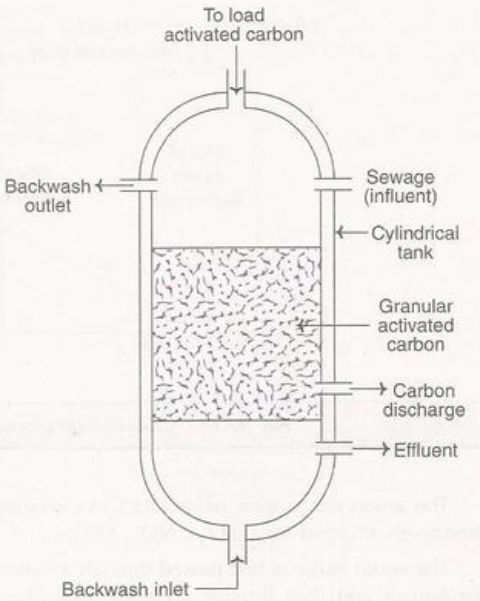


Fig. 57.13 : A diagrammative representation of granular activated carbon contactor (GAC).

Powdered activated carbon (PAC), instead of granulated carbon, is sometimes used for absorption process. PAC is added to an aeration tank (of secondary treatment) which can adsorb several organic compounds.

Ion-exchange for dissolved solids removal

As the name indicates, ion-exchange involves the displacement of one ion by another. The exchange occurs between the ions of insoluble exchange material (ion-exchange materials) and the ions of different species in solution (i.e. waste water for advanced treatment).

The ion-exchange process is carried out by employing two types of ion-exchange materials—cation exchangers and anion exchangers (***Fig. 57.14***). The synthetic resins with strong acidic (H^+) and basic (OH^-) functional groups serve as ion exchangers. The cation exchangers (with H^+ or Na^+) can replace the positively charged ions (Ca^{2+}, Mg^{2+}) in water by hydrogen ions. This is what is done for removing the hardness of water.

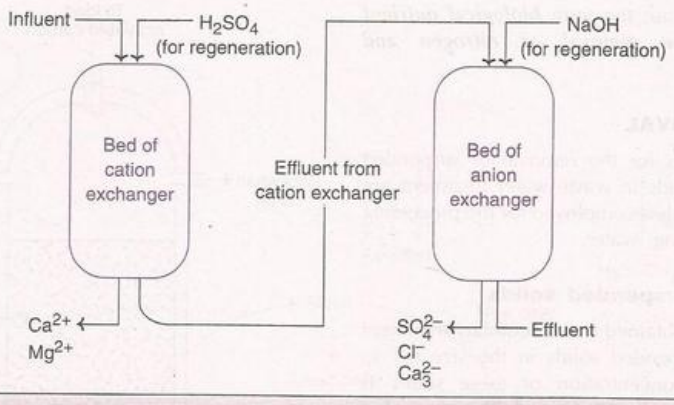


Fig. 57.14 : Ion-exchange process for the removal of dissolved solids.

The anion exchangers (with OH^-) can remove negatively charged ions (SO_4^{2-}, NO_3, CO_3^{2-}).

The waste water is first passed through a cation exchanger and then through an anion exchanger packed in two separate columns. When the ion exchange capacity of the resin is exhausted, it has to be regenerated for further use. For cation-exchange resins, regeneration can be done with strong acids (H_2SO_4, HCl), while for anion exchange resins, alkali (NaOH) is used.

For an effective removal of dissolved solids by ion exchange, the waste water should not contain high concentration of suspended solids as they block the ion exchange beds.

BIOLOGICAL NITROGEN REMOVAL

Decomposition products of proteins and the urea present in sewage are the major constituents of biological nitrogen. Although, nitrogen is a nutrient, its excess concentration causes eutrophication, and thus its removal is required. Biological nitrogen removal (***BNR***) is carried out by the methods based on the following principles (***Fig. 57.15***).

1. Assimilation of nitrogen
2. Nitrification and denitrification.

Assimilation of nitrogen

Since nitrogen is a nutrient, the microorganisms in the sewage can assimilate ammonia nitrogen, and grow. As some of these cells die, a portion of this ammonia nitrogen will be returned to the sewage.

Nitrification and denitrification

Nitrification : Ammonia nitrogen first gets oxidized to nitrite (NO_2^-) by the bacteria *Nitrosomonas* sp. This is followed by further oxidation nitrite (NO_2^-) to nitrate (NO_3^-) by *Nitrobacter* sp.

NH_4^+ (ammonia N)
↓ *Nitrosomonas*
NO_2^- (nitrite)
↓ *Nitrobacter*
NO_3^- (nitrate)

The bacteria involved in nitrification are auxotrophs. The nitrification process is accomplished by aerobic suspended growth and aerobic attached-growth systems. In the general practice, nitrification is carried out along with the BOD removal in the secondary treatment with suitable modifications. Trickling filters, rotating biological contactors and packed towers can be used for nitrification process.

Denitrification : The ***removal of nitrogen*** in the form of nitrate by converting to nitrogen gas is referred to as denitrification. This process ***occurs under anaerobic conditions*** and is brought out by certain genera of bacteria-*Aerobacter, Bacillus, Brevibacterium, Lactobacillus, Micrococcus, Pseudomonas* and *Spirillum*. These bacteria are heterotrophs and require no oxygen, but the

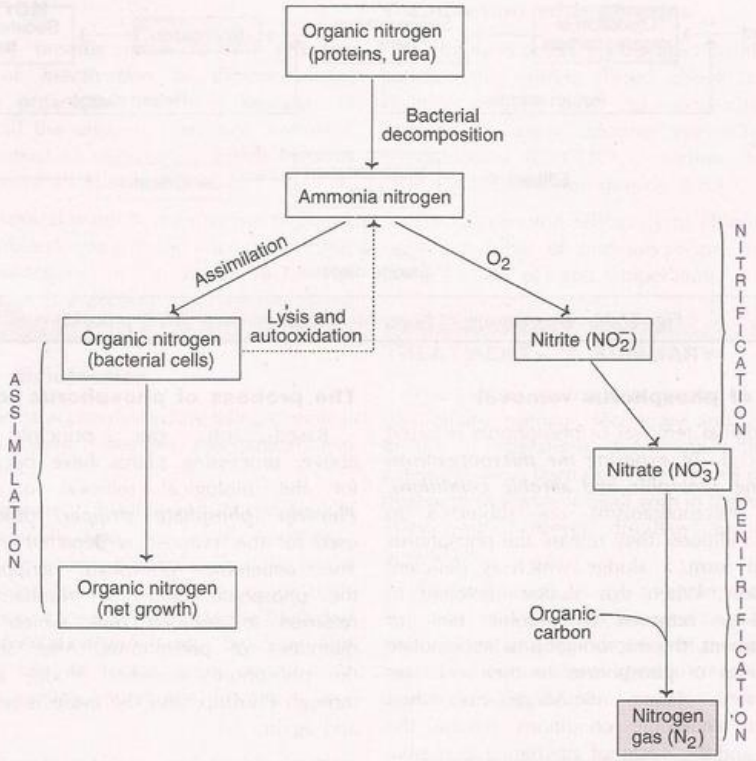


Fig. 57.15 : Biological nitrogen removal.

presence of organic carbon is essential. The presence of even minute quantities of O_2 suppresses denitrification. The heterotrophic bacteria can reduce nitrate in the following stages, to finally nitrogen gas.

NO_3^- (nitrate)
↓
NO_2^- (nitrite)
↓
NO (nitric oxide)
↓
N_2O (nitrous oxide)
↓
N_2 (nitrogen gas)

The process denitrification can be carried out by suspended growth and attached growth systems. The plug flow type of activated sludge system is commonly used.

The flow diagram for the ***removal of biological nitrogen*** is depicted in ***Fig. 57.16***. The process ***primarily consists of the removal of organic carbon (aerobic), followed by nitrification (aerobic) and denitrification (anaerobic).***

BIOLOGICAL PHOSPHORUS REMOVAL

Phosphorus in the sewage is mostly present in the form of orthophosphate (PO_4^{3-}), polyphosphate (P_2O_7) and organic bound phosphorus. In fact, phosphorus is an essential nutrient for microorganisms. Thus, during the normal secondary treatment process, 10–30% of the sewage phosphorus is utilized by the microorganisms for growth and energy purposes.

Phosphorus removal from waste water is required to control eutrophication and to maintain water quality.

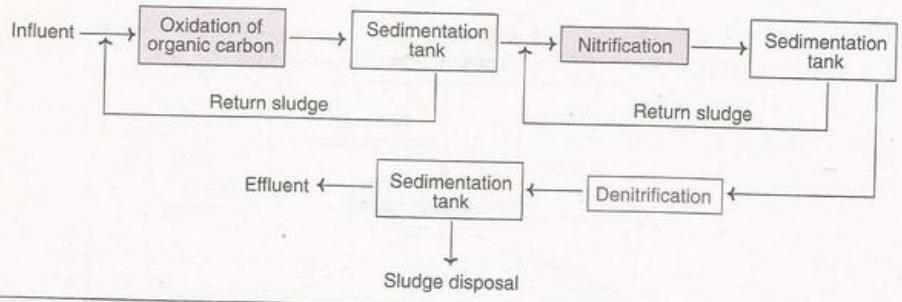


Fig. 57.16 : An overview of biological nitrogen removal (BNR) process.

Principle of phosphorus removal

The biological removal of phosphorus is based on the principle of ***exposing the microorganisms to alternating anaerobic and aerobic conditions.*** When the microorganisms are subjected to anaerobic conditions, they release the phosphorus content and form a sludge which is deficient in phosphorus. When this sludge (deficient in phosphorus) is returned to aerobic tank of sewage treatment, the microorganisms accumulate large quantities of phosphorus in their cells (as polyphosphates). These microorganisms when subjected to anaerobic conditions release the phosphorus and this cycle (of alternating anaerobic and aerobic processes) can be continued for the biological removal of phosphorus.

Acinetobacter sp are mainly responsible for the biological removal of phosphorus.

The process of phosphorus removal

Based on the principle described above, processing plants have been developed for the biological removal of phosphorus. ***Phostrip*** (***phosphate stripper***) process system, used for this purpose is depicted in ***Fig. 57.17.*** The anaerobic phosphate stripper removes the phosphate, and the resultant sludge is returned to aeration tank which takes large quantities of phosphorus. After sedimentation, the phosphorus enriched sludge again passes through Phostrip, and the cycle is repeated again and again.

The phosphorus enriched supernatant that comes out of the phosphorus stripper is treated with lime to precipitate the phosphorus. The resultant liquid supernatant can be returned to the aeration tank for further treatment.

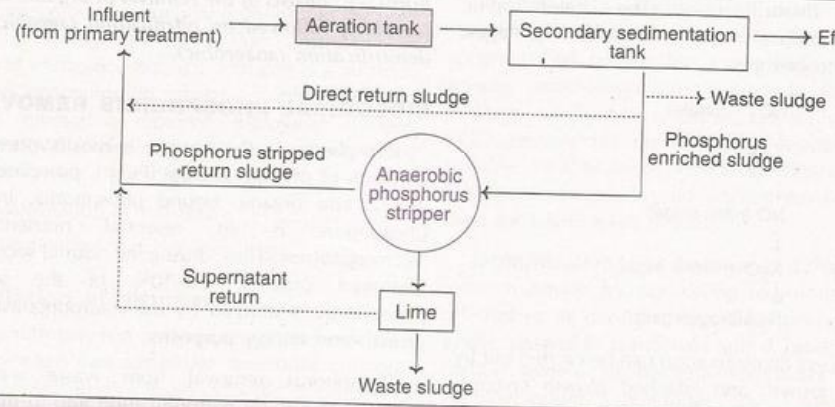


Fig. 57.17 : Biological phosphorus removal by Phostrip (phosphate stripper) process.

DISINFECTION

Disinfection broadly refers to the ***selective destruction*** or ***inactivation of disease-causing*** (pathogenic) ***organisms***. In the process of disinfection, all the organisms are not destroyed. This is in contrast to sterilization which involves the destruction of all the organisms.

There are several water borne diseases (typhoid, cholera, dysentery) caused by bacteria, viruses and other pathogens (Refer ***Table 56.1***). The very purpose of disinfection is to control these diseases.

Agents for disinfection

Disinfection is accomplished by using chemical and physical agents, besides mechanical and radiation means.

Chemical agents : Chlorine and its compounds are most commonly used. The other chemicals—bromine, iodine, ozone, alcohols, phenols, heavy metals, hydrogen peroxide, alkalies and acids are sometimes employed. After ***chlorine*** (regarded as ***most universal disinfectant***), bromine and iodine are in use. In recent years, ozone as a disinfectant is gaining importance, since it is very effective.

Physical agents : Heat and light can be effectively used as disinfectants. Sunlight (particularly ultra-violet rays) is in fact a good disinfectant.

Mechanical means : The pathogenic organisms can also be removed by mechanical means, during the course of waste water treatment. The processes involving screens (coarse and fine), grit chambers and sedimentation can partly remove the disease-causing organisms.

Radiation means : The gamma rays emitted from radioisotopes can serve as effective disinfectants.

Characteristics of an ideal disinfectant

An ideal disinfectant should possess the following characteristics.

- Toxic to pathogens at low concentration
- Soluble and stable in water
- Non-toxic to man and higher organisms
- Cheap and easily available.

Disinfection with chlorine

Chlorine is a very widely used disinfectant, as it satisfies the criteria (listed above) of an ideal disinfectant. The most commonly used chlorine compounds are — chlorine gas (Cl_2), calcium hypochloride [$Ca(COCl_2)$], sodium hypochlorite (NaOCl) and chlorine dioxide (ClO_2).

The disinfection efficiency of chlorine depends on the number of microorganisms in the water being treated, pH and temperature.

SEWAGE/WASTE WATER TREATMENT — A SUMMARY

The various treatment processes of sewage i.e. preliminary, primary, secondary and tertiary (with details of operations processing units in each category), have been described in this chapter. The treatment of sludge is dealt with in the next chapter. They were, however, discussed as independent operations.

A conventional sewage treatment plant has the requisite operating units arranged one after another for treatment and final disposal of sewage. The flow chart of a conventional sewage treatment plant is depicted in ***Fig. 57.18***.

WASTE WATER TREATMENT OF SOME INDUSTRIES

The characteristics of waste water are highly variable depending on the industrial source, which in turn depends on the industrial activity. Thus, treatment of waste water differs, and this is largely dependent on the industry. The treatment schemes of waste waters of dairy, distillery, tannery, sugar and antibiotic industries are briefly described.

WASTE WATER TREATMENT FOR DAIRIES

Dairy waste water is polluted at various stages by the activities of dairy operation. This has to be appropriately treated before being discharged. This is essential to avoid water pollution. A schematic represntation of dairy waste water treatment is depicted in ***Fig. 57.19***, and briefly described below.

The processing of dairy waste water is carried out ***by a two-stage pond system***. The treatment is first done in an anaerobic pond, and then in an

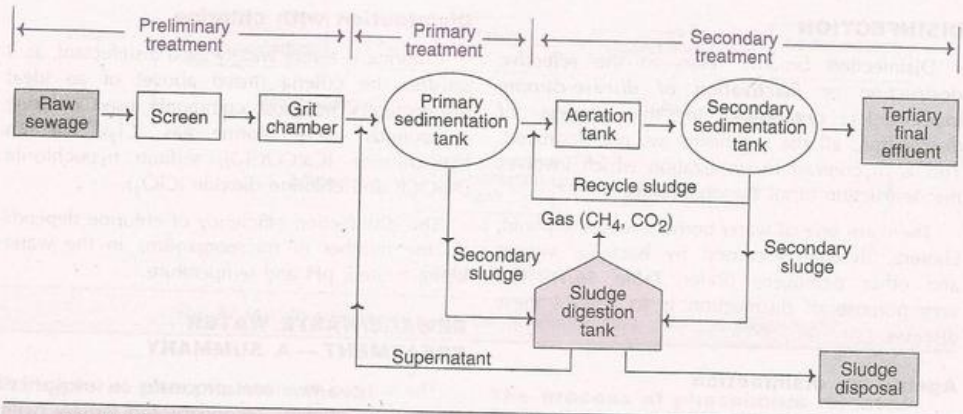


Fig. 57.18 : A diagrammatic representation of a flow chart of conventional sewage treatment plant.

aerobic pond. The treated water coming out of the anaerobic pond is rich in nutrients, hence is useful for irrigation purposes. At the end of the two-stage treatment, about 95% of the BOD material is removed from the waste water.

WASTE WATER TREATMENT FOR DISTILLERY

The waste water from distilleries mostly consists of dissolved O_2 and organic wastes (predominantly fermented starches and organic nitrogen). The treatment is carried out by a ***two-stage bacterial oxidation in anaerobic reactors***. The complex organic wastes are first degraded to organic molecules (acids, aldehydes, alcohols, ketones etc.). These molecules are further oxidized to smaller organic (CH_4) and inorganic (CO_2, NH_3, H_2S, H_2O) compounds (***Fig. 57.20***).

The products formed in the first stage of bacterial oxidation will promote the growth and multiplication of new bacterial cells. This is advantageous for more efficient bacterial oxidation.

WASTE WATER TREATMENT FOR TANNERY

Tanneries are industries involved in the processing of animal skins. For the transformation of skins to leathers, a large number of chemicals, dyes, tanning agents etc., are used. Tannery waste water poses a serious environmental threat since many of the ingradients have low biodegradability. The treatment process is carried out in two phases (***Fig. 57.21***).

Phase I : This phase involves ***biological acidification*** and production of sulfides from sulfates present in the waste water. The sulfide recovered in the form of sodium hydrogen sulfide is reused in tanneries.

Phase II : This is carried out in an anaerobic sludge blanket reactor. In this phase, the ***organic***

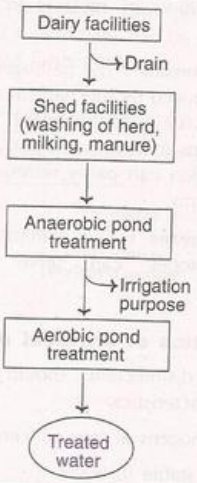


Fig. 57.19 : A diagrammatic representation of dairy waste water treatment.

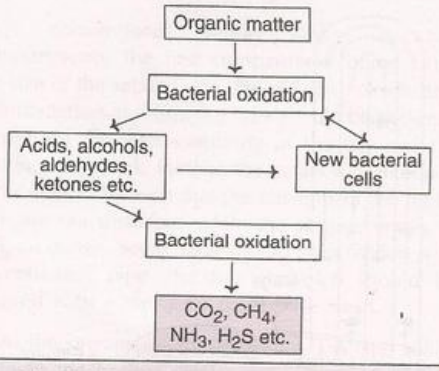


Fig. 57.20 : A diagrammatic representation of distillery organic waste water treatment.

acids are converted into gases. The left over sulfide from phase I is converted into solution, and processed to produce sulfur which will be useful in chemical industries.

With the two phase systems for waste water treatment of tanneries, it is possible to remove 20% of chemical oxygen demand (COD) and 90% of the sulfur compounds. Sometimes, additional aerobic bioreactors are used to remove more COD.

WASTE WATER TREATMENT FOR SUGAR INDUSTRY

The waste water from sugar industries contains high COD materials which are biodegradable. The treatment process is comparable to that described for waste water treatment of distillery (***Fig.* 57.20**).

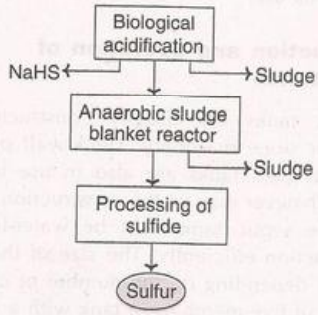


Fig. 57.21 : A diagrammatic representation of tannery waste water treatment.

TREATMENT OF ANTIBIOTICS IN WASTE WATER

There is a widespread use of antibiotics in human healthcare, animal husbandry and agriculture. In addition, the drug and antibiotic industries also directly contribute to pollution. Antibiotic pollution has become a major environmental threat. This is mostly due to the development of antibiotic resistance by the pathogenic bacteria. The conventional waste water treatment processes are ineffective in removing the antibiotics.

Special tertiary treatment processes are required to remove the antibiotic pollutants. The treatment involves the degradation of target molecules by oxidation, exposure to ultraviolet light etc. Some pharmaceutical industries have introduced a new technique namely ***membrane filtration*** to remove even trace amounts of antibiotics from the waste water.

WATER RECYCLING

There is shortage of water throughout the world. Therefore, there is a need for recycling of waste water (i.e. the use of reclaimed waste water) wherever, possible.

Approximately two thirds of world's water is utilized for irrigation purposes. In the developing countries at some places, the raw domestic sewage (completely untreated) is directly used for irrigation of crop land. Such practices expose the people to risk of infections by pathogenic bacteria, viruses and prions. It is therefore desirable to develop low-cost ***technologies to produce irrigation water from waste waters***.

Large quantities of water is also utilized in industries. For instance, to make one ton of steel, about 300 tones of water is needed. Consumption of water is also high for many other industries such as food, paper and textiles. At some places, and for some industries, processes for recycling of water have been developed. The technology for the recycling of water is variable, and this mainly depends on the industry.

Recycling of waste water is useful for agriculture, horticulture and watering of lawns and golf courses, besides supplying to lakes (fishing and

boating purposes). However, the use of recycled water for drinking purposes should be avoided.

SMALL SEWAGE/WASTE WATER TREATMENT SYSTEMS

The large scale sewage/waste water of treatment processes have been described. They are not suitable for the treatment of small sewage or waste water, particularly coming from small communities.

The on-site systems of sewage treatment for individual residences (or communities) with special reference to oxidation ditch, septic tanks and Imhoff tanks are briefly described.

OXIDATION DITCH

Oxidation ditch method, first developed in Netherlands, is a suitable method for the treatment of sewage in small communities. This is basically an ***aeration type of activated sludge process*** (See p. 690) with a mechanical system of aeration. However, there is no primary sedimentation of sewage, consequently the problem of handling and treatment of primary sludge is eliminated.

Oxidation ditch consists of aeration units, namely ditch channels (2 or more) constructed side by side (***Fig. 57.22***). The sizes of the ditch channels are variable-length 150–1000 m, width 1-5 m and depth 1-5 m. They are constructed with brick or stone masonary. A special type of rotors (cage rotors) are fitted into each ditch channel for continuously agitating and circulating the sludge, besides supplying O_2. The sludge is allowed to settle down in a separate sedimentation tank. The activated sludge is returned to ditch channels.

Instead of using a separate sedimentation tank, the sewage may be allowed to settle down in the ditch channels by stopping the rotors (usually during night). The supernatant from the ditch channels can be taken out. The excess sludge collected (in the sedimentation tank or at the bottom of ditch channels) can be stabilized.

SEPTIC TANKS

Septic tanks are recommended for individual houses and for small communities and institutions with a contributing population around 300.

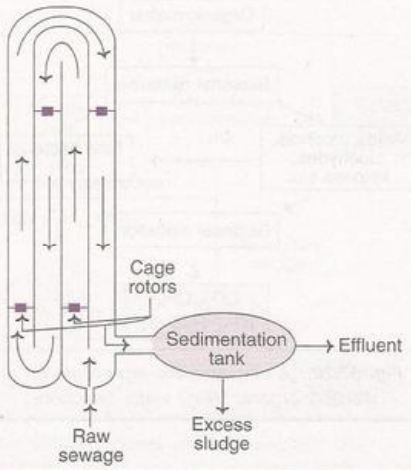


Fig. 57.22 : A diagrammatic representation of oxidation ditch.

Septic tanks work on the principle of ***anaerobic digestion***. This occurs as the solids of the sewage settle at the bottom of the tank. Under anaerobic conditions, the biodegradable organic matter is converted to gases (CH_4, CO_2, H_2S etc.) and liquid compounds. This results in a drastic reduction in the volume of sludge. A thick crust of scum formed on the surface of septic tank maintains anaerobic conditions. The effluent coming out of the septic tank contains some organic solids and pathogens. Therefore, disposal of the effluent from septic tank has to be dealt with very carefully. The septic tanks are desludged and cleaned at regular intervals, usually once in 2-5 years (depending on the tank size and its use).

Construction and operation of septic tanks

Septic tanks are usually constructed with bricks, or stone masonary. Thick-wall polythene and fibre glass tanks are also in use in recent years. Whatever may be the construction material used, the septic tank must be water-tight and must function efficiently. The size of the tank is variable, depending on the number of users. For a family of five members, a tank with a length of 1.5 m and a breadth of 0.75 m is recommended. For such a tank, the cleaning interval is 2-3 years.

A conventional septic tank has two compartments, the first compartment being twice the size of the second one (***Fig. 57.23***). For effective sedimentation of solids, the tank should be designed to prevent the short-circuiting at the top and the bottom of the tank. Further, the location of inlet and outlet should be such that the contents of the septic tank are not disturbed while the sewage enters or effluent leaves. Septic tank should be provided with a ventilation pipe, the top of which should be covered with a mosquito proof wire mesh.

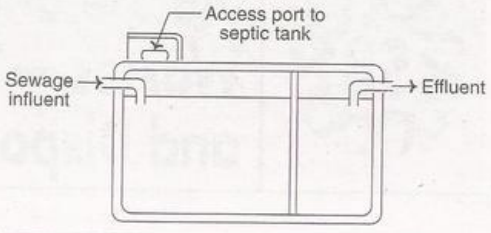


Fig. 57.23 : Schematic representation of a conventional two-compartment septic tank.

As the sewage enters the septic tank, the solids settle to the bottom while grease and other light materials float on the surface, and form a scum. The ***bottom-settled organic material undergoes facultative and anaerobic decomposition to form more stable compounds and gases*** (CH_4, CO_2, H_2S). In this way, there is a continuous reduction of solids of sewage entering the septic tank. However, sludge accumulates at the bottom of the tank. Desludging of septic tank has to be carried out periodically (once in 2-5 years).

IMHOFF TANKS

Imhoff tank is an ***improved septic tank.*** It basically consists of a two storey tank in which the sedimentation (settling) occurs in the upper tank while the digestion of the settled solids takes place in the lower compartment (***Fig. 57.24***).

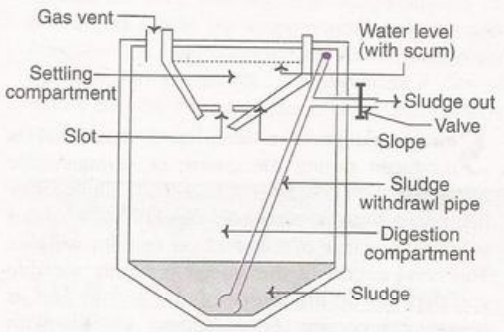
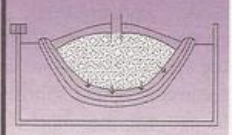


Fig. 57.24 : A diagrammatic view of imhoff tank.

As the sewage enters the sedimentation tank, the solids settle down to the bottom and the sewage flows into the digestion tank through slopes and slot of the sedimentation tank. Gas produced in the digestion process escapes through gas vent. The sedimentation tank is designed in such a way that the gases and the gas-buoyed sludge particles raising from the sludge layer do not enter into it. The sludge collection at the bottom can be withdrawn periodically.

58 Sludge and Solid Wastes — Treatment and Disposal



Sewage sludge is a semi-liquid mass that is produced during the course of sewage/waste water treatment processes (Chapter 57). Sludge may be regarded as a ***semisolid liquid*** that contains solids in the range of 0.5 to 12 per cent (by weight). The composition of the sludge is highly variable and depends on the source of raw sewage and its treatment processes. Some authors use the term ***organic slurries*** for sewage sludge.

SOURCES AND CHARACTERISTICS OF SLUDGE

The following unit operations are the ***major sources*** of sludge.

- Screening
- Grit removal
- Primary sedimentation
- Secondary sedimentation
- Sludge-processing units.

The details on the above processes have already been described. The ***important sludges*** and their ***characteristics*** are given in ***Table 58.1***. The characters of sludge are variable and depend on the origin of the sludge, and the aging after it is produced.

Chemical composition of sludge

The chemical composition of sludge is variable. The major constituents of sludge are proteins and other nitrogen containing compounds, grease and fats, cellulose, hydrocarbons, phosphorus, iron, silica, organic acids, heavy metals, pathogens and pesticides.

The various ***methods of sludge treatment and disposal*** are given ***in Table 58.2***. The important aspects of selected processes are briefly described.

PRELIMINARY OPERATIONS

The preliminary operations are carried out to provide a constant and homogeneous sludge for further processing.

Sludge grinding

Grinding of sludge is carried out to cut a large mass of sludge into small pieces. For this purpose, hardended steel cutters and overload sensors are used.

Sludge degritting

Sometimes, removal of grit (degritting) is necessary for the effective treatment of sludge. Degritting can be carried out by applying centrifugal force to the flowing sludge. Cyclone degritters are usually employed for this purpose to separate grit particles from the organic sludge.

Sludge blending

Sludge produced in ***primary*** (settleable solids), ***secondary*** (settleable solids and biological solids) and ***tertiary*** (biological solids and chemical solids) treatments is mixed (blended) to produce a uniform mixture.

TABLE 58.1 Major sludges and their characteristics

Sludge	*Characteristics*
Screenings	Contains organic and inorganic materials that can be easily removed.
Grit	Rich in heavier inorganic solids and a good quantity of organic matter (fats, grease).
Primary sludge	Obtained from the primary settling tanks, grey in colour and offensive in odour.
Activated sludge	Brownish to dark in colour with flocculant appearance, usually has earthy odour and can be digested easily.
Trickling-filter sludge	Brownish in colour with inoffensive odour (fresh sludge), easily digestible.
Scum	Floatable materials skimmed from the surfaces of primary and secondary settling tanks. Scum contains oils, fats, food wastes, plastic materials, cotton etc.
Aerobically digested sludge	Brownish to black in colour with flocculant appearance. No offensive odour.
Anaerobically digested sludge	Dark brown to black in colour, contains large quantities of gas. On drying the sludge, the gases escape.
Septage	Sludge of septic tanks, black in colour. Usually well digested due to long storage. Offensive odour due to hydrogen sulfide.

Sludge storage

Storage of sludge becomes necessary when the processing units are not in operations (night shifts, weekends). Short-term storage of sludge is accomplished in settling tanks while for long-term storage, specially designed tanks have to be used.

SLUDGE THICKENING (CONCENTRATION)

The solid content of the sludge is variable and may range from 0.5 to 12%. It is necessary to concentrate the sludge (particularly with low solid content) by removing some amount of liquid. Physical methods are employed for sludge thickening.

Gravity thickening

Gravity thickening is the most commonly used method for the thickening of primary sludge. It can be carried out in the conventional sedimentation tank (circular tank is preferred). As the sludge gets concentrated by gravity, the supernatant can be returned to the treatment plant (i.e. primary settling tank).

TABLE 58.2 Sludge treatment and disposal methods

Type of process	*Treatment methods*
Preliminary operations	Grinding, degritting, blending, storage
Thickening	Gravity thickening Flotation thickening Centrifugal thickening Rotary drum thickening
Stabilization	Lime stabilization Heat treatment Anaerobic digestion Aerobic digestion Composting
Conditioning	Chemical method Heat treatment method
Disinfection	Pasteurization
Dewatering	Vacuum filter Centrifuge Sludge drying beds
Heat drying	Flash dryer Slow dryer Rotary dryer
Thermal reduction	Multiple-hearth incinerator Fluidized bed incinerator
Ultimate disposal	Land application Distribution and marketing Landfill Lagooning

Flotation thickening

Flotation thickening is the technique of choice for the treatment of sludges from suspended-growth biological treatment processes. There are three types of flotation thickening processes — ***dissolved air flotation***, ***dispersed-air flotation***, and ***vacuum flotation***. Among these, dissolved-air flotation is widely used. In this technique, air is passed through the sludge that is held at high pressure. The air raises to the top of the sludge in the form of bubbles along with attached particles to form a sludge blanket. This can be skimmed off at regular intervals. The efficiency of flotation thickening depends on the air to solids ratio.

Centrifugal thickening

Centrifugal thickening is based on the principle of settlement of sludge particles under the influence of centrifugal forces. Two types of centrifuges — ***solid bowl centrifuge*** and ***basket centrifuge*** are used for this purpose. Although concentration of sludge by centrifugal thickening is efficient, it involves high cost of maintenance and power consumption.

Rotary drum thickening

Thickening of the sludge can be carried out by use of rotary drums. As the sludge is passed over the rotating screen drums, the solids separate from the water and the thickened sludge rolls out at the drum ends.

SLUDGE STABILIZATION

Sludge requires stabilization to achieve the following objectives.

- To reduce the load of disease-causing organisms (pathogens).
- To inhibit the potential for putrefaction.
- To eliminate offensive odours.

Sludge stabilization can be carried out by appropriate biological and chemical means. The ***techniques used for sludge stabilization are — lime stabilization, heat treatment, anaerobic digestion, aerobic digestion and composting***. They are briefly described.

Lime stabilization

When lime, in the form of hydrated lime [$Ca(OH)_2$] or quick lime (CaO) is added to the sludge, the pH raises to 12 or higher. At high pH, the microorganisms cannot grow. Consequently, the sludge will not putrify, create unpleasant odours and does not pose health hazards.

In the ***lime pretreatment***, addition of lime is done prior to dewatering, while in ***lime post-treatment***, it is done after dewatering.

Heat treatment

Heat treatment involves heating of sludge for a short period (250°C for about 30 minutes) under pressure. By this process, sludge is dewatered and sterilized. Heat treatment helps in stabilization and conditioning (discussed later) of the sludge. However, due to high cost, it is less frequently used.

Anaerobic sludge digestion

Anaerobic digestion of sewage is one of the oldest forms of biological treatment processes. The same operating unit is equally useful for the stabilization of sludge. There are different ways of carrying out anaerobic sludge digestion — standard rate digestion, single-stage high rate digestion and two-stage digestion.

Standard rate digestion : This is a single-stage digestion process (Refer ***Fig. 57.9***). The sludge is added and heated by external heat exchangers. As the sludge gets digested, the gas released comes out to the surfaces. Along with the gas, certain sludge particles (fats, oils, grease) also raise to the top to form a scum which can be removed.

Single-stage high rate digestion : This is characterized by high rate of sludge loading, and its digestion under anaerobic conditions. The sludge is continuously mixed by gas, heated and recirculated for optimal digestion.

Two-stage digestion : There are two digestion tanks used in this process of high-rate digestion. The first tank is used for digestion proper (with mixing and heating facilities) while the second digester is employed for storage and concentration.

Thermophilic anaerobic digestion : Sometimes, sludge digestion can be carried out at relatively ***higher temperature*** (45–60°C) by ***thermophilic bacteria***. In fact, the digestion process is faster with increased bacterial destruction and improved dewatering at higher temperature. But the major limitation is the requirement of high energy for

continuous heating. For this reason, thermophilic anaerobic digestion process is not widely used.

Aerobic sludge digestion

Aerobic digestion of sludges from various sources (primary, secondary and tertiary treatments) is becoming popular in recent years for the following reasons.

- The digestion of volatile solids is as effective as in anaerobic digestion.
- Fertilizer value of the sludge is much higher.
- Lower capital and operational costs.

However, the major limitation in aerobic digestion is the continuous supply of O_2 which is a costly affair.

The process, aerobic digestion of sludge is comparable to activated sludge process (Refer p. 690).

Principle of aerobic digestion : When the nutrient (substrates) supply to the microorganisms is depleted, they start oxidizing their own cellular (protoplasmic) materials for energy supply and maintenance. About 70–80% of the cellular organic matter can be oxidized aerobically to produce carbon dioxide, nitrate (NO_3^-) and water. Approximately 20–30% of the cellular organic matter cannot be oxidized, as it is not biodegradable.

A diagrammatic representation of an ***aerobic digester*** is depicted in ***Fig. 58.1***. The process can be carried out by batch or continuous flow system of operation. With regard to the aeration process, aerobic digestion is of two types—conventional aerobic digestion and high purity oxygen aerobic digestion.

Conventional aerobic digestion : Atmospheric air is used in the conventional digester for the process of aerobic digestion.

High-purity oxygen aerobic digestion : In this case, a pure grade of oxygen instead of air is used. The digestion process is usually carried out in closed tanks.

Thermophilic aerobic digestion : This is refinement of the above two techniques, and can effectively digest up to 20% of the biodegradable organics. The procers is carried out by thermophilic bacteria at a temperature around 45°C. There are certain advantages of thermophilic aerobic digestion. These include killing of more pathogenic organisms and digestion of more solids, with minimal unpleasent odours in stabilized sludge.

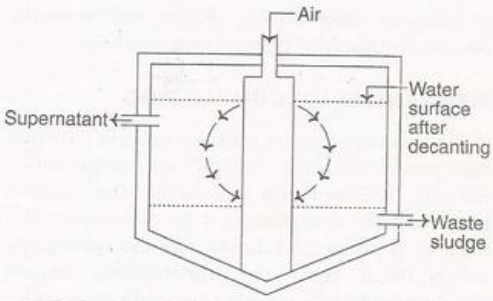


Fig. 58.1 : Diagrammatic representation of an aerobic digester.

COMPOSTING

Composting basically involves ***the process of biological degradation of solid organic waste material to stable end products***. Although composting can be carried out under aerobic and anaerobic conditions, aerobic composting is more frequently used. Thus, composting may be considered as an ***aerobic microbiological process*** for converting solid organic wastes into stable sanitary, nuisance free, humus like material that can be safely disposed into the environment.

Composting is a cost-effective and ***environment friendly process for stabilization and ultimate disposal of sludge***. The product of composting is useful for soil improvement and the production of mushrooms. Thus, composting is ultimately helpful for reuse and recycling of organic waste materials from domestic, agriculture and industry.

Organisms involved in composting

A wide variety of organisms (both unicellular and multicellular) are involved in the process composting. The bacteria make up 80–90% of the microorganisms found in the compost. These bacteria possess a broad range of enzymes to degrade a wide range of organic compounds.

The other organisms actively involved in composting are ***actinomycetes*** (a filamentous type

of bacteria), ***fungi*** (molds, yeasts), and ***protozoa***, besides earthworms, insects, mites and ants.

MECHANISM OF COMPOSTING

Composting is a very complex process involving the participation of several microorganisms—bacteria, actinomycetes and fungi. The bacteria bring out the decomposition of macromolecules namely proteins and lipids, besides generating energy (heat). Fungi and actinomycetes degrade cellulose and other complex organic compounds.

Composting may be divided into three stages with reference to changes in temperature—mesophilic, thermophilic and cooling.

Mesophilic stage : The fungi and acid-producing bacteria are active in this stage, and the temperature increases from ambient to about 40°C.

Thermophilic stage : As the composting proceeds, the temperature raises from 40°C to 70°C. Thermophilic bacteria, thermophilic fungi and actinomycetes are active in this stage. Thermophilic stage is associated with high rate and maximum degradation of organic materials.

Cooling stage : The microbial degradative activity slows down and the thermophilic organisms are replaced by mesophilic bacteria and fungi. Cooling stage is associated with formation of water, pH stabilization and completion of humeic acid formation.

METHODS OF COMPOSTING

The operation of composting involves the following steps.

- Mixing of dewatered sludge with a bulking agent (saw dust, rice hulls, straw or recycle compost). The bulking agent improves the porosity of the mixture for good aeration.
- Creating aerobic conditions (aeration) by mechanical or other means. This is needed for the supply of oxygen, to control temperature, and for the removal of water (moisture).
- Removal of the bulking agent, if possible.
- Storage and disposal of the compost.

There are three major methods of composting—***aerated static pile, windrow, and in-vessel systems***. As per a recent survey, the approximate distribution of different composting methods is given.

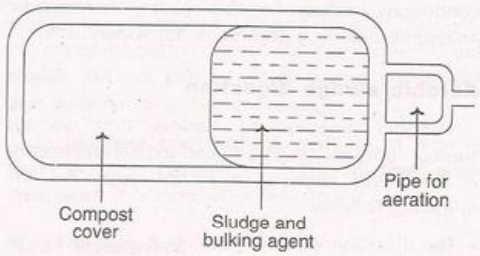


Fig. 58.2 : A diagrammatic representation of aerated static pole of composting.

Aerated static pile	– 55%
Windrow	– 30%
In-vessel	– 15%

Aerated static pile system

The ***dewatered sludge*** is mixed with a bulking agent (wood chips) and ***placed over a grid of aeration*** or ***exhaust piping***. Air is supplied through blowers for efficient aeration. A layer of compost is kept over the top of the aerated static pile for insulation and good aeration (***Fig. 58.2***). It takes about 3-4 weeks for composting, and another 4-5 weeks for curing. The cured compost is screened to reduce the quantity, besides the recovery of the bulking agent.

Windrow system

Windrows are a type of static piles with periodical turning and mixing of sludge during the composting period (3-4 weeks). The mixing is usually carried out at weekly intervals and is associated with release of unpleasant and offensive odours. The windrows may be open or covered and the aeration is carried out by mechanical means.

In-vessel system

In the in-vessel composting system, the process is ***carried out in a closed vessel*** or container. This is an advanced method and is designed to control the environmental conditions (temperature, air flow and O_2 supply), besides minimizing the release of offensive odours. The advantages of in-vessel system include higher efficiency of composting, with lower labour costs, and smaller area for the plant.

There are two major systems of in-vessel composting — plug flow and dynamic.

Plug-flow in-vessel composting reactors : There are two types of plug-flow in-vessel reactors— ***cylindrical tower*** and ***tunnel*** (***Fig. 58.3A*** and ***58.3B***). In both the cases, the relationship between the particles in the composting mass is maintained the same throughout the process. As the sewage is fed, the composting occurs on the principle of first in and first out.

Dynamic in-vessel composting reactors : These are also known as ***agitated bed in-vessel composting reactors***. In these units, the composting material is mechanically mixed during the period of processing. ***Dynamic circular reactors*** and ***dynamic rectangular reactors*** (***Fig. 58.3C and 58.3D***) are in common use. In circular reactors, the ***augurs*** rotate around the centre of the reaction vessel and mix the composting material. As regards the rectangular reactors, ***extraction conveyor*** is responsible for mixing the compost, and for the discharge of the compost to the outlet conveyor.

Factors affecting aerobic composting

There are several factors that influence the sludge-composting process. The important ones are briefly discussed.

1. **Type of sludge :** Both untreated and digested sludges can be composted. However, untreated sludge requires more oxygen, and emits unpleasant odours.

2. **Carbon nitrogen ratio :** For efficient composting, the carbon nitrogen ratio should be in the range of 25 : 1 to 35 : 1. It is desirable to periodically check and maintain this ratio.

3. **Bulking agents :** Cheap and readily available bulking agents (saw dust, wood chips) are used. Their particle size and moisture content influence composting.

4. **Moisture content :** A moisture content of sludge less than 60% is ideal for composting.

5. **Aeration :** It is desirable that the oxygen supply is constantly maintained, and it properly reaches the composting material.

6. **Temperature and pH :** Best results of composting are seen when the temperature is

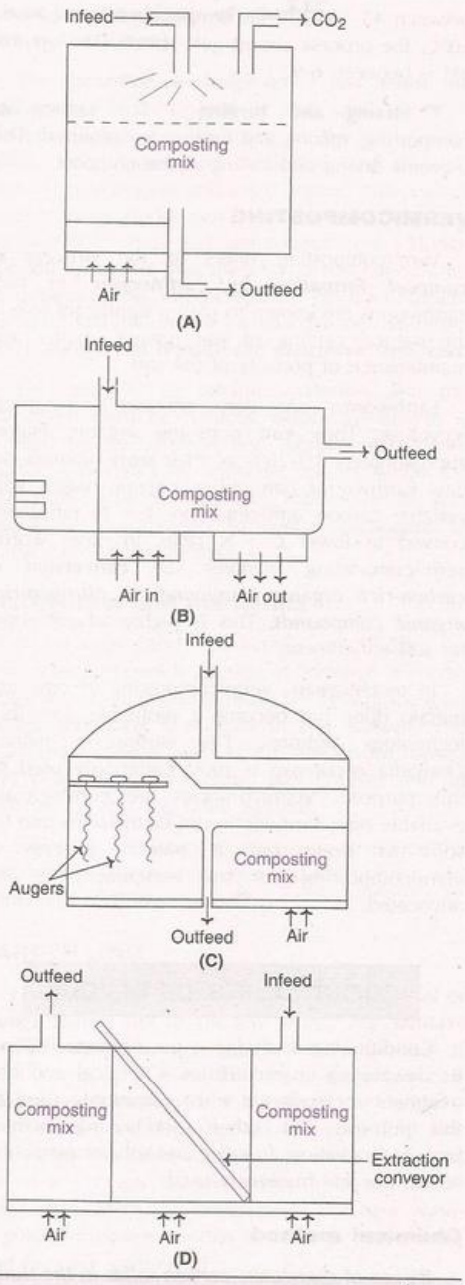


Fig. 58.3 : Diagrammatic representation of composting reactors (A) Plug-flow cylindrical tower (B) Plug-flow tunnel (C) Dynamic circular reactor (D) Dynamic rectangular reactor.

between 45–55°C. If the temperature goes beyond 60°C, the process almost gets halted. The optimal pH is between 6-9.

7. **Mixing and turning :** For appropriate composting, mixing and turning are required. This prevents drying and caking of the compost.

VERMICOMPOSTING

Vermicomposting refers to the ***process of compost formation by earthworms***. In fact, earthworms are known to play a significant role in the natural cycling of soil organic matter and maintenance of porosity of the soil.

Earthworms are very efficient in nutrient recycling. They can consume organic matter, approximately 10–20% of their own biomass per day. Earthworms can utilize organic matter with variable carbon nitrogen ratio (C : N ratio) and convert to lower C : N ratio. In other words, vermicomposting involves the ***conversion of carbon-rich organic compounds to nitrogen-rich organic compounds***. This is highly advantageous for soil enrichment.

In recent years, vermicomposting of cow and buffalo dung has become a profitable, low (bio) technology industry. The earthworm namely *Drawidia nepalensis* is most commonly used for this purpose. Vermicomposts are commercially available now. Introduction of earthworms into the soils to bring out a natural process of vermicomposting (for soil enrichment) is also advocated.

CONDITIONING OF SLUDGE

Conditioning of sludge is necessary to improve its dewatering characteristics. Chemical and heat treatment methods are most commonly used for this purpose. The other conditioning methods include irradiation, freezing and solvent extraction, which are less frequently used.

Chemical method

By use of chemicals, ***certain solids in the sludge can be coagulated*** with a release of absorbed water. Chemical conditioning can reduce the moisture content of the sludge by about 15–30% (i.e. from about 95% to about 65%). The most commonly used chemicals are alum, lime, ferric chloride and organic polymers.

Heat treatment method

Heat treatment ***can stabilize and condition the sludge***. This process is carried out for a short period under pressure. Heat treatment results in the coagulation of solids, besides reducing the water affinity of sludge solids.

DISINFECTION OF SLUDGE

In recent years, disinfection of sludge is becoming significant due to its reuse or its application on the land. This is because the ***pathogenic organisms of the sludge*** should be ***destroyed*** to protect the health of the inhabitants who are exposed to it. There are a large number of disinfection methods to choose.

- Pasteurization
- Heat drying
- Irradiation
- High pH treatment
- Addition of chlorine
- Long-term storage of digested sludge
- Complete composting.

DEWATERING

Dewatering basically involves the process of ***reducing the moisture content of sludge***. Dewatering can be carried out by vacuum filters, centrifuges and sludge drying beds.

Vacuum filtration

Dewatering by vacuum filtration is rather old, but has been discontinued in recent years due to high operating and maintenance costs, besides the complexicity of the process. In fact, improved and more efficient alternative methods have been developed during the past ten years.

Centrifugation

Centrifugation is commonly used for the dewatering of sludges obtained from industries. By

the process of centrifugation, it is possible to remove liquids from solids. Different types of centrifuges (solid bowel centrifuge, basket centrifuge) are commercially available for sludge dewatering.

Sludge drying beds

Drying beds are widely used in developed countries (USA, Britain) to dewater the digested sludge. This method produces high solid containing dried product which can be easily disposed off (in a landfill or as soil conditioner). Sludge drying beds are cost-effective.

HEAT DRYING

When the sludge is subjected to mechanical heat drying, the water content can be substantially reduced. The ultimate purpose of heat drying is to prepare a sludge, free from moisture that can be incinerated efficiently. Mechanical heat drying can be carried out by flash dryers, spray dryers, rotary dryers and multiple hearth dryers.

THERMAL REDUCTION OF SLUDGE

Thermal reduction basically involves the total or partial conversion of organic solids to oxidized end products such as carbon dioxide and water. This may be carried out by incineration or by wet-air oxidation. Thermal reduction of sludge is associated with destruction of pathogenic organisms, detoxification of toxic compounds and reducing the volume of disposable sludge.

The major processes employed for thermal reduction of sludge are multiple-hearth incineration, fluidized-bed incineration and wet-air oxidation.

ULTIMATE DISPOSAL OF SLUDGE

While considering the final disposal of sludge, its beneficial uses are first taken into account. Sludge is useful for the supply of nutrients to the soil, besides possessing the properties of soil conditioning. Thus, attempts are made to dispose the sludge in a beneficial manner. If this is not possible, alternates are considered.

Land applications of sludge (as a fertilizer)

The ***spreading of sludge on*** or ***just below the soil surface*** is considered as land applications of sludge. Sludge may be used in the agricultural lands, forest lands, and dedicated land disposal sites. The pathogens and toxic organic compounds present in the sludge can be respectively destroyed by sunlight and soil microorganisms. Sludge applied to land is thus useful as a soil conditioner to improve the characteristics of land-nutrient transport facilitation and increased water retention. Thus, ***sludge can replace the expensive fertilizers.***

The quantity of organic materials and the pathogens must be reduced before the sludge is applied on the land. A high content of organic matter will result in offensive odours while the pathogens spread diseases. There are in fact regulatory requirements to control pathogens of sludge by various means.

Distribution and marketing

Distribution and marketing of sludge for beneficial purposes is gaining importance in recent years. It is estimated that about 10–20% of the total sludge produced is utilized in this fashion. The marketed sludge is used as substitution for topsoil and peat on parks, lawns, golf courses and in ornamental and vegetable gardens.

There are regulatory requirements to reduce the pathogenic organisms for distribution and marketing of sludge.

LANDFILLING

Landfilling is a method for the ***final disposal of sludge*** that is not useful any more. The ***sanitary landfill method*** is most suitable for the disposal of solid domestic wastes. This involves a low-cost ***anaerobic technology***. In this method, the sludge or solid wastes are deposited in low-lying and low value sites. The deposition is done almost daily and the deposits are covered with a layer of soil (***Fig. 58.4***). With the coverage of the new waste deposits, nuisance conditions such as bad odours and flies are minimized.

It is desirable that the sludge is dewatered so that its transportation becomes easy. Further, the generation of ***leachate*** (***liquid that percolates out due to leaching***) is minimal from a dewatered sludge.

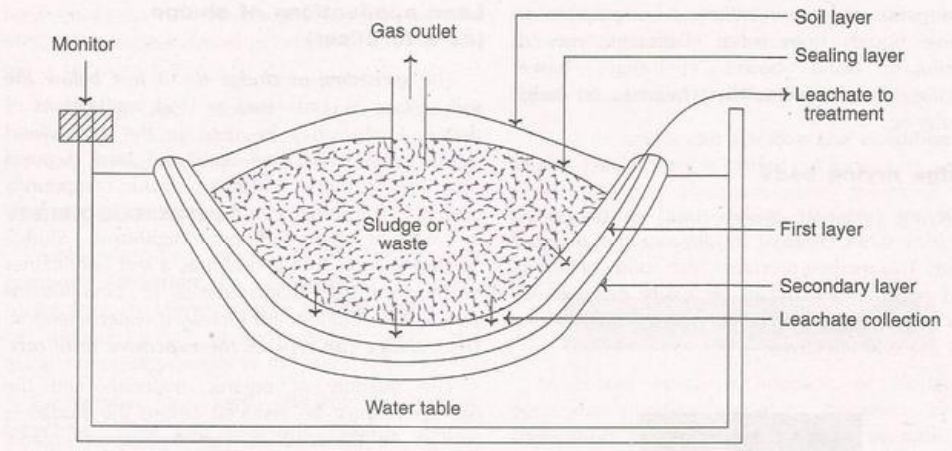


Fig. 58.4 : A diagrammatic representation of a landfill.

At lease two impermeable layers are built below the landfill to prevent the leakage of leachate to the surrounding lands. The accumulated leachate can be taken out and treated by appropriate methods.

The complete filling of landfills may take several months or even years, depending on the size of the site and the quantity of waste being deposited. Landfills can be used for the ***generation of methane gas*** for commercial use. However, methane production usually commences several months after the landfill is completely filled.

In some countries, there are strict regulations to use landfills for sewage disposal. These include the air- and water tight sites to protect the environment.

LAGOONING

A lagoon is a ***shallow lake*** (or earth basin) ***usually located near a river or a sea***. Lagooning (disposal of sludge into lagoons) is a convenient method of sludge disposal if the treatment plant is located at a remote place. In lagooning, the sludge is stabilized by anaerobic and aerobic decomposition which is accompanied by release of objectionable odours. For this reason, lagoons should be located away from dwelling areas and high ways to avoid nuisance conditions.

The stabilized solids of the sludge settle to the bottom of lagoon and accumulate. The sludges can be stored indefinitely in lagoons or may be removed periodically.

SEPTAGE AND SEPTAGE DISPOSAL

Septage is a ***combination of sludge, scum and liquid coming out from a septic tank*** (Refer Chapter 57). It must be disposed under controlled conditions to avoid environmental pollution.

The most commonly used methods of septage disposal are listed below.

- ***Land application*** (surface or subsurface).
- ***Co-treatment with waste water*** (biological or chemical treatment processes).
- ***Co-disposal with solid wastes*** (composting and landfilling).
- ***Independent processing facilities*** (composting, biological treatment, chemical oxidation, lime stabilization).

For more details on these treatment processes, the reader must refer the preceeding pages.

TREATMENT AND DISPOSAL OF SOLID WASTES

Solid wastes are mostly being treated by incineration and landfilling, although these methods have certain limitations.

Incineration : This requires costly equipment and high power consumption. Incinerators do not allow recovery of any useful materials. Further, it is frequently associated with environmental pollution.

Landfilling : The various aspects of landfilling have been described (Refer p. 715). The major limitation of landfills is the problem of leachates and gas emissions which pollute the environment. Further, they are not efficient producers of ***biogas***. It is not possible to recycle the reusable products (paper, plastic, construction materials etc.) in landfilling.

SEPARATION AND COMPOSTING PLANTS

The industrial and municipal solid wastes can be effectively treated by separation, followed by composting. In fact, huge plants are constructed to serve the dual purposes.

Separation

It is possible to recover several useful materials from the solid wastes by employing physical processes.

- Plastic materials for reuse.
- Sand and gravel for construction purposes.
- Paper and cardboard for use in paper industry.
- Iron and aluminium for metallurgy.

After the separation of the important reusable materials, the left out in the solid wastes is mainly the ***biodegradable organic matter.***

Composting

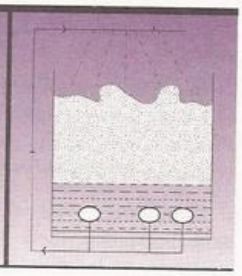
A good account on the principles and processes of composting have already been described (See p. 711). The solid wastes are mostly composted by anaerobic process and to a lesser extent by aerobic means.

Dry anaerobic composting (DRANCO) process

In recent years, some companies have developed specific anaerobic digesters for composting solid wastes. The designing of digester is mainly based on the solid content of the waste, the temperature (from 35°C to 55°C) and the number of stages (1 or 2).

Dry anaerobic composting process is a commercially designed anaerobic digester. It is employed for composting of high solid (200–400 g/l) wastes at thermophilic temperature (around 55°C) by using a single stage reactor. The main advantage of DRANCO process is the ***high rate of composting under controlled conditions***. This is evident from the fact that the composting of solids can be completed in about two weeks time. This is in contrast to a landfill which takes several years, sometimes even decades (10–30 years)!

59 Biodegradation and Bioremediation



Biodegradation or biological degradation is the ***phenomenon of biological transformation of organic compounds*** by ***living organisms, particularly the microorganisms***. Biodegradation basically involves the conversion of complex organic molecules to simpler (and mostly non-toxic) ones. The term ***biotransformation*** is used for ***incomplete biodegradation*** of organic compounds involving one or a few reactions. Biotransformation is employed for the synthesis of commercially important products by microorganisms. For details on biotransformation, refer Chapter 22.

Bioremediation refers to the ***process of using microorganisms to remove the environmental pollutants*** i.e. the toxic wastes found in soil, water, air etc. The ***microbes serve as scavengers*** in bioremediation. The removal of organic wastes by microbes for ***environmental clean-up*** is the essence of bioremediation. The other names used (by some authors) for bioremediation are ***biotreatment***, ***bioreclamation*** and ***biorestoration***.

It is rather difficult to show any distinction between biodegradation and bioremediation, hence they are considered together in this chapter. Further, in biotechnology, most of the reactions of biodegradation/bioremediation involve xenobiotics. Thus, this chapter may be appropriately considered as microbial degradation of xenobiotics. For the bioremediation of metal pollutants, the reader must Refer Chapter 32).

Xenobiotics

Xenobiotics (xenos-foregin) broadly refer to the unnatural, foreign and synthetic chemicals such as pesticides, herbicides, refrigerants, solvents and other organic compounds. Microbial degradation of xenobiotics assumes significance, since it provides an effective and economic means of disposing of toxic chemicals, particularly the environmental pollutants.

PSEUDOMONAS — THE PREDOMINANT MICROORGANISM FOR BIOREMEDIATION

Members of the genus *Pseudomonas* (a soil microorganism) are the most predominant microorganisms that degrade xenobiotics. Different strains of *Pseudomonas*, that are capable of detoxifying more than 100 organic compounds, have been identified. The examples of organic compounds are several hydrocarbons, phenols, organophosphates, polychlorinated biphenyls (PCBs) and polycylic aromatics and naphthalene.

About 40–50 microbial strains of micro-organisms, capable of degrading xenobiotics have been isolated. Besides *Pseudomonas*, other good examples are *Mycobacterium*, *Alcaligenes*, and *Nocardia*. A selected list of microorganisms and the xenobiotics degraded is given in ***Table 59.1***.

Consortia of microorganisms for biodegradation : A particular strain of microorganism may degrade one or more compounds. Sometimes, for the

TABLE 59.1 A selected list of microorganisms and the pollutants (xenobiotics) that are degraded by bioremediation

Microorganism	*Pollutant chemicals*
Pseudomonas sp	Aliphatic and aromatic hydrocarbons—alkylaminoxides, alkylammonium benzene, naphthalene, anthracene xylene, toluene, polychlorinated biphenyls (PCBs), malathion, parathion, organophosphates.
Mycobacterium sp	Benzene, branched hydrocarbons, cycloparaffins
Alcaligenes sp	Polychlorinated biphenyls, alkyl benzene, halogenated hydrocarbons.
Nocardia sp	Naphthalene, alkylbenzenes, phenoxyacetate.
Arthrobacter sp	Benzene, polycyclic aromatics, phenoxyacetate, pentachlorophenol.
Corynebacterium sp	Halogenated hydrocarbons, phenoxyacetate.
Bacillus sp	Long chain alkanes, phenylurea.
Candida sp	Polychlorinated biphenyls
Aspergillus sp	Phenols
Xanthomonas sp	Polycyclic hydrocarbons
Streptomyces sp	Halogenated hydrocarbons, phenoxyacetate.
Fusarium sp	Propanil
Cunninghamella sp	Polycyclic aromatics, polychlorinated biphenyls.

degradation of a single compound, the synergetic action of a few microorganisms (i.e. a consortium or cocktail of microbes) may be more efficient. For instance, the insecticide parathion is more efficiently degraded by the combined action of *Pseudomonas aeruginosa* and *Psudomonas stulzeri*.

Co-metabolism in biodegradation : In general, the metabolism (breakdown) of xenobiotics is not associated with any advantage to the microorganism. That is the pollutant chemical cannot serve as a source of carbon or energy for the organism. The term co-metabolism is often used to indicate the ***non-beneficial (to the microorganism) biochemical pathways concerned with the biodegradation of xenobiotics***. However, co-metabolism depends on the presence of a suitable substrate for the microorganism. Such compounds are referred to ***co-substrates***.

Factors affecting biodegradation

Several factors influence biodegradation. These include the chemical nature of the xenobiotic, the capability of the individual microorganism, nutrient and O_2 supply, temperature, pH and redox potential. Among these, ***the chemical nature of the substrate that has to be degraded is very important***. Some of the relevant features are given hereunder.

- In general, aliphatic compounds are more easily degraded than aromatic ones.
- Presence of cyclic ring structures and length chains or branches decrease the efficiency of biodegradation.
- Water soluble compounds are more easily degraded.
- Molecular orientation of aromatic compounds influences biodegradation i.e. ortho > para > meta.
- The presence of halogens (in aromatic compounds) inhibits biodegradation.

Besides the factors listed above, there are two recent developments to enhance the biodegradation by microorganisms.

Biostimulation : This is a process by which the microbial activity can be enhanced by ***increased supply of nutrients*** or by addition of certain ***stimulating agents*** (electron acceptors, surfactants).

Bioaugmentation : It is possible to increase biodegradation ***through manipulation of genes***. More details on this genetic manipulation i.e. genetically engineered microorganisms (GEMs), are described later. Bioaugmentation can also be achieved by employing a consortium of micro-organisms.

Enzyme systems for biodegradation

Several enzyme systems (with independent enzymes that work together) are in existence in the microorganisms for the degradation of xenobiotics.

TABLE 59.2 A selected list of xenobiotics and the plasmids containing genes (in *Plasmodium*) for biodegradation

Xenobiotic	*Name of plasmid in* Pseudomonas
Naphthalene	NAH
Xylene	XYL
Xylene and toluene	TOL, pWWO, XYL-K
Salicylate	SAL
Camphor	CAM
3-Chlorobenzene	pAC25

The genes coding for the enzymes of biodegradative pathways may be present in the chromosomal DNA or more frequently on the plasmids. In certain microorganisms, the genes of both chromosome and plasmid contribute for the enzymes of biodegradation.

As already described, the microorganism *Pseudomonas* occupies a special place in biodegradation. A selected list of xenobiotics and the plasmids containing the genes for their degradation is given in ***Table 59.2***.

RECALCITRANT XENOBIOTICS

There are certain compounds that ***do not easily undergo biodegradation*** and therefore ***persist in the environment for a long period*** (sometimes in years). They are labeled as recalcitrant. There may be several reasons for the resistance of xenobiotics to microbial degradation.

- They may chemically and biologically inert (highly stable).
- Lack of enzyme system in the microorganisms for biodegradation.
- They cannot enter the microorganisms being large molecules or lack of transport systems.
- The compounds may be highly toxic or result in the formation highly toxic products that kill microorganisms.

There are a large number of racalcitrant xenobiotic compounds e.g. chloroform, freons, insecticides (DDT, lindane), herbicides (dalapon) and synthetic polymers (plastics e.g. polystyrene, polyethylene, polyvinyl chlorine).

It takes about 4-5 years for the degradation of DDT (75–100%) in the soil. A group of microorganisms (*Aspergillus flavus, Mucor aternans, Fusarium oxysporum* and *Trichoderma viride*) are associated with the slow biodegradation of DDT.

Biomagnification : The phenomenon of ***progressive increase in the concentration of a xenobiotic*** compound, as the substance is passed ***through the food chain*** is referred to as biomagnification or ***bioaccumulation***. For instance, the insecticide DDT is absorbed repeatedly by plants and microorganism. When they are eaten by fish and birds, this pesticide being recalcitrant, accumulates, and enters the food chain. Thus, DDT may find its entry into various animals, including man. DDT affects the nervous systems, and it has been banned in some countries.

TYPES OF BIOREMEDIATION

The most important aspect of environmental biotechnology is the effective management of hazardous and toxic pollutants (xenobiotics) by bioremediation. The environmental clean-up process through bioremediation can be achieved in two ways—*in situ* and *ex situ* bioremediation.

In situ bioremediation

In situ bioremediation involves a direct approach for the microbial degradation of xenobiotics at the sites of pollution (soil, ground water). Addition of adequate quantities of nutrients at the sites promotes microbial growth. When these microorganisms are exposed to xenobiotics (pollutants), they develop metabolic ability to degrade them. The growth of the microorganisms and their, ability to bring out biodegradation are dependent on the supply of essential nutrients (nitrogen, phosphorus etc.). *In situ* bioremediation has been successfully applied for clean-up of oil spillages, beaches etc. There are two types of *in situ* bioremediation-intrinsic and engineered.

Intrinsic bioremediation : The inherent metabolic ability of the microorganisms to degrade certain pollutants is the intrinsic bioremediation. In fact, the microorganisms can be tested in the laboratory for their natural capability of biodegradation and appropriately utilized.

Engineered *in situ* bioremediation : The inherent ability of the microorganisms for bioremediation is

generally slow and limited. However, by using suitable physico-chemical means (good nutrient and O_2 supply, addition of electron acceptors, optimal temperature), the bioremediation process can be engineered for more efficient degradation of pollutants.

Advantages of *in situ* bioremediation

1. Cost-effective, with minimal exposure to public or site personnel.
2. Sites of bioremediation remain minimally disrupted.

Disadvantages of *in situ* bioremediation

1. Very time consuming process.
2. Sites are directly exposed to environmental factors (temperature, O_2 supply etc.).
3. Microbial degrading ability varies seasonally.

Ex situ bioremediation

The waste or toxic materials can be collected from the polluted sites and the bioremediation with the requisite microorganisms (frequently a consortium of organisms) can be carried out at designed places. This process is certainly an improvement over *in situ* bioremediation, and has been successfully used at some places.

Advantages of *ex situ* bioremediation

1. Better controlled and more efficient process.
2. Process can be improved by enrichment with desired microorganisms.
3. Time required in short.

Disadvantages of *ex situ* bioremediation

1. Very costly process.
2. Sites of pollution are highly disturbed.
3. There may be disposal problem after the process is complete.

METABOLIC EFFECTS OF MICROORGANISMS ON XENOBIOTICS

Although it is the intention of the biotechnologist to degrade the xenobiotics by microorganisms to the advantage of environment and ecosystem, it is not always possible. This is evident from the different types of metabolic effects as shown below.

Detoxification : This process involves the microbial conversion of toxic compound to a non-toxic one. Biodegradation involving detoxification is highly advantageous to the environment and population.

Activation : Certain xenobiotics which are not toxic or less toxic may be converted to toxic or more toxic products. This is dangerous.

Degradation : The complex compounds are degraded to simpler products which are generally harmless.

Conjugation : The process of conjugation may involve the conversion of xenobiotics to more complex compounds. This is however, not very common.

TYPES OF REACTIONS IN BIOREMEDIATION

Microbial degradation of organic compounds primarily involves aerobic, anaerobic and sequential degradation.

Aerobic bioremediation

Aerobic biodegradation involves the utilization of O_2 for the oxidation of organic compounds. These compounds may serve as substrates for the supply of carbon and energy to the microorganisms. Two types of enzymes namely monooxygenases and dioxygenases are involved in aerobic biodegradation. Monooxygenases can act on both aliphatic and aromatic compounds while dioxygenases oxidize aliphatic compounds.

Anaerobic bioremediation

Anaerobic biodegradation does not require O_2 supply. The growth of anaerobic microorganisms (mostly found in solids and sediments), and consequently the degradation processes are slow. However, anaerobic biodegradation is cost-effective, since the need for continuous O_2 supply is not there. Some of the important anaerobic reactions and examples of organic compounds degraded are listed below.

Hydrogenation and dehydrogenation — benzoate, phenol, catechol.

Dehalogenation — Polychlorinated biphenyls (PCBs), chlorinated ethylenes. The term ***dechlorination*** is frequently used for dehalogenation of chlorinated compounds.

Carboxylation and decarboxylation — toluene, cresol and benzoate.

Sequential bioremediation

In the degradation of several xenobiotics, both aerobic and anaerobic processes are involved. This is often an effective way of reducing the toxicity of a pollutant. For instance, tetrachloromethane and tetrachloroethane undergo sequential degradation.

BIODEGRADATION OF HYDROCARBONS

Hydrocarbon are mainly the pollutants from oil refineries and oil spills. These pollutants can be degraded by a consortium or cocktail of microorganisms e.g. *Pseudomonas, Corynebacterium, Arthrobacter, Mycobacterium* and *Nocardia.*

Biodegradation of aliphatic hydrocarbons

The uptake of aliphatic hydrocarbons is a slow process due to their low solubility in aqueous medium. Both aerobic and anaerobic processes are operative for the degradation of aliphatic hydrocarbons. For instance, unsaturated hydrocarbons are degraded in both anaerobic and aerobic environments, while saturated ones are degraded by aerobic process.

Some aliphatic hydrocarbons which are reclacitrant to aerobic process are effectively degraded in anaerobic environment e.g. chlorinated aliphatic compounds (carbon tetrachloride, methyl chloride, vinyl chloride).

Biodegradation of aromatic hydrocarbons

Microbial degradation of aromatic hydrocarbons occurs through aerobic and anaerobic processes. The most important microorganism that participates in these processes is *Pseudomonas.*

The biodegradation of aromatic compounds basically involves the following sequence of reactions.

1. Removal of the side chains.
2. Opening of the benzene ring.

Most of the non-halogenated aromatic compounds undergo a series of reactions to produce ***catechol*** or ***protocatechuate***. The bioremediation of toluene, L-mandelate, benzoate, benzene, phenol, anthracene, naphthalene, phenanthrene and salicylate to produce catechol is shown in ***Fig. 59.1***. Likewise, ***Fig. 59.2***, depicts the bioremediation of quinate, p-hydroxymandelate, p-hydroxybenzoyl formate, p-toluate, benzoate and vanillate to produce protocatechuate. Catechol and protocatechuate can undergo oxidative cleavage pathways. In ***ortho-cleavage pathway***, catechol and protocatechuate form acetyl CoA (***Fig. 59.3***), while in ***metacleavage pathway*** (***Fig. 59.4***), they are converted to pyruvate and acetaldehyde. The degraded products of catechol and protocatechuate are readily metabolised by almost all the organisms.

BIODEGRADATION OF PESTICIDES AND HERBICIDES

Pesticides and herbicides are regularly used to contain various plant diseases and improve the crop yield. In fact, they are a part of the modern agriculture, and have significantly contributed to ***green revolution***. The common herbicides and pesticides are propanil (anilide), propham (carbamate), atrazine (triazine), picloram (pyridine), dichlorodiphenyl trichloroethane (DDT) monochloroacetate (MCA), monochloropropionate (MCPA) and glyphosate (organophosphate).

Most of the pesticides and herbicides are toxic and are recalcitrant (resistant to biodegradation). Some of them are ***surfactants*** (active on the surface) and retained on the surface of leaves.

Biodegradation of halogenated aromatic compounds

Most commonly used herbicides and pesticides are aromatic halogenated (predominantly chlorinated) compounds. The biodegradative pathways of halogenated compounds are comparable with that described for the degradation of non-halogenated aromatic compounds (***Figs. 59.1, 59.2, 59.3*** and ***59.4***). The rate of degradation of halogenated compounds is inversely related to the number of halogen atoms that are originally present on the target molecule i.e. compounds with higher number of halogens are less readily degraded.

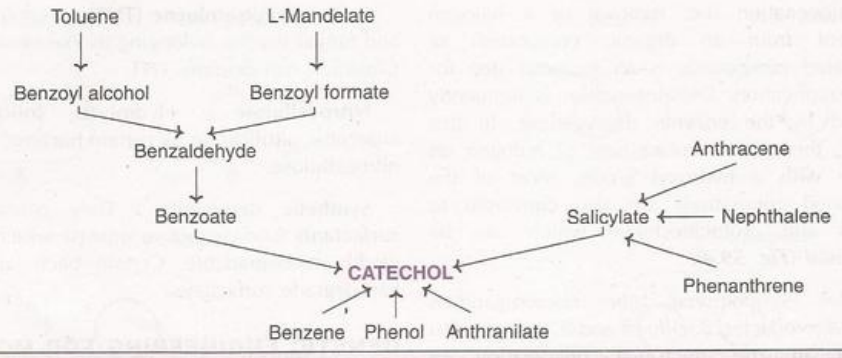


Fig. 59.1 : Bioremediation of certain aromatic compounds by bacteria to produce catechol.

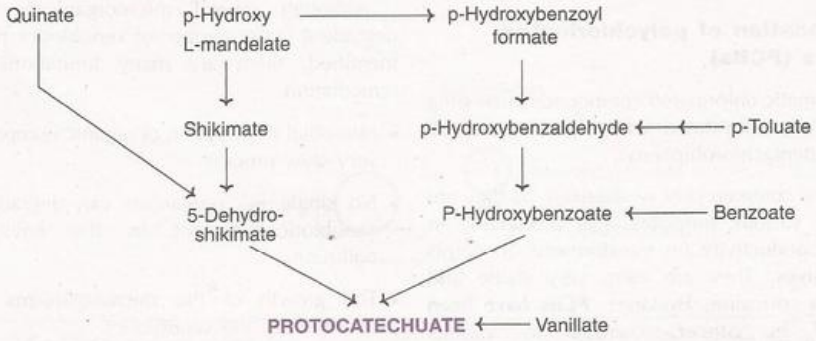


Fig. 59.2 : Bioremediation of certain organic compounds by bacteria to produce protocatechuate.

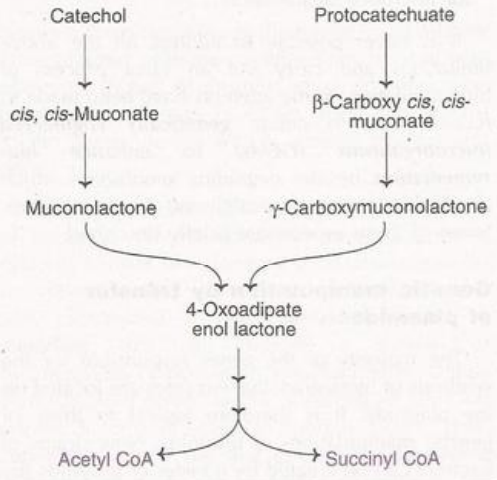


*Fig. 59.3 : Conversion of catechol and protocatechuate to acetyl CoA and succinate by **ortho–cleavage pathway**.*

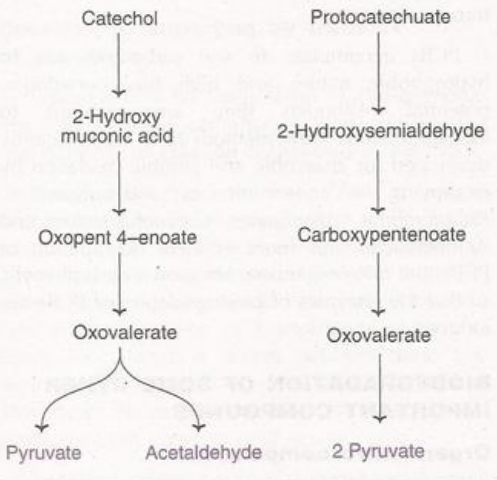


*Fig. 59.4 : Conversion of catechol and protocatechuate to pyruvate and acetaldehyde by **meta–cleavage pathway.***

Dehalogenation (i.e. removal of a halogen substituent from an organic compound) of halogenated compounds is an essential step for their detoxification. Dehalogenation is frequently catalysed by the enzyme dioxygenase. In this reaction, there is a replacement of halogen on benzene with a hydroxyl group. Most of the halogenated compounds are also converted to catechol and protocatechuate which can be metabolised (***Fig. 59.4***).

Besides *Pseudomonas,* other microorganisms such as *Azotobacter, Bacilluefs* and *E. coli* are also involved in the microbial degradation of halogenated aromatic compounds.

Biodegradation of polychlorinated biphenyls (PCBs)

The aromatic chlorinated compounds possessing biphenyl ring (substituted with chlorine) are the PCBs e.g. pentachlorobiphenyl.

PCBs are commercially synthesized, as they are useful for various purposes — as pesticides, in electrical conductivity (in transformers), in paints and adhesives. They are inert, very stable and resistant to corrosion. However, ***PCBs have been implicated in cancer,*** damage to various organs and impaired reproductive function. Their commercial use has been restricted in recent years, and are now used mostly in electrical transformers.

PCBs accumulate in soil sediments due to hydrophobic nature and high bioaccumulation potential. Although they are resistant to biodegradation, some methods have been recently developed for anaerobic and aerobic oxidation by employing a consortium of microorganisms. *Pseudomonas, Alkaligenes, Corynebacterium* and *Acinetobacter.* For more efficient degradation of PCBs, the microorganisms are grown on biphenyls, so that the enzymes of biodegradation of PCBs are induced.

BIODEGRADATION OF SOME OTHER IMPORTANT COMPOUNDS

Organo-nitro compounds

Some of the toxic organo-nitro compounds can be degraded by microorganisms for their detoxification.

2, 4, 6-Trinitrotoluene (TNT) : Certain bacteria and fungal species belonging to *Pseudomonas* and *Clostrium* can detoxify TNT.

Nitrocellulose : Hydrolysis, followed by anaerobic nitrification by certain bacteria, degrades nitrocellulose.

Synthetic detergents : They contain some ***surfactants*** (surface active agents) which are not readily biodegradable. Certain bacterial plasmid can degrade surfactants.

GENETIC ENGINEERING FOR MORE EFFICIENT BIOREMEDIATION

Although several microorganisms that can degrade a large number of xenobiotics have been identified, there are many limitations in bio-remediation.

- Microbial degradation of organic compounds is a very slow process.
- No single microorganism can degrade all the xenobiotics present in the environmental pollution.
- The growth of the microorganisms may be inhibited by the xenobiotics.
- Certain xenobiotics get adsorbed on to the particulate matter of soil and become unavailable for microbial degradation.

It is never possible to address all the above limitations and carry out an ideal process of bioremediation. Some attempts have been made in recent years to create ***genetically engineered microorganisms (GEMs) to enhance bio-remediation,*** besides degrading xenobiotics which are highly resistant (recalcitrant) for breakdown. Some of these aspects are briefly described.

Genetic manipulation by transfer of plasmids

The majority of the genes responsible for the synthesis of biodegradative enzymes are located on the plasmids. It is therefore logical to think of genetic manipulations of plasmids. New strains of bacteria can be created by transfer of plasmids (by conjugation) carrying genes for different degradative pathways. If the two plasmids contain homologous regions of DNA, recombination occurs

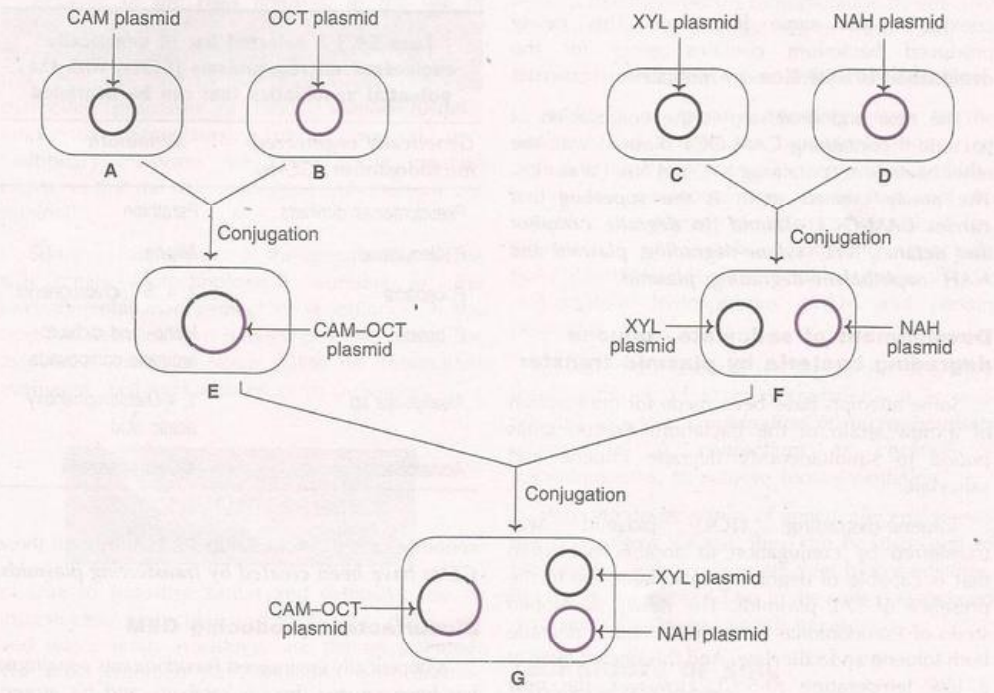


Fig. 59.5 : Creation of the superbug by transfer of plasmids (A, B, C, D, E, F and G are the different strains of bacteria containing the plasmids shown. Strain G is the superbug.)

between them, resulting in the formation of a larger fused plasmid (with the combined functions of both plasmids). In case of plasmids which do not possess homologous regions of DNA, they can coexist in the bacterium (to which plasmid transfer was done).

The first successful development of a new strain of bacterium (*Psuedomonas*) by manipulations of plasmid transfer was done by Chakrabarty and his co-workers in 1970s. They used different plasmids and constructed a ***new bacterium*** called as ***superbug***, that can degrade a number of hydrocarbons of petroleum simultaneously. United States granted patent to this superbug in 1981 (as per the directive of American Supreme Court). Thus, ***superbug*** became ***the first genetically engineered microorganism to be patented***. Superbug has played a significant role in the development of biotechnology industry, although it has not been used for large scale degradation of oil spills.

Creation of superbug by transfer of plasmids

Superbug is a bacterial strain of *Pseudomonas* that can degrade camphor, octane, xylene and naphthalene. Its creation is depicted in ***Fig. 59.5***.

The bacterium containing CAM (camphor-degrading) plasmid was conjugated with another bacterium with OCT (octane-degrading) plasmid. These plasmids are not compatible and therefore, cannot coexist in the same bacterium. However, due to the presence of homologous regions of DNA, recombination occurs between these two plasmids resulting in a single CAM-OCT plasmid. This new bacterium possesses the degradative genes for both camphor and octane.

Another bacterium with XYL (xylene-degrading) plasmid is conjugated with NAH (naphthalene-degrading) plasmid containing bacterium. XYL and NAH plasmids are compatible and therefore can

coexist in the same bacterium. This newly produced bacterium contains genes for the degradation of xylene and naphthalene.

The next and final step is the conjugation of bacterium containing CAM-OCT plasmid with the other bacterium containing XYL and NAH plasmids. The newly created strain is the ***superbug*** that ***carries CAM-OCT plasmid*** (***to degrade camphor and octane***), ***XYL*** (***xylene-degrading***) ***plasmid and NAH*** (***naphthalene-degrading***) ***plasmid***.

Development of salicylate—toluene degrading bacteria by plasmid transfer

Some attempts have been made for the creation of a new strain of the bacterium *Pseudomonas putida* to simultaneously degrade toluene and salicylate.

Toluene-degrading (TOL) plasmid was transferred by conjugation to another bacterium that is capable of degrading salicylate (due to the presence of SAL plasmid). The newly developed strain of *Pseudomonas* can simultaneously degrade both toluene and salicylate. And this occurs even at a low temperature (0-5°C). However, the new bacterium is not in regular use, as more research is being conducted on its merits and demerits.

Genetic manipulation by gene alteration

Work is in progress to manipulate the genes for more efficient biodegradation. The plasmid pWWO of *Pseudomonas* codes for 12 different enzymes responsible for the meta-cleavage pathway (for the conversion of catechol and protocatechuate to pyruvate and acetaldehyde, See ***Fig. 59.4***) for degradation of certain aromatic compounds. Some success has been reported to alter the genes of plasmid pWWO for more efficient degradation of toluene and xylene.

GENETICALLY ENGINEERED MICROORGANISMS (GEMs) IN BIOREMEDIATION

Superbug (described already) is the first genetically engineered microorganism. Several workers worldover have been working for the creation of GEMs, specifically designed for the detoxification of xenobiotics. A selected list of GEMs with a potential for the degradation of xenobiotics is given in ***Table 59.3***. Almost all these ***GEMs have been created by transferring plasmids***.

TABLE 59.3 A selected list of genetically engineered microorganisms (GEMs) with the potential xenobiotics that can be degraded

Genetically engineered microorganism (GEMs)	*Xenobiotic*
Pseudomonas diminuta	Parathion
P. oleovorans	Alkane
P. cepacia	2, 4, 5-Trichlorophenol
P. putida	Mono- and dichloro-aromatic compounds
Alcaligenes sp	2, 4-Dichlorophenoxy acetic acid
Acinetobacter sp	4-Chlorobenzene

Biosurfactant producing GEM

A genetically engineered *Pseudomonas aeruginosa* has been created (by Chakarabarty and his group). This new strain can produce a ***glycolipid emulsifier*** (a biosurfactant) which can reduce the surface tension of an oil water interface. The reduced interfacial tension promotes biodegradation of oils.

GEM for degradation of vanillate and SDS

A new strain of *Pseudomonas* sp (strain ATCC 1915) has been developed for the degradation of vanillate (waste product from paper industry) and sodium dodecyl sulfate (SDS, a compound used in detergents).

GEMs and environmental safety

The genetically engineered microorganisms (GEMs) have now become handy tools of biotechnologists. The risks and health hazards associated with the use of GEMs are highly controversial and debatable issues. The fear of the biotechnologists, and even the general public is that the new organism (GEM), once it enters the environment, may disturb the ecological balance and cause harm to the habitat. Some of the GEMs may turn virulant and become ***genetic bombs***, causing great harm to humankind.

Because of the risks involved in the use of GEMs, so far no GEM has been allowed to enter the environmental fields. Thus, the use of GEMs has been confined to the laboratories, and fully controlled processes of biodegradation (usually employing bioreactors). Further, several precautionary measures are taken while creating GEMs, so that the risks associated with their use are minimal.

Some researchers are of the opinion that GEMs will create biotechnological wonders for the environmental management of xenobiotics, in the next few decades. This may be possible only if the associated risks of each GEM is thoroughly evaluated, and fully assured of its biosafety.

BIOREMEDIATION OF CONTAMINATED SOILS AND WASTE LANDS

Due to industrialization and extensive use of insecticides, herbicides and pesticides, the solids and waste lands worldover are getting polluted. The most common pollutants are hydrocarbons, chlorinated solvents, polychlorobiphenyls and metals.

Bioremediation of soils and waste lands by the use of microorganisms is gaining importance in the recent years. In fact, some success has been reported for the detoxification of certain pollutants (e.g. hydrocarbons) in the soil by microorganisms.

Bioremediation of soils can be done by involving two principles-biostimulation and bioaugmentation.

Biostimulation in soil bioremediation

Biostimulation basically involves the ***stimulation of microorganisms already present in the soil,*** by various means. This can be done by many ways.

- Addition of nutrients such as nitrogen and phosphorus.
- Supplementation with co-substrates e.g. methane added to degrade trichloroethylene.
- Addition of surfactants to disperse the hydrophobic compounds in water.

Addition of nutrient and co-substrates promote microbial growth while surfactants expose the hydrophobic molecules. In all these situations, the result is that there occurs biostimulation by effective bioremediation of polluted soil or waste land.

Bioaugmentation in soil bioremediation

Addition of specific microorganisms to the polluted soil constitutes bioaugmentation. The pollutants are very complex molecules and the native soil microorganisms alone may not be capable of degrading them effectively. The examples of such pollutants include polychlorobiphenyls (PCBs), trinitrotoluene (TNT), polyaromatic hydrocarbons (PAHs) and certain pesticides.

Based on the research findings at the laboratory level (with regard to biodegradation), it is now possible to add a ***combination of microorganisms*** referred to as ***consortium or cocktail*** of microorganisms, to achieve bioaugmentation.

With the development of genetically engineered microorganisms (GEMs), they can be also used to bioaugment soils for very efficient bioremediation. But the direct use of GEMs in the soils is associated with several risks and health hazards.

TECHNIQUES OF SOIL BIOREMEDIATION

The most commonly used methods for the bioremediation are soils are *in situ* bioremediation, land farming and slurry phase bioreactors.

In situ bioremediation of soils

In situ bioremediation broadly involves the biological clean-up of soils without excavation. This technique is used for the bioremediation of sub-surfaces of soils, buildings and road ways that are polluted. Sometimes, water (oxygenated) is cycled through the sub-surfaces for increasing the efficiency of microbial degradation. There are two types of *in situ* soil bioremediation techniques-bioventing and phytoremediation.

Bioventing : This is very efficient and cost-effective technique for the bioremediation of petroleum contaminated soils. Bioventing involves ***aerobic biodegradation of pollutants by circulating air through sub-surfaces of soil.*** Although, it takes some years, bioventing can be used for the degradation of soluble paraffins and polyaromatic hydrocarbons. The major limitation of this technique is air circulation which is not always practicable.

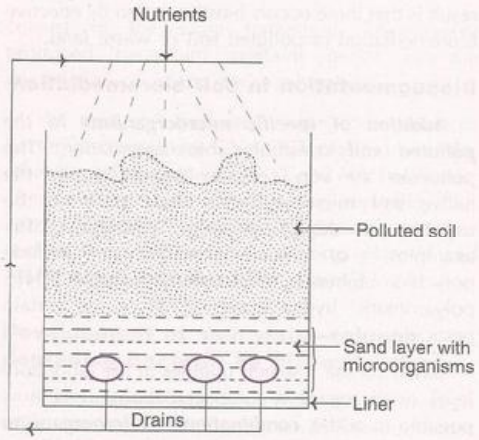


Fig. 59.6 : A diagrammatic representation of landfarming system.

Phytoremediation : Bioremediation by ***use of plants*** constitutes phytoremediation. Specific plants are cultivated at the sites of polluted soil. These plants are capable of stimulating the biodegradation of pollutants in the soil adjacent to roots (rhizosphere). although phytoremediation is a cheap and environmental friendly clean-up process for the biodegradation of soil pollutants, it takes several years.

Landfarming in soil bioremediation

Landfarming is a technique for the bioremediation of hydrocarbon contaminated soils. A diagrammatic representation of land farming system (also referred to as ***solid phase soil reactor***) is depicted in ***Fig. 59.6***.

The soil is excavated, mixed with microorganisms and nutrients and spread out on a liner, just below the polluted soil. The soil has to be regularly ploughed for good mixing and aeration. If the soil is mixed with compost and/or temperature is increased the efficiency of biodegradation increases. Addition of co-substrates, and anaerobic pretreatment of polluted soils also increases the degradation process.

Landfarming has been successfully used for the bioremediation of soils polluted with chloroethane ***b***enzene, ***t***oluene and ***x***ylene. The last three compounds are often referred to as ***BTX aromatics***.

Slurry-phase bioreactors in soil bioremediation

Slurry-phase bioreactors are improved land farming systems. In these cases, the ***excavated polluted soil is subjected to bioremediation under optimally controlled conditions*** in specifically designed bioreactors. Due to a close contact between the xenobiotics and the microorganisms, and the optimal conditions (nutrient supply, temperature, aeration etc.), the degradation is very rapid and efficient. Slurry-phase bioreactors, however, are not suitable for widespread use due to high cost.

BIOREMEDIATION OF GROUND WATER

Environmental pollution also results in the contamination of ground water at several places. The commonly found pollutants are the petroleum hydrocarbons (aliphatic, aromatic, cyclic and substituted molecules). Bioremediation of ground water can be carried out by two methods pump- and -treat technique and biofencing technique.

Pump- and -treat technique for bioremediation of ground water

Bioremediation of underground water by pump- and -treat technology is mostly based on physico-chemical principles to remove the pollutants. The treatment units are set up above the ground. Strip columns and activated carbon filters can remove most of the ground water pollutants. Treated water is recycled through injection well several times so that the pollutants are effectively removed.

For removal of certain organic pollutants, biological reactors (bioreactors) have to be installed (***Fig. 59.7A***). For instance, for the biodegradation of tetrachloroethane, a bioreactor with granular methogenic sludge is found to be effective. In recent years, bioreactors with both aerobic and anaerobic bacteria have been developed for better bioremediation of highly polluted ground waters.

It is however, not possible to achieve good clean up of ground water by pump- and -treat technology, for various reasons (sub-surface heterogenecities, strongly adsorbed compounds, low permeability of pollutants etc.).

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